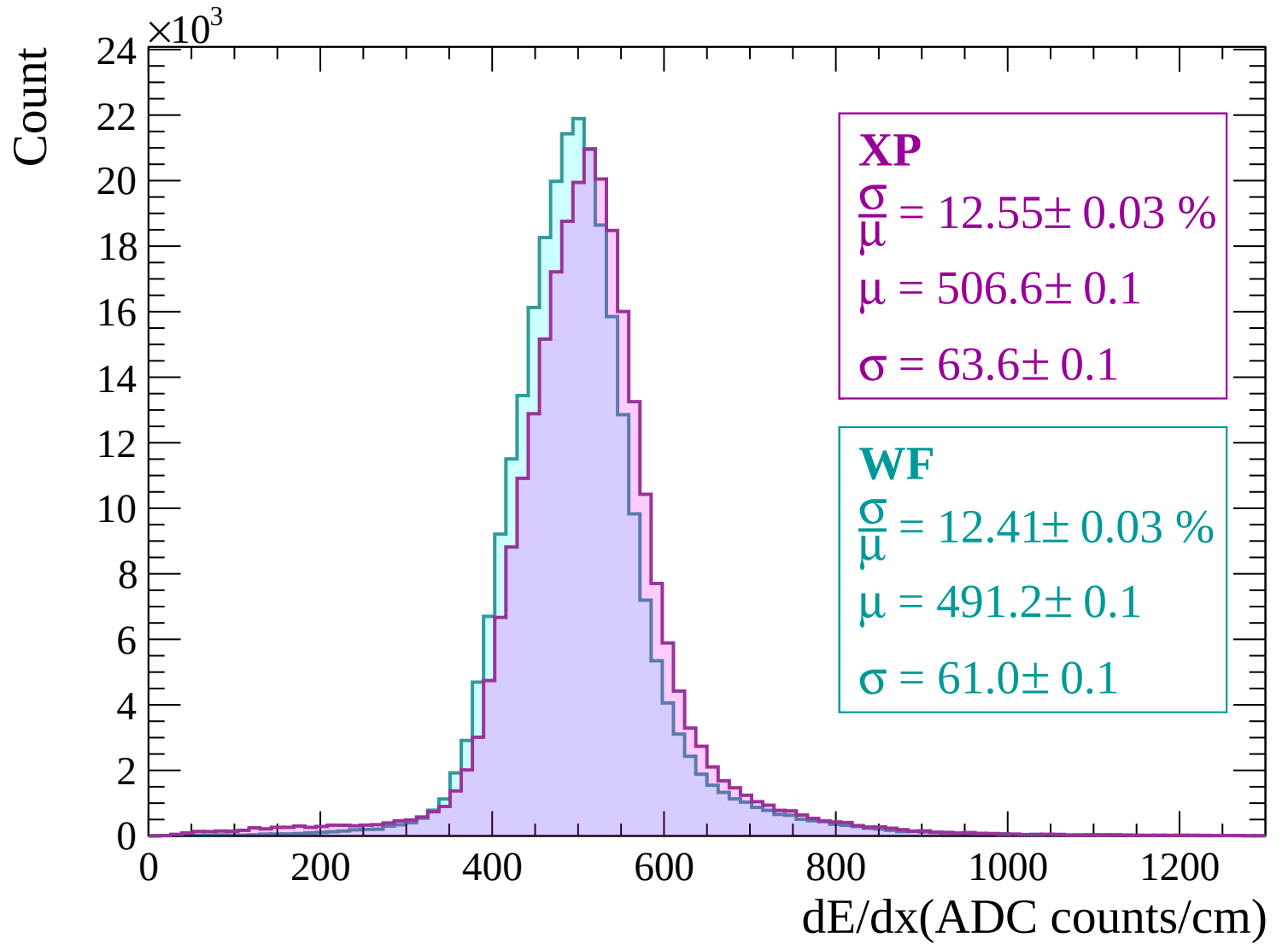
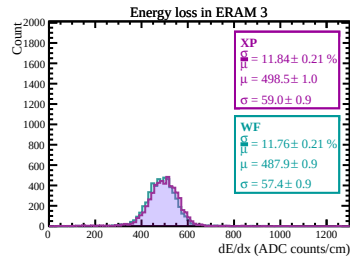
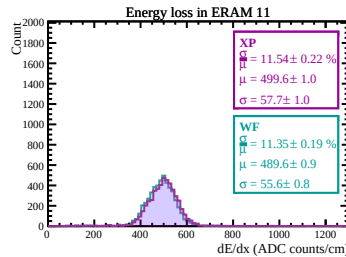
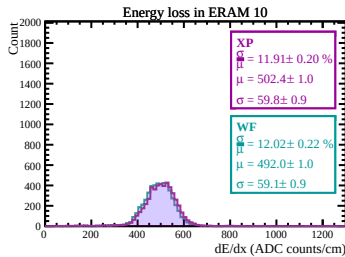
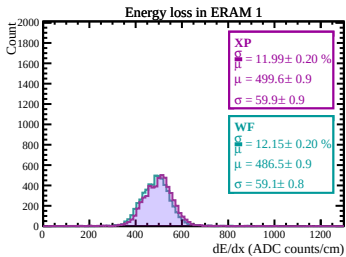
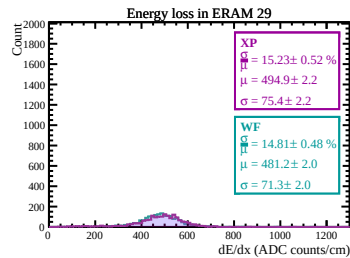
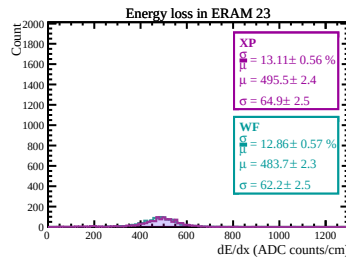
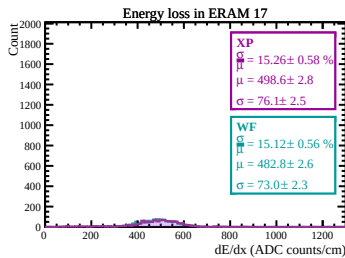
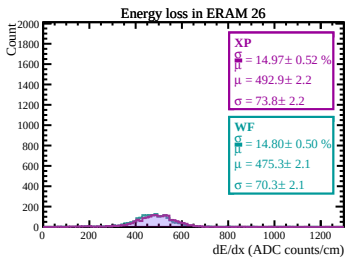
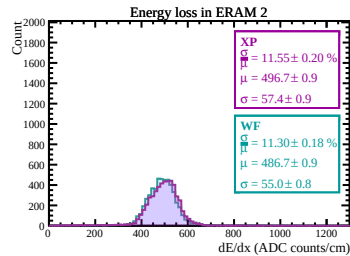
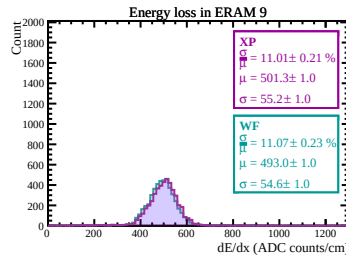
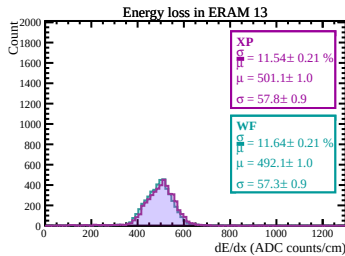
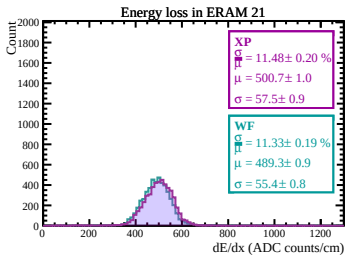
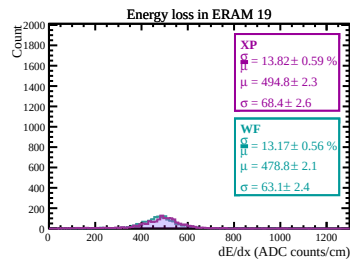
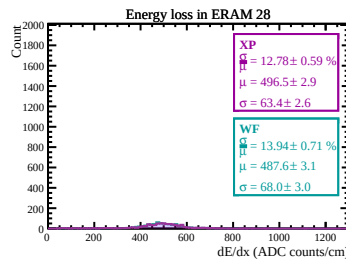
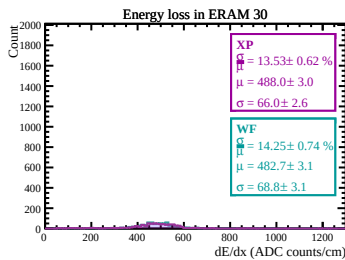
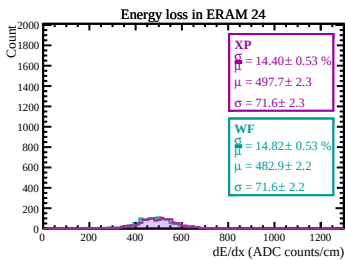
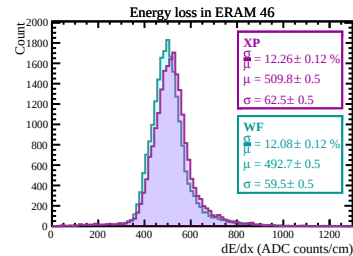
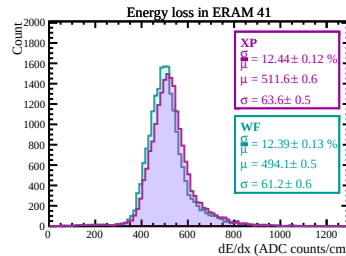
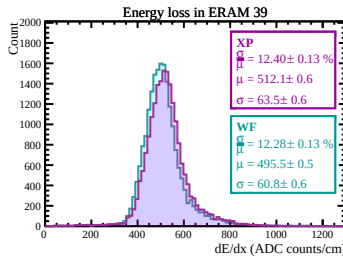
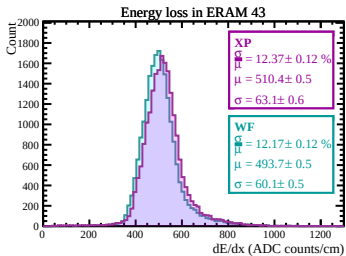
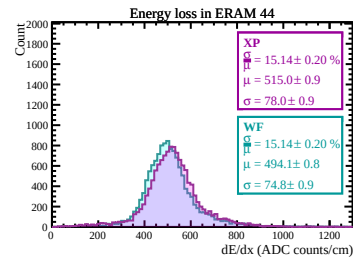
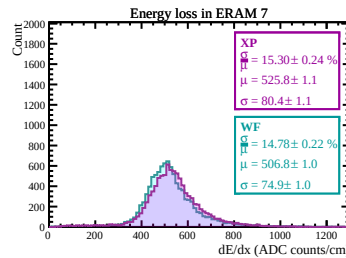
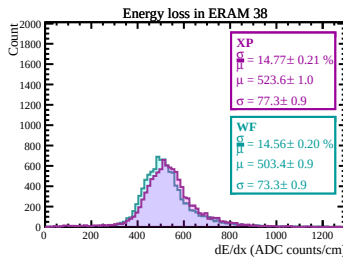
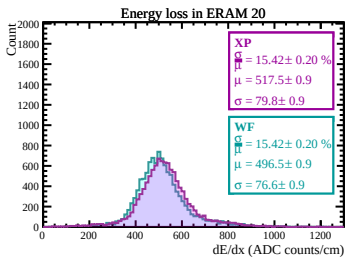
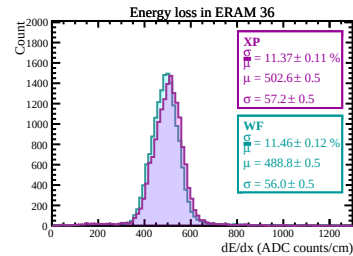
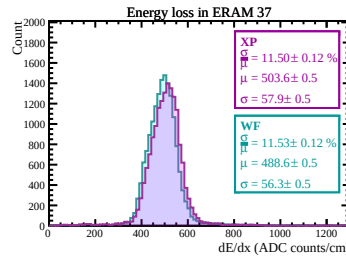
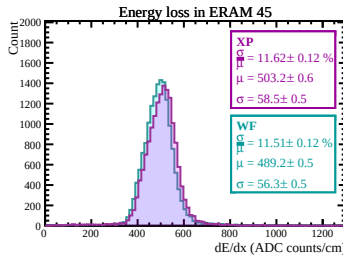
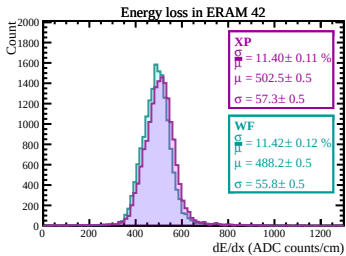
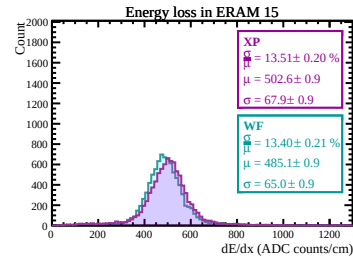
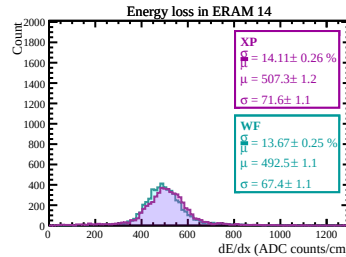
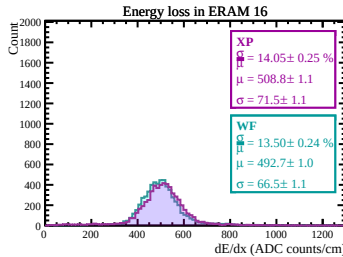
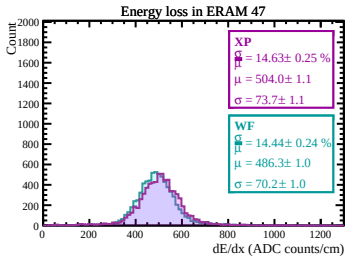
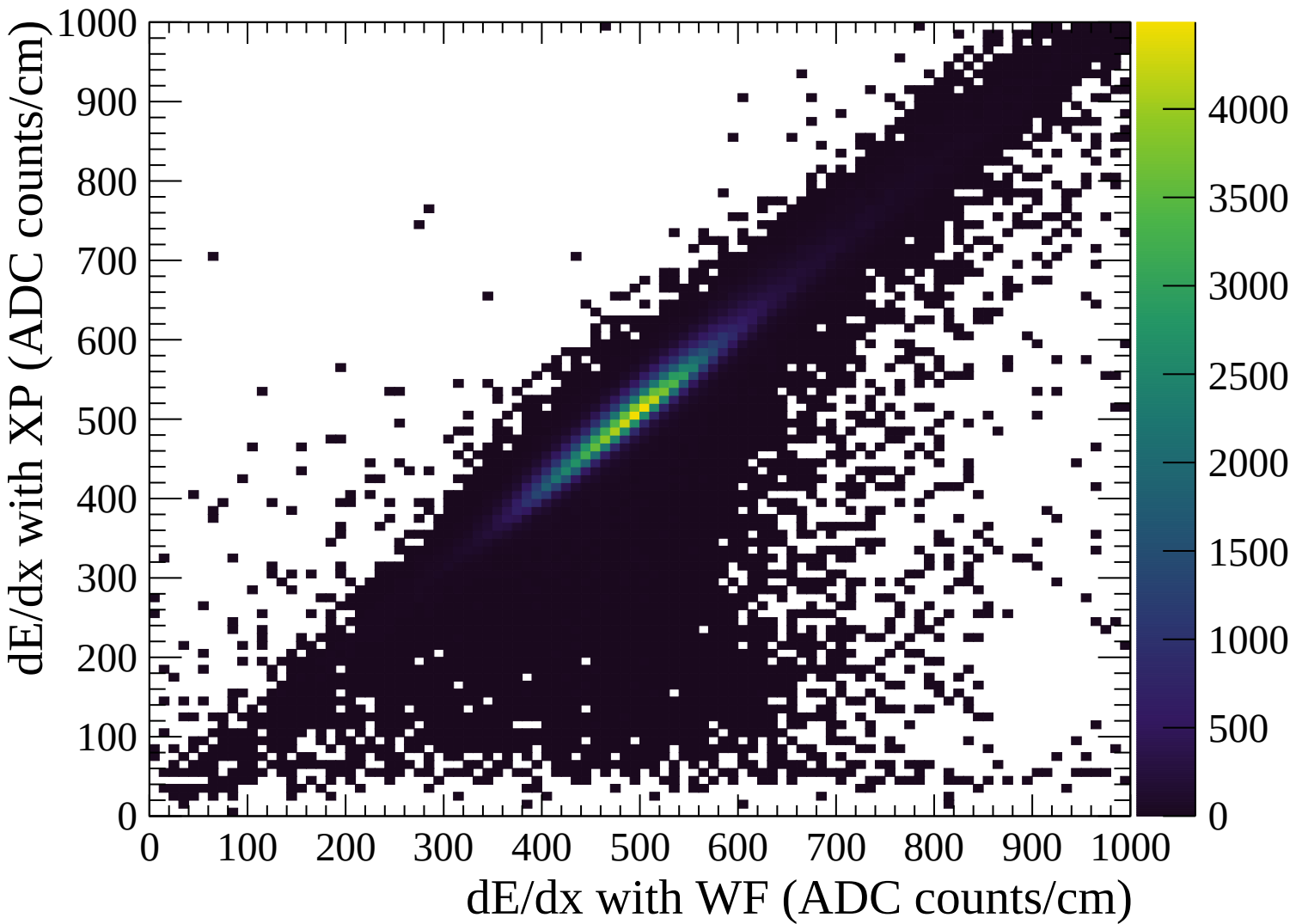
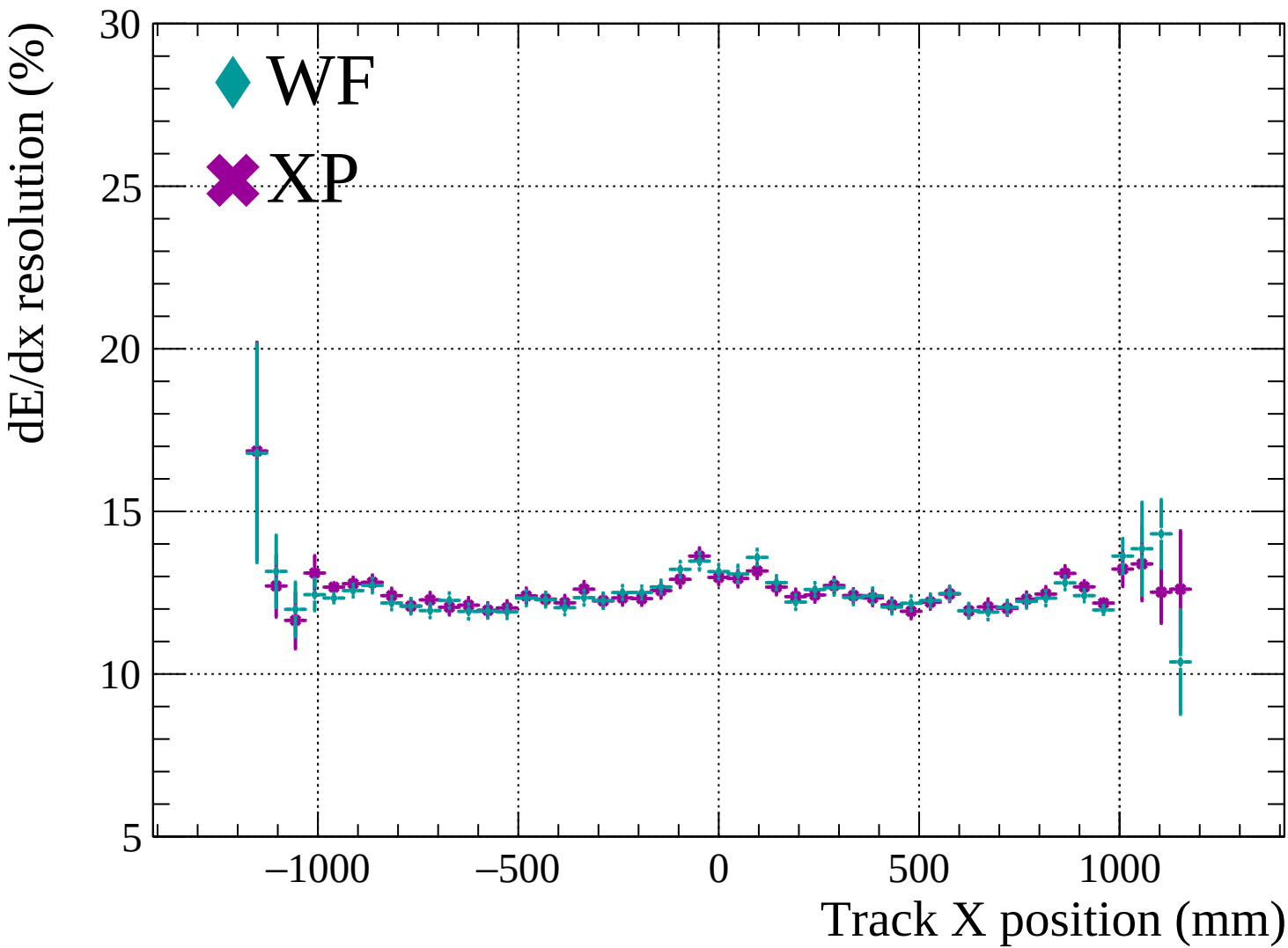


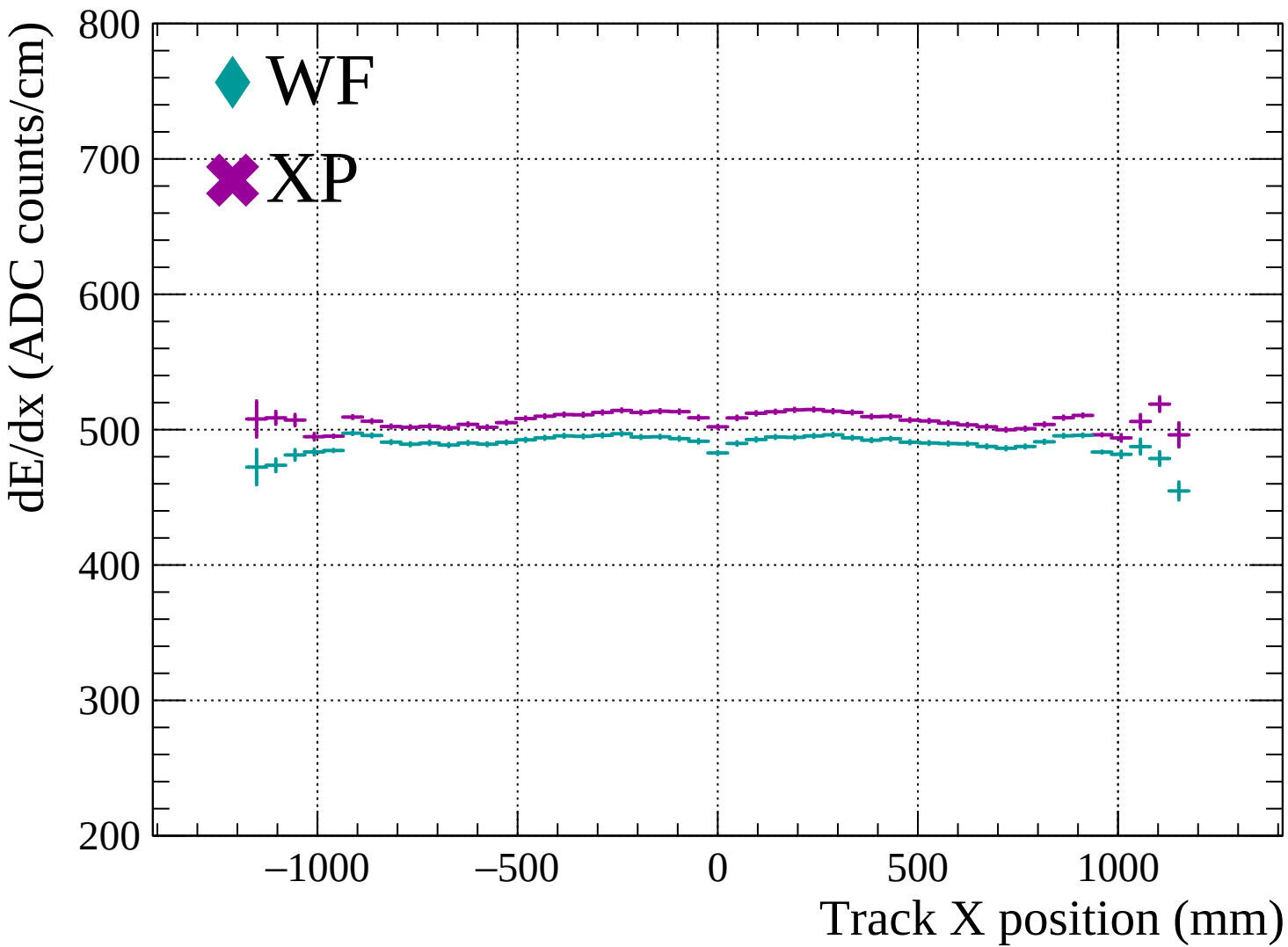


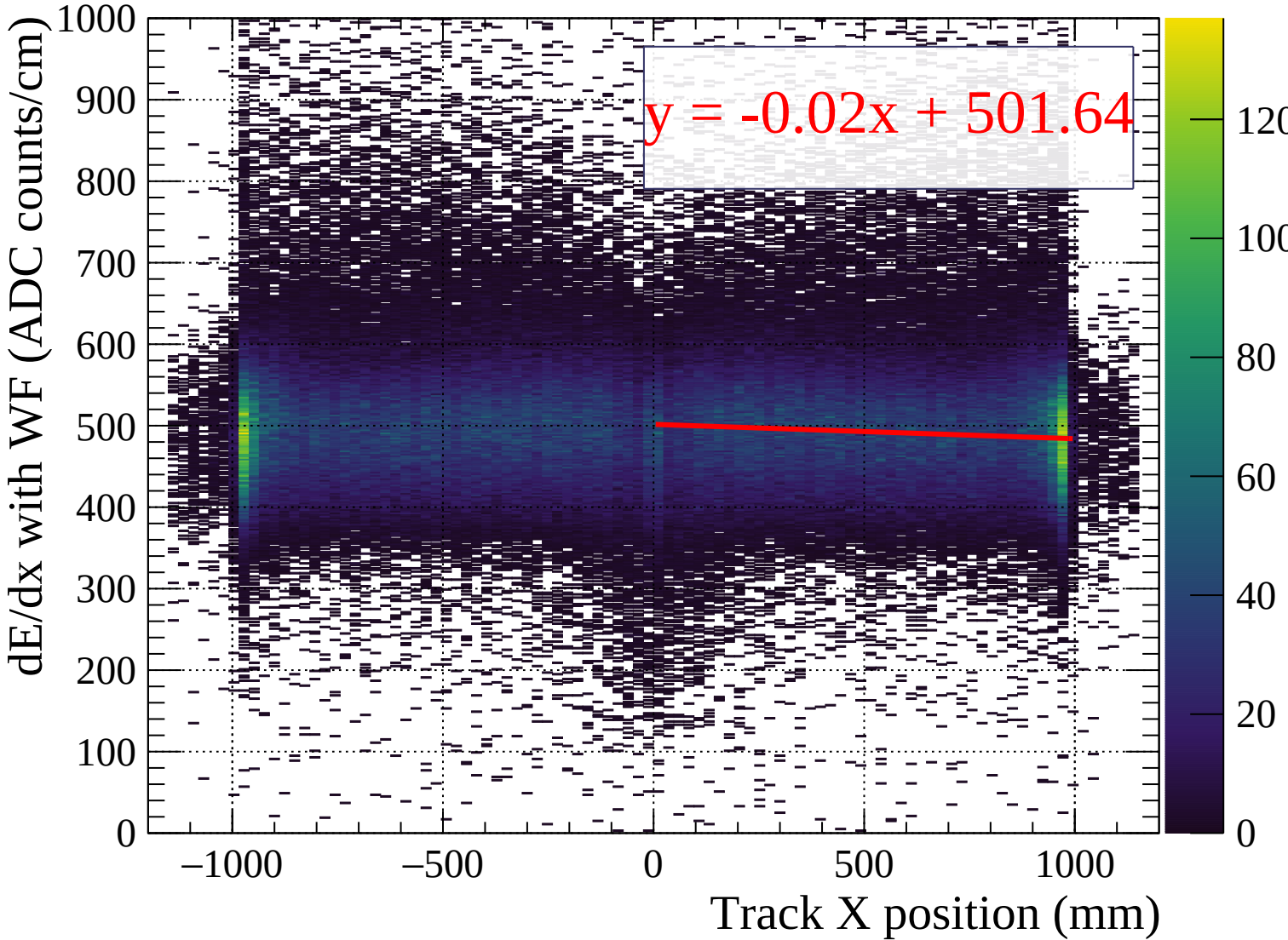


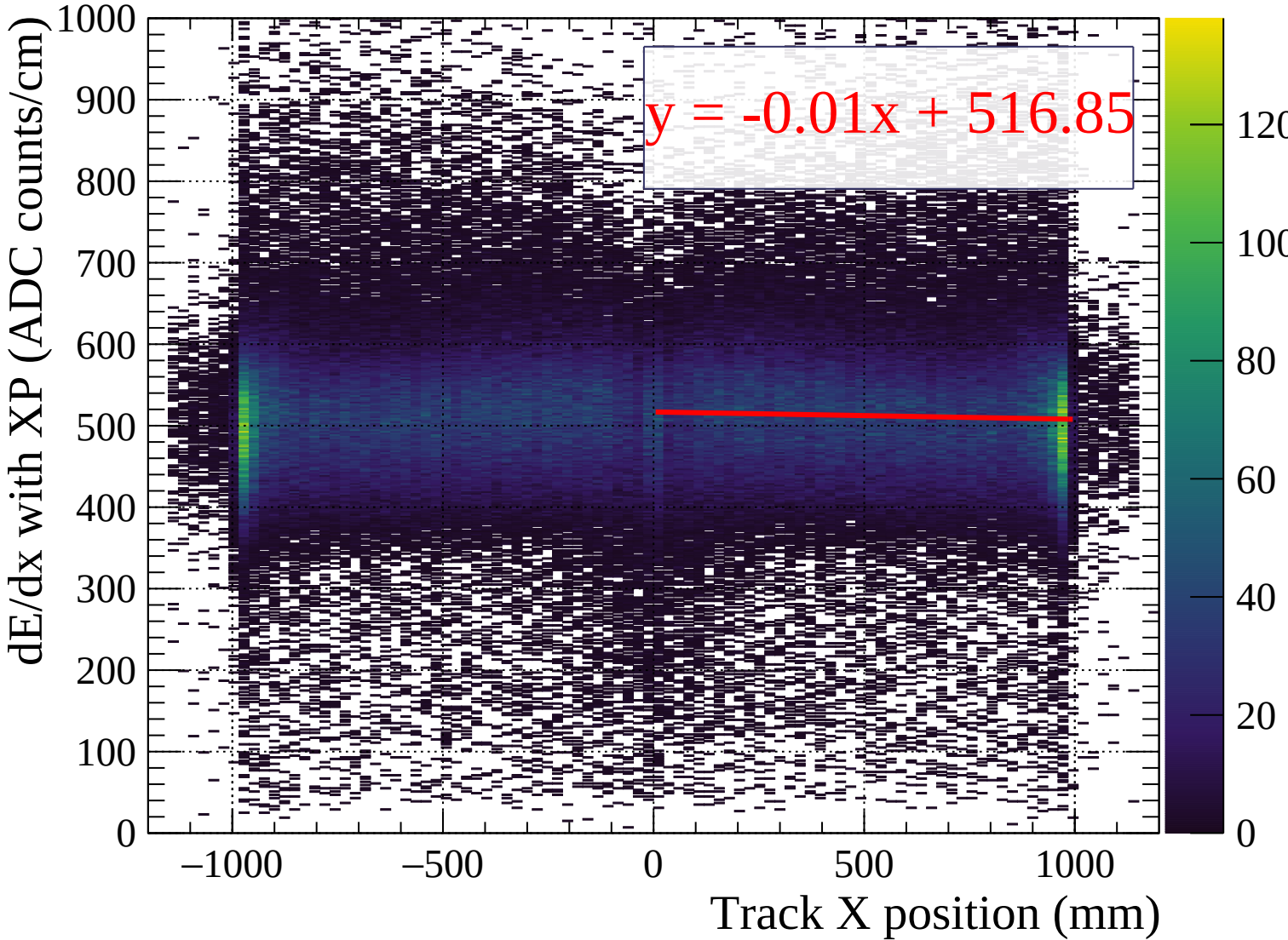


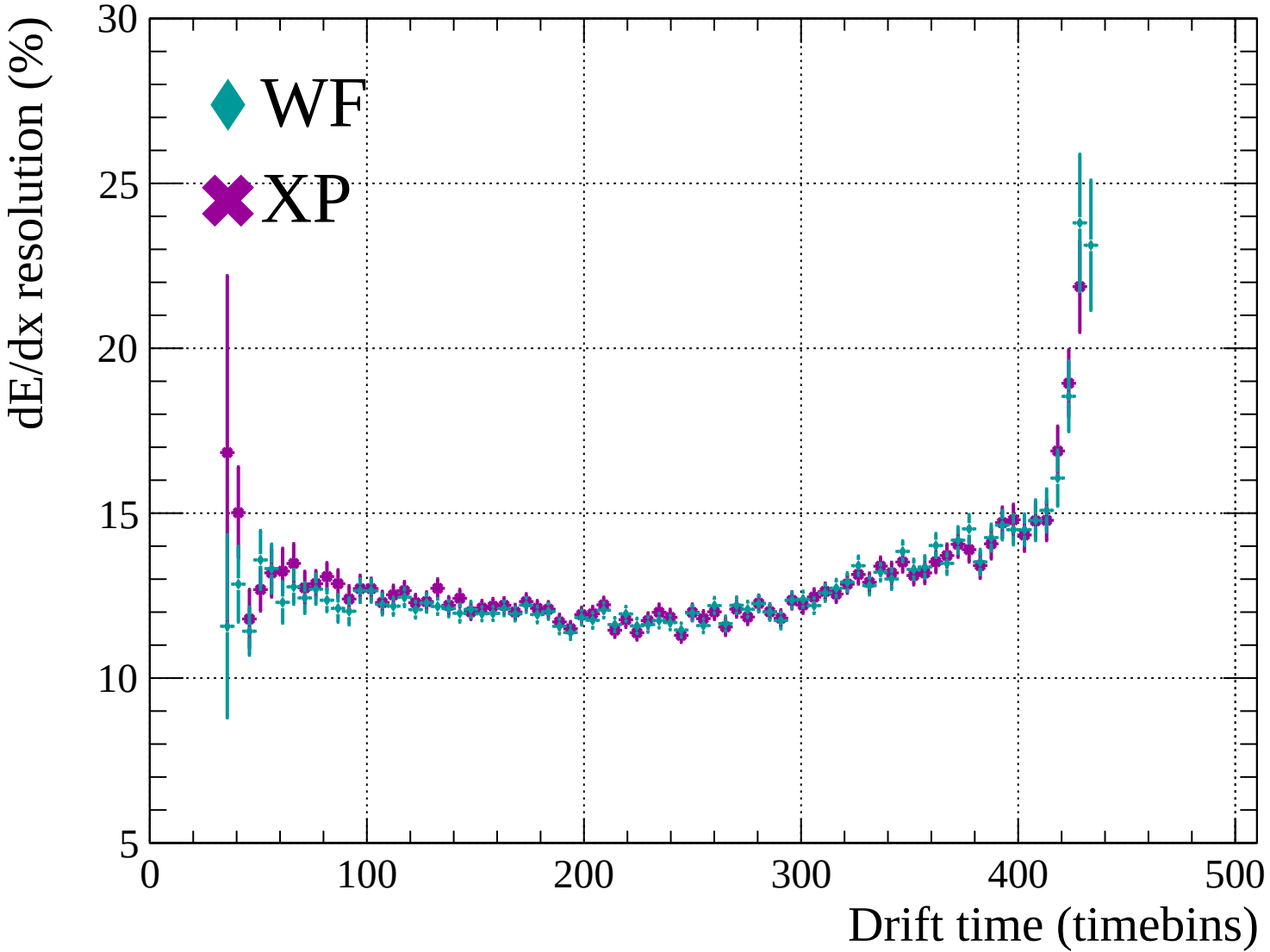


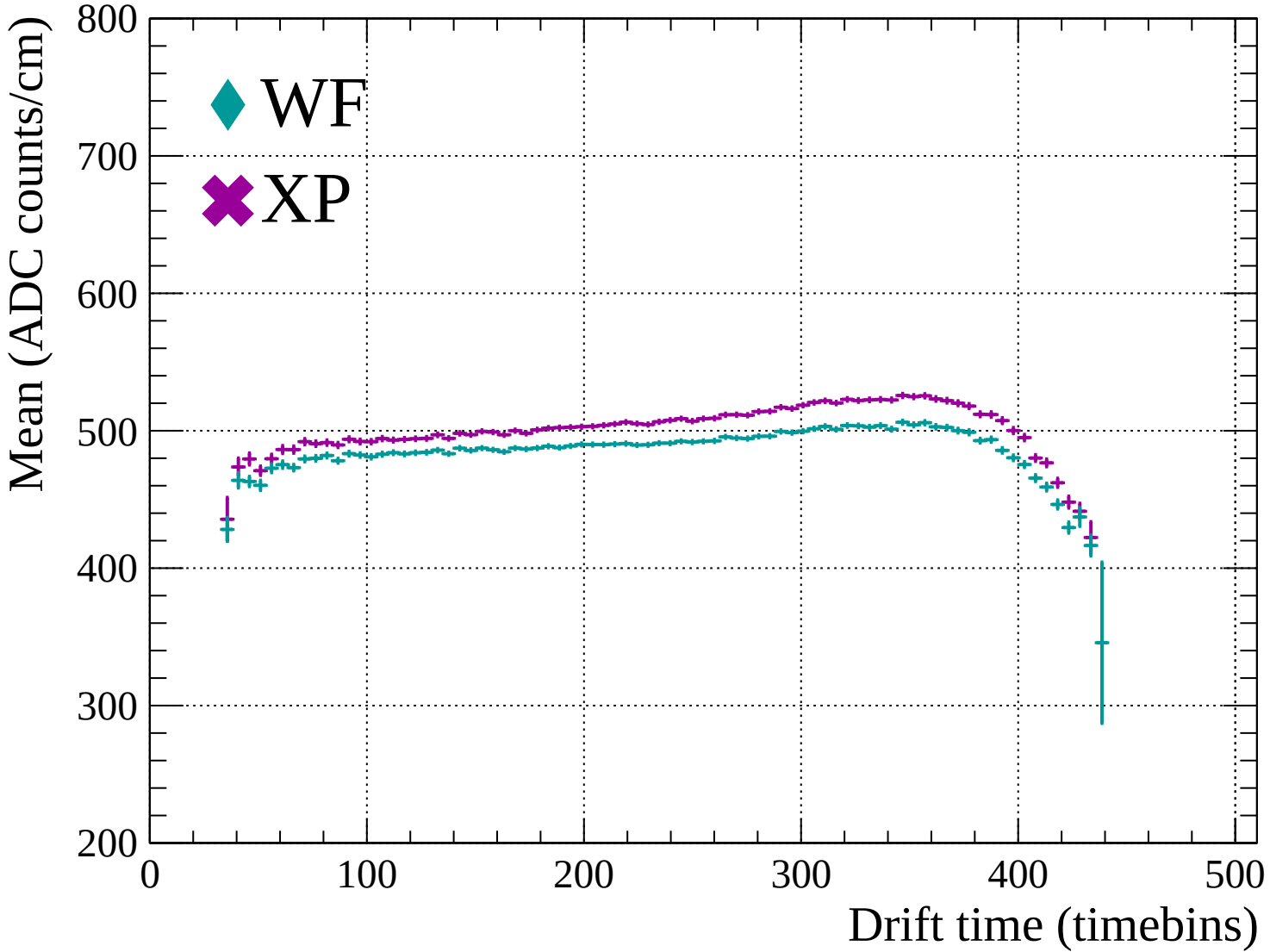


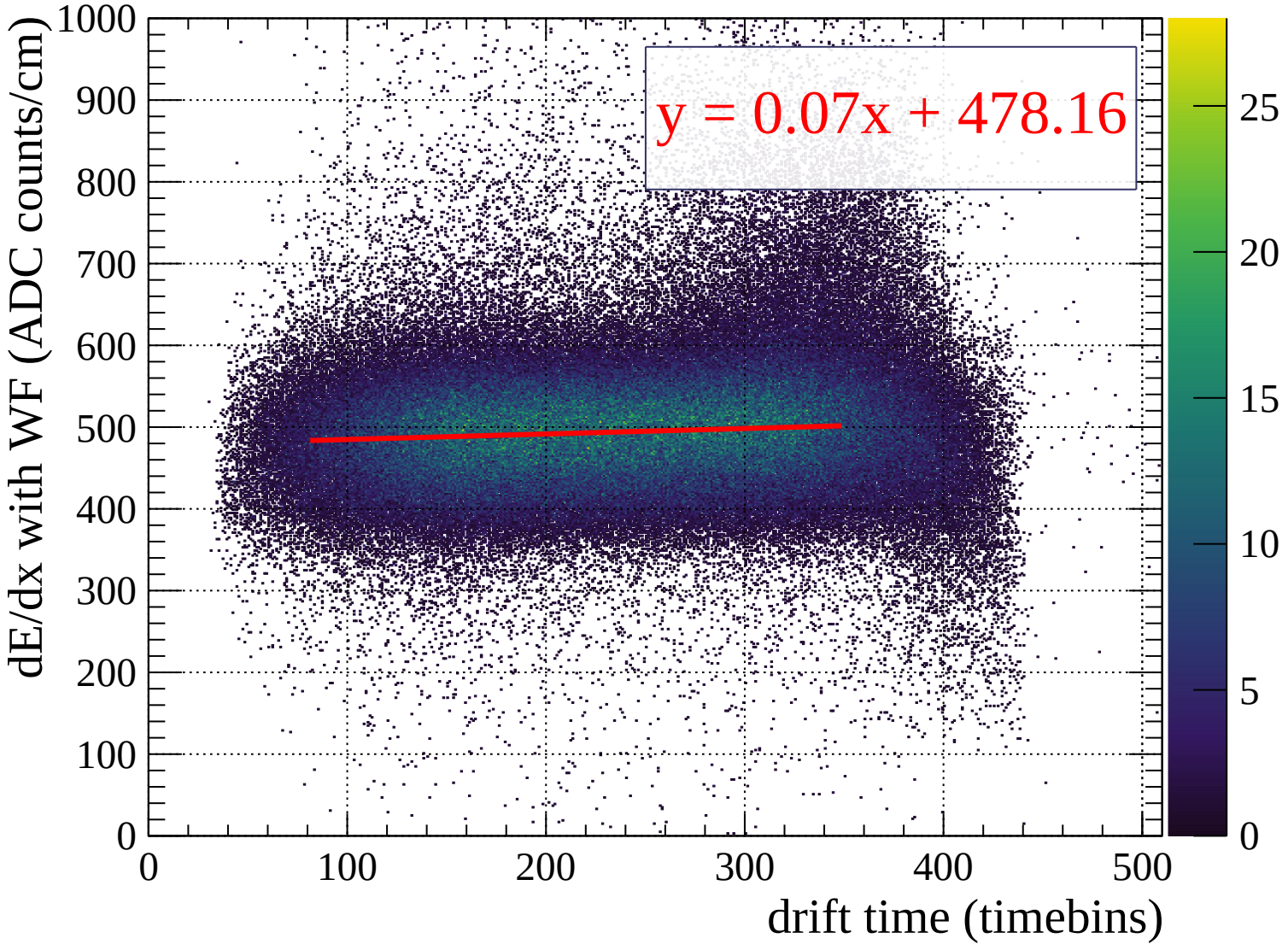


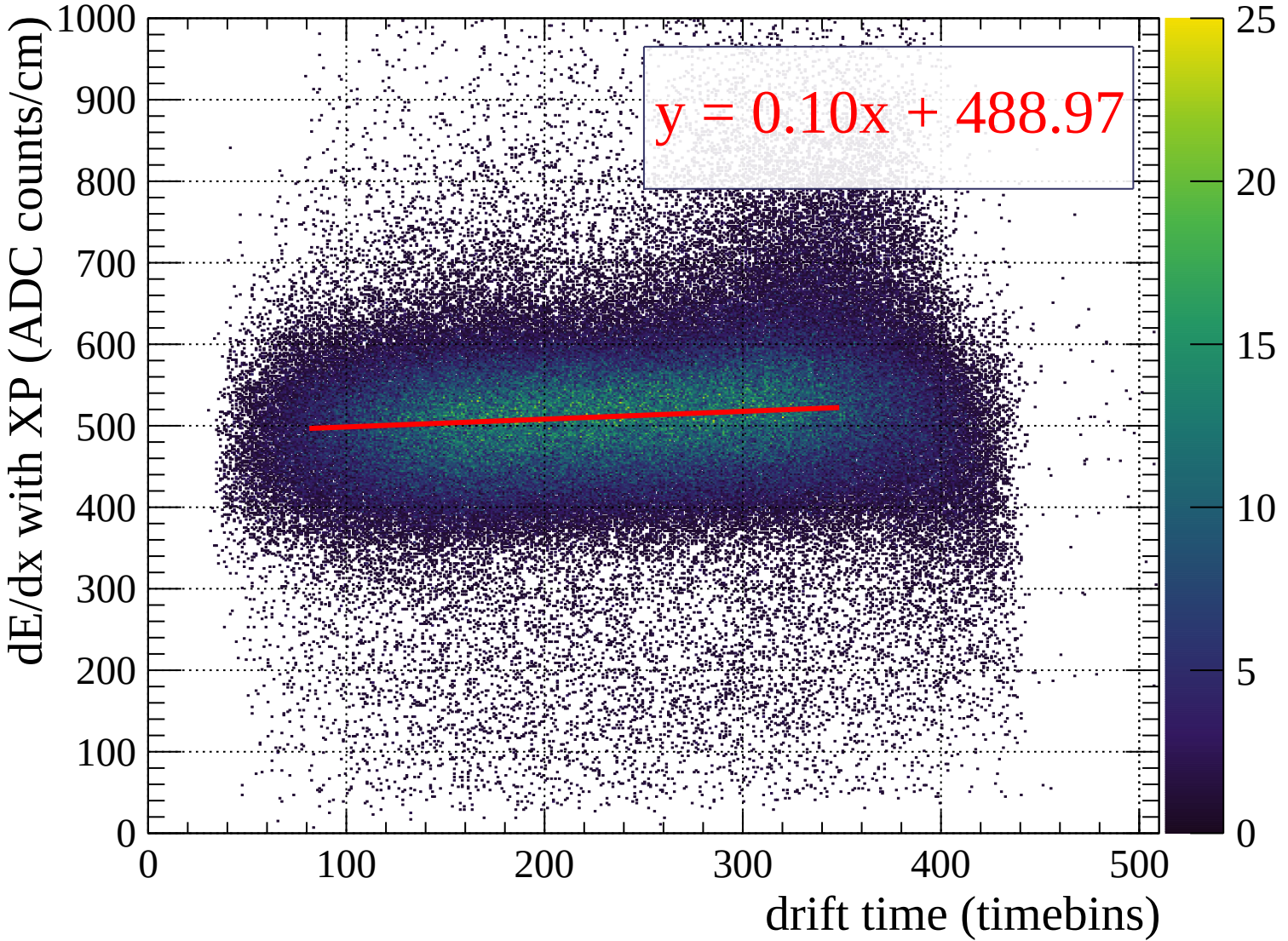


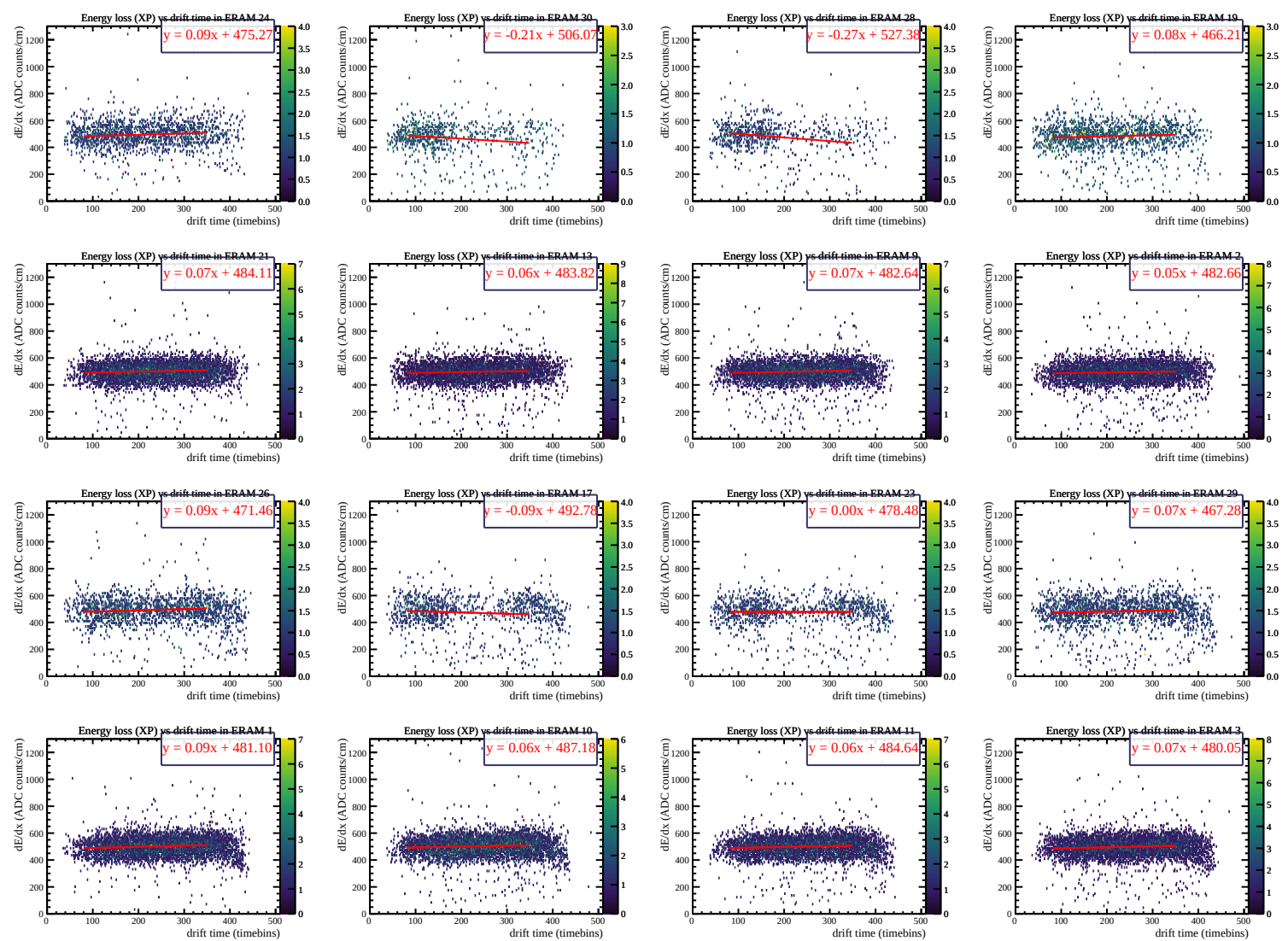


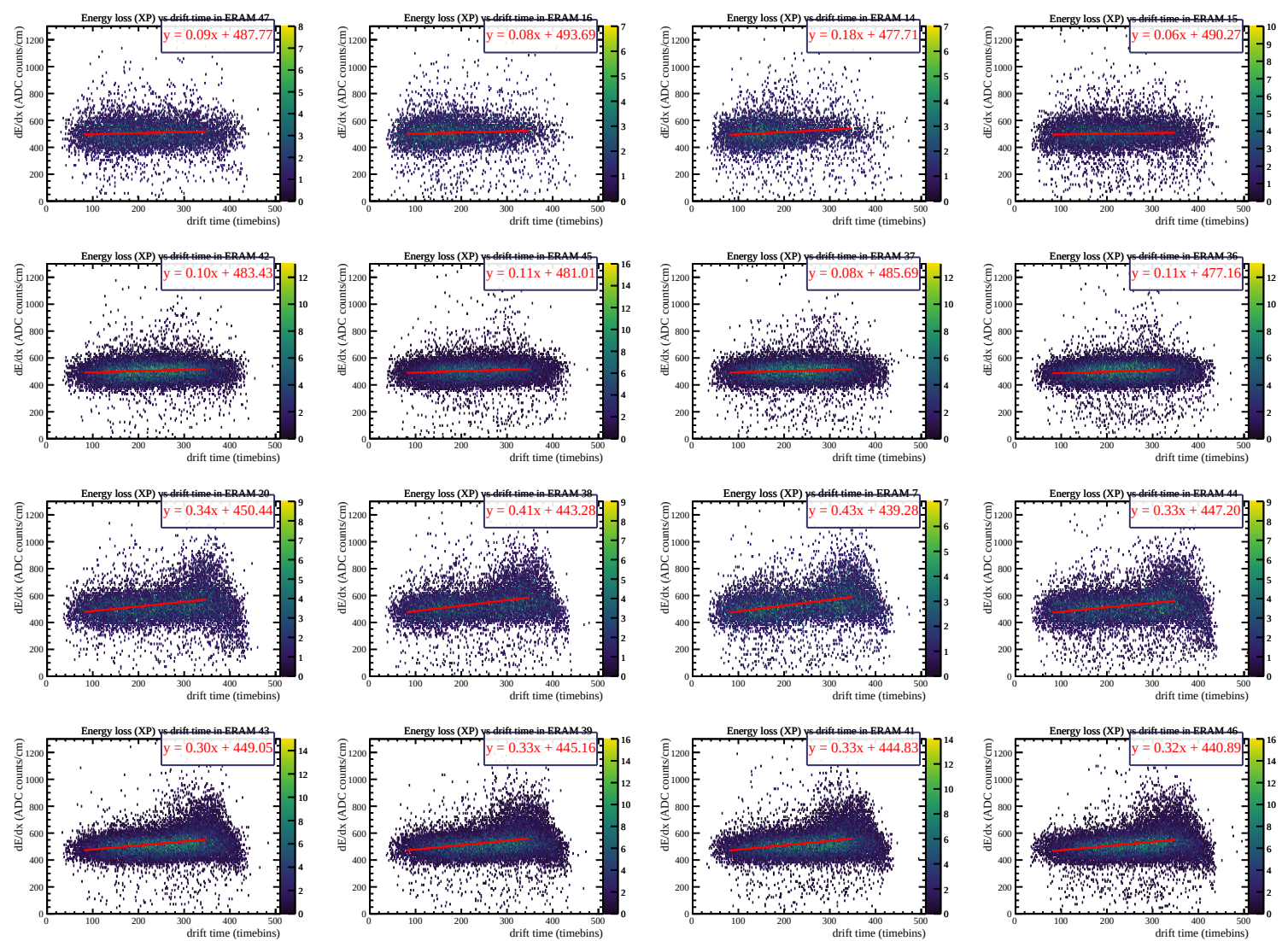


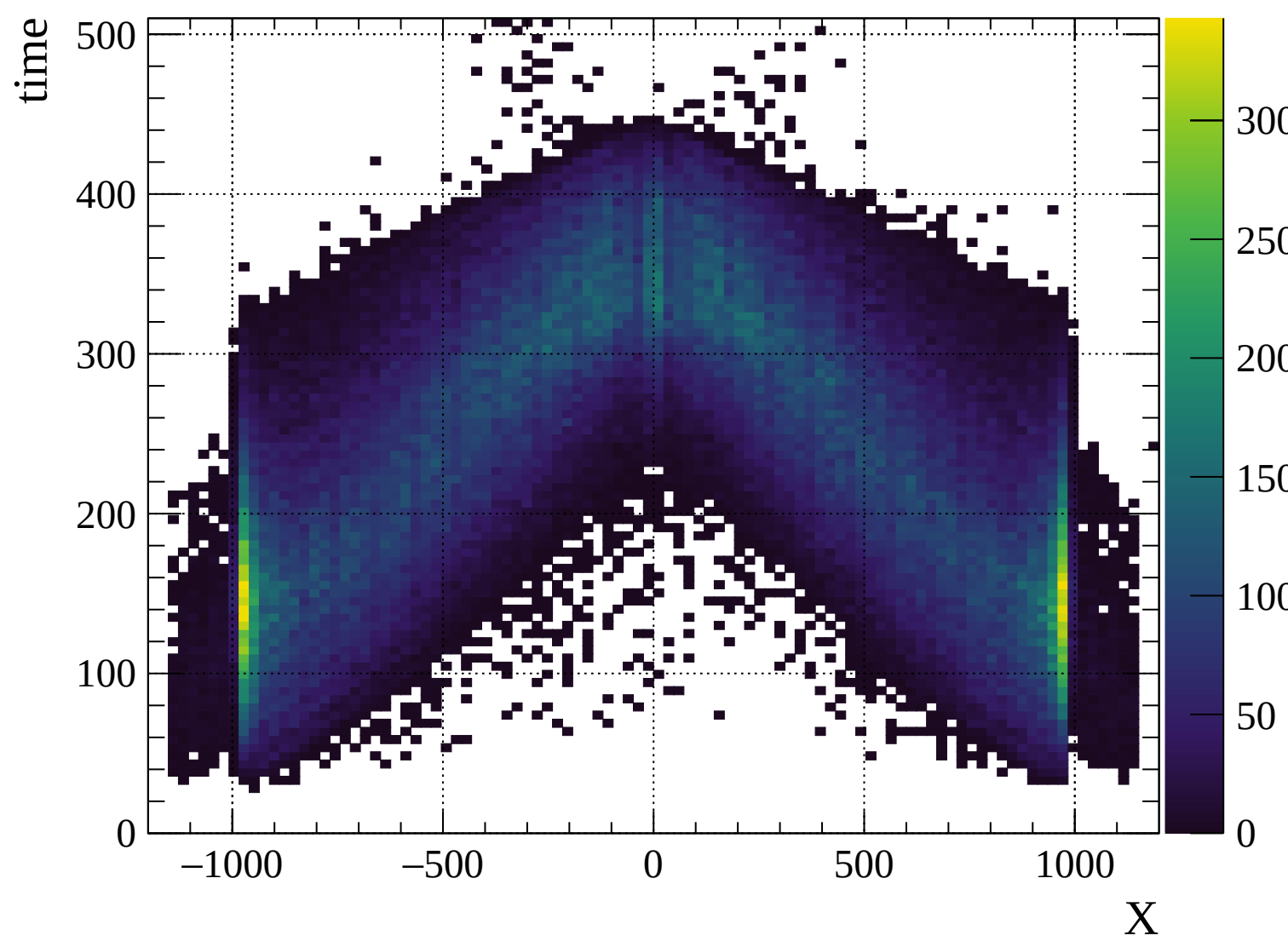


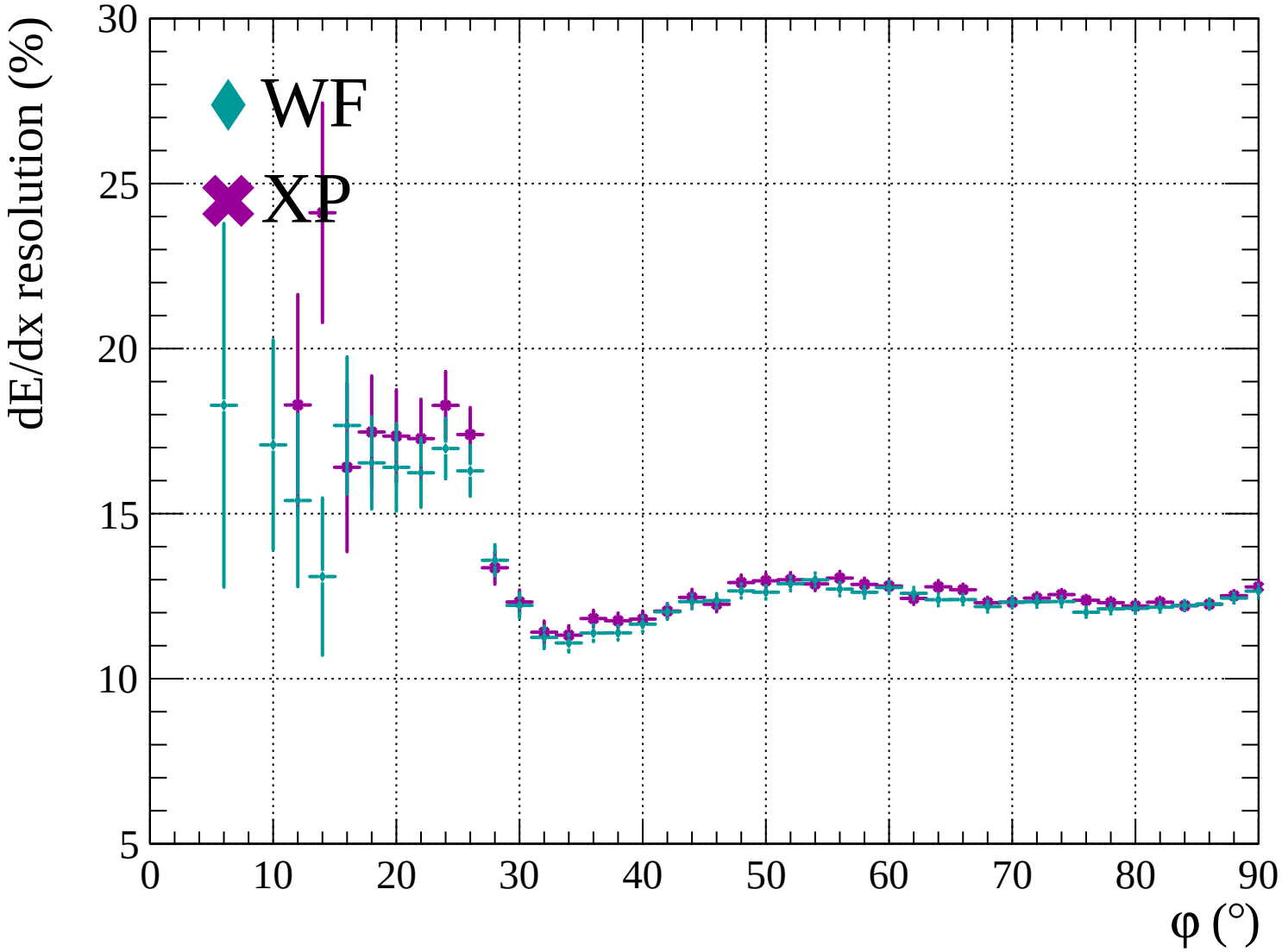


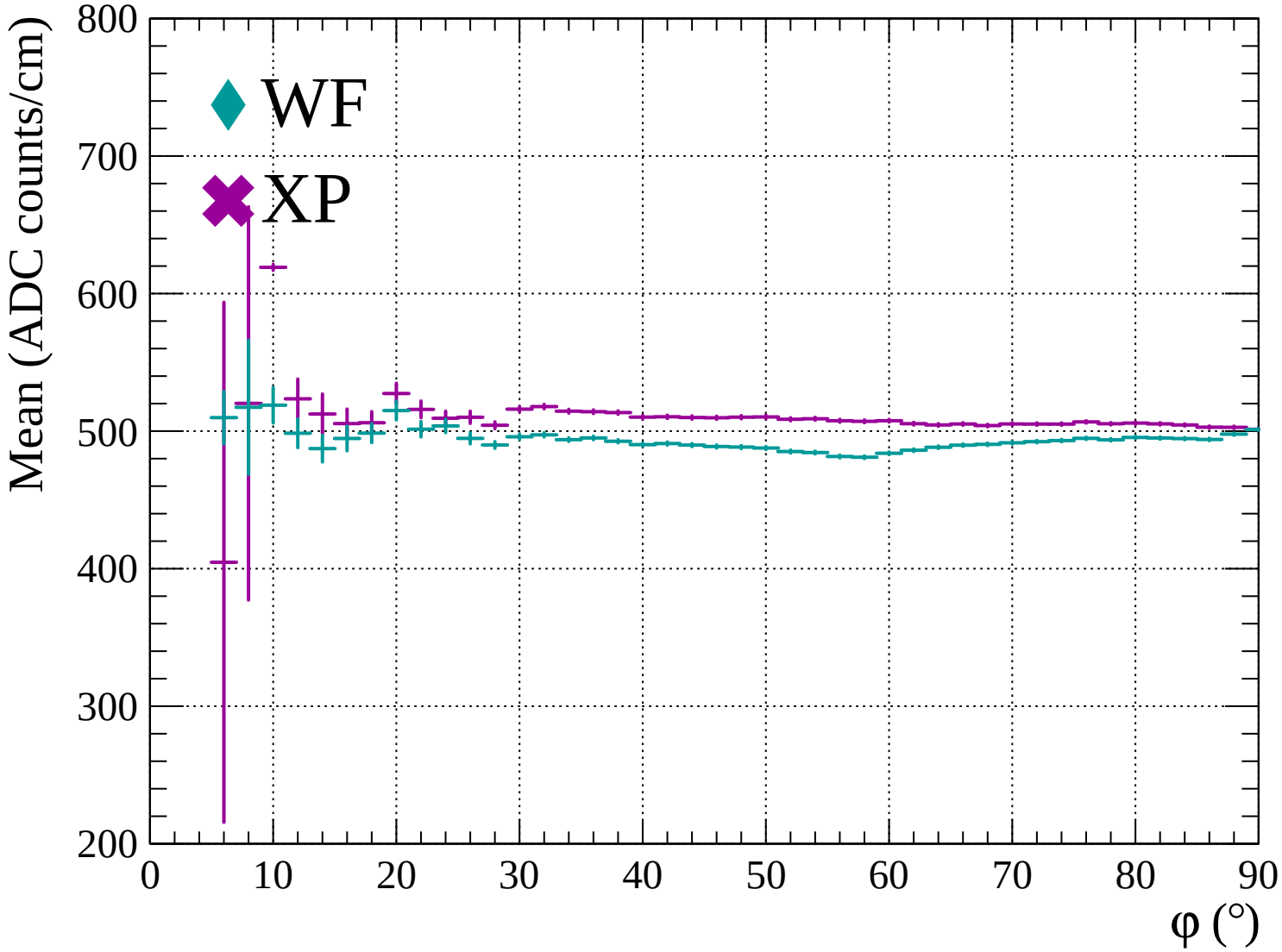


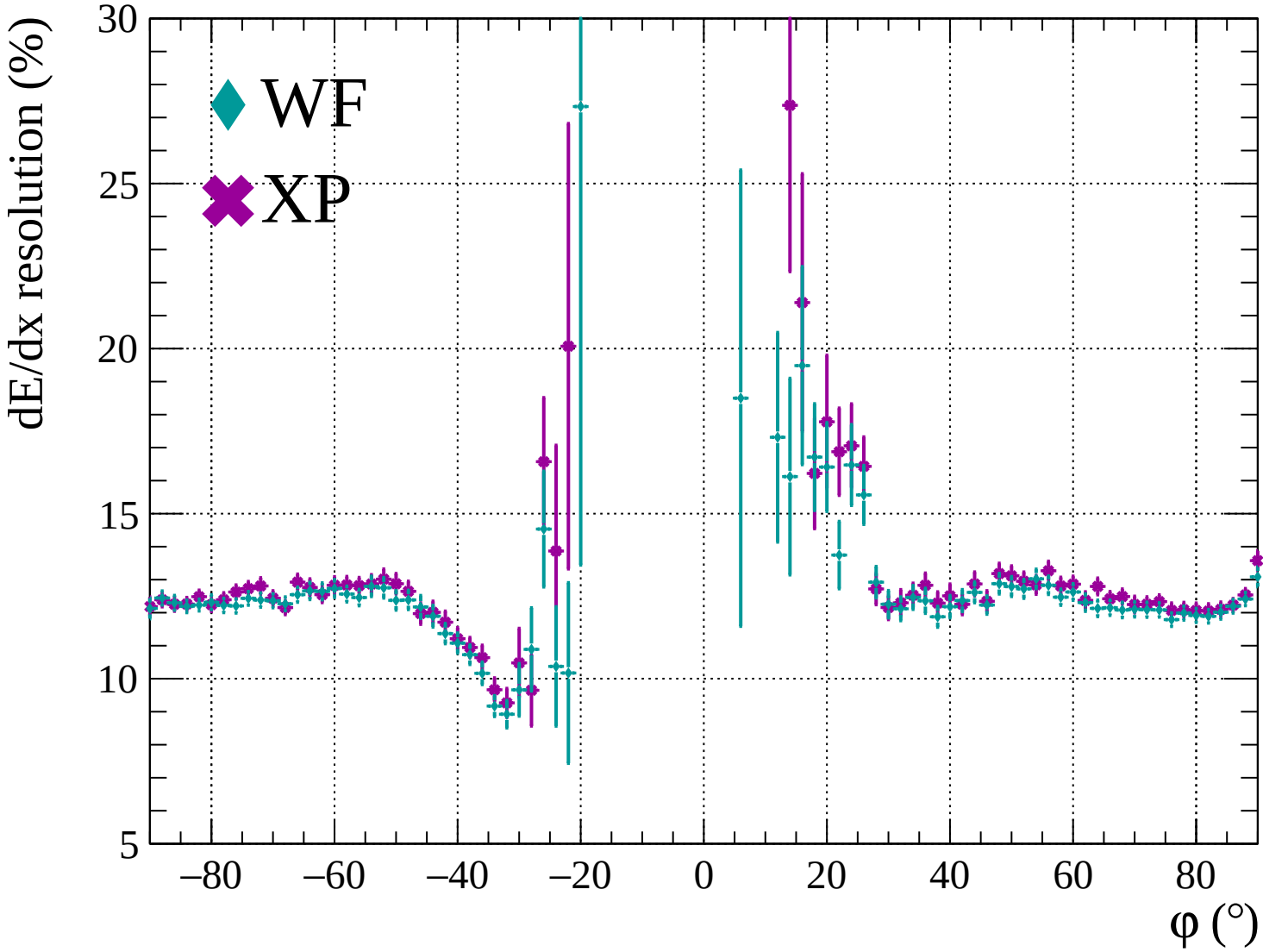


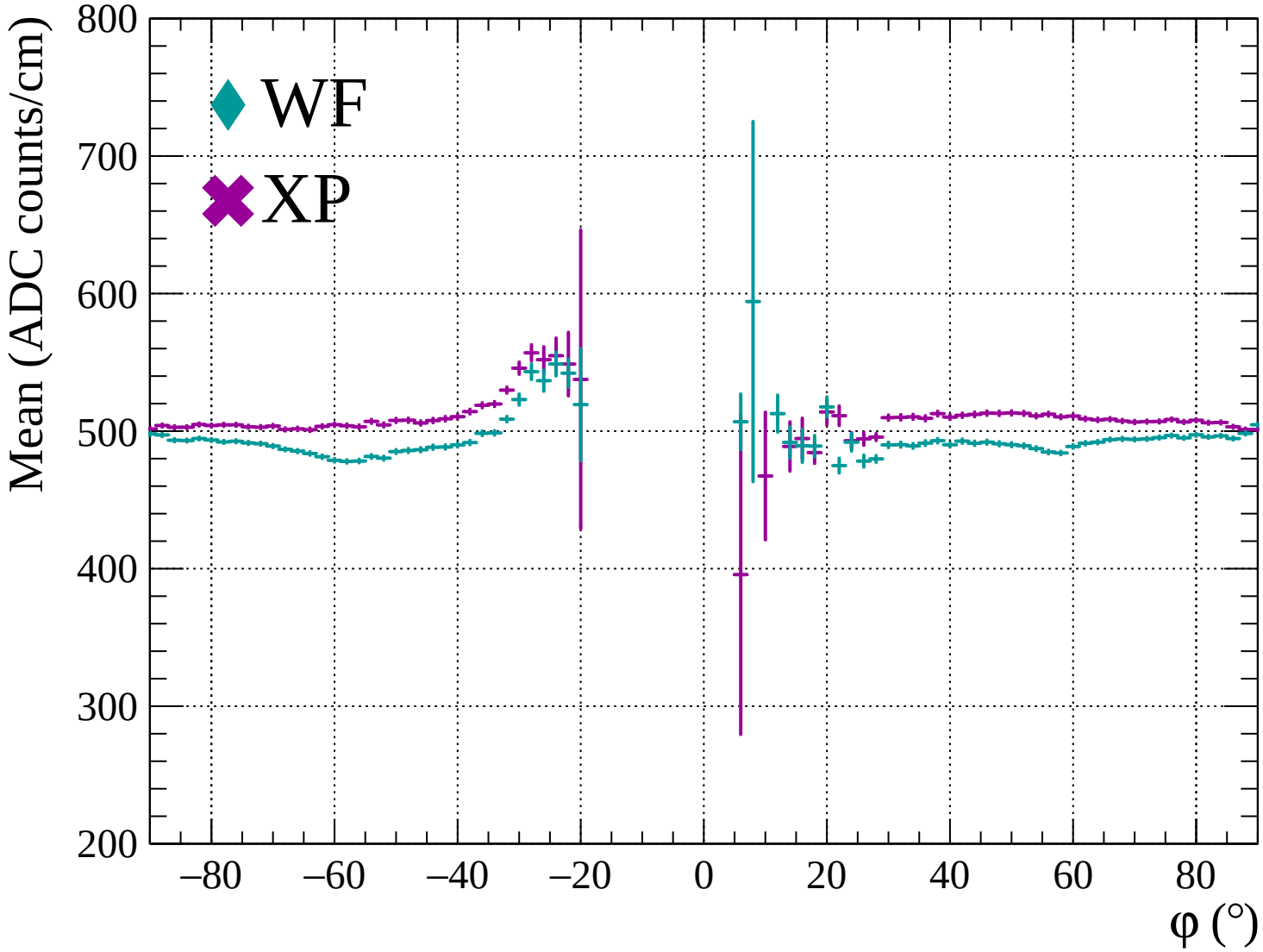


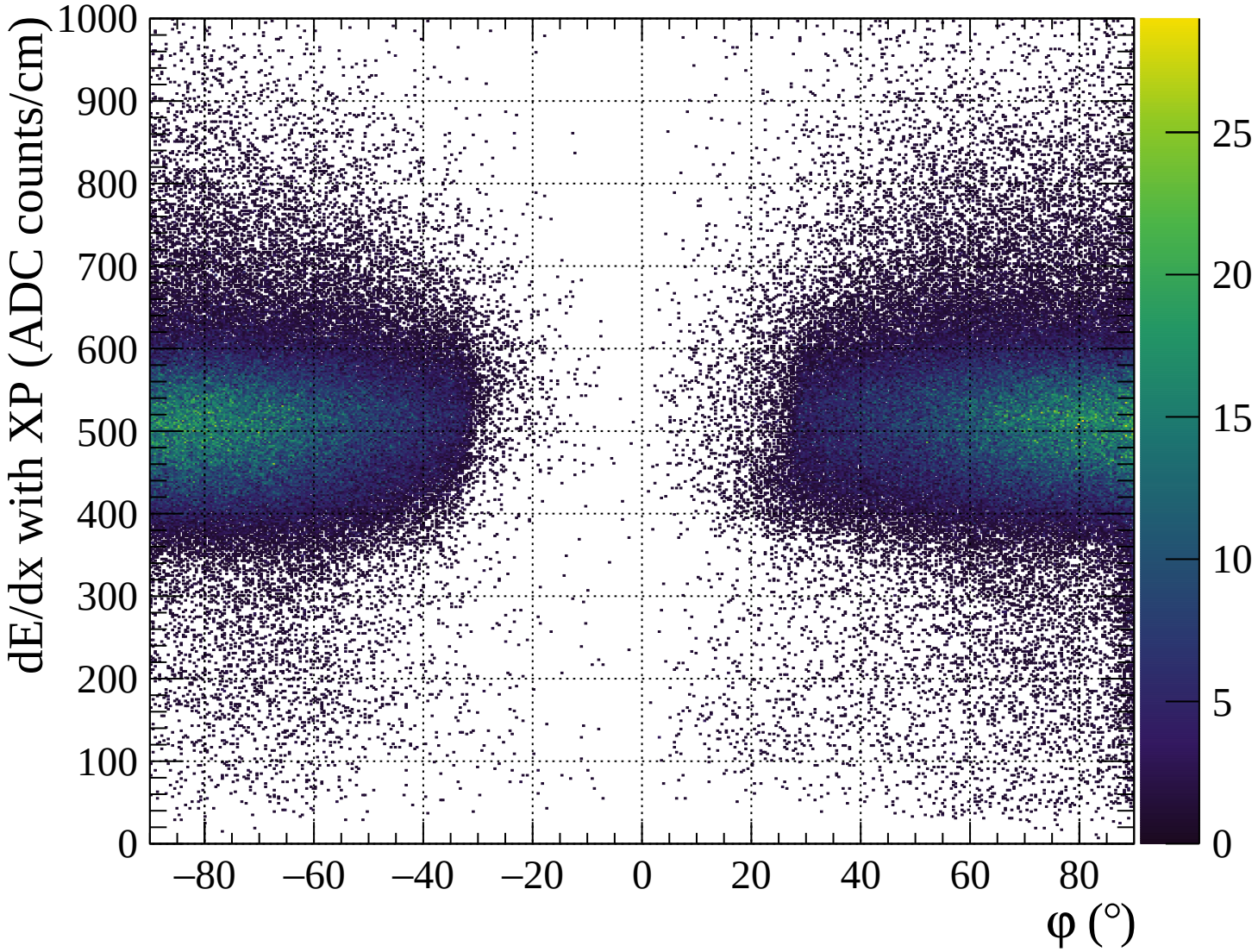


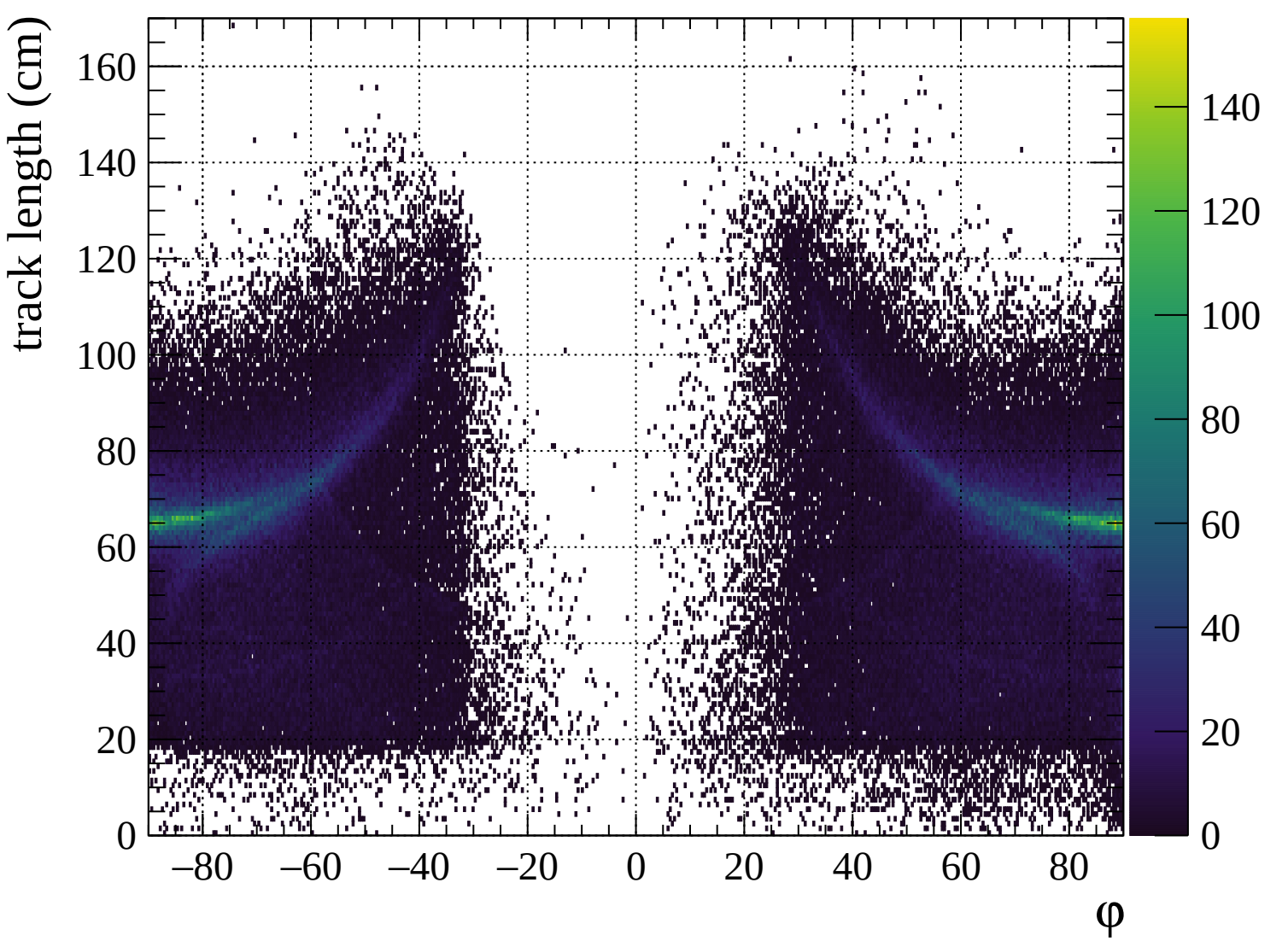


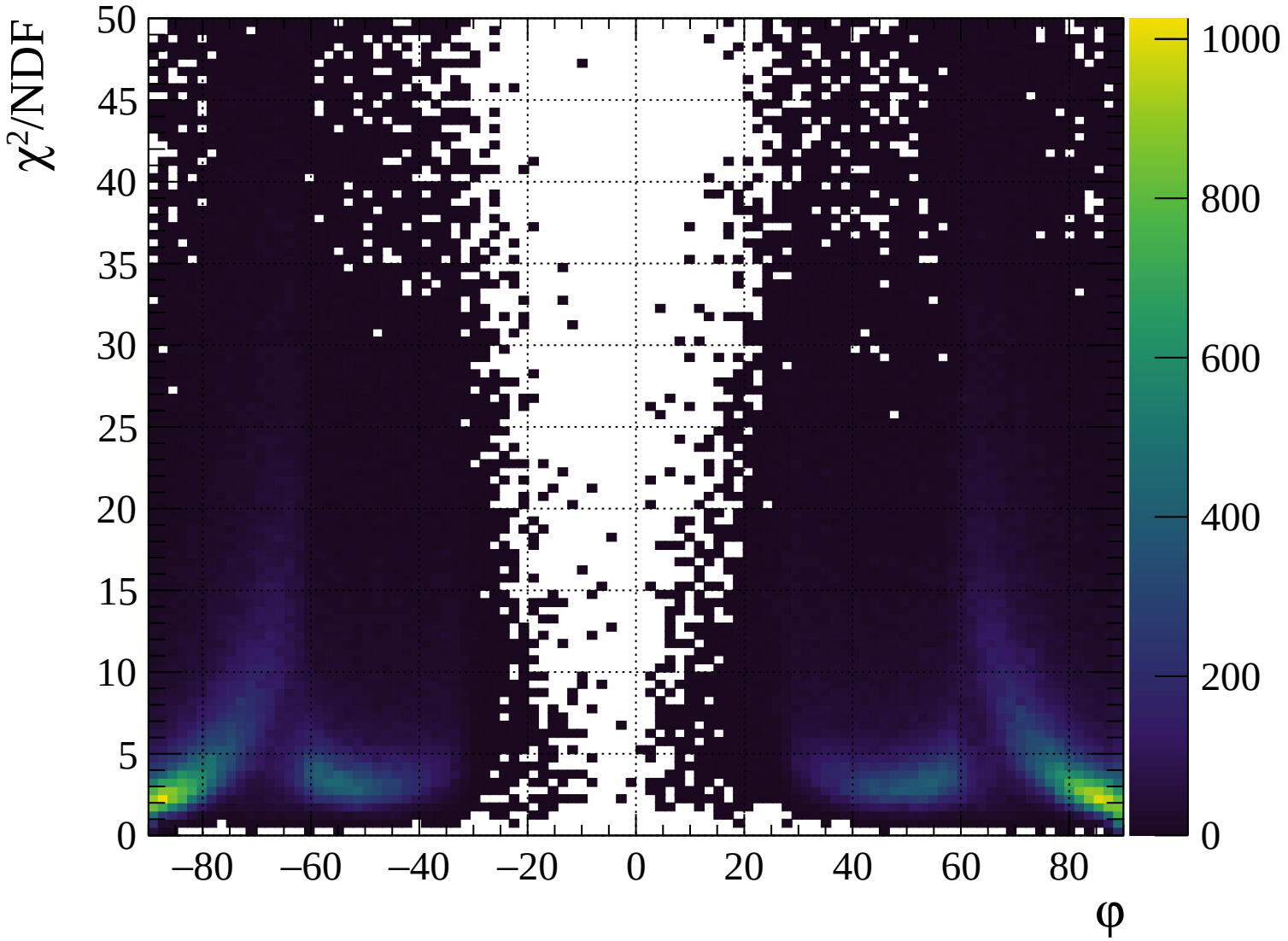


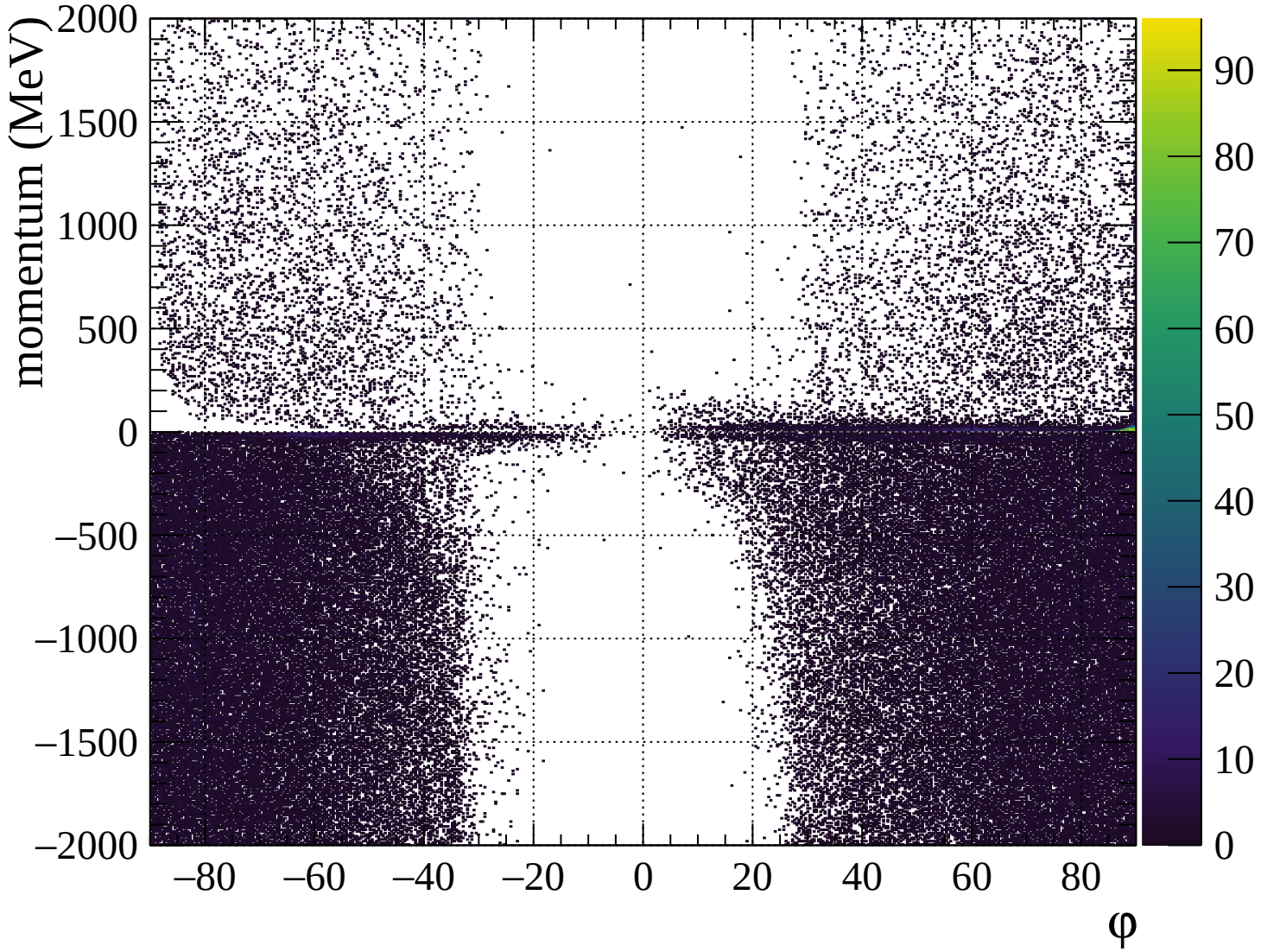


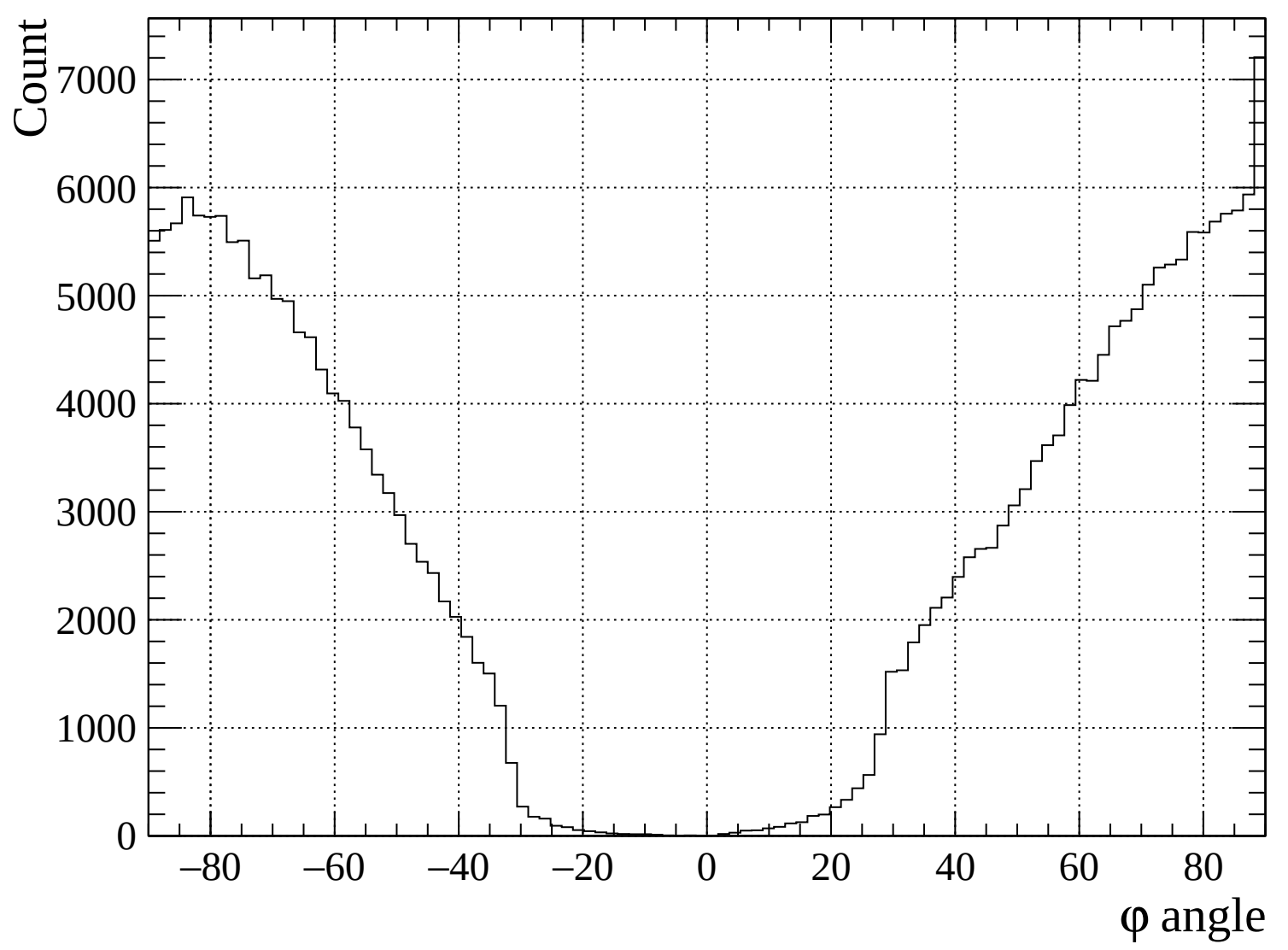


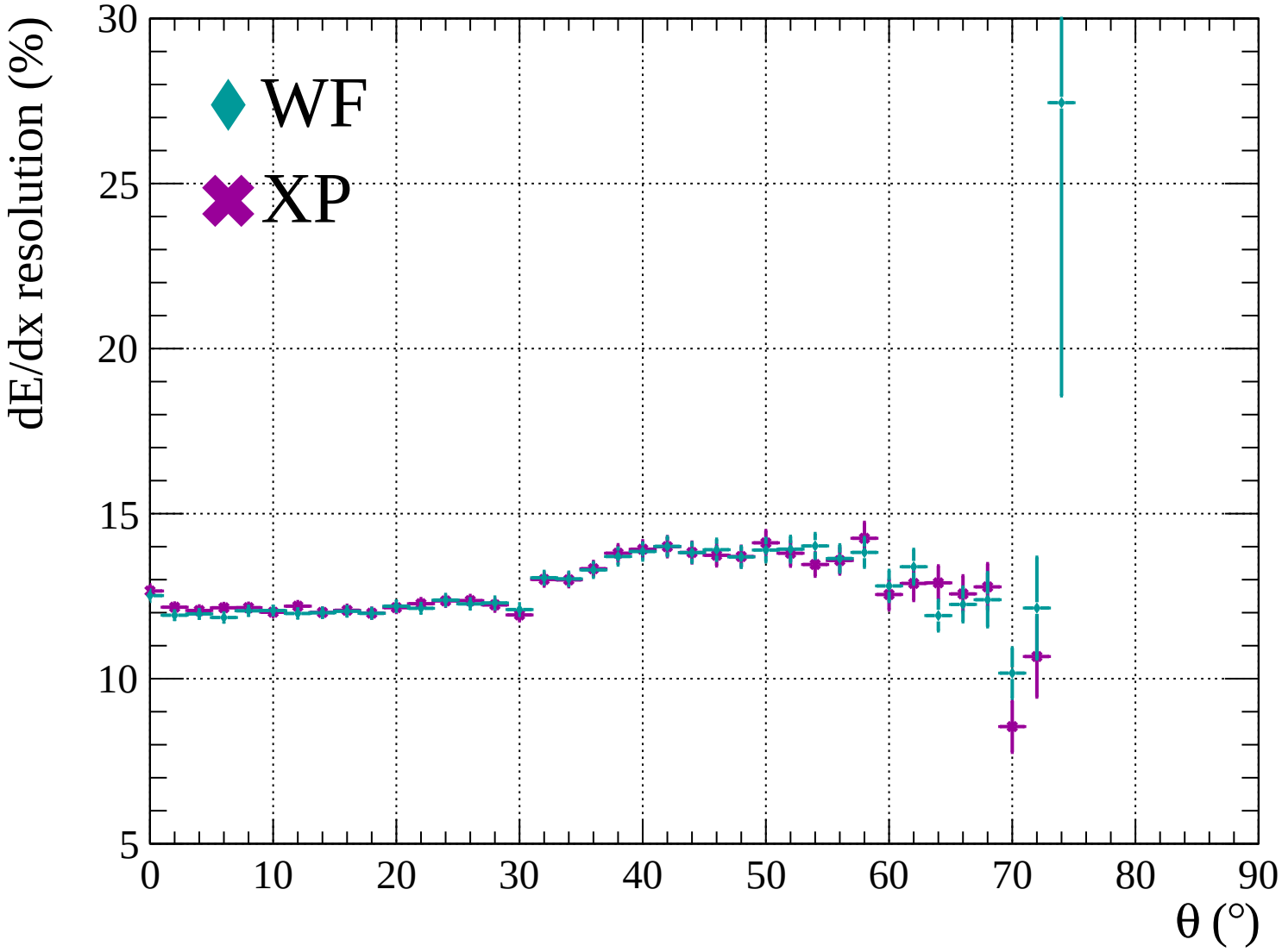


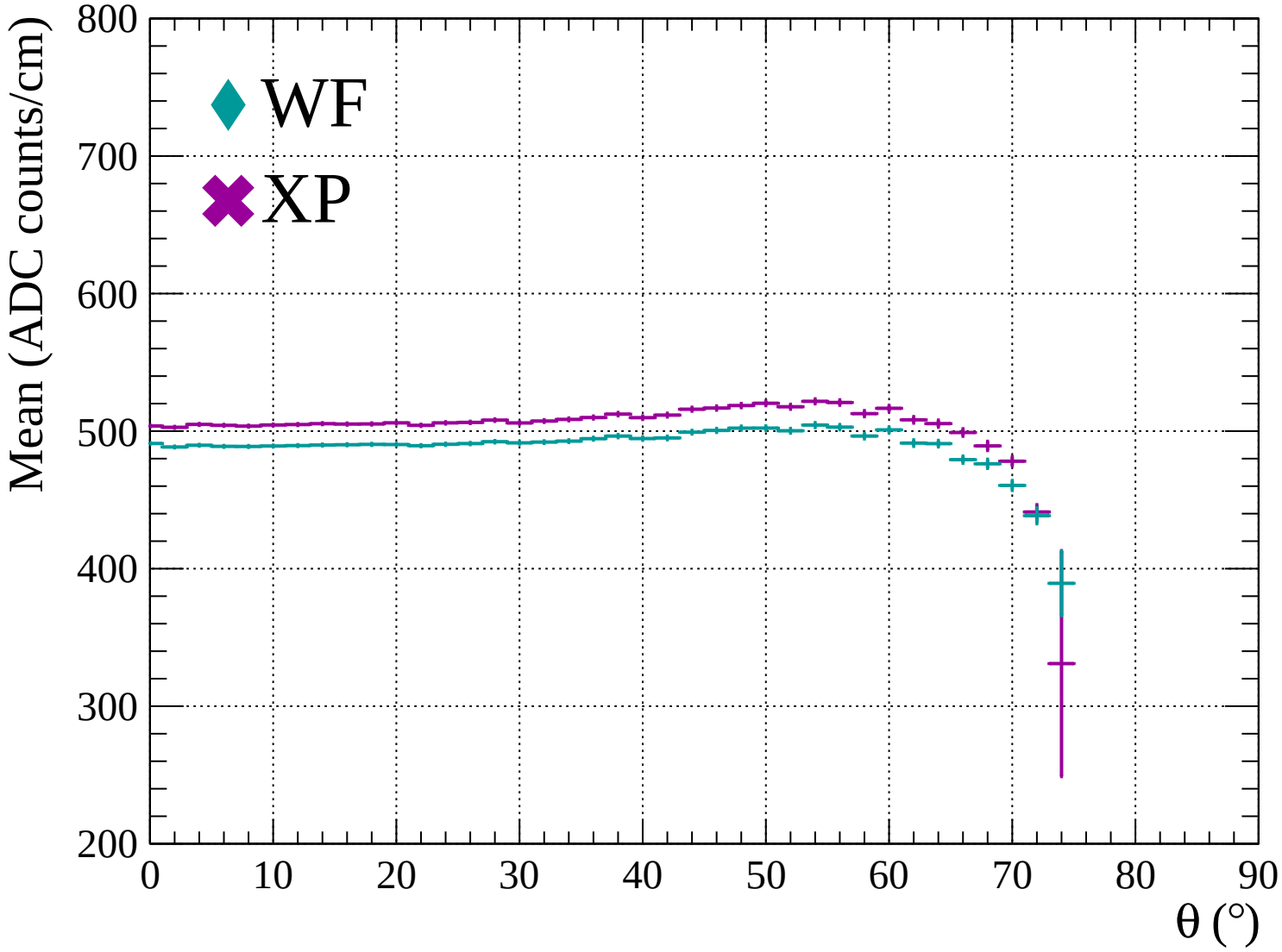


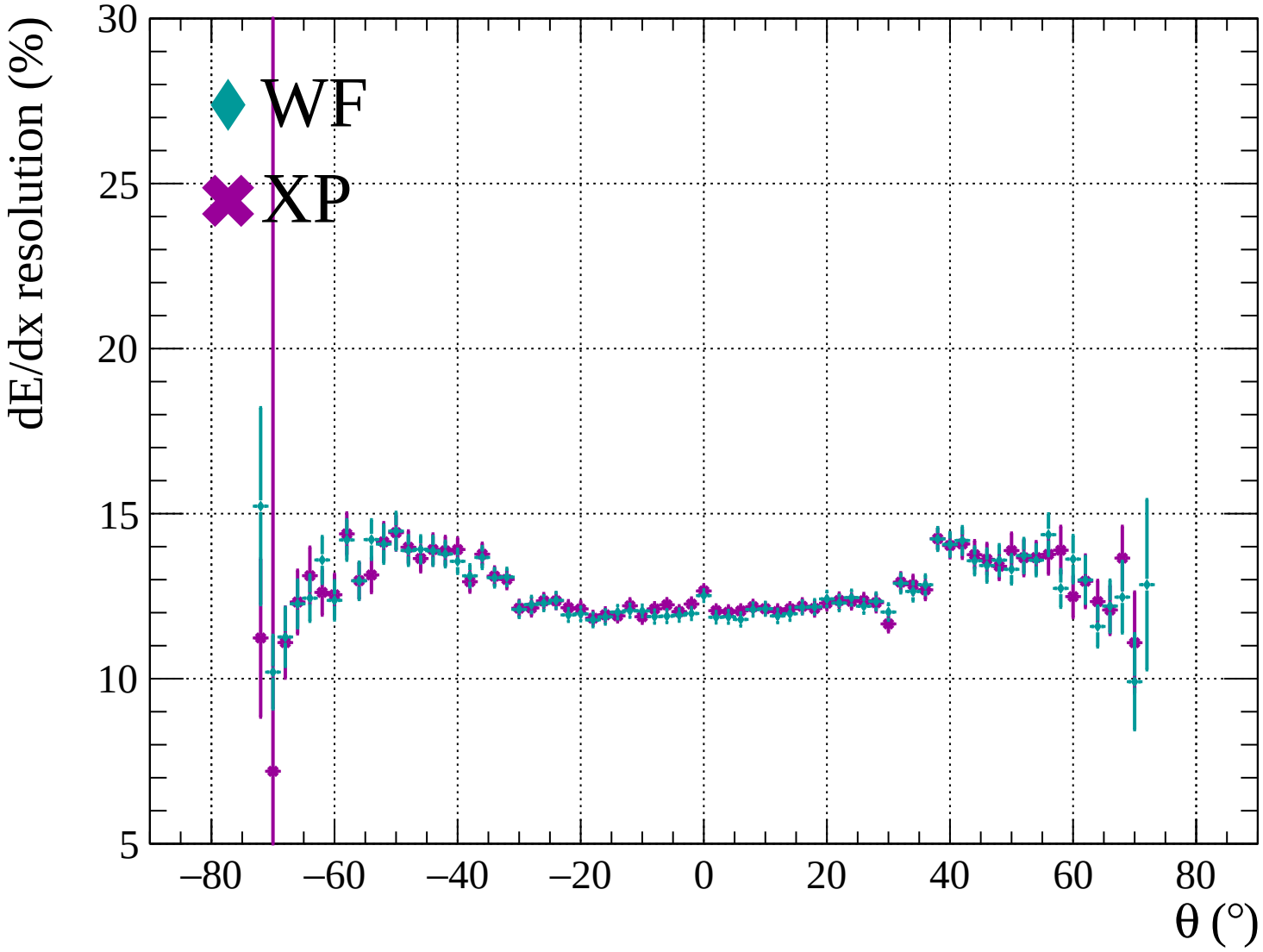


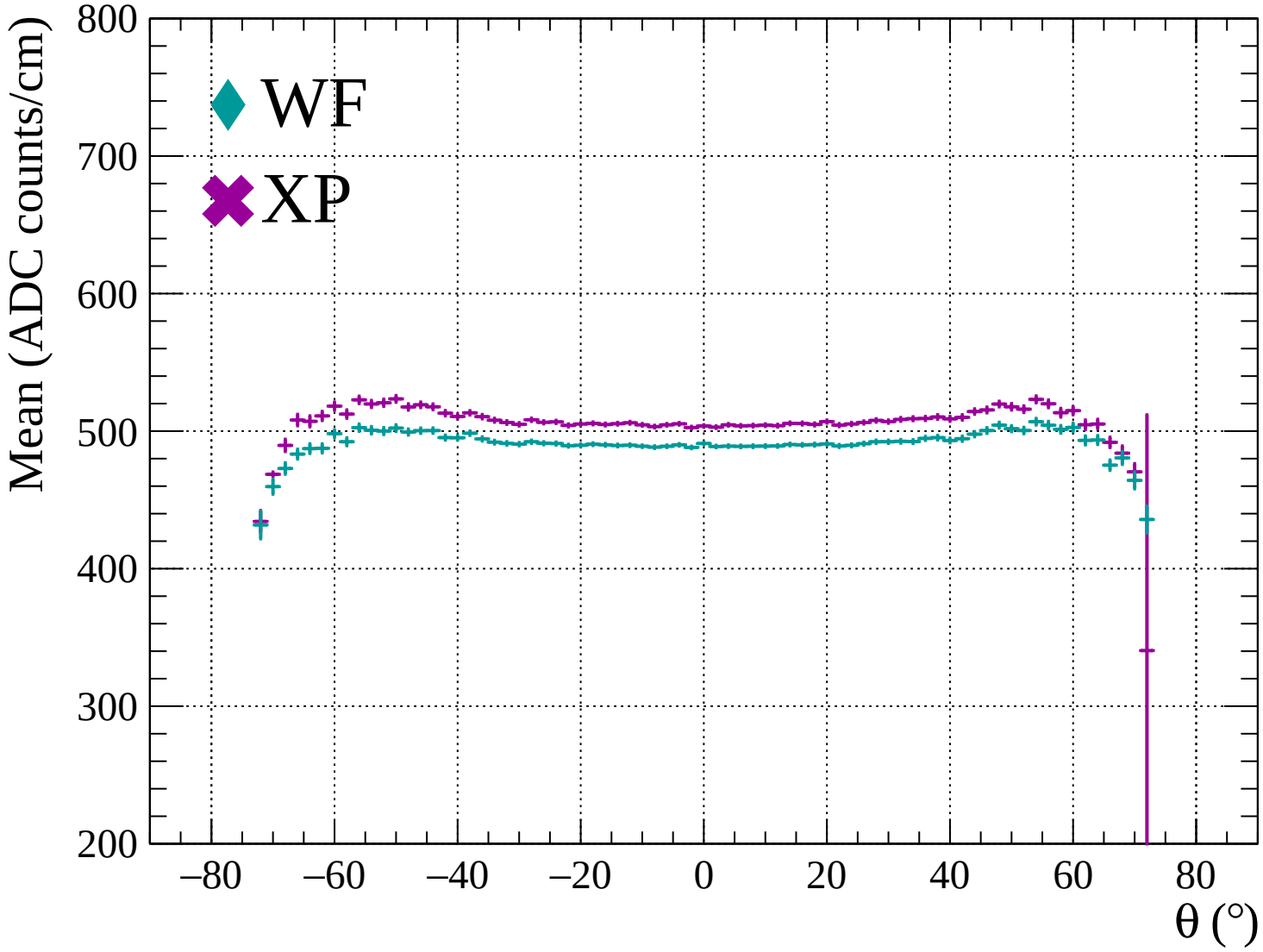


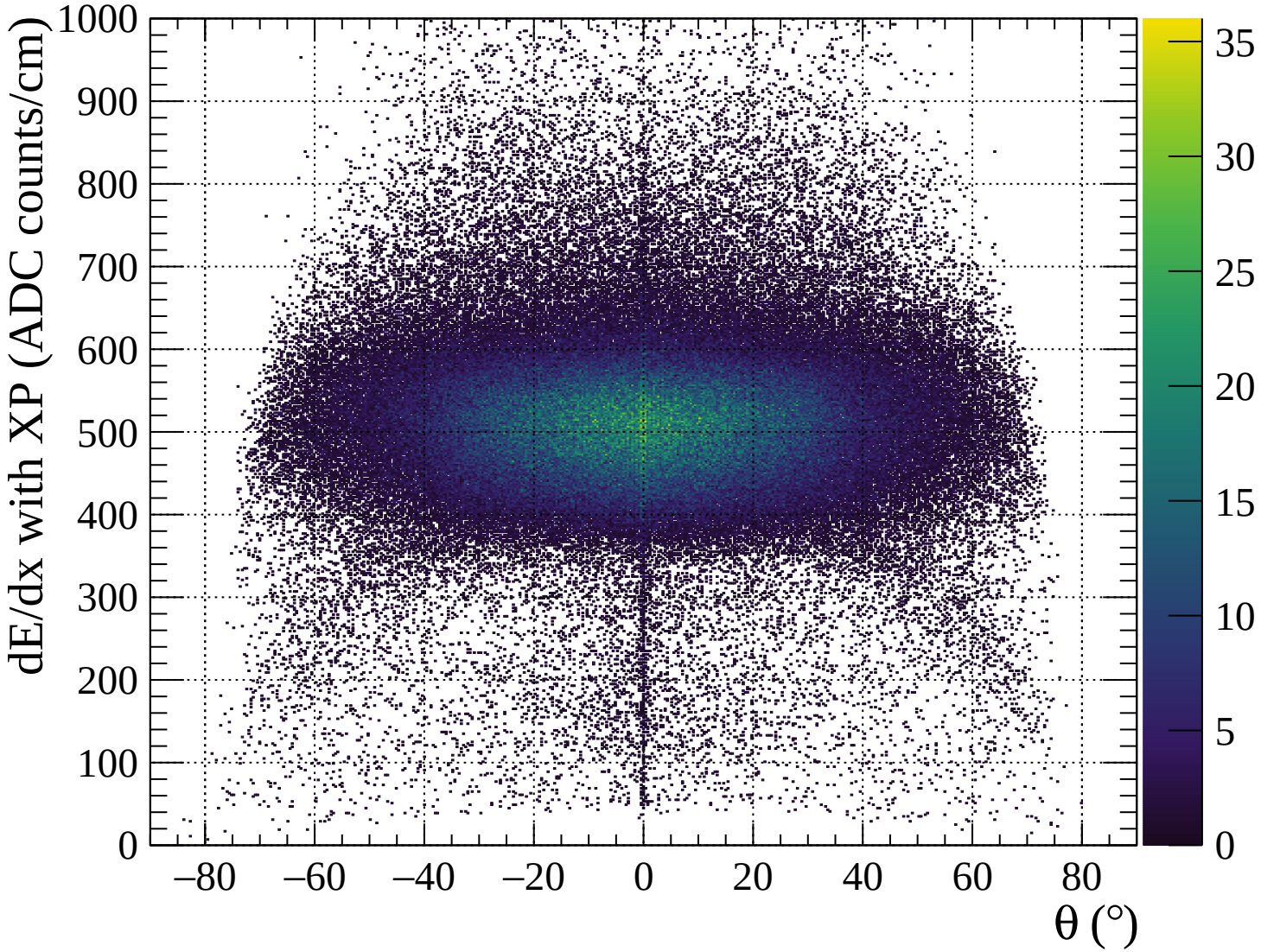


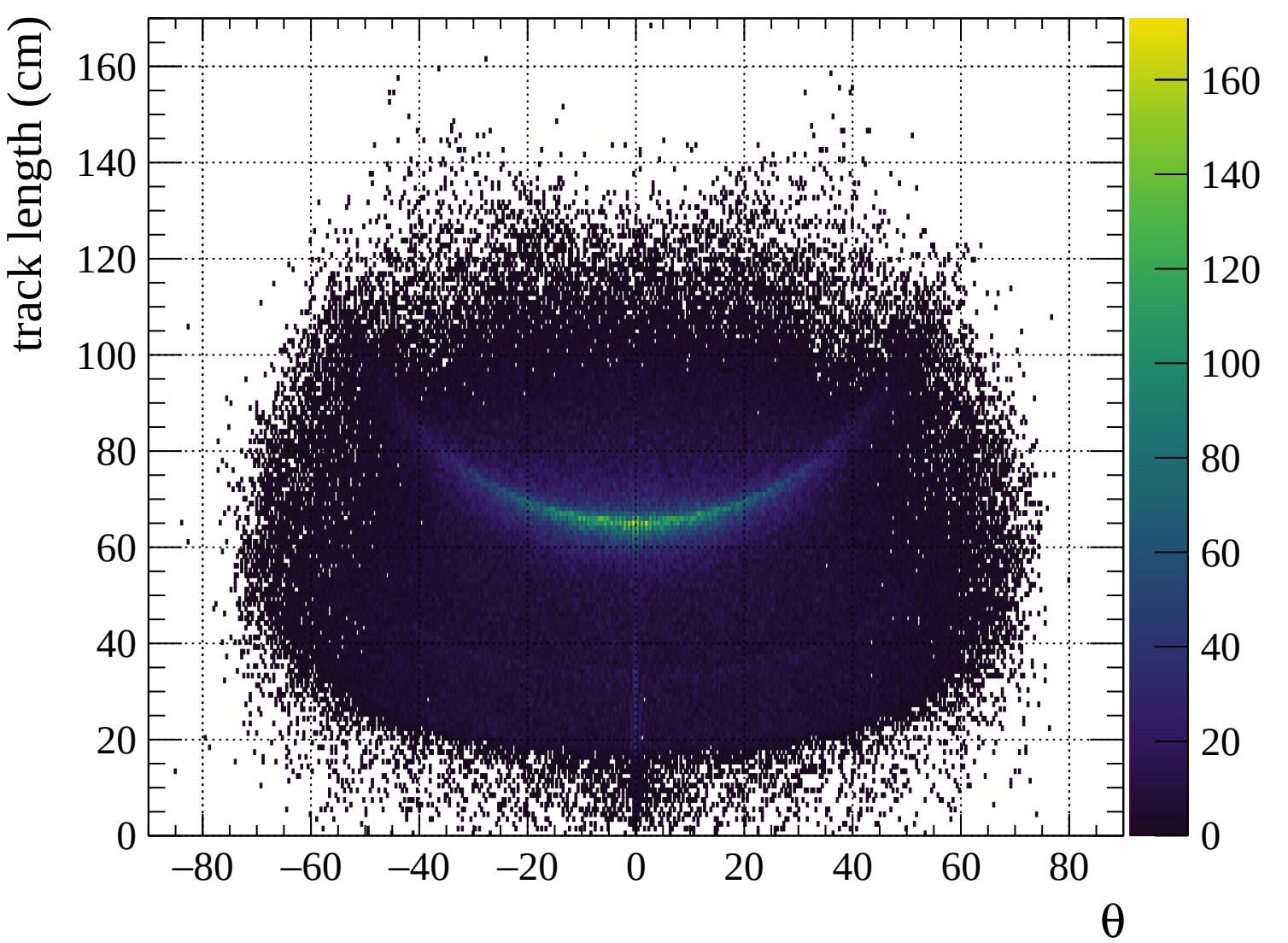


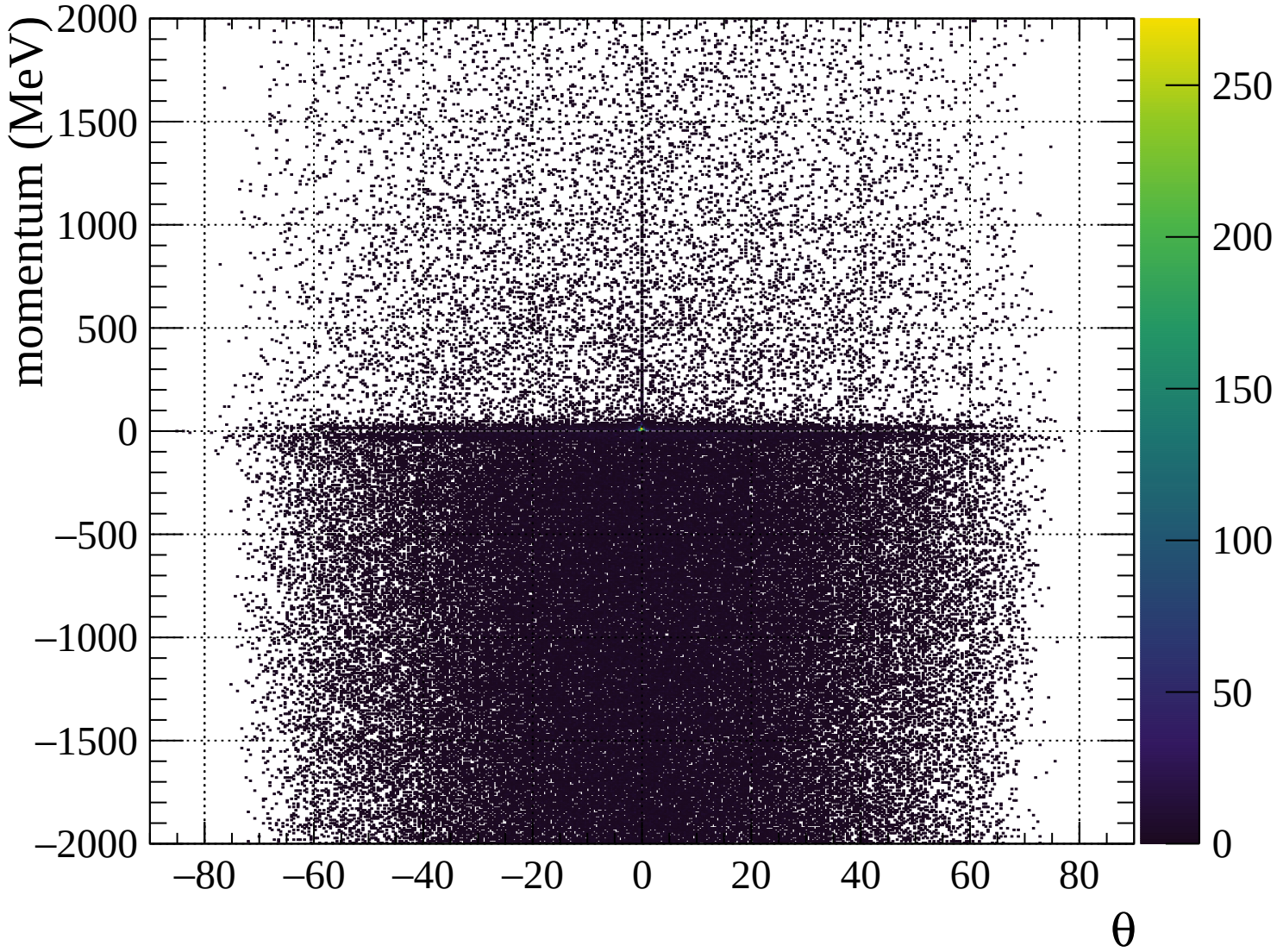


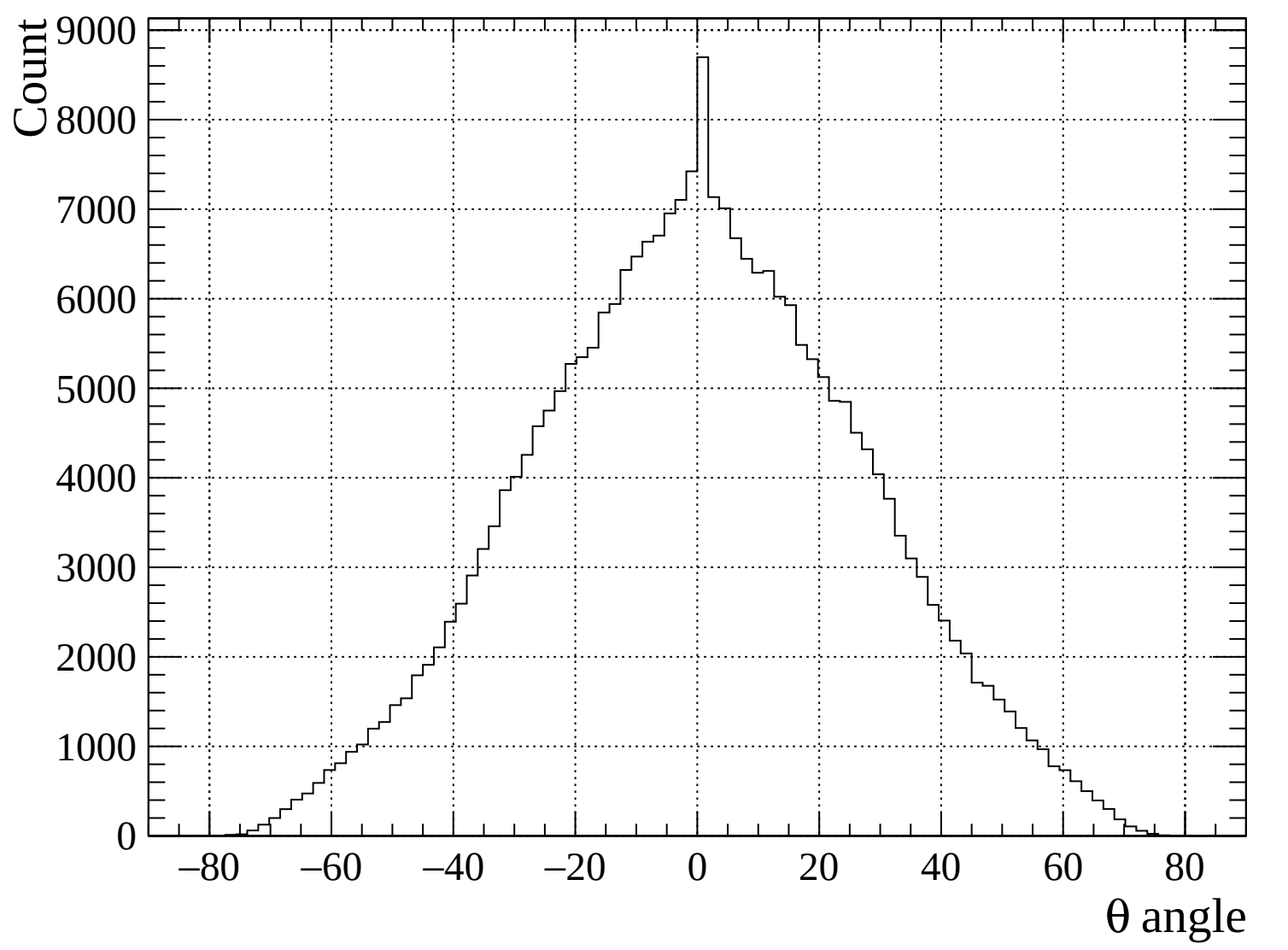


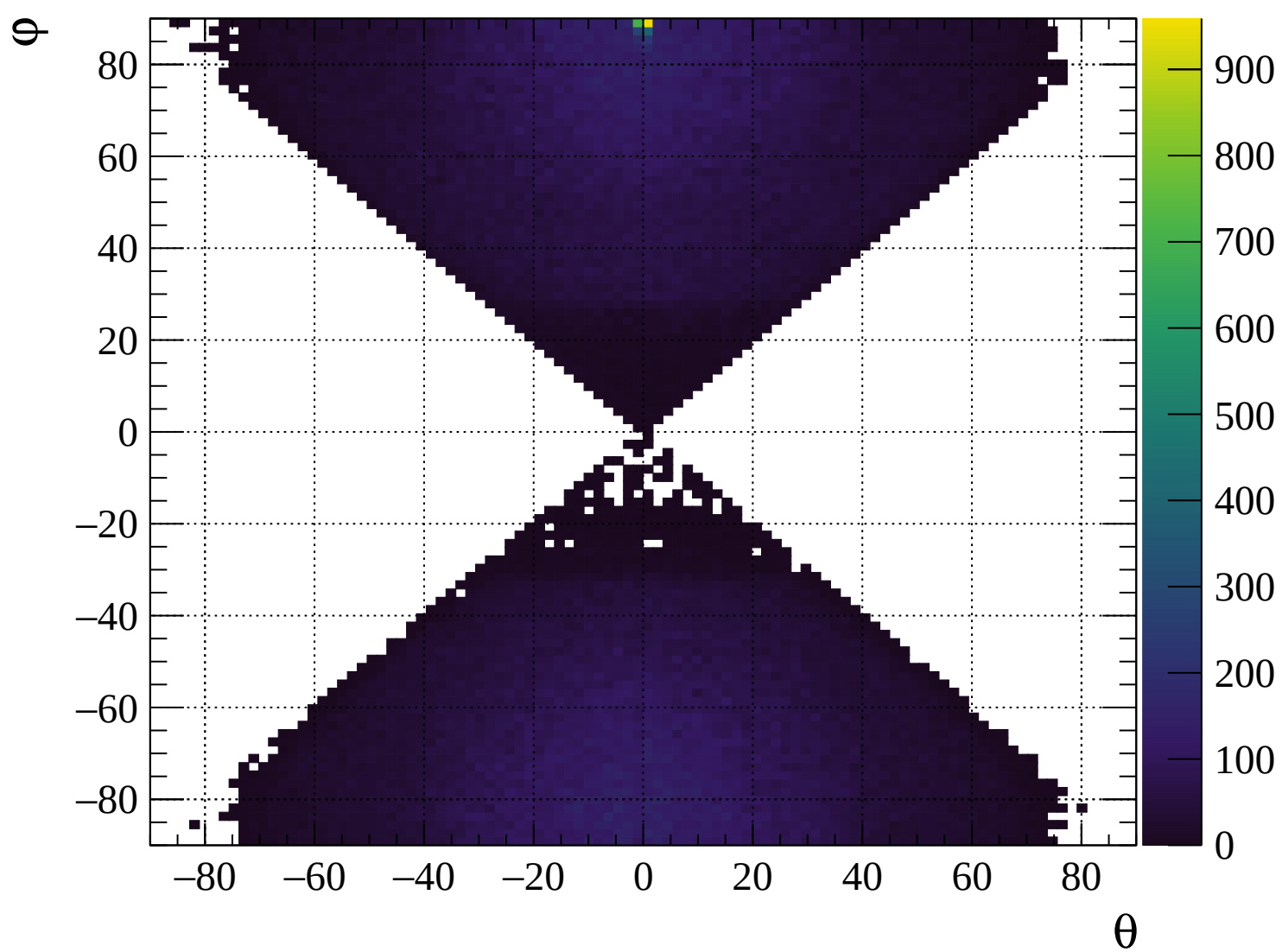


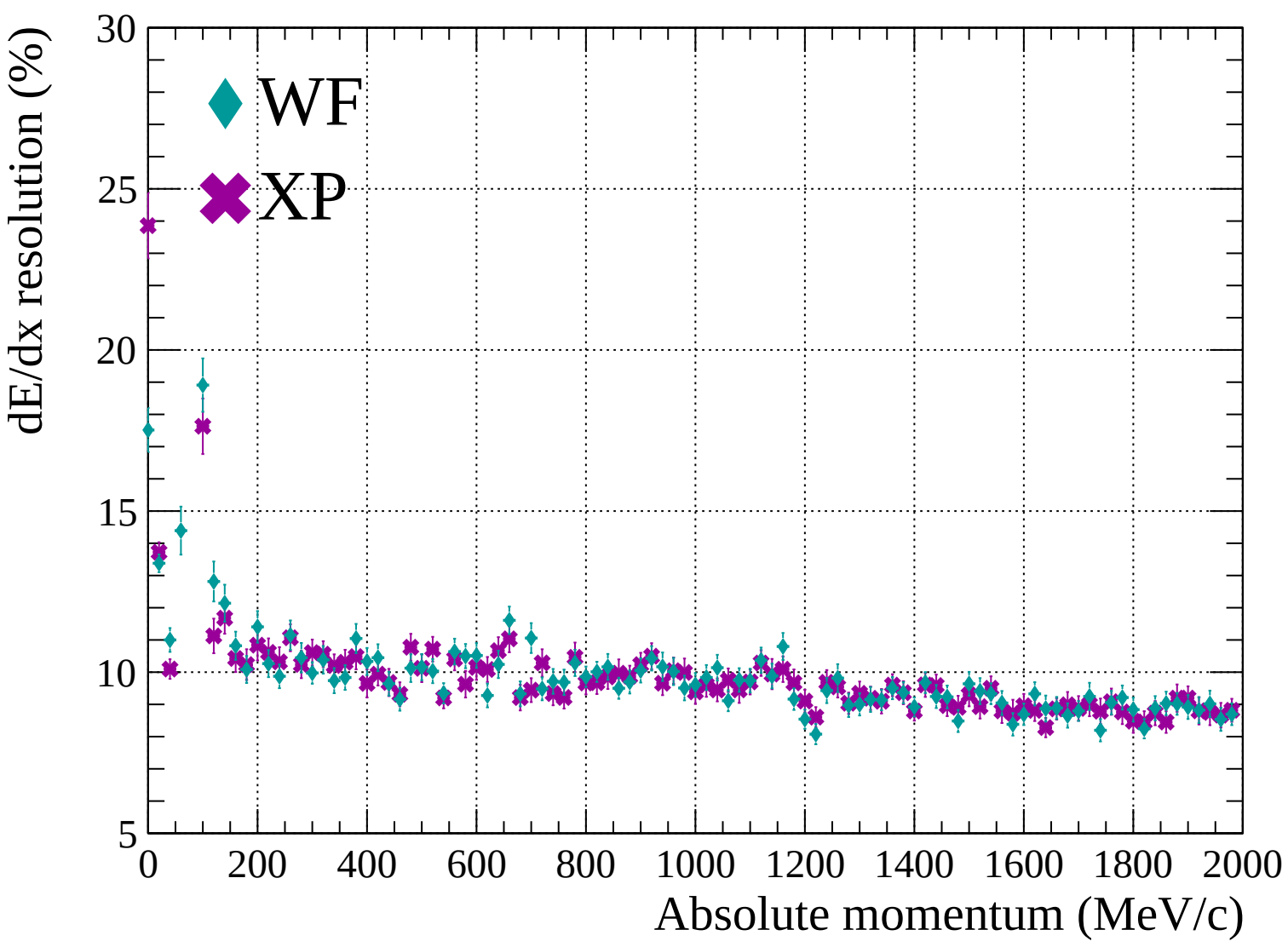


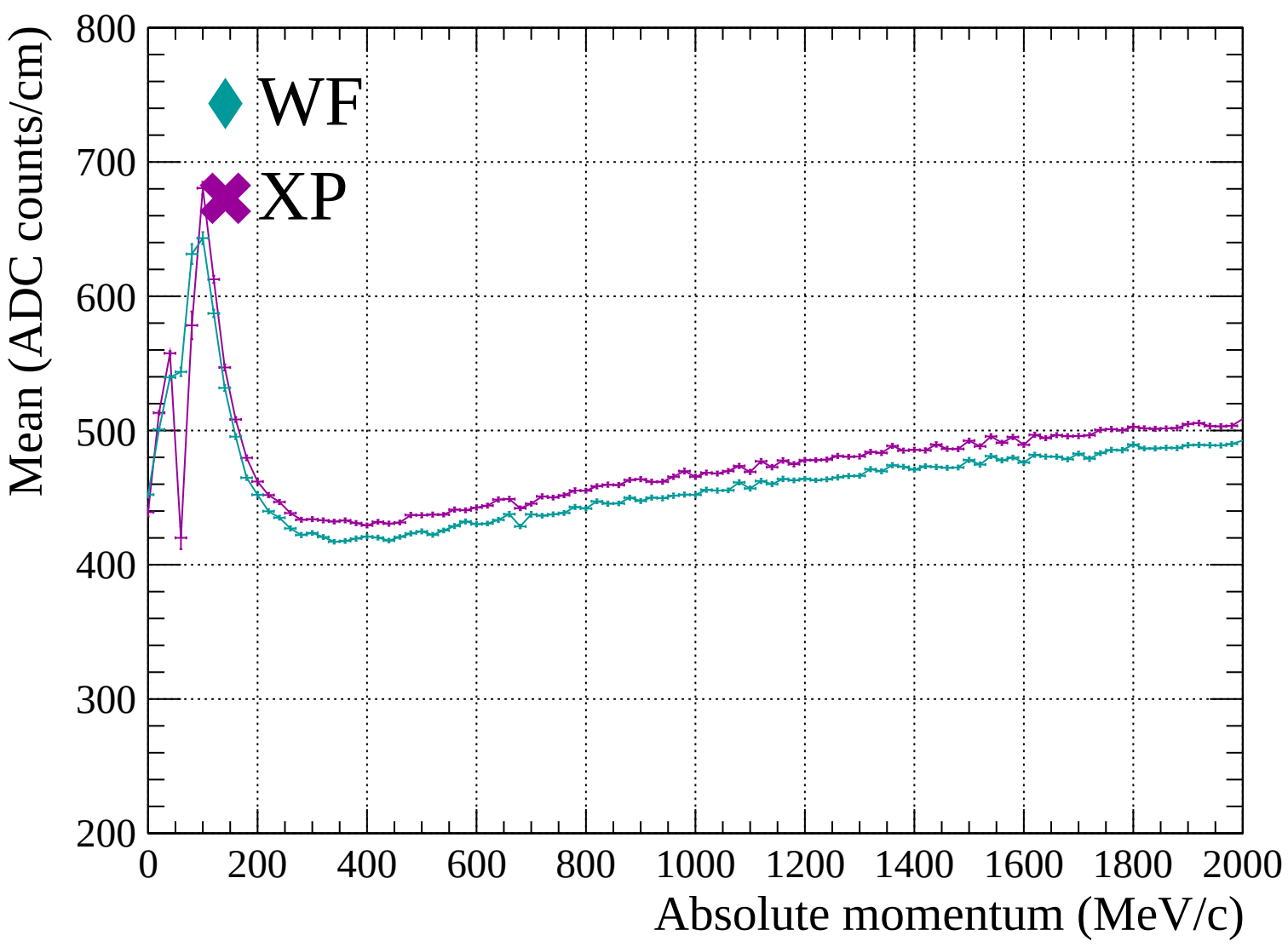


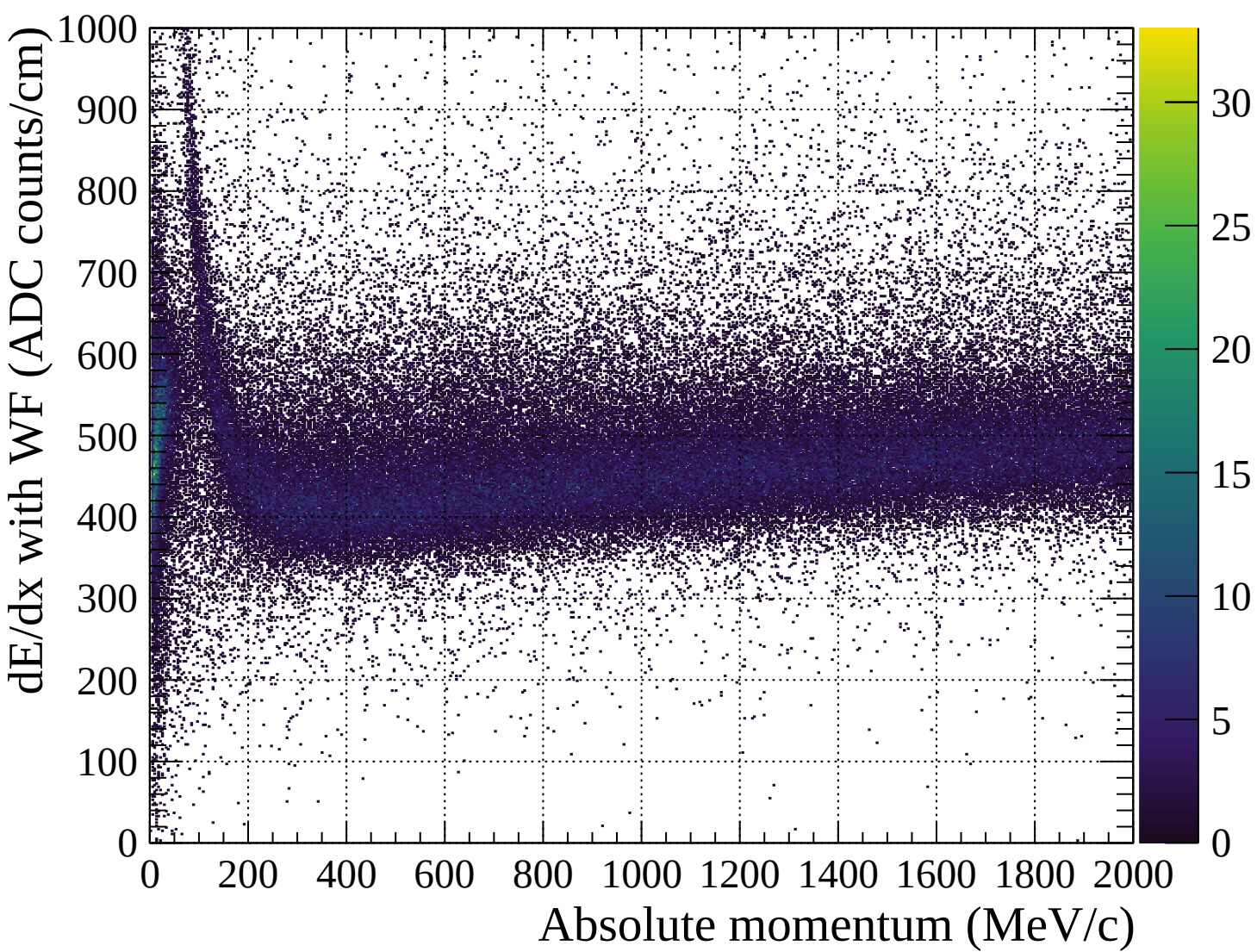


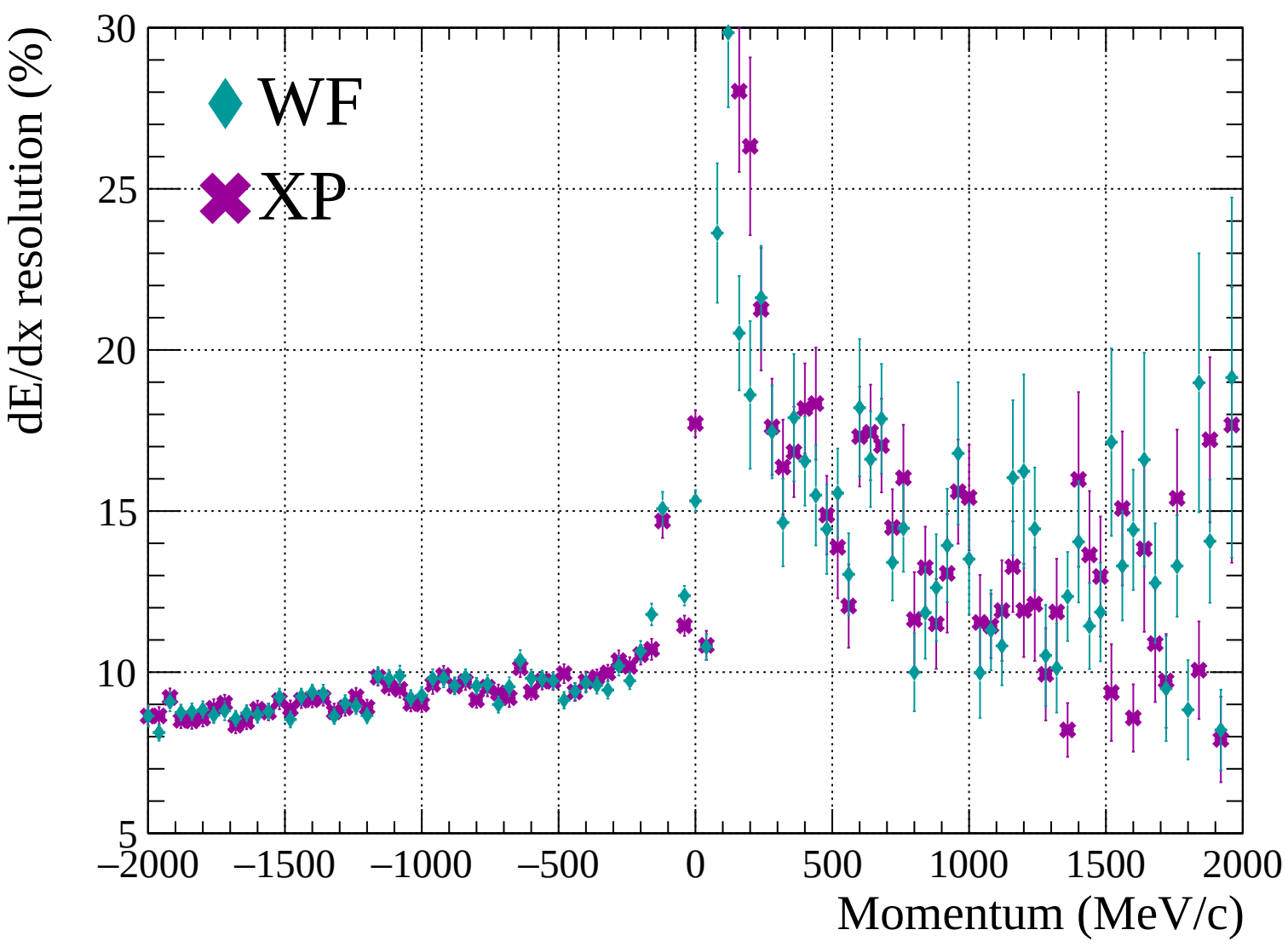


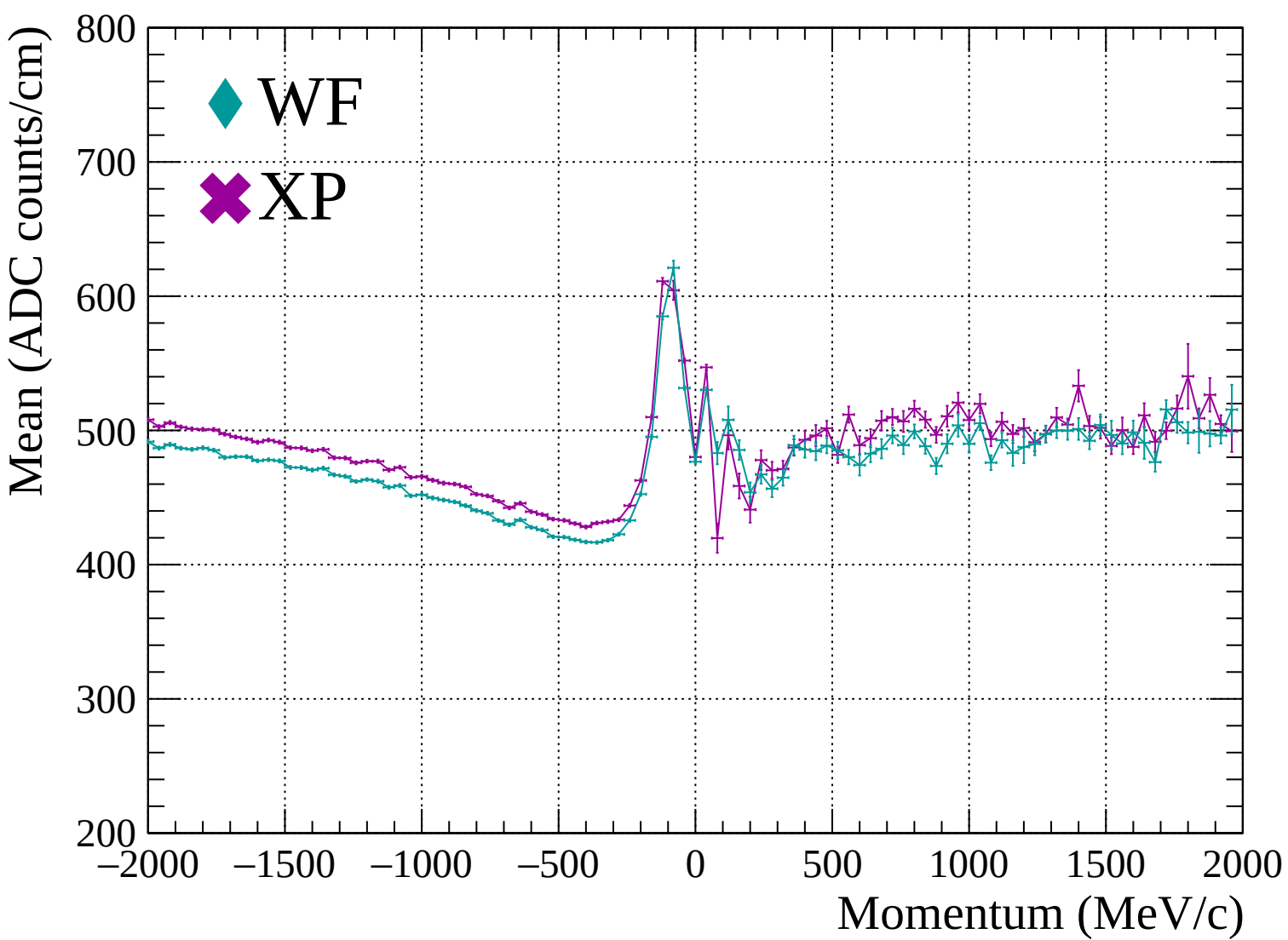


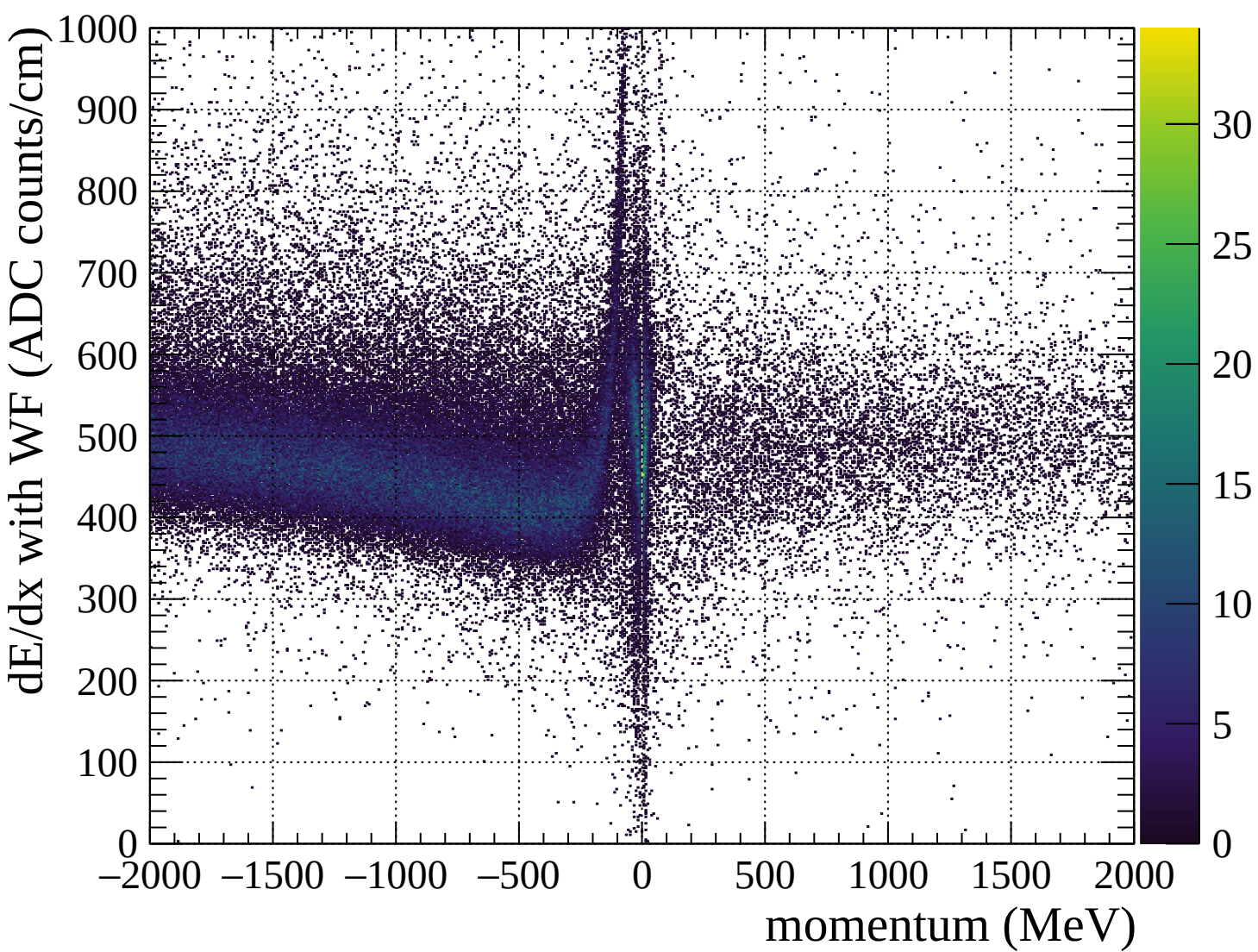


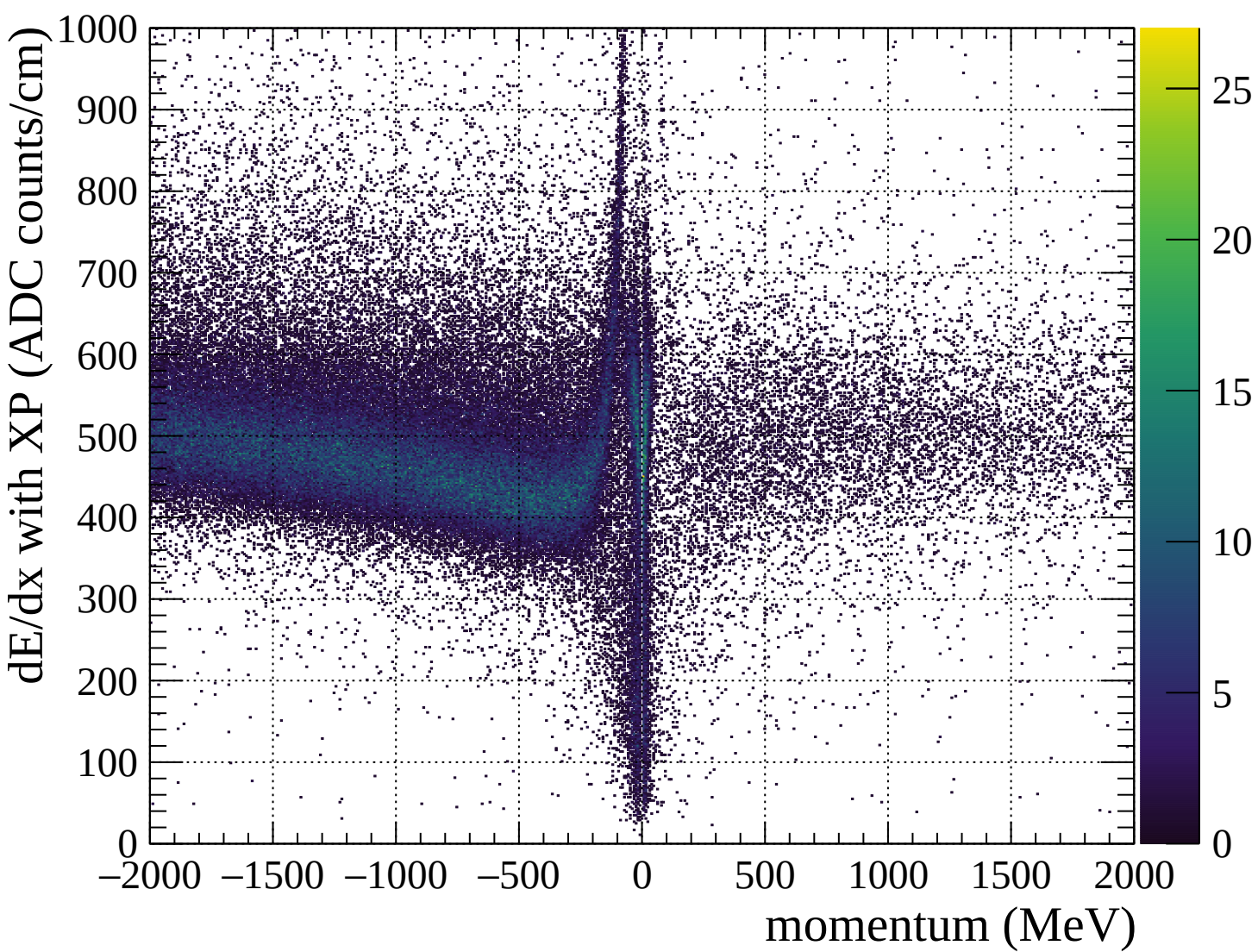


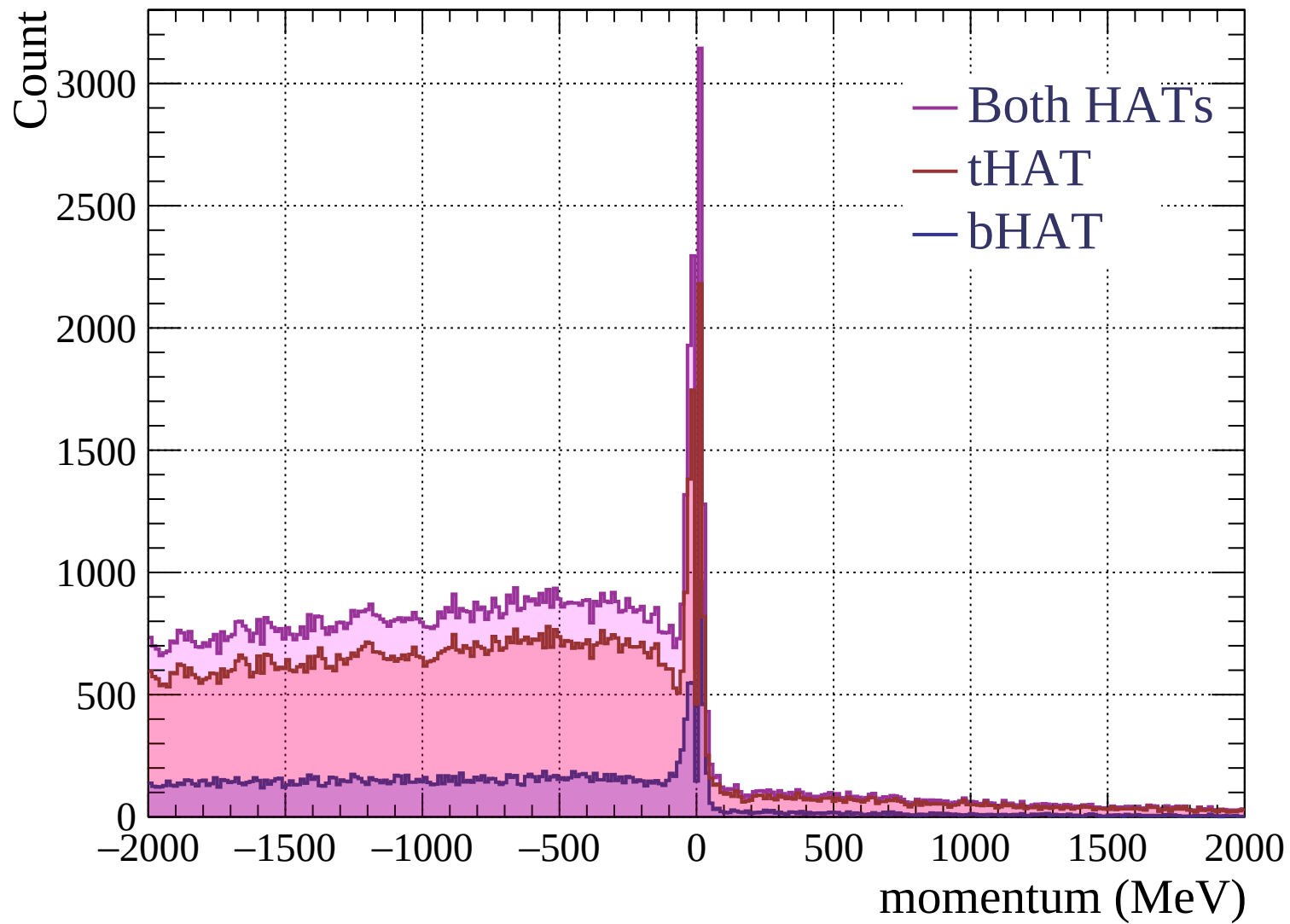


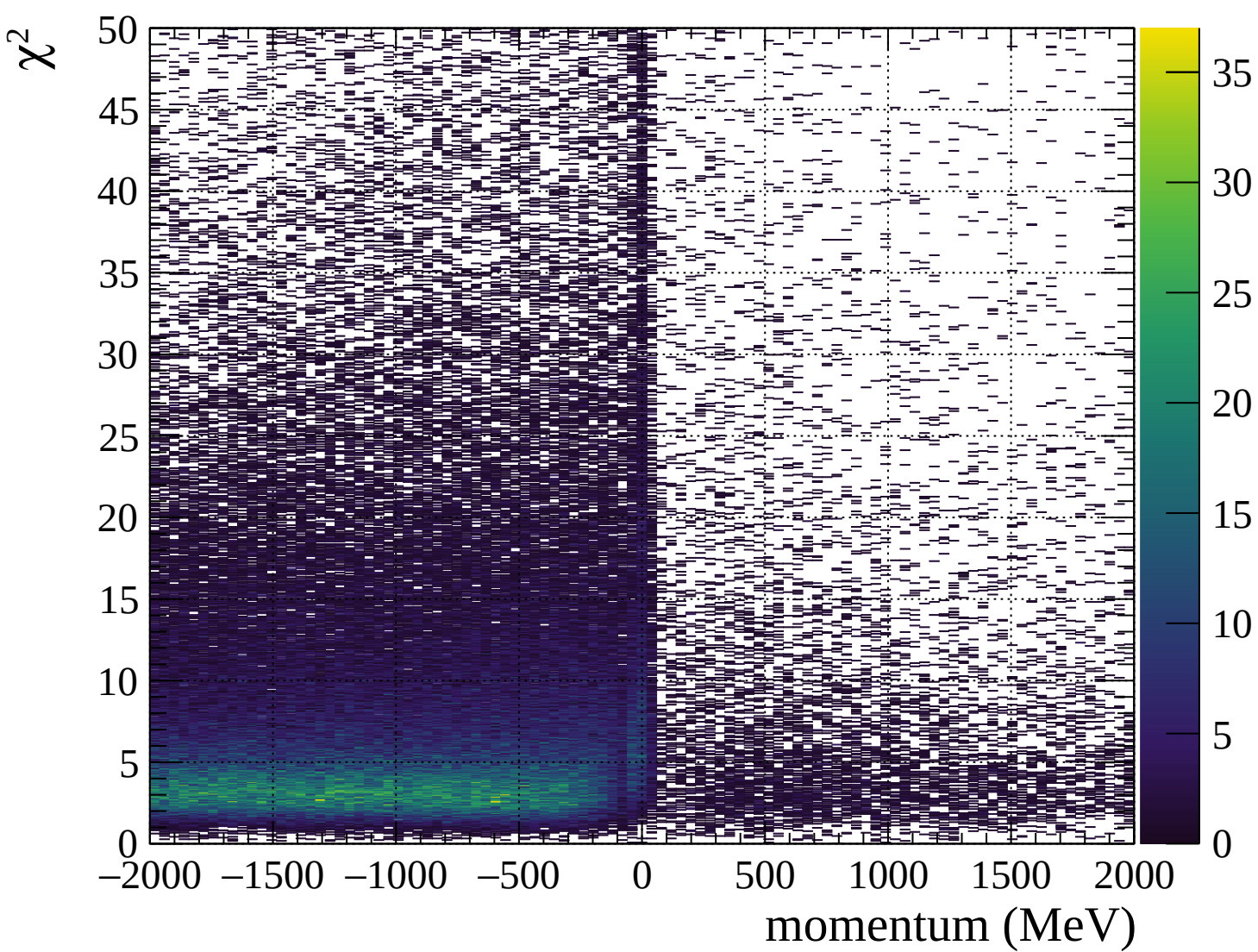




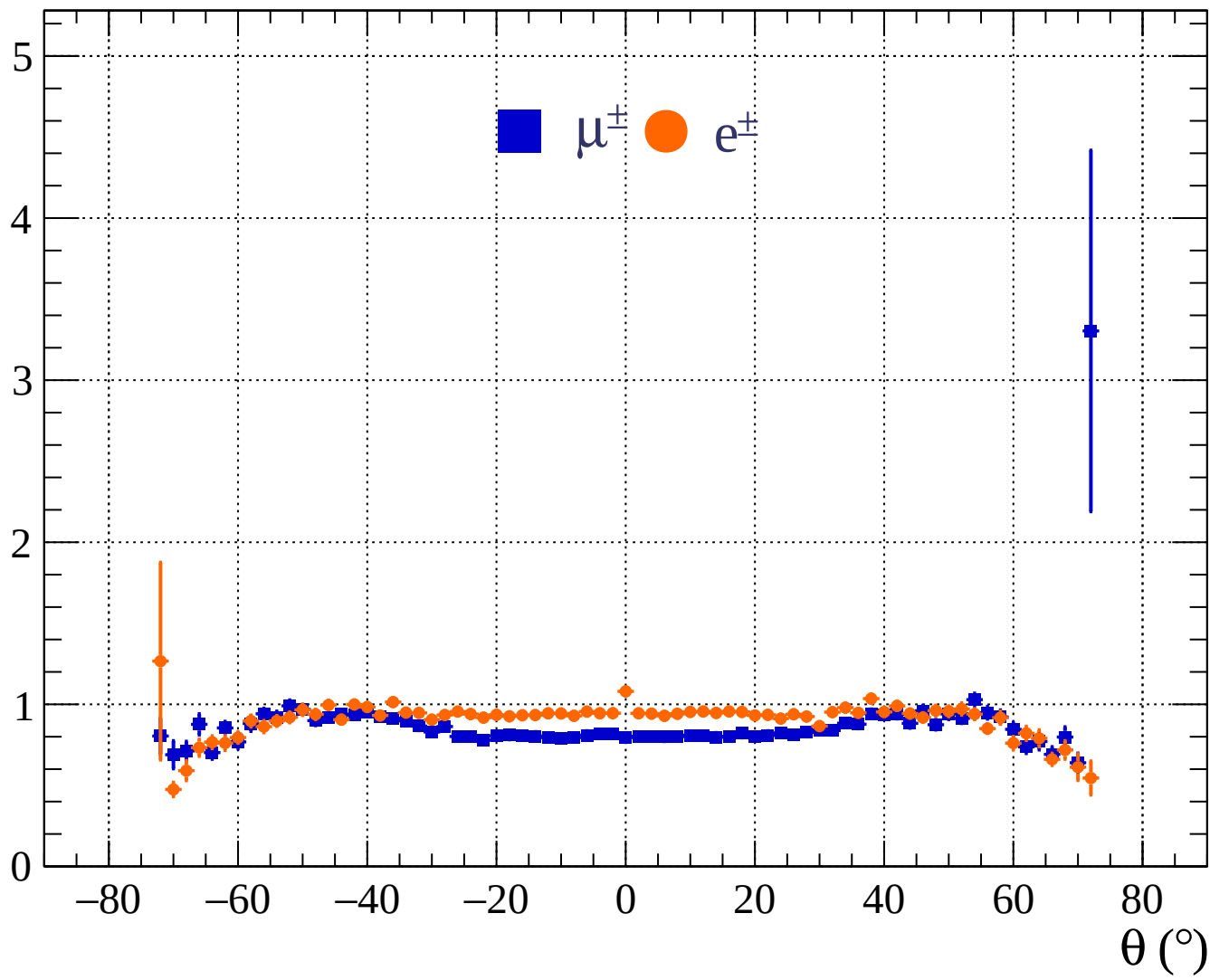


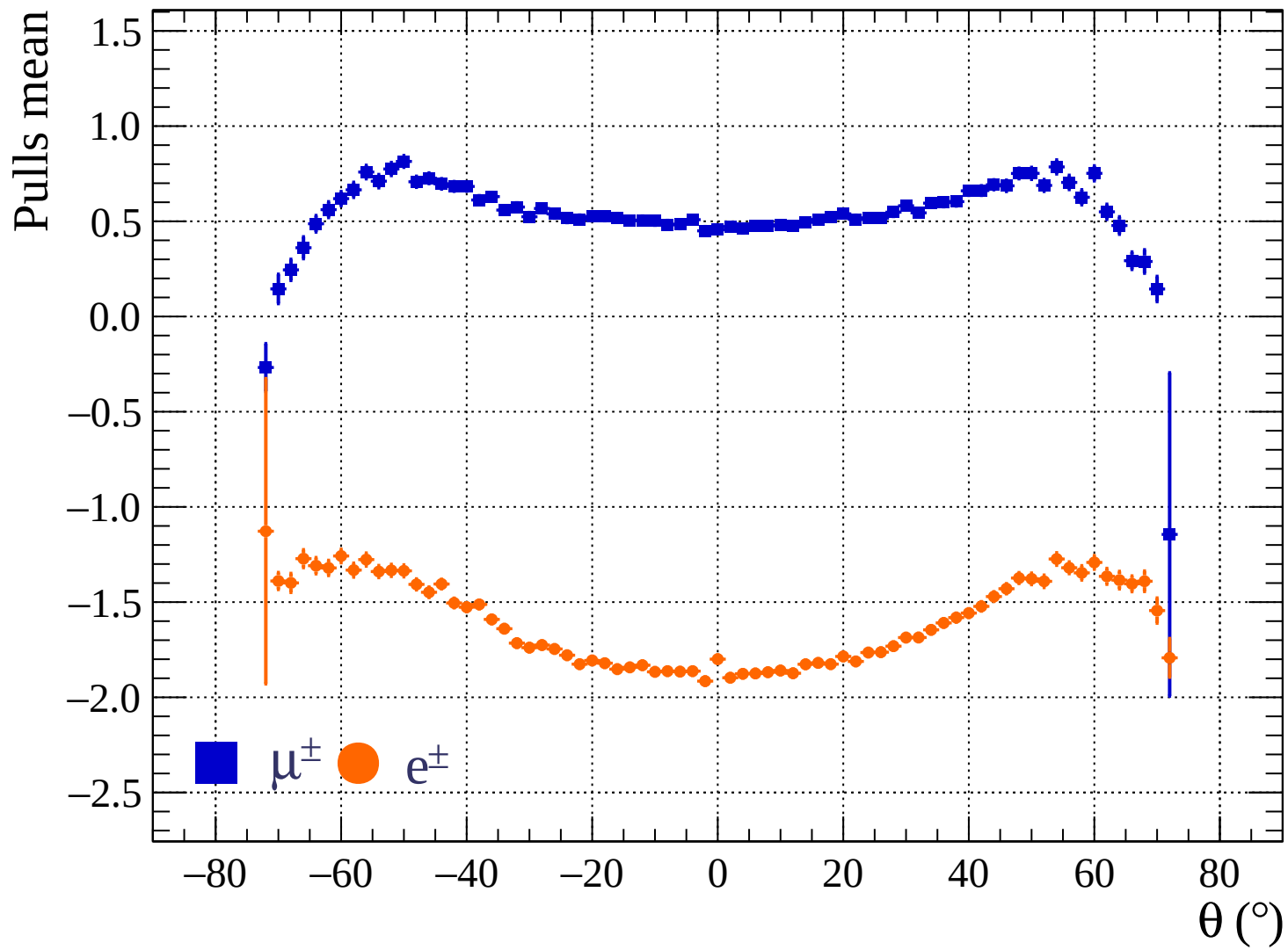






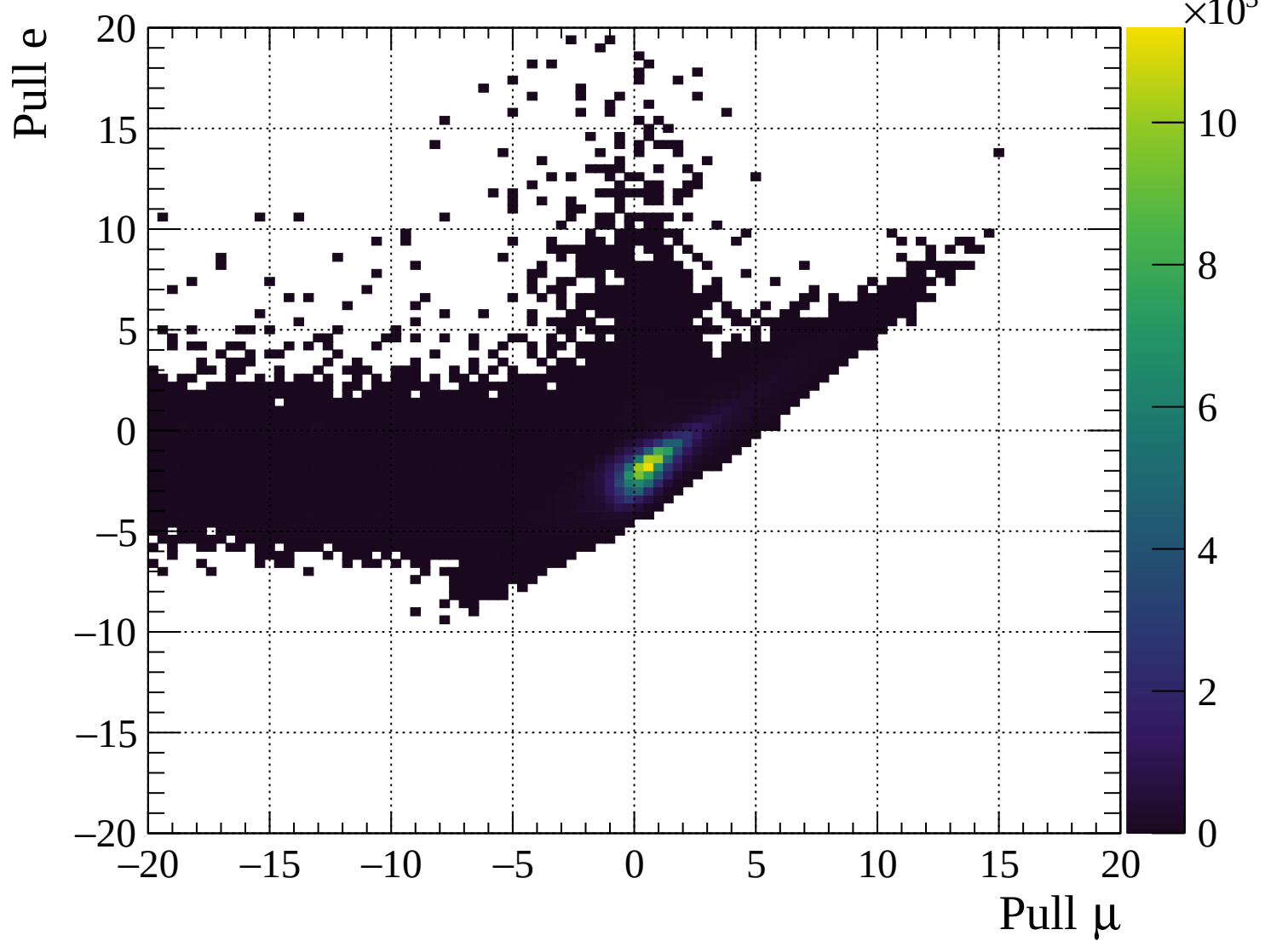
Pulls standard deviation

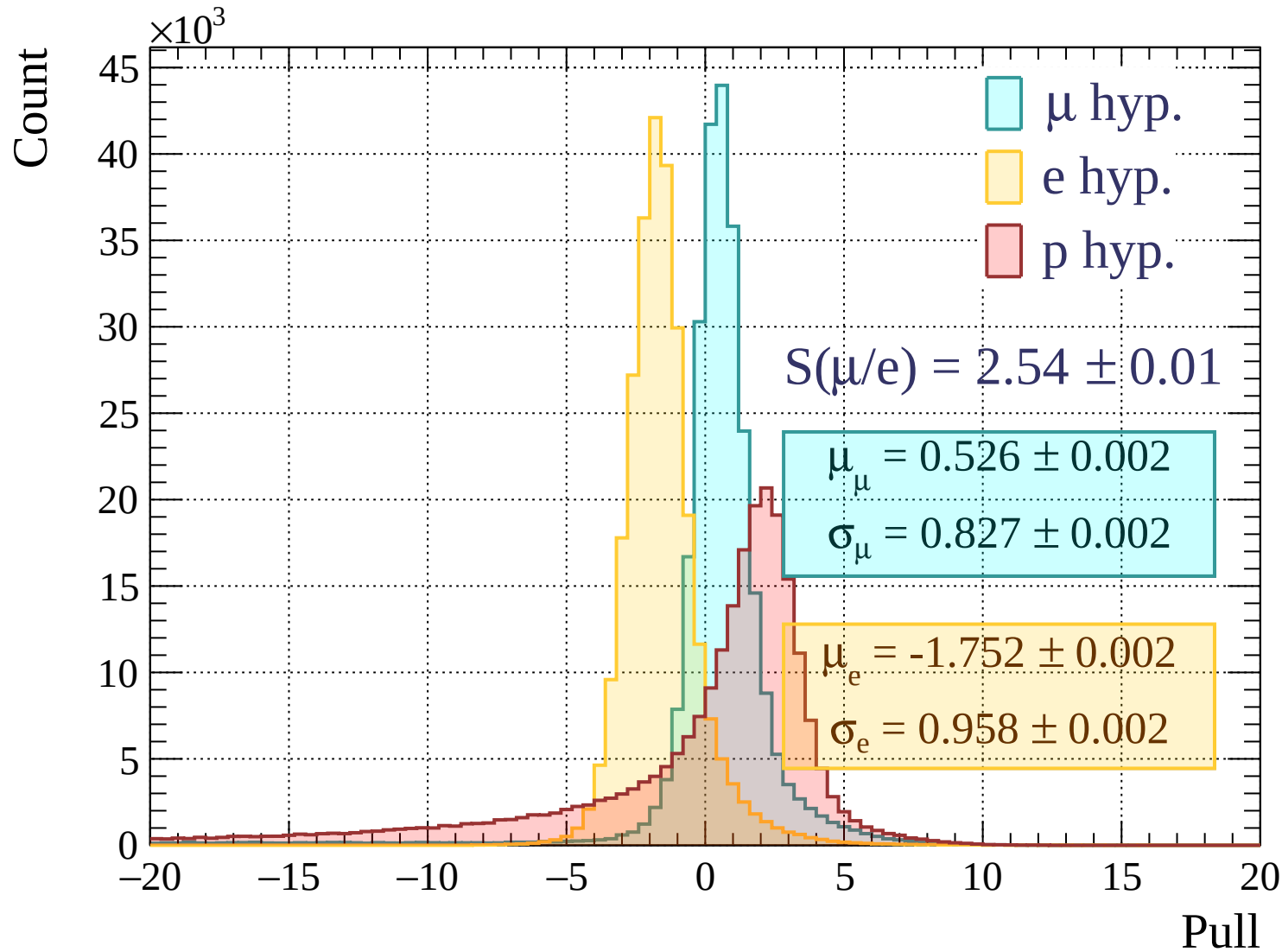


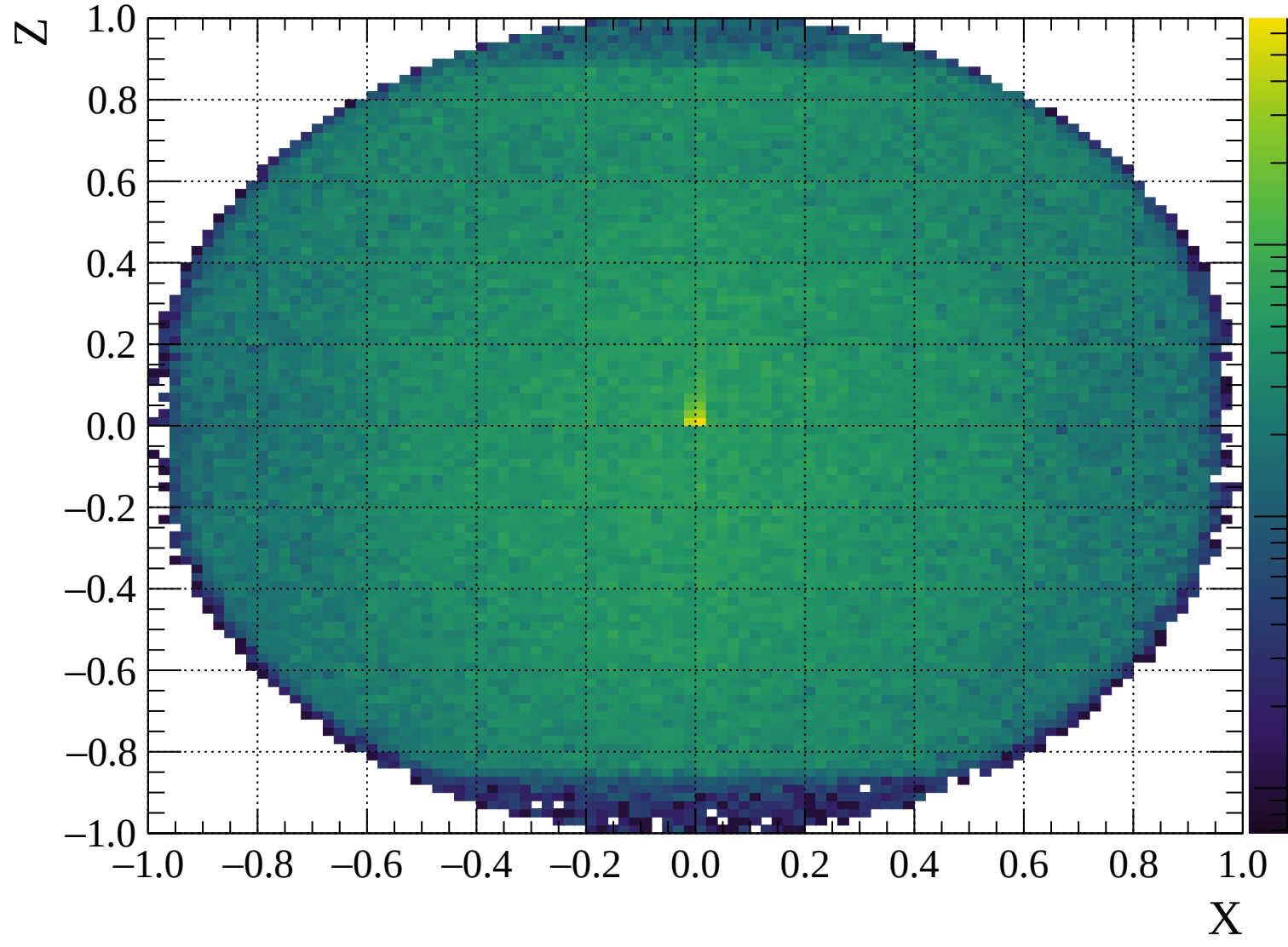


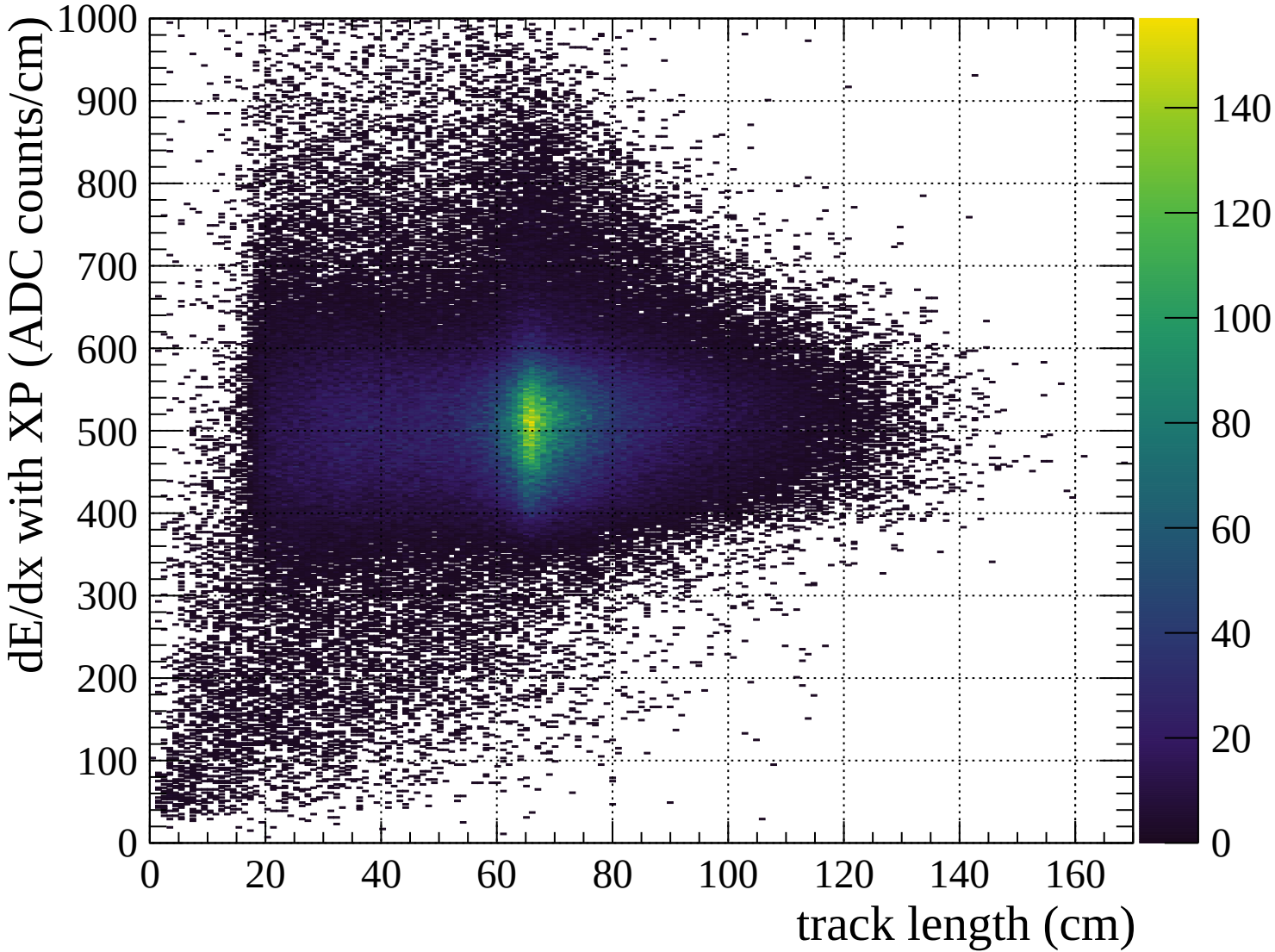
■ $\mu^\pm$ ● $e^\pm$

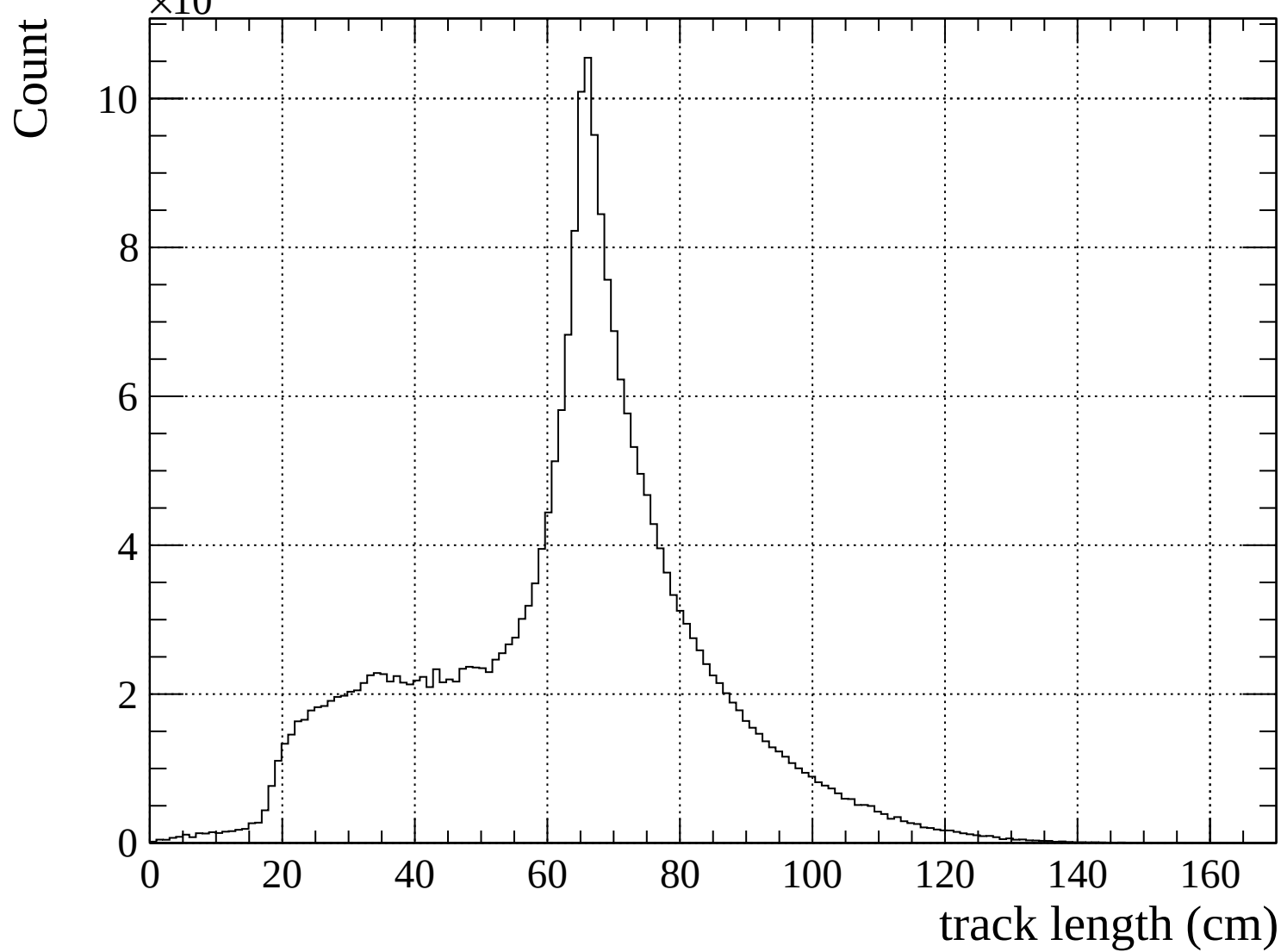
■ $\mu^\pm$ ● $e^\pm$

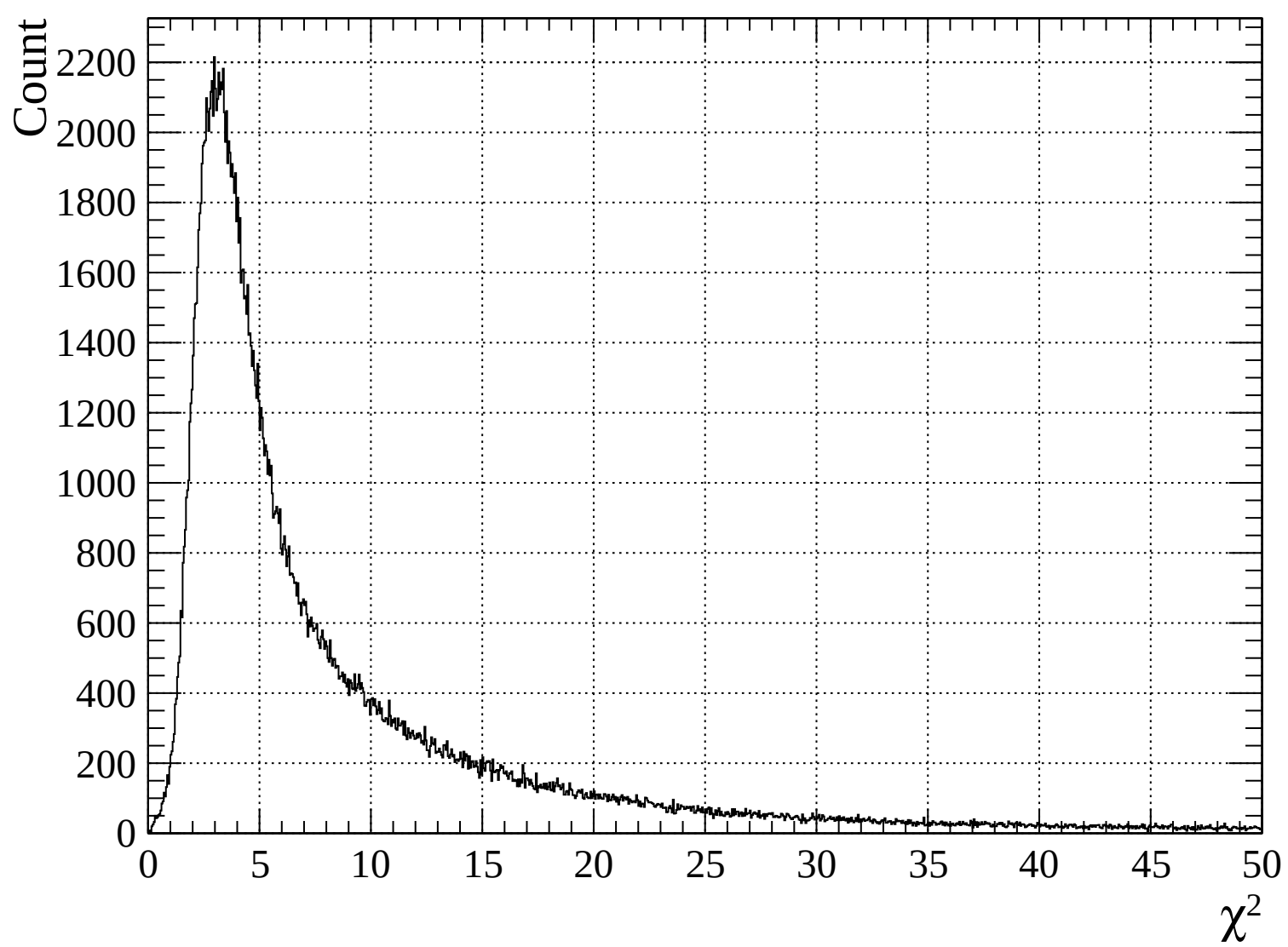


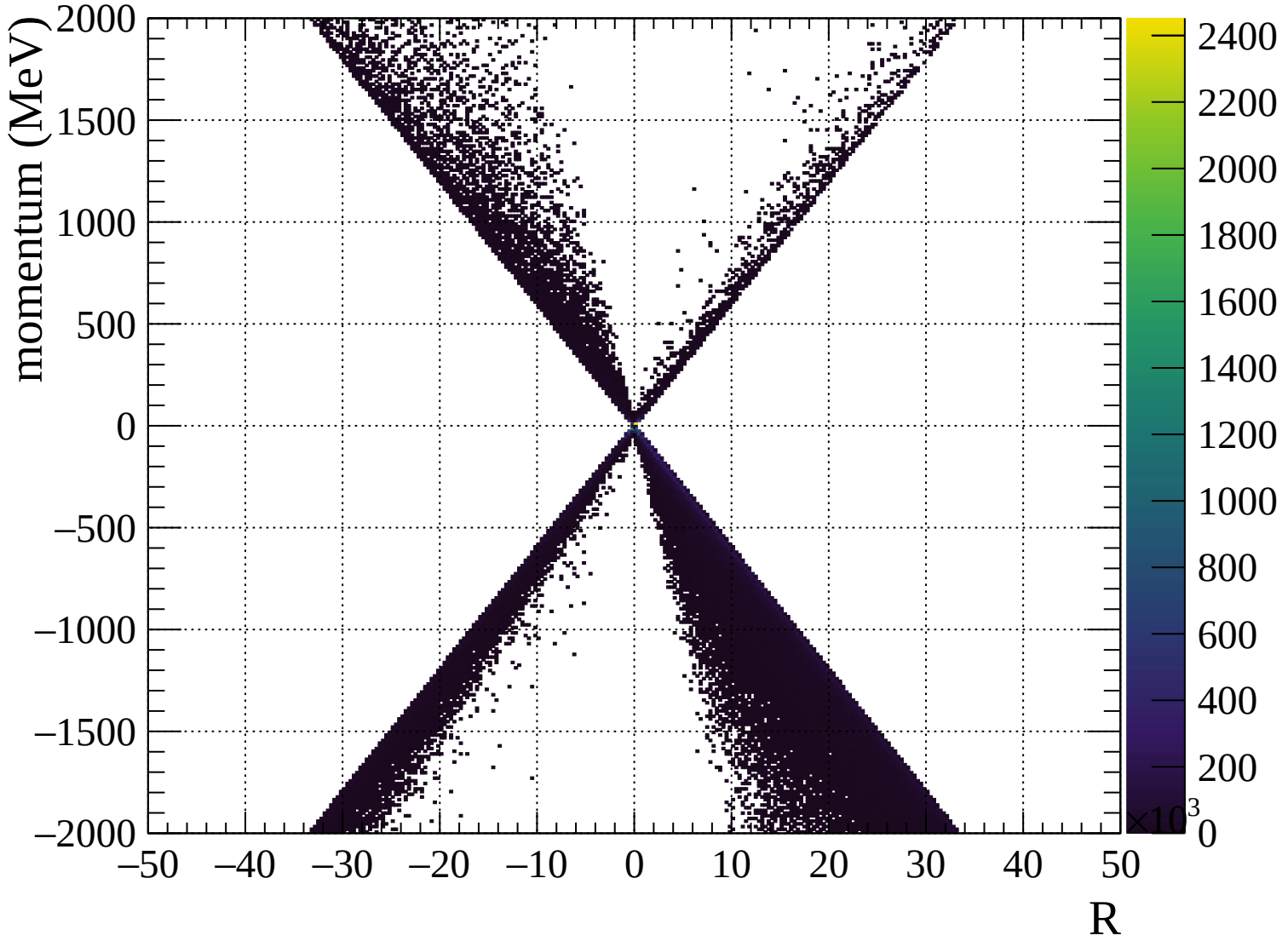


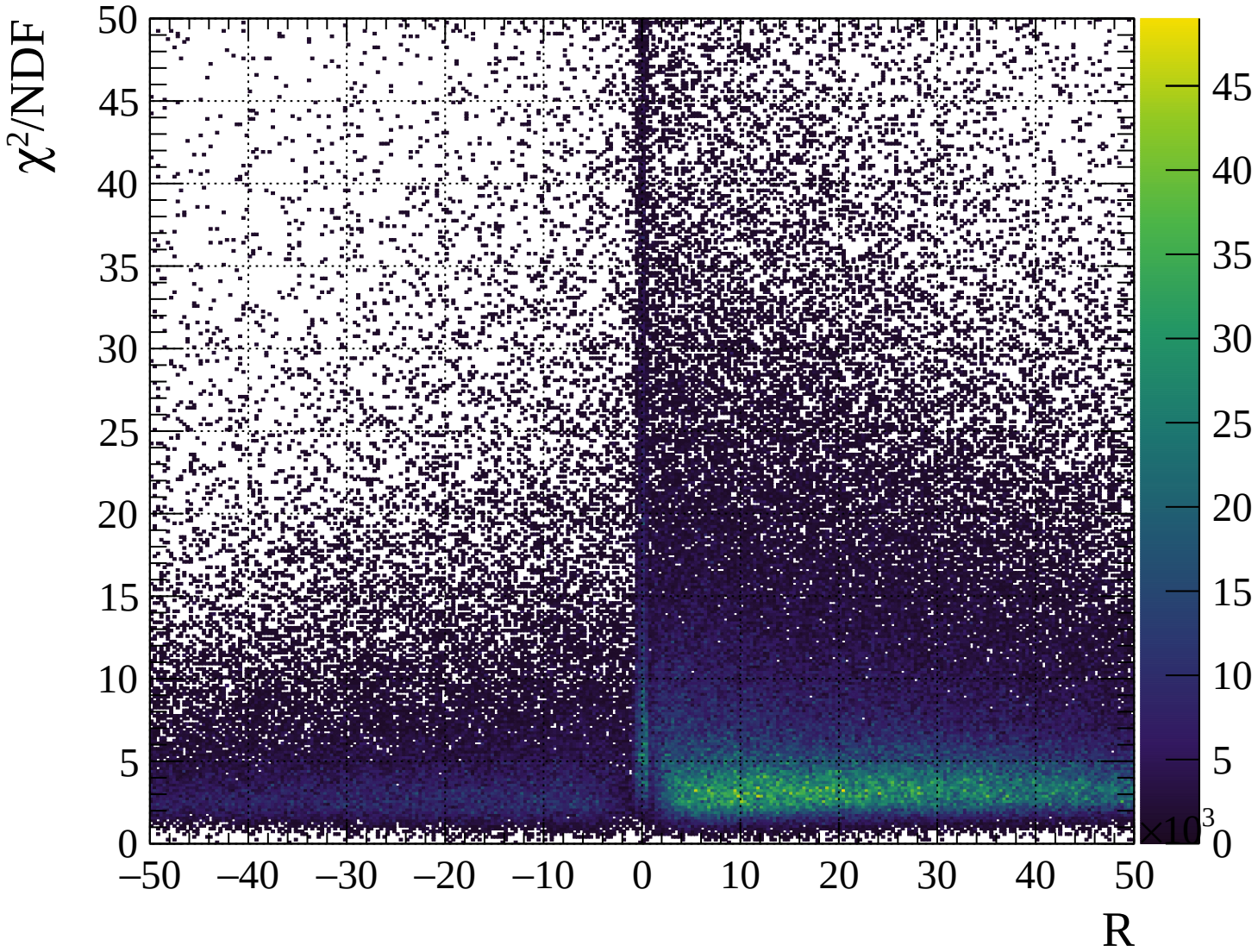




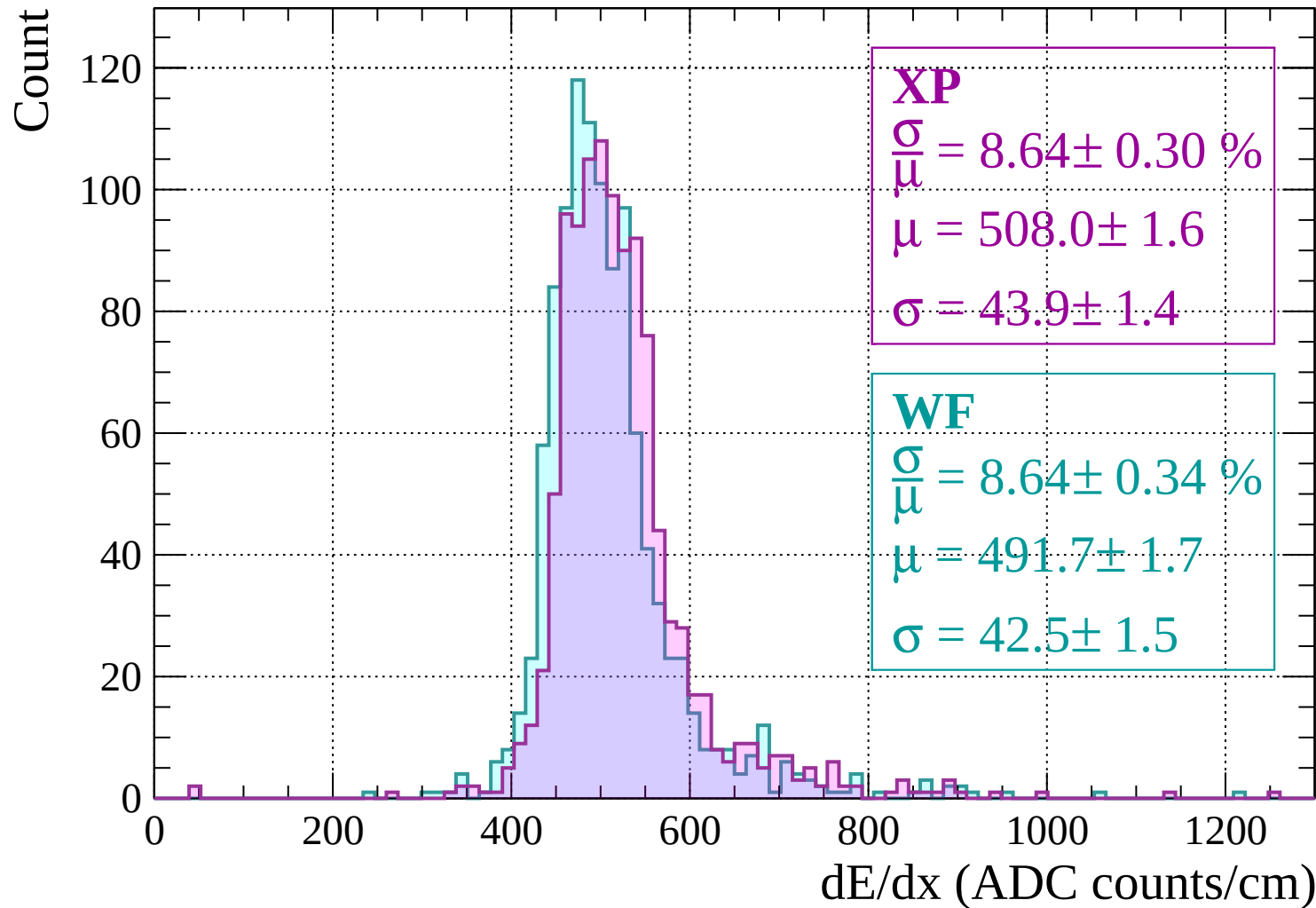






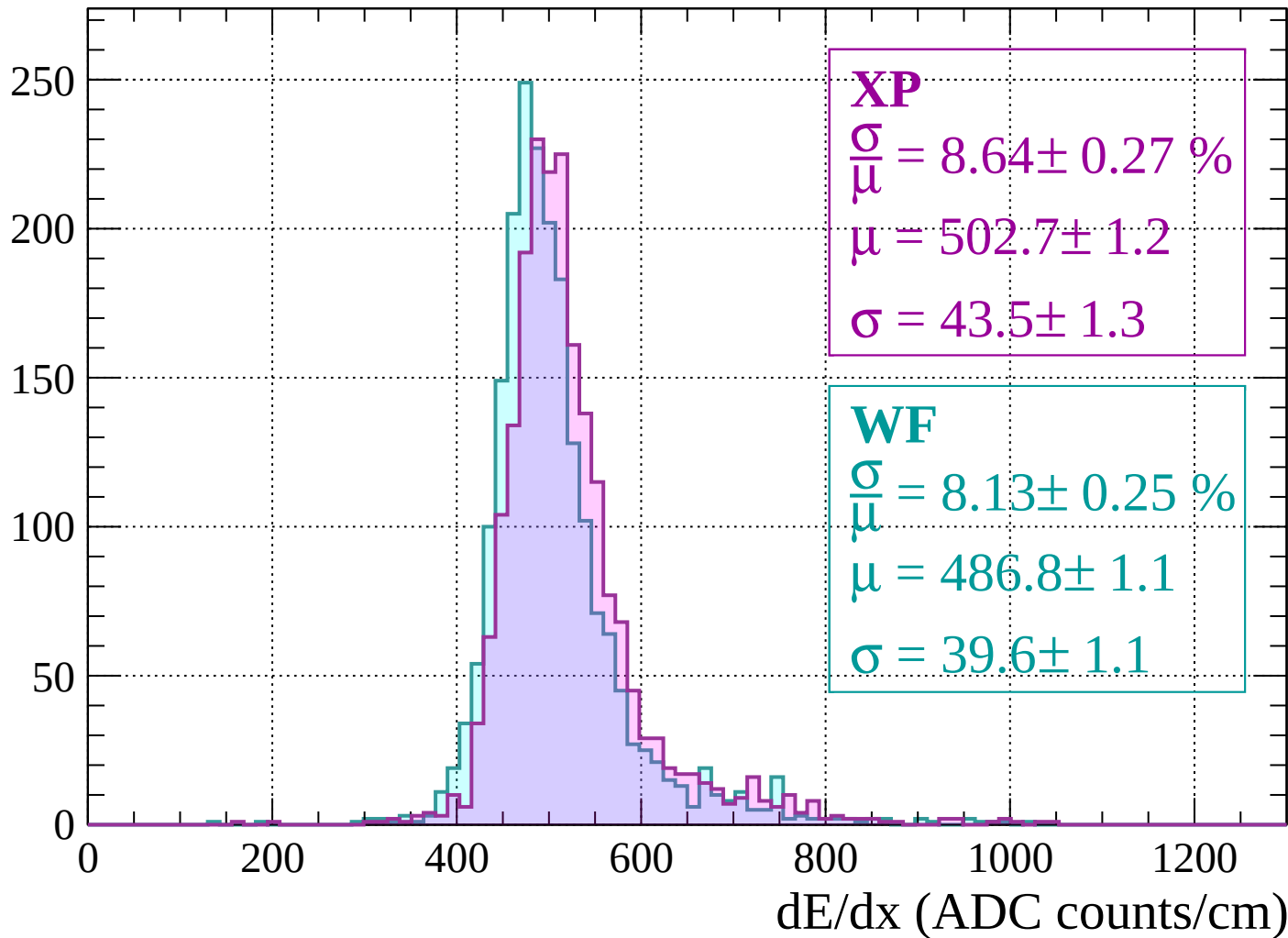


Energy loss | $-2000 < p < -1960$

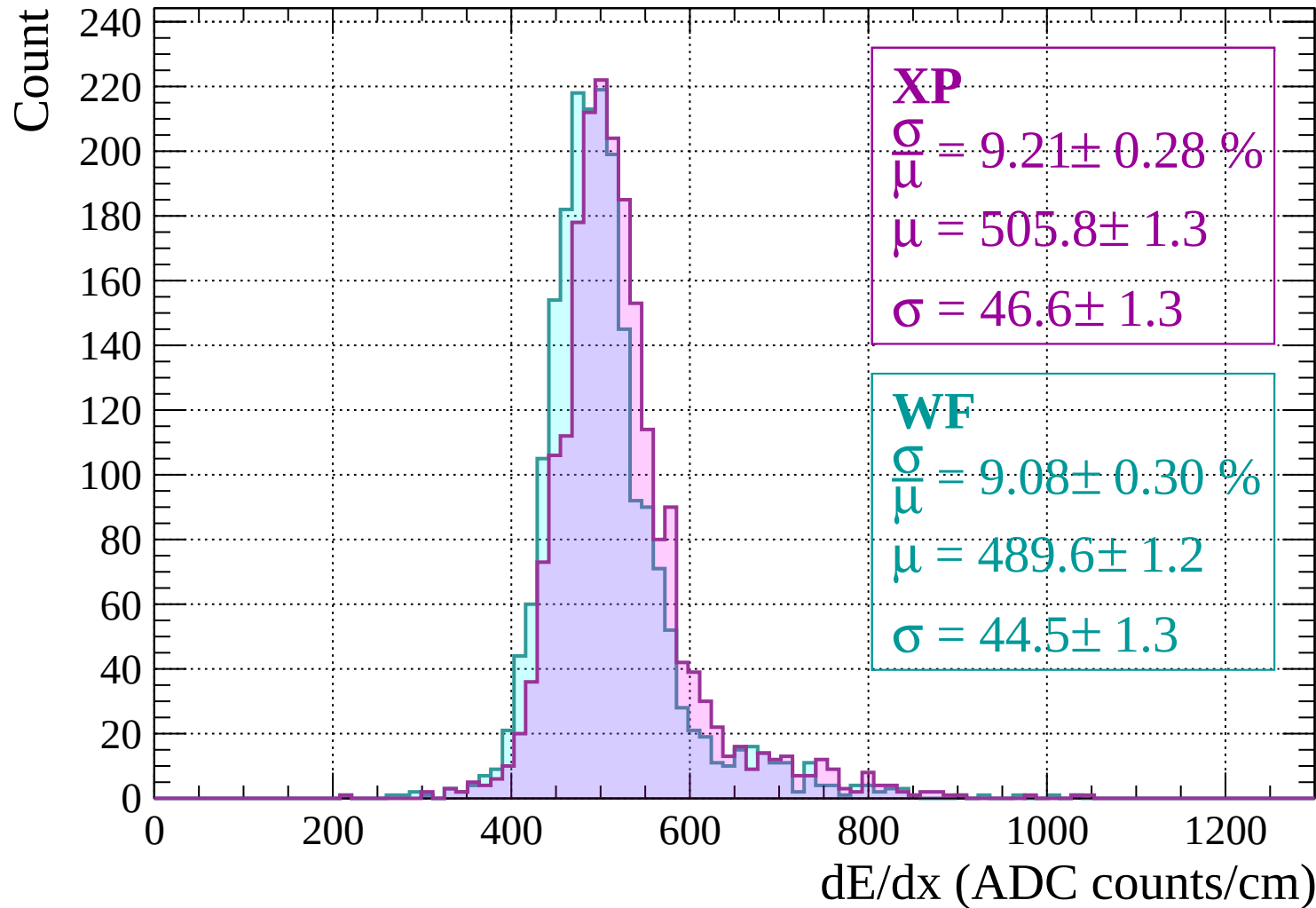


Energy loss | -1960 < p < -1920

Count

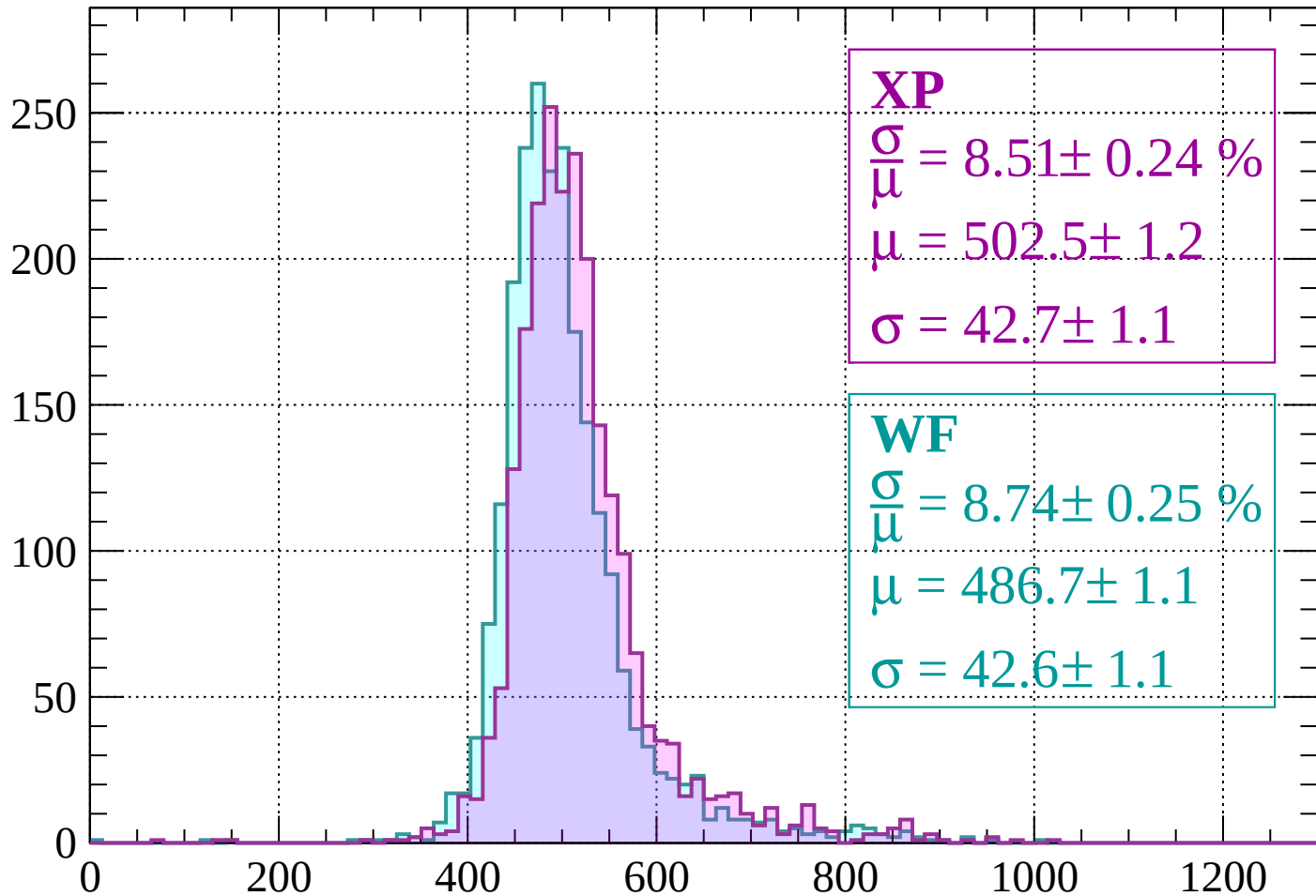


Energy loss | -1920 < p < -1880



Energy loss | $-1880 < p < -1840$

Count



XP

$$\frac{\sigma}{\mu} = 8.51 \pm 0.24 \%$$

$$\mu = 502.5 \pm 1.2$$

$$\sigma = 42.7 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 8.74 \pm 0.25 \%$$

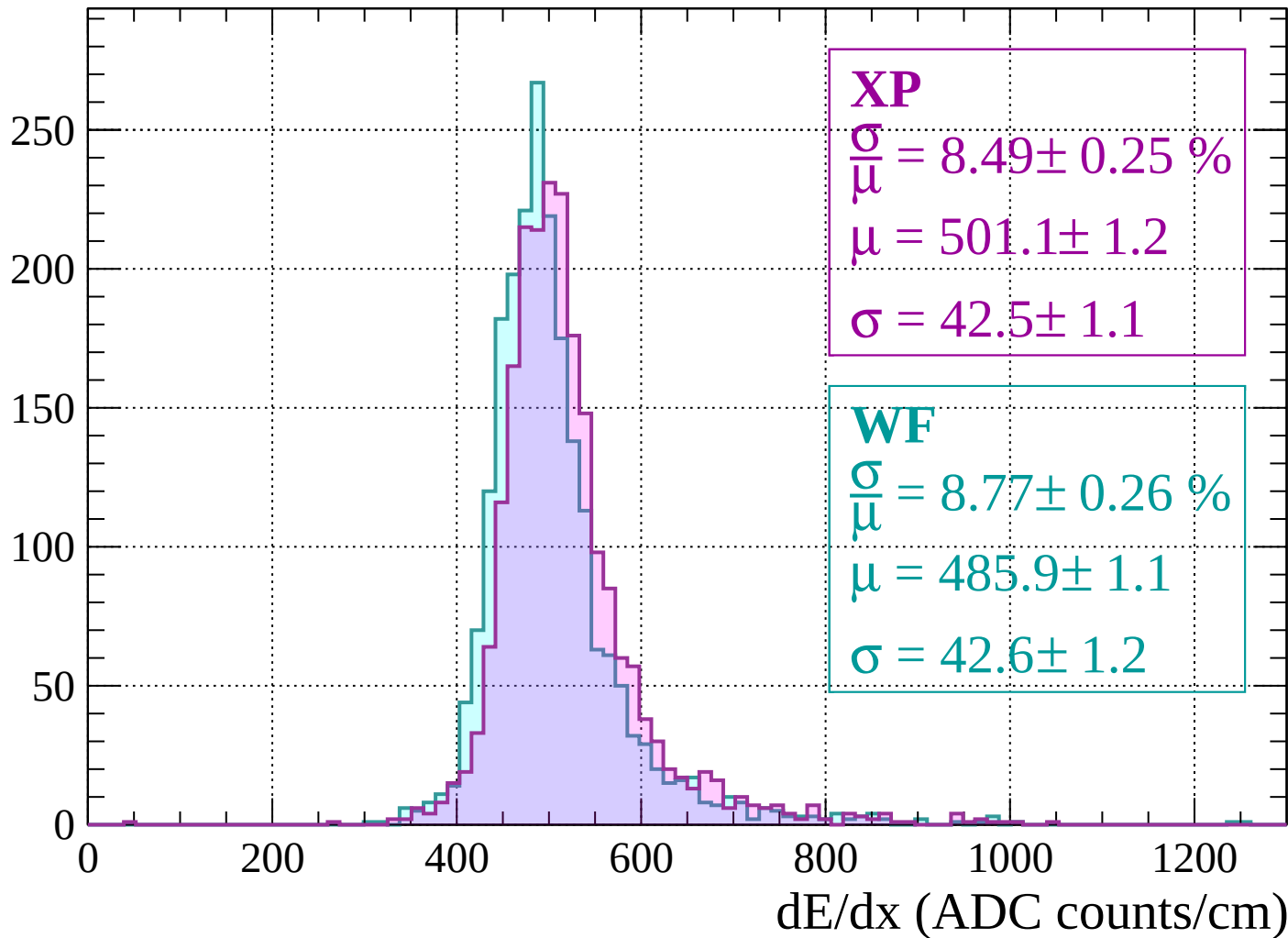
$$\mu = 486.7 \pm 1.1$$

$$\sigma = 42.6 \pm 1.1$$

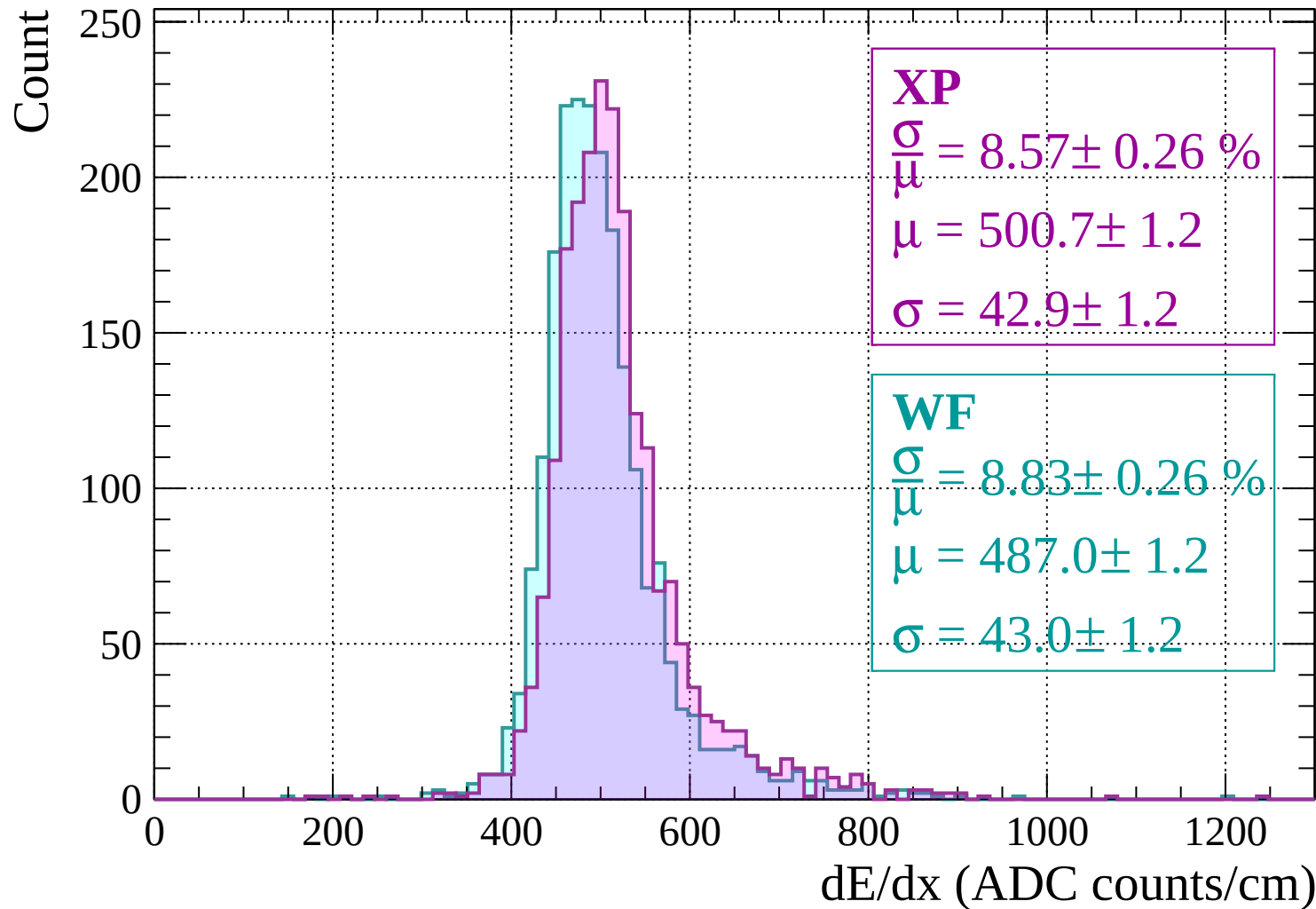
dE/dx (ADC counts/cm)

Energy loss | $-1840 < p < -1800$

Count

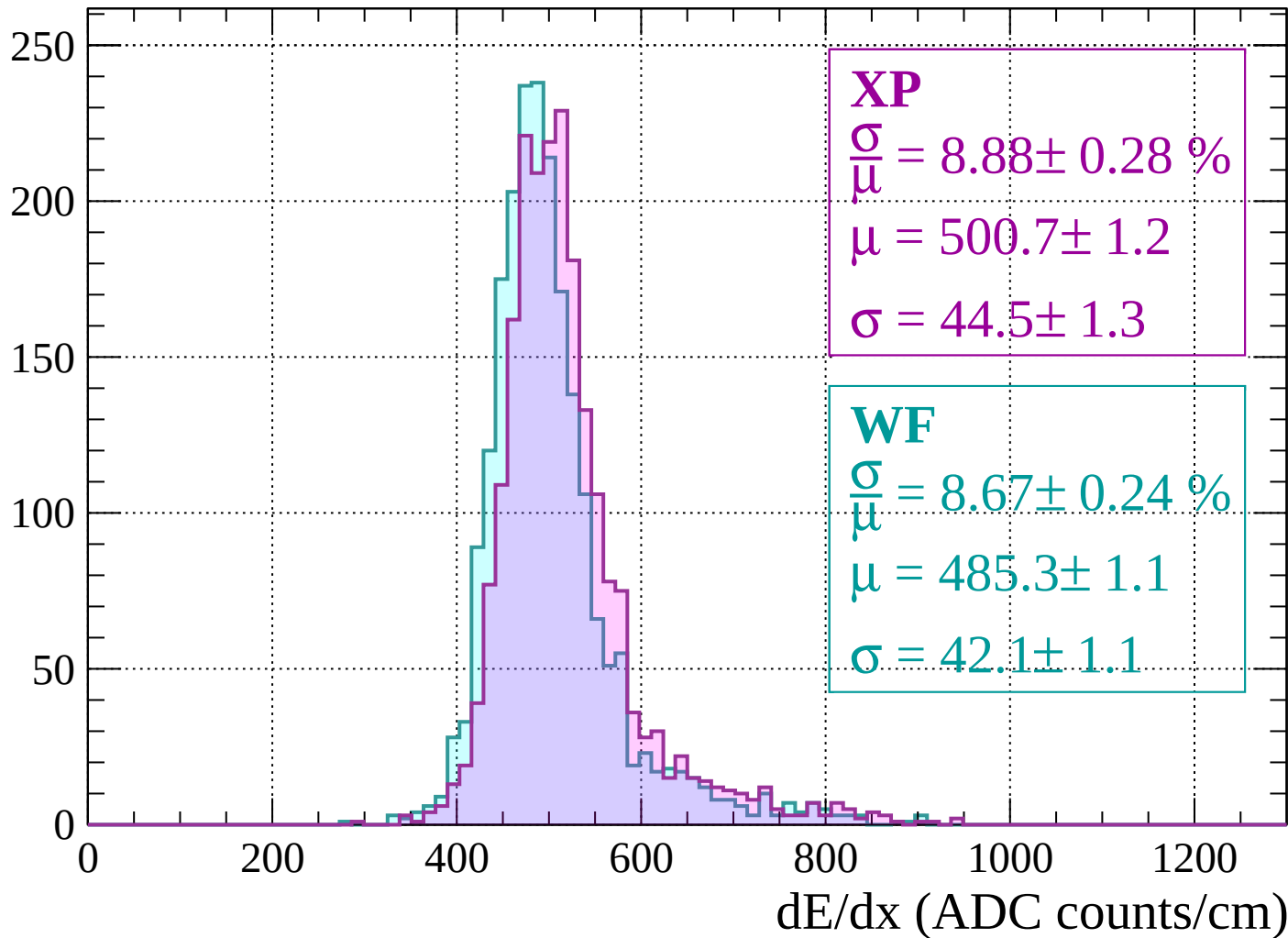


Energy loss | -1800 < p < -1760



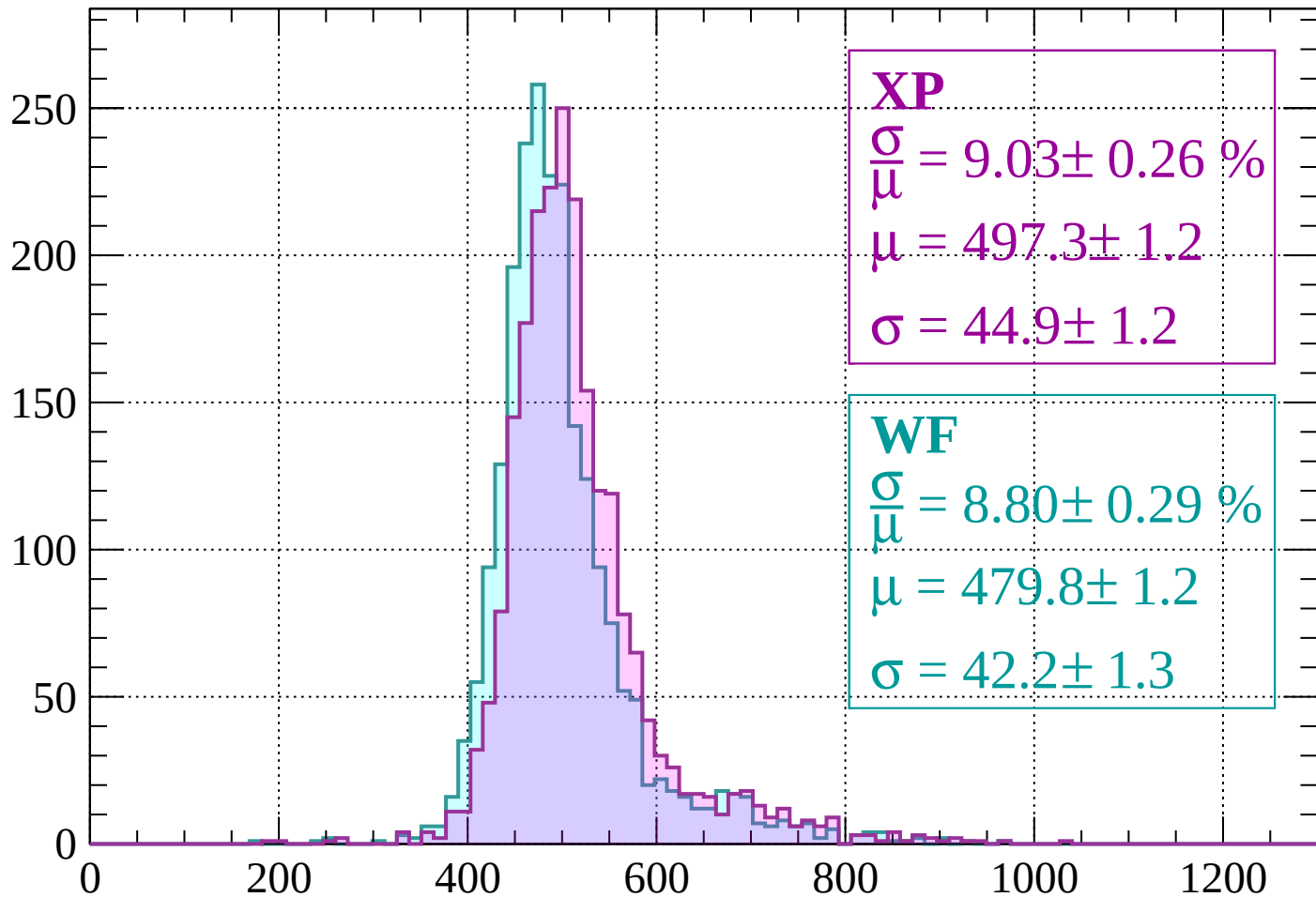
Energy loss | $-1760 < p < -1720$

Count



Energy loss | $-1720 < p < -1680$

Count



XP

$$\frac{\sigma}{\mu} = 9.03 \pm 0.26 \%$$

$$\mu = 497.3 \pm 1.2$$

$$\sigma = 44.9 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 8.80 \pm 0.29 \%$$

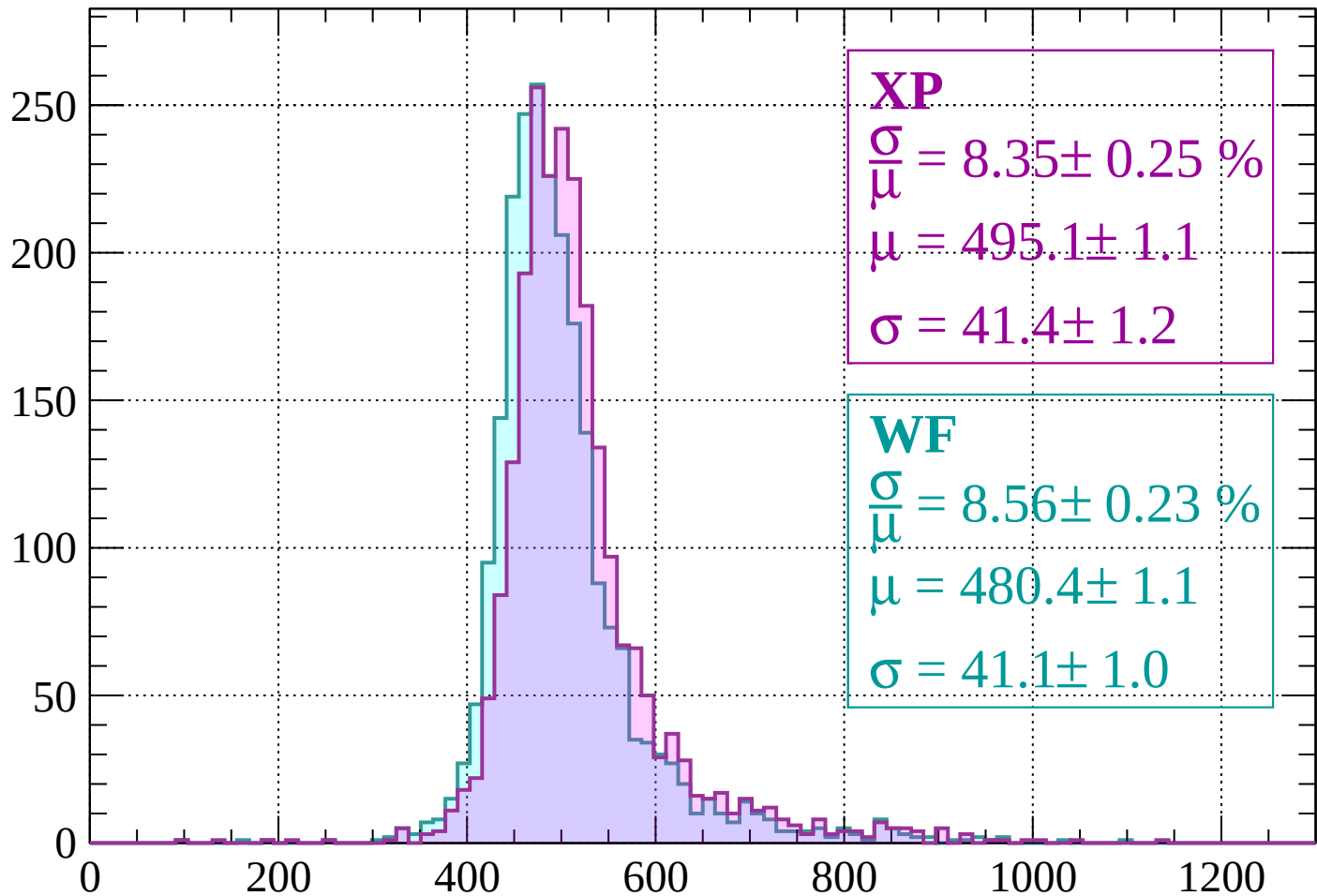
$$\mu = 479.8 \pm 1.2$$

$$\sigma = 42.2 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $-1680 < p < -1640$

Count



XP

$$\frac{\sigma}{\mu} = 8.35 \pm 0.25 \%$$

$$\mu = 495.1 \pm 1.1$$

$$\sigma = 41.4 \pm 1.2$$

WF

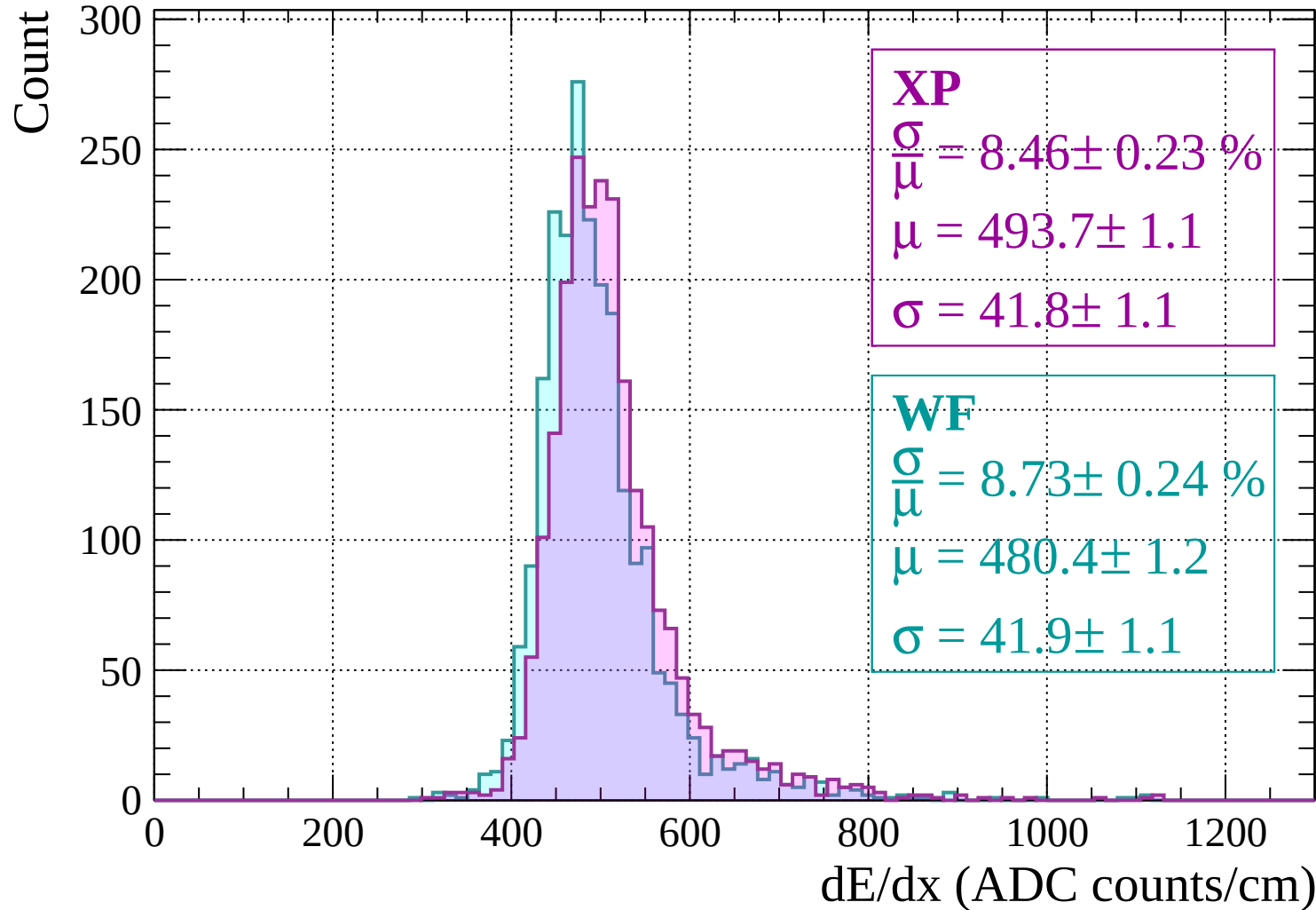
$$\frac{\sigma}{\mu} = 8.56 \pm 0.23 \%$$

$$\mu = 480.4 \pm 1.1$$

$$\sigma = 41.1 \pm 1.0$$

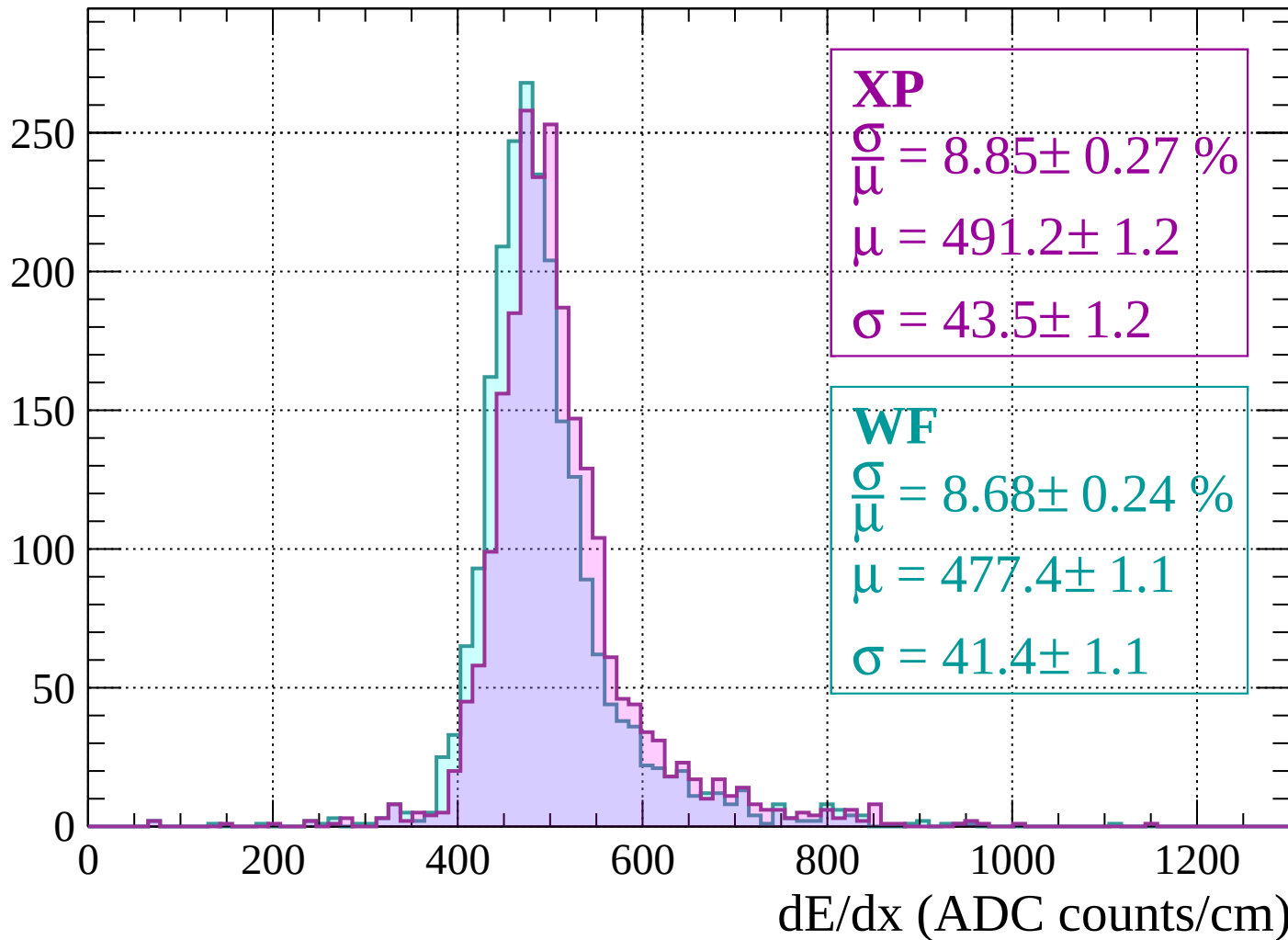
dE/dx (ADC counts/cm)

Energy loss | $-1640 < p < -1600$



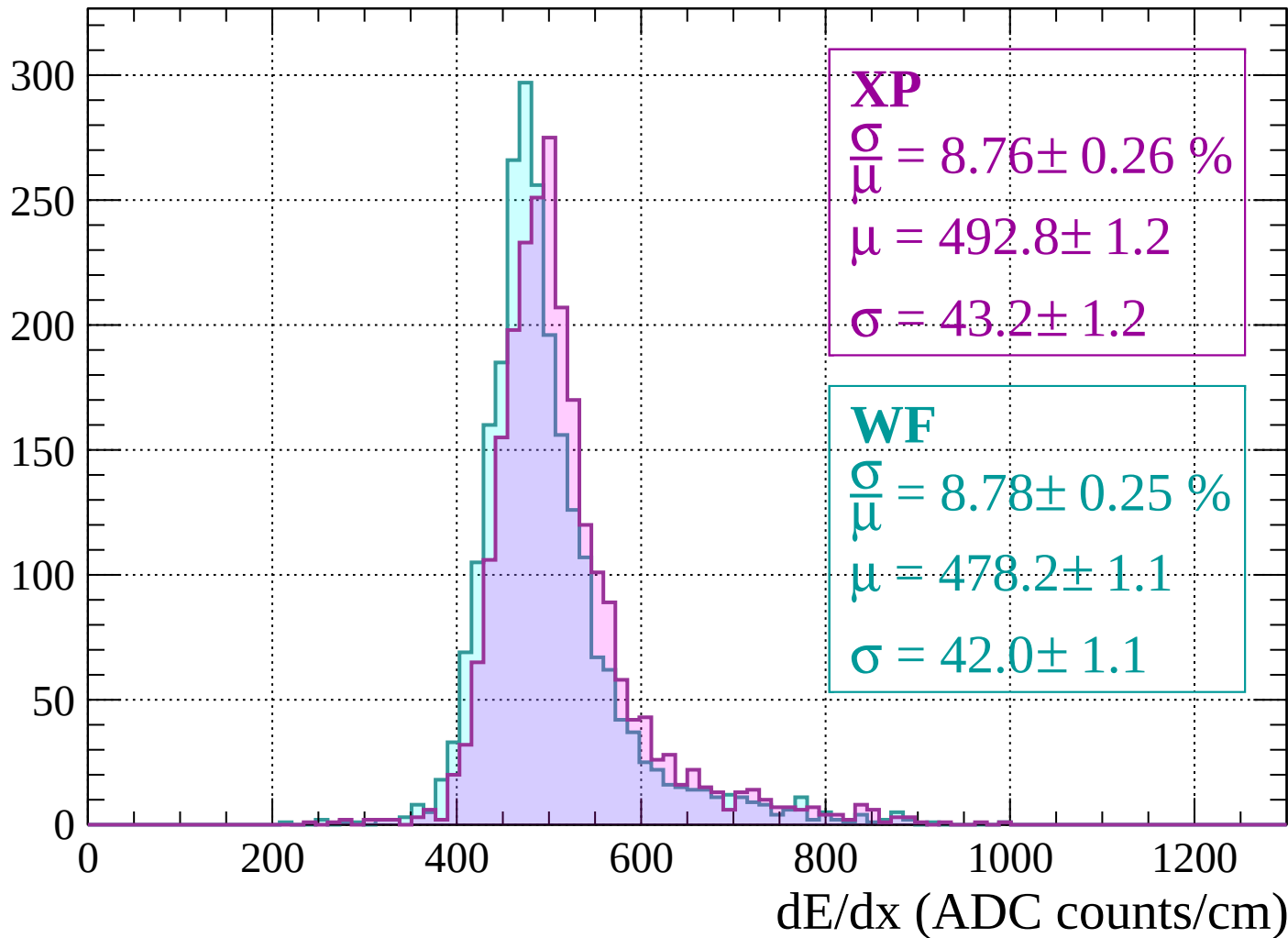
Energy loss | $-1600 < p < -1560$

Count



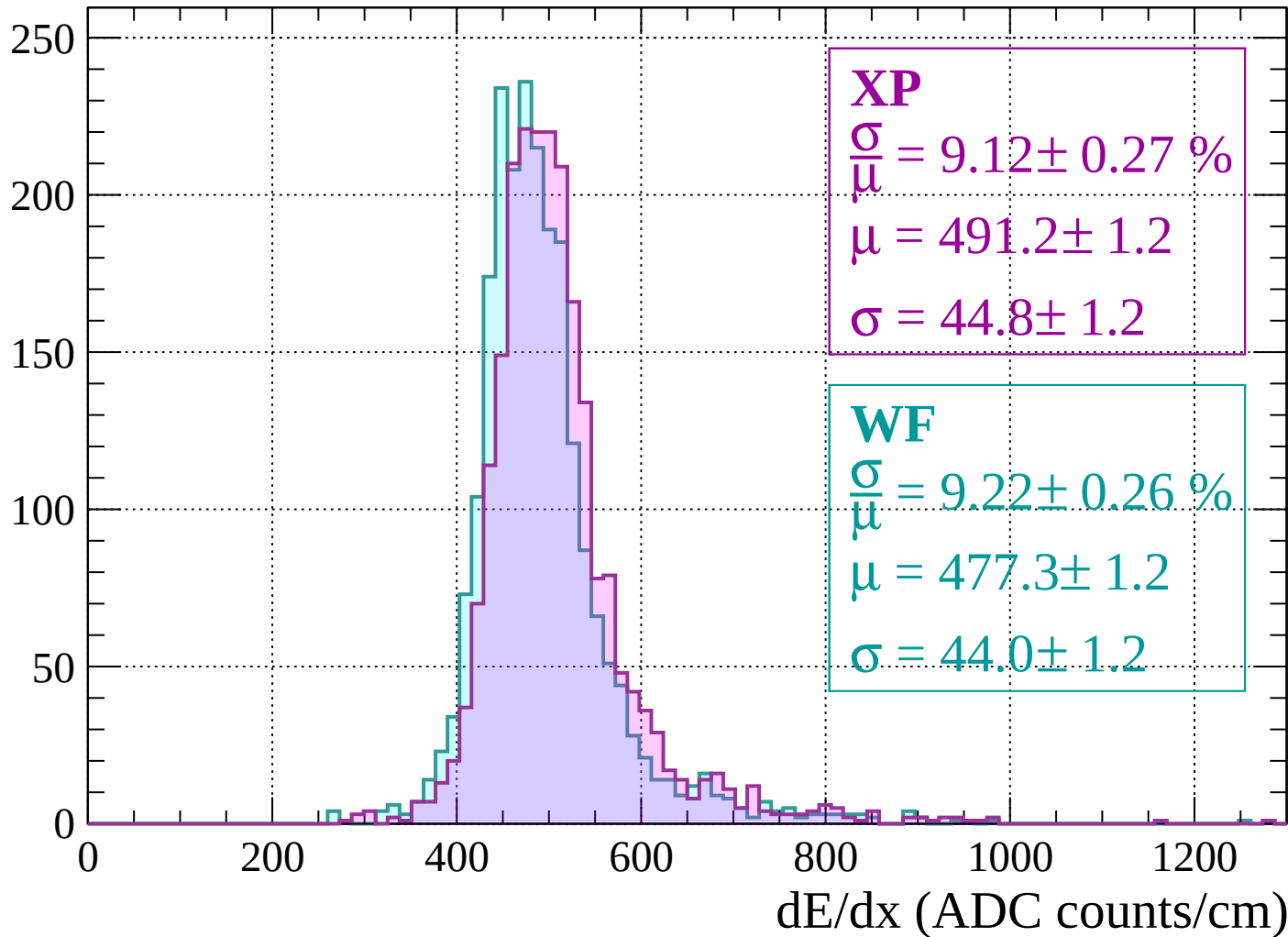
Energy loss | $-1560 < p < -1520$

Count



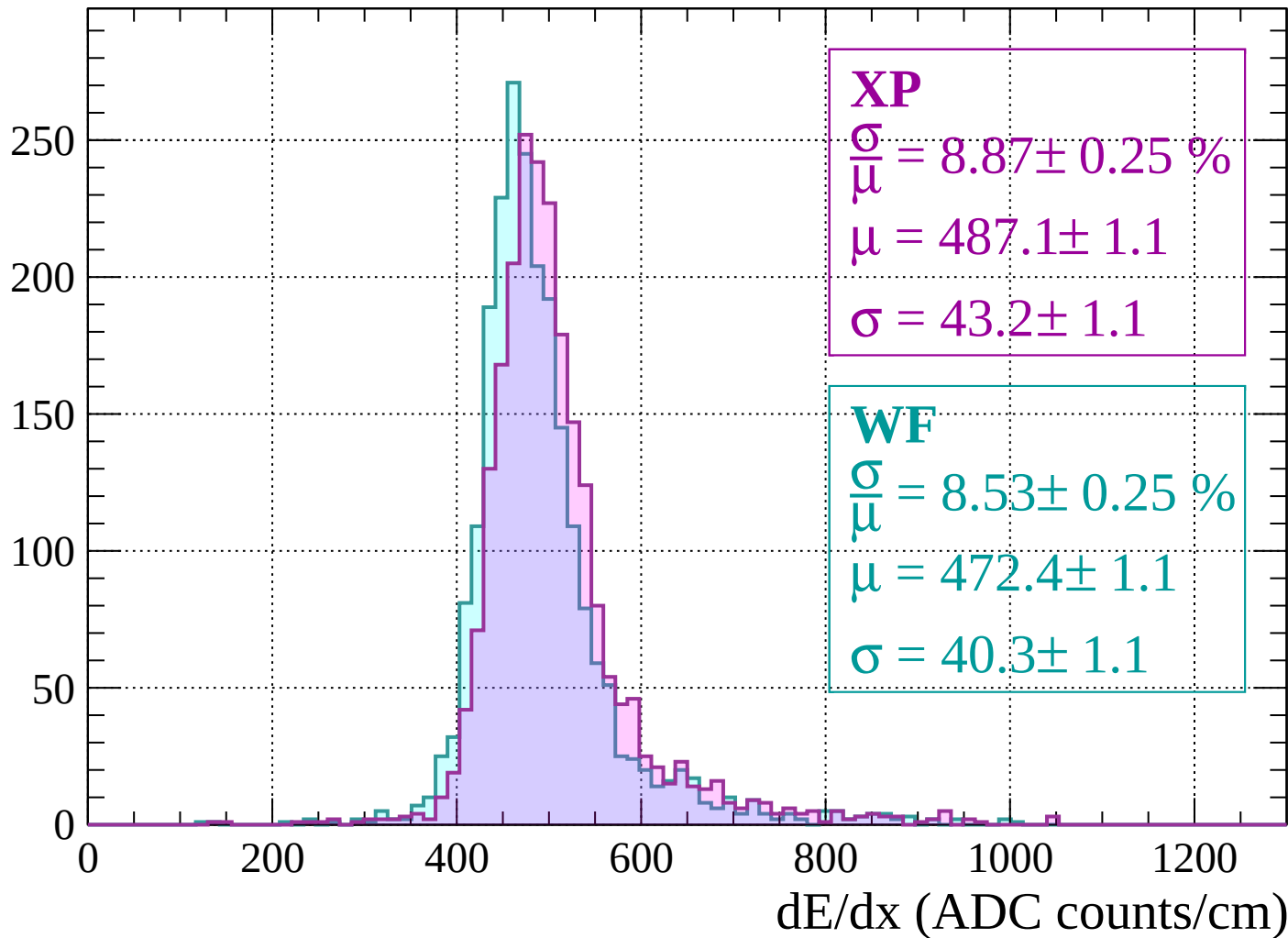
Energy loss | $-1520 < p < -1480$

Count



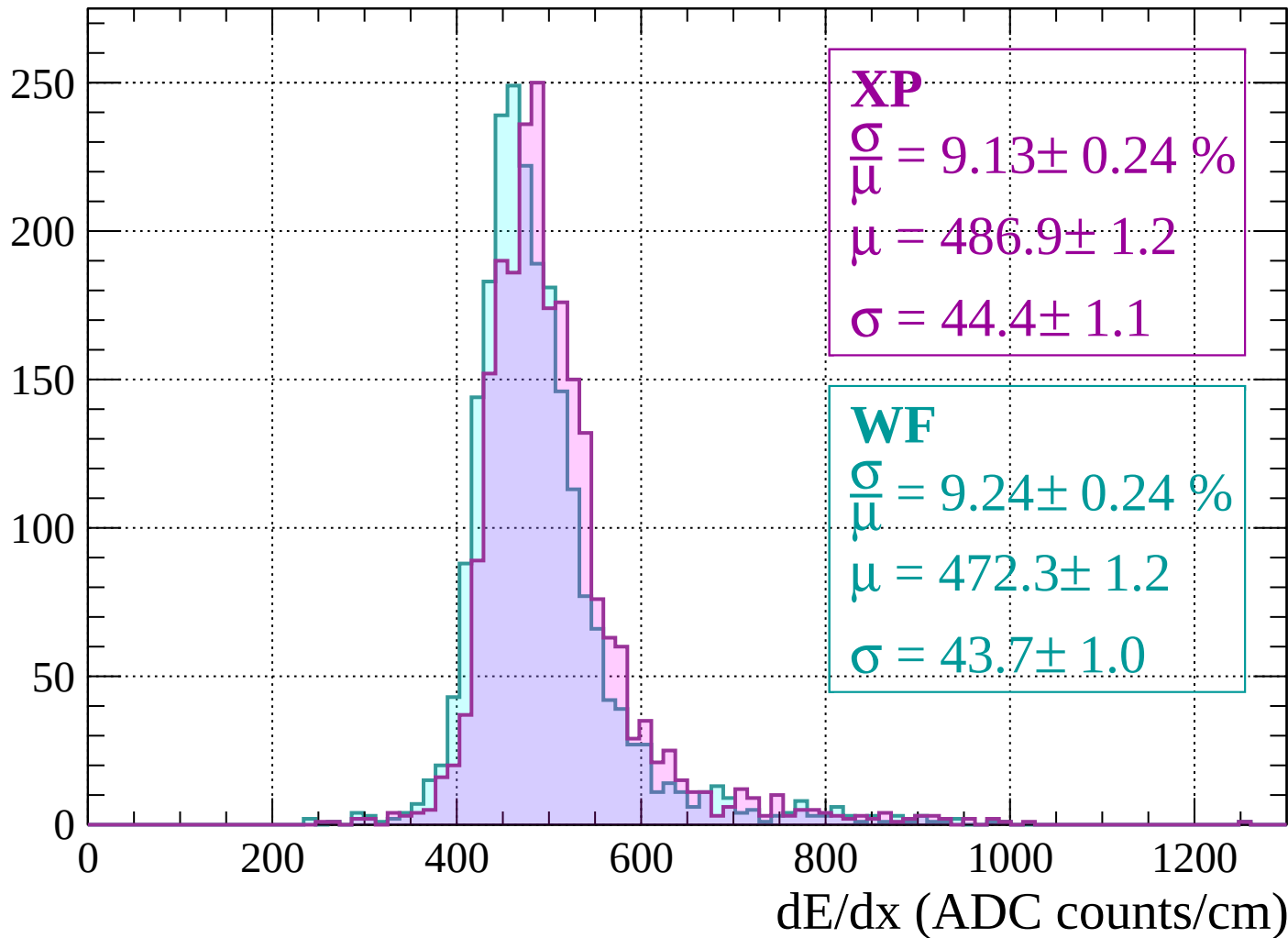
Energy loss | $-1480 < p < -1440$

Count



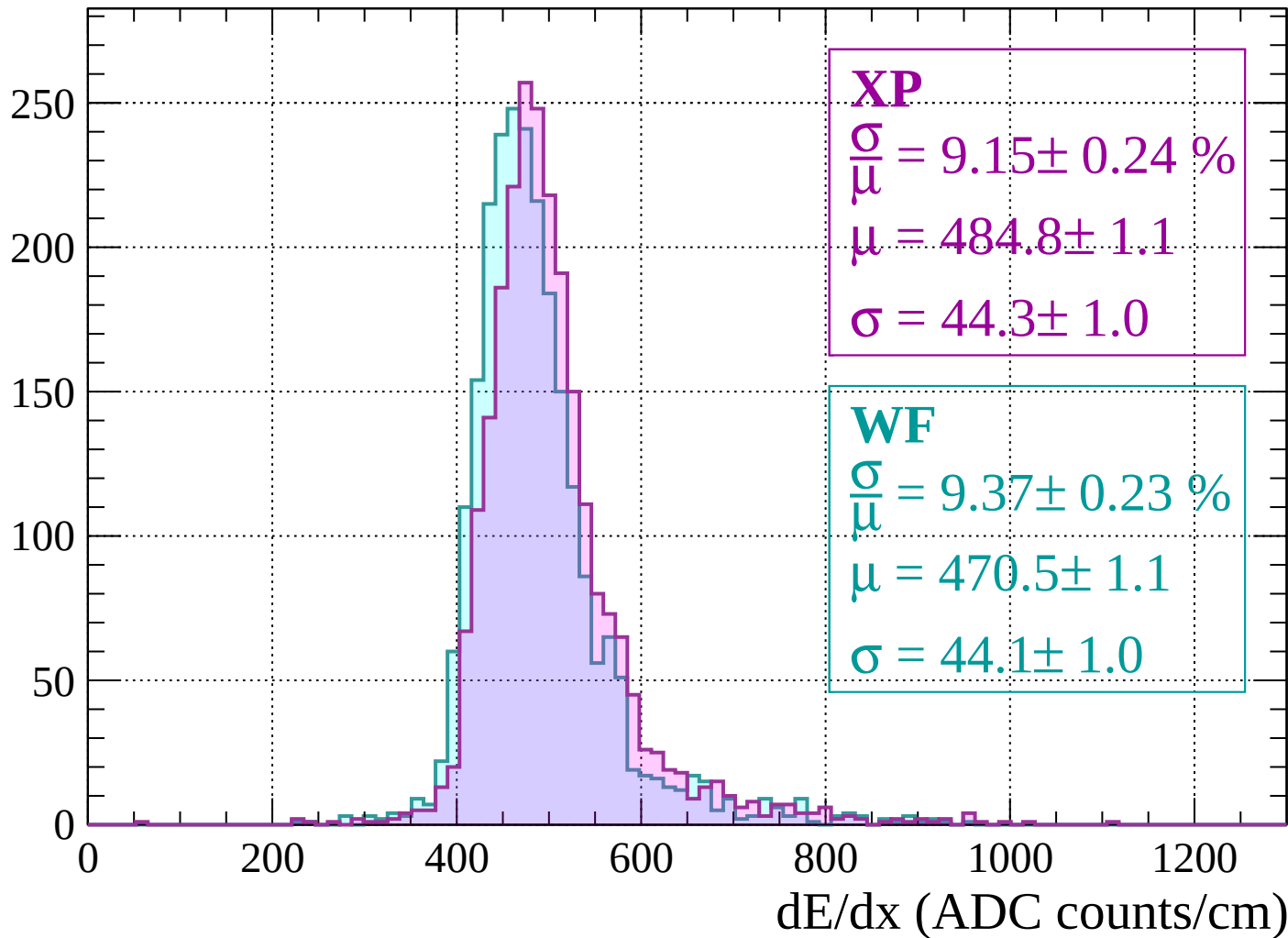
Energy loss | $-1440 < p < -1400$

Count



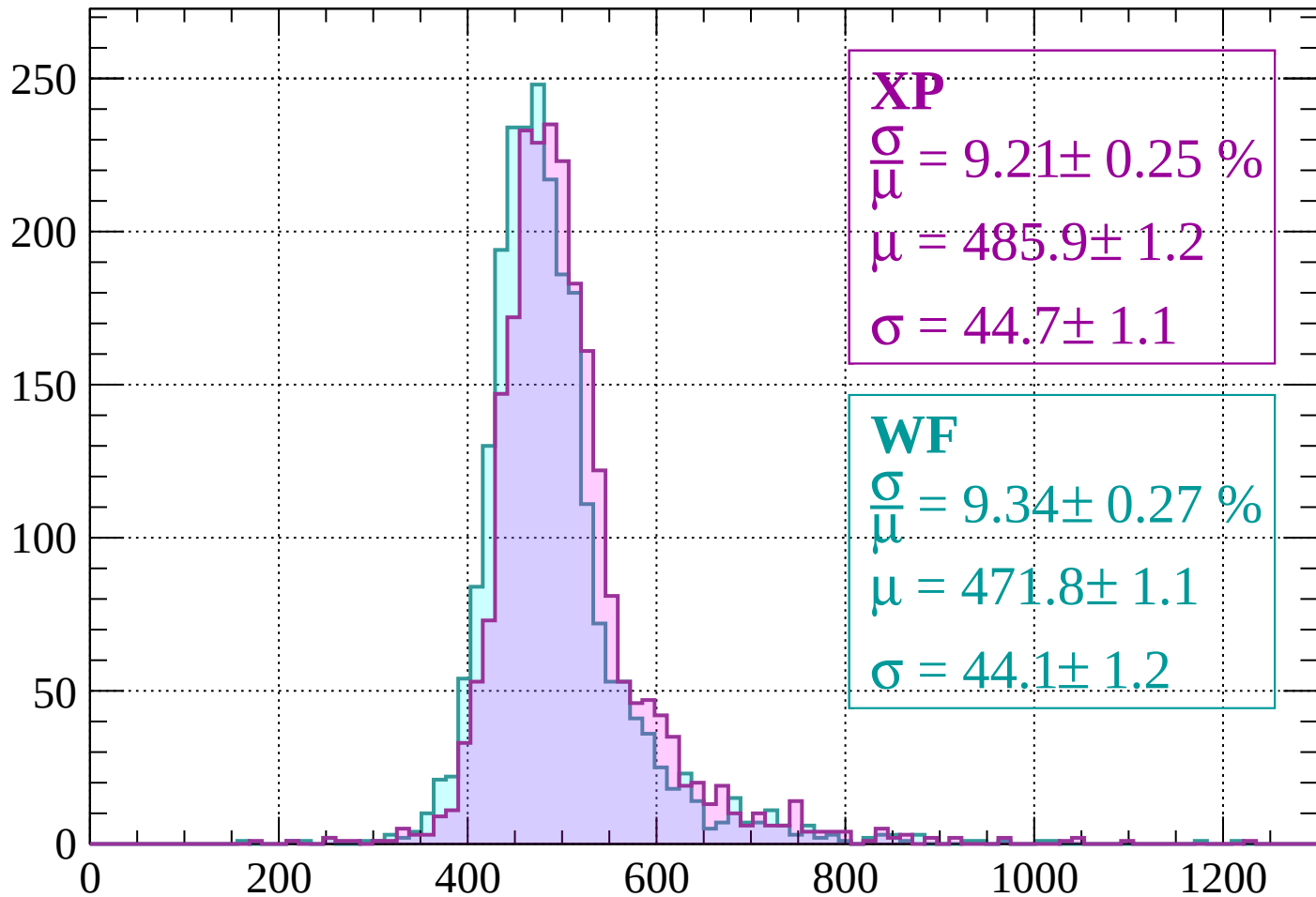
Energy loss | $-1400 < p < -1360$

Count



Energy loss | $-1360 < p < -1320$

Count



XP

$$\frac{\sigma}{\mu} = 9.21 \pm 0.25 \%$$

$$\mu = 485.9 \pm 1.2$$

$$\sigma = 44.7 \pm 1.1$$

WF

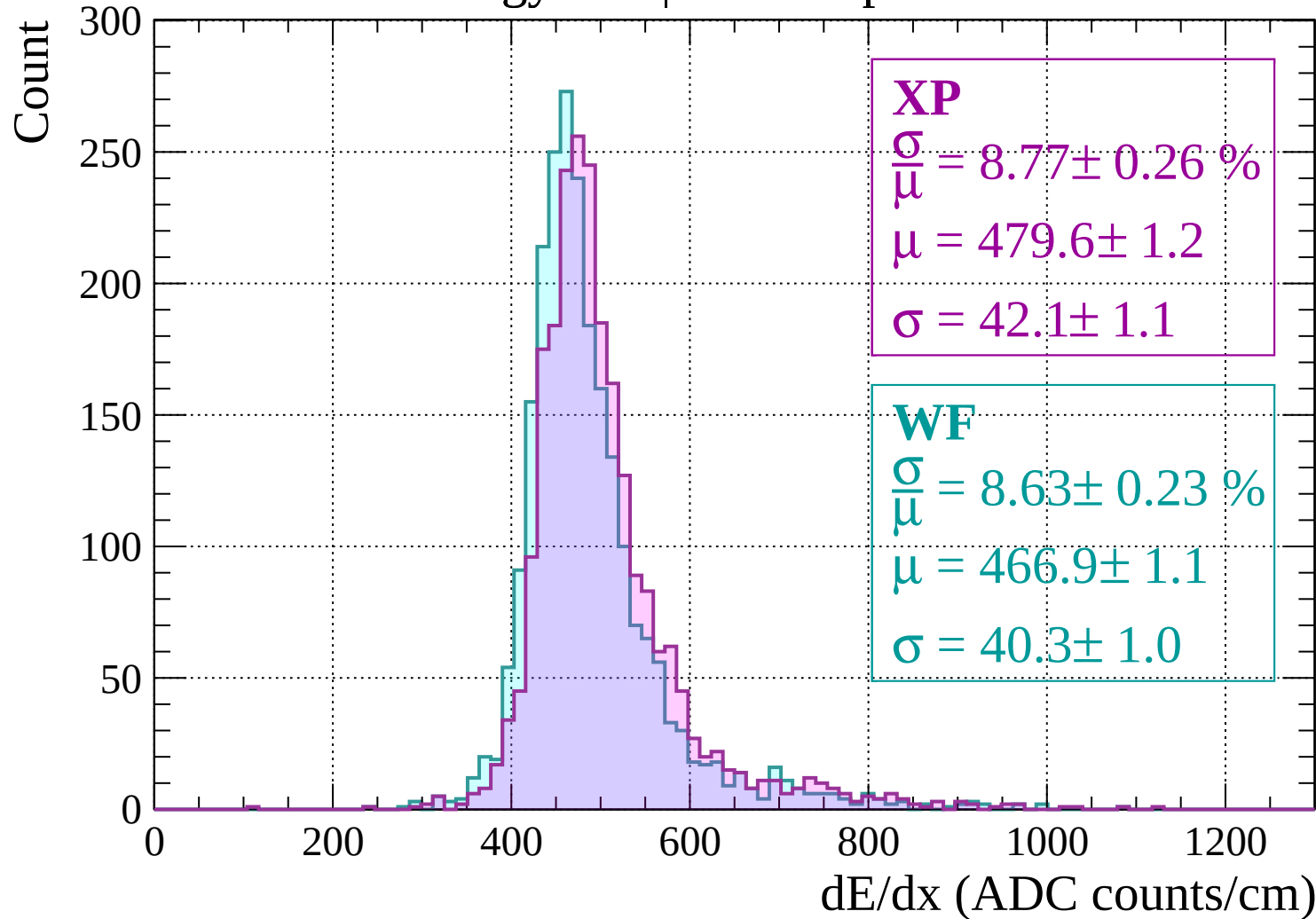
$$\frac{\sigma}{\mu} = 9.34 \pm 0.27 \%$$

$$\mu = 471.8 \pm 1.1$$

$$\sigma = 44.1 \pm 1.2$$

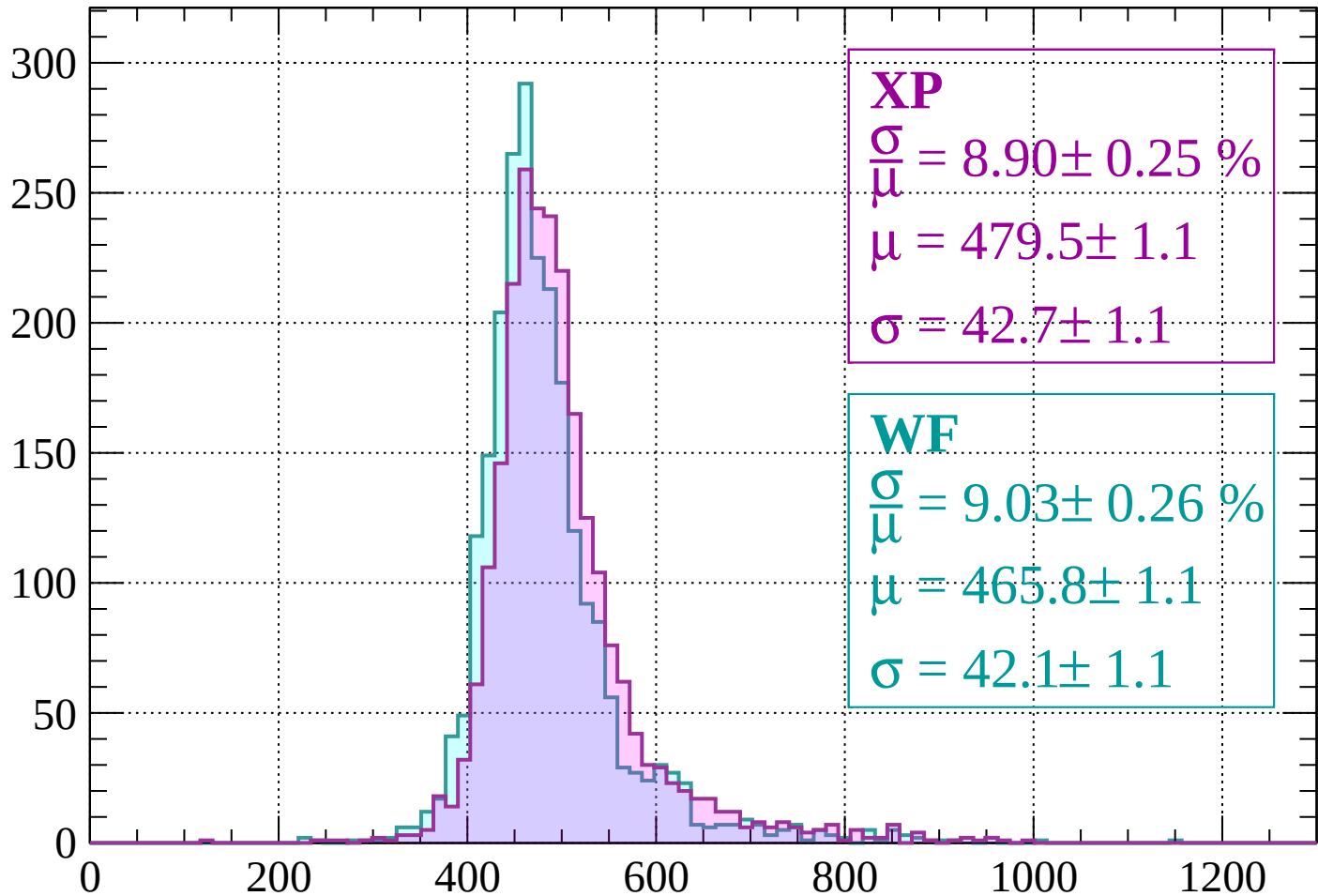
dE/dx (ADC counts/cm)

Energy loss | $-1320 < p < -1280$



Energy loss | $-1280 < p < -1240$

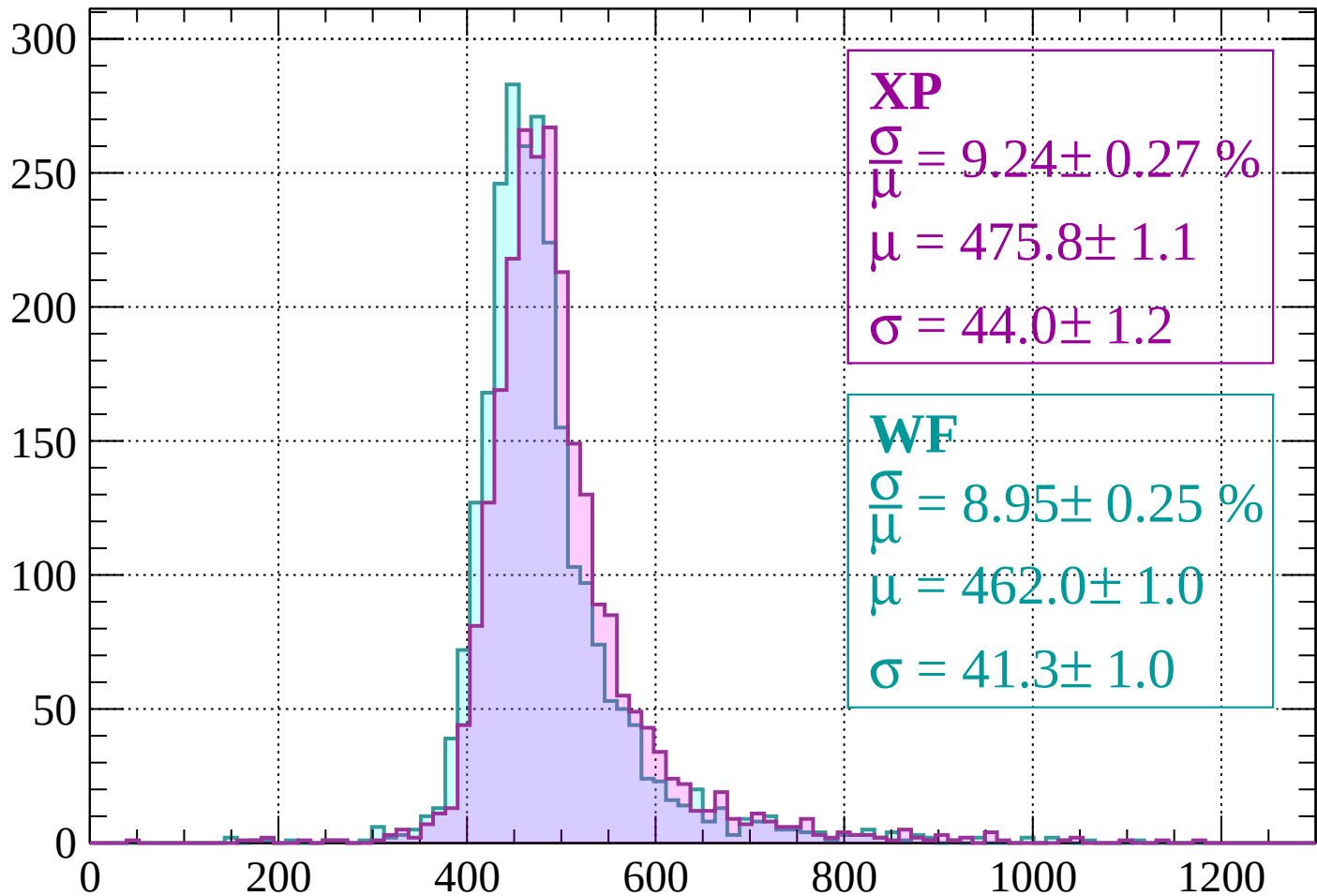
Count



dE/dx (ADC counts/cm)

Energy loss | $-1240 < p < -1200$

Count



XP

$$\frac{\sigma}{\mu} = 9.24 \pm 0.27 \%$$

$$\mu = 475.8 \pm 1.1$$

$$\sigma = 44.0 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 8.95 \pm 0.25 \%$$

$$\mu = 462.0 \pm 1.0$$

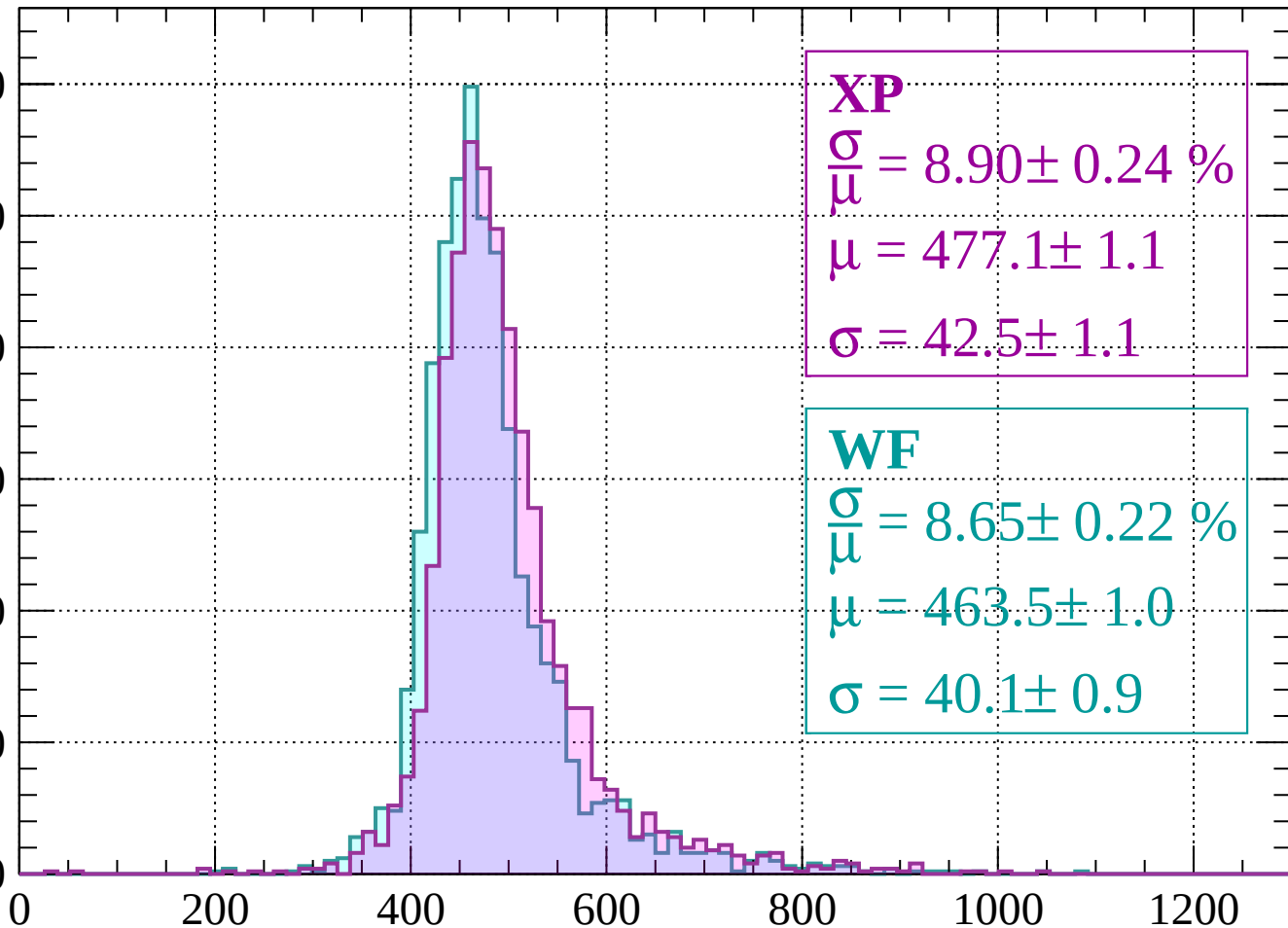
$$\sigma = 41.3 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | -1200 < p < -1160

Count

300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 8.90 \pm 0.24 \%$$

$$\mu = 477.1 \pm 1.1$$

$$\sigma = 42.5 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 8.65 \pm 0.22 \%$$

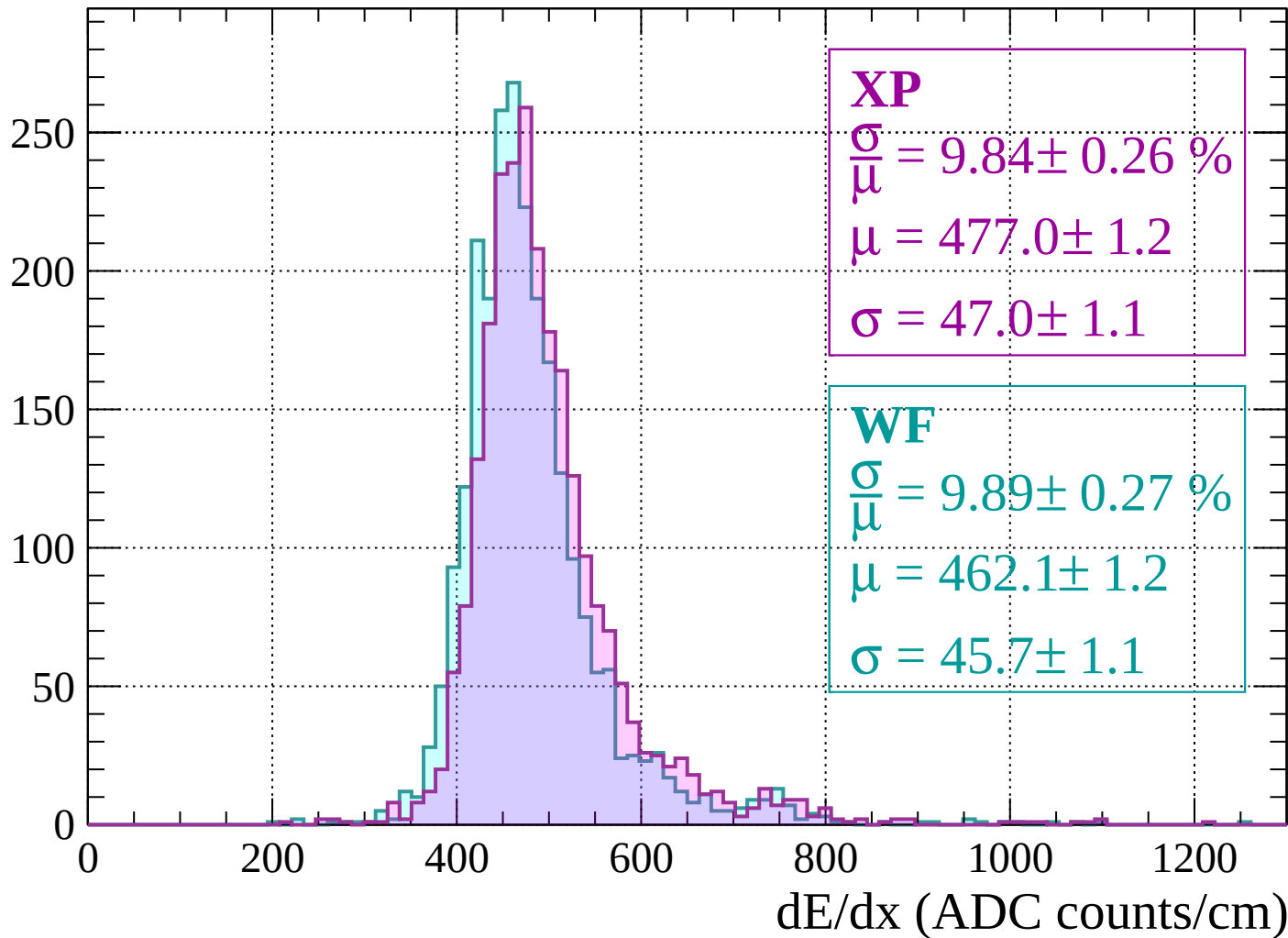
$$\mu = 463.5 \pm 1.0$$

$$\sigma = 40.1 \pm 0.9$$

dE/dx (ADC counts/cm)

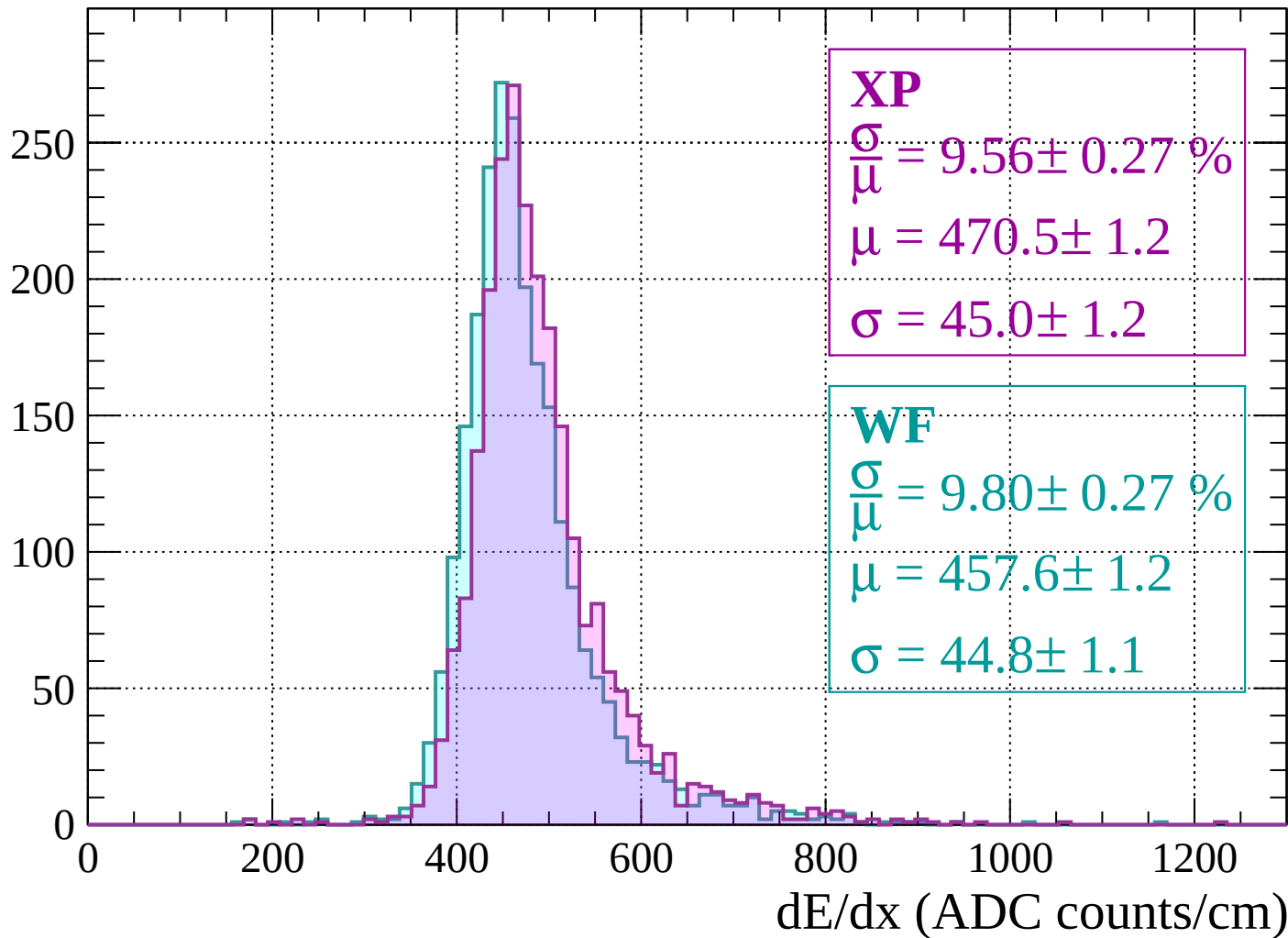
Energy loss | $-1160 < p < -1120$

Count

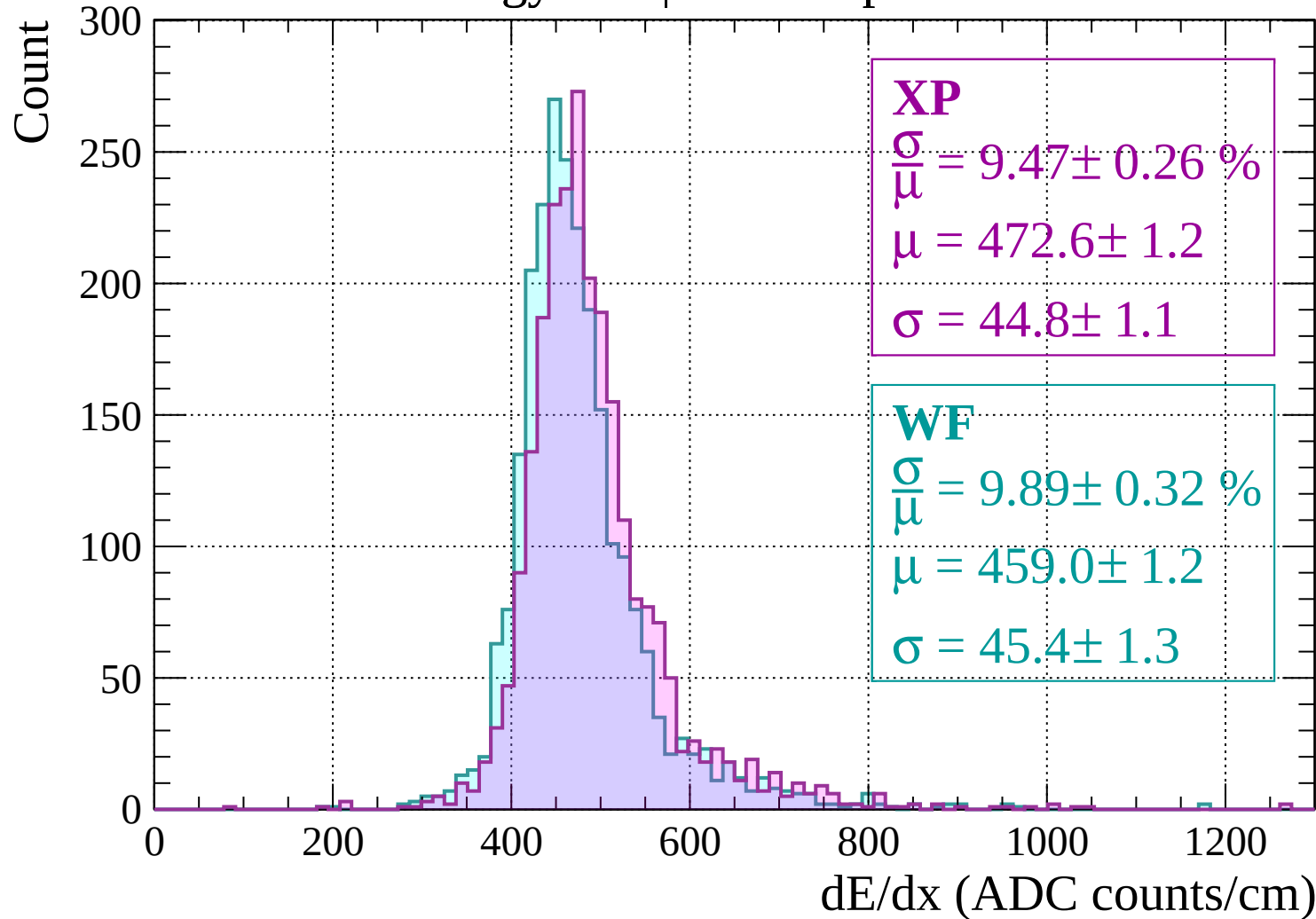


Energy loss | $-1120 < p < -1080$

Count

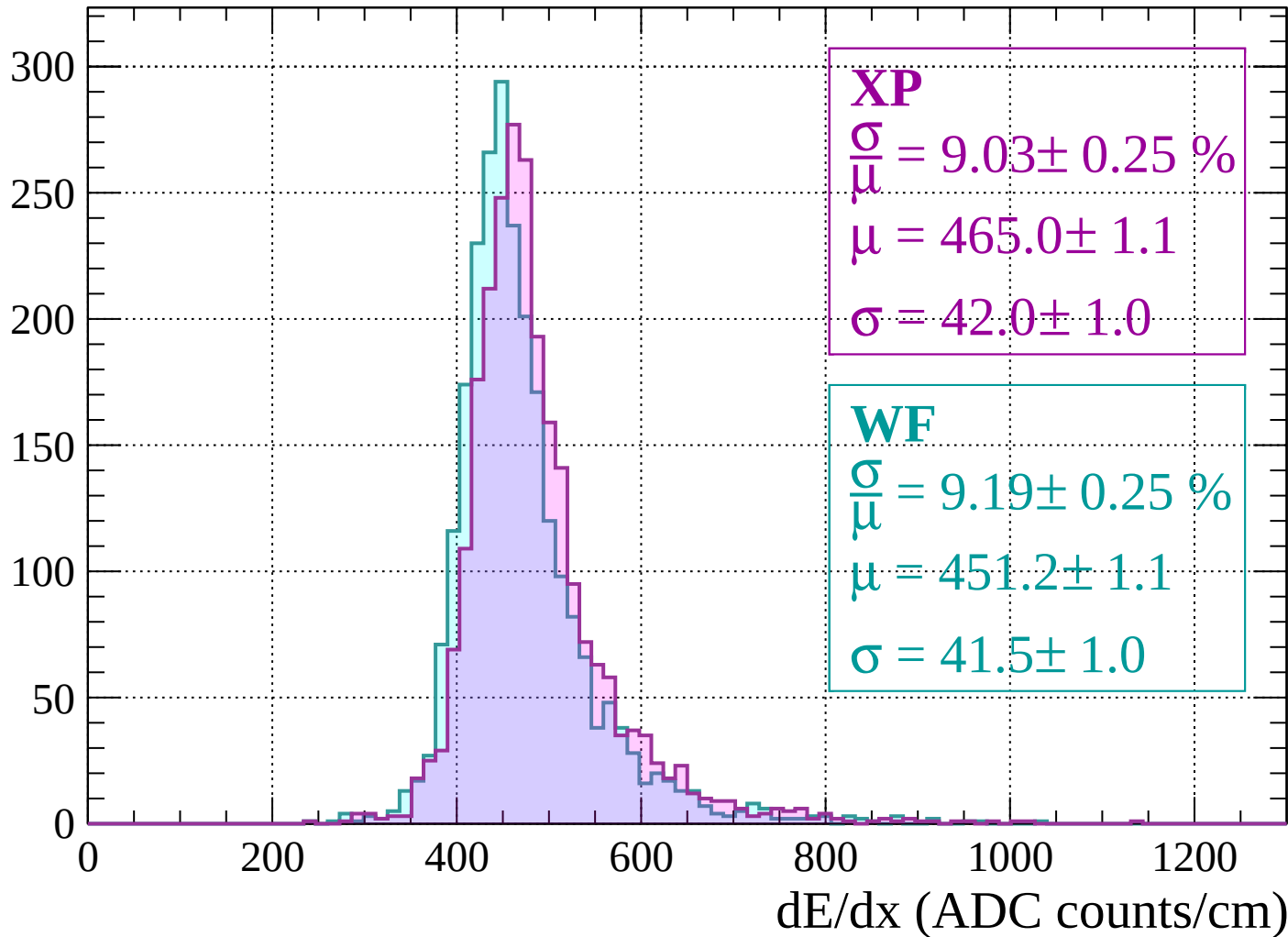


Energy loss | $-1080 < p < -1040$



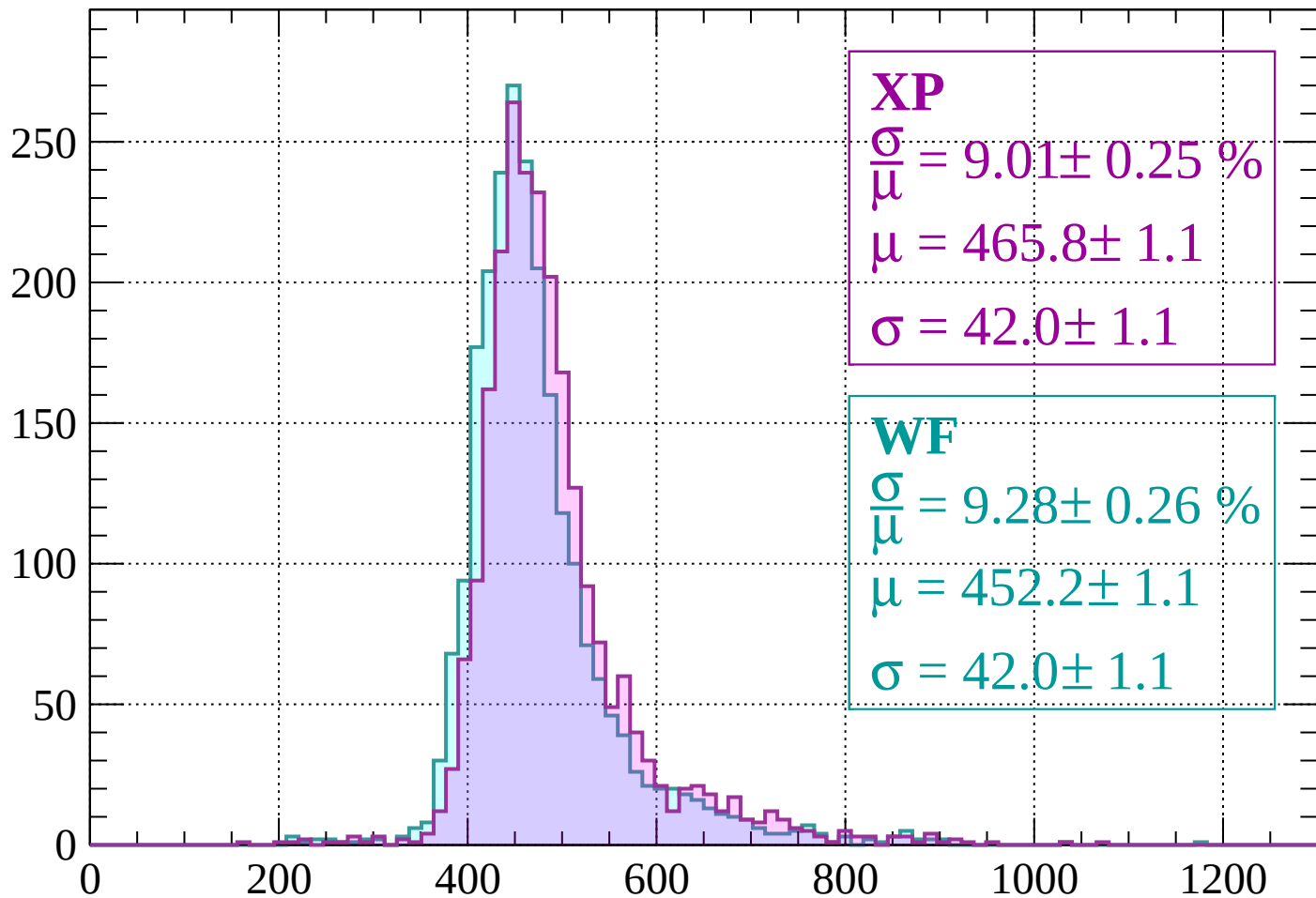
Energy loss | $-1040 < p < -1000$

Count



Energy loss | $-1000 < p < -960$

Count



XP

$$\frac{\sigma}{\mu} = 9.01 \pm 0.25 \%$$

$$\mu = 465.8 \pm 1.1$$

$$\sigma = 42.0 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 9.28 \pm 0.26 \%$$

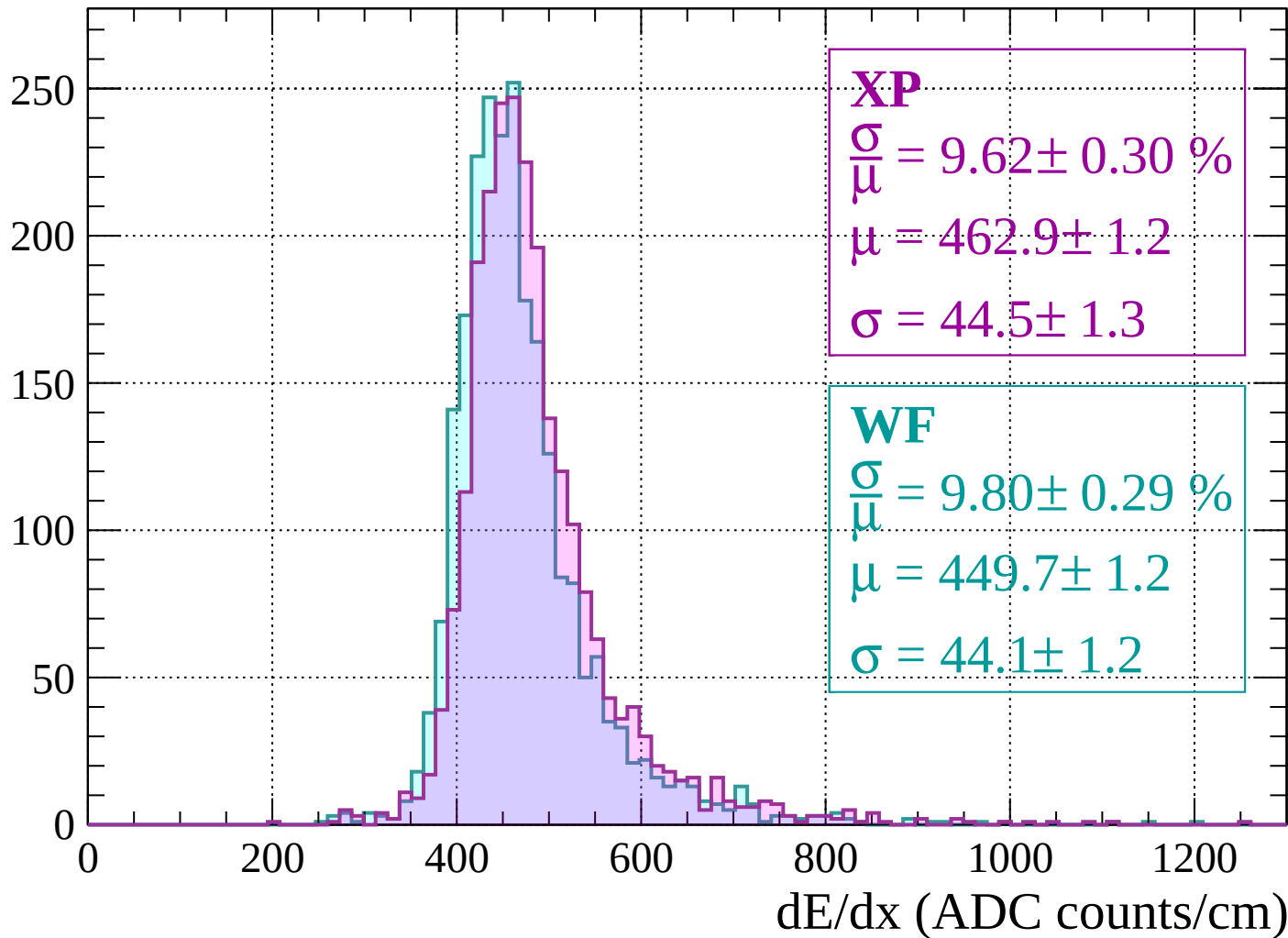
$$\mu = 452.2 \pm 1.1$$

$$\sigma = 42.0 \pm 1.1$$

dE/dx (ADC counts/cm)

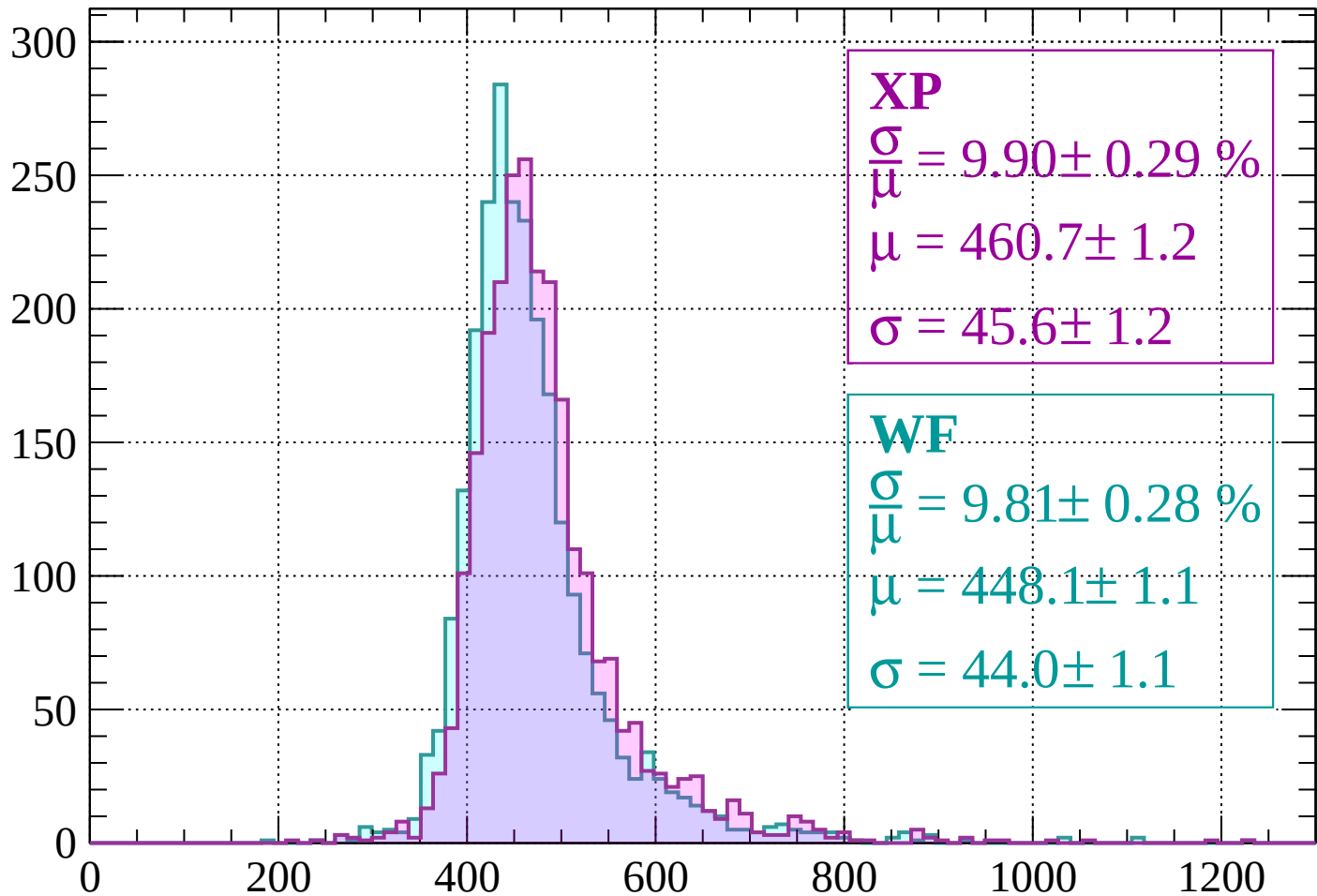
Energy loss | $-960 < p < -920$

Count



Energy loss | $-920 < p < -880$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 9.90 \pm 0.29 \%$$

$$\mu = 460.7 \pm 1.2$$

$$\sigma = 45.6 \pm 1.2$$

WF

$$\frac{\sigma}{\bar{\mu}} = 9.81 \pm 0.28 \%$$

$$\mu = 448.1 \pm 1.1$$

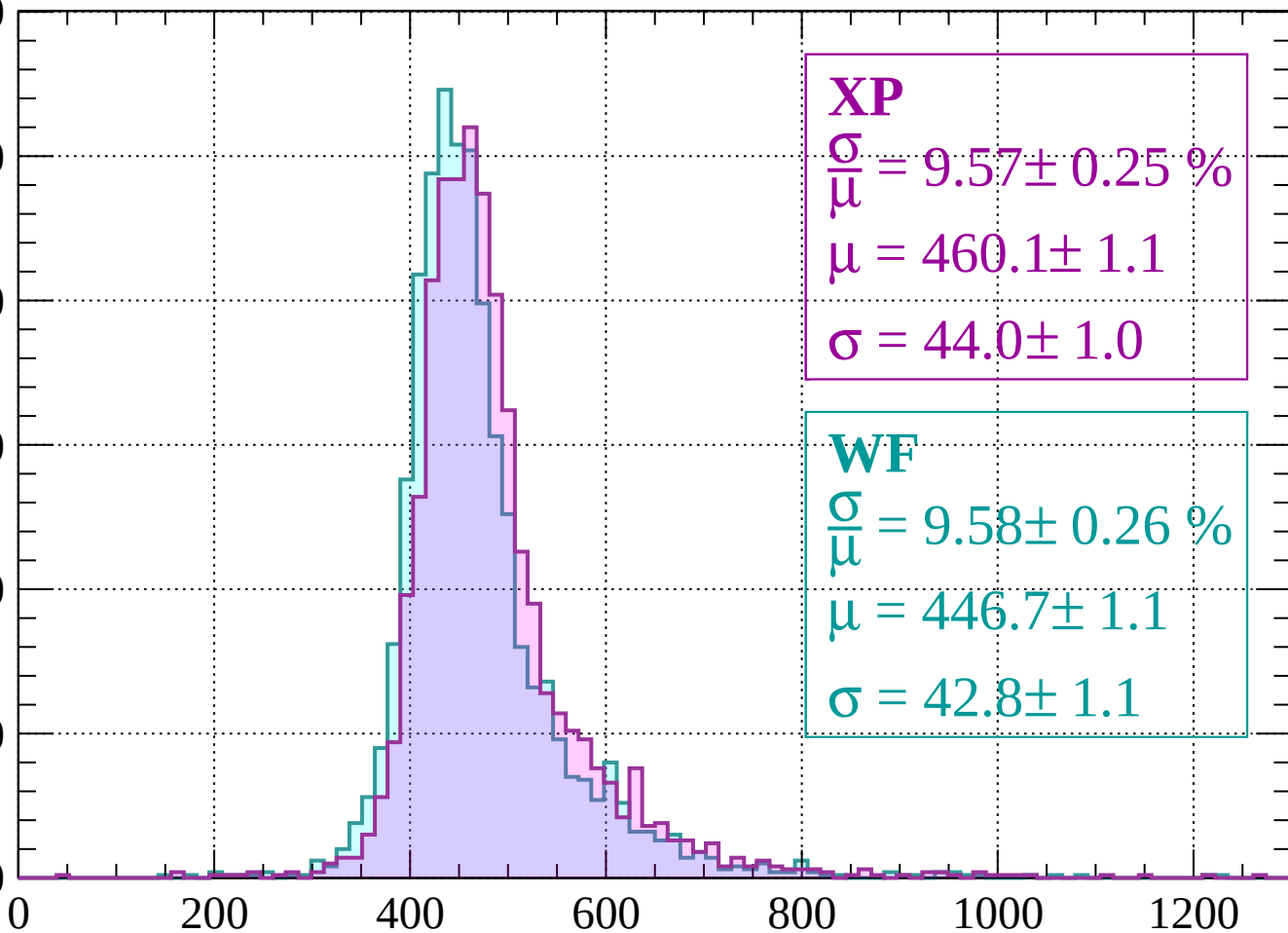
$$\sigma = 44.0 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $-880 < p < -840$

Count

300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 9.57 \pm 0.25 \%$$

$$\mu = 460.1 \pm 1.1$$

$$\sigma = 44.0 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 9.58 \pm 0.26 \%$$

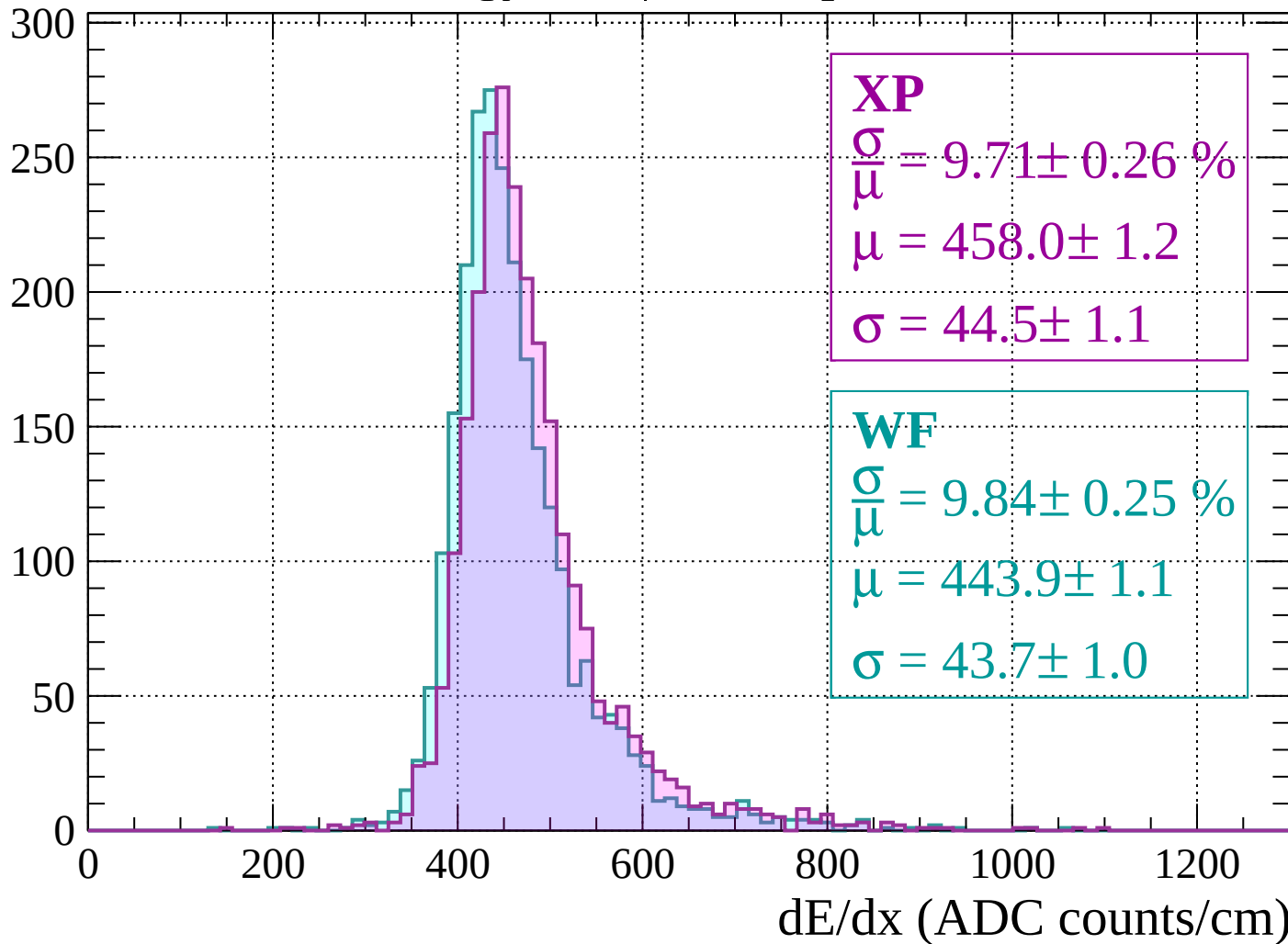
$$\mu = 446.7 \pm 1.1$$

$$\sigma = 42.8 \pm 1.1$$

dE/dx (ADC counts/cm)

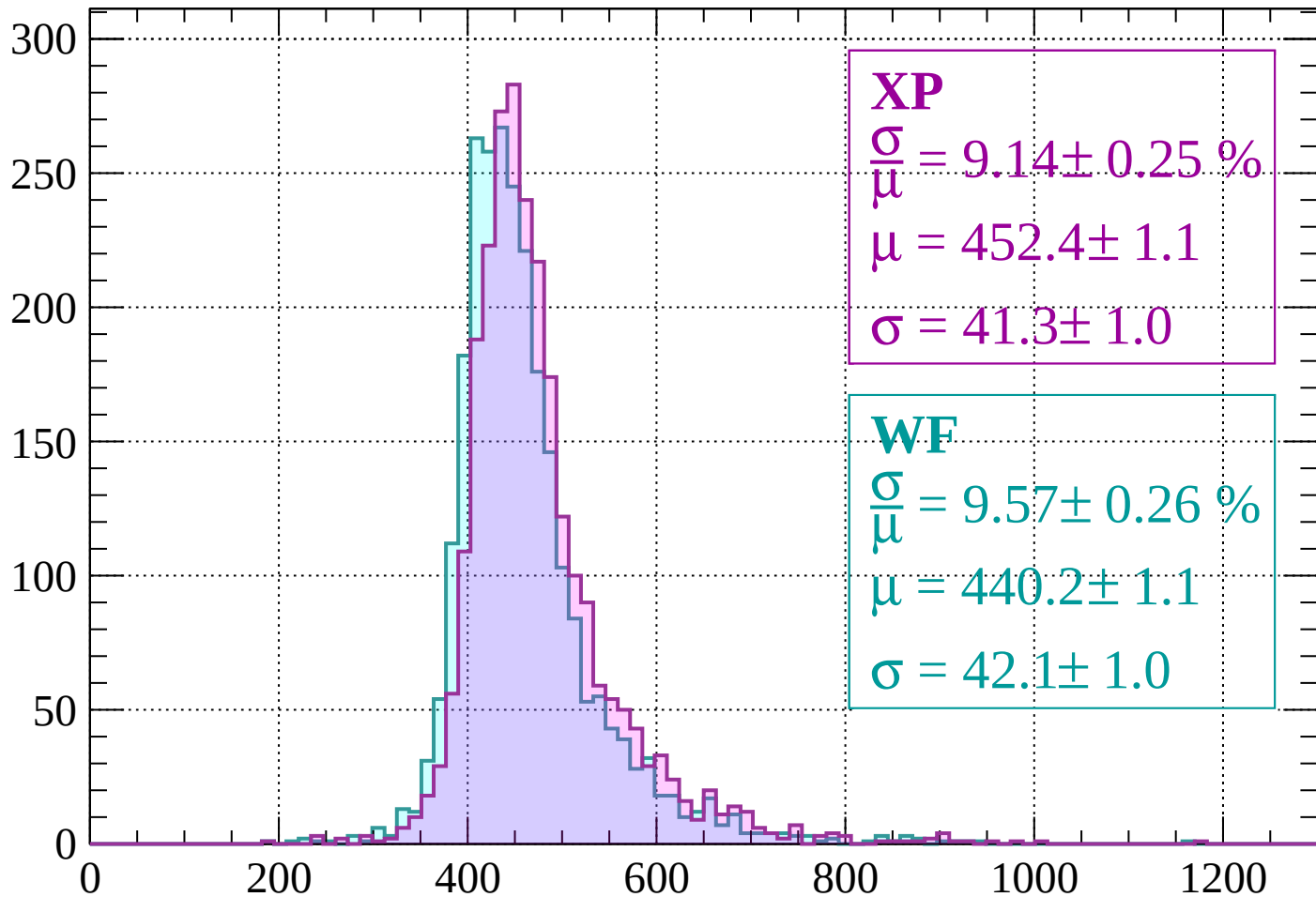
Energy loss | $-840 < p < -800$

Count



Energy loss | $-800 < p < -760$

Count



XP

$$\frac{\sigma}{\mu} = 9.14 \pm 0.25 \%$$

$$\mu = 452.4 \pm 1.1$$

$$\sigma = 41.3 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 9.57 \pm 0.26 \%$$

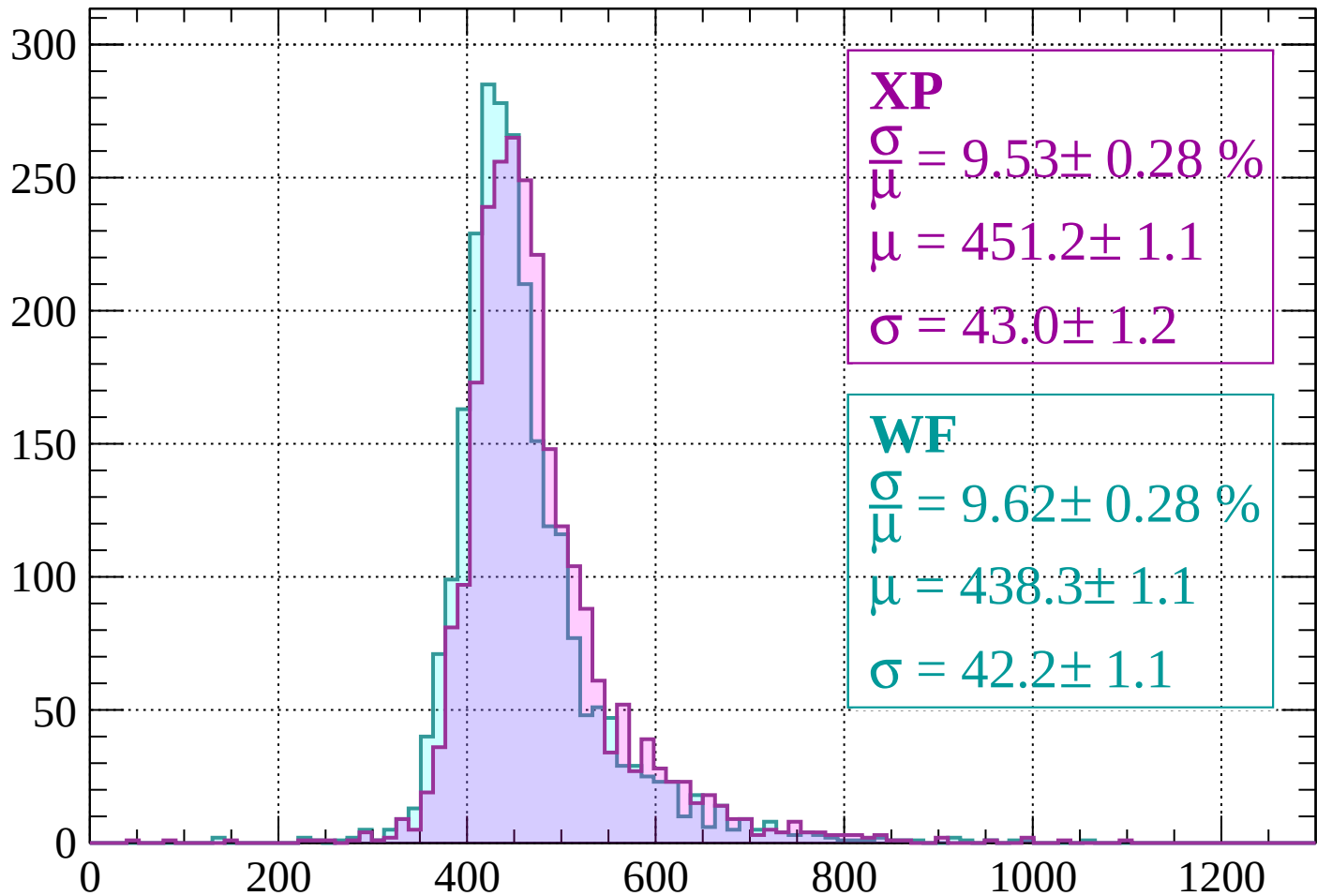
$$\mu = 440.2 \pm 1.1$$

$$\sigma = 42.1 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-760 < p < -720$

Count



XP

$$\frac{\sigma}{\mu} = 9.53 \pm 0.28 \%$$

$$\mu = 451.2 \pm 1.1$$

$$\sigma = 43.0 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 9.62 \pm 0.28 \%$$

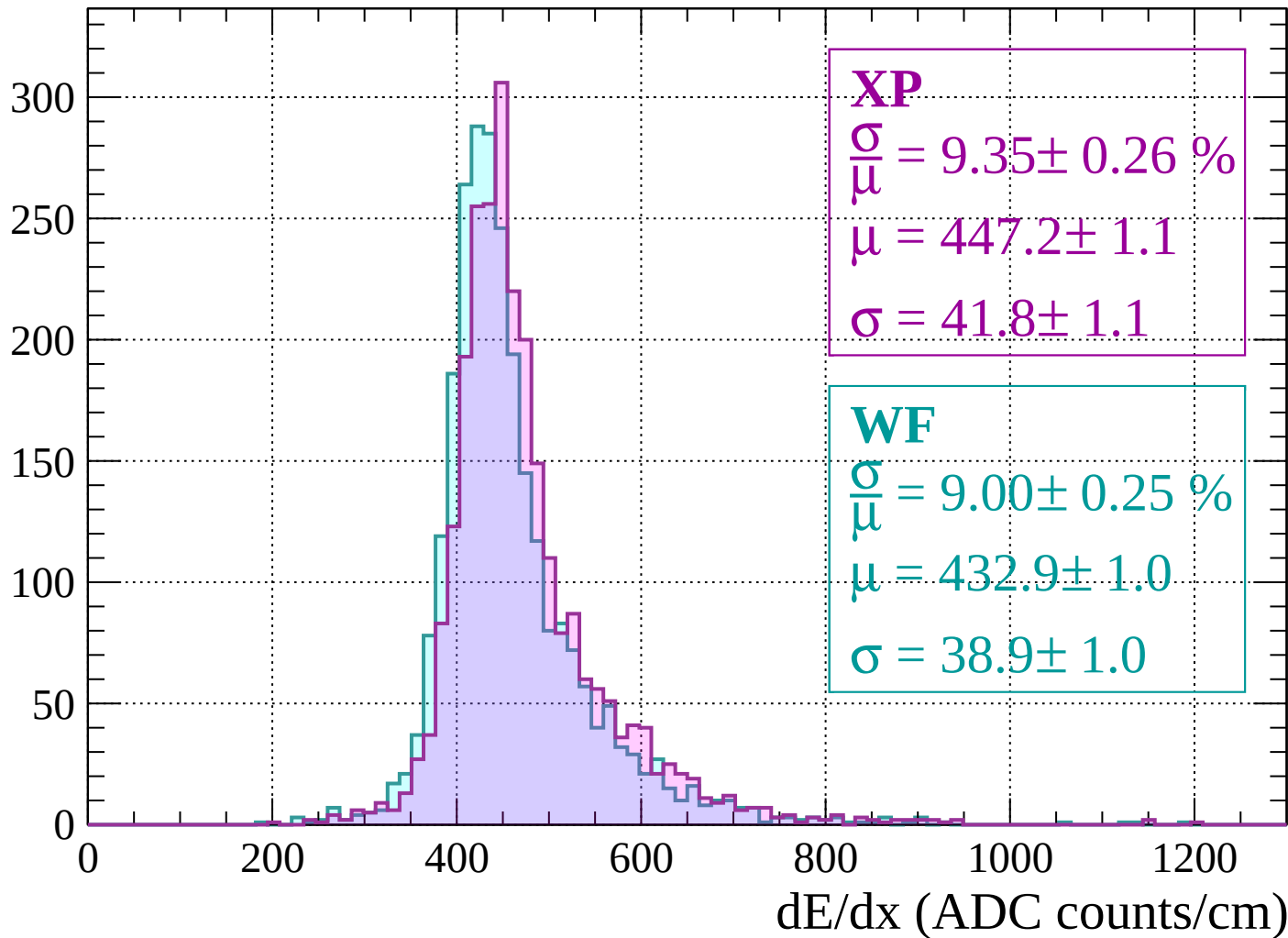
$$\mu = 438.3 \pm 1.1$$

$$\sigma = 42.2 \pm 1.1$$

dE/dx (ADC counts/cm)

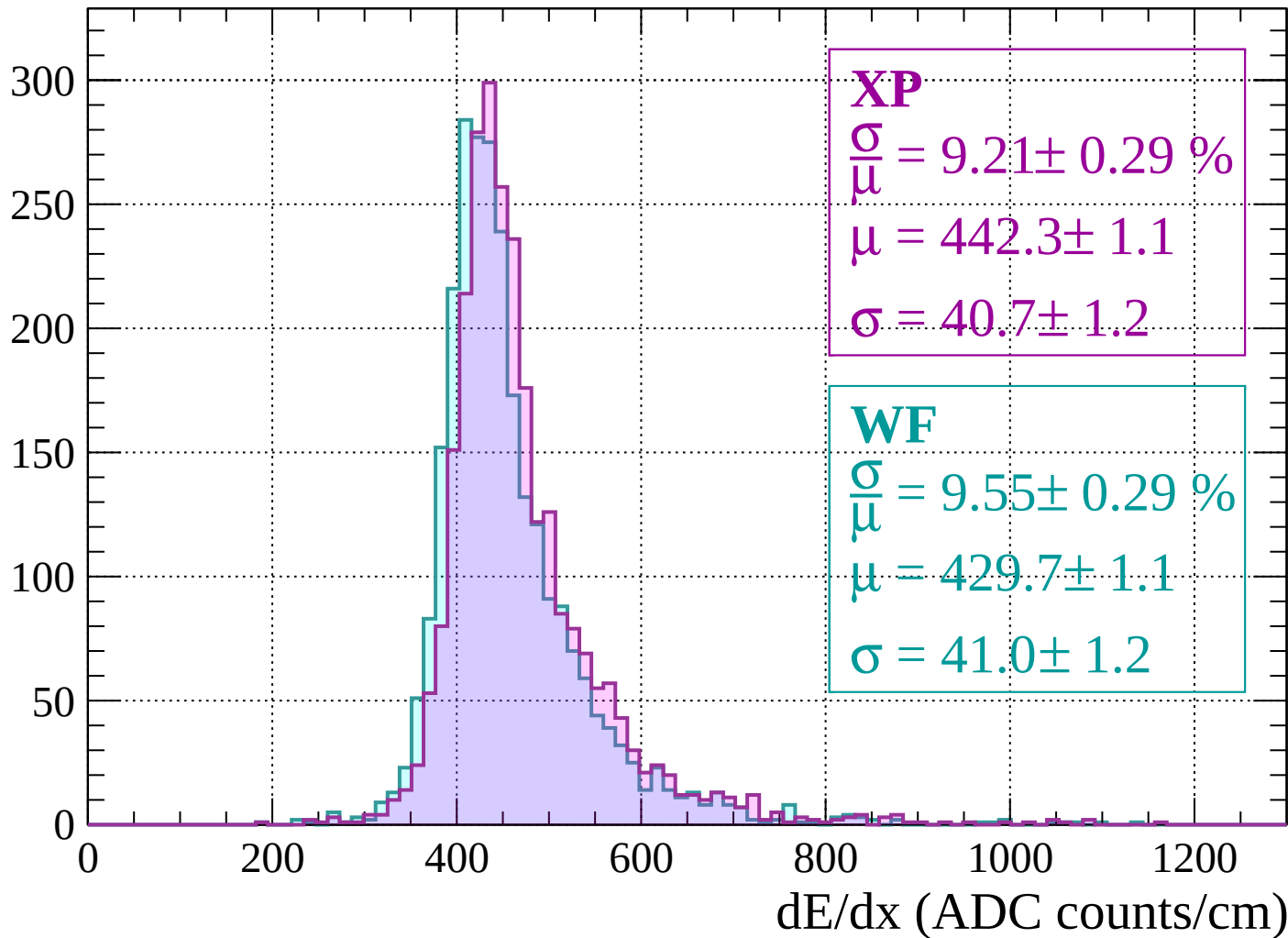
Energy loss | $-720 < p < -680$

Count



Energy loss | $-680 < p < -640$

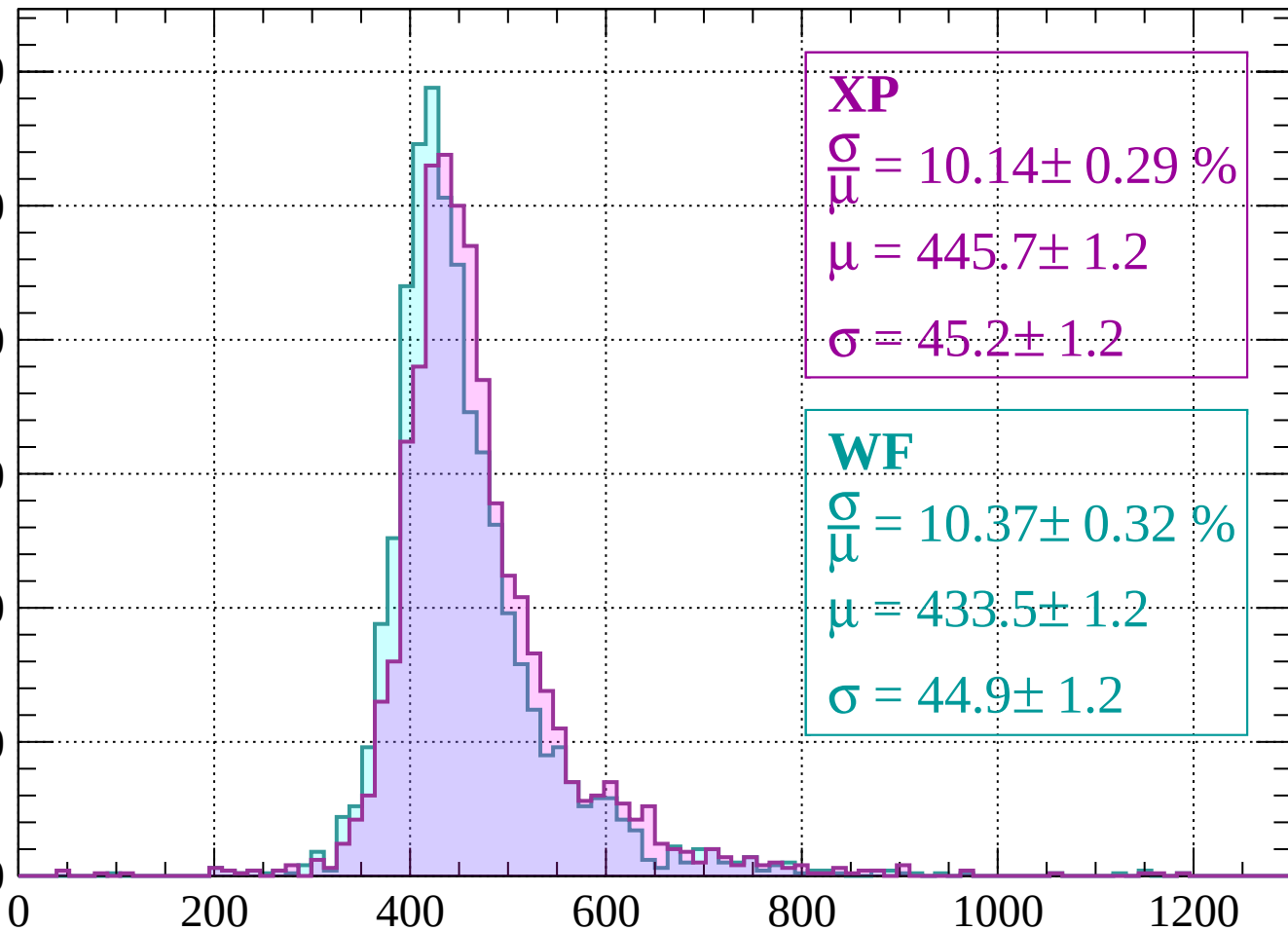
Count



Energy loss | $-640 < p < -600$

Count

300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 10.14 \pm 0.29 \%$$

$$\mu = 445.7 \pm 1.2$$

$$\sigma = 45.2 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 10.37 \pm 0.32 \%$$

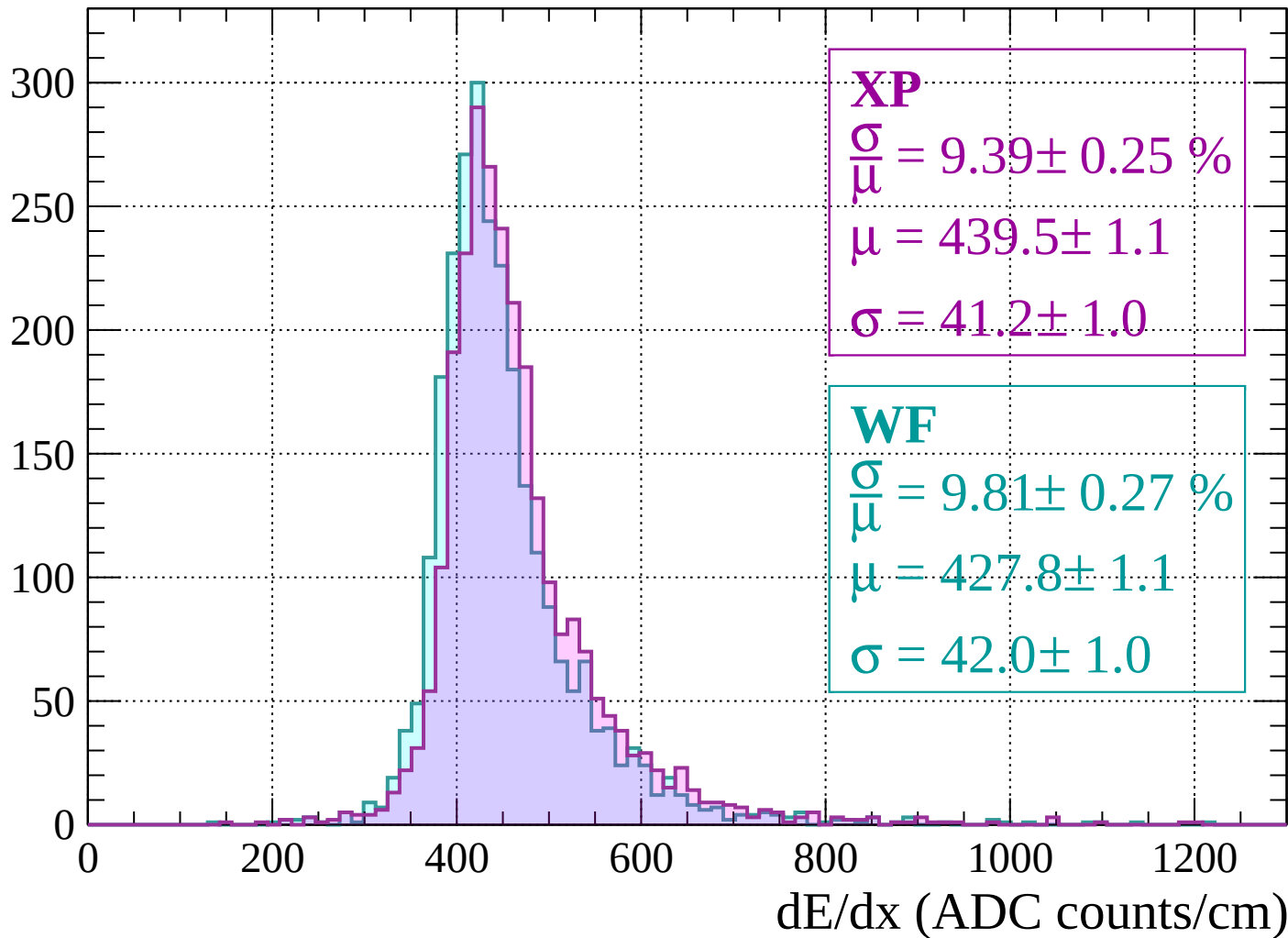
$$\mu = 433.5 \pm 1.2$$

$$\sigma = 44.9 \pm 1.2$$

dE/dx (ADC counts/cm)

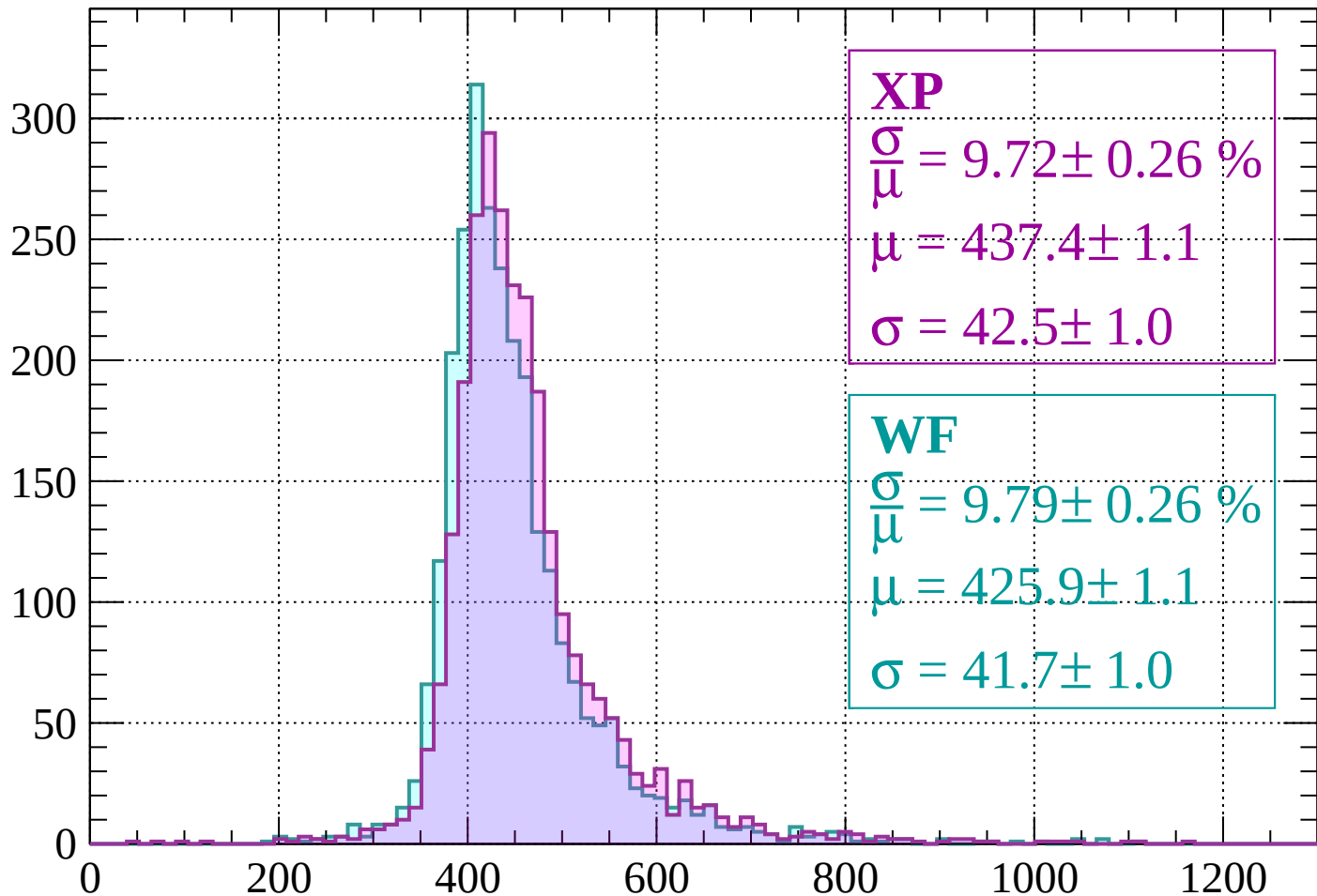
Energy loss | $-600 < p < -560$

Count



Energy loss | $-560 < p < -520$

Count



XP

$$\frac{\sigma}{\mu} = 9.72 \pm 0.26 \%$$

$$\mu = 437.4 \pm 1.1$$

$$\sigma = 42.5 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 9.79 \pm 0.26 \%$$

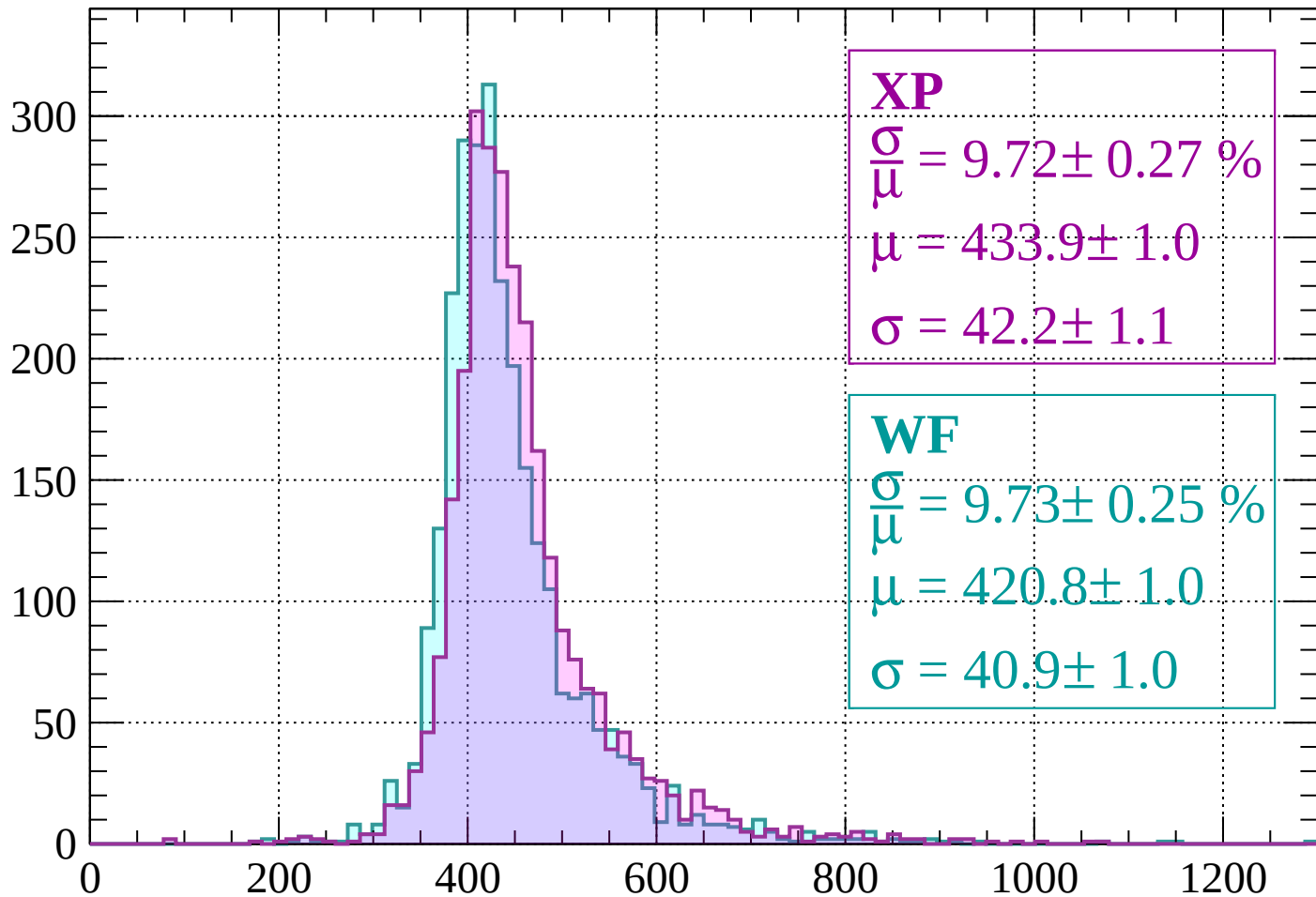
$$\mu = 425.9 \pm 1.1$$

$$\sigma = 41.7 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-520 < p < -480$

Count



XP

$$\frac{\sigma}{\mu} = 9.72 \pm 0.27 \%$$

$$\mu = 433.9 \pm 1.0$$

$$\sigma = 42.2 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 9.73 \pm 0.25 \%$$

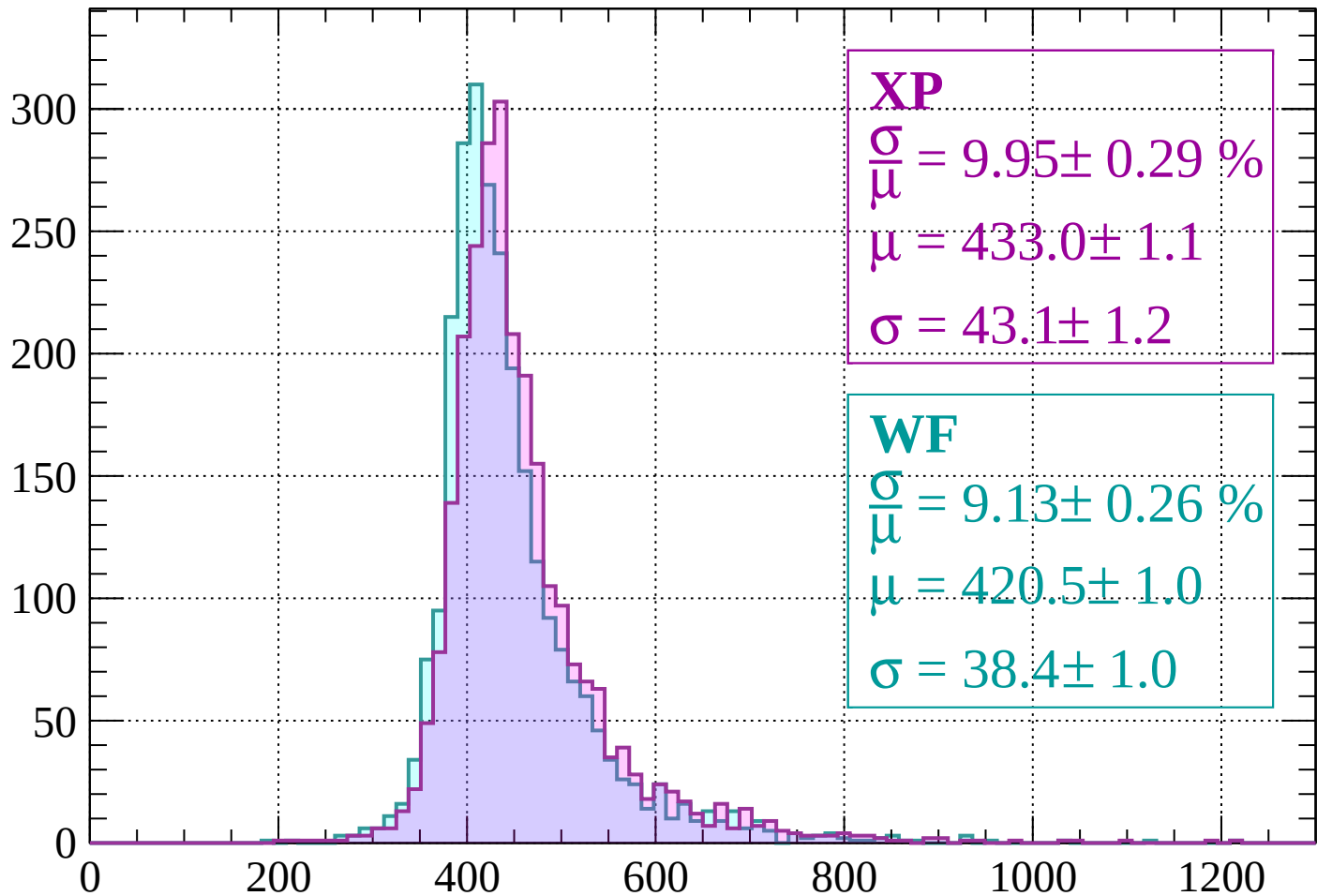
$$\mu = 420.8 \pm 1.0$$

$$\sigma = 40.9 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-480 < p < -440$

Count



XP

$$\frac{\sigma}{\mu} = 9.95 \pm 0.29 \%$$

$$\mu = 433.0 \pm 1.1$$

$$\sigma = 43.1 \pm 1.2$$

WF

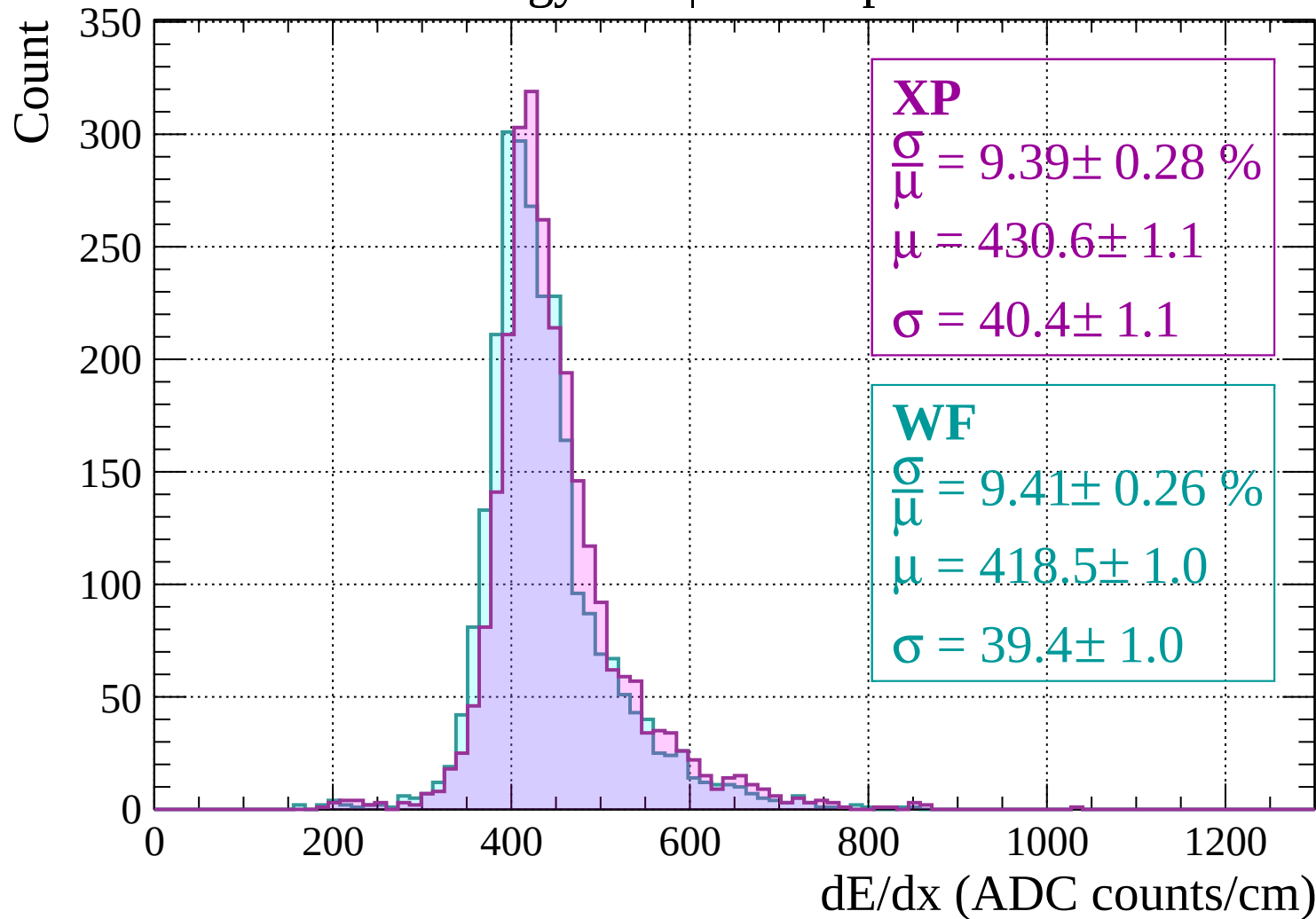
$$\frac{\sigma}{\mu} = 9.13 \pm 0.26 \%$$

$$\mu = 420.5 \pm 1.0$$

$$\sigma = 38.4 \pm 1.0$$

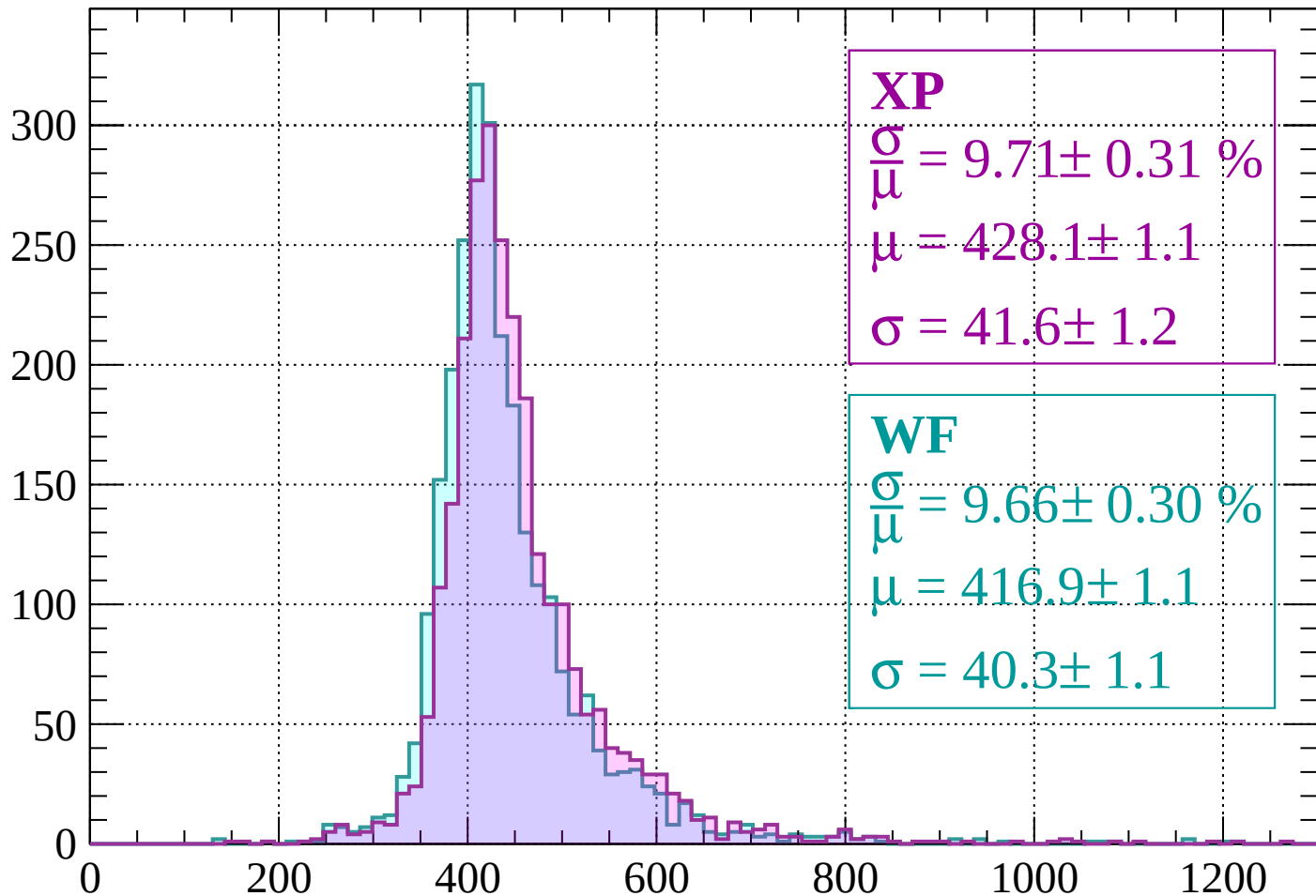
dE/dx (ADC counts/cm)

Energy loss | $-440 < p < -400$



Energy loss | $-400 < p < -360$

Count



XP

$$\frac{\sigma}{\mu} = 9.71 \pm 0.31 \%$$

$$\mu = 428.1 \pm 1.1$$

$$\sigma = 41.6 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 9.66 \pm 0.30 \%$$

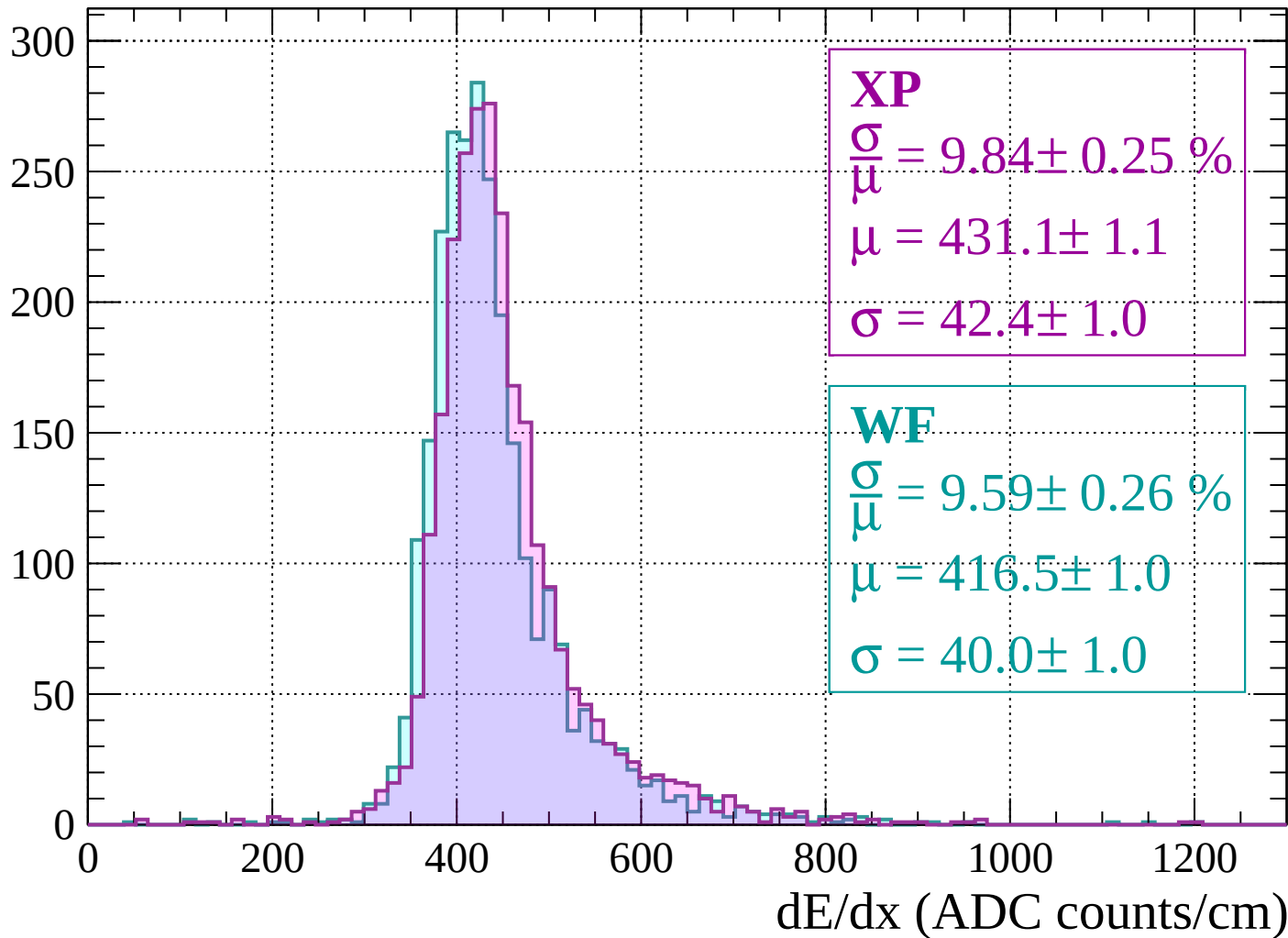
$$\mu = 416.9 \pm 1.1$$

$$\sigma = 40.3 \pm 1.1$$

dE/dx (ADC counts/cm)

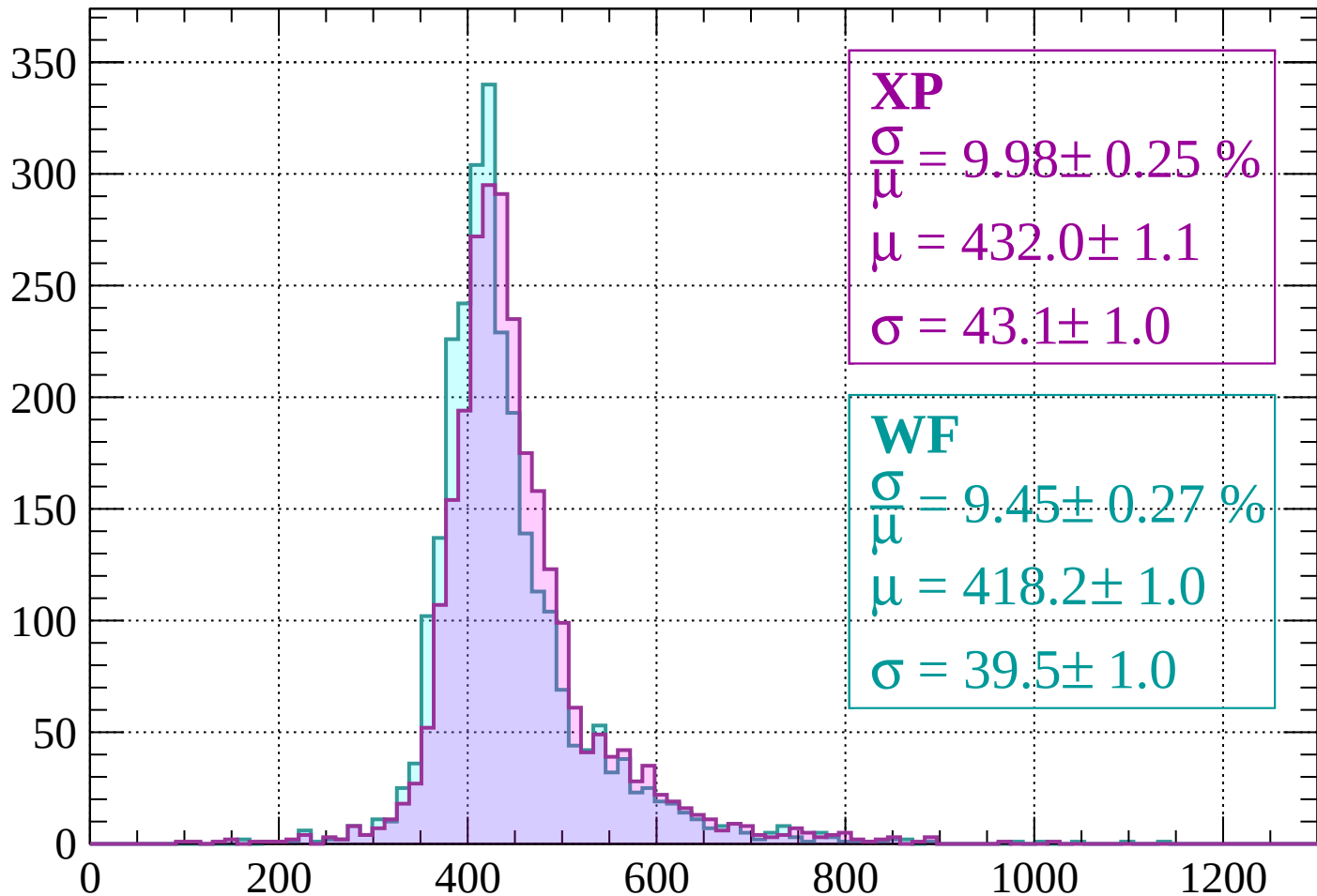
Energy loss | $-360 < p < -320$

Count



Energy loss | $-320 < p < -280$

Count



XP

$$\frac{\sigma}{\mu} = 9.98 \pm 0.25 \%$$

$$\mu = 432.0 \pm 1.1$$

$$\sigma = 43.1 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 9.45 \pm 0.27 \%$$

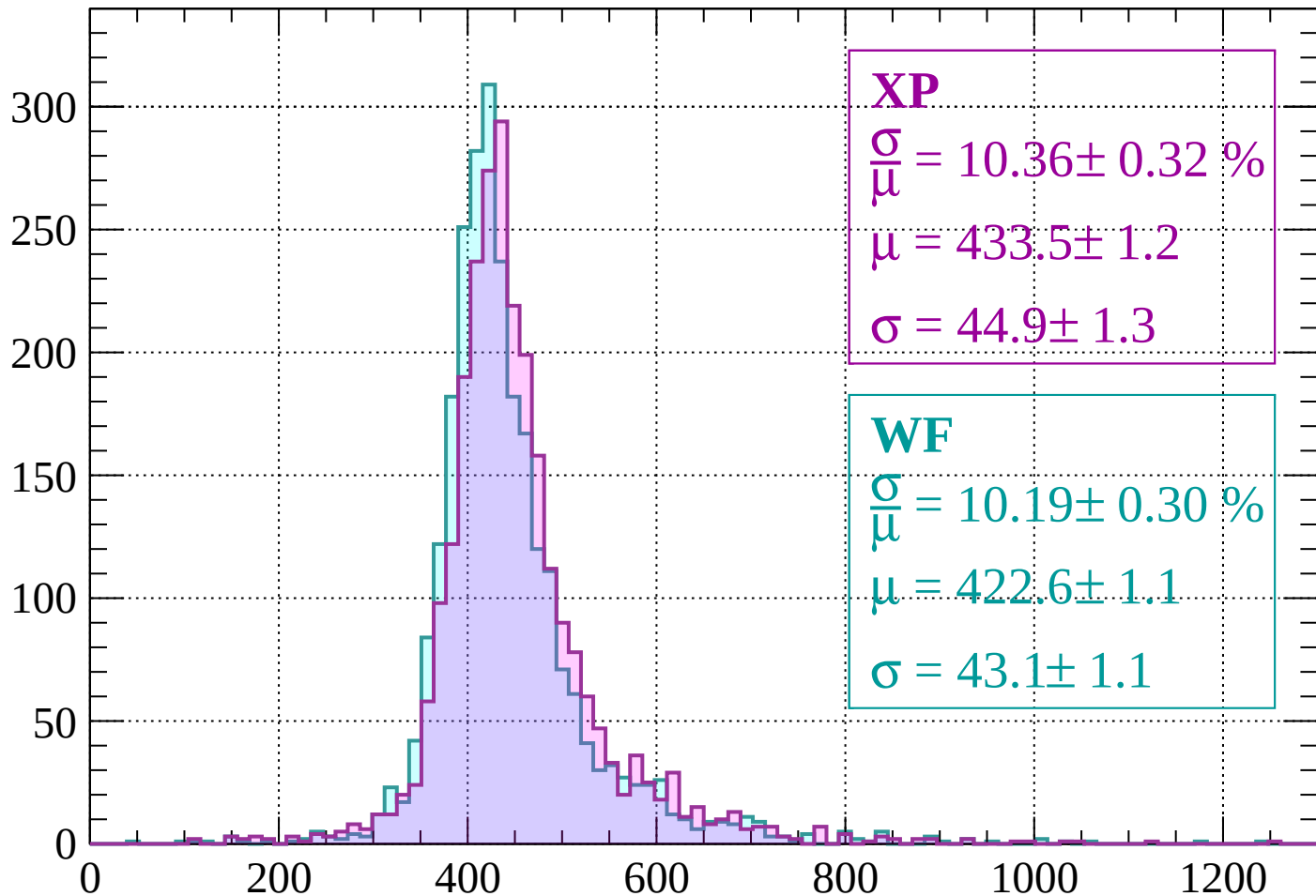
$$\mu = 418.2 \pm 1.0$$

$$\sigma = 39.5 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-280 < p < -240$

Count



XP

$$\frac{\sigma}{\mu} = 10.36 \pm 0.32 \%$$

$$\mu = 433.5 \pm 1.2$$

$$\sigma = 44.9 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 10.19 \pm 0.30 \%$$

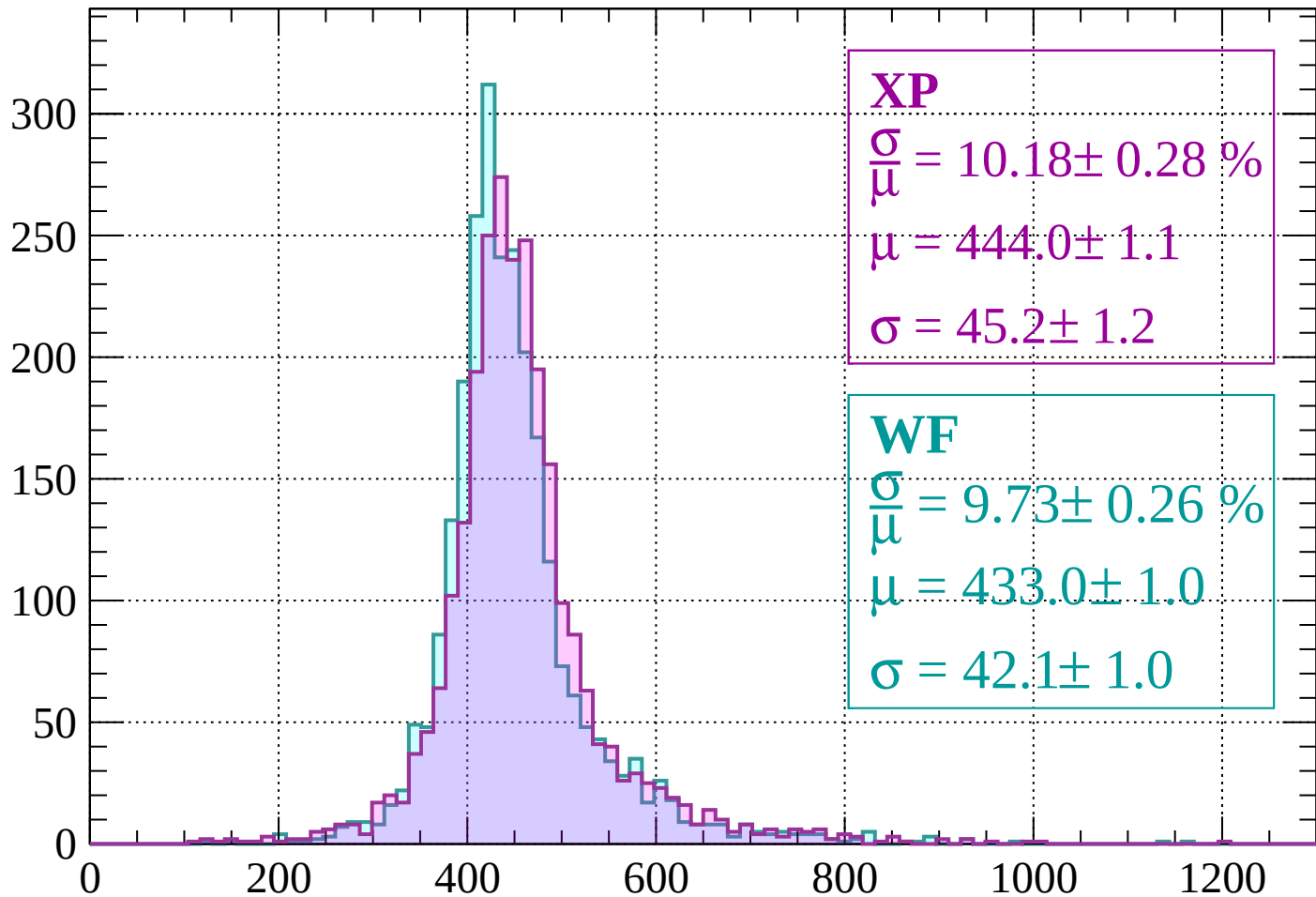
$$\mu = 422.6 \pm 1.1$$

$$\sigma = 43.1 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $-240 < p < -200$

Count



XP

$$\frac{\sigma}{\mu} = 10.18 \pm 0.28 \%$$

$$\mu = 444.0 \pm 1.1$$

$$\sigma = 45.2 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 9.73 \pm 0.26 \%$$

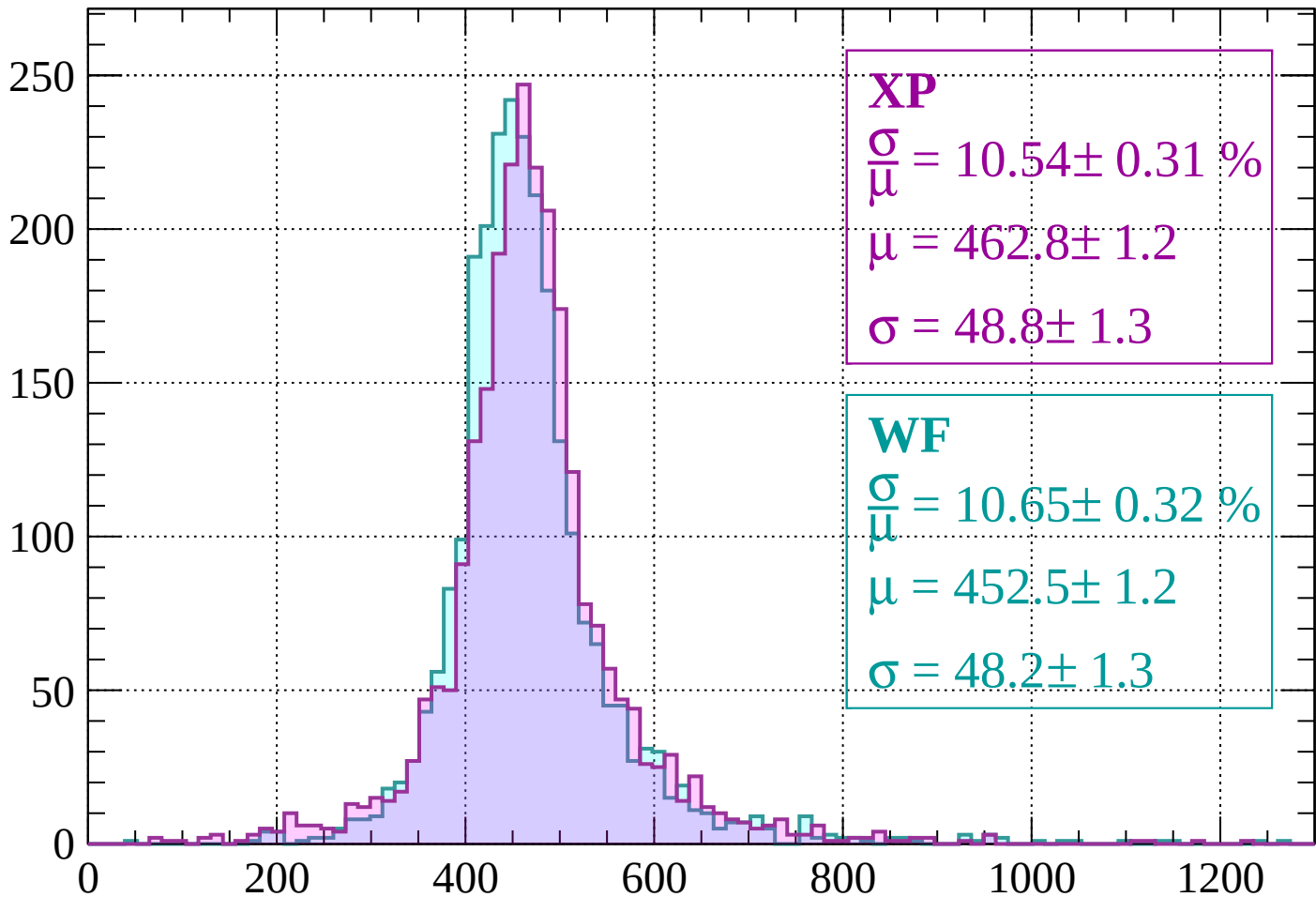
$$\mu = 433.0 \pm 1.0$$

$$\sigma = 42.1 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-200 < p < -160$

Count



XP

$$\frac{\sigma}{\mu} = 10.54 \pm 0.31 \%$$

$$\mu = 462.8 \pm 1.2$$

$$\sigma = 48.8 \pm 1.3$$

WF

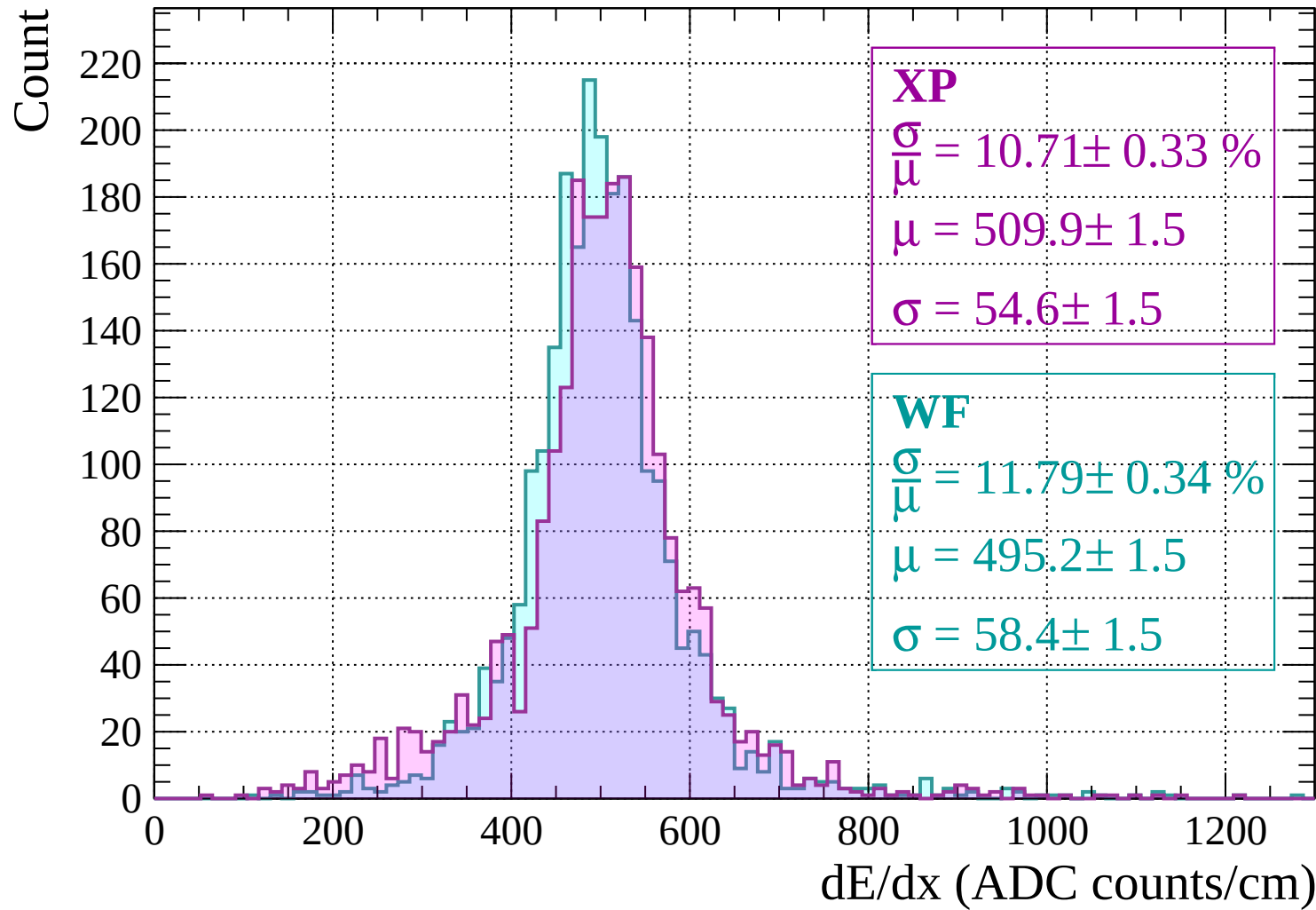
$$\frac{\sigma}{\mu} = 10.65 \pm 0.32 \%$$

$$\mu = 452.5 \pm 1.2$$

$$\sigma = 48.2 \pm 1.3$$

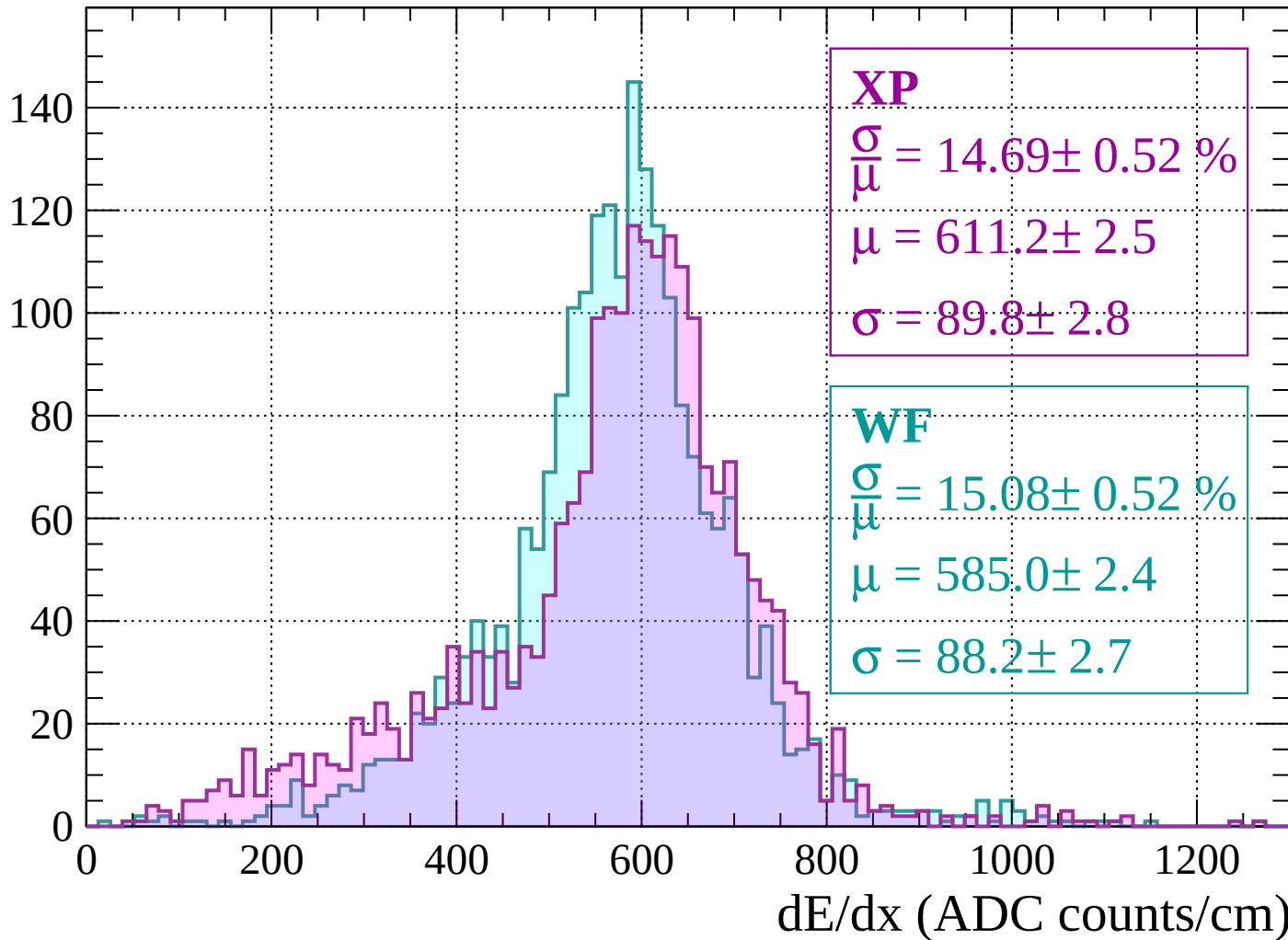
dE/dx (ADC counts/cm)

Energy loss | -160 < p < -120



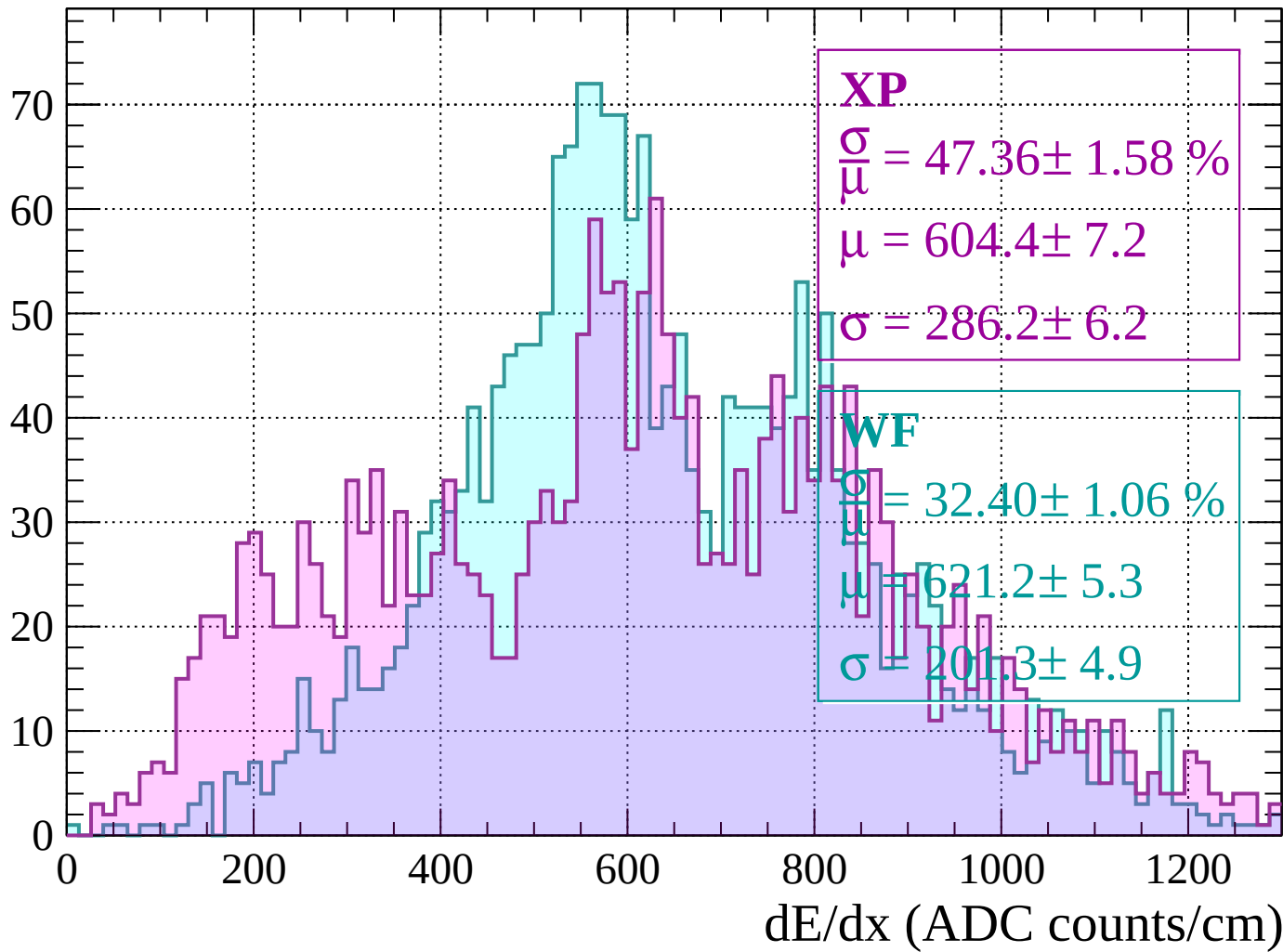
Energy loss | $-120 < p < -80$

Count

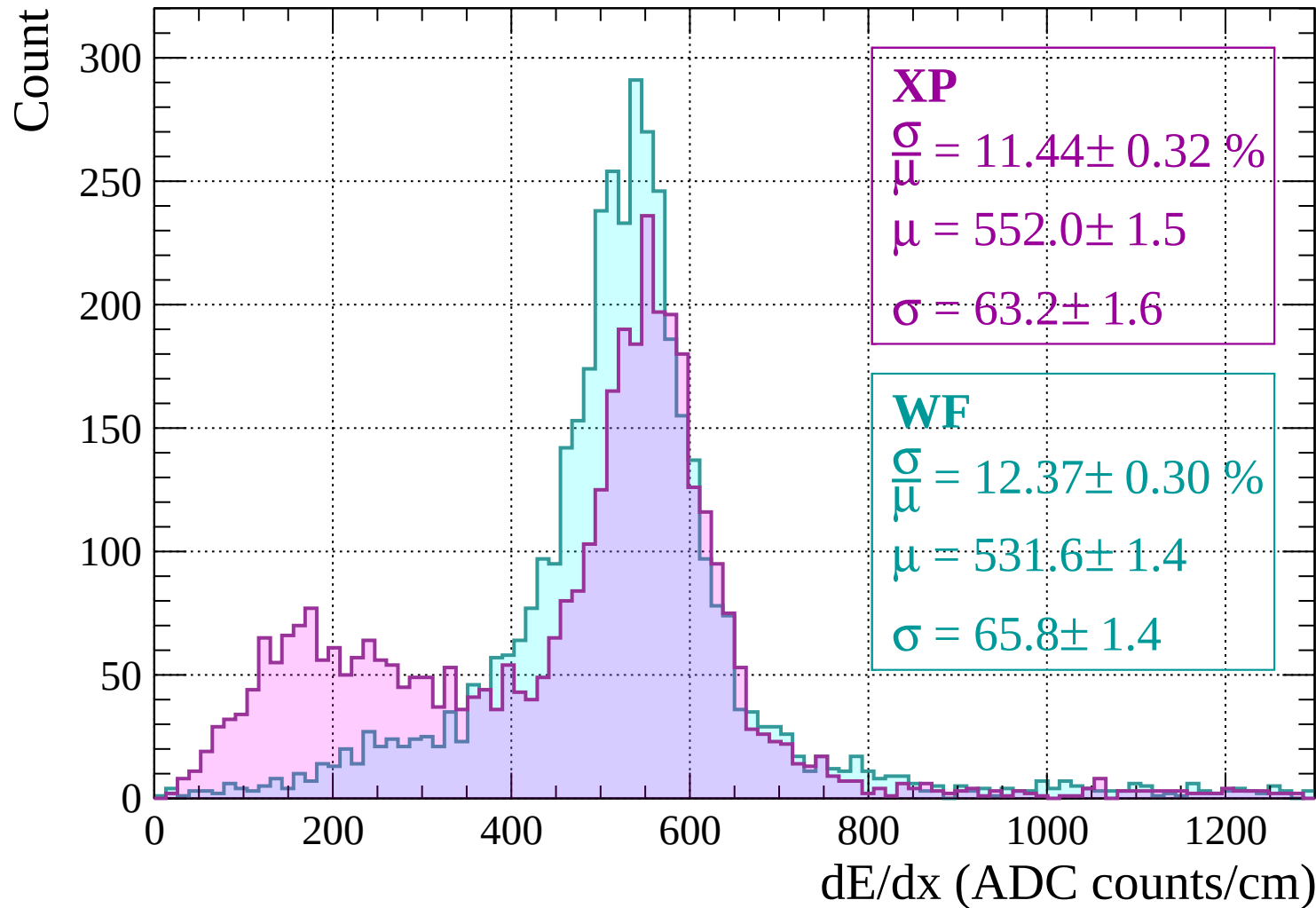


Energy loss | $-80 < p < -40$

Count

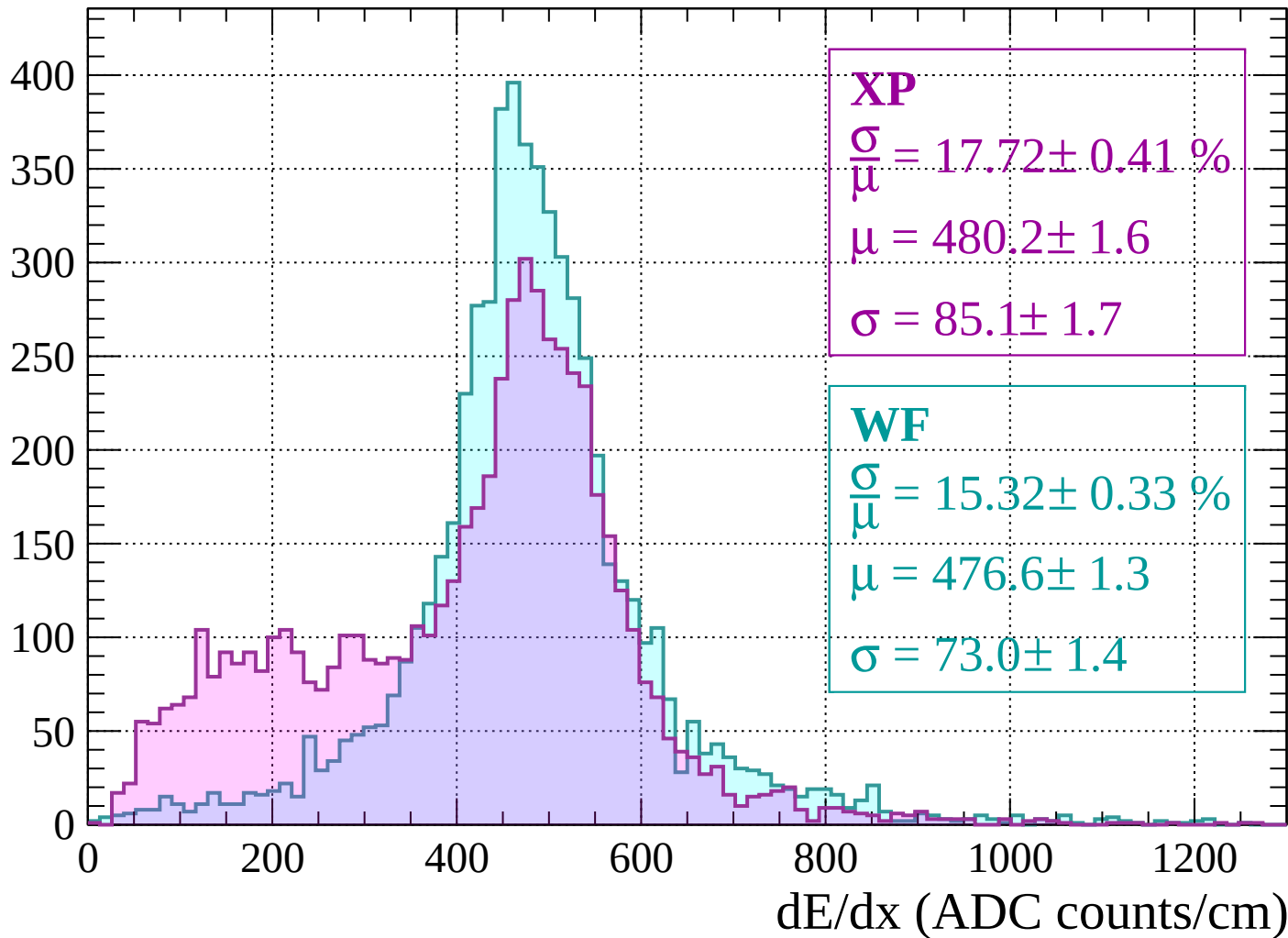


Energy loss | $-40 < p < 0$



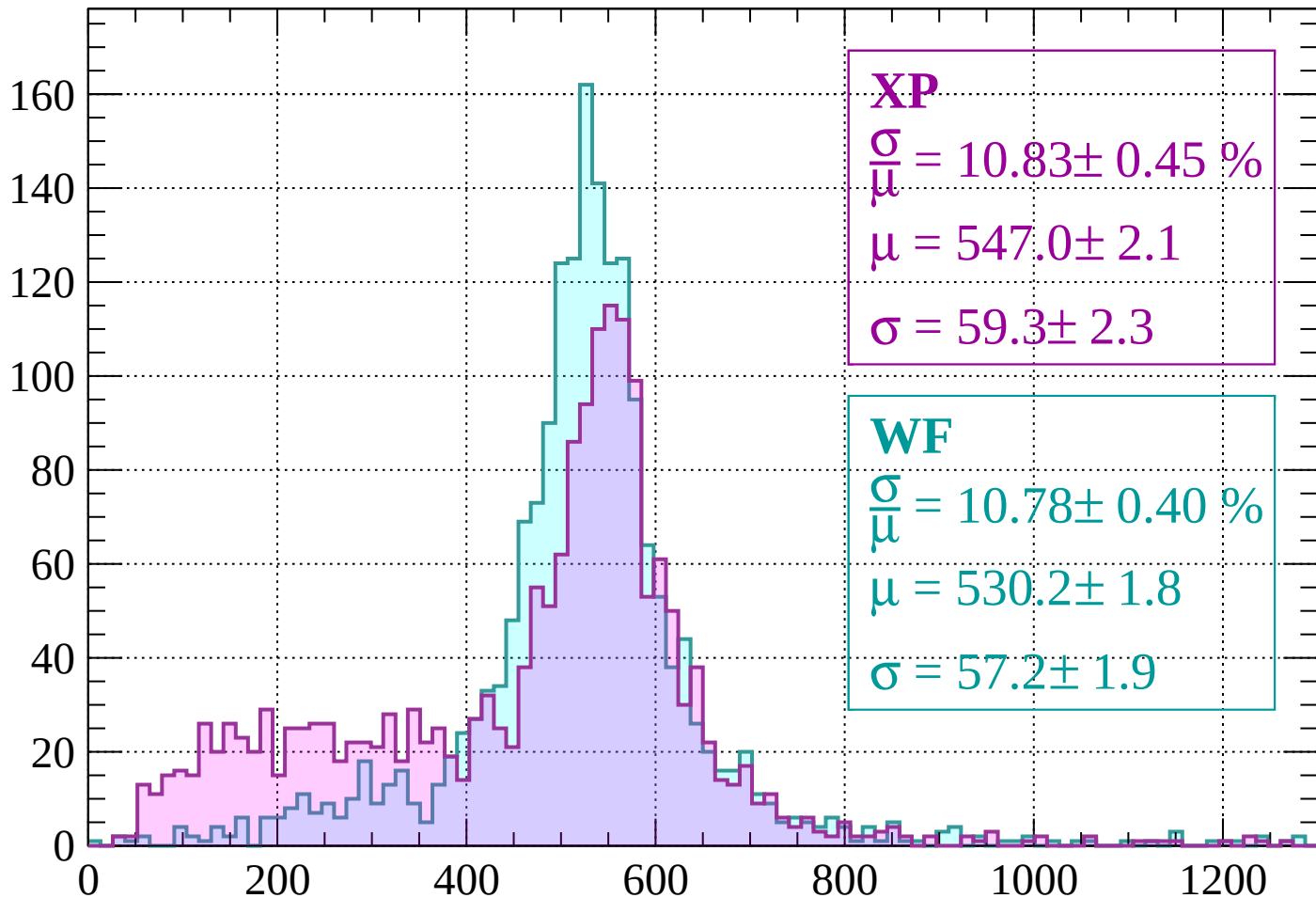
Energy loss | $0 < p < 40$

Count



Energy loss | $40 < p < 80$

Count



XP

$$\frac{\sigma}{\mu} = 10.83 \pm 0.45 \%$$

$$\mu = 547.0 \pm 2.1$$

$$\sigma = 59.3 \pm 2.3$$

WF

$$\frac{\sigma}{\mu} = 10.78 \pm 0.40 \%$$

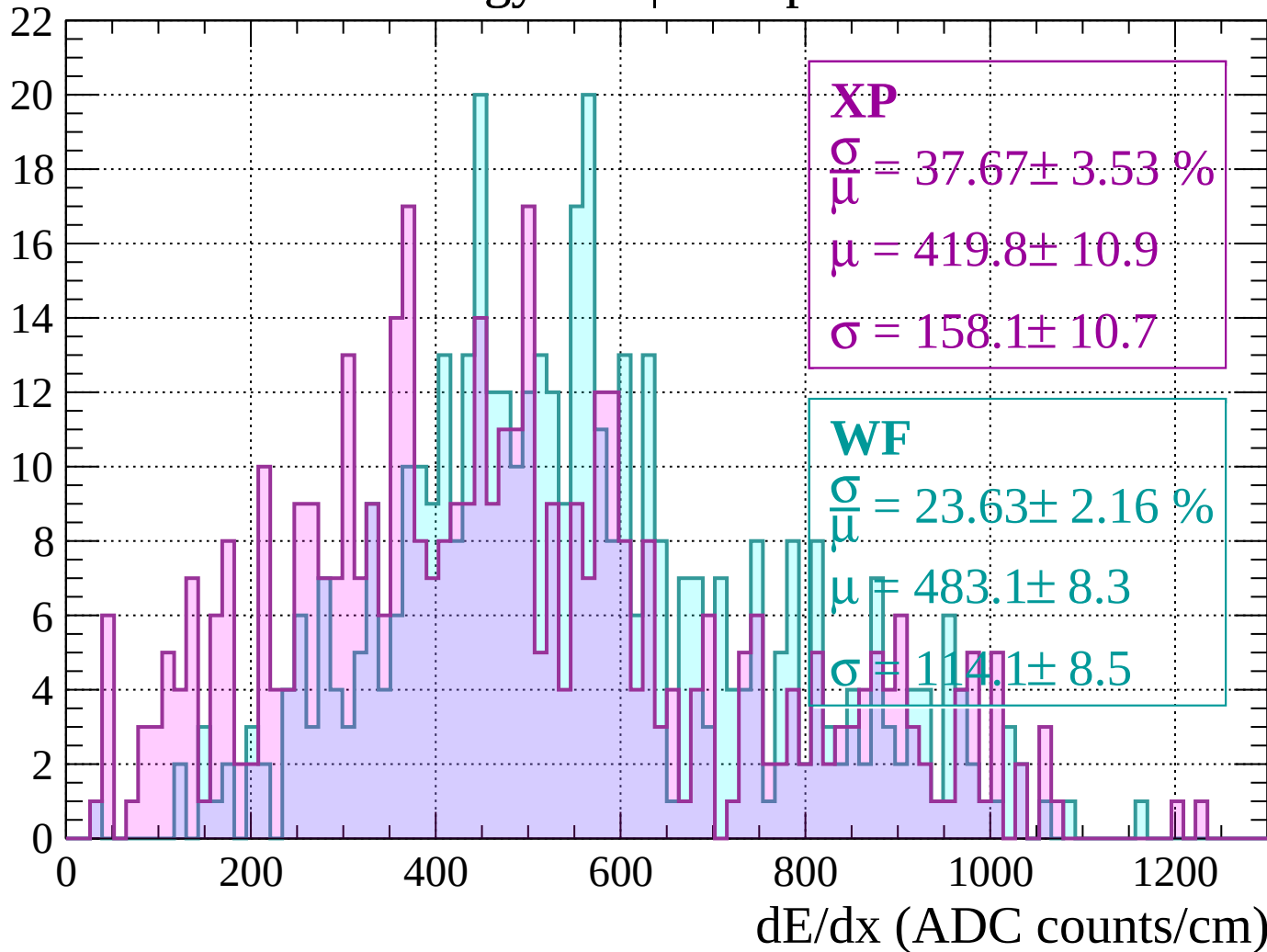
$$\mu = 530.2 \pm 1.8$$

$$\sigma = 57.2 \pm 1.9$$

dE/dx (ADC counts/cm)

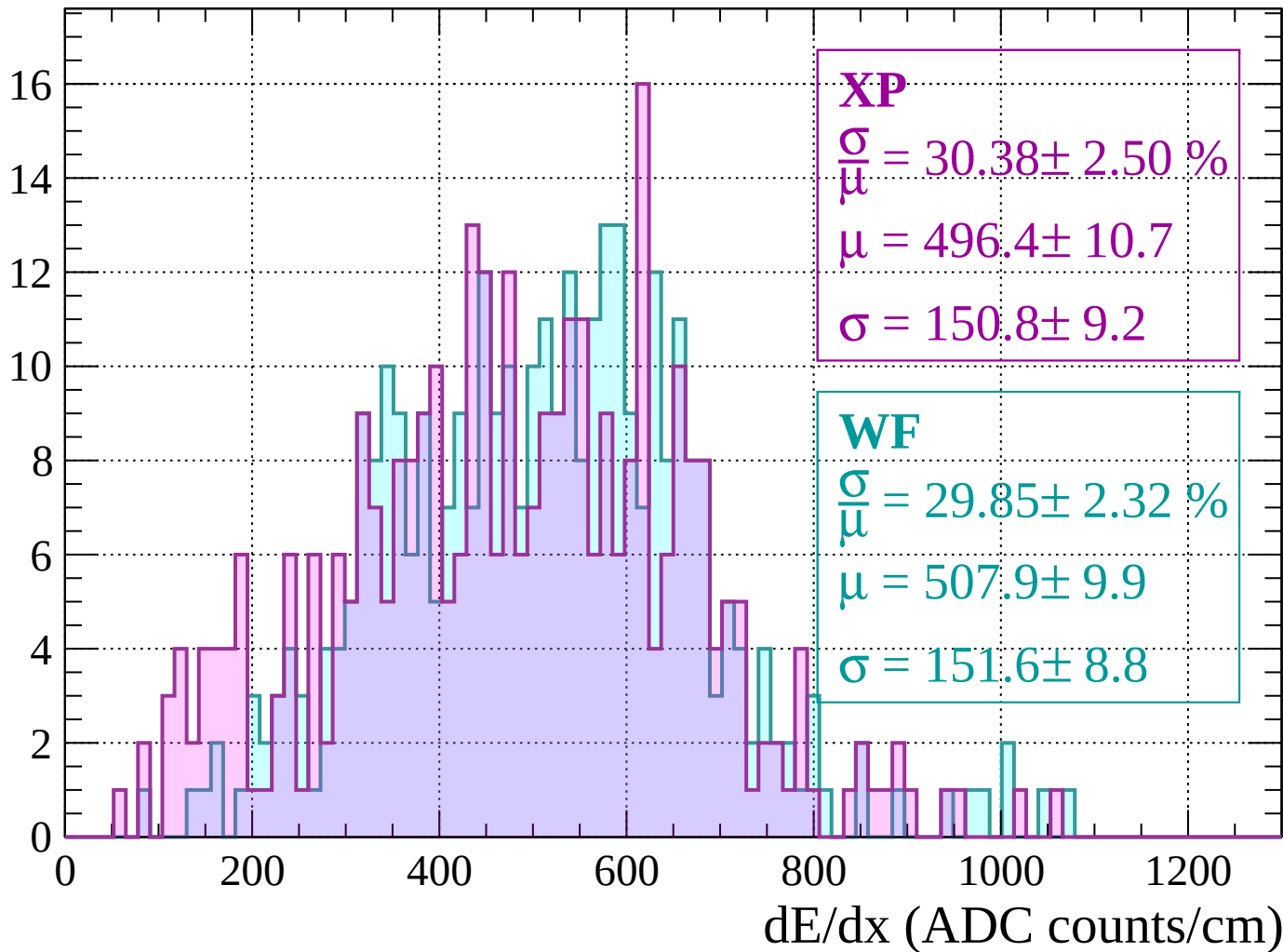
Energy loss | $80 < p < 120$

Count



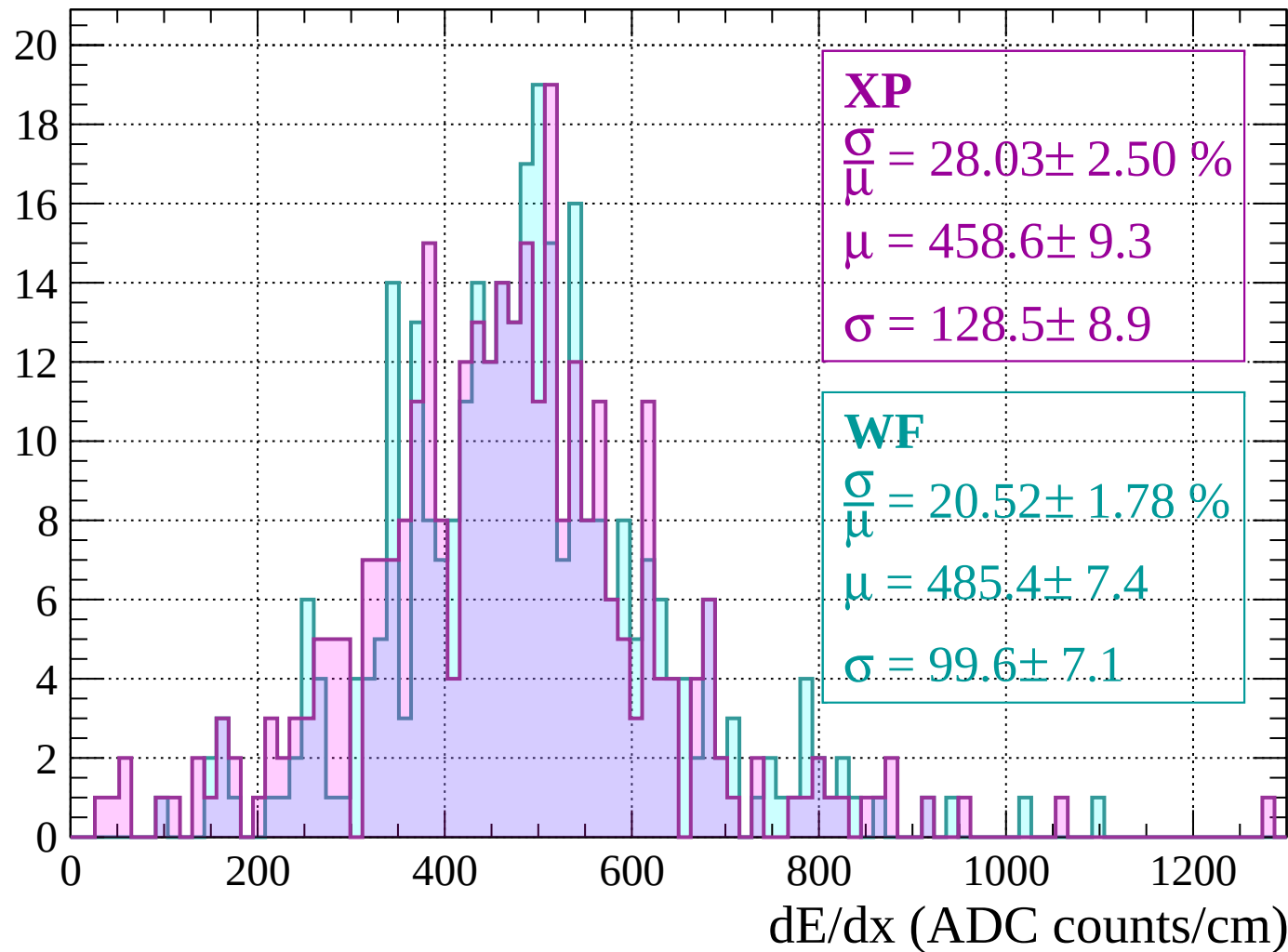
Energy loss | $120 < p < 160$

Count



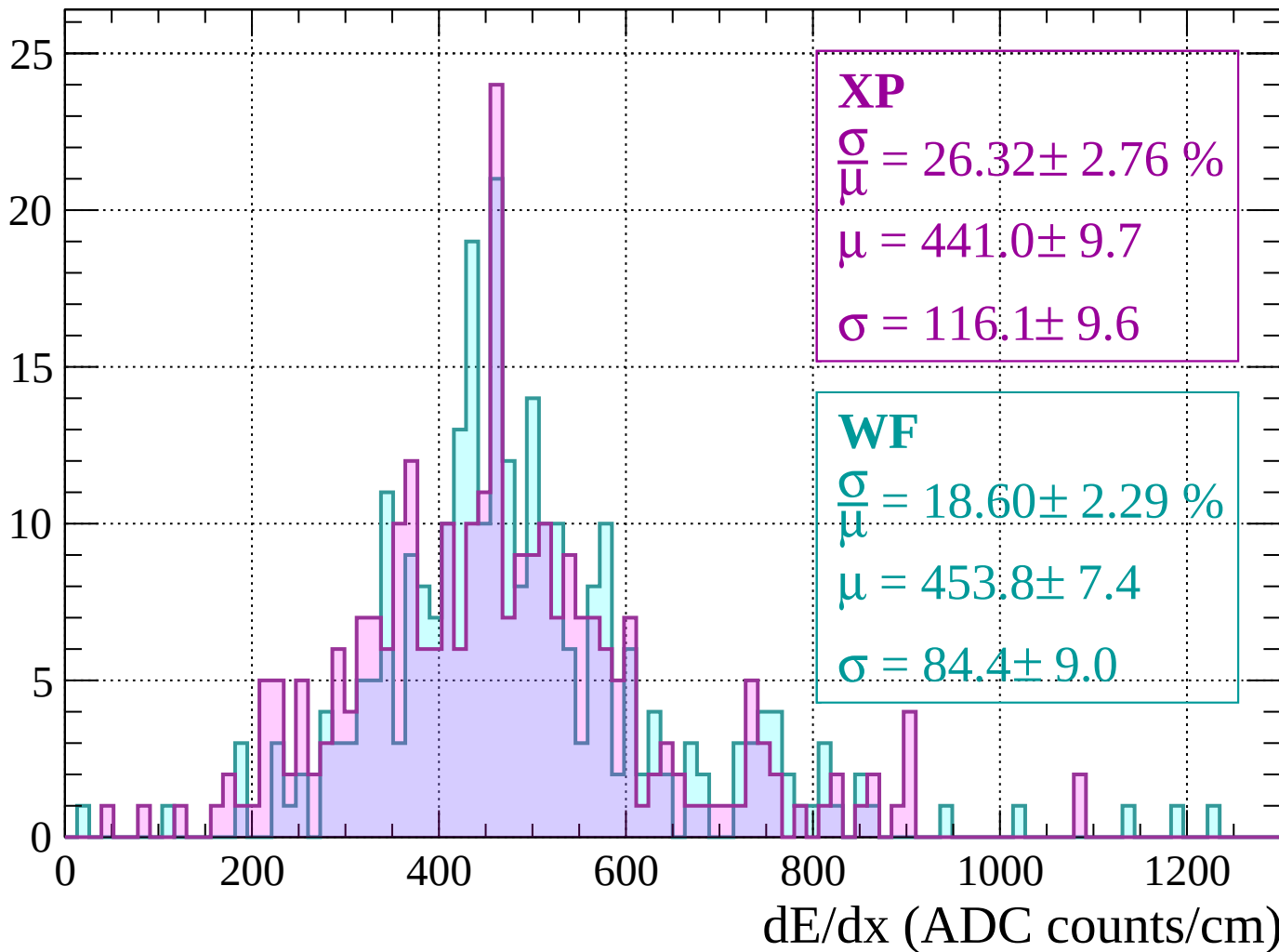
Energy loss | $160 < p < 200$

Count



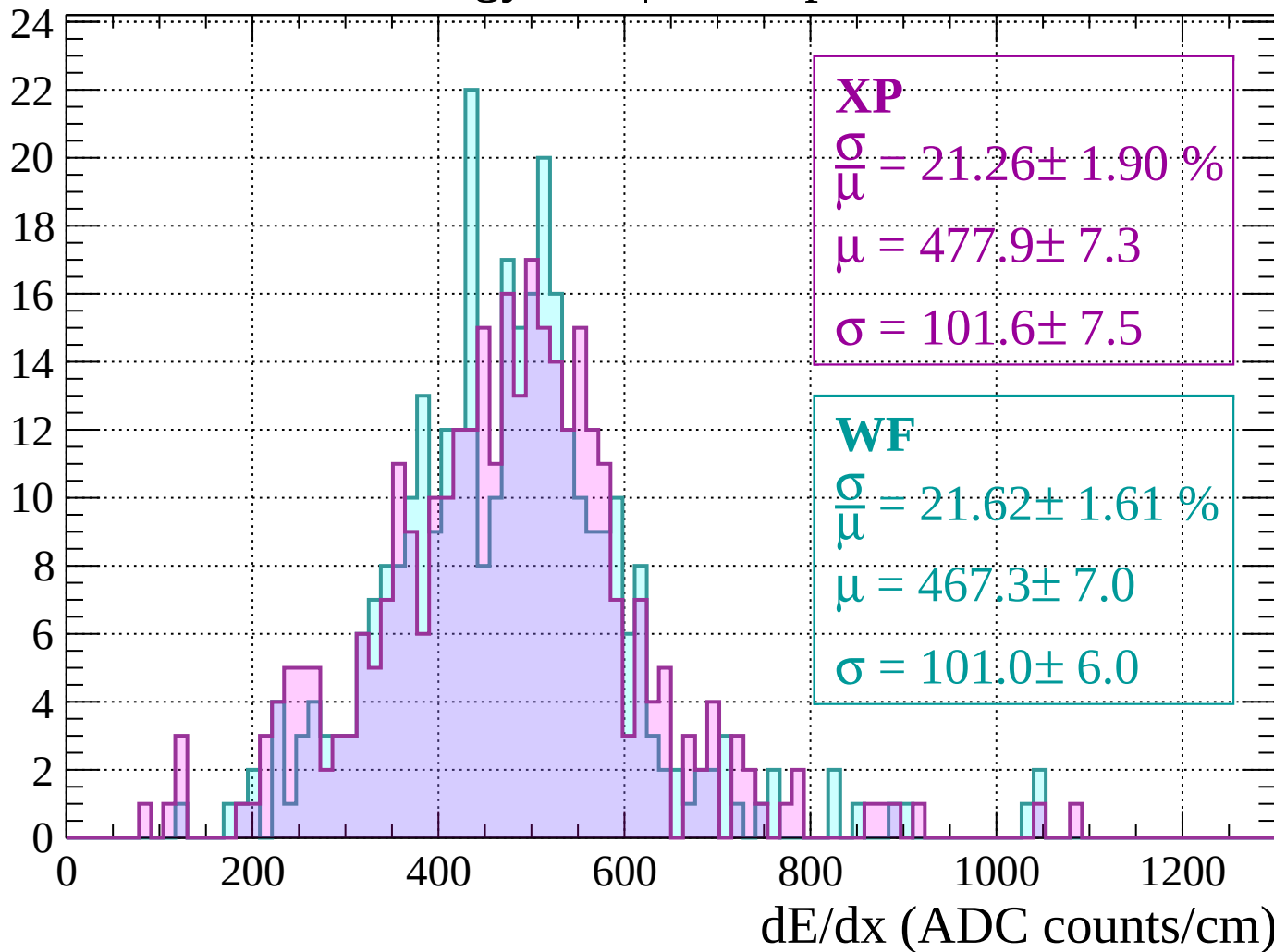
Energy loss | $200 < p < 240$

Count



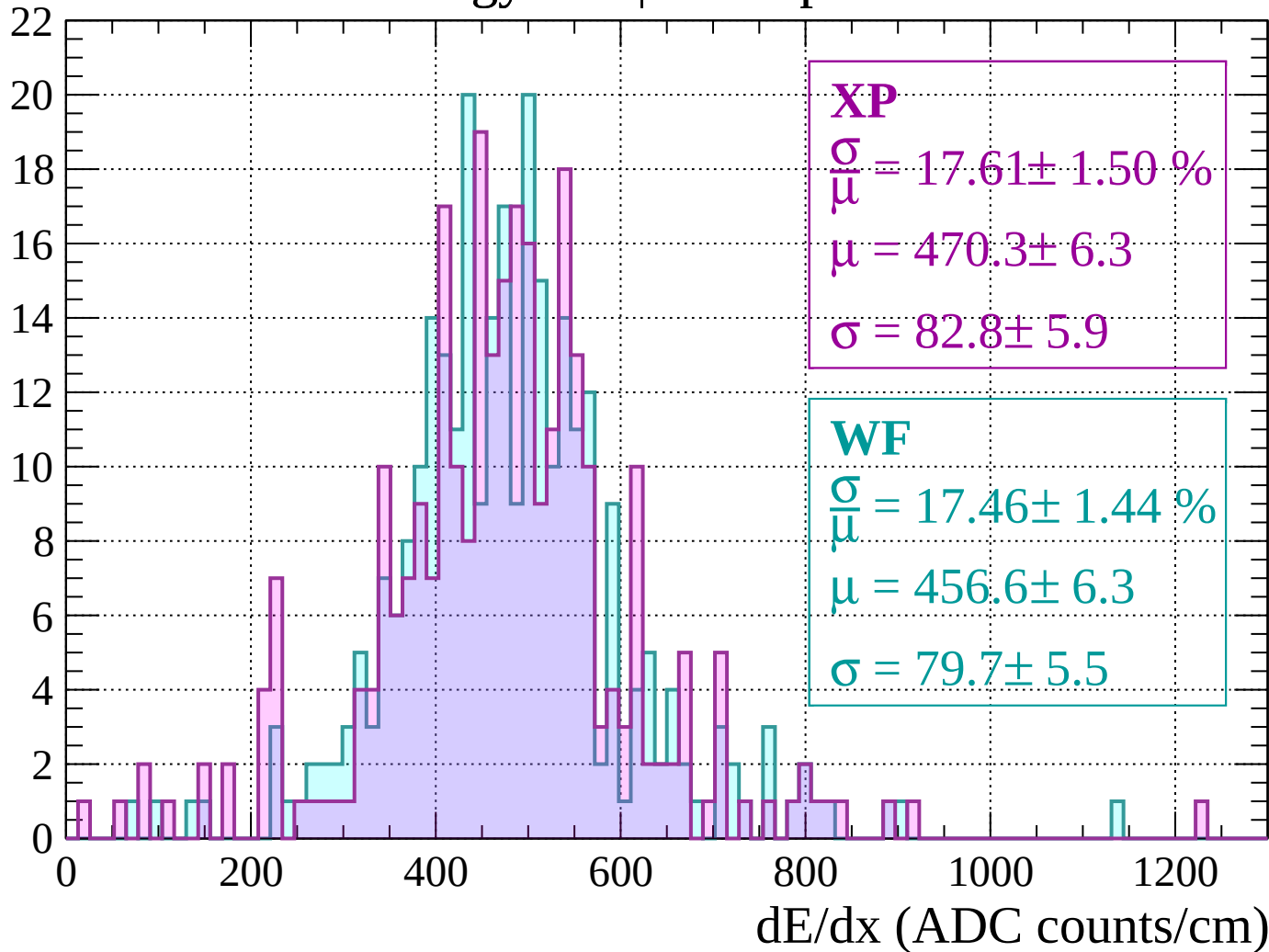
Energy loss | $240 < p < 280$

Count



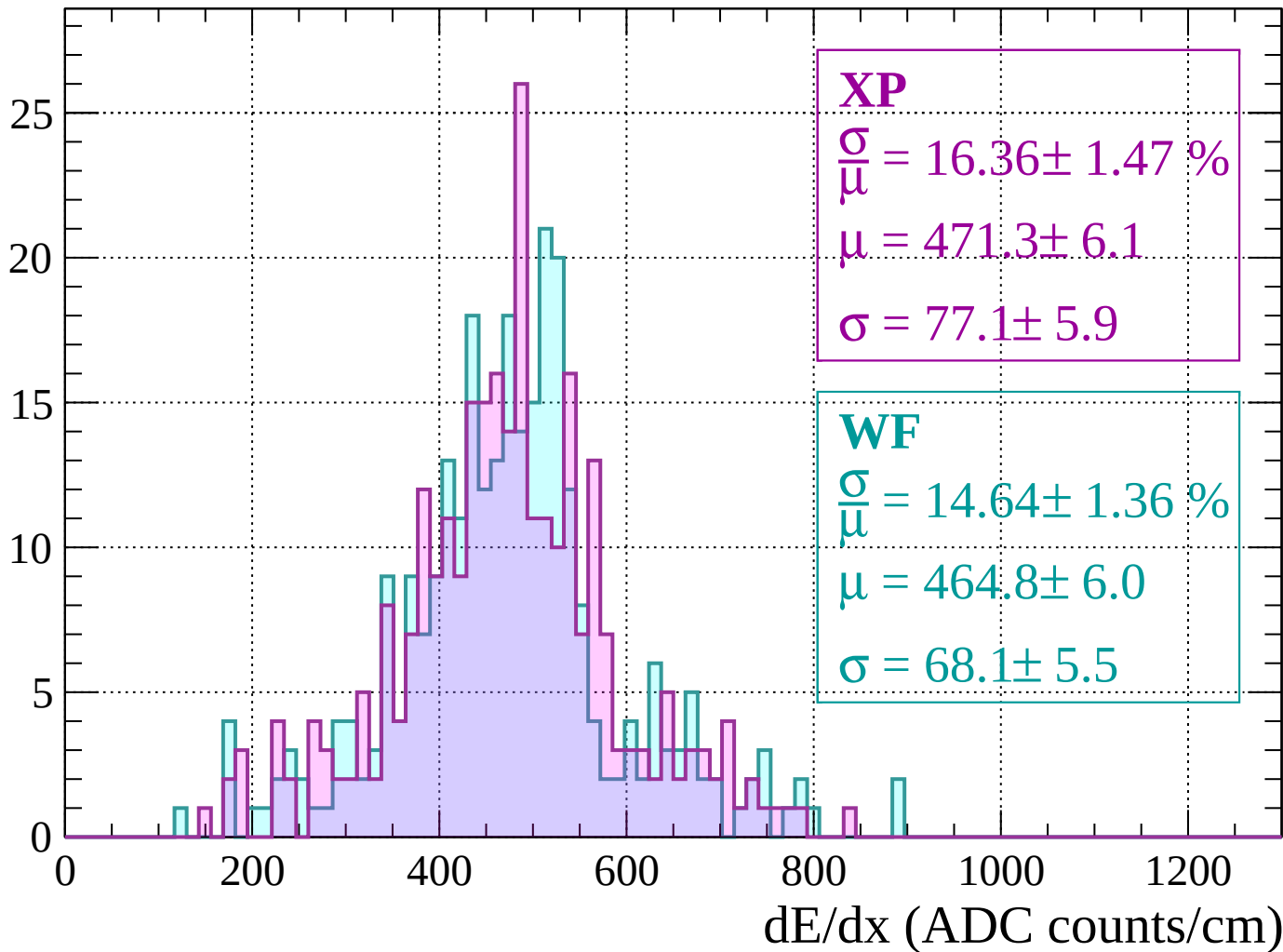
Energy loss | $280 < p < 320$

Count



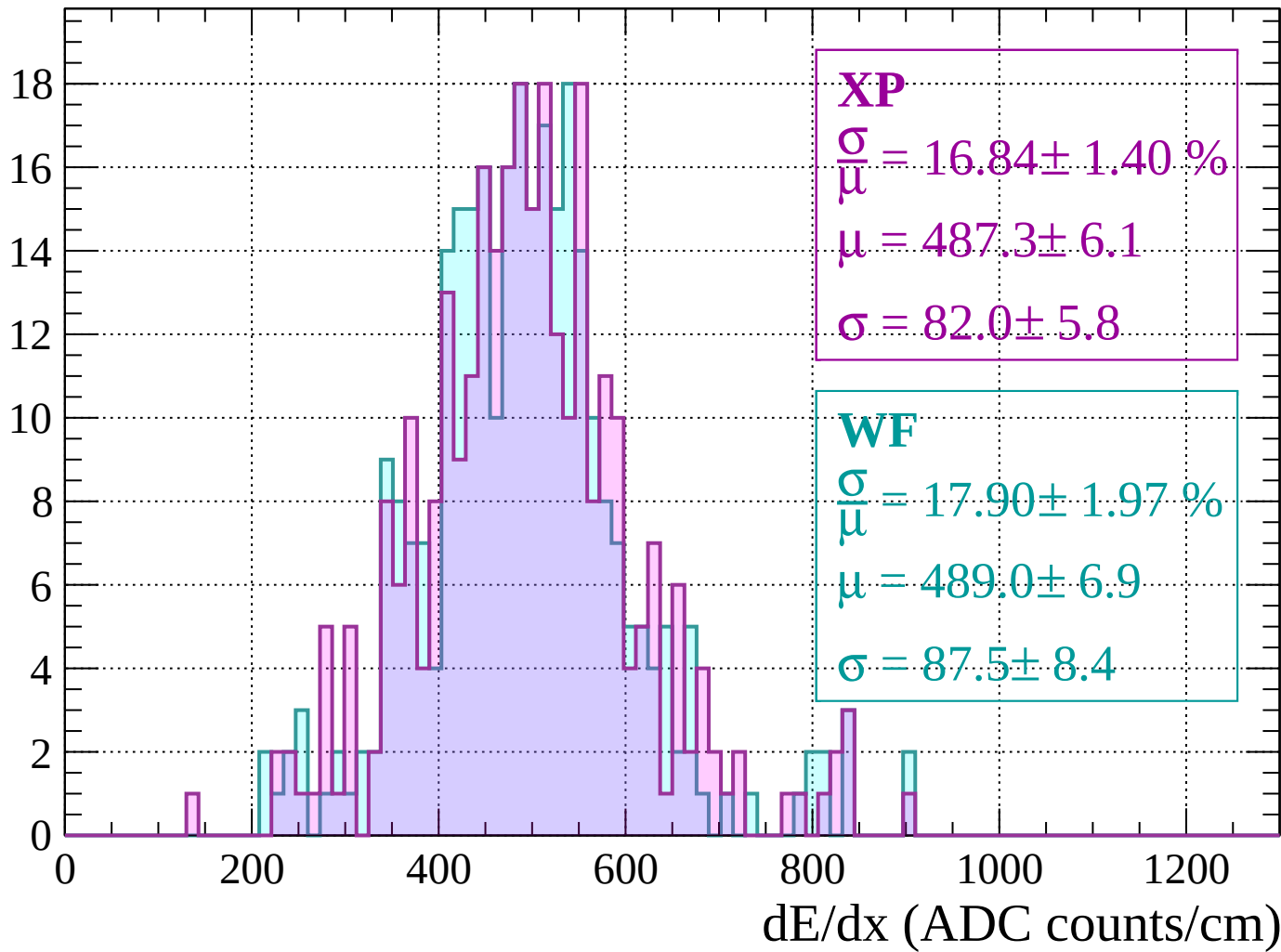
Energy loss | $320 < p < 360$

Count



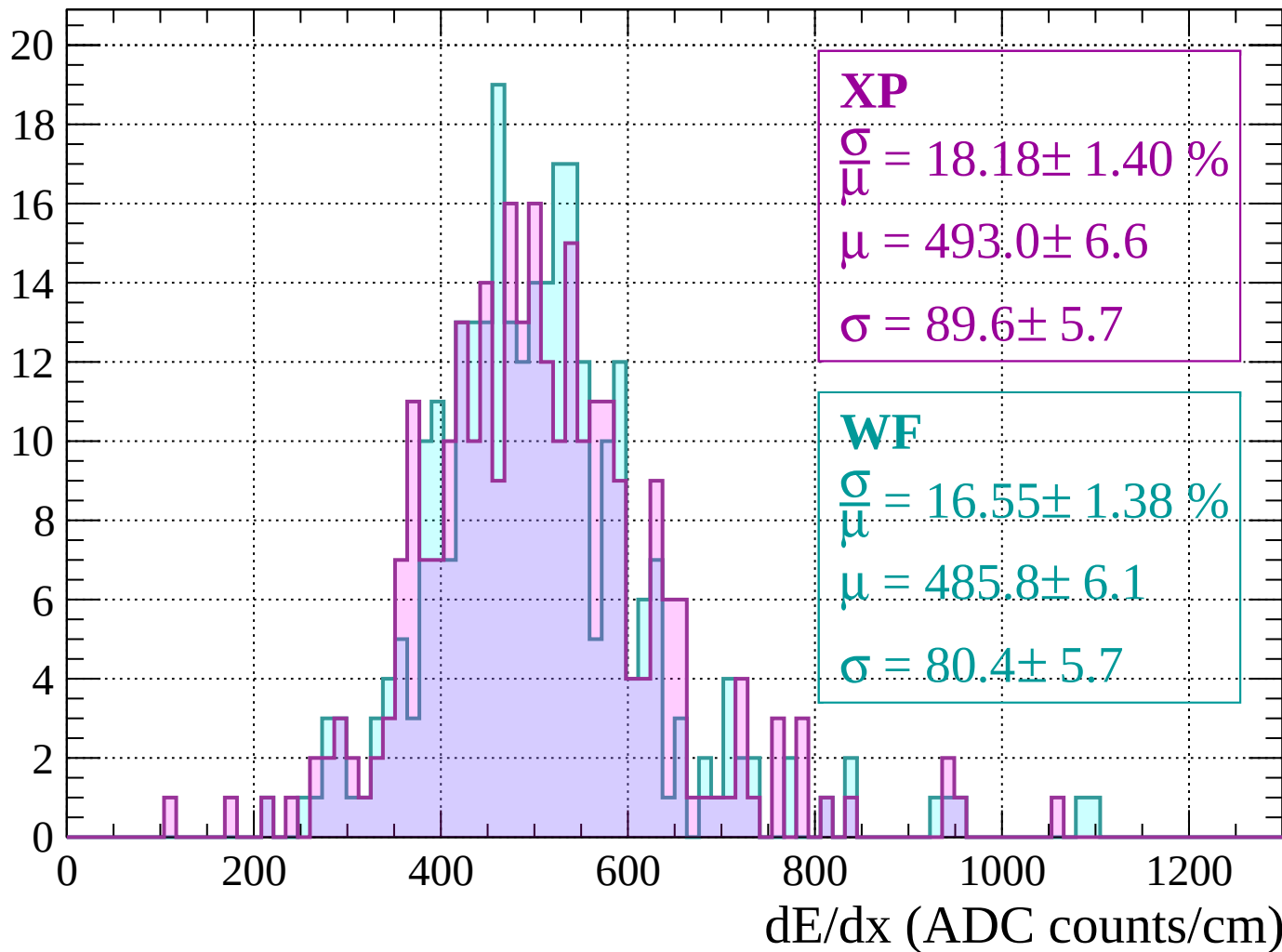
Energy loss | $360 < p < 400$

Count



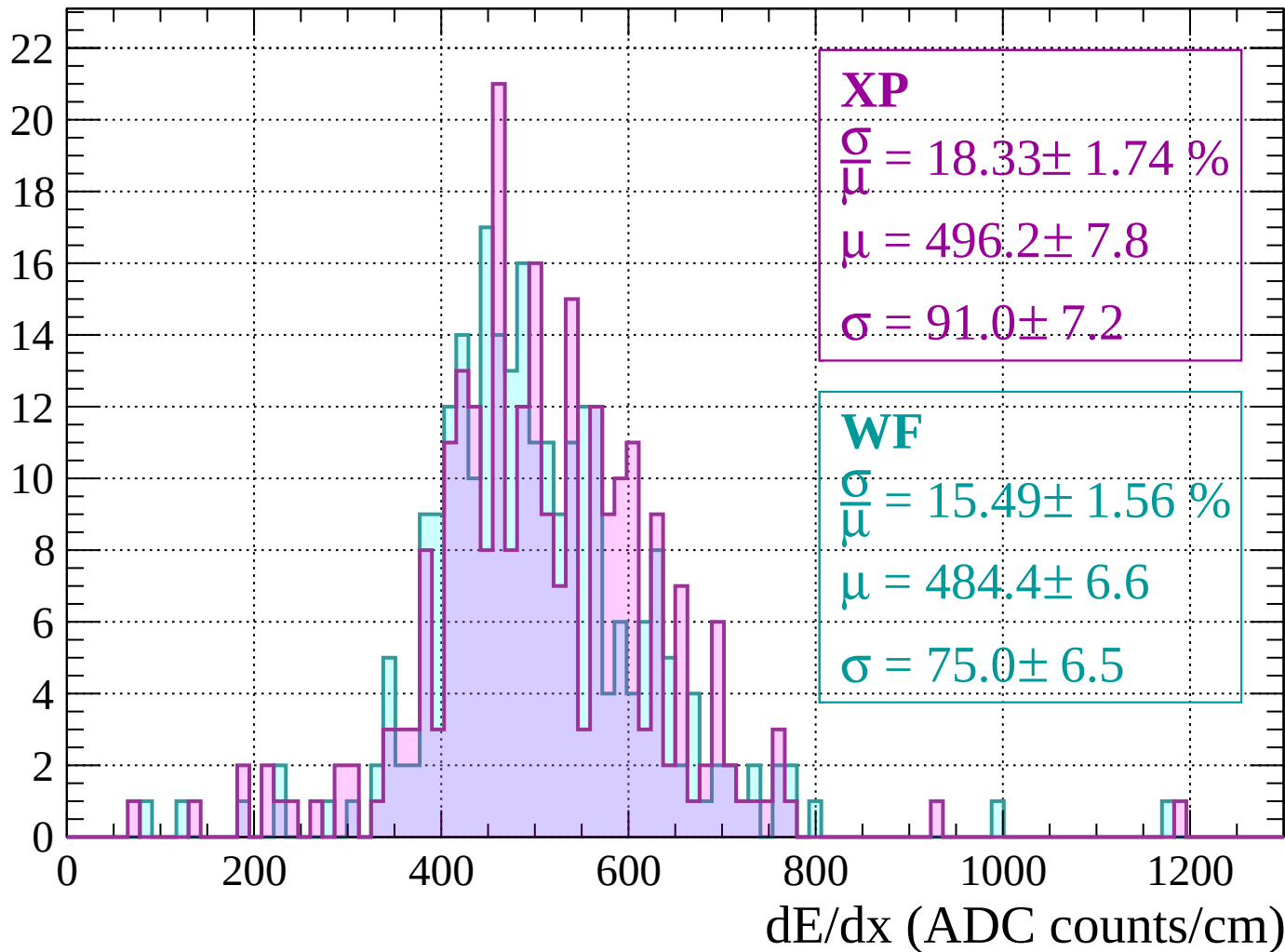
Energy loss | $400 < p < 440$

Count



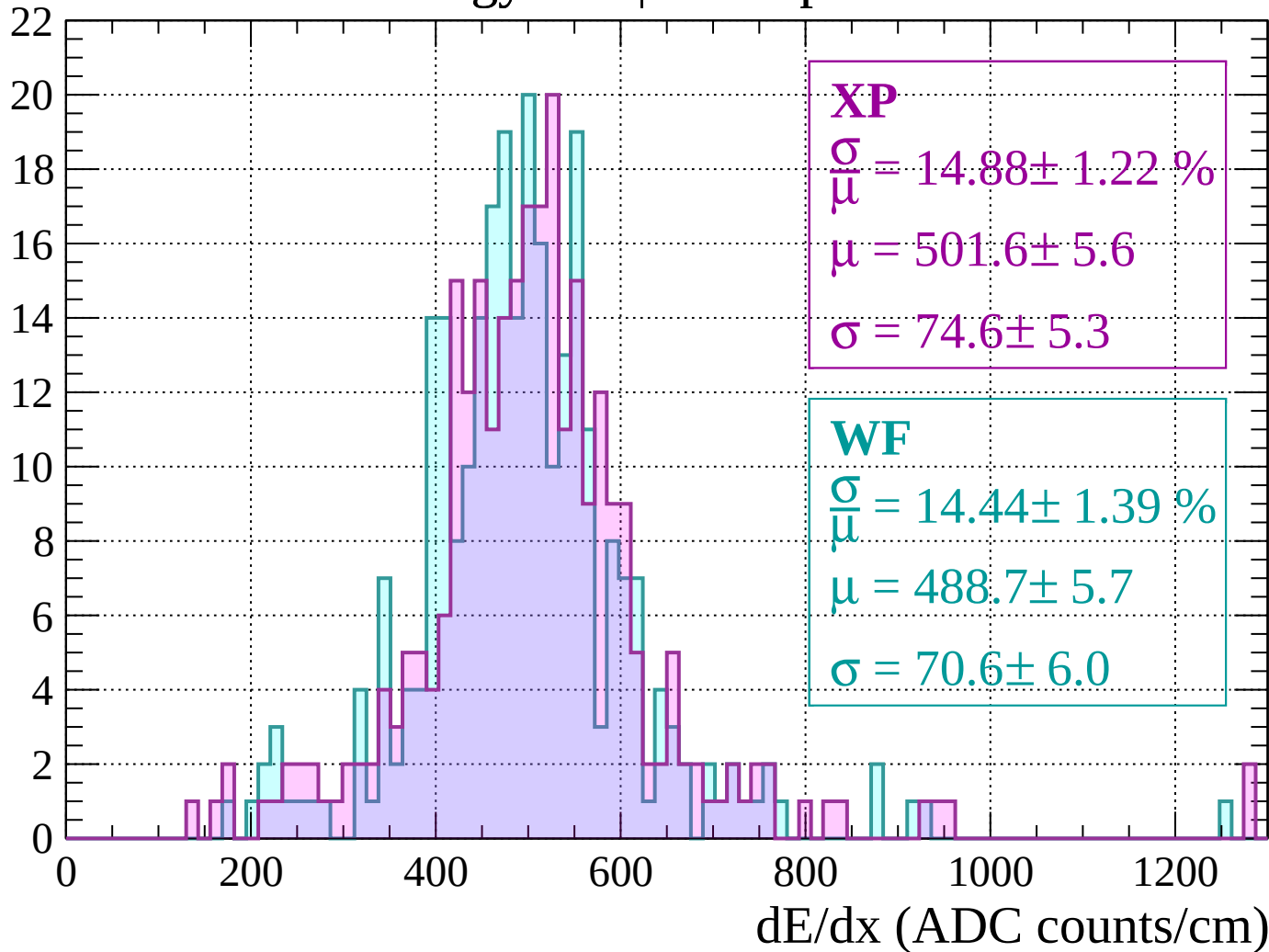
Energy loss | $440 < p < 480$

Count



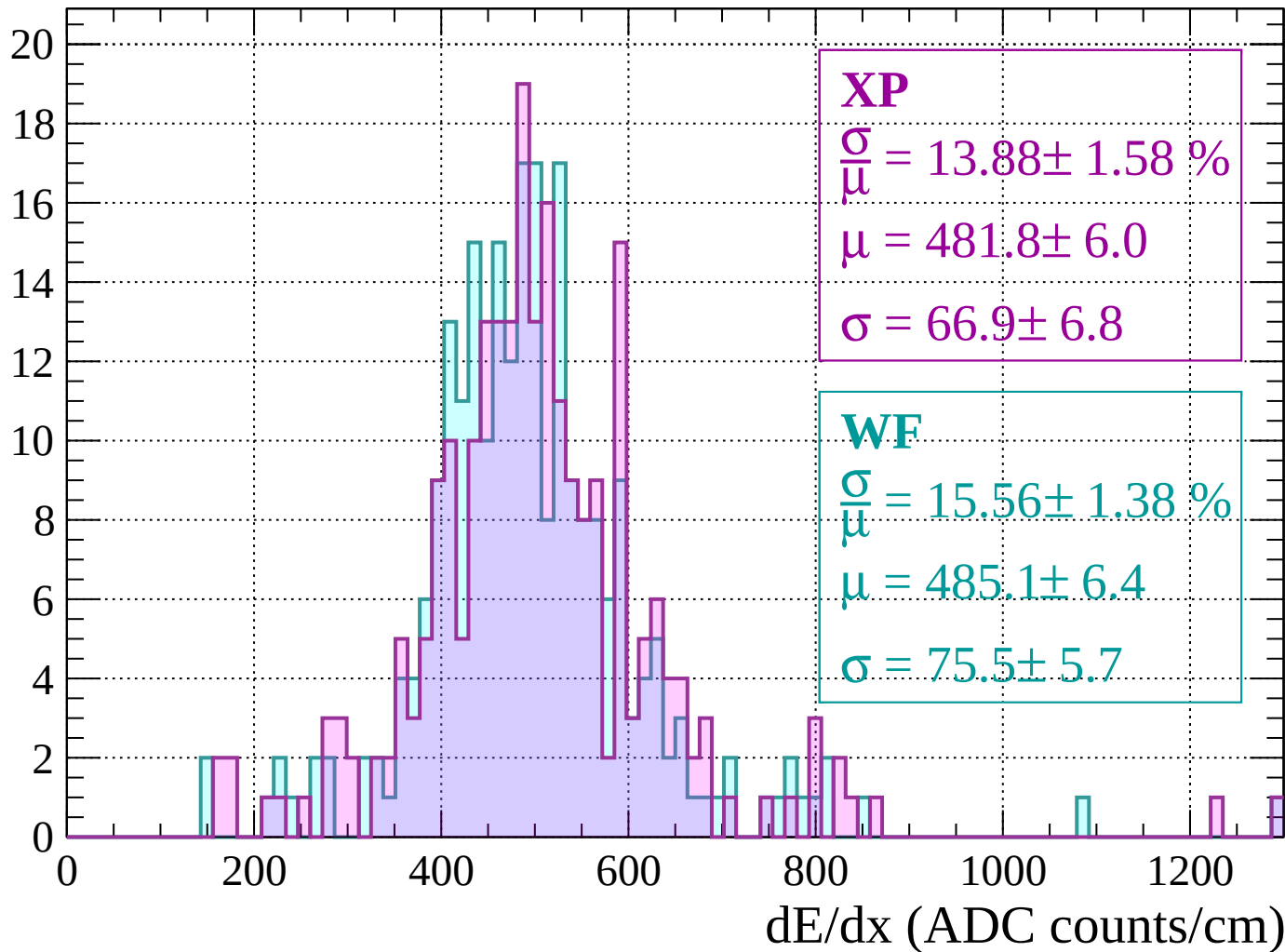
Energy loss | $480 < p < 520$

Count



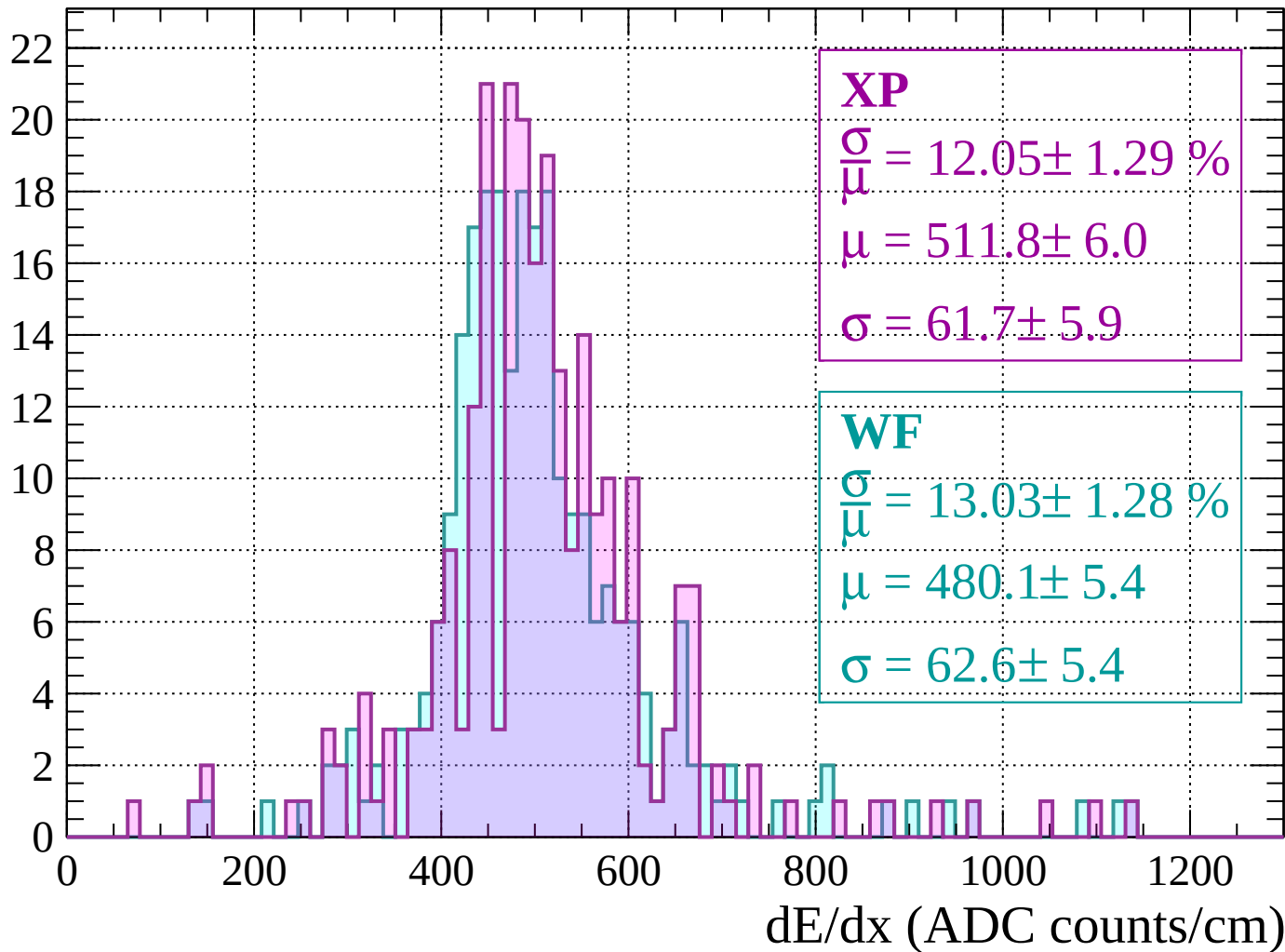
Energy loss | $520 < p < 560$

Count



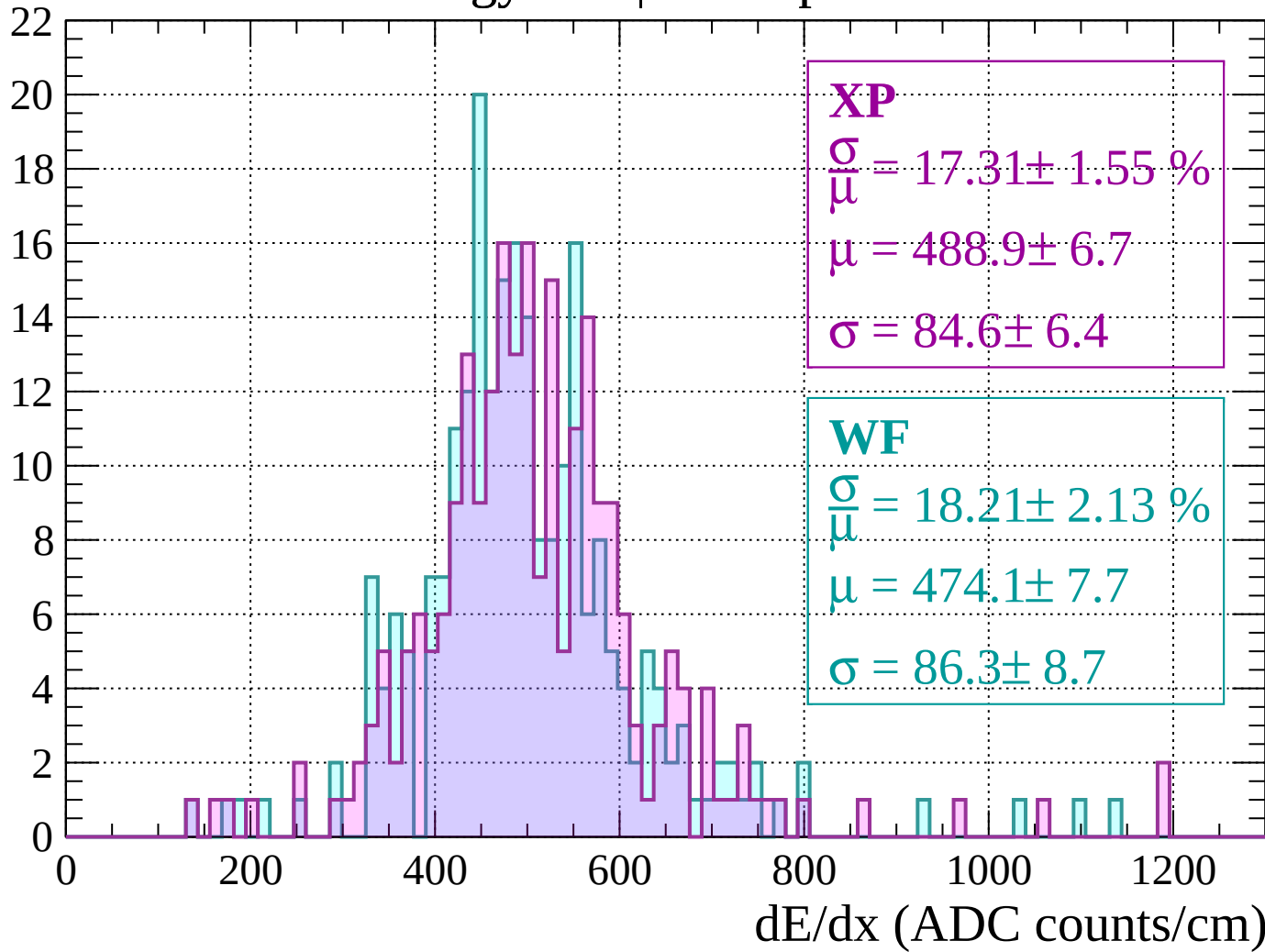
Energy loss | $560 < p < 600$

Count



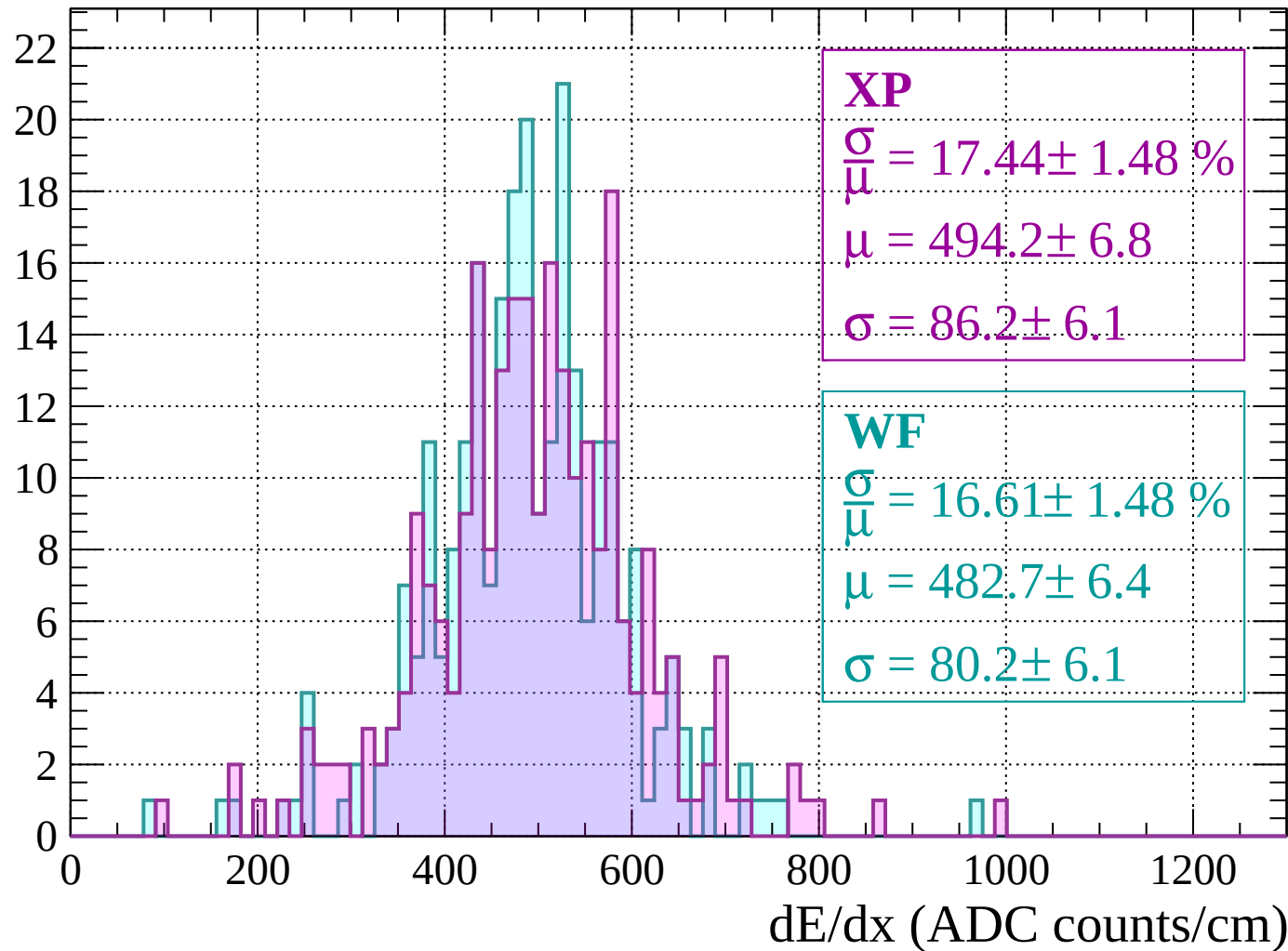
Energy loss | $600 < p < 640$

Count



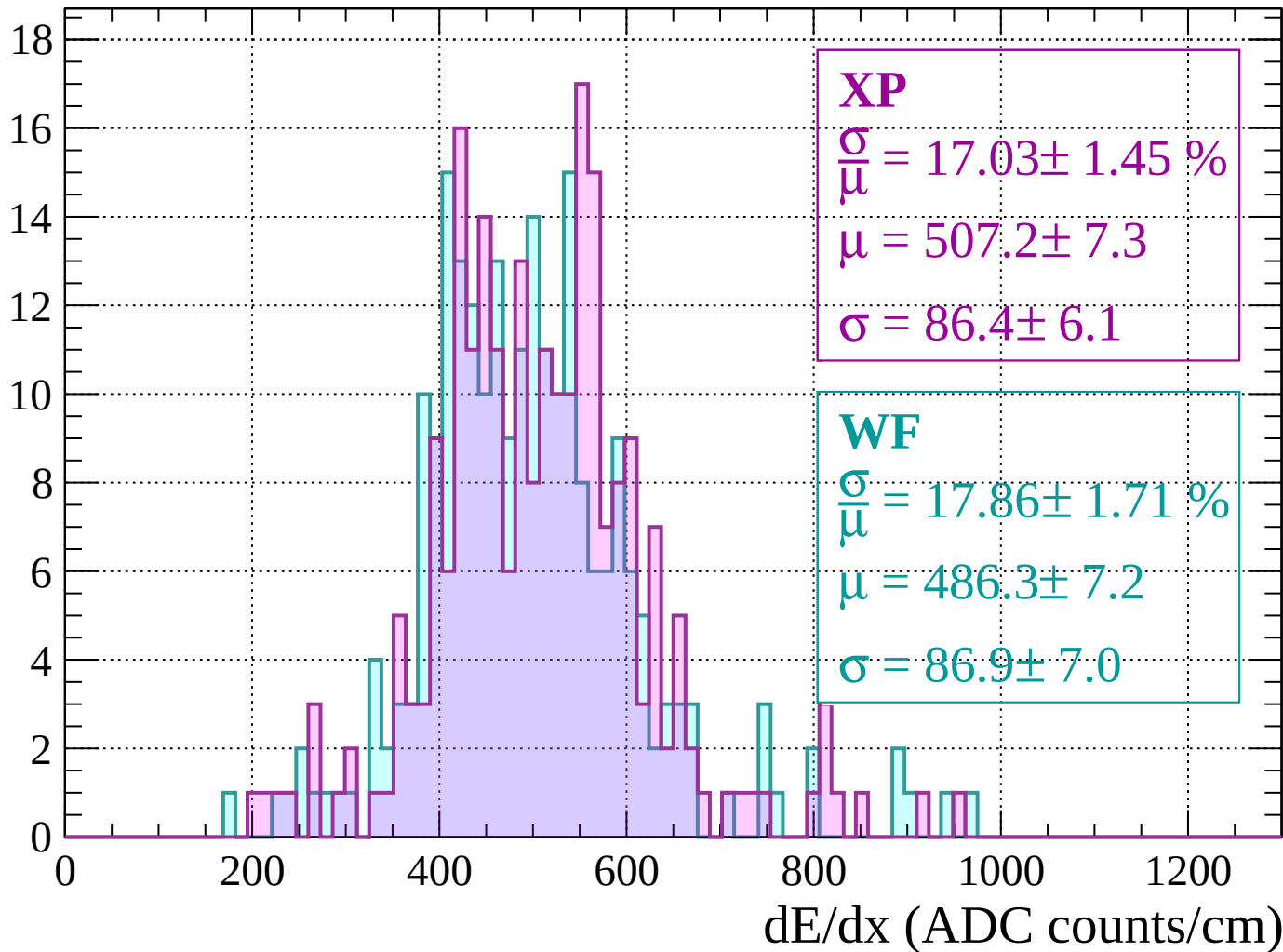
Energy loss | $640 < p < 680$

Count



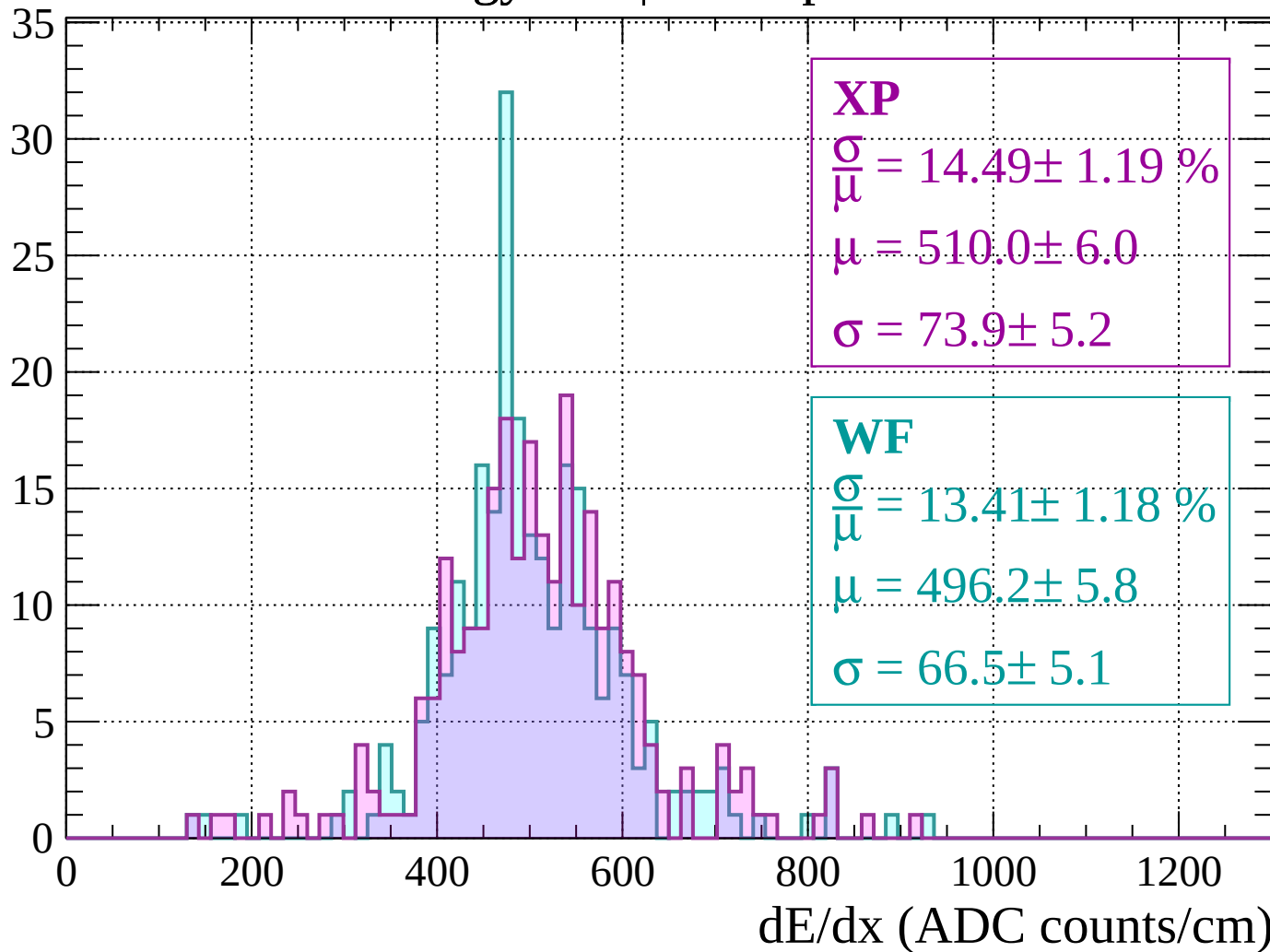
Energy loss | $680 < p < 720$

Count



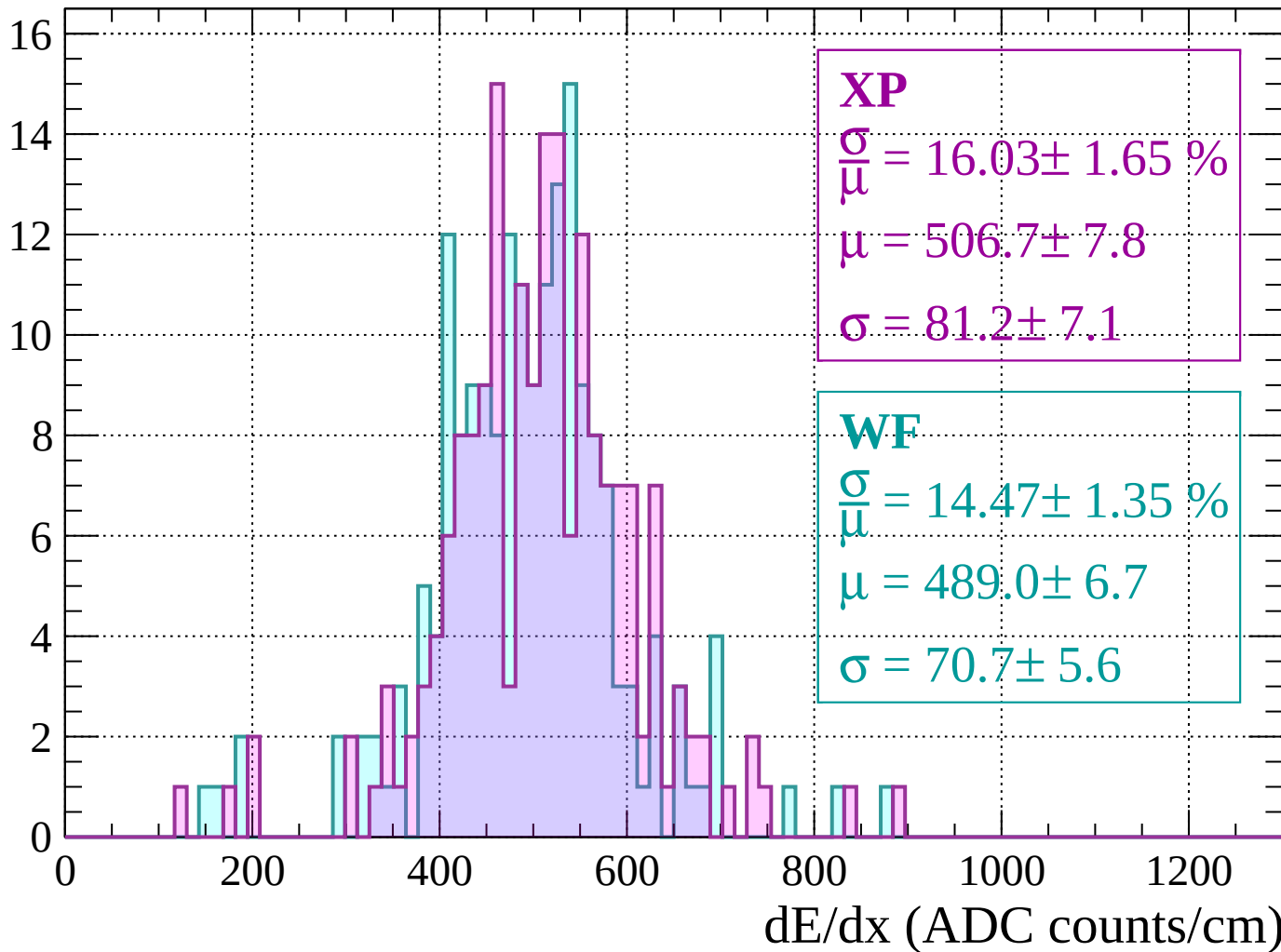
Energy loss | $720 < p < 760$

Count



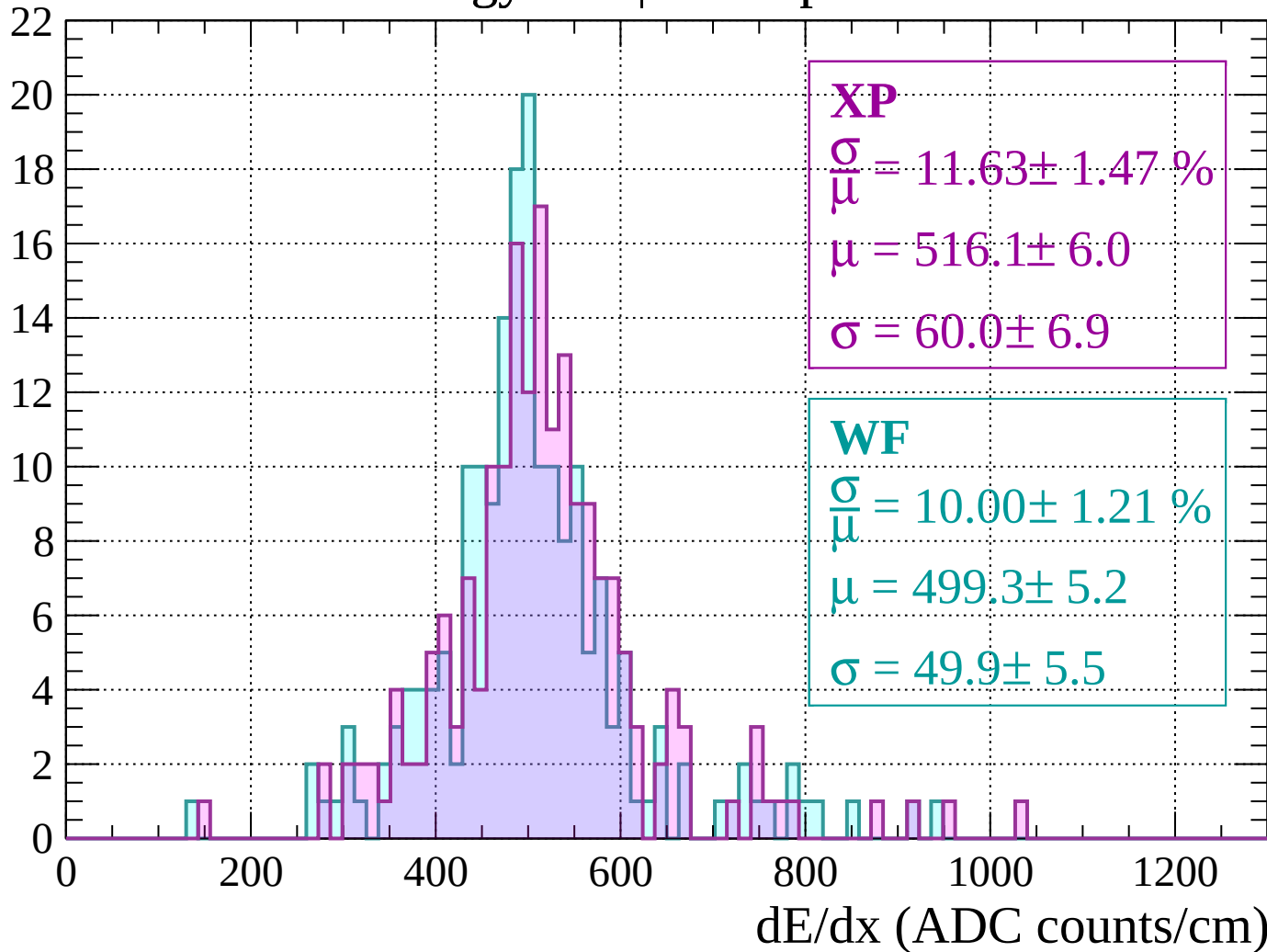
Energy loss | $760 < p < 800$

Count



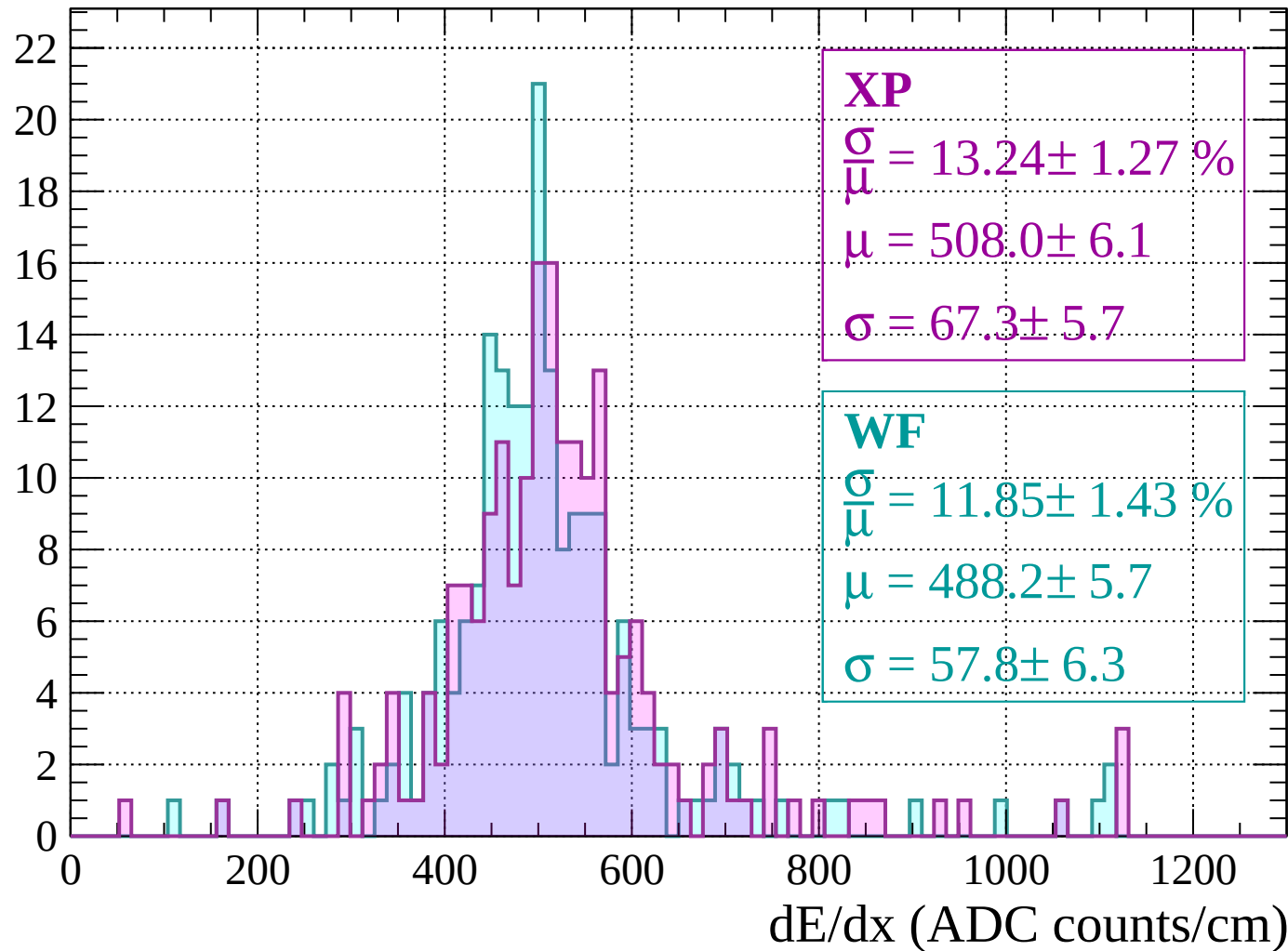
Energy loss | $800 < p < 840$

Count



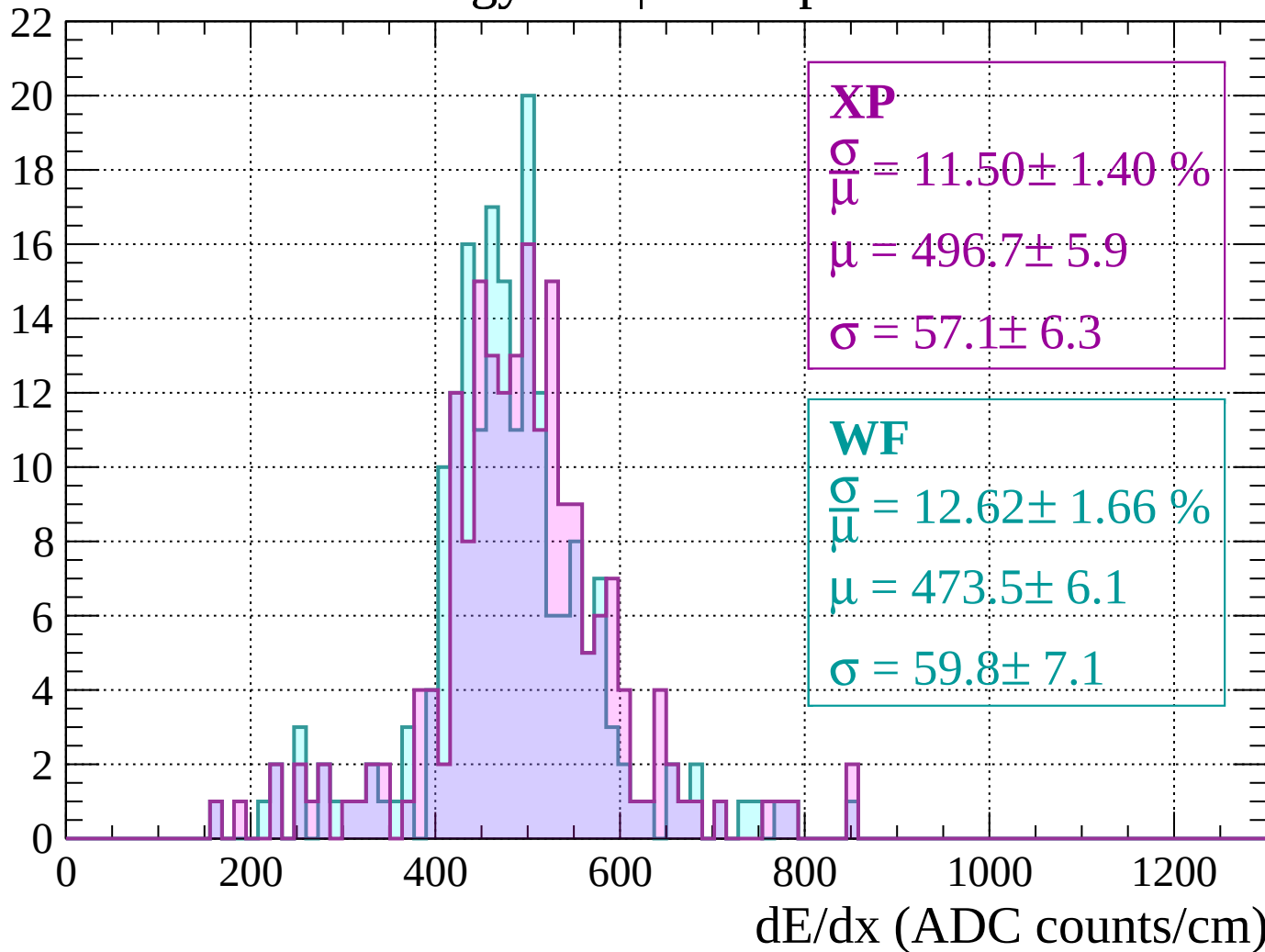
Energy loss | $840 < p < 880$

Count



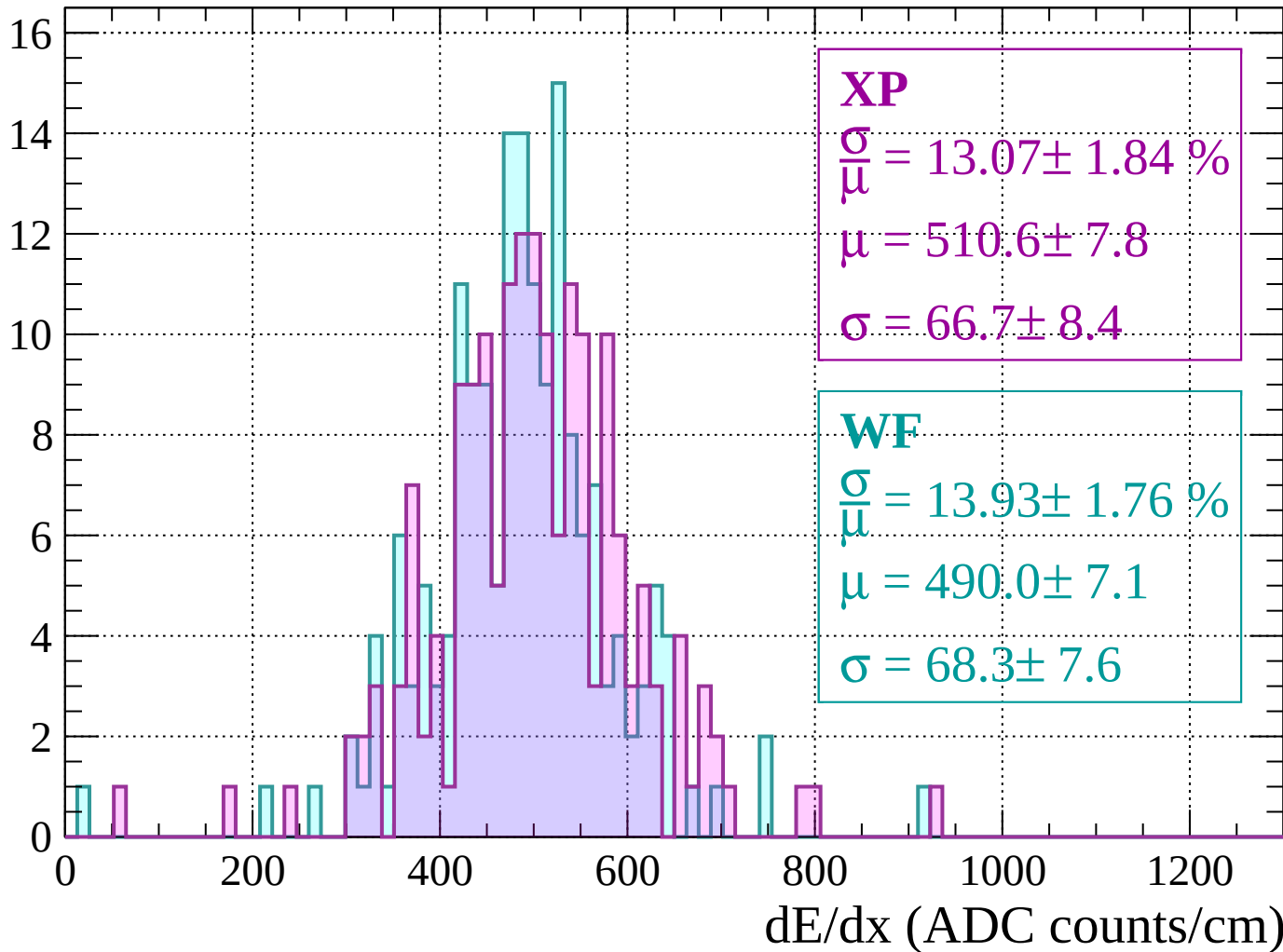
Energy loss | $880 < p < 920$

Count



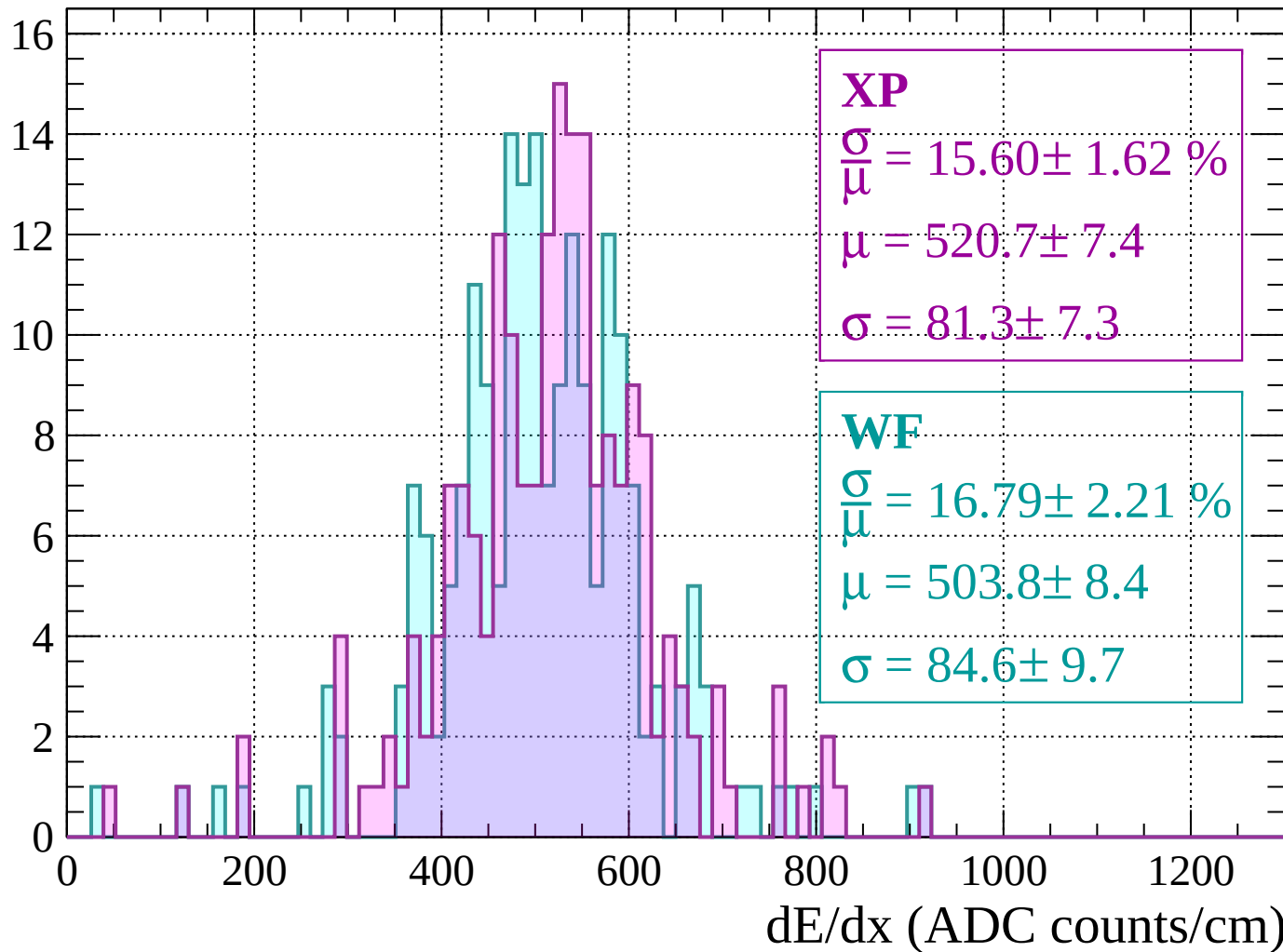
Energy loss | $920 < p < 960$

Count



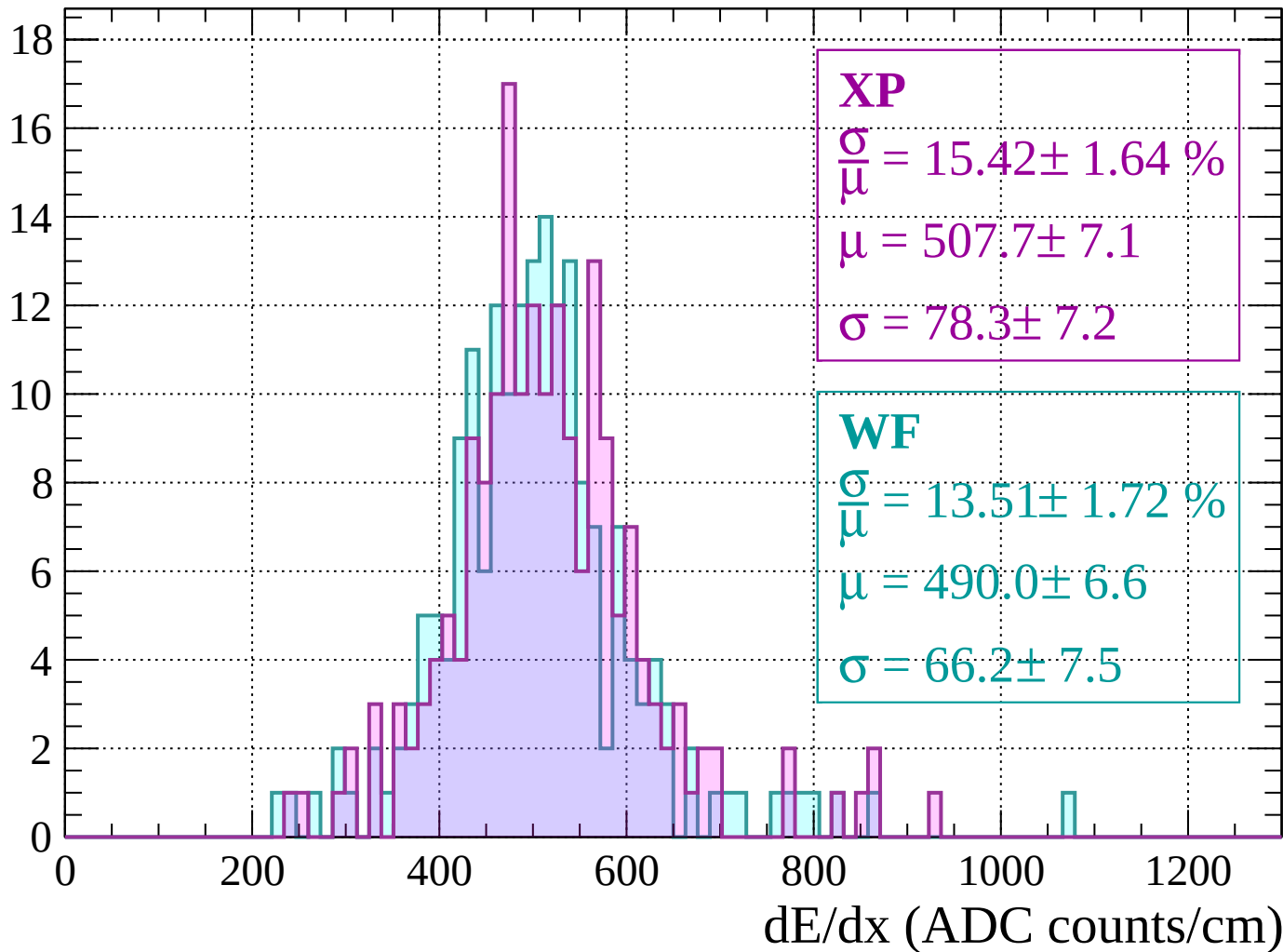
Energy loss | $960 < p < 1000$

Count



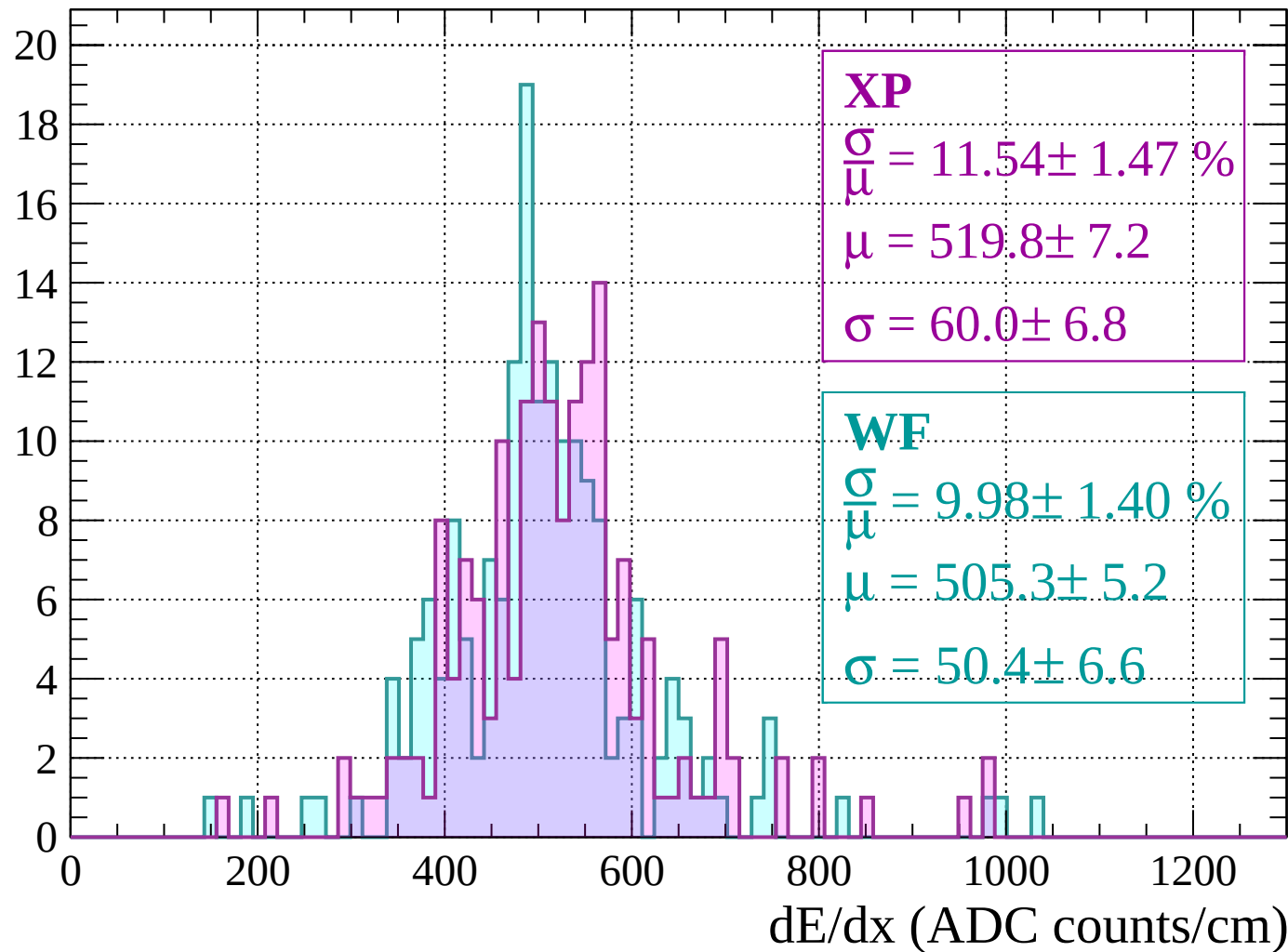
Energy loss | $1000 < p < 1040$

Count



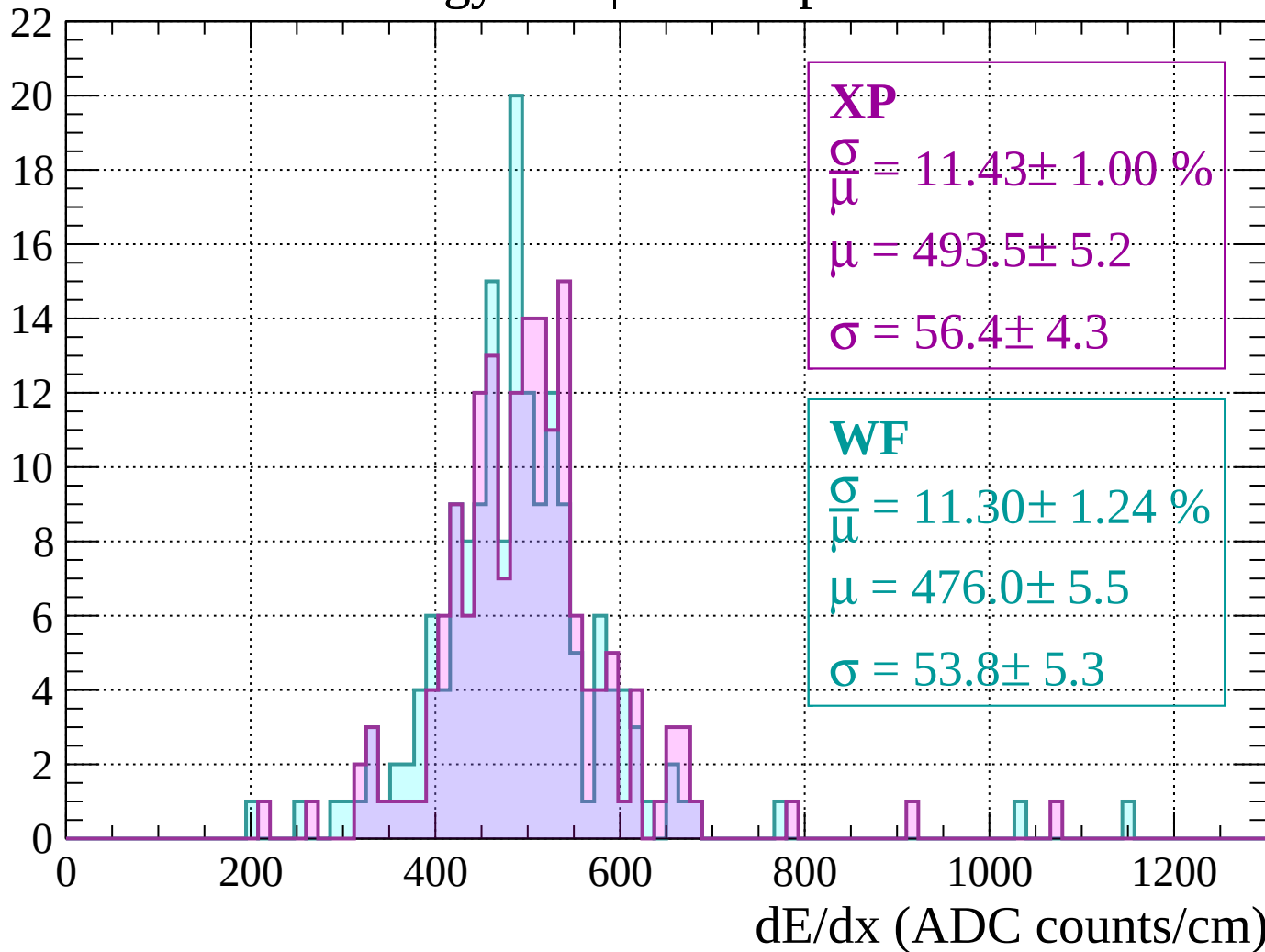
Energy loss | $1040 < p < 1080$

Count



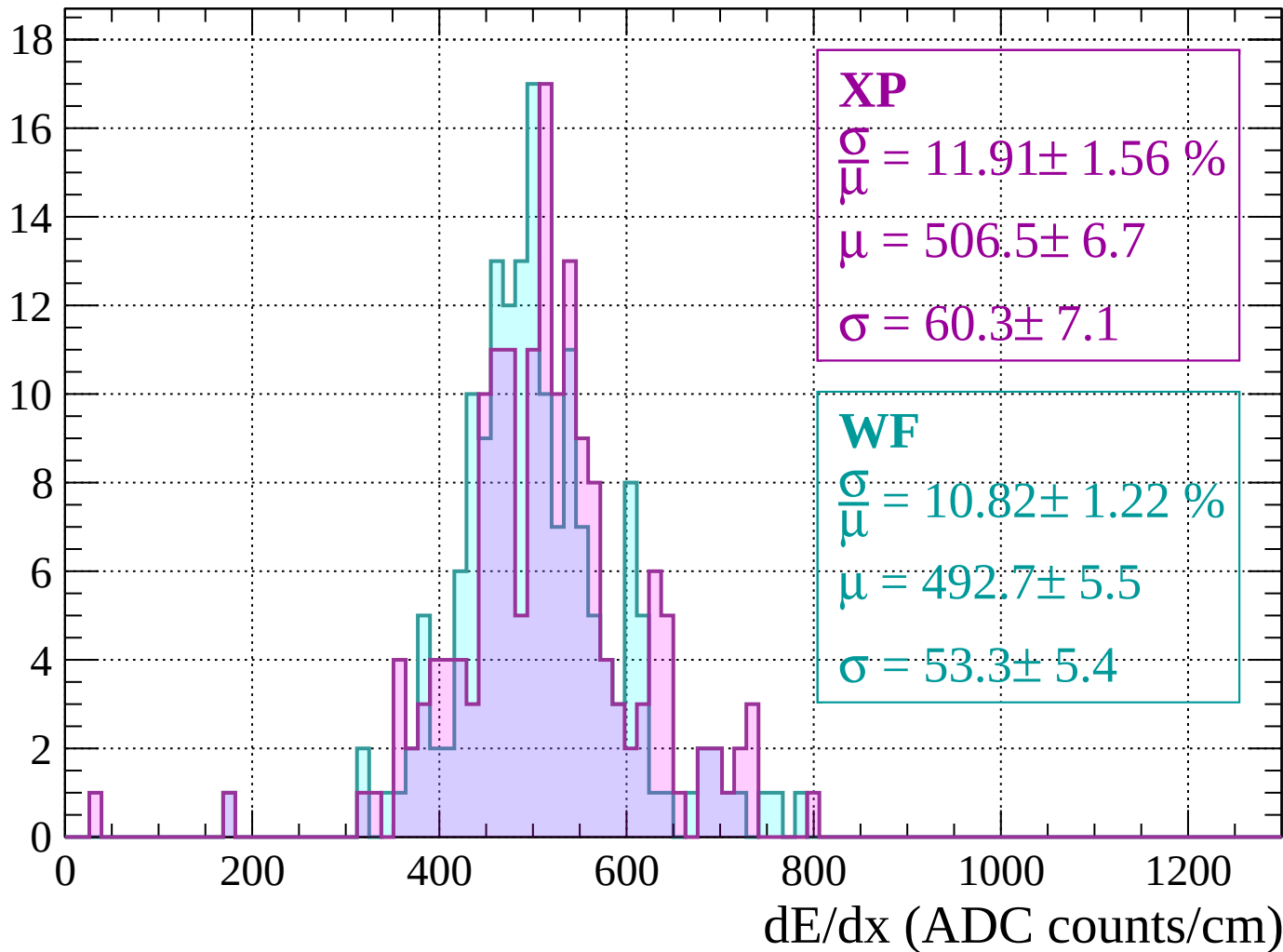
Energy loss | $1080 < p < 1120$

Count



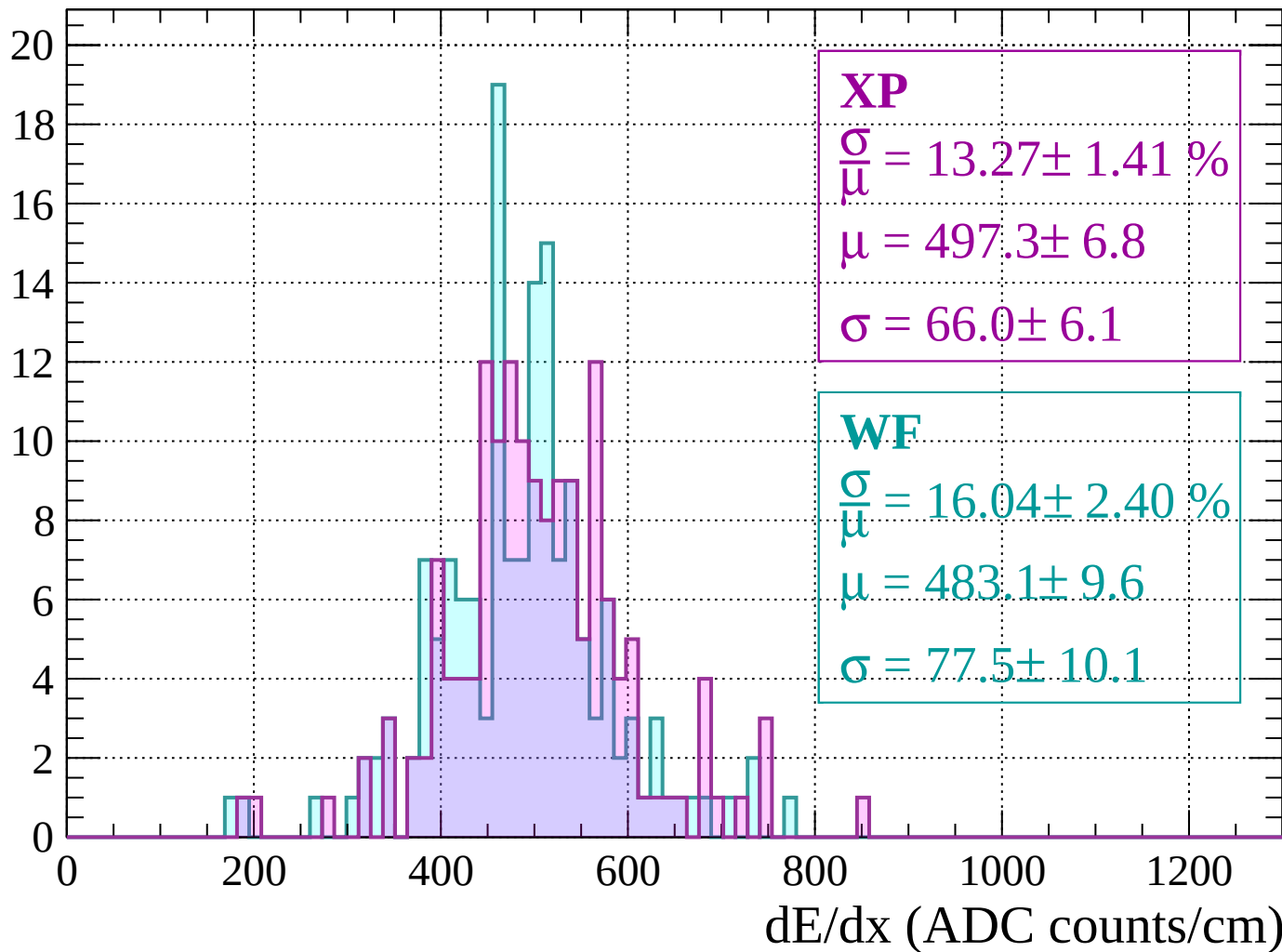
Energy loss | $1120 < p < 1160$

Count



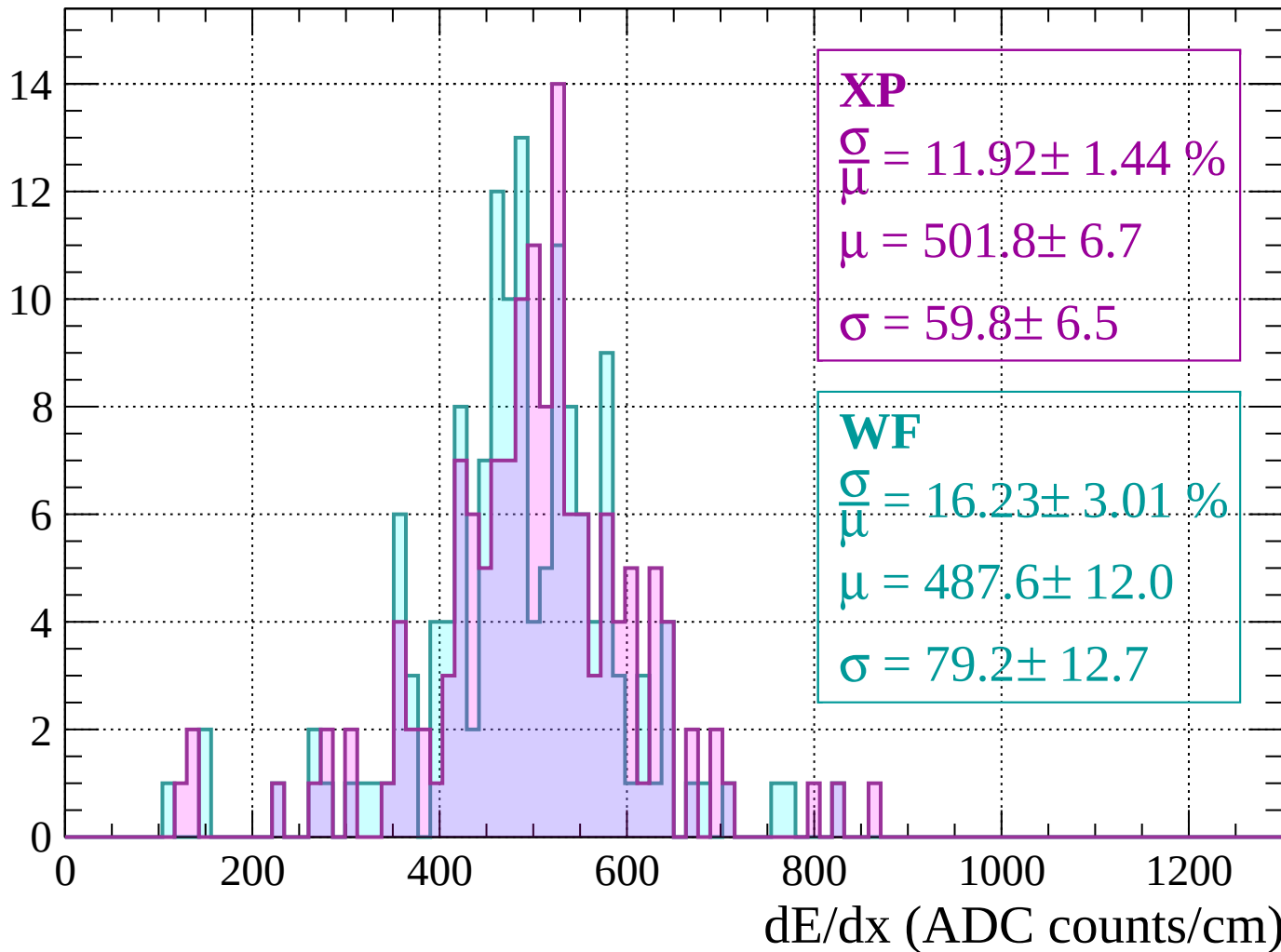
Energy loss | $1160 < p < 1200$

Count



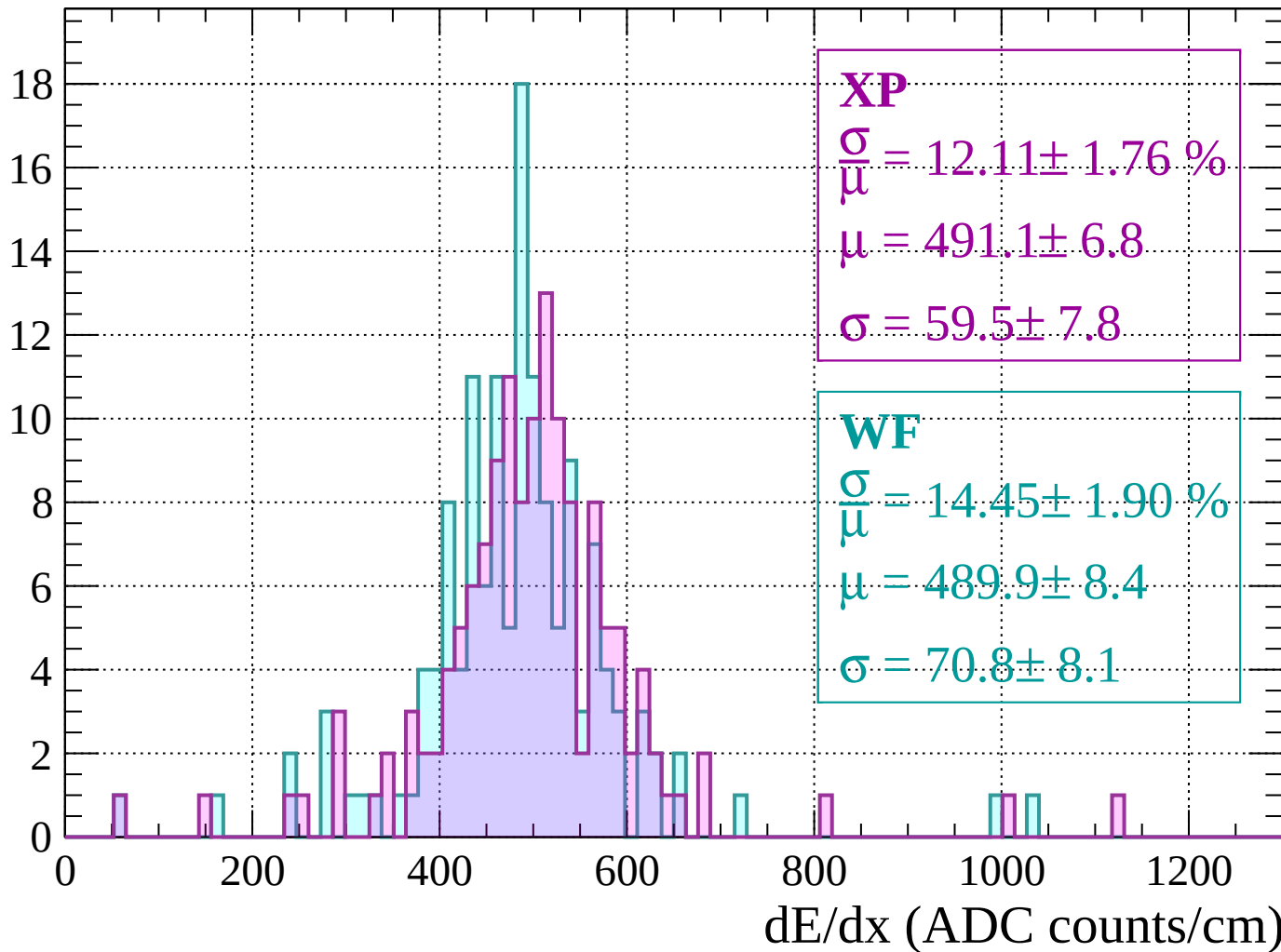
Energy loss | $1200 < p < 1240$

Count



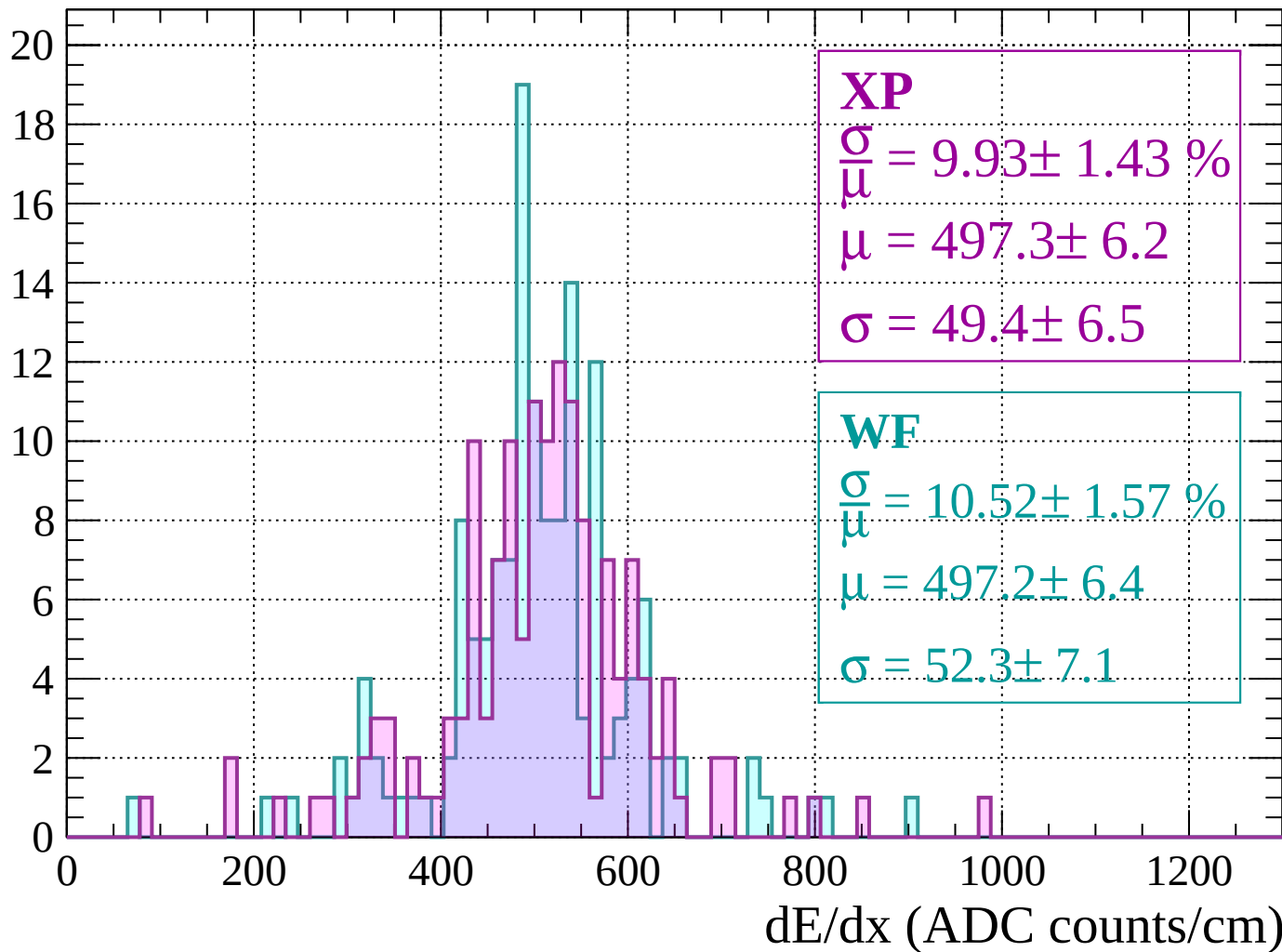
Energy loss | $1240 < p < 1280$

Count



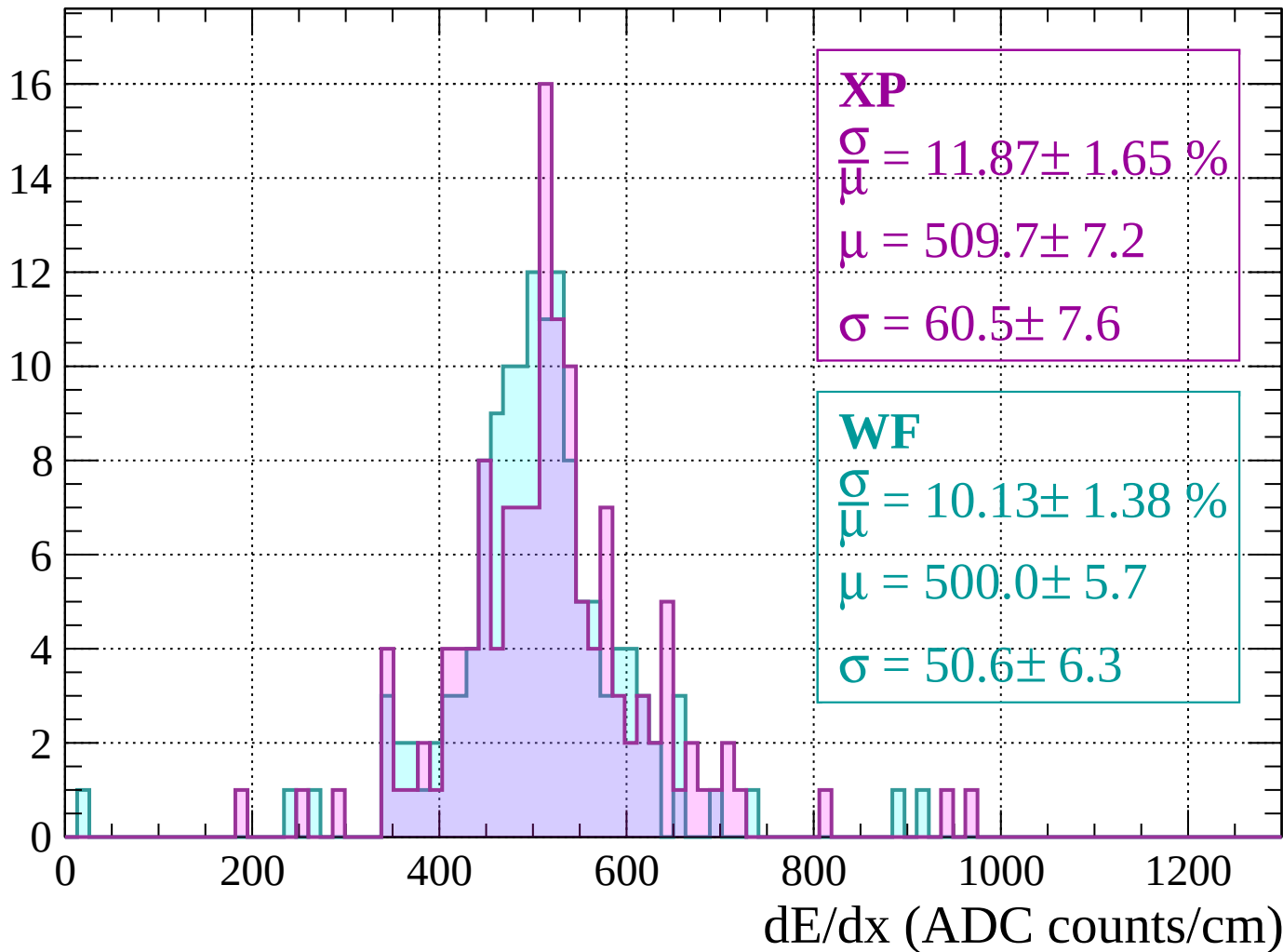
Energy loss | $1280 < p < 1320$

Count



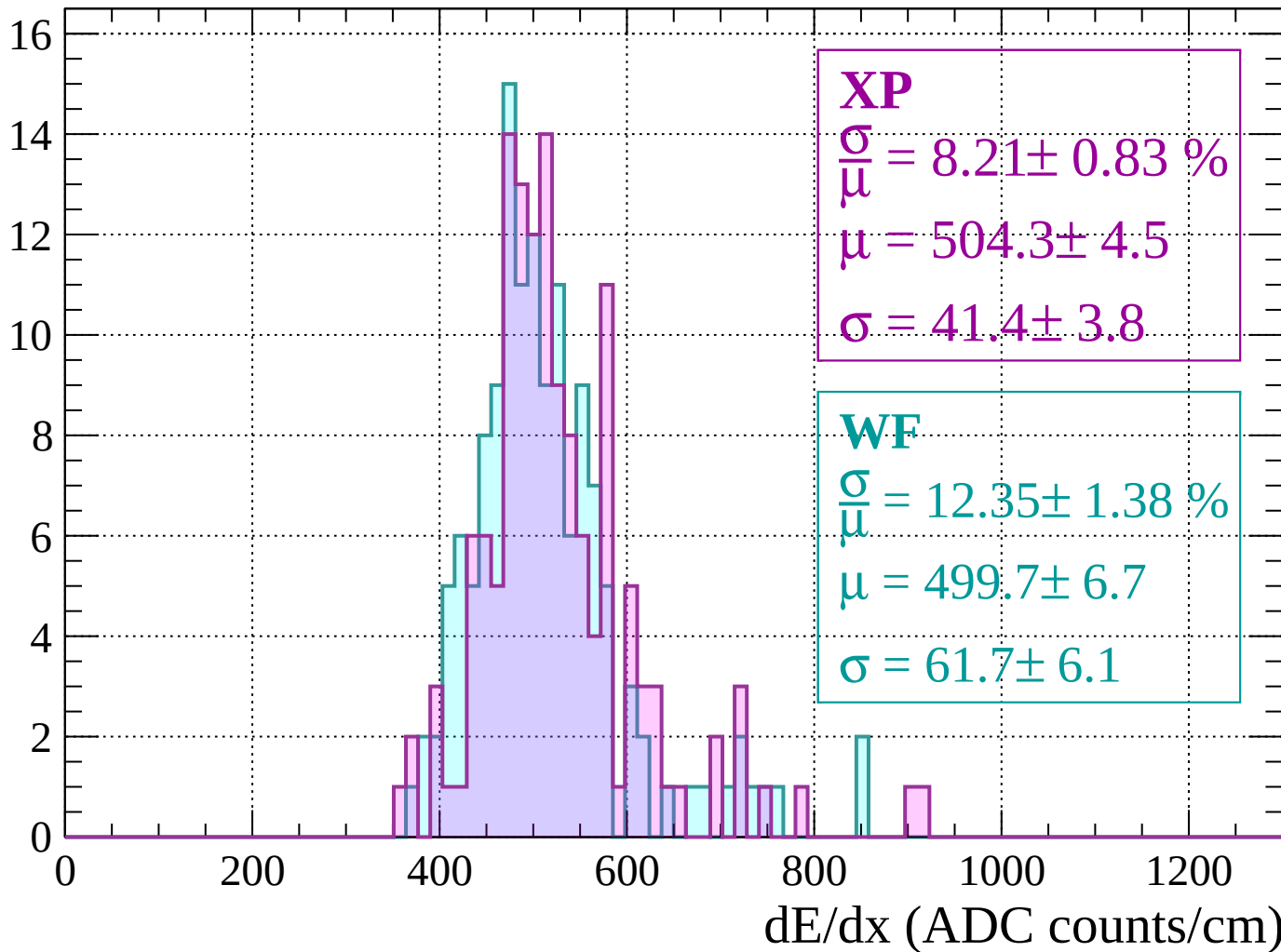
Energy loss | $1320 < p < 1360$

Count



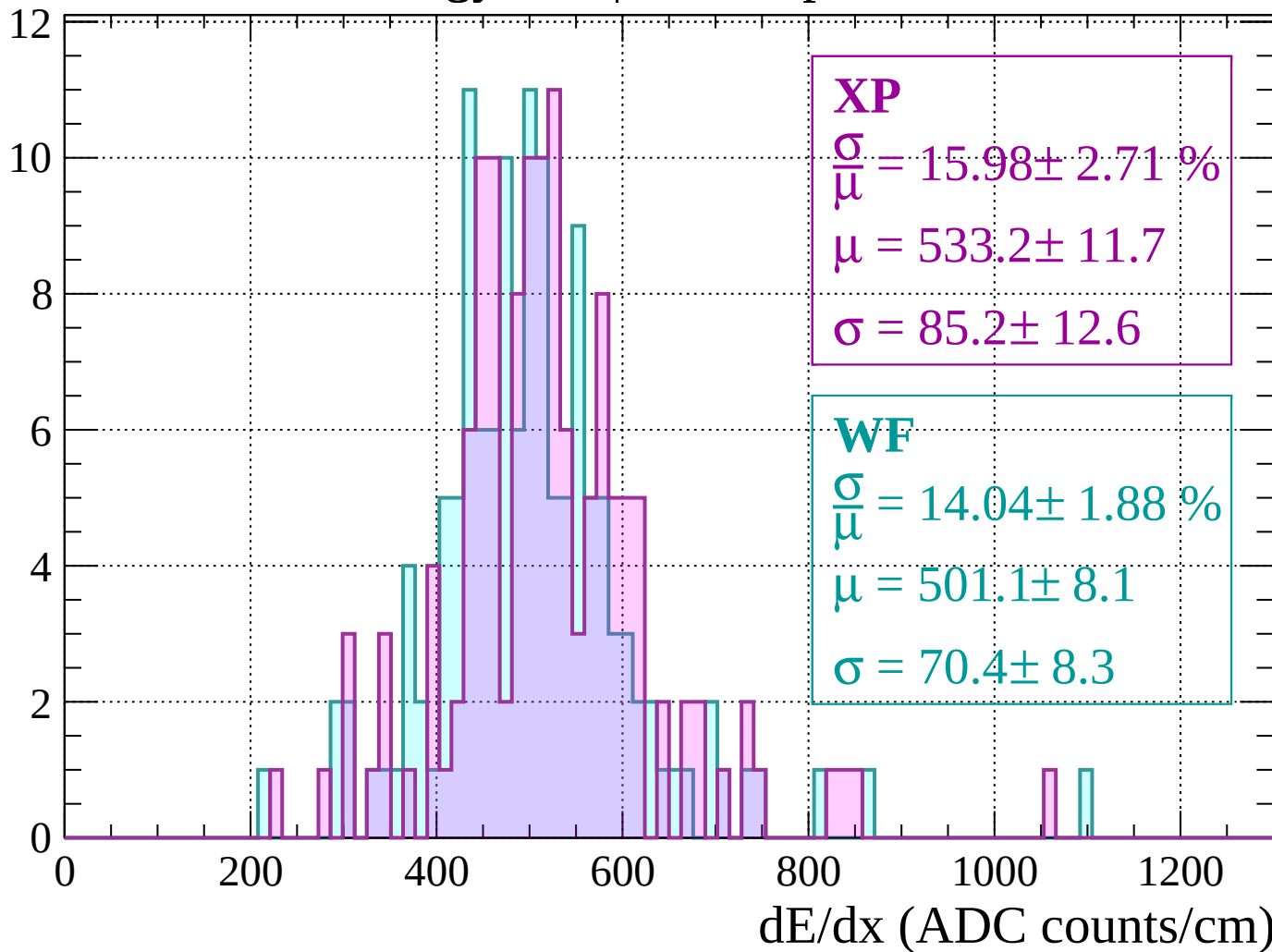
Energy loss | $1360 < p < 1400$

Count



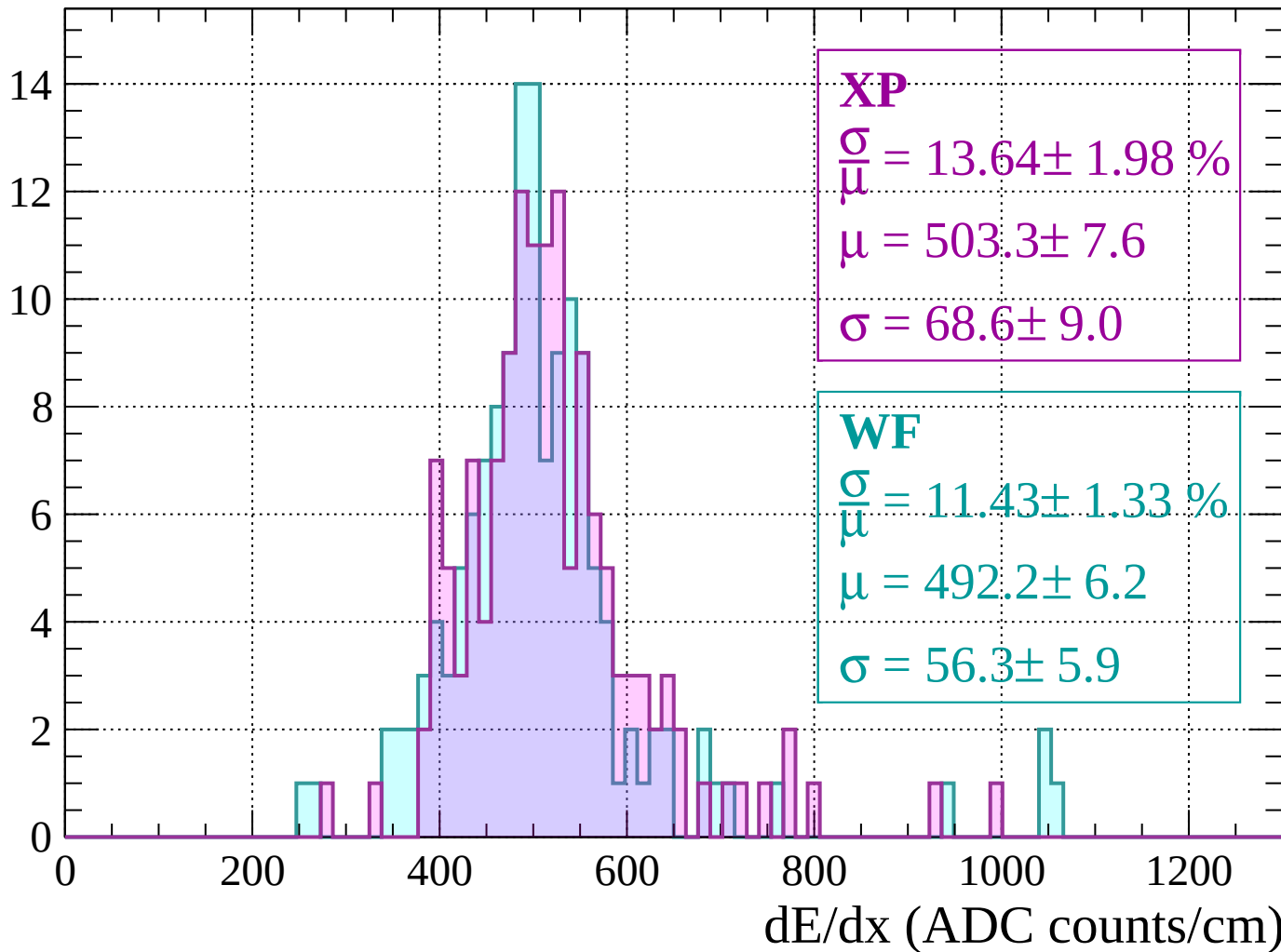
Energy loss | $1400 < p < 1440$

Count



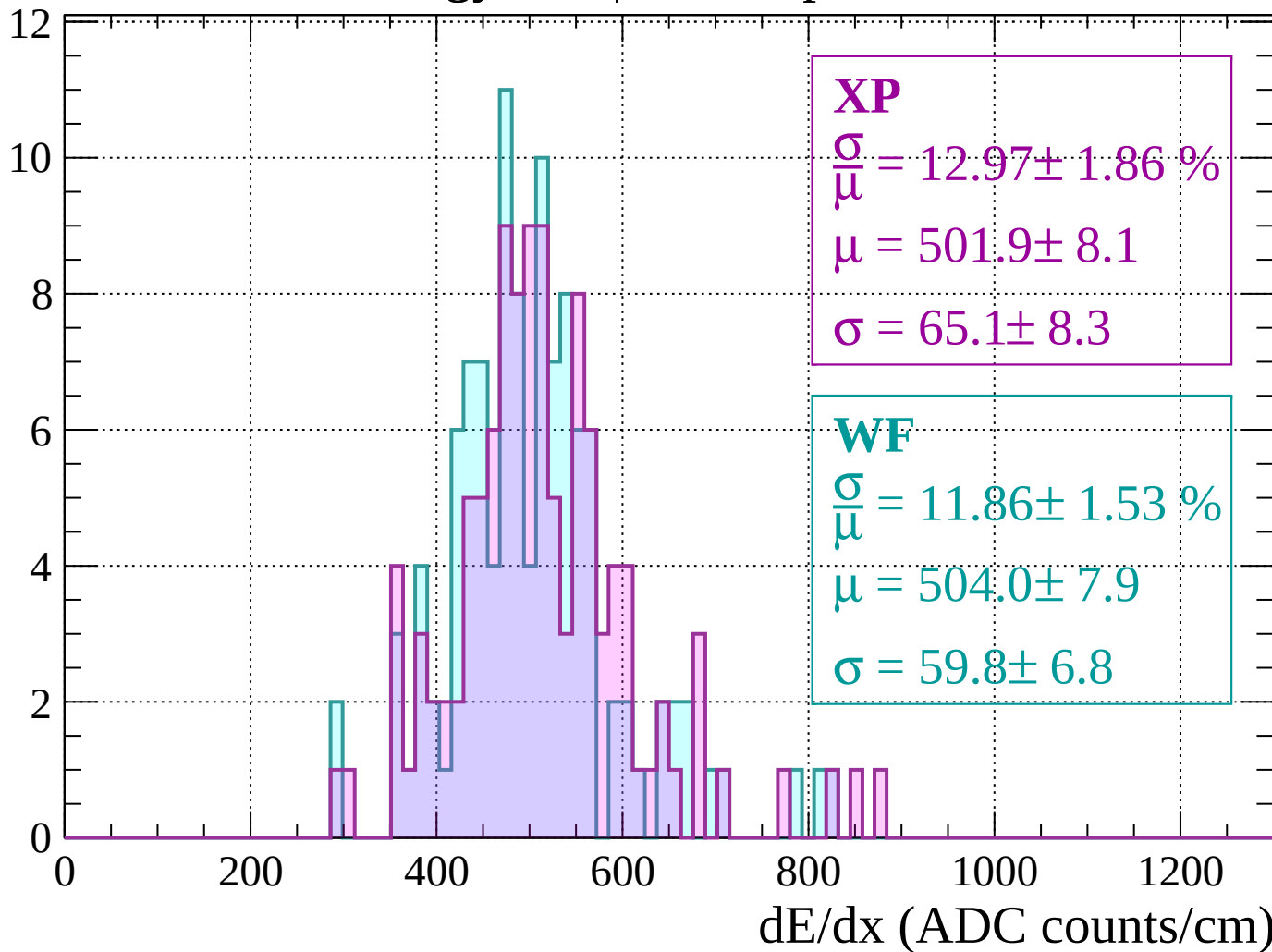
Energy loss | $1440 < p < 1480$

Count



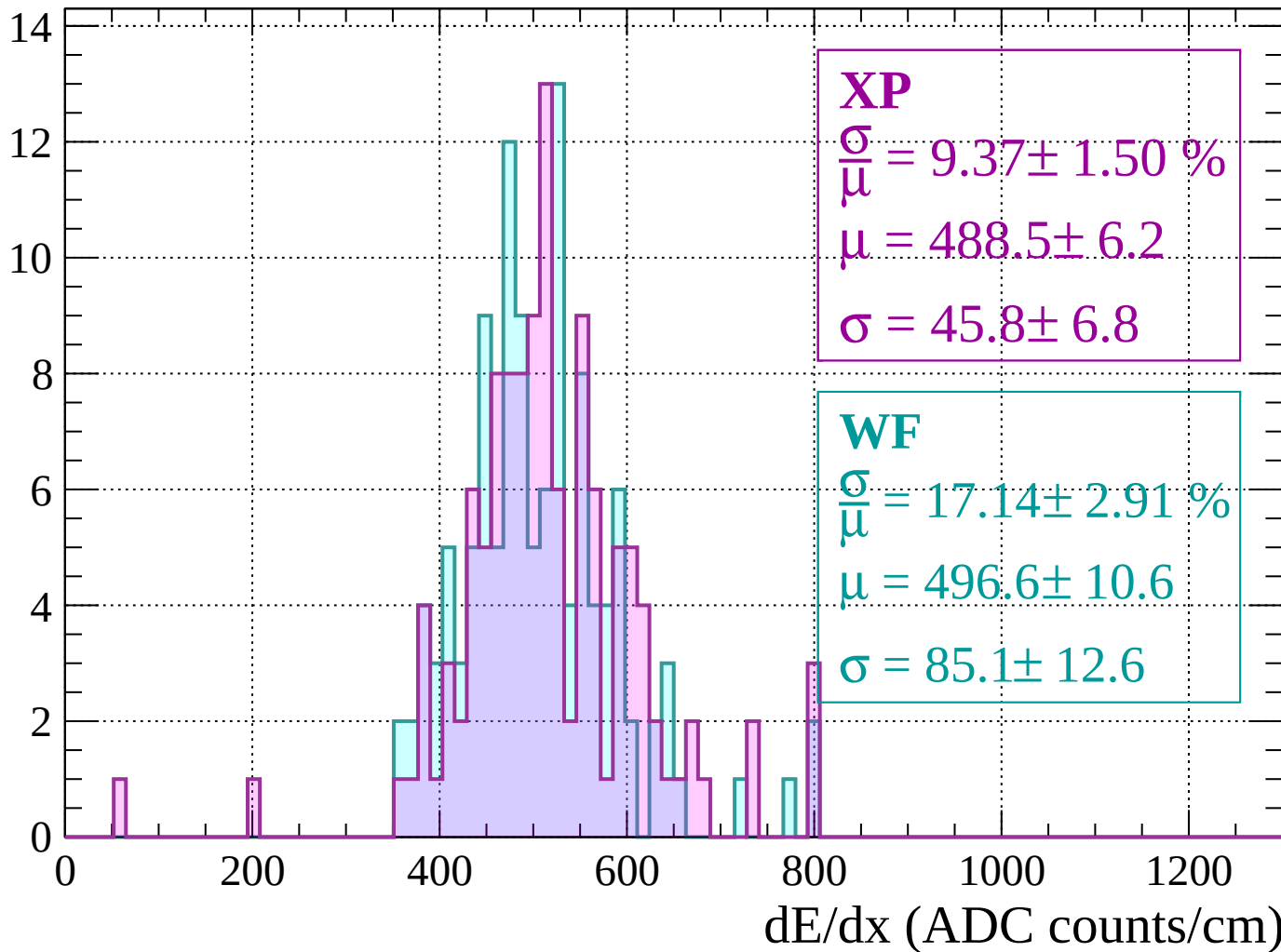
Energy loss | $1480 < p < 1520$

Count



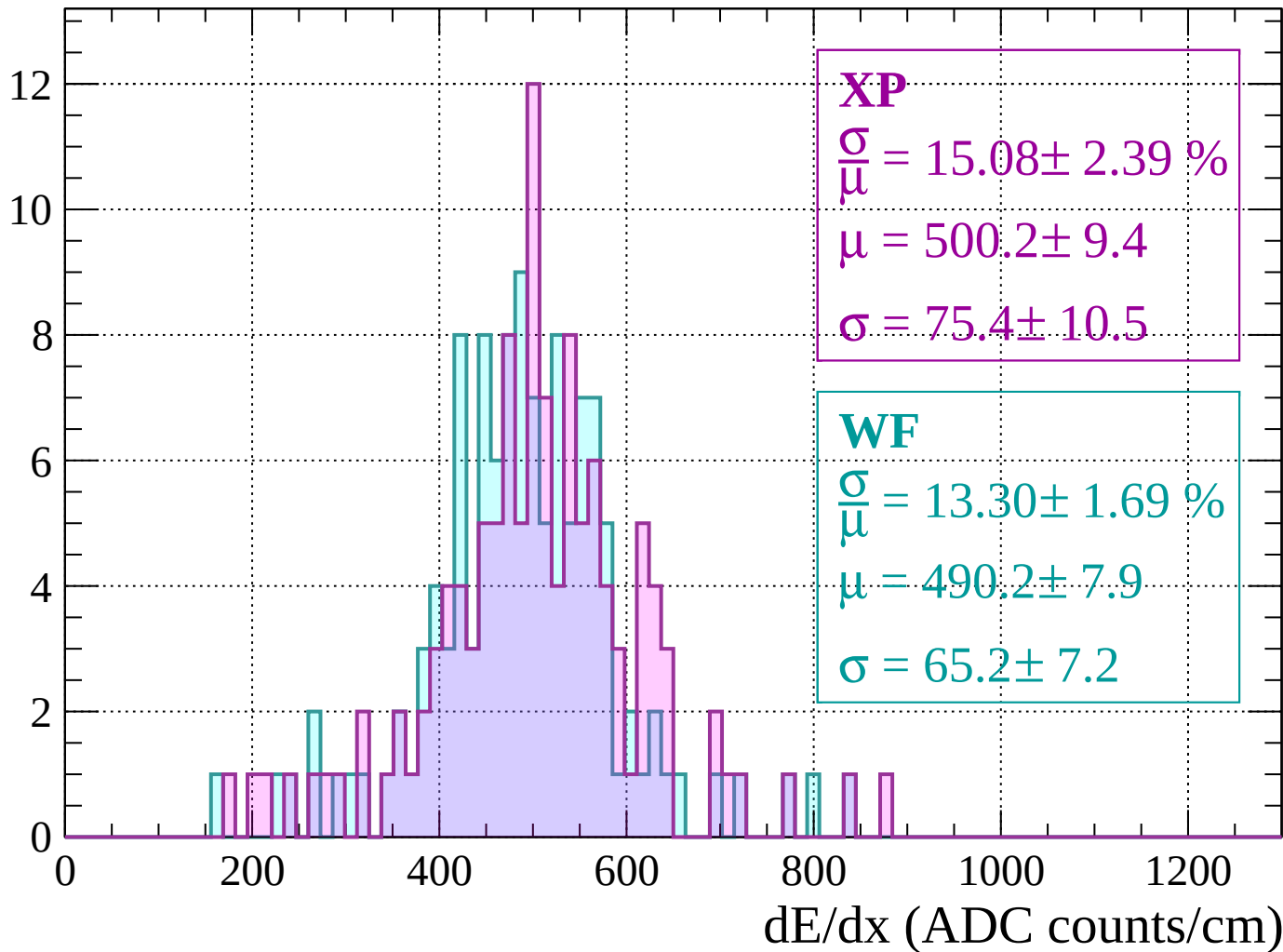
Energy loss | $1520 < p < 1560$

Count



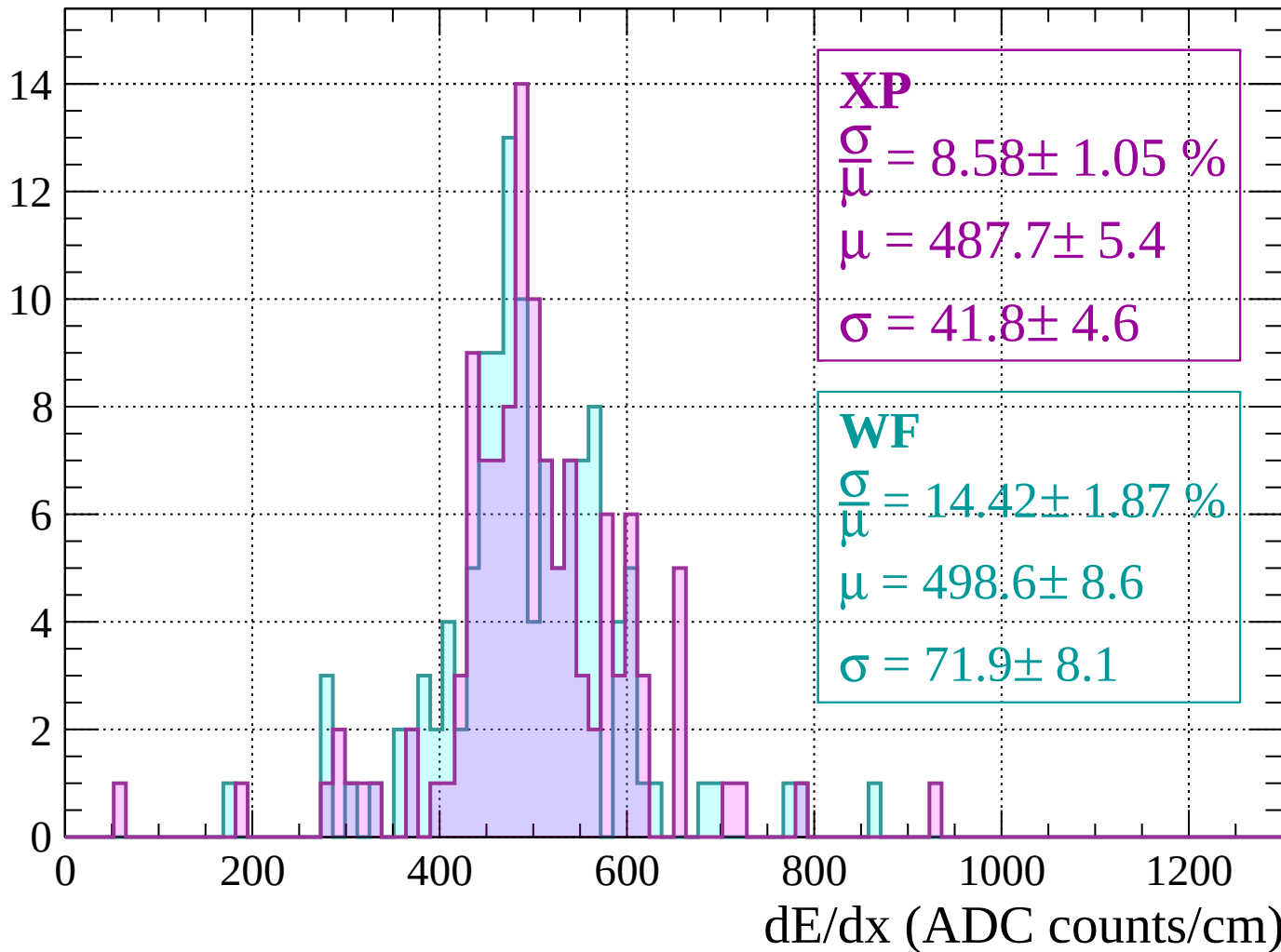
Energy loss | $1560 < p < 1600$

Count



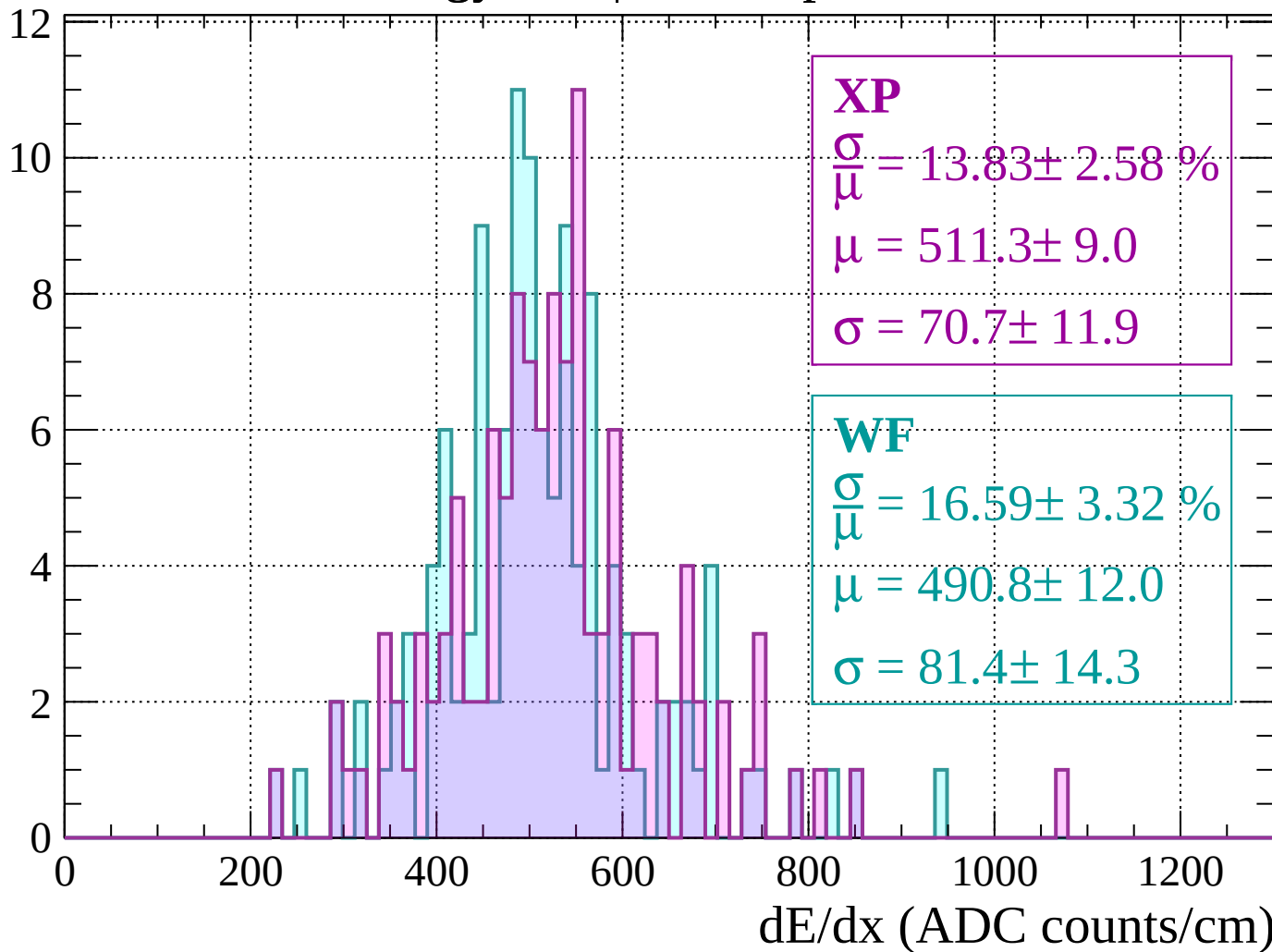
Energy loss | $1600 < p < 1640$

Count



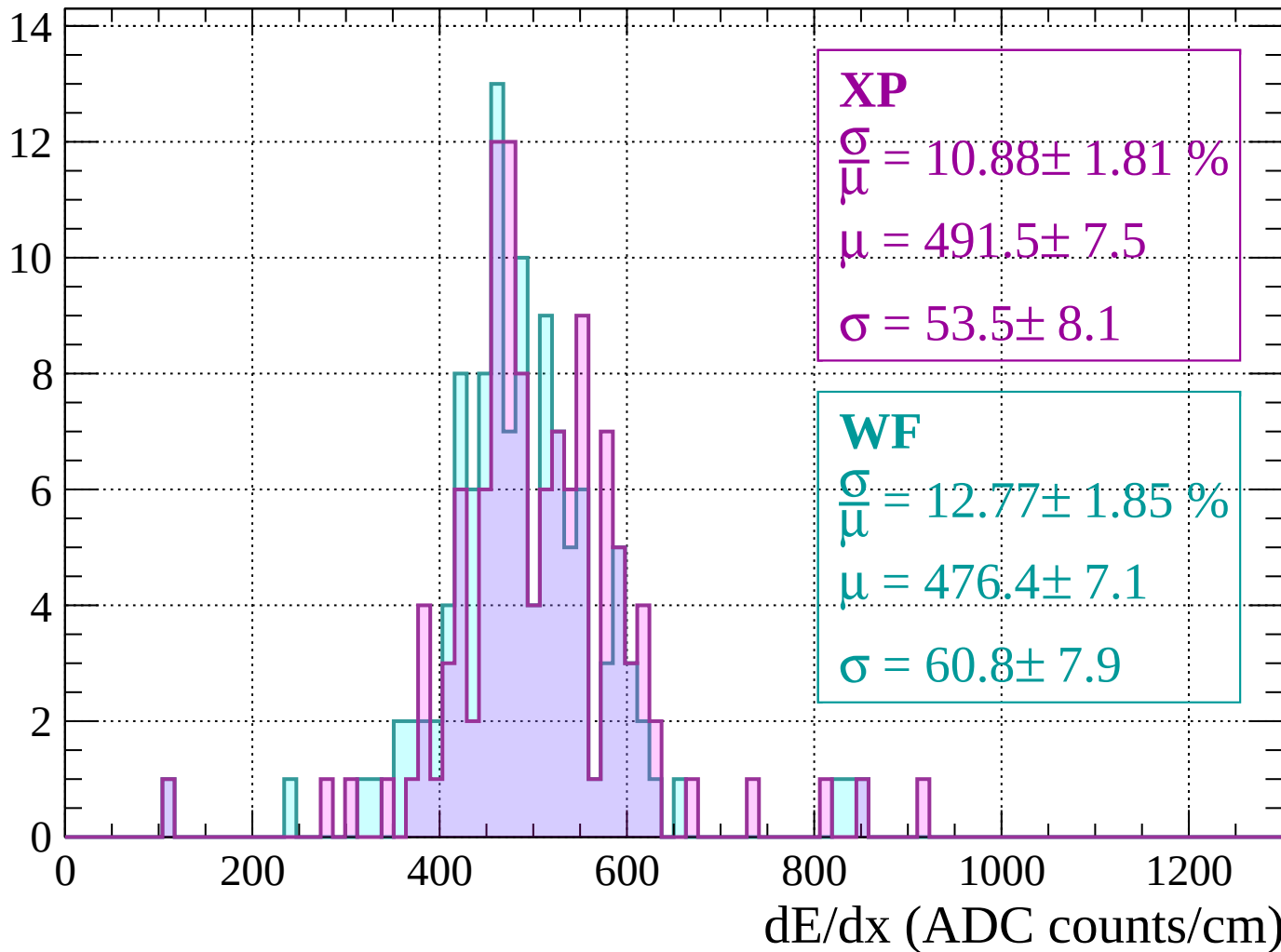
Energy loss | $1640 < p < 1680$

Count



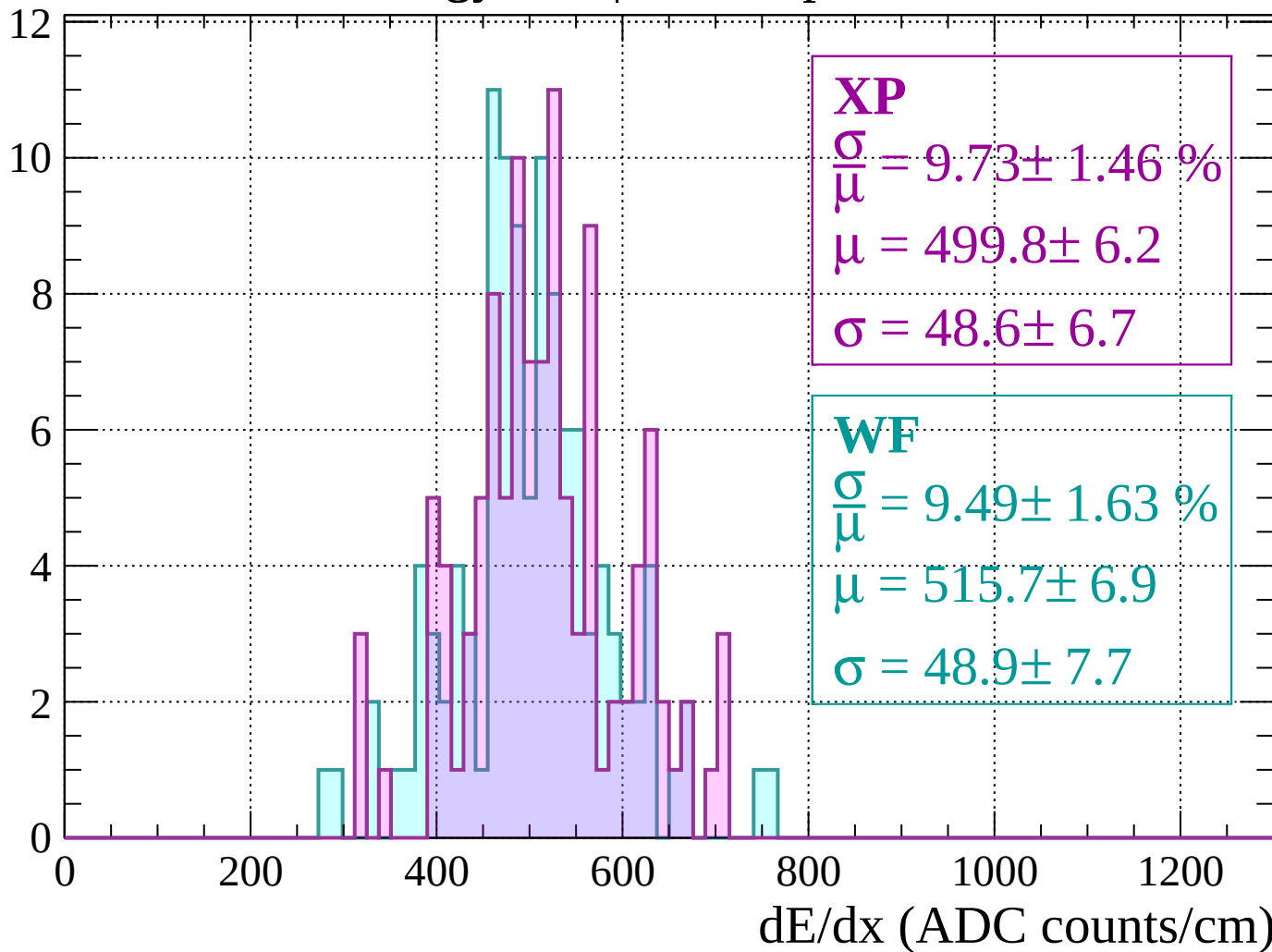
Energy loss | $1680 < p < 1720$

Count



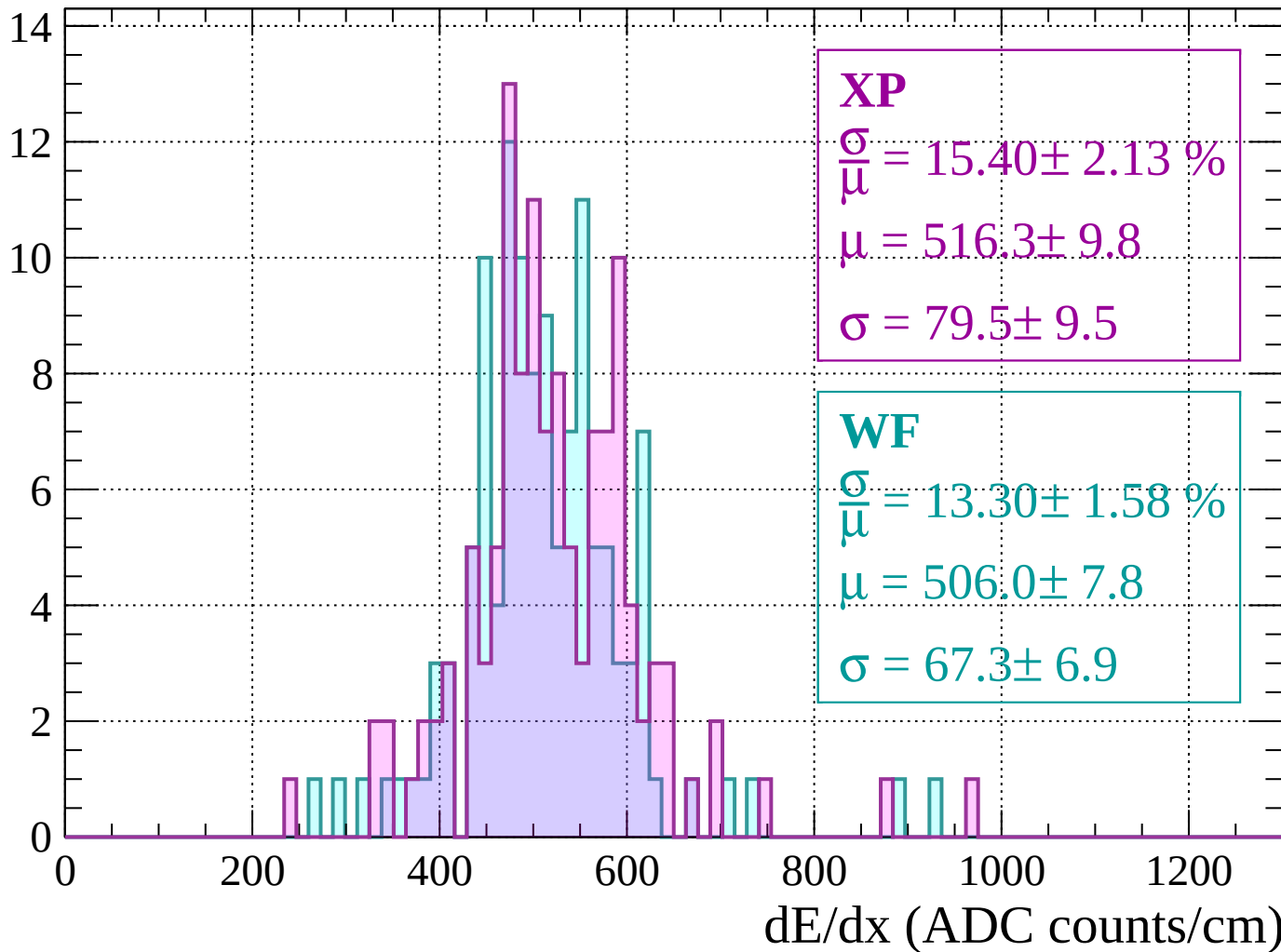
Energy loss | $1720 < p < 1760$

Count



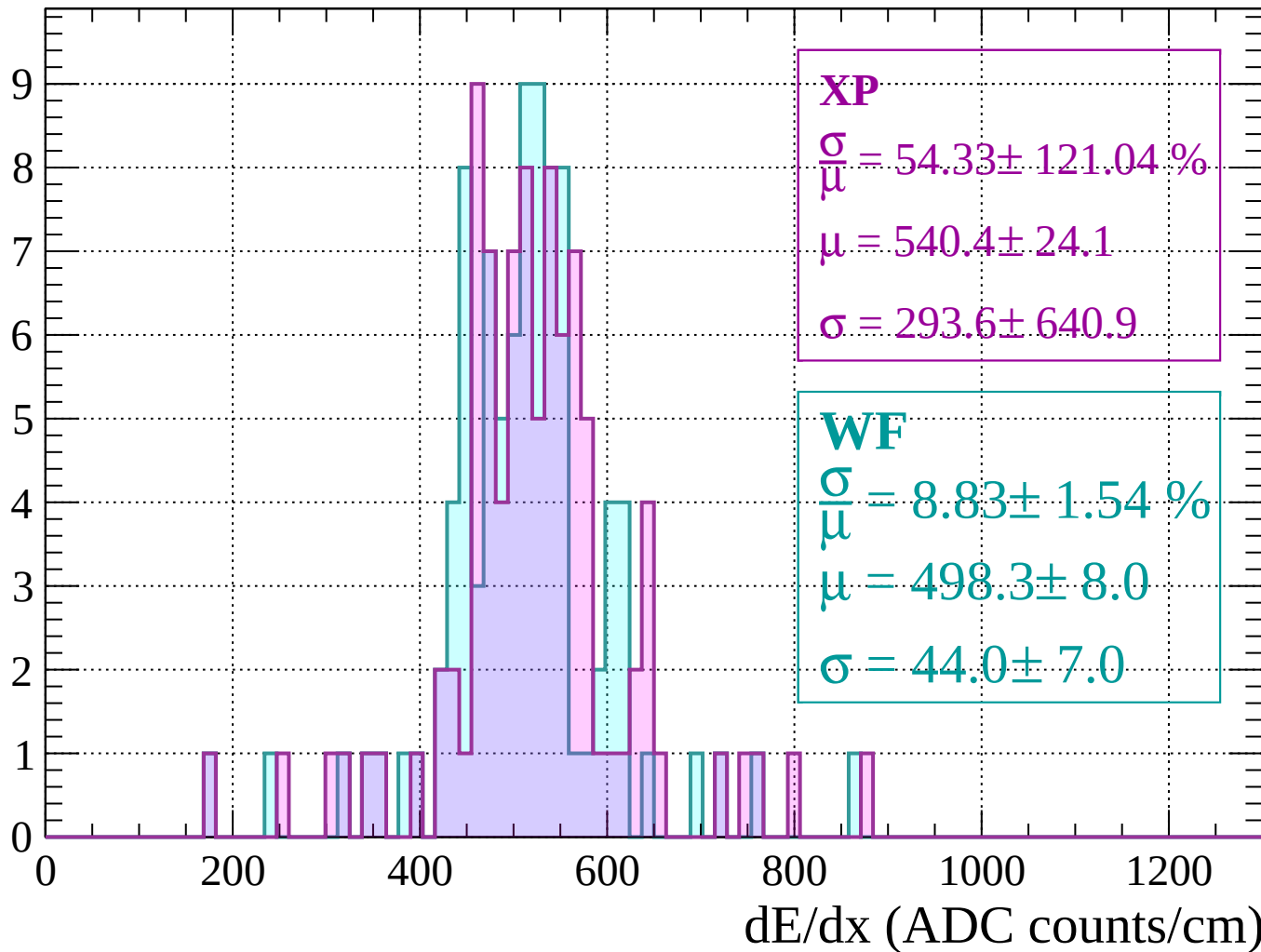
Energy loss | $1760 < p < 1800$

Count



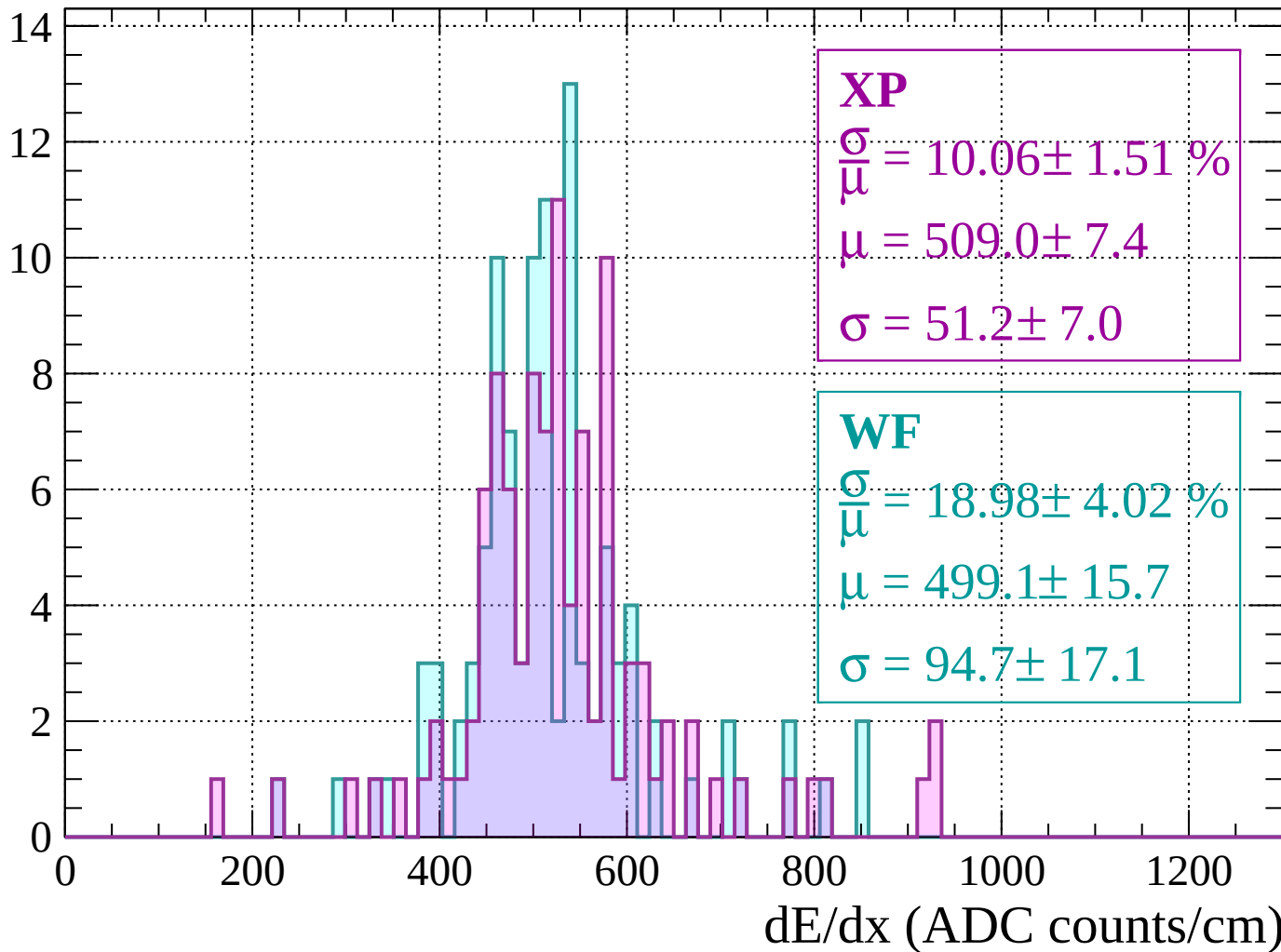
Energy loss | $1800 < p < 1840$

Count



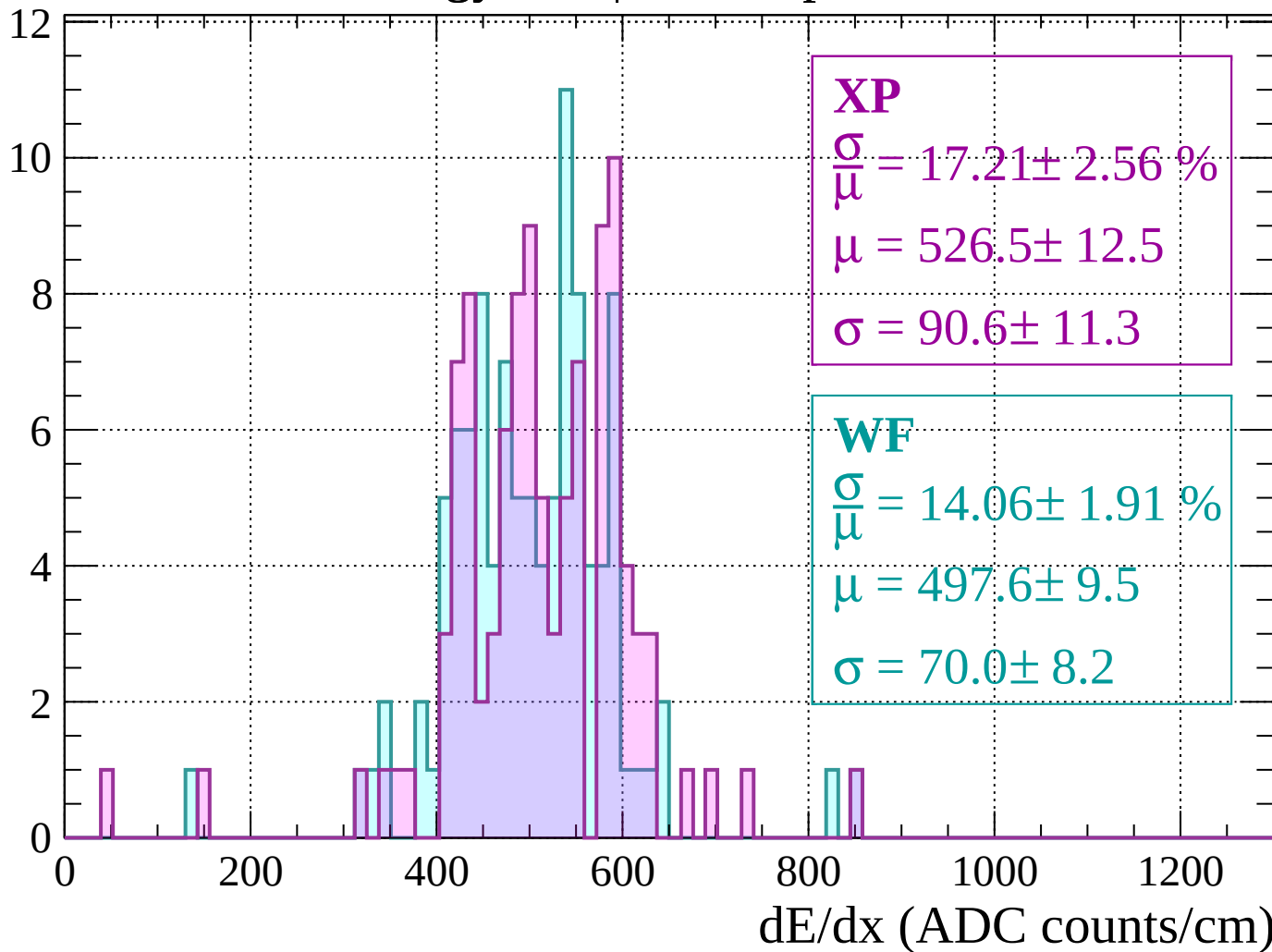
Energy loss | $1840 < p < 1880$

Count



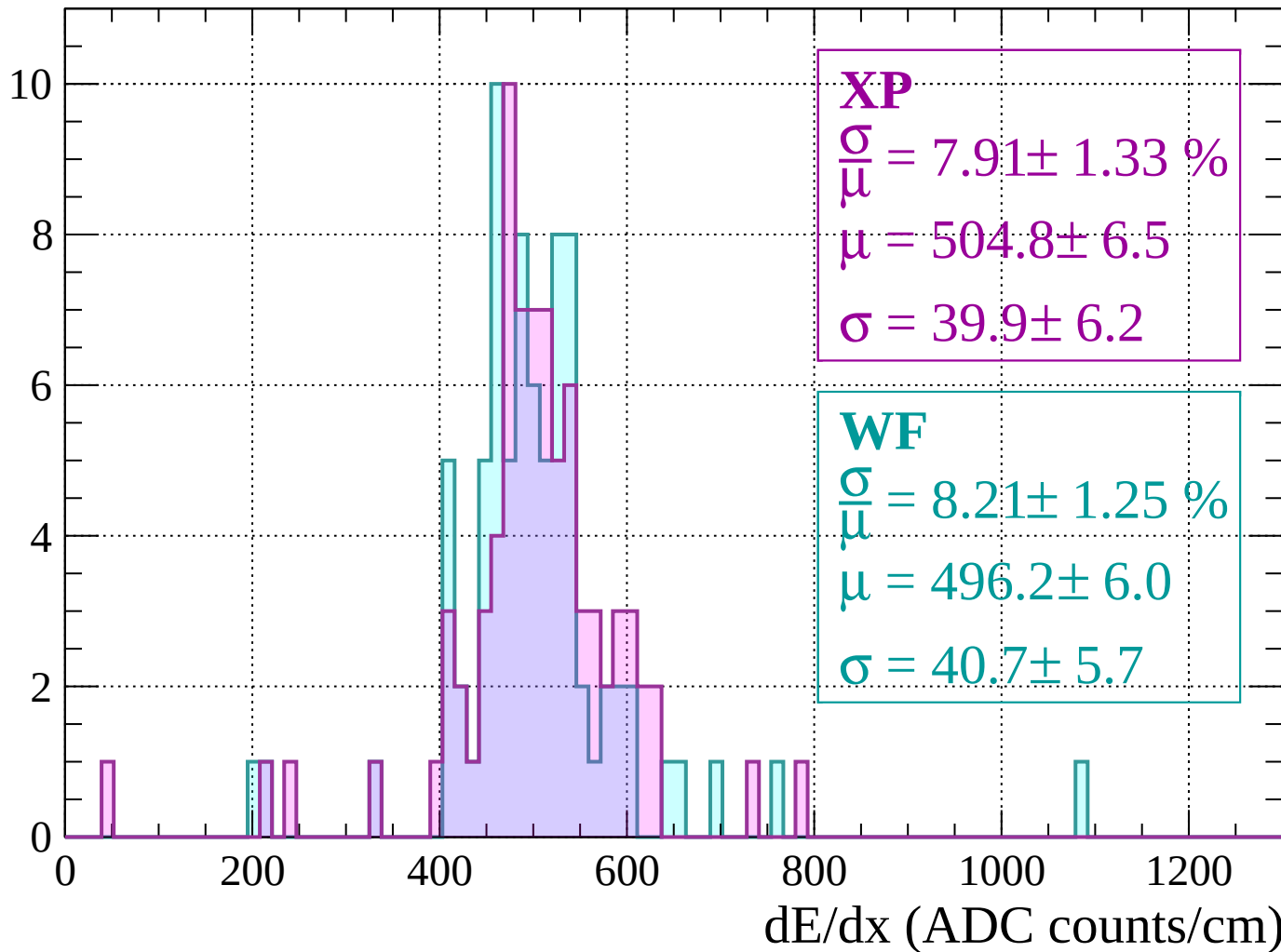
Energy loss | $1880 < p < 1920$

Count



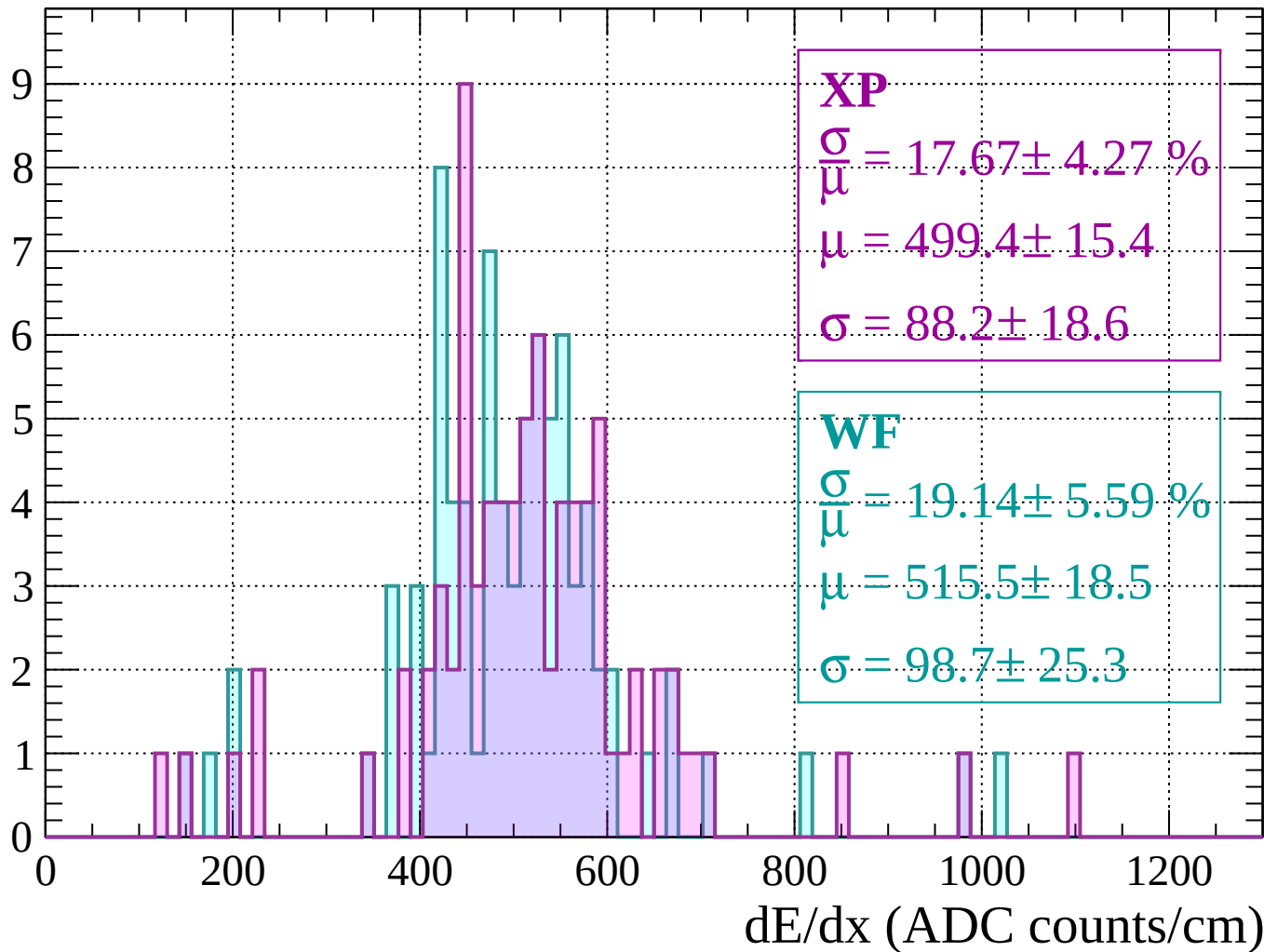
Energy loss | $1920 < p < 1960$

Count



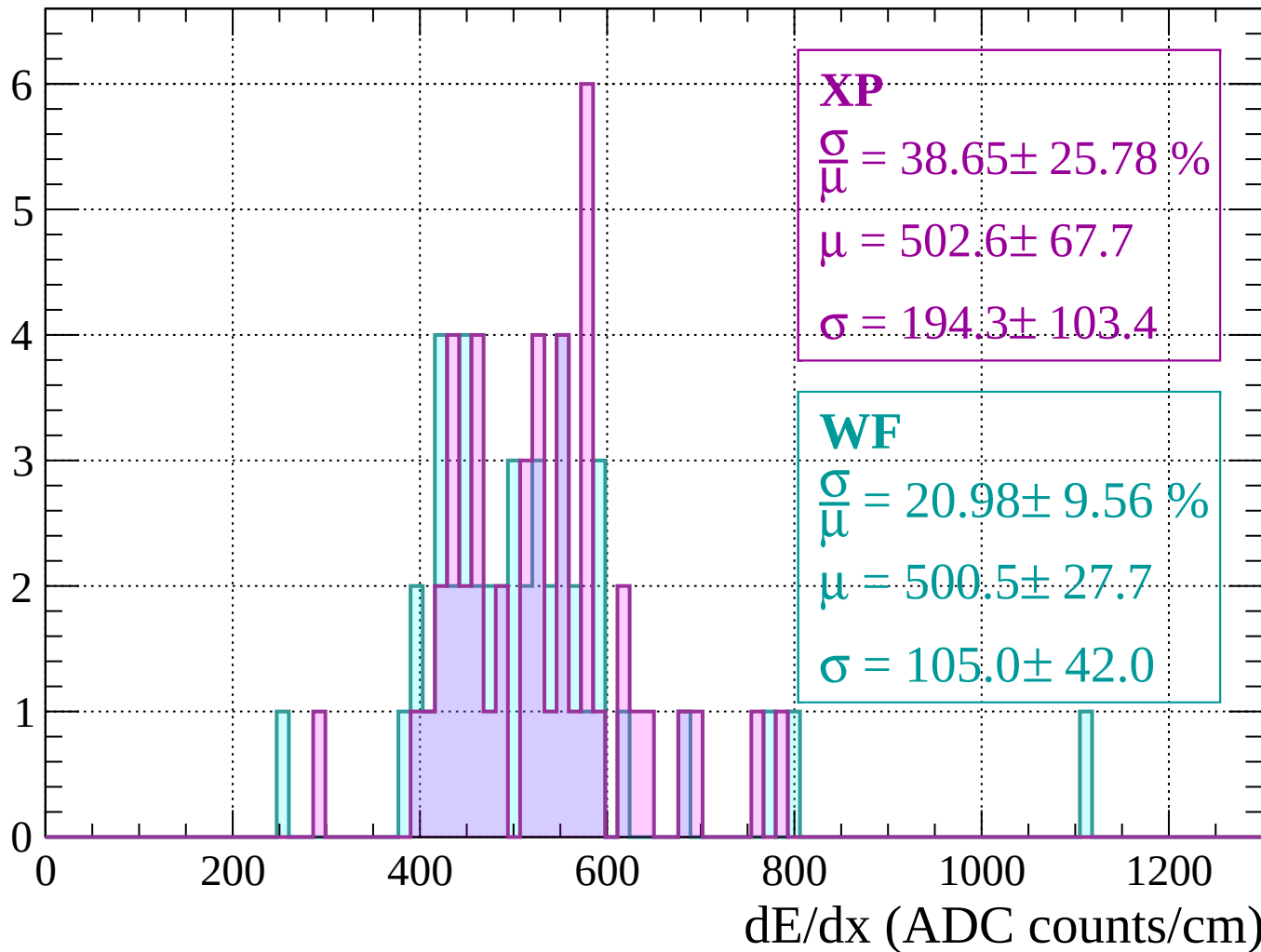
Energy loss | 1960 < p < 2000

Count



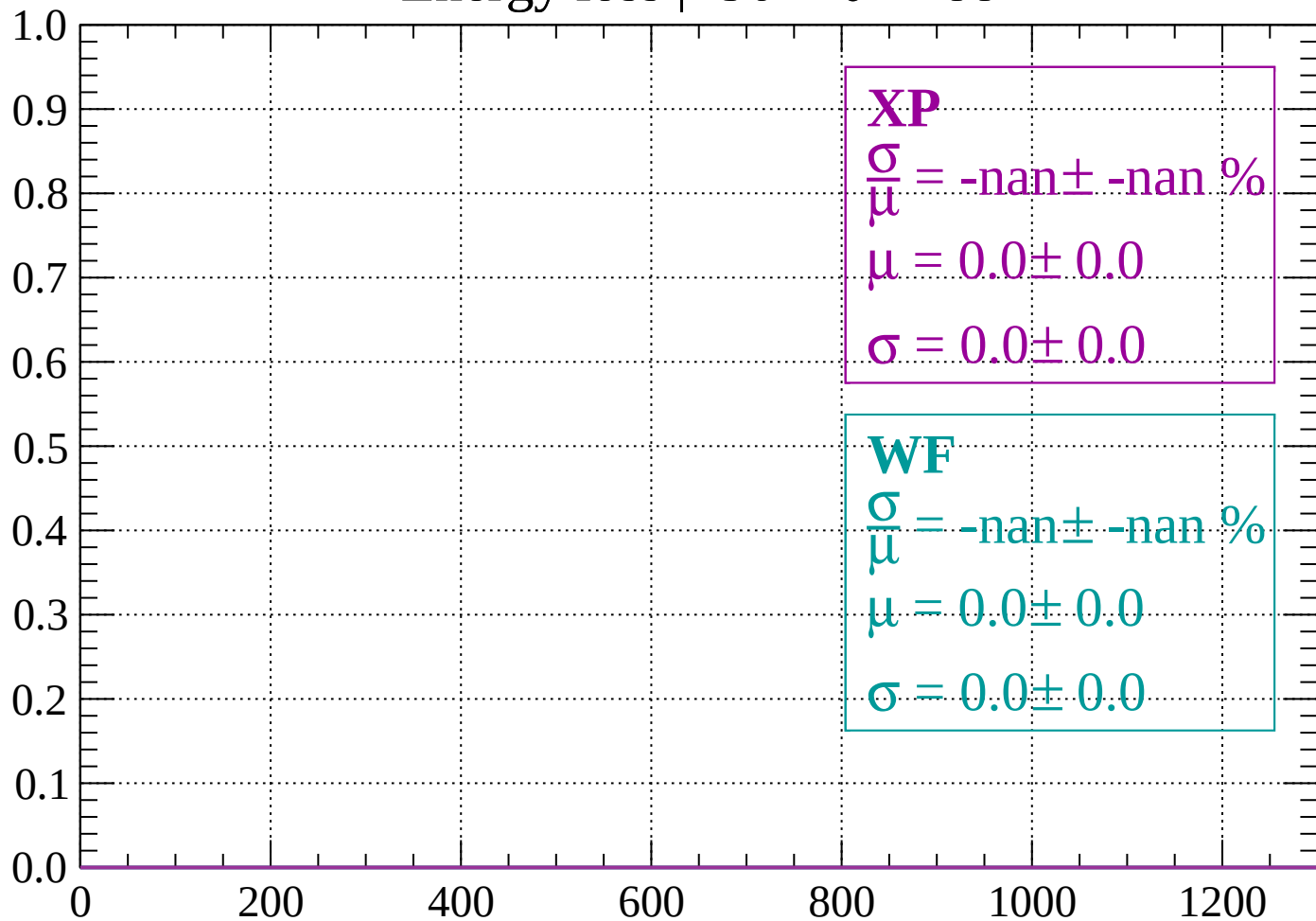
Energy loss | $2000 < p < 2040$

Count



Energy loss | $-90 < \theta < -88$

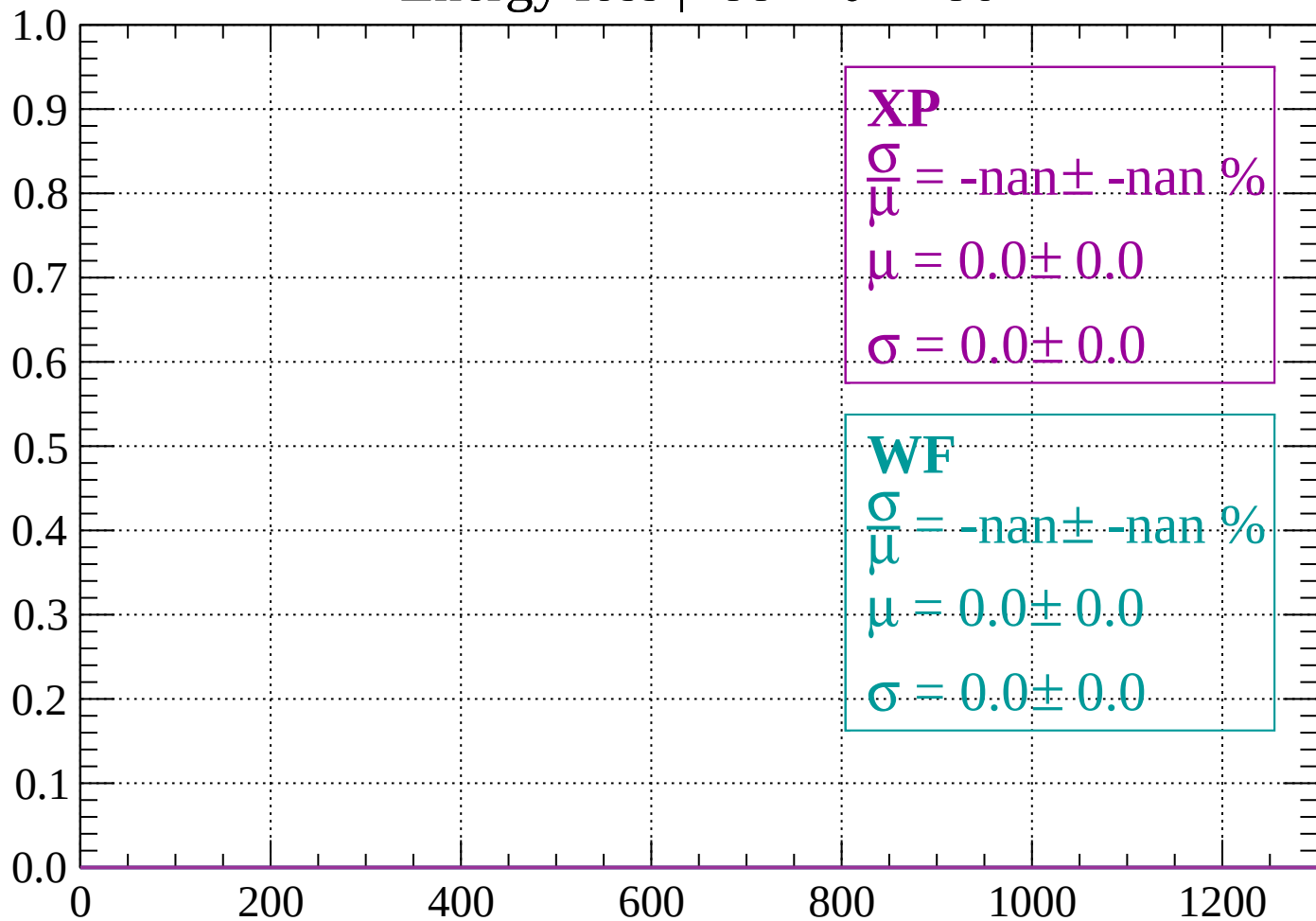
Count



dE/dx (ADC counts/cm)

Energy loss | $-88 < \theta < -86$

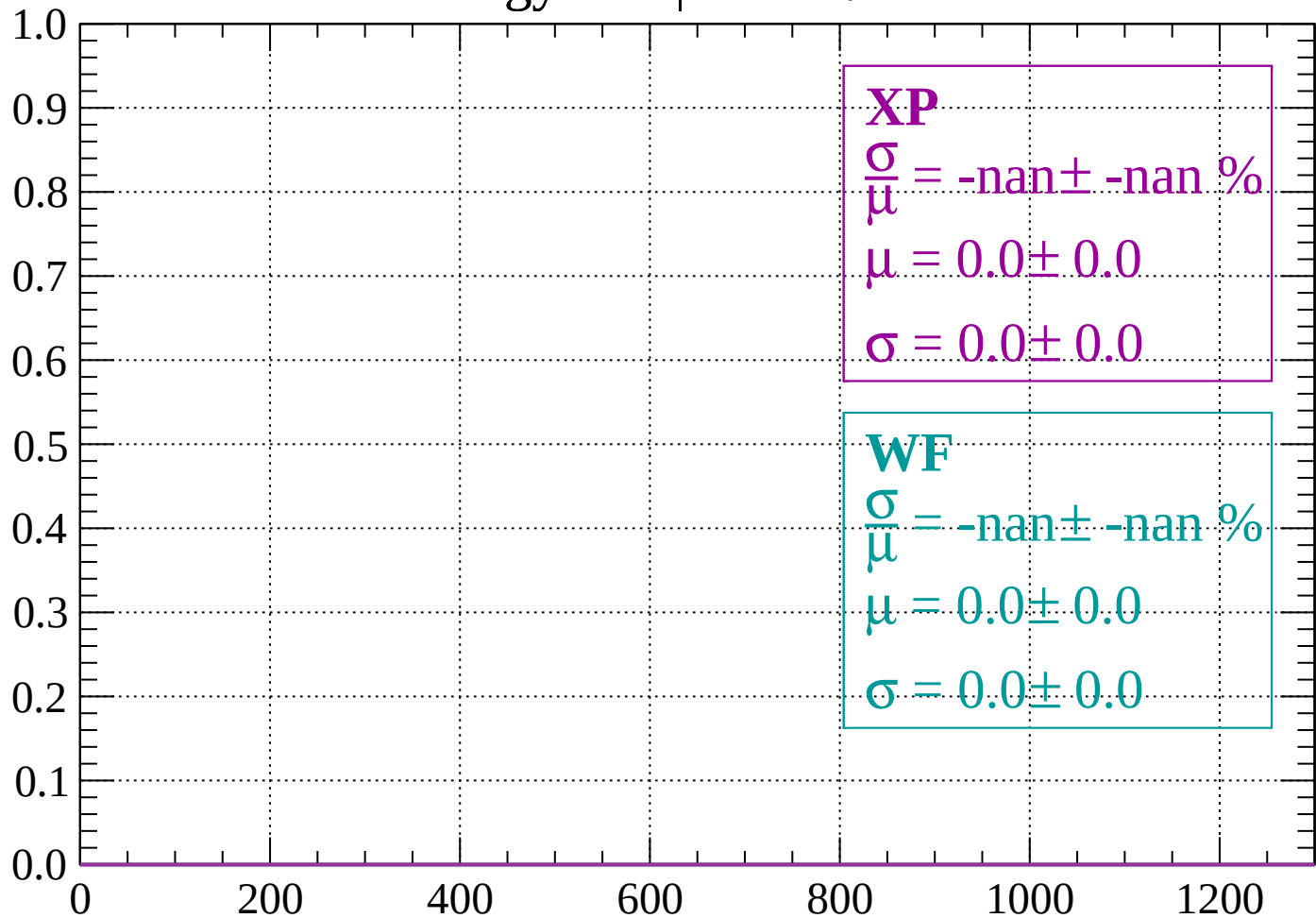
Count



dE/dx (ADC counts/cm)

Energy loss | $-86 < \theta < -84$

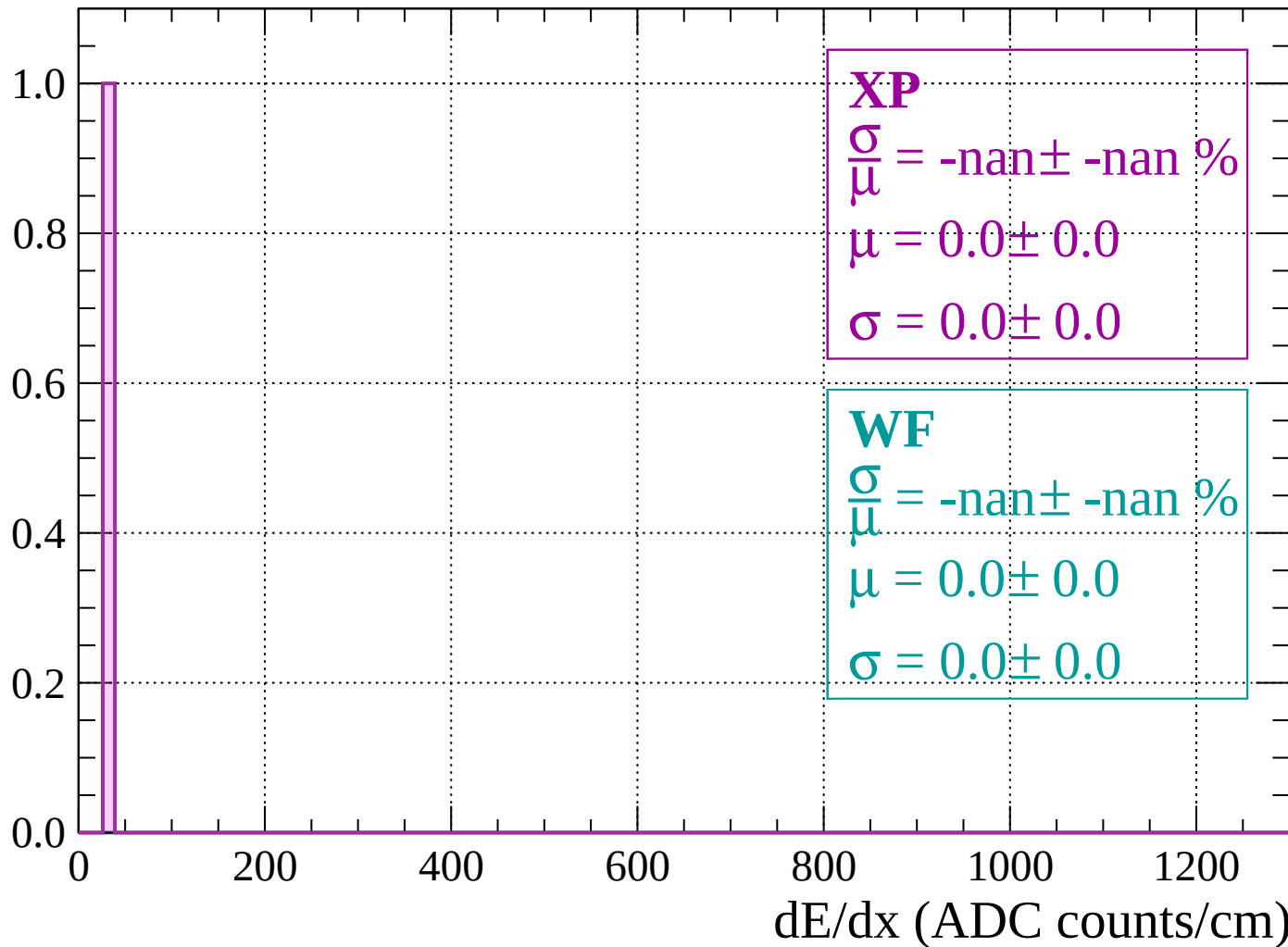
Count



dE/dx (ADC counts/cm)

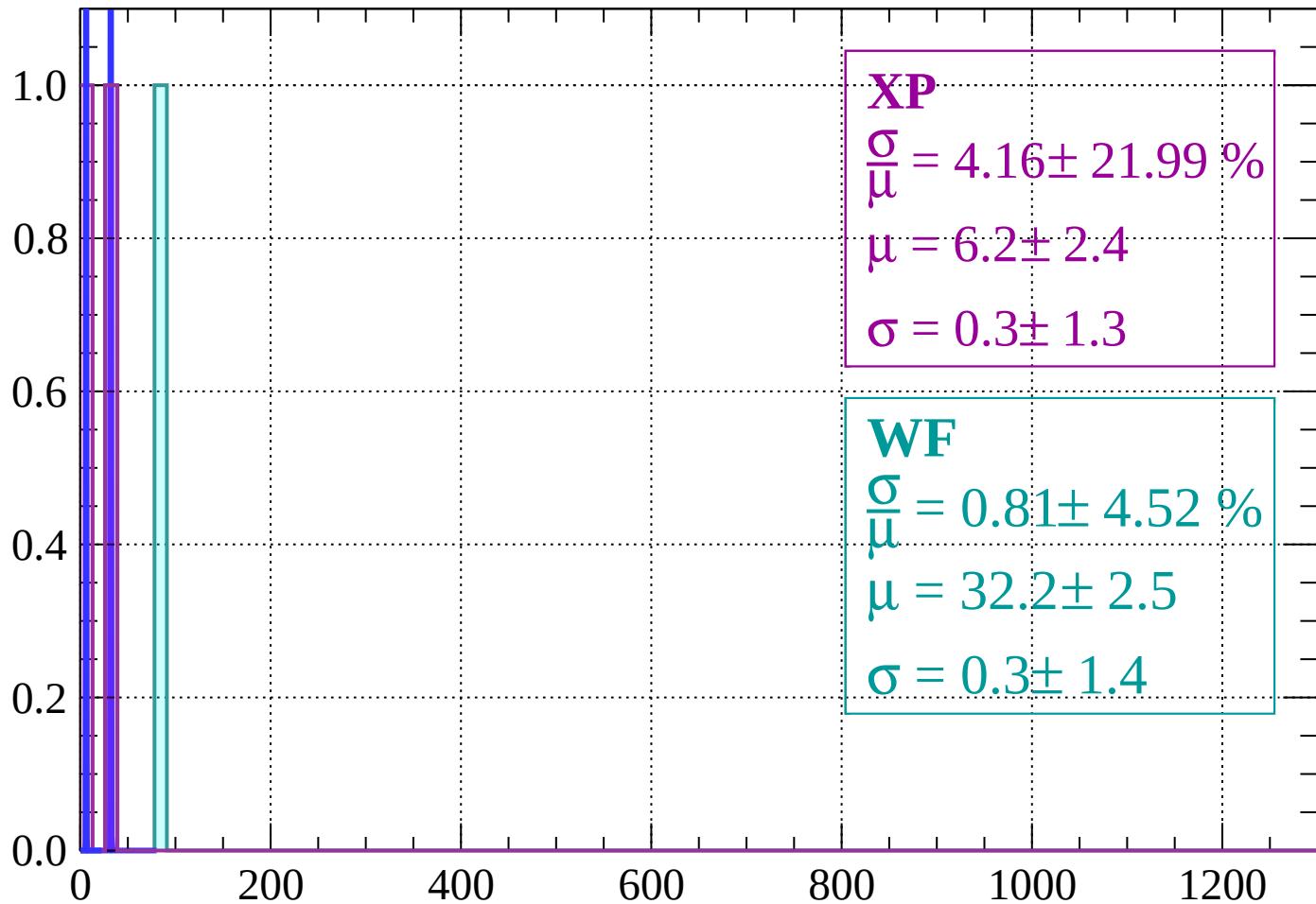
Energy loss | $-84 < \theta < -82$

Count



Energy loss | $-82 < \theta < -80$

Count



XP

$$\frac{\sigma}{\mu} = 4.16 \pm 21.99 \%$$

$$\mu = 6.2 \pm 2.4$$

$$\sigma = 0.3 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 0.81 \pm 4.52 \%$$

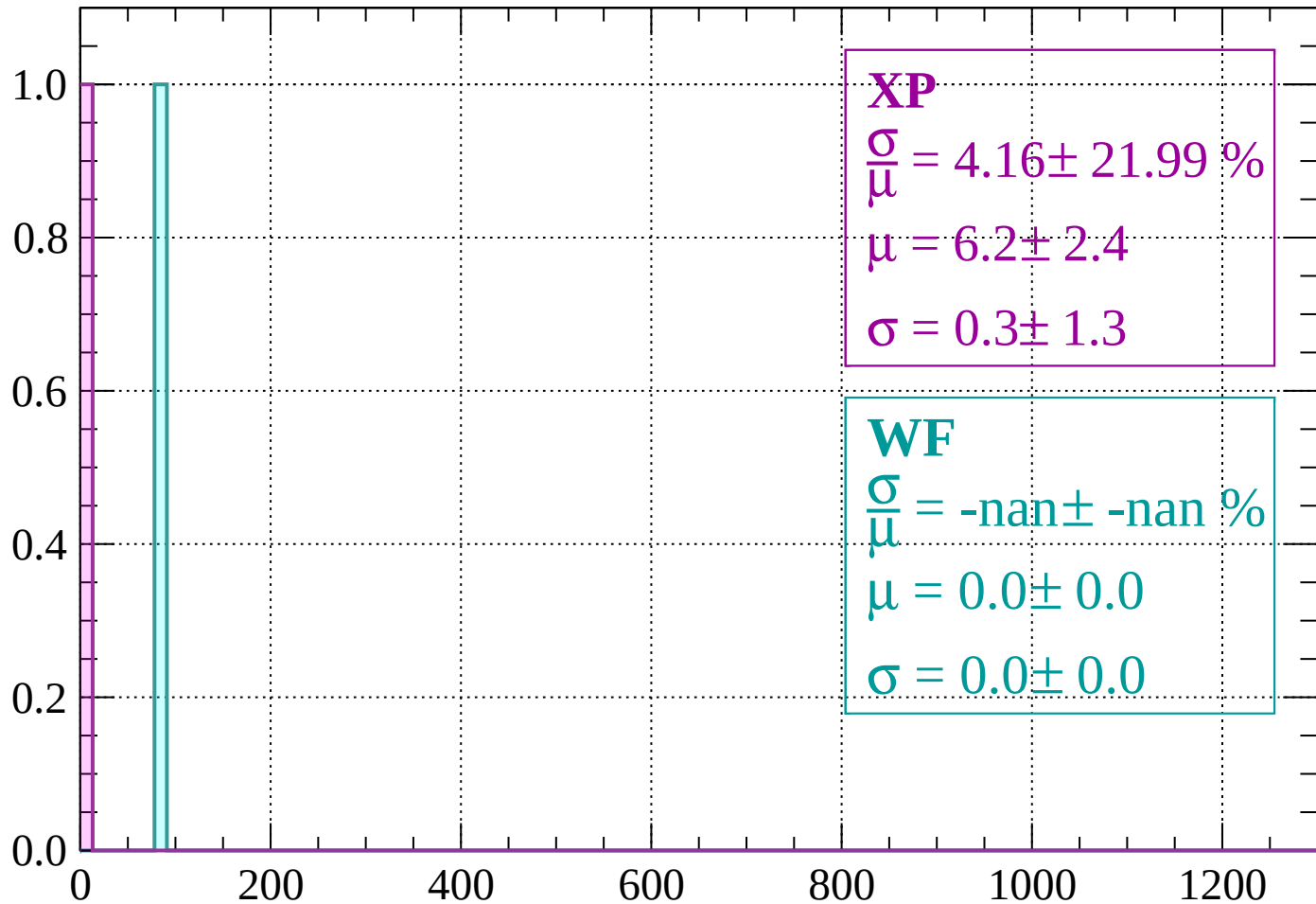
$$\mu = 32.2 \pm 2.5$$

$$\sigma = 0.3 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $-80 < \theta < -78$

Count



XP

$$\frac{\sigma}{\mu} = 4.16 \pm 21.99 \%$$

$$\mu = 6.2 \pm 2.4$$

$$\sigma = 0.3 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$$

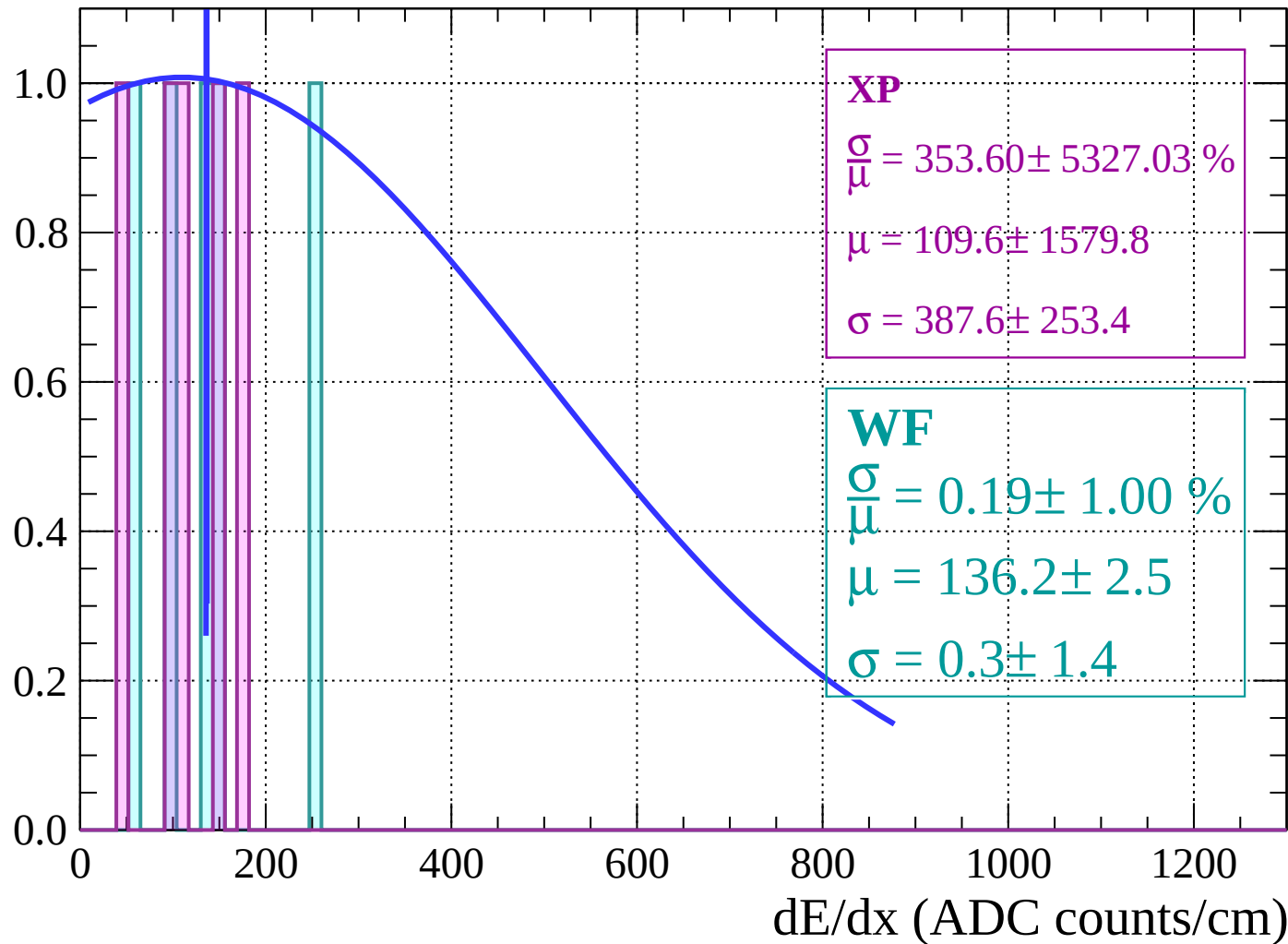
$$\mu = 0.0 \pm 0.0$$

$$\sigma = 0.0 \pm 0.0$$

dE/dx (ADC counts/cm)

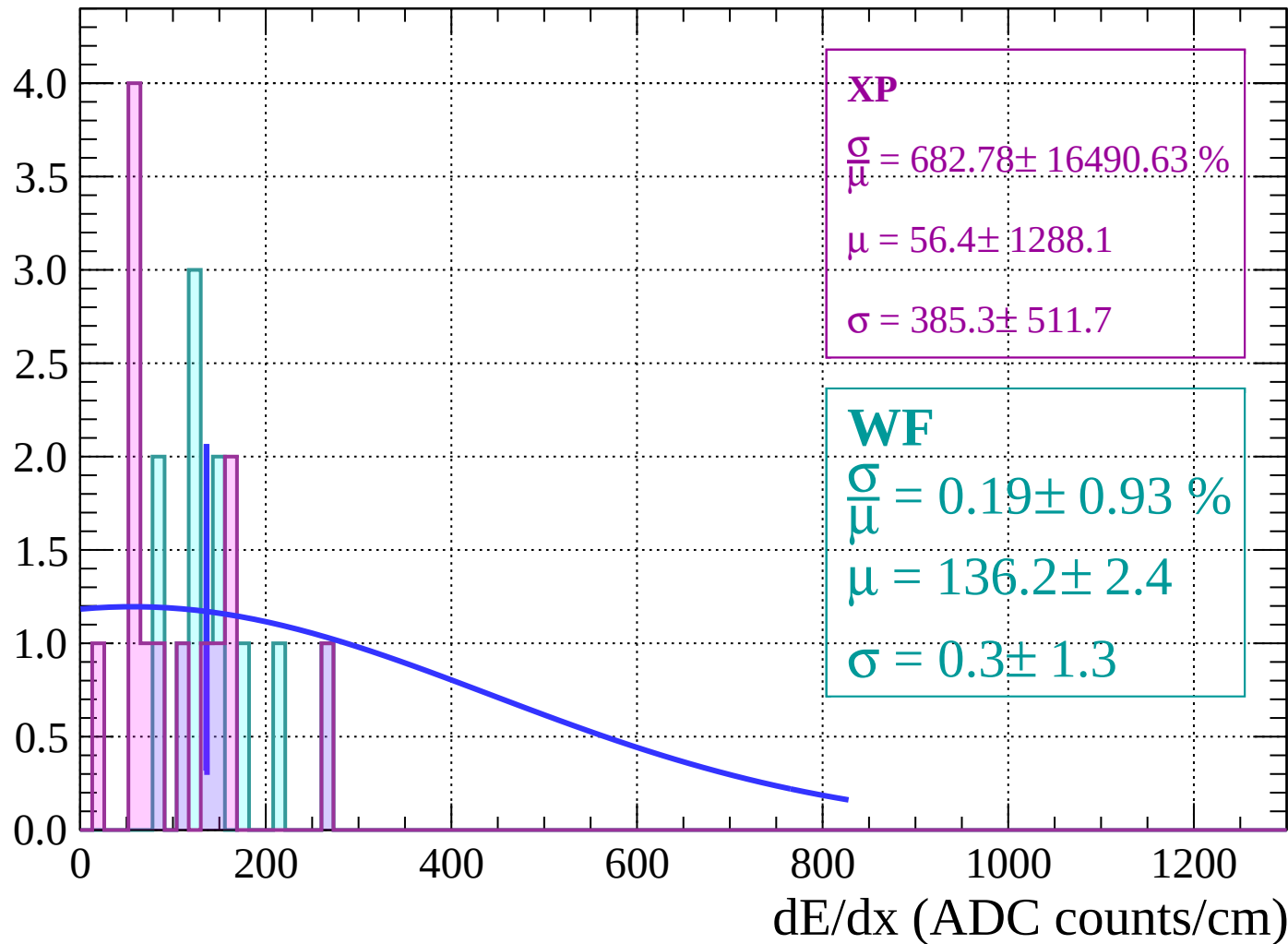
Energy loss | $-78 < \theta < -76$

Count



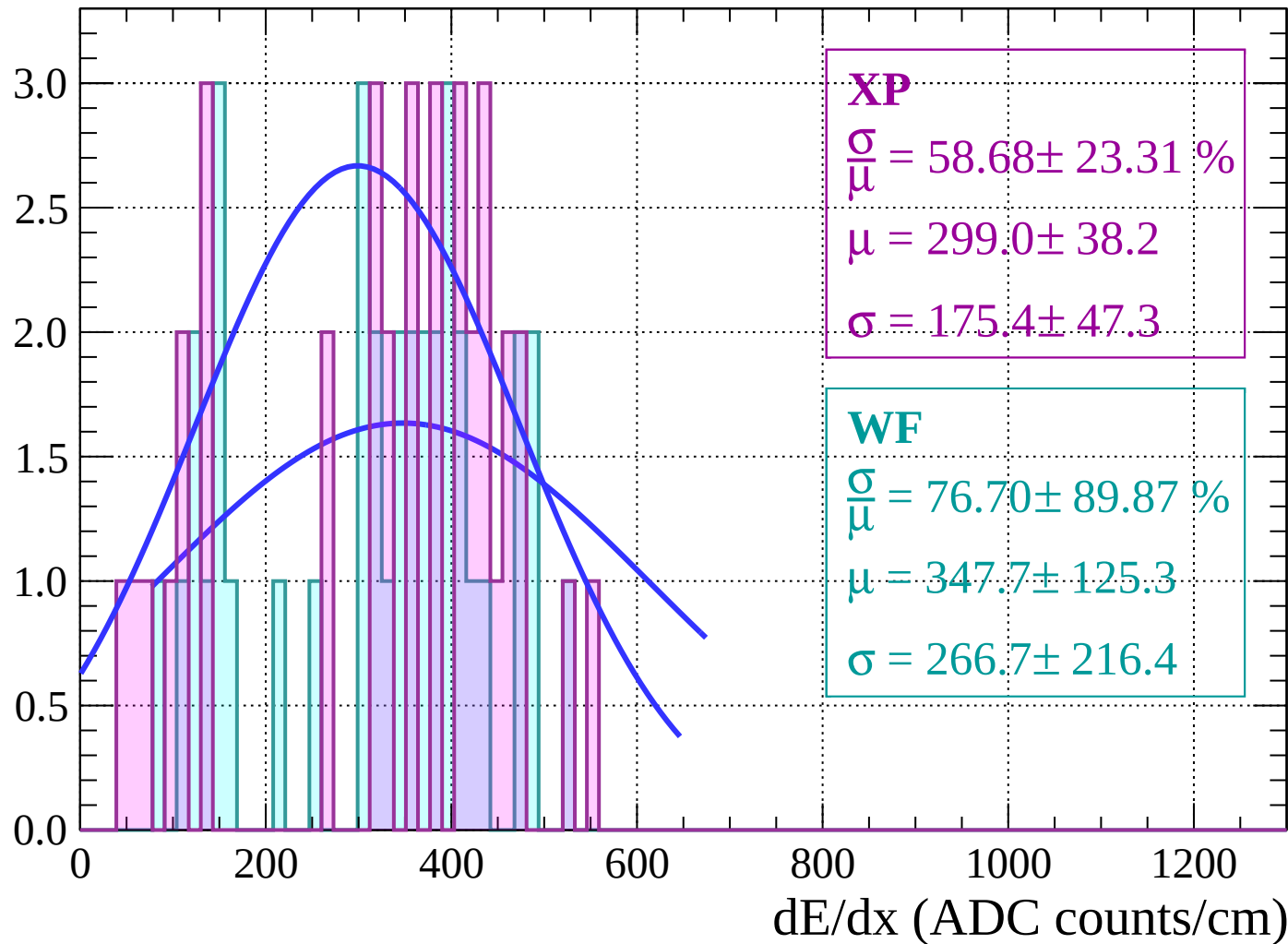
Energy loss | $-76 < \theta < -74$

Count



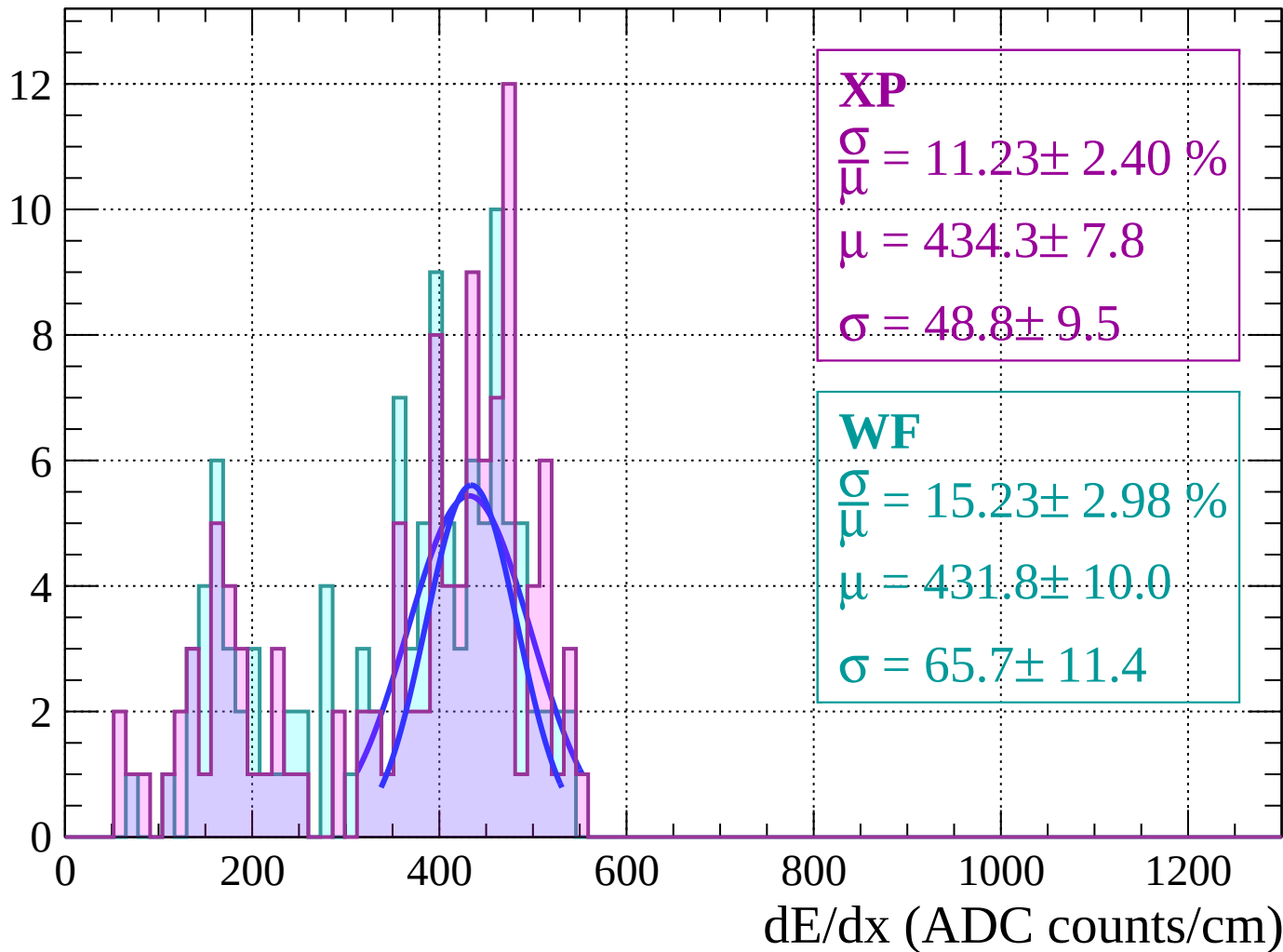
Energy loss $|-74 < \theta < -72$

Count



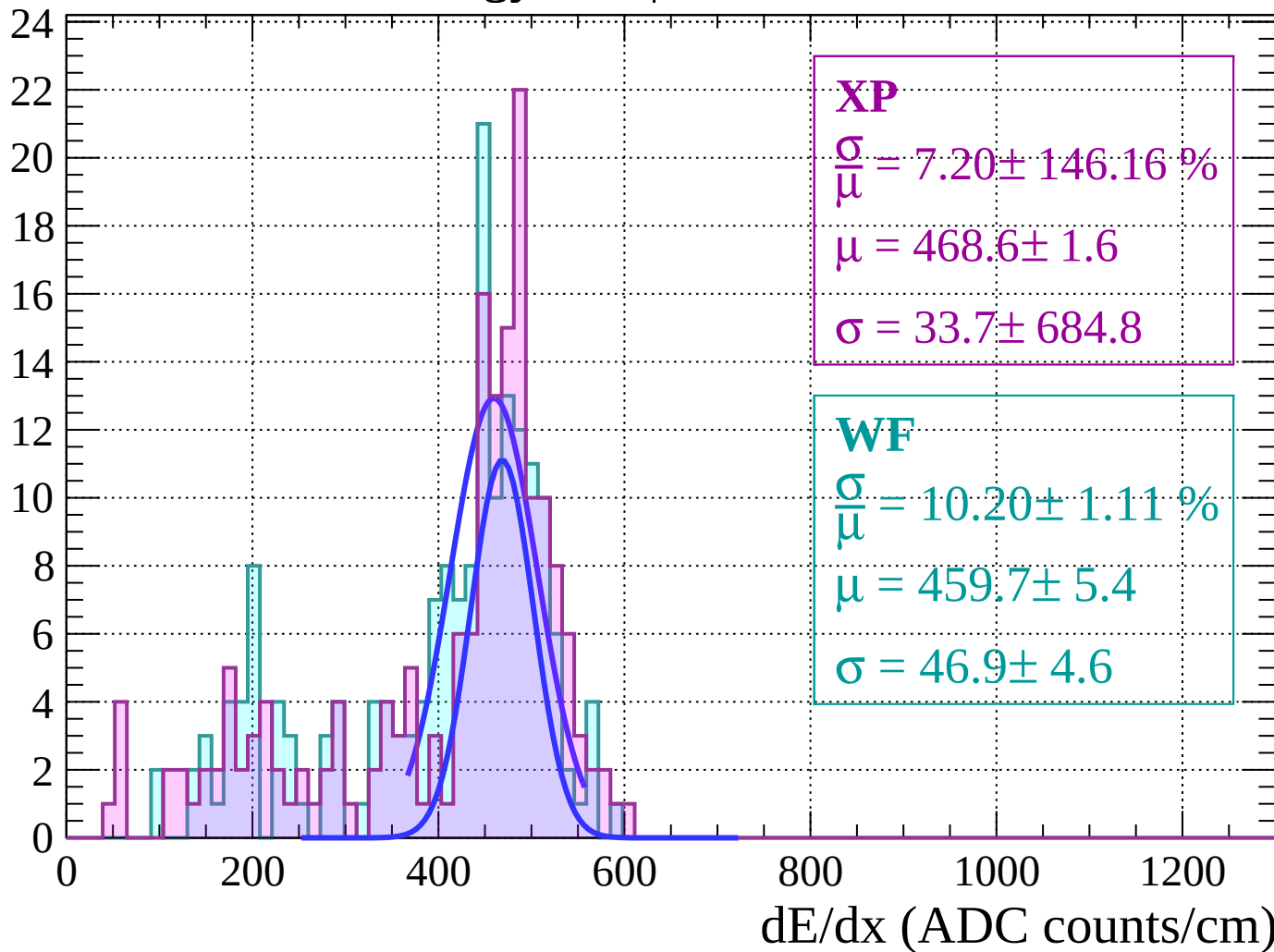
Energy loss | $-72 < \theta < -70$

Count



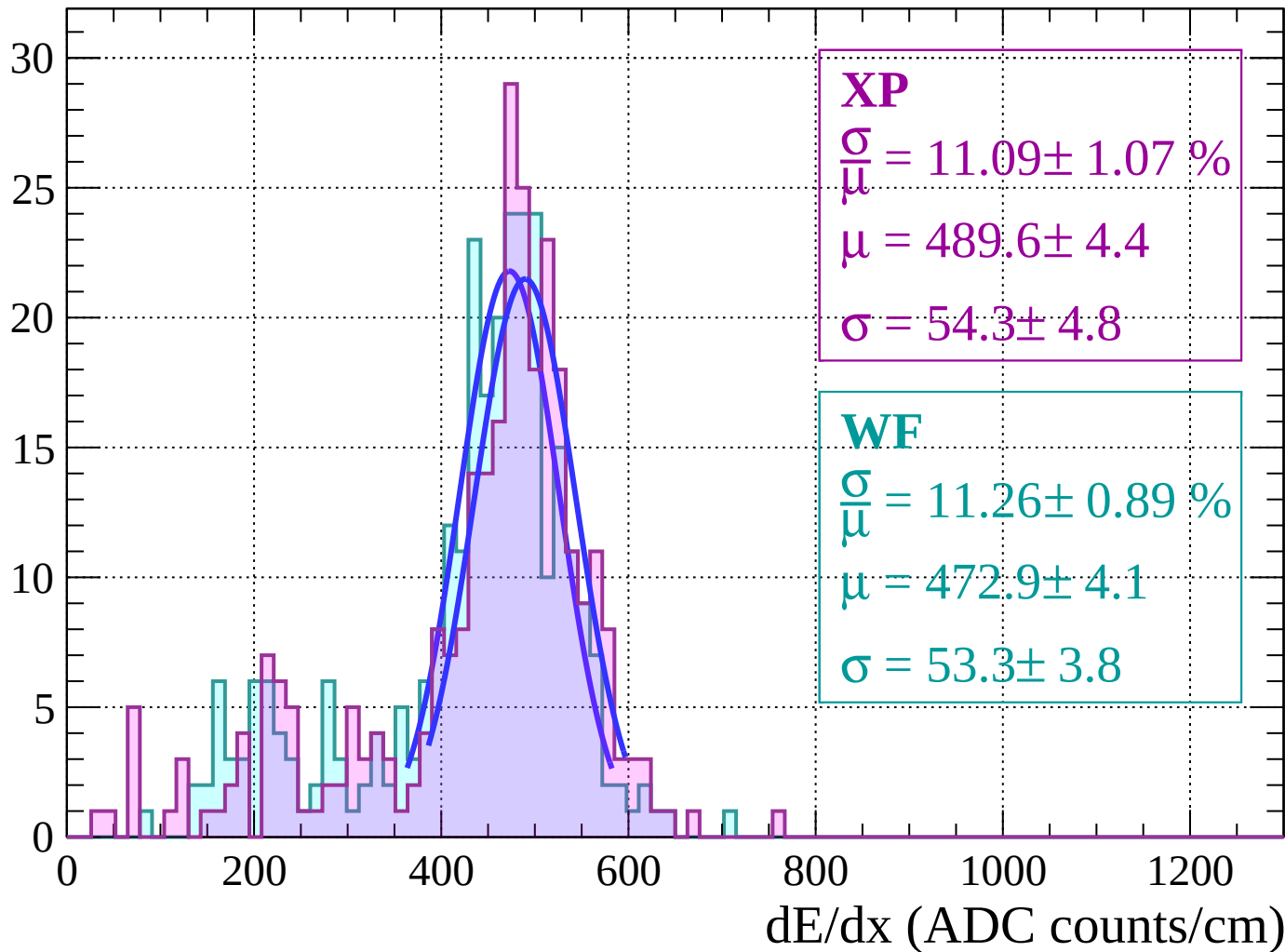
Energy loss | $-70 < \theta < -68$

Count



Energy loss | $-68 < \theta < -66$

Count



Energy loss $|-66 < \theta < -64$

Count

XP

$$\frac{\sigma}{\mu} = 12.32 \pm 0.96 \%$$

$$\mu = 508.1 \pm 4.1$$

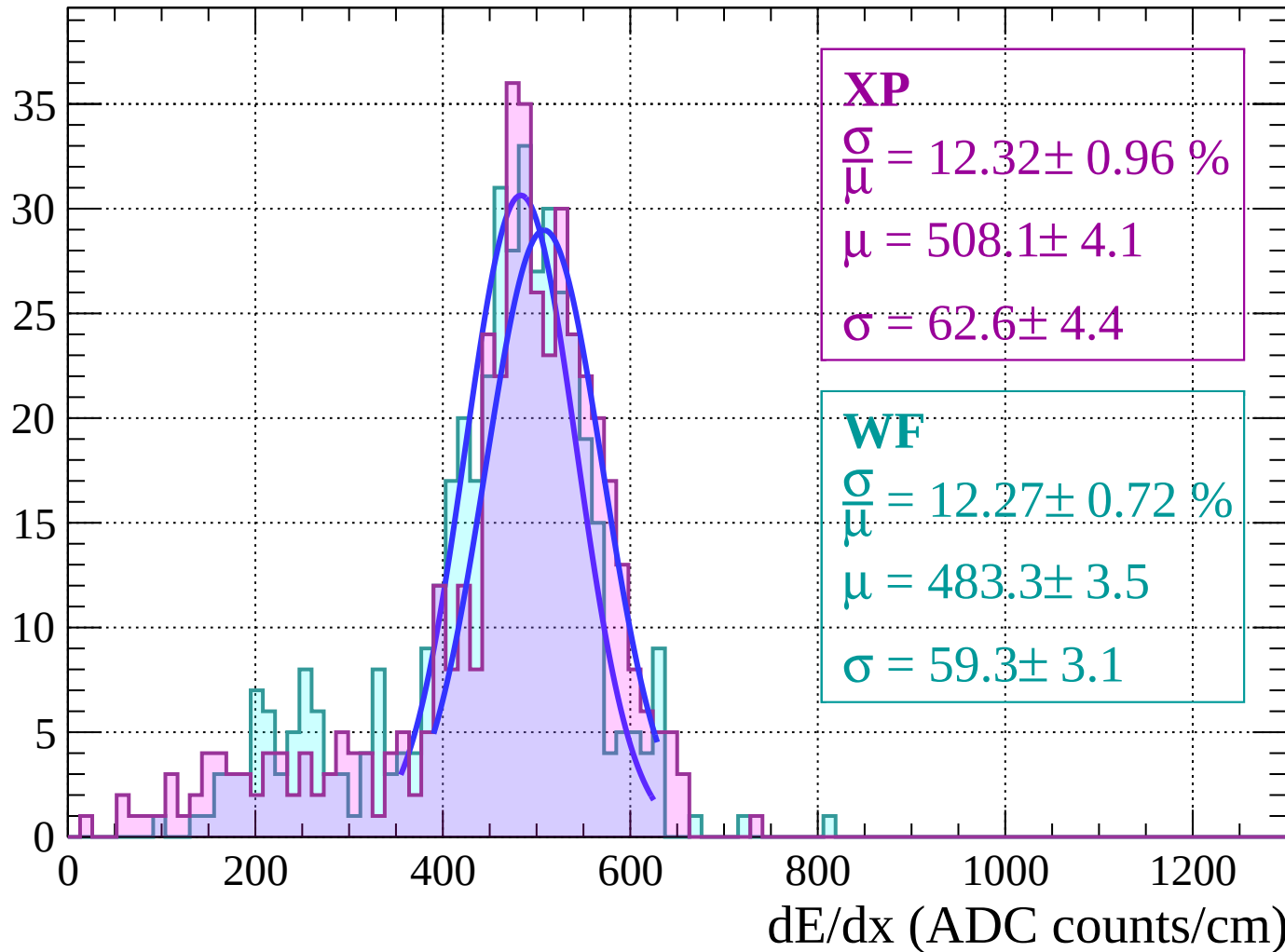
$$\sigma = 62.6 \pm 4.4$$

WF

$$\frac{\sigma}{\mu} = 12.27 \pm 0.72 \%$$

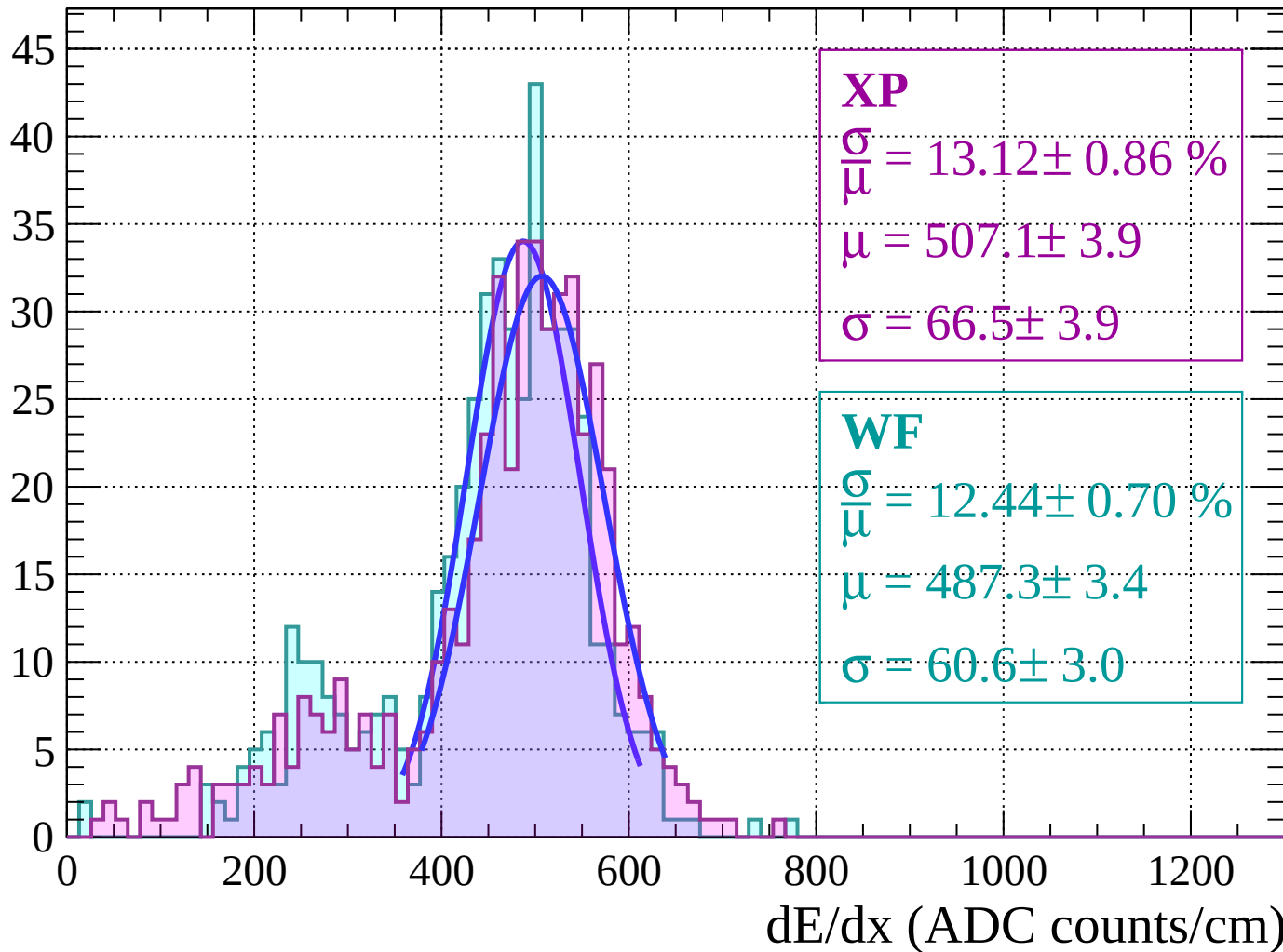
$$\mu = 483.3 \pm 3.5$$

$$\sigma = 59.3 \pm 3.1$$



Energy loss $|-64 < \theta < -62$

Count



Energy loss $|-62 < \theta < -60|$

Count

XP

$$\frac{\sigma}{\mu} = 12.61 \pm 0.69 \%$$

$$\mu = 511.1 \pm 3.2$$

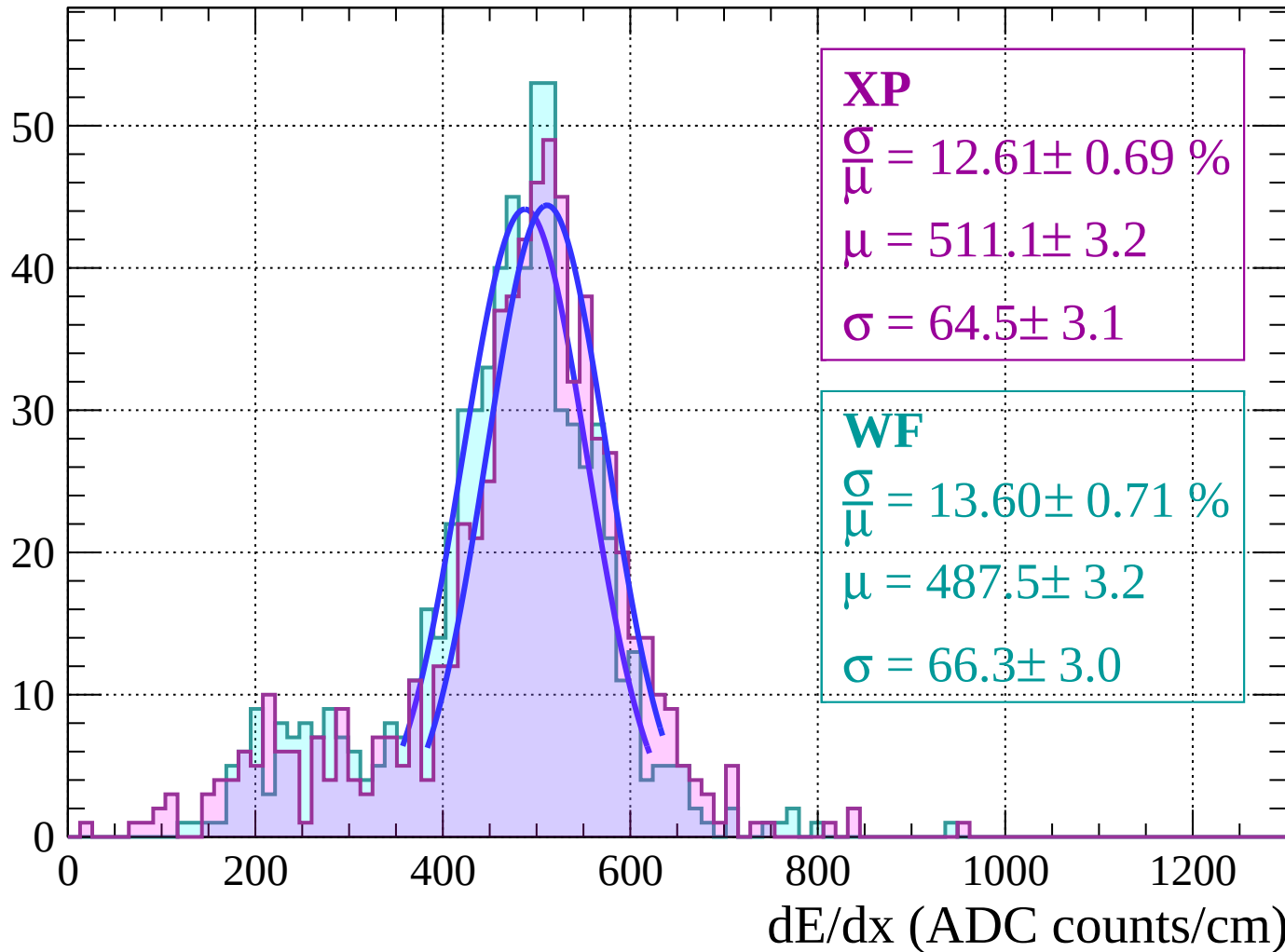
$$\sigma = 64.5 \pm 3.1$$

WF

$$\frac{\sigma}{\mu} = 13.60 \pm 0.71 \%$$

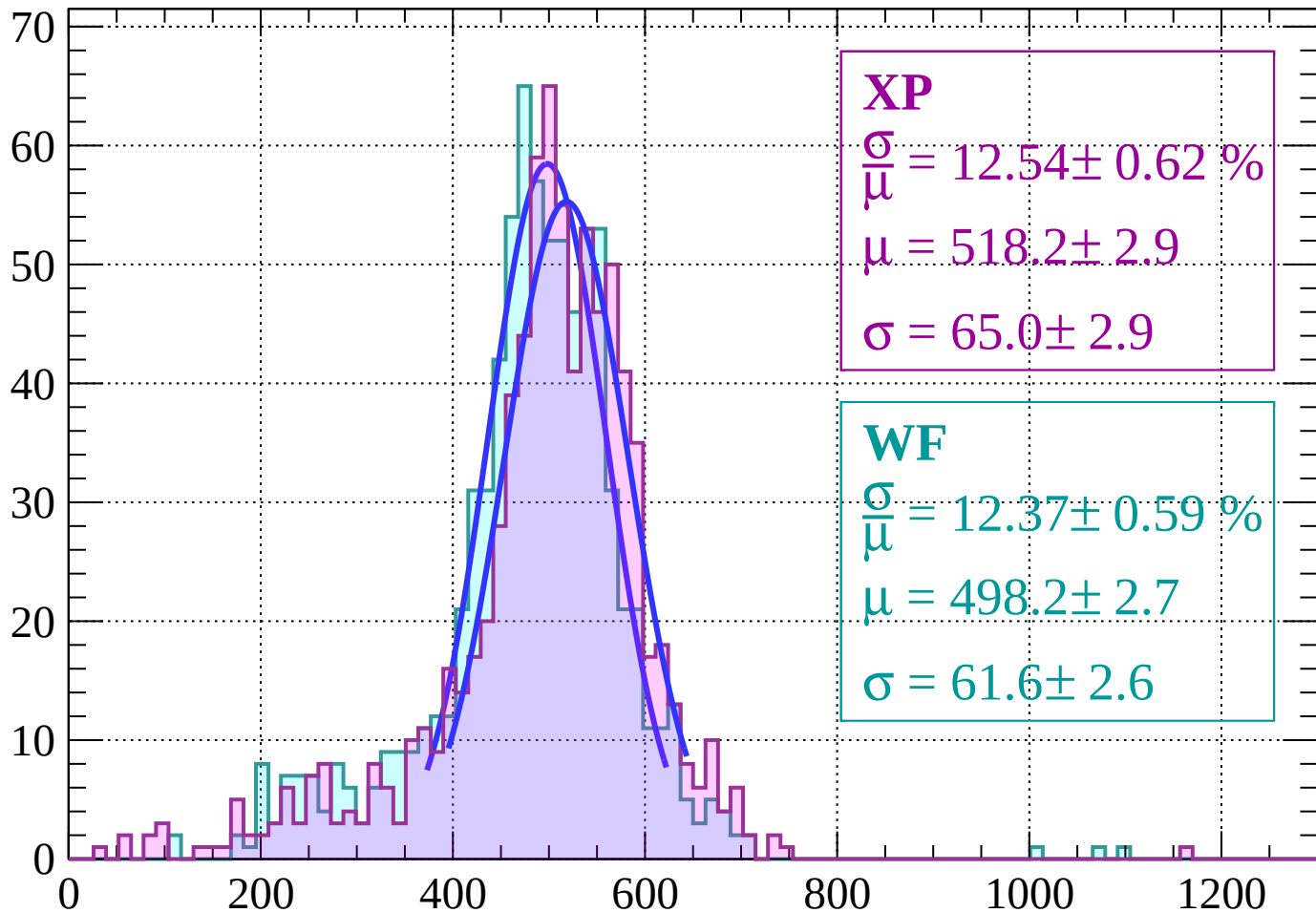
$$\mu = 487.5 \pm 3.2$$

$$\sigma = 66.3 \pm 3.0$$



Energy loss | $-60 < \theta < -58$

Count



XP

$$\frac{\sigma}{\mu} = 12.54 \pm 0.62 \%$$

$$\mu = 518.2 \pm 2.9$$

$$\sigma = 65.0 \pm 2.9$$

WF

$$\frac{\sigma}{\mu} = 12.37 \pm 0.59 \%$$

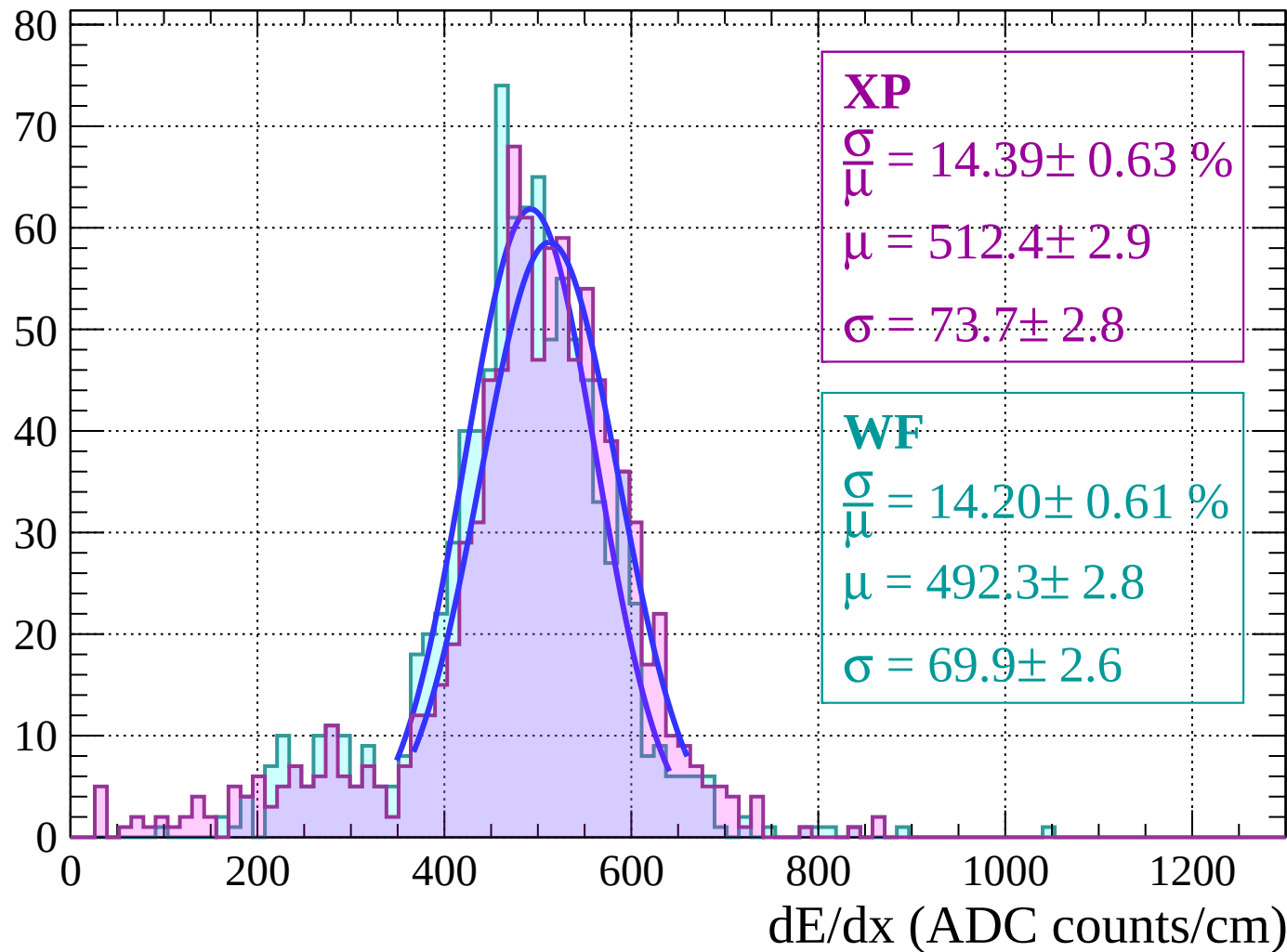
$$\mu = 498.2 \pm 2.7$$

$$\sigma = 61.6 \pm 2.6$$

dE/dx (ADC counts/cm)

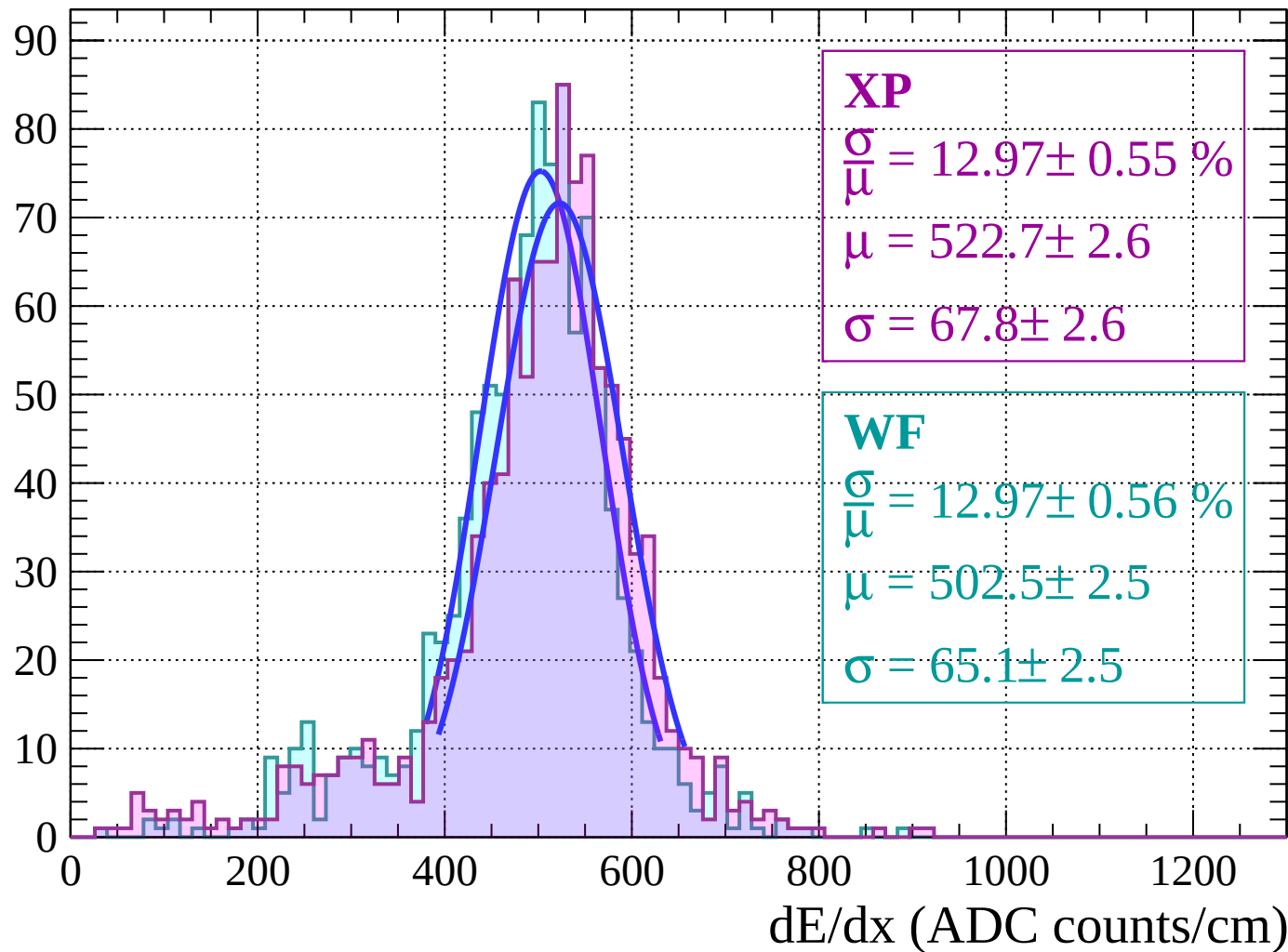
Energy loss | $-58 < \theta < -56$

Count



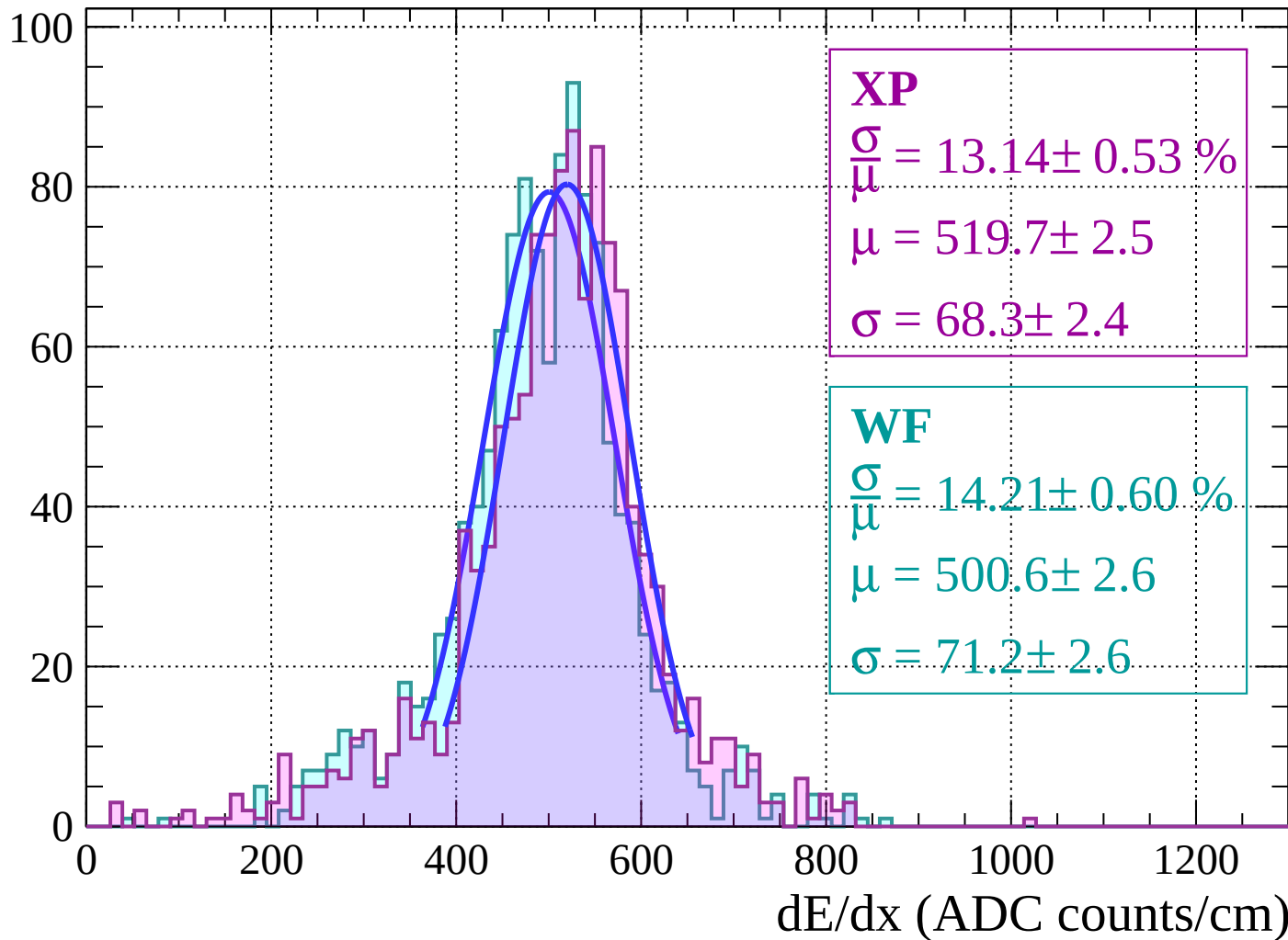
Energy loss $|-56 < \theta < -54$

Count



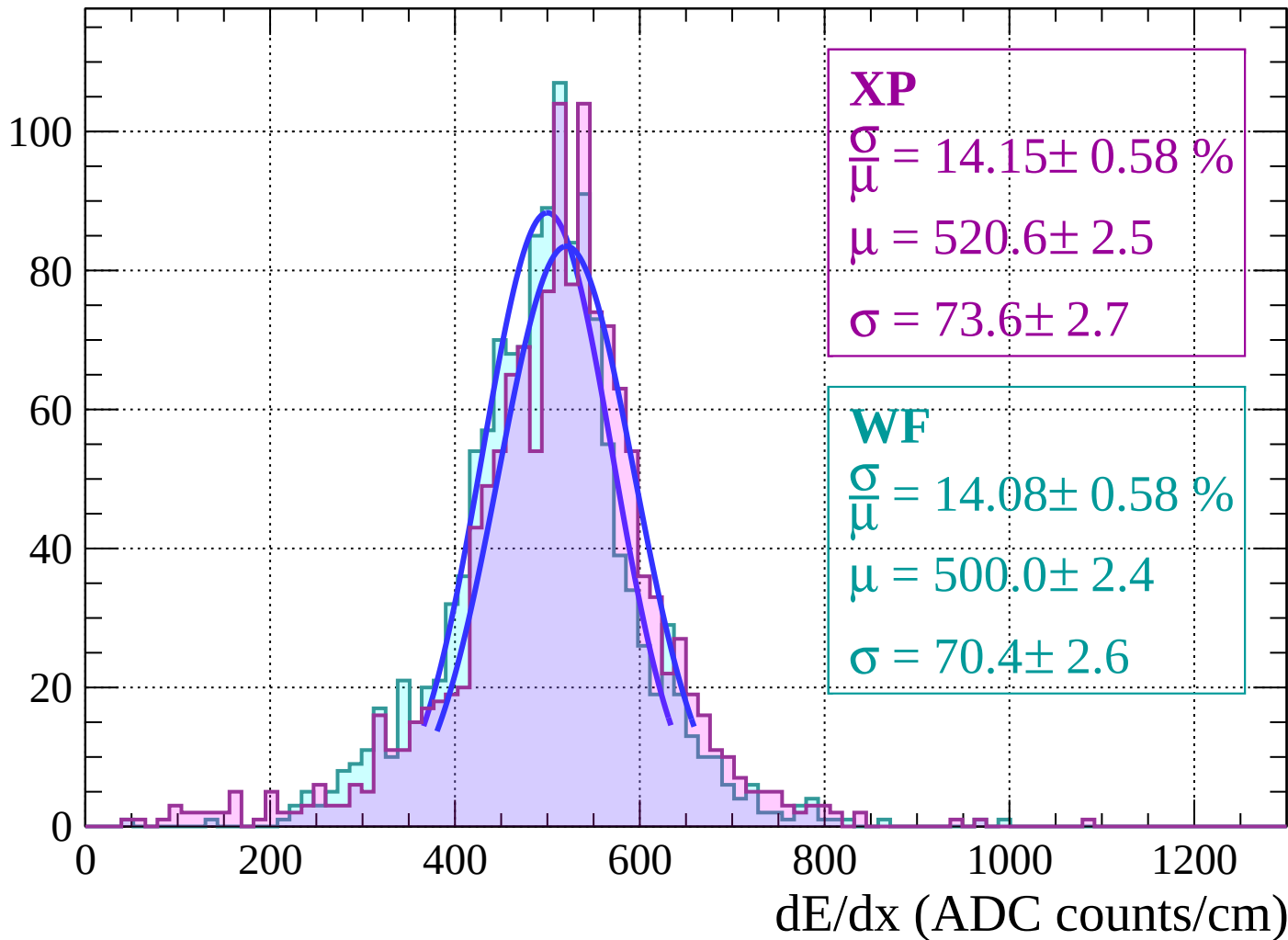
Energy loss | $-54 < \theta < -52$

Count



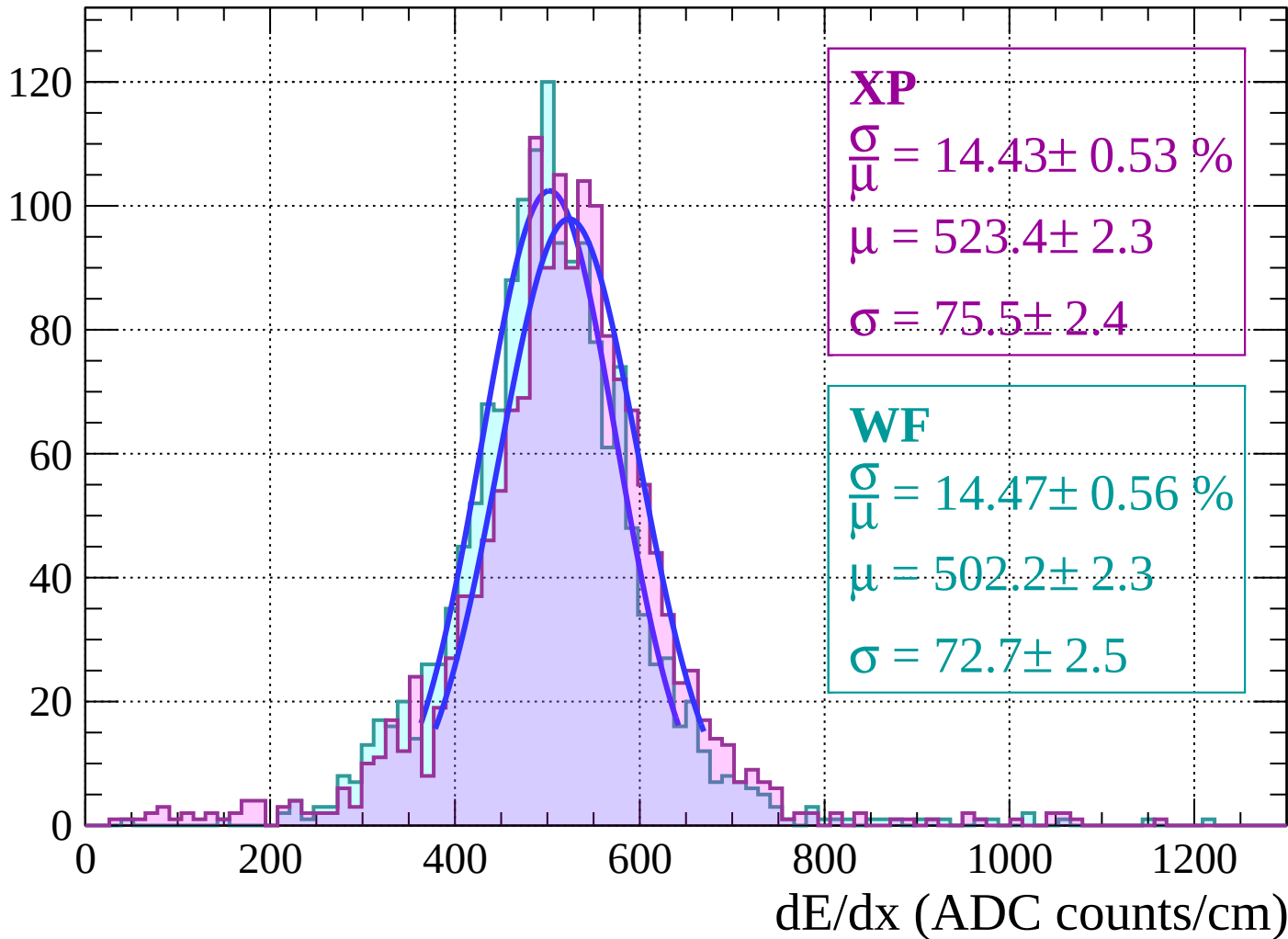
Energy loss | $-52 < \theta < -50$

Count



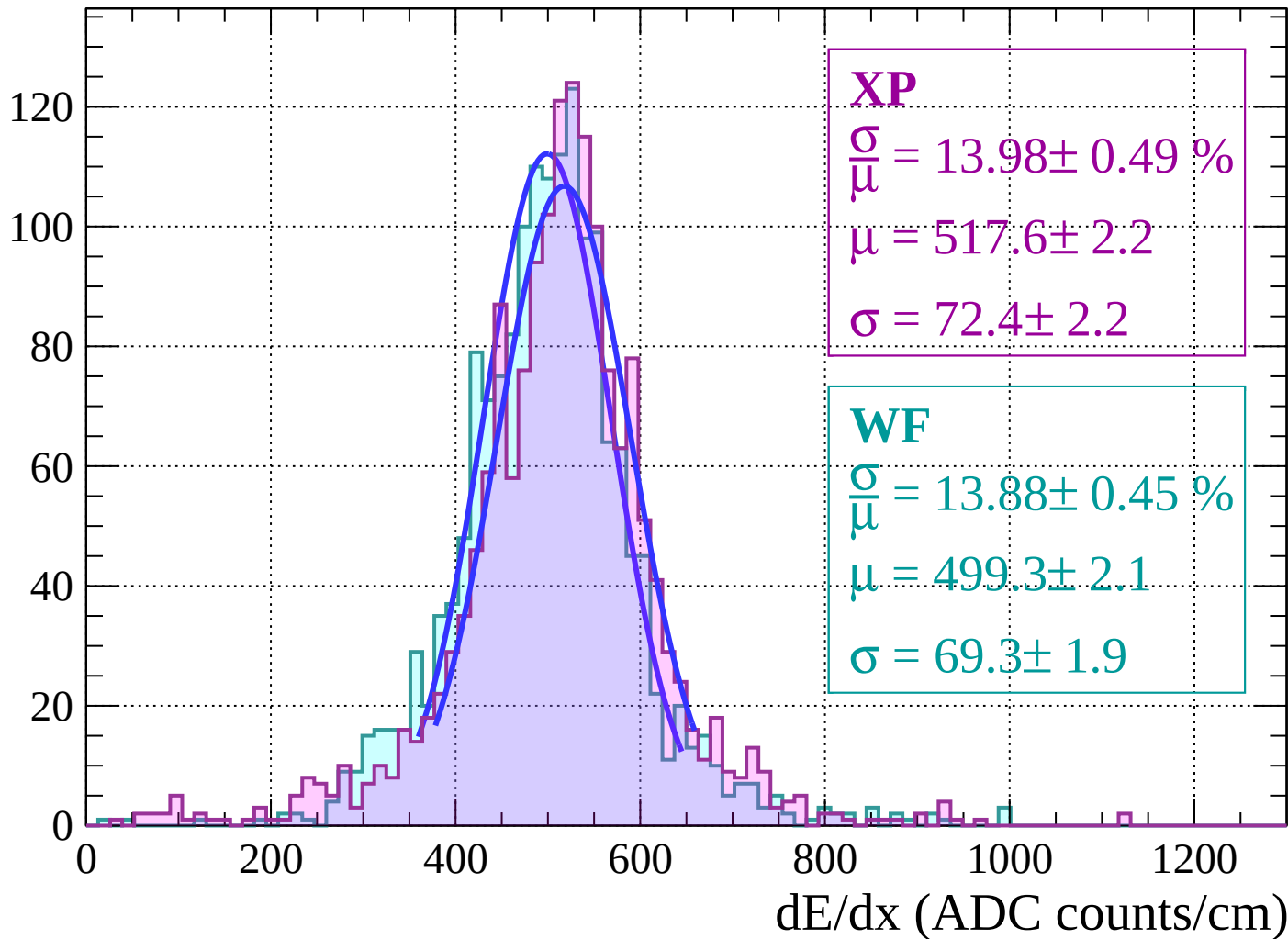
Energy loss | $-50 < \theta < -48$

Count



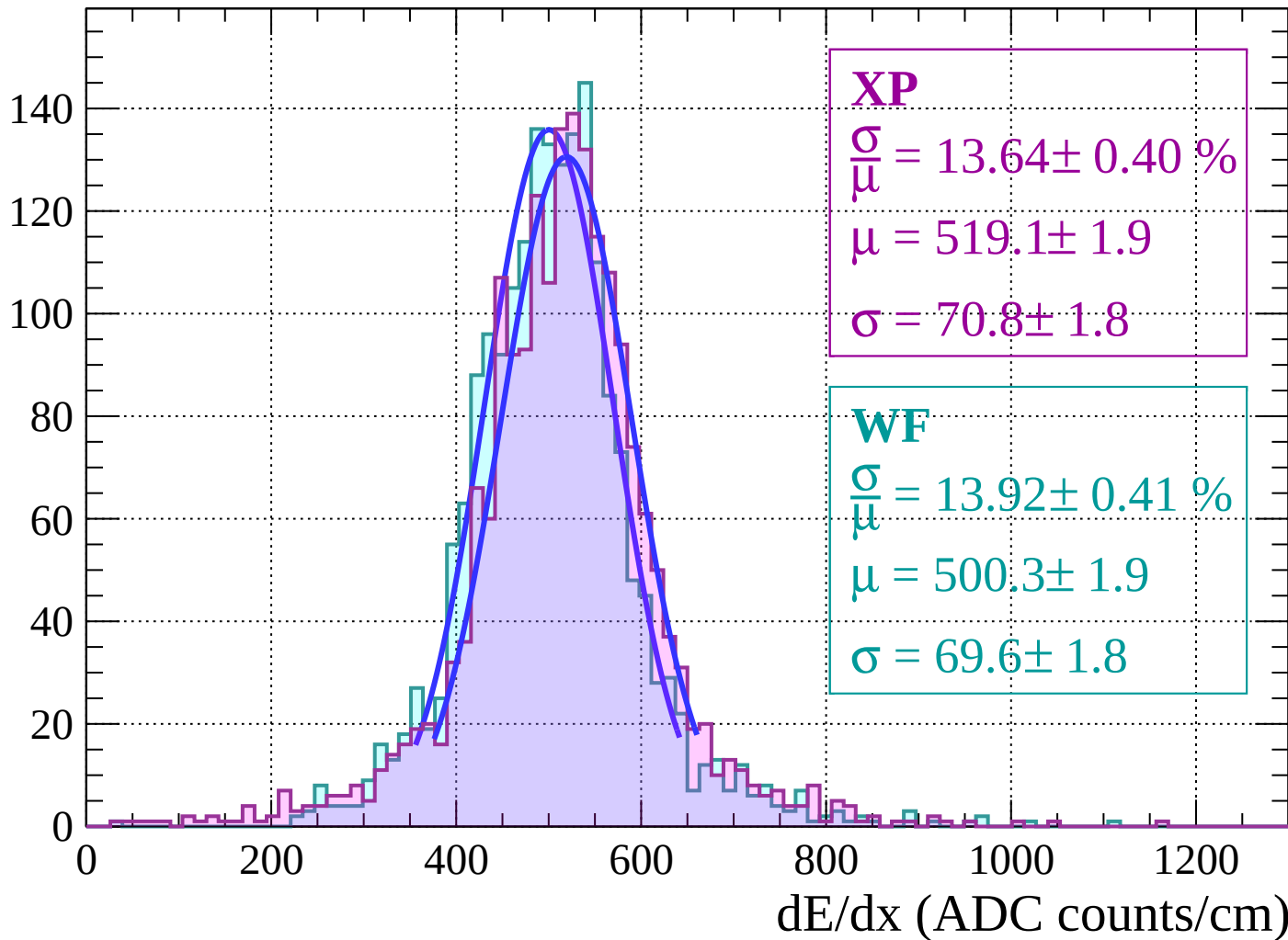
Energy loss $|-48 < \theta < -46$

Count



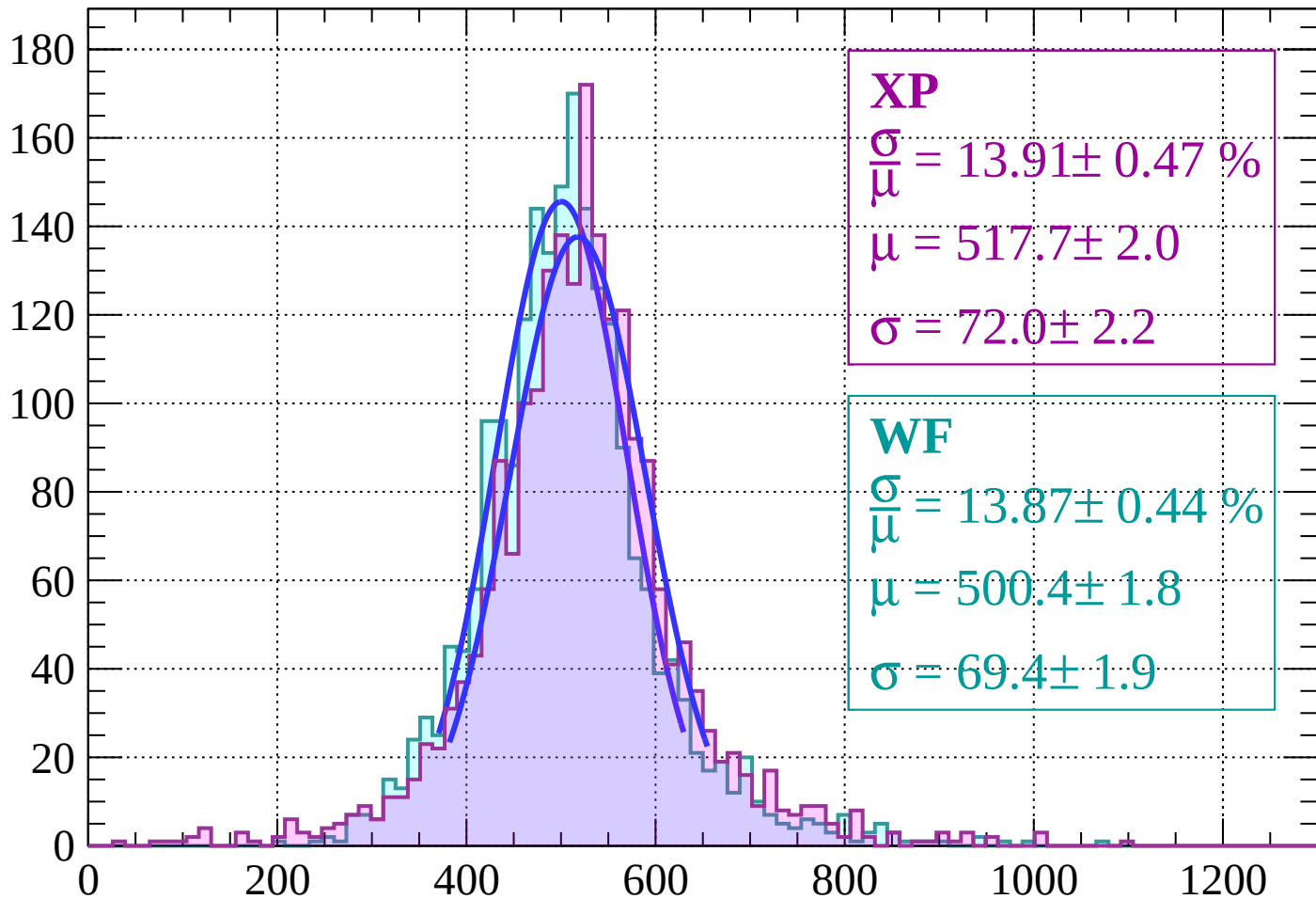
Energy loss $|-46 < \theta < -44$

Count



Energy loss $|-44 < \theta < -42$

Count



XP

$$\frac{\sigma}{\mu} = 13.91 \pm 0.47 \%$$

$$\mu = 517.7 \pm 2.0$$

$$\sigma = 72.0 \pm 2.2$$

WF

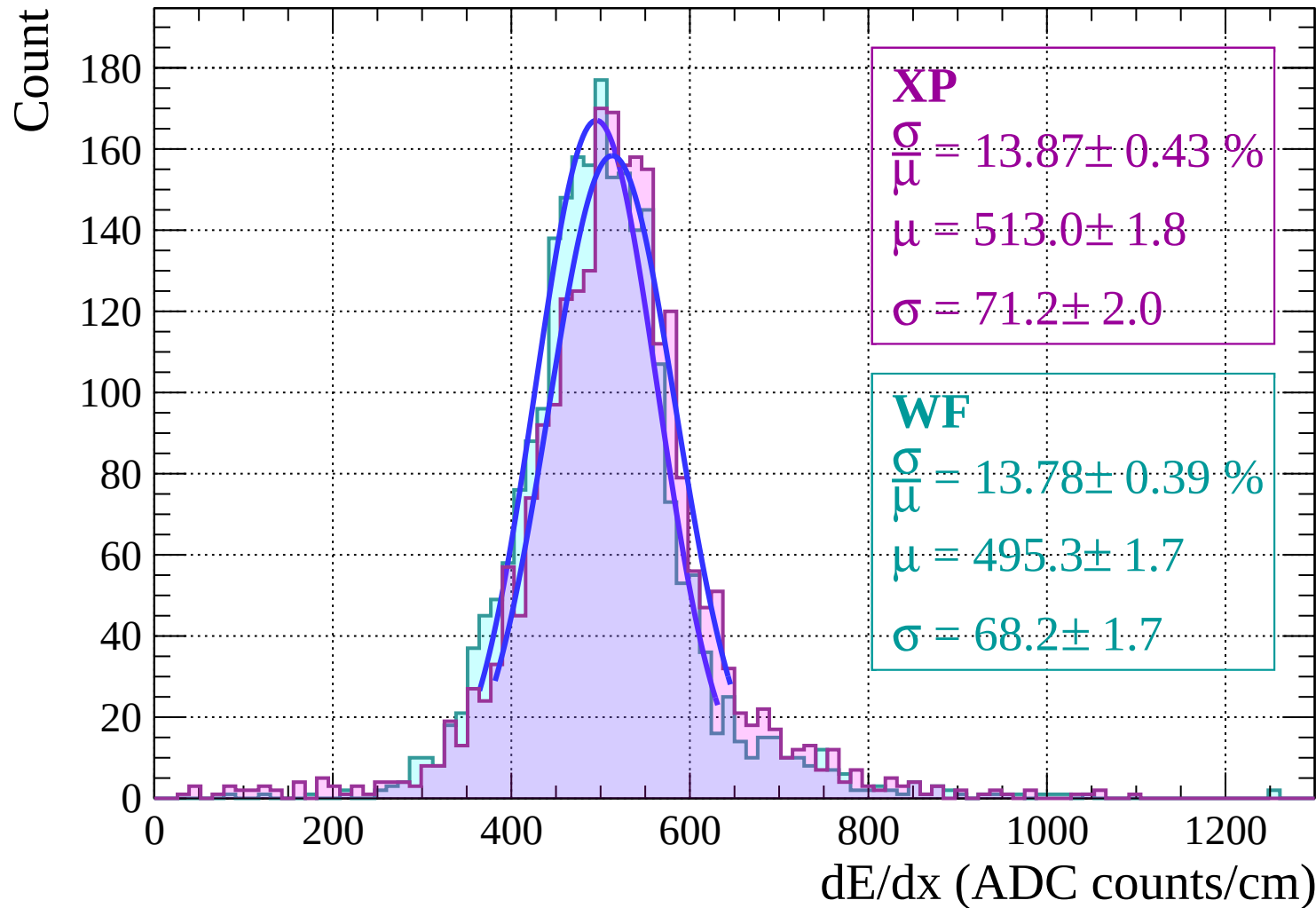
$$\frac{\sigma}{\mu} = 13.87 \pm 0.44 \%$$

$$\mu = 500.4 \pm 1.8$$

$$\sigma = 69.4 \pm 1.9$$

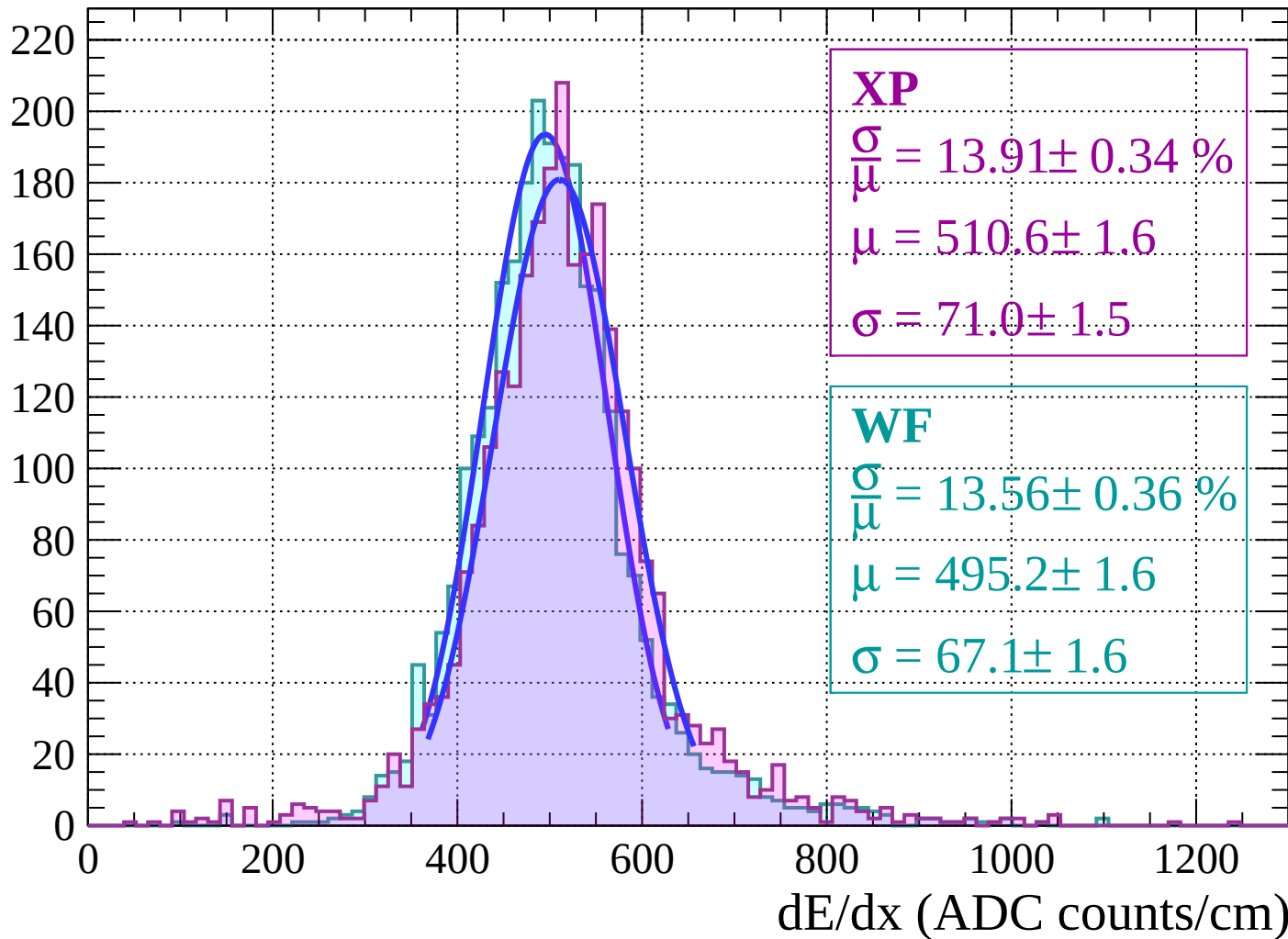
dE/dx (ADC counts/cm)

Energy loss | $-42 < \theta < -40$

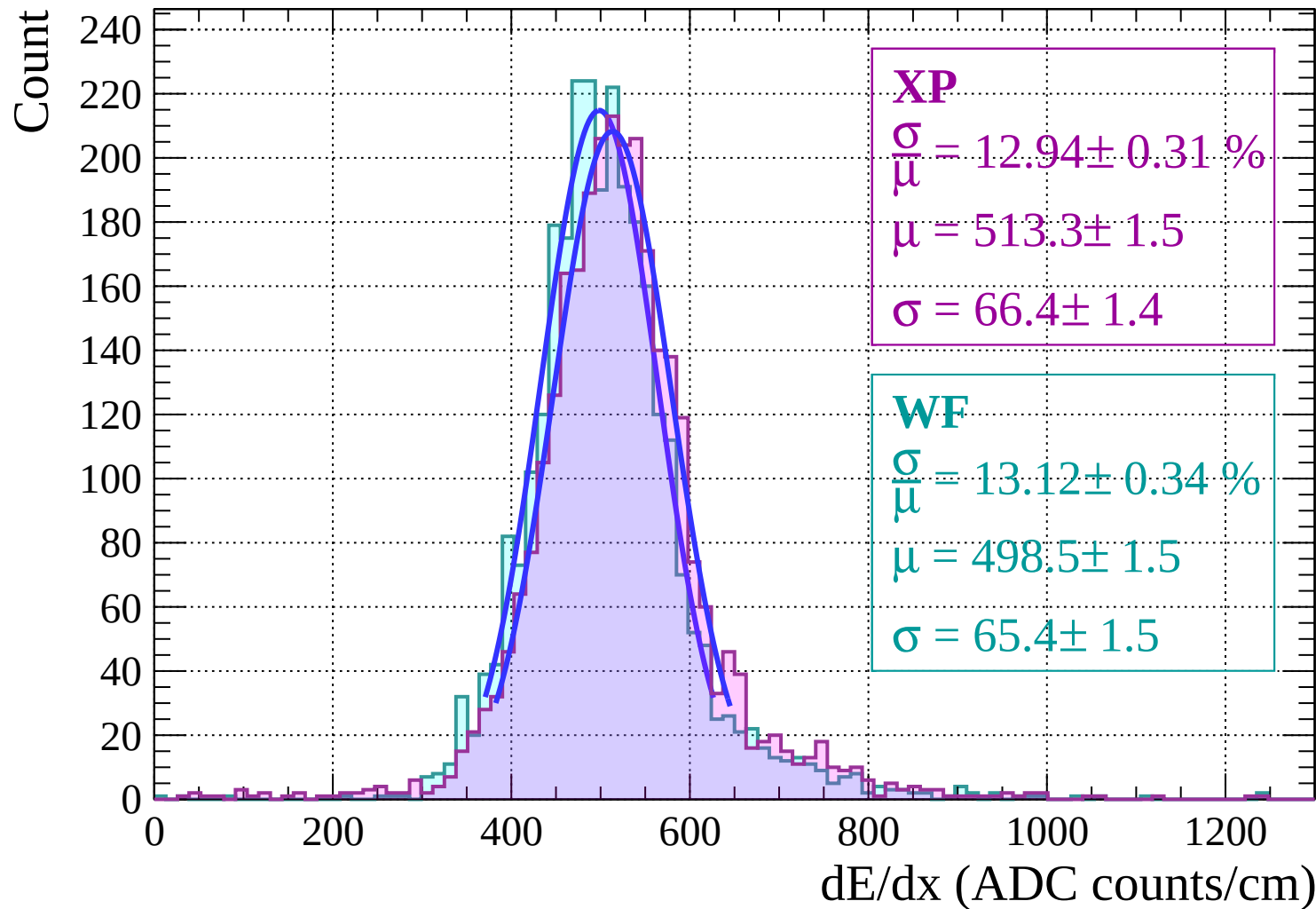


Energy loss | $-40 < \theta < -38$

Count

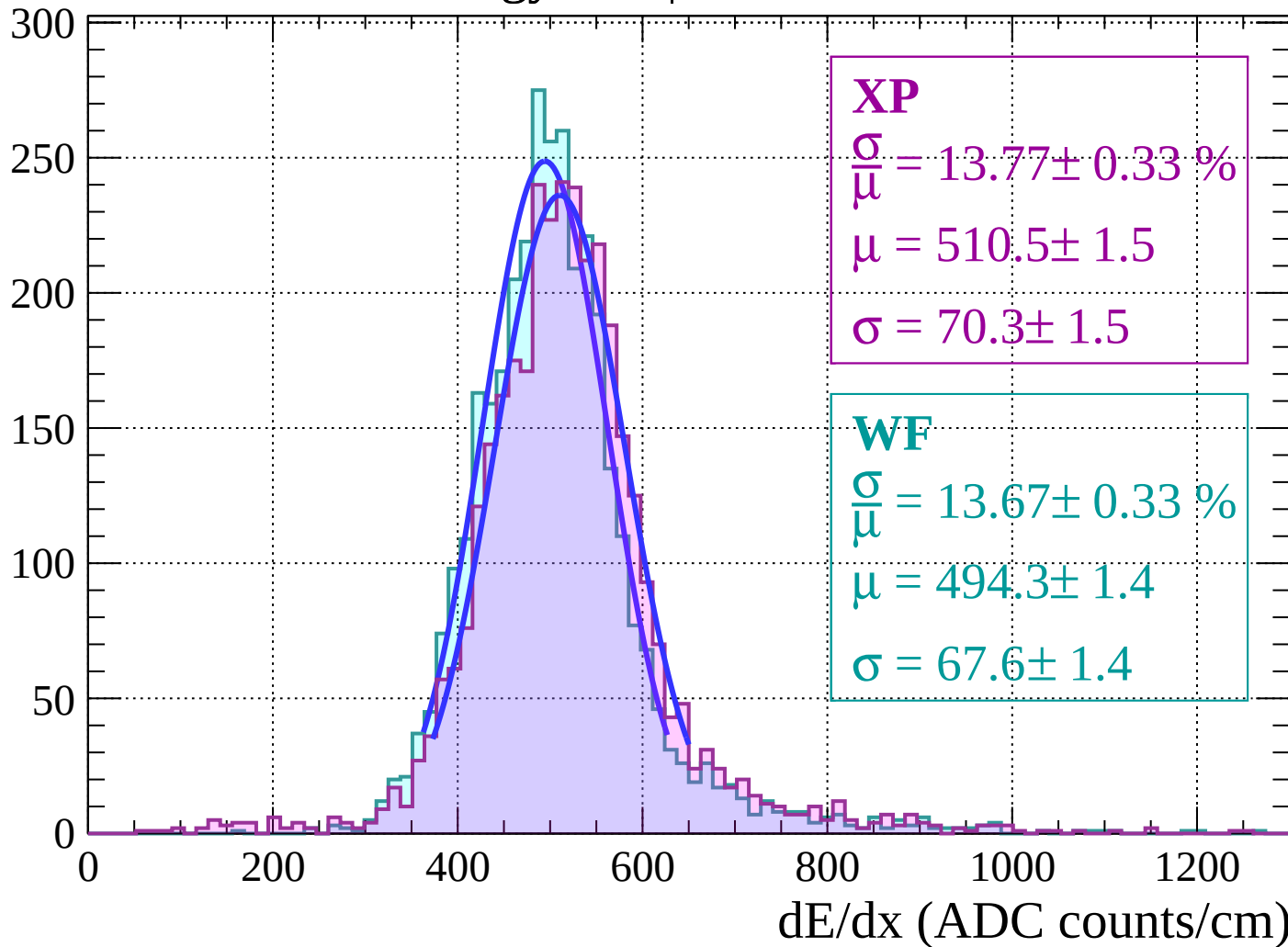


Energy loss $|-38 < \theta < -36$



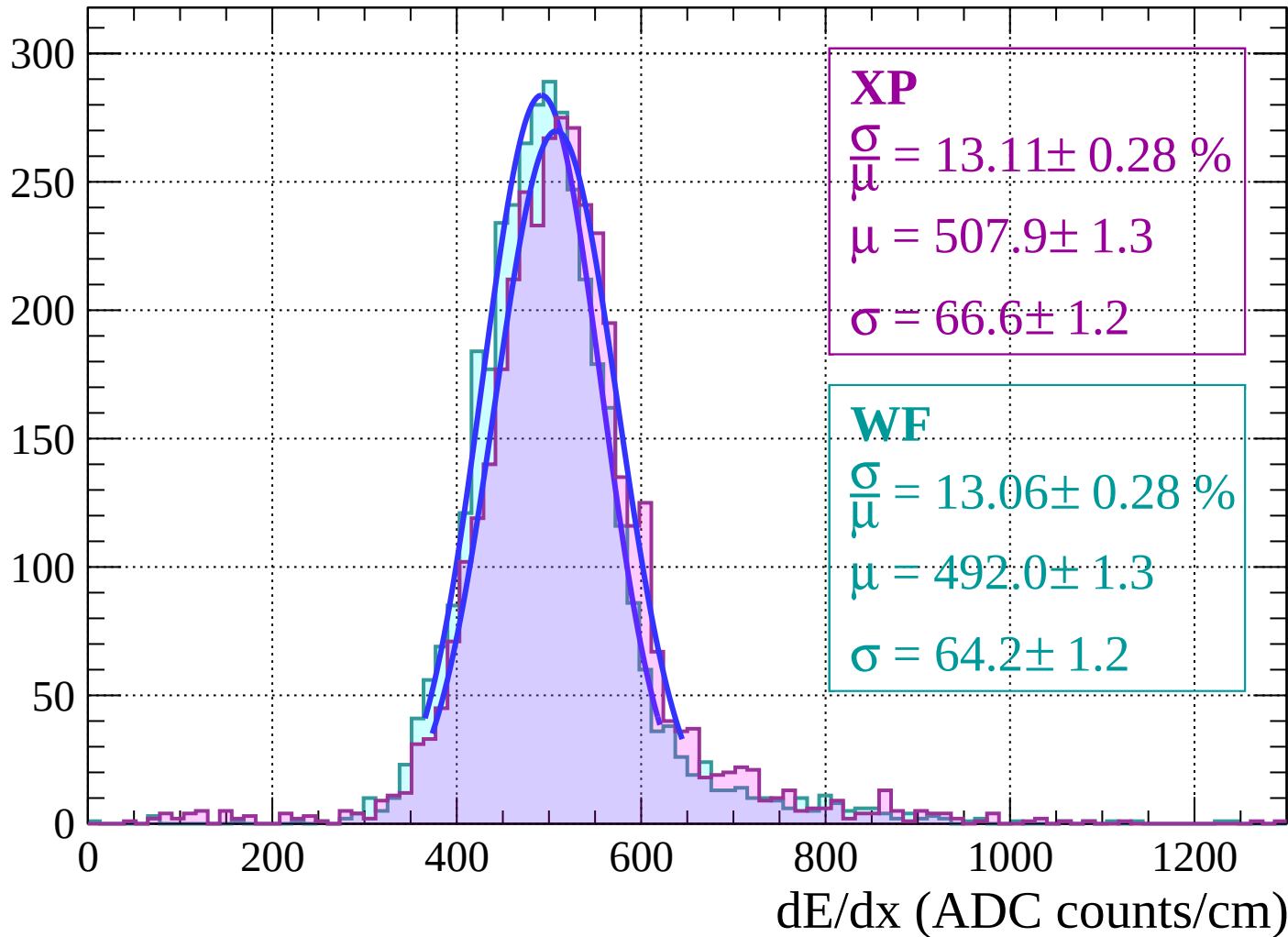
Energy loss $|-36 < \theta < -34$

Count



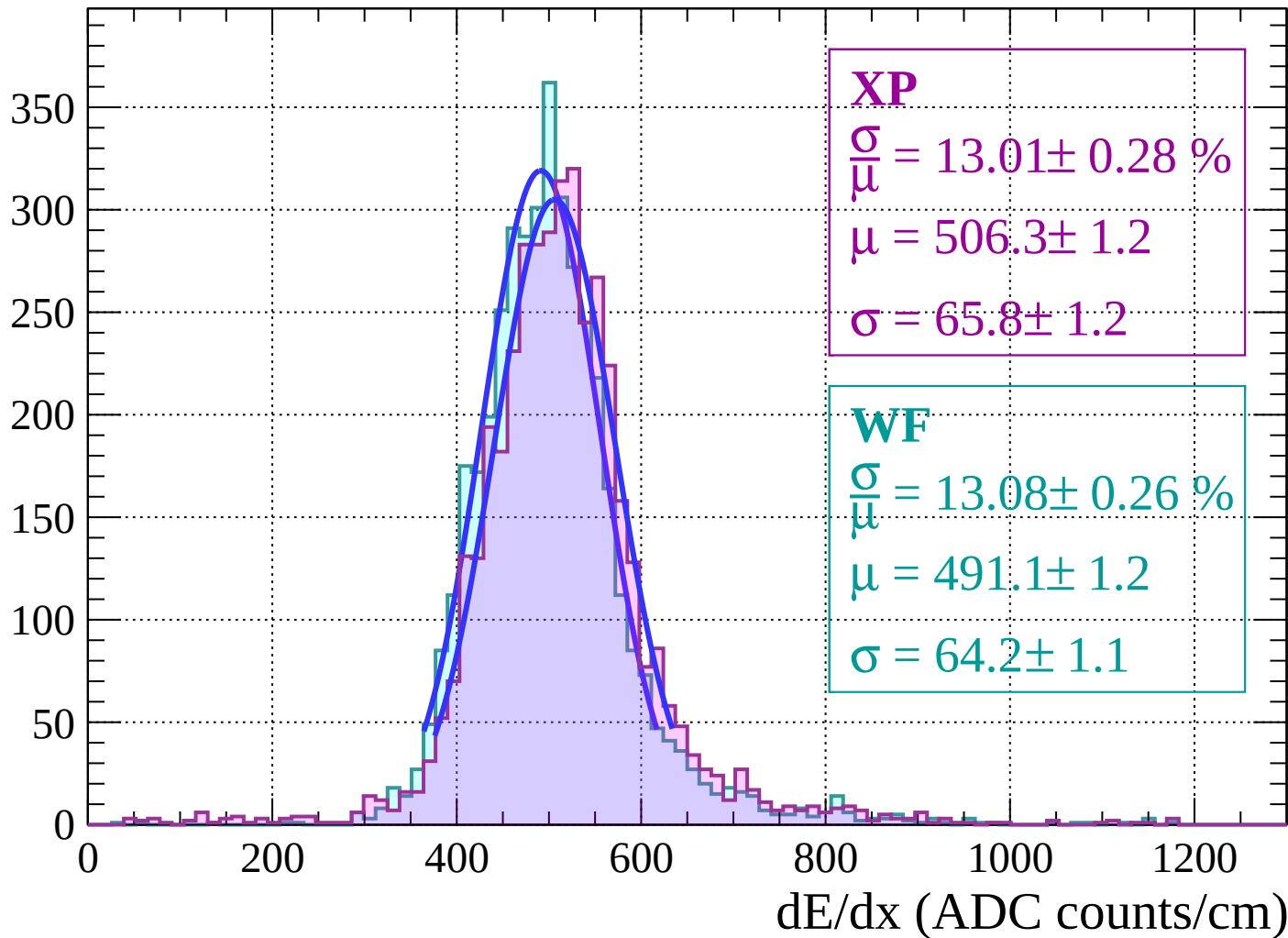
Energy loss $|-34 < \theta < -32$

Count



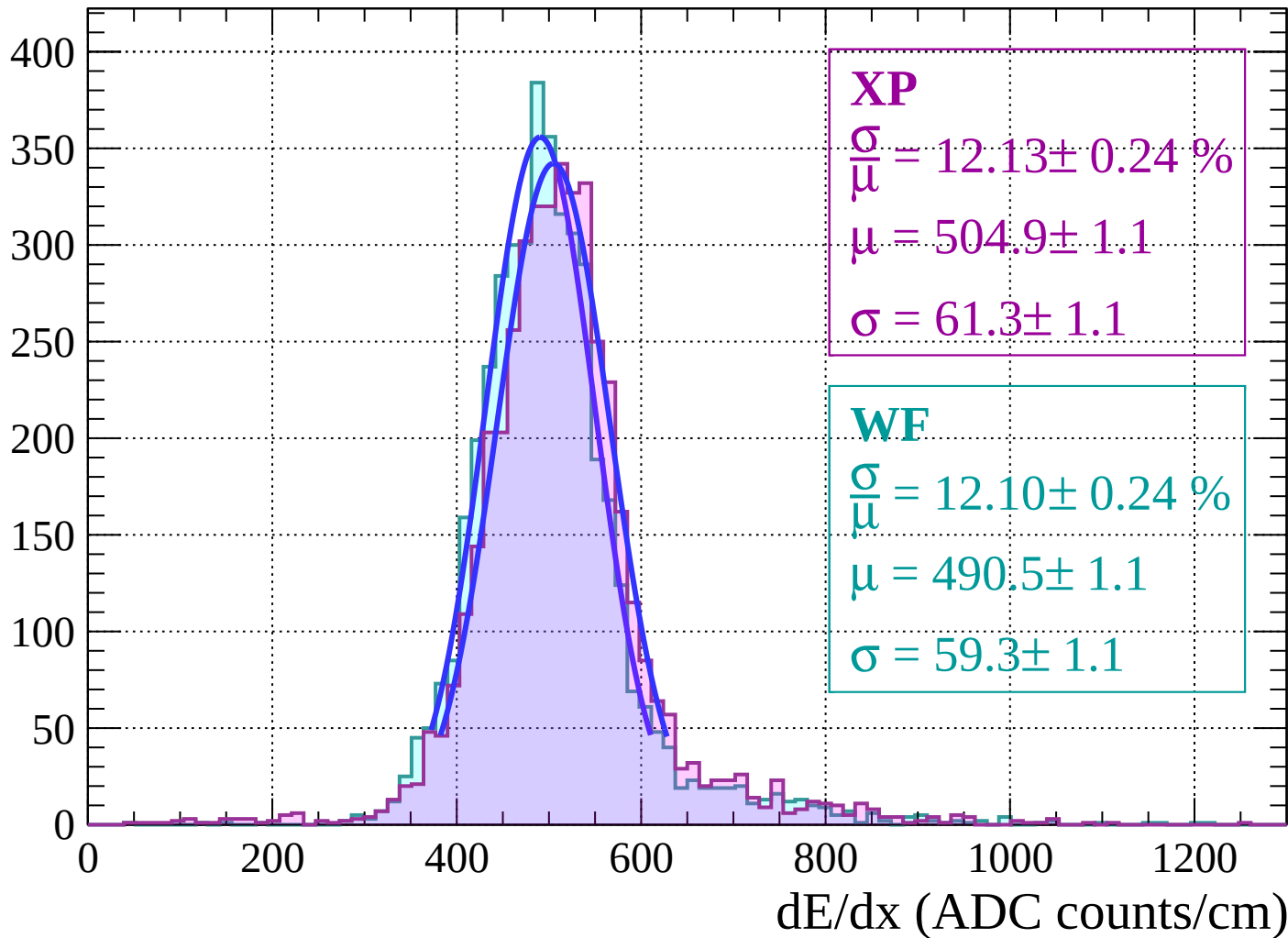
Energy loss | $-32 < \theta < -30$

Count



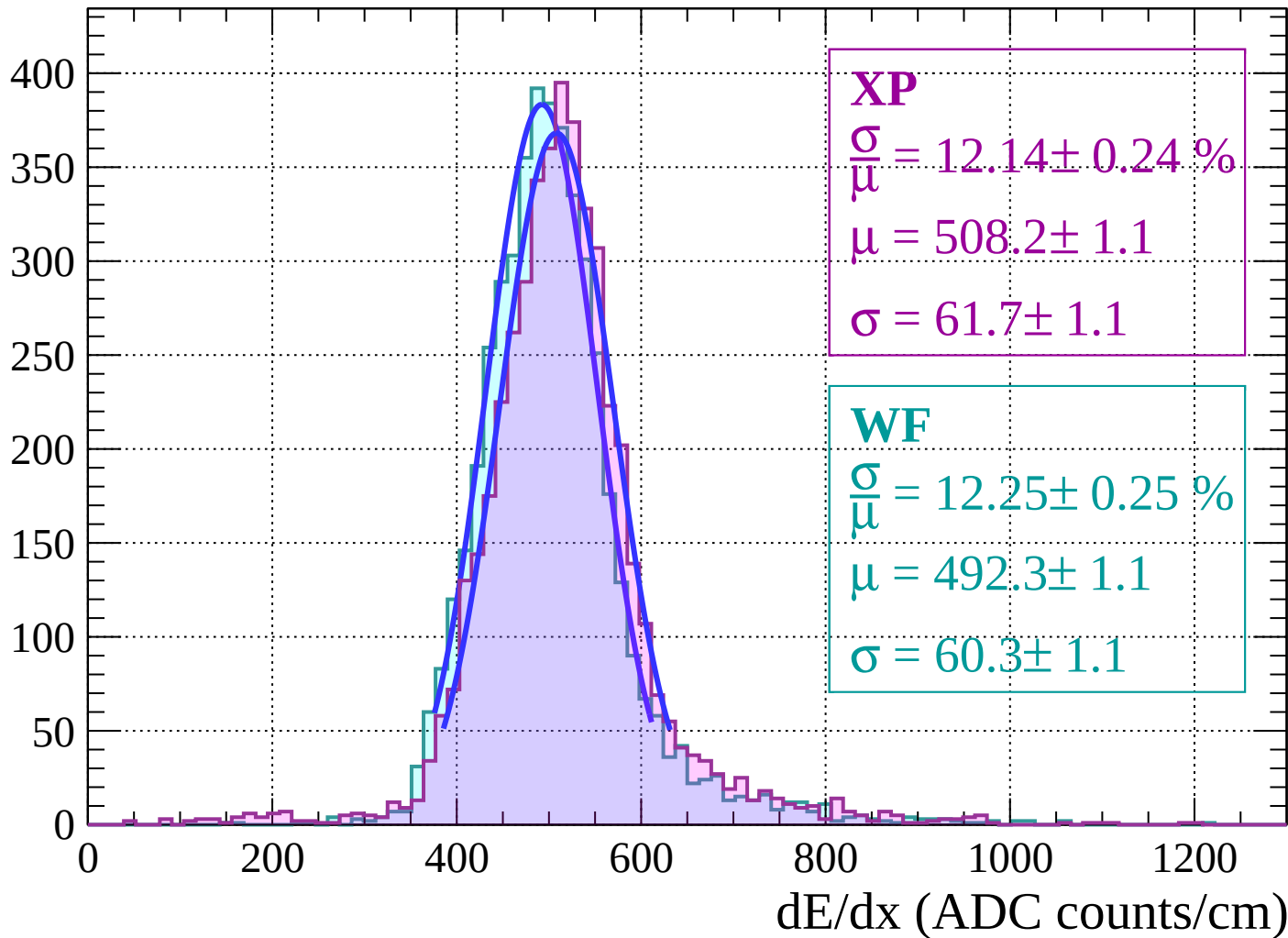
Energy loss | $-30 < \theta < -28$

Count

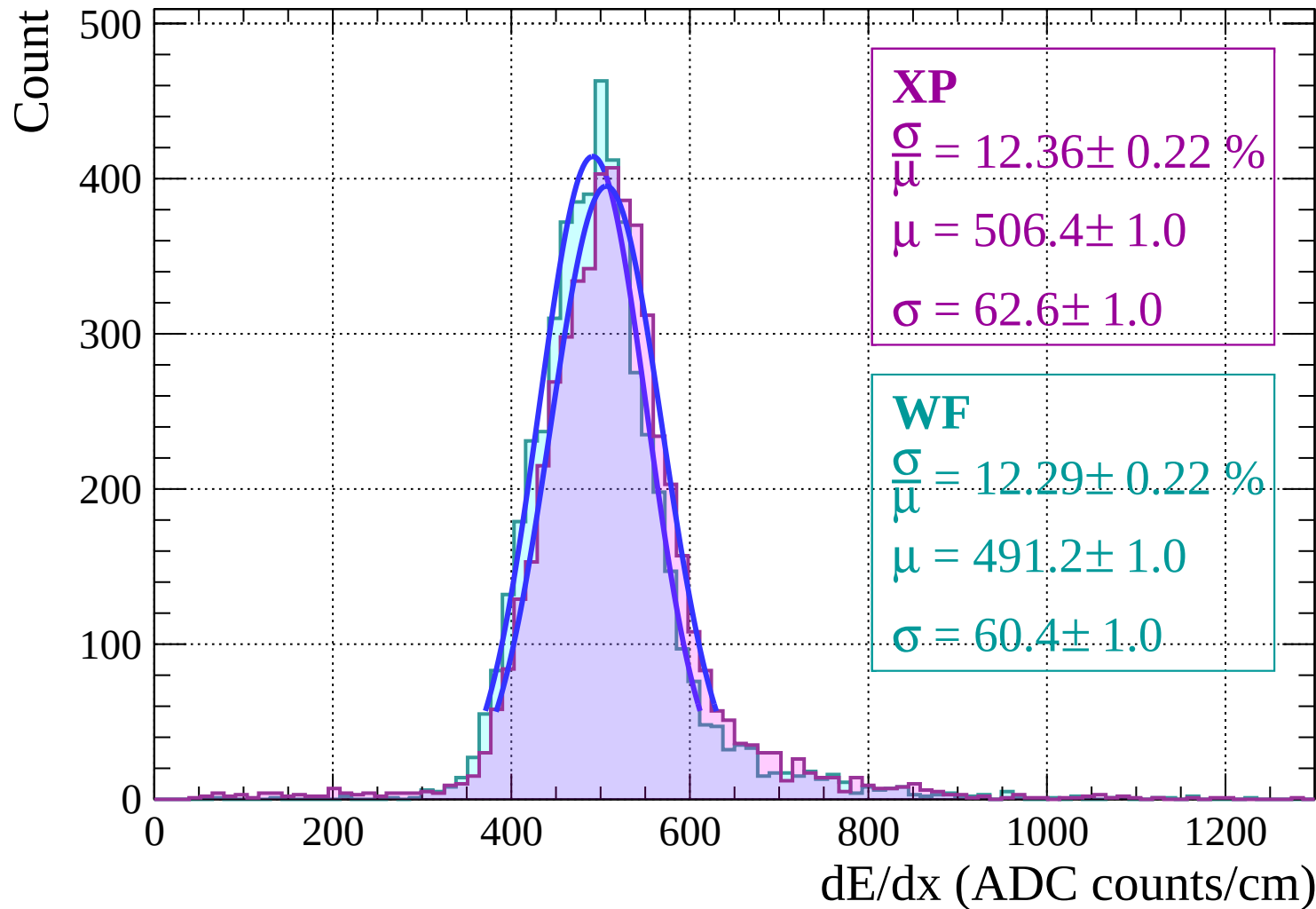


Energy loss $|-28 < \theta < -26$

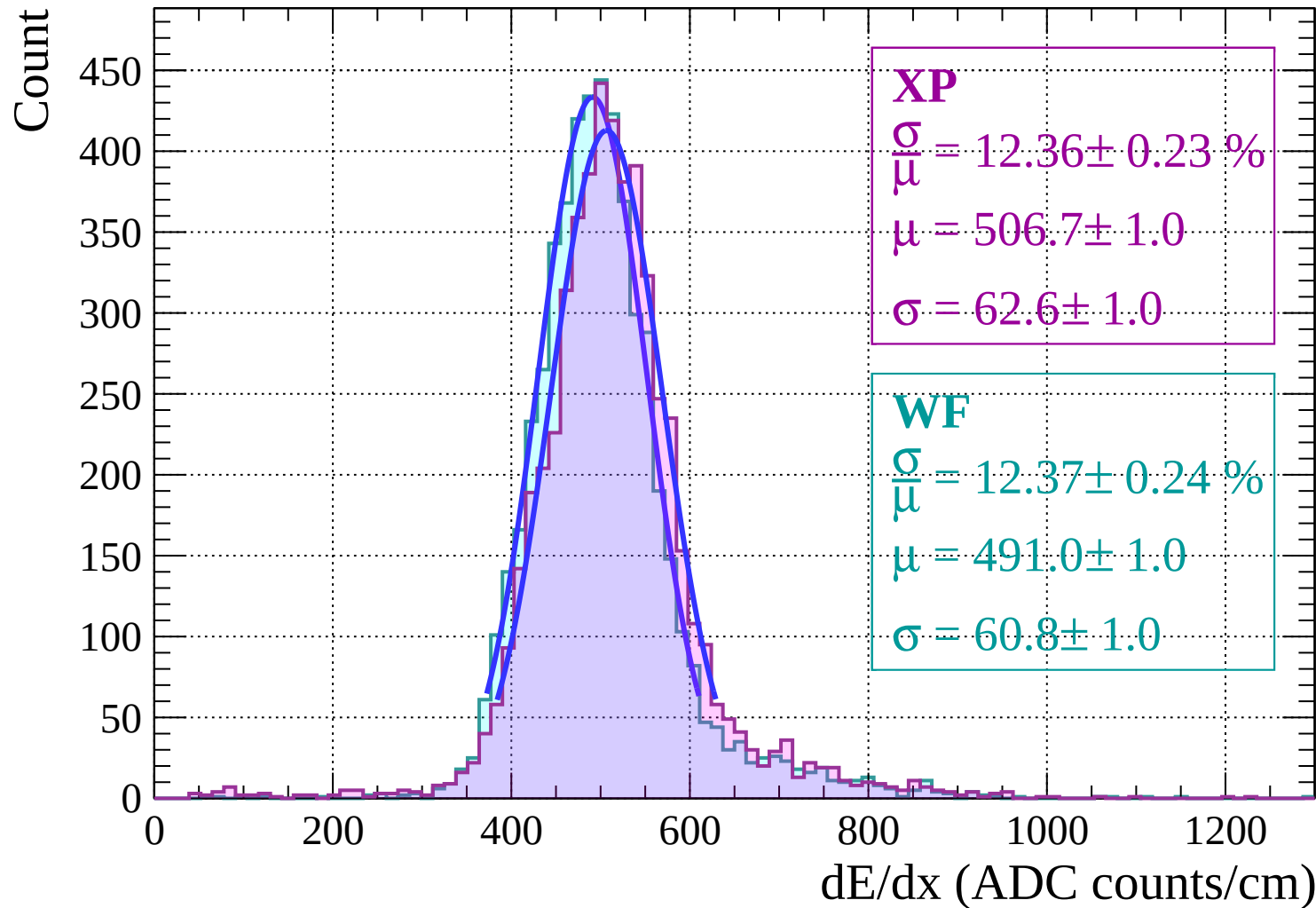
Count



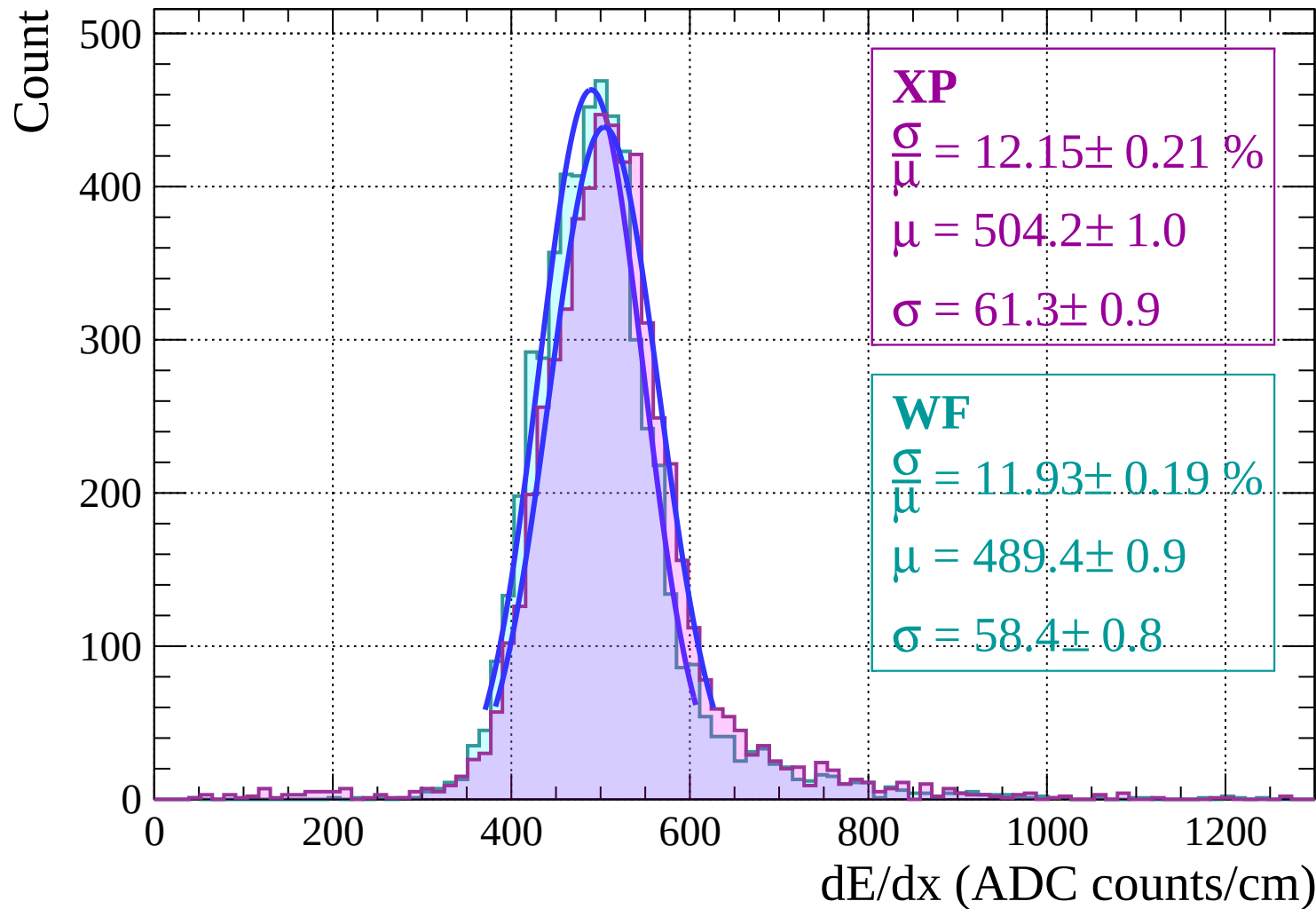
Energy loss | $-26 < \theta < -24$



Energy loss | $-24 < \theta < -22$

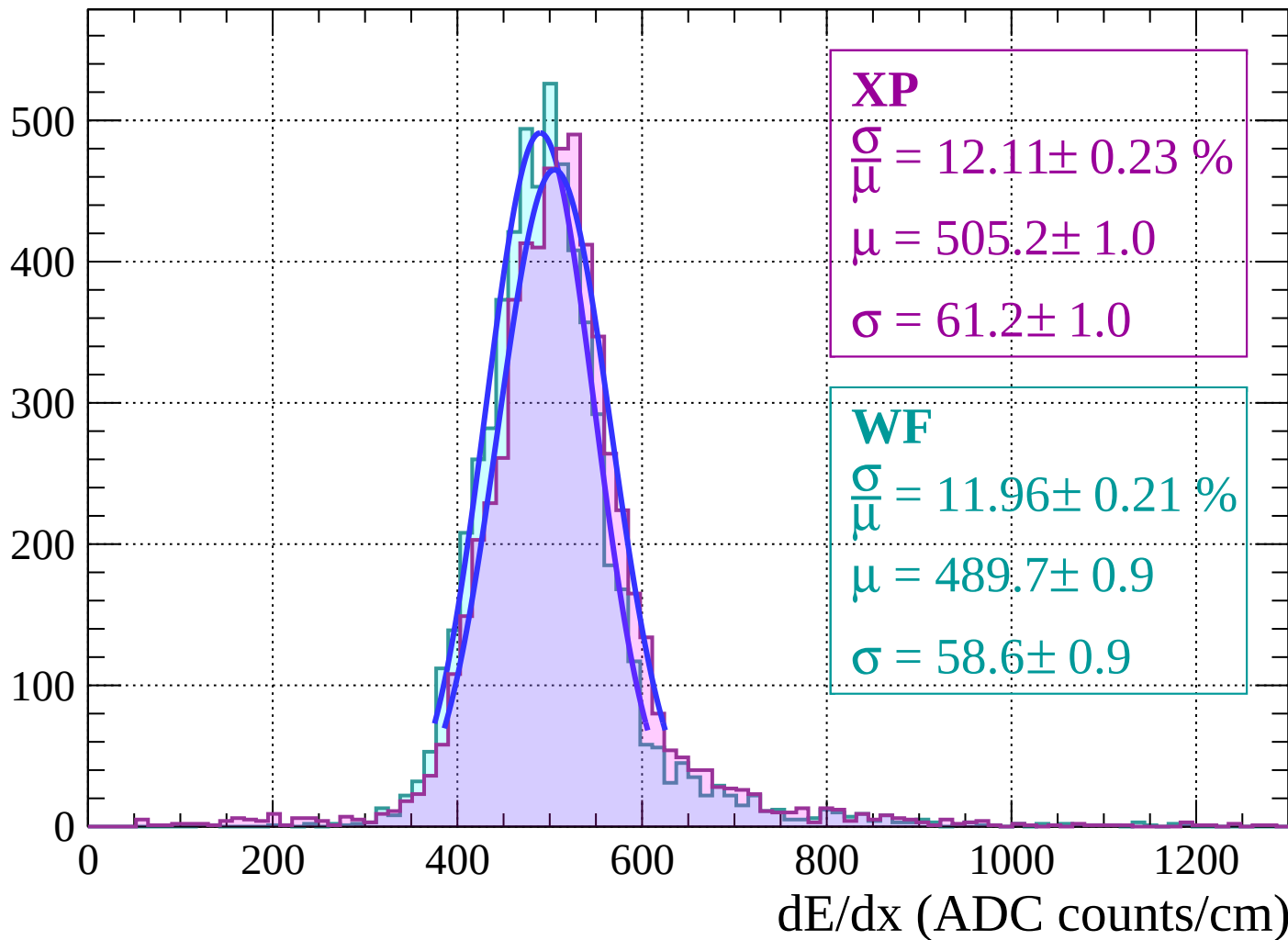


Energy loss | $-22 < \theta < -20$



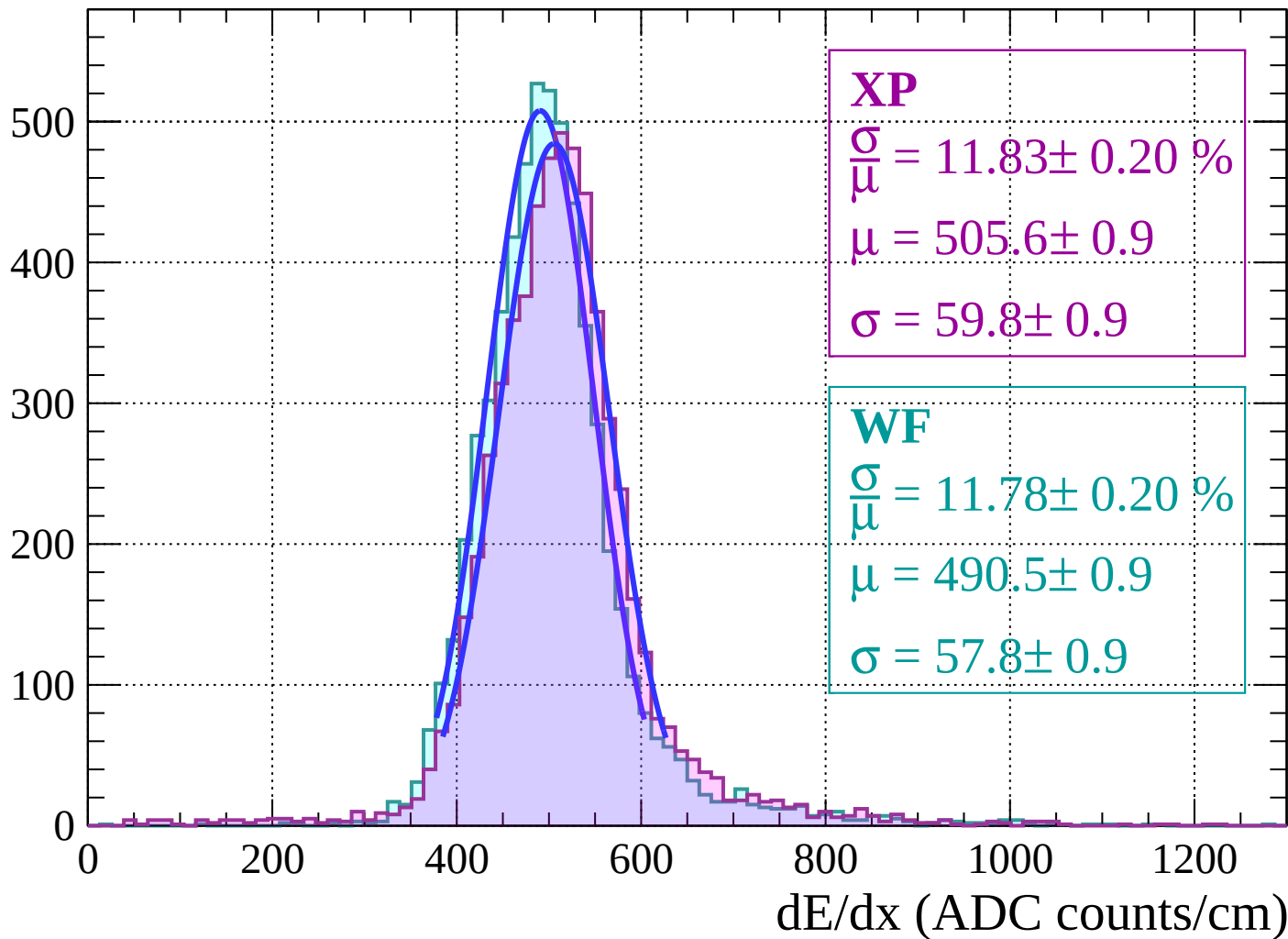
Energy loss | $-20 < \theta < -18$

Count



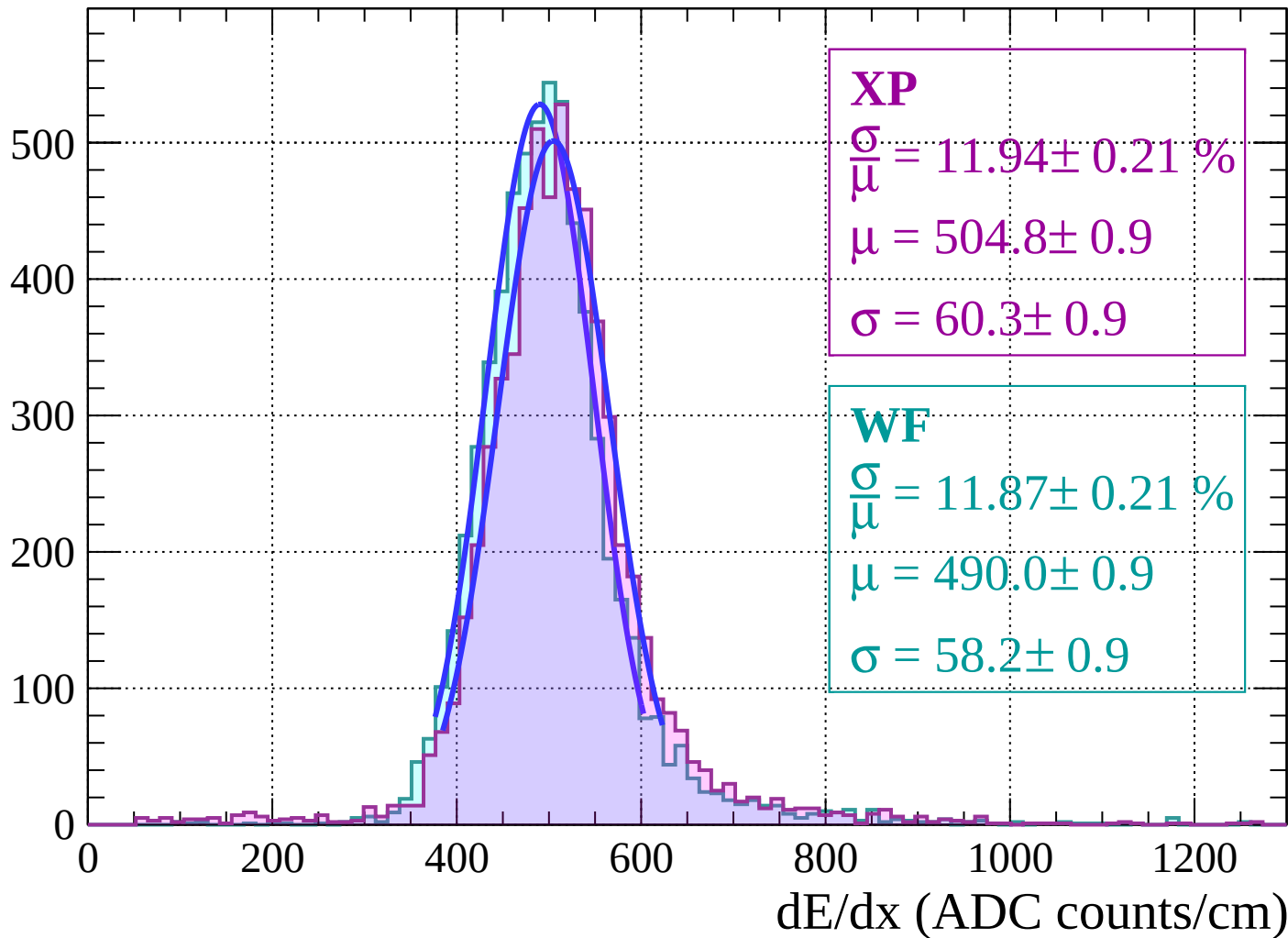
Energy loss $|-18 < \theta < -16$

Count

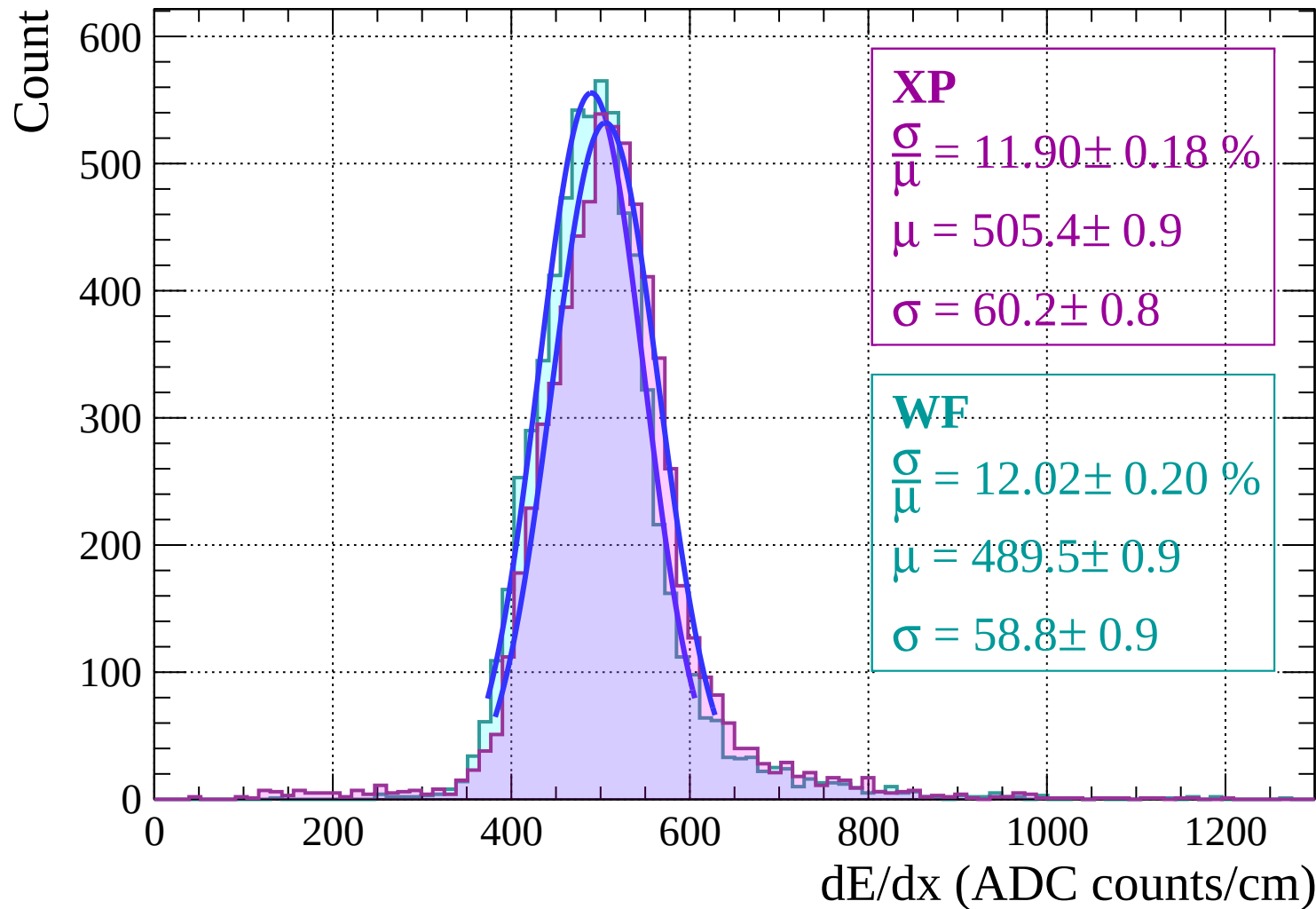


Energy loss $|-16 < \theta < -14$

Count

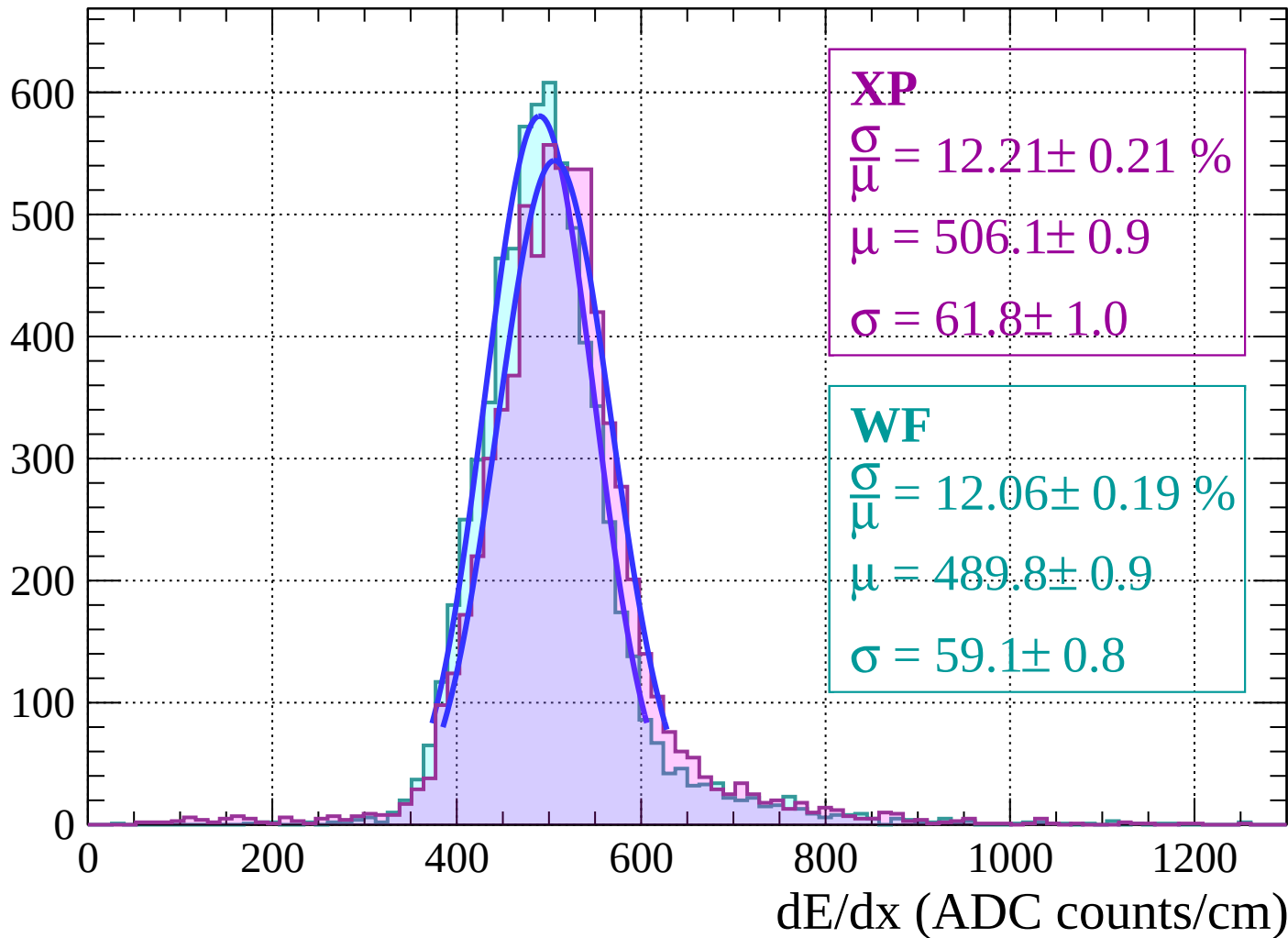


Energy loss $|-14 < \theta < -12$



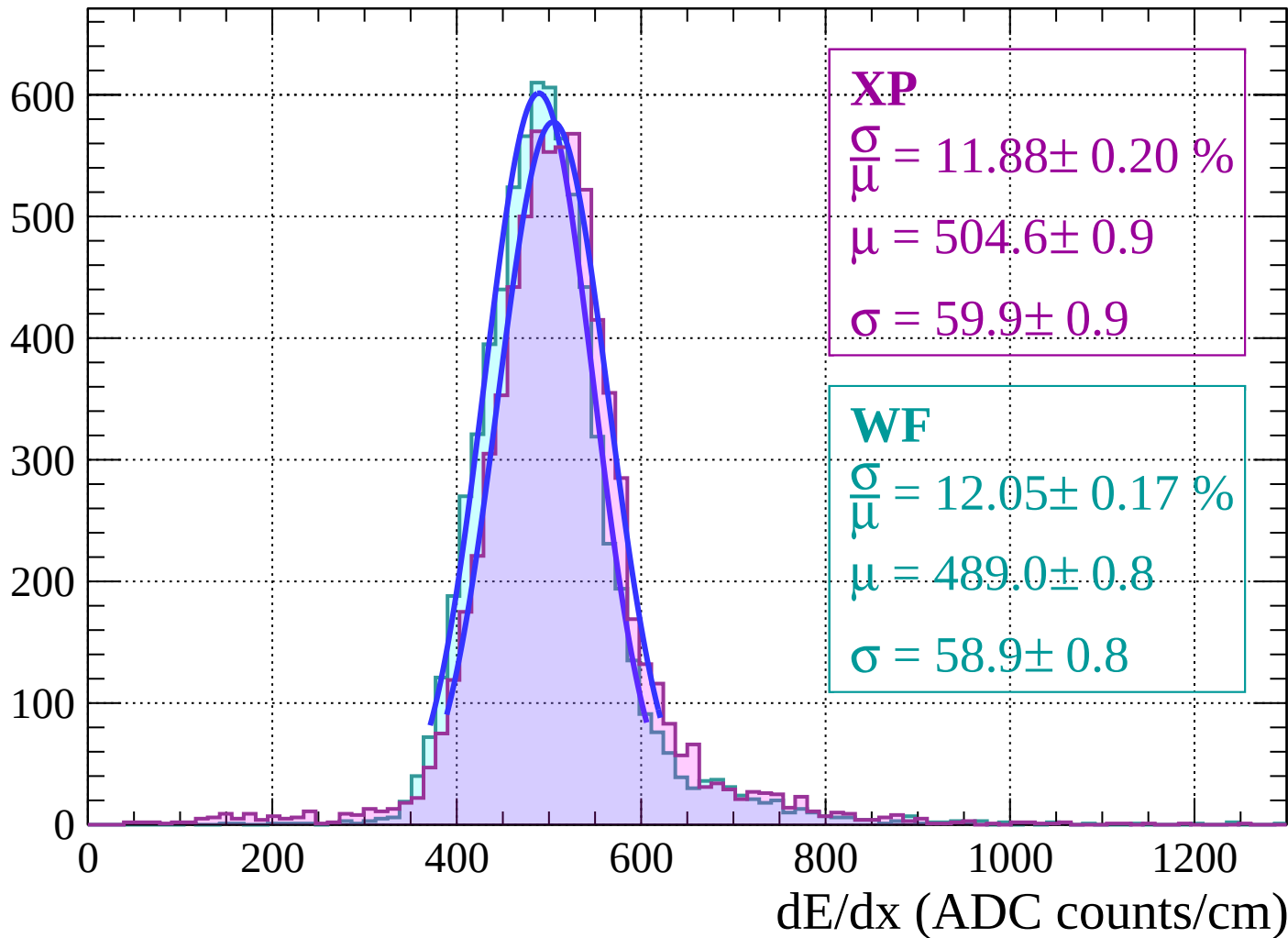
Energy loss | $-12 < \theta < -10$

Count

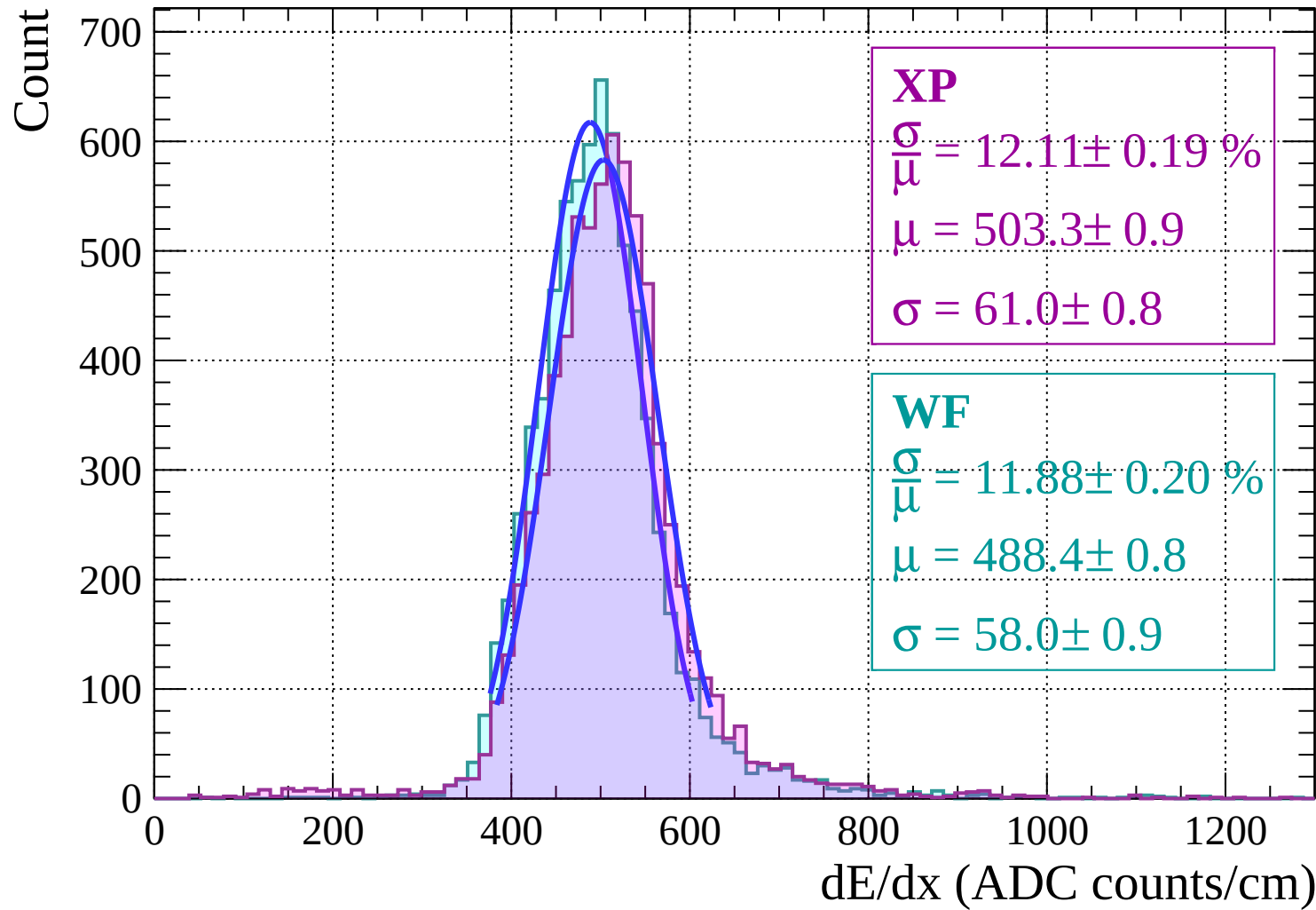


Energy loss | $-10 < \theta < -8$

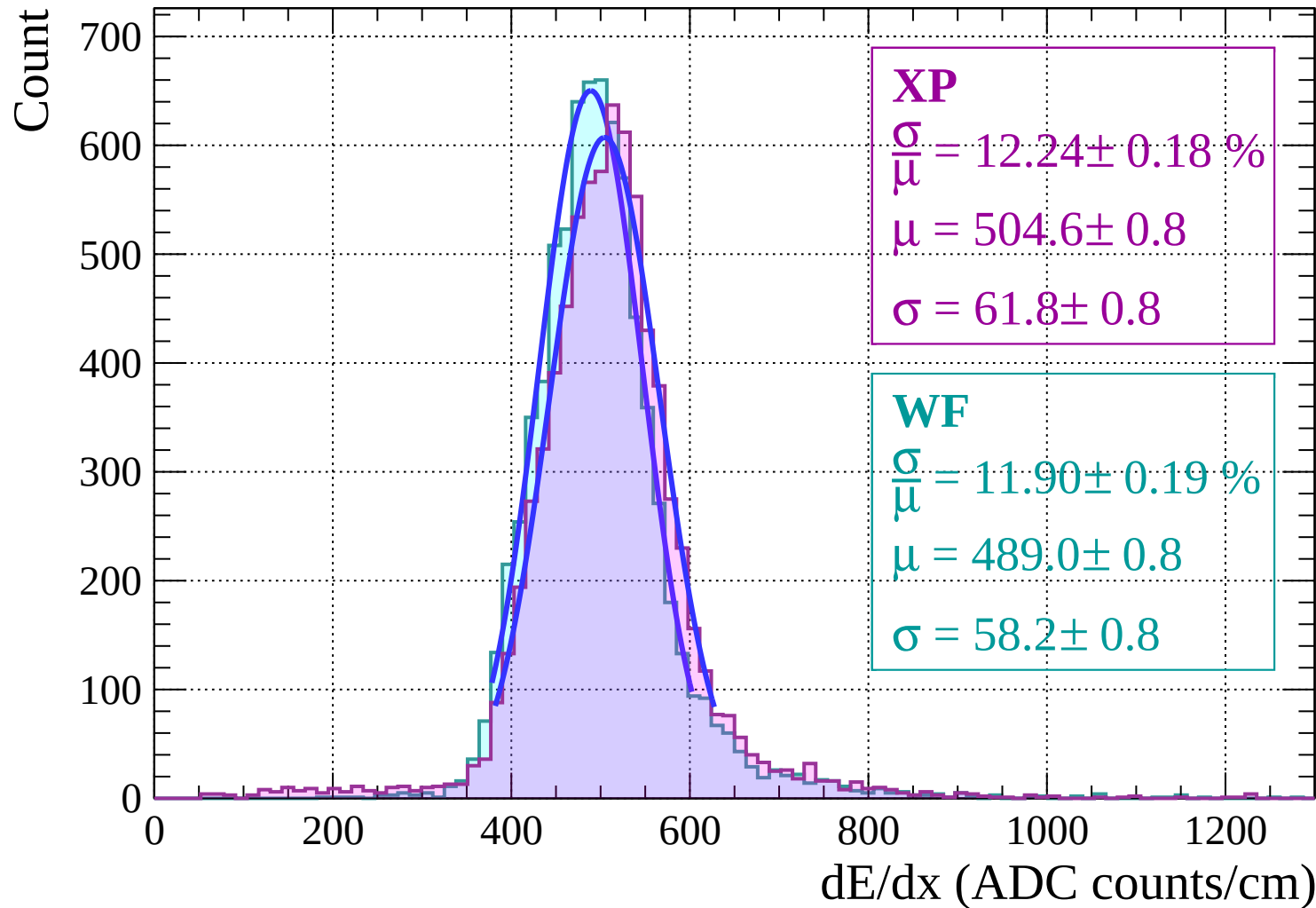
Count



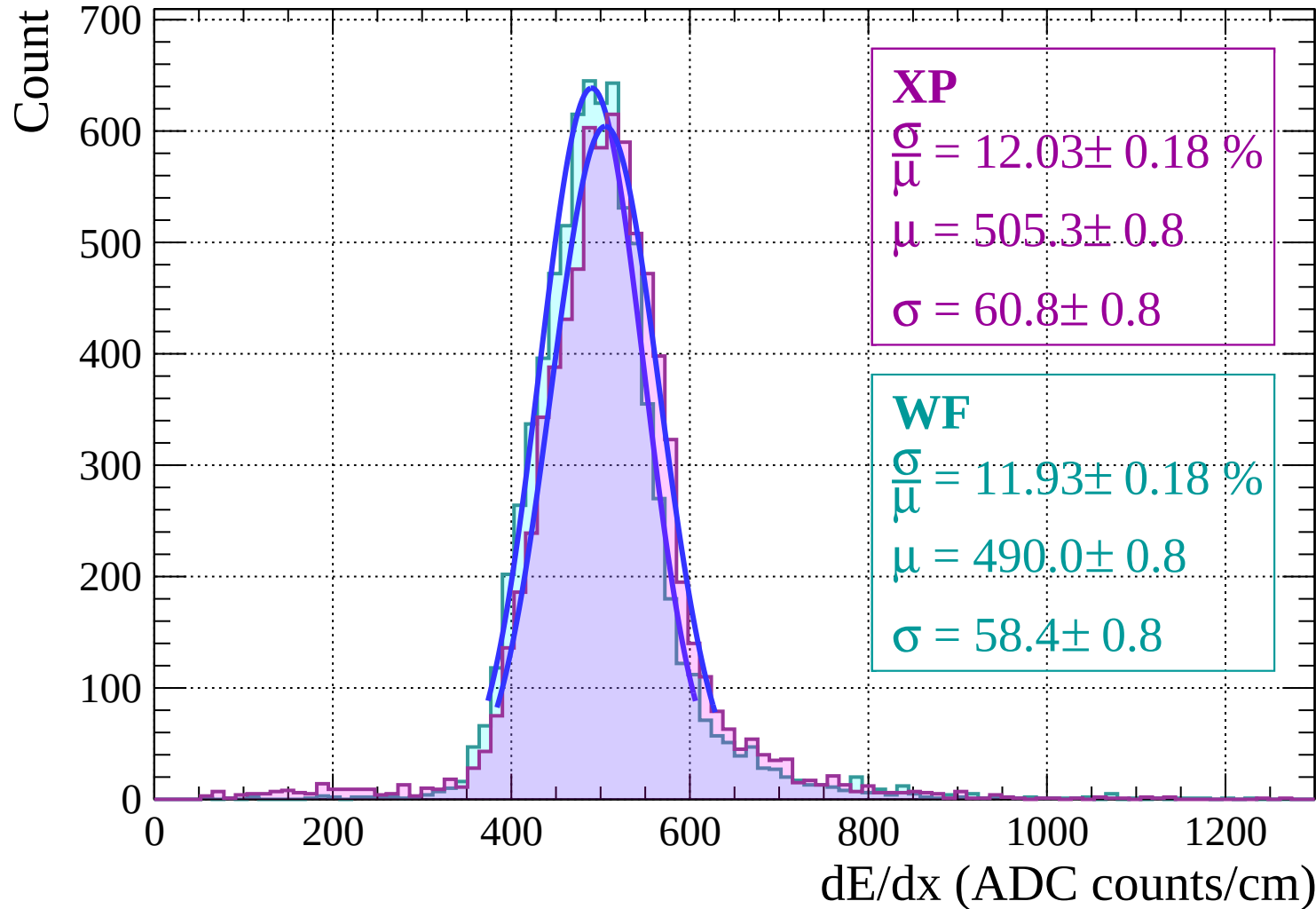
Energy loss | $-8 < \theta < -6$



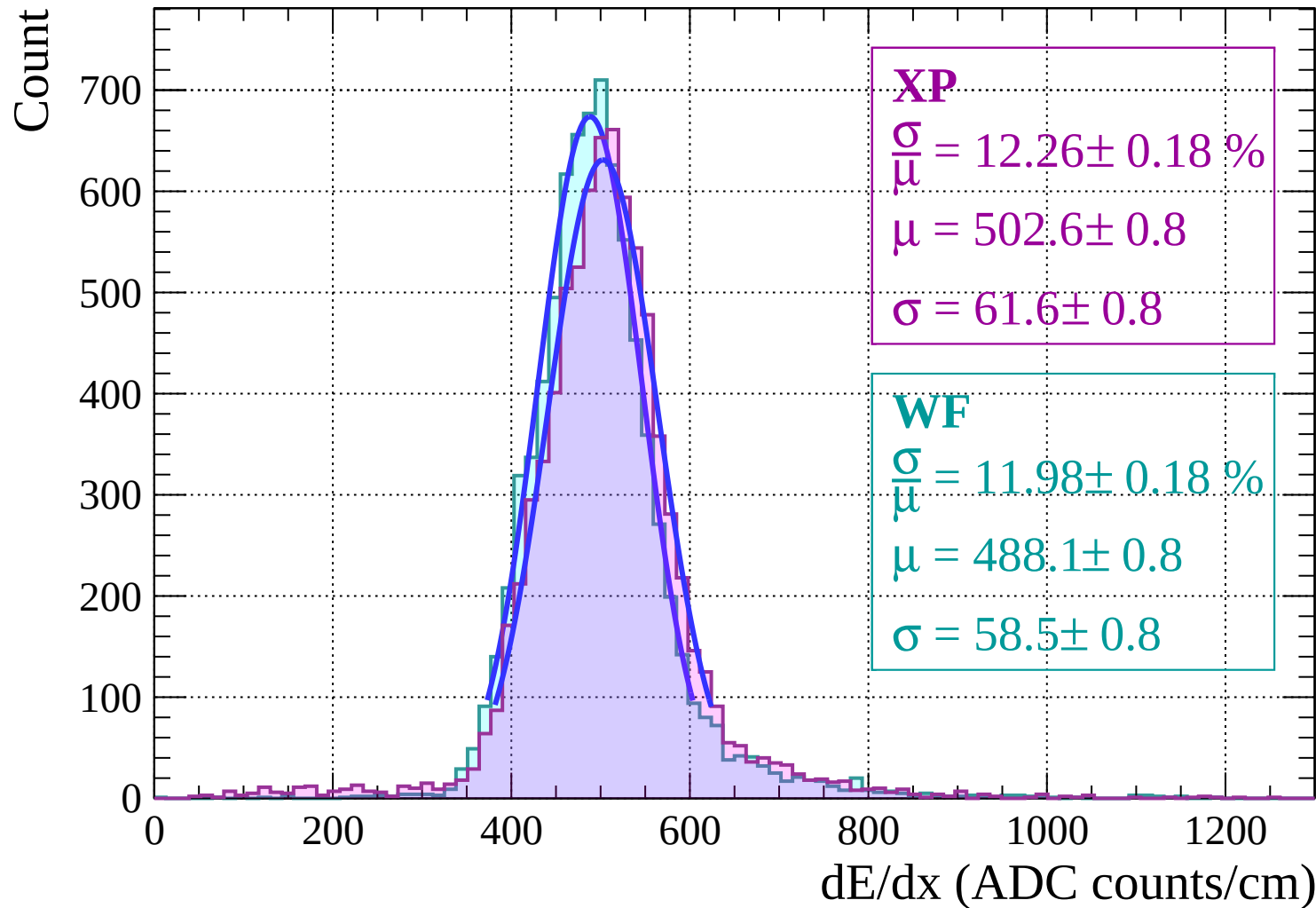
Energy loss | $-6 < \theta < -4$



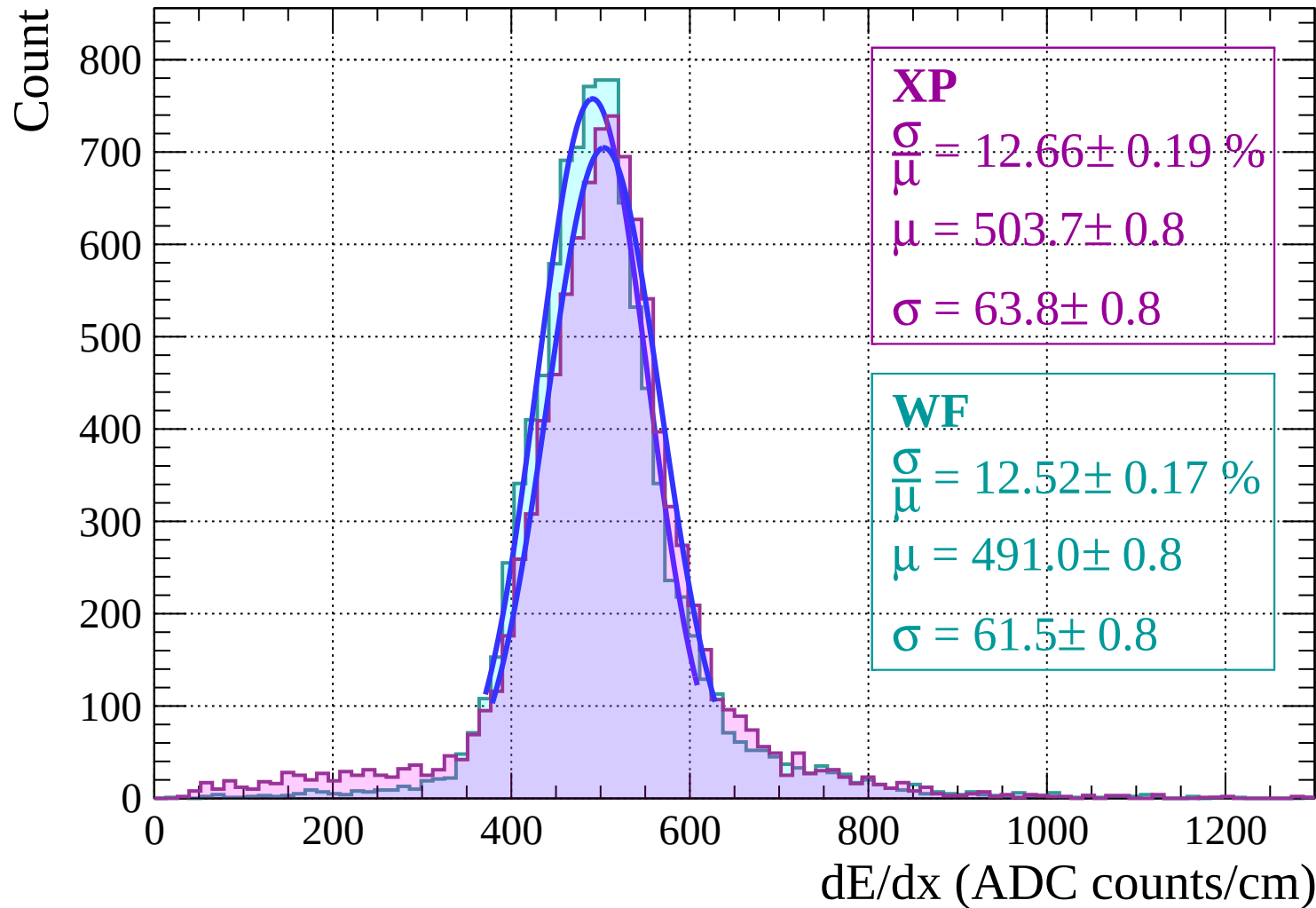
Energy loss $-4 < \theta < -2$



Energy loss | $-2 < \theta < 0$

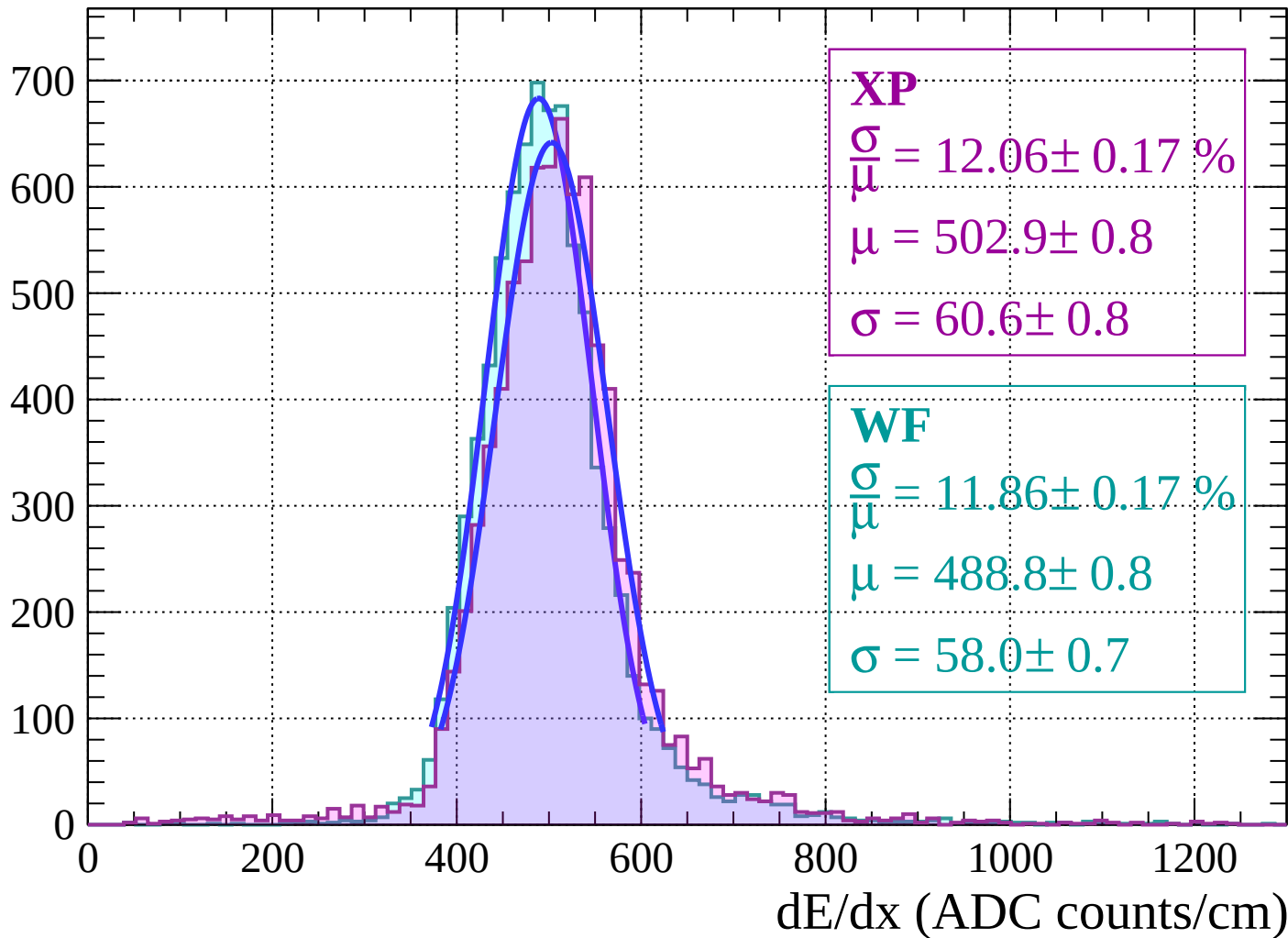


Energy loss | $0 < \theta < 2$



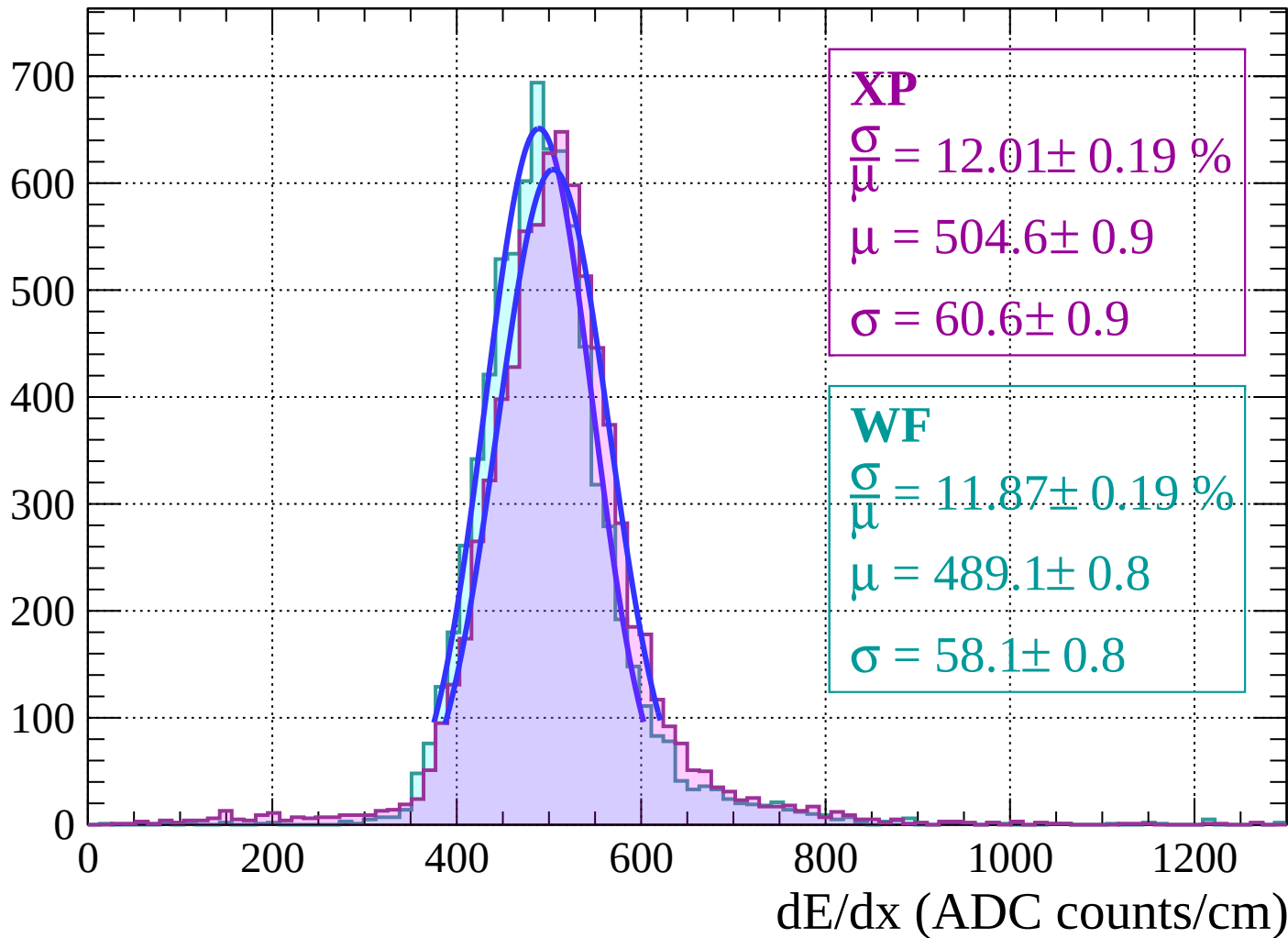
Energy loss | $2 < \theta < 4$

Count

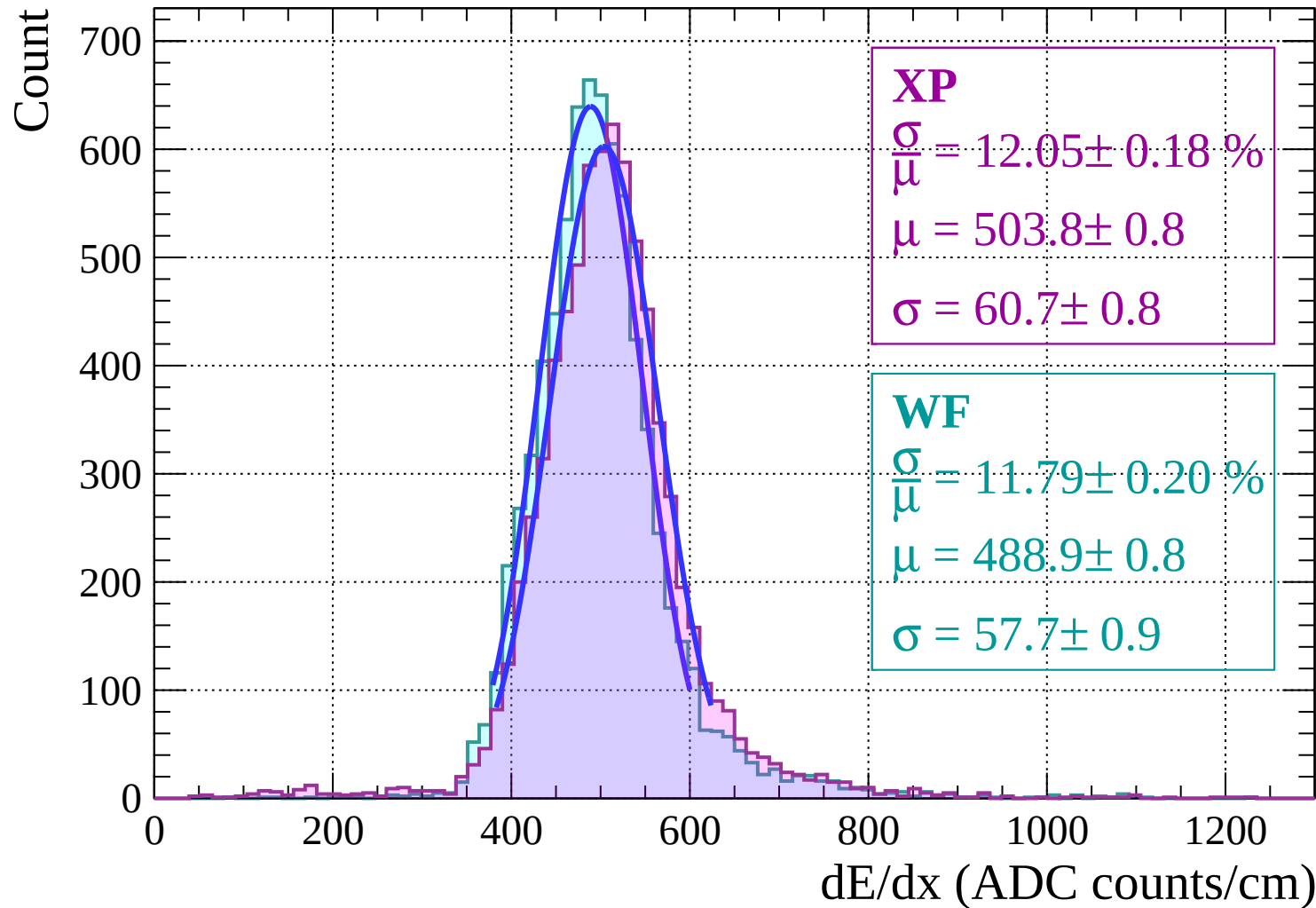


Energy loss | $4 < \theta < 6$

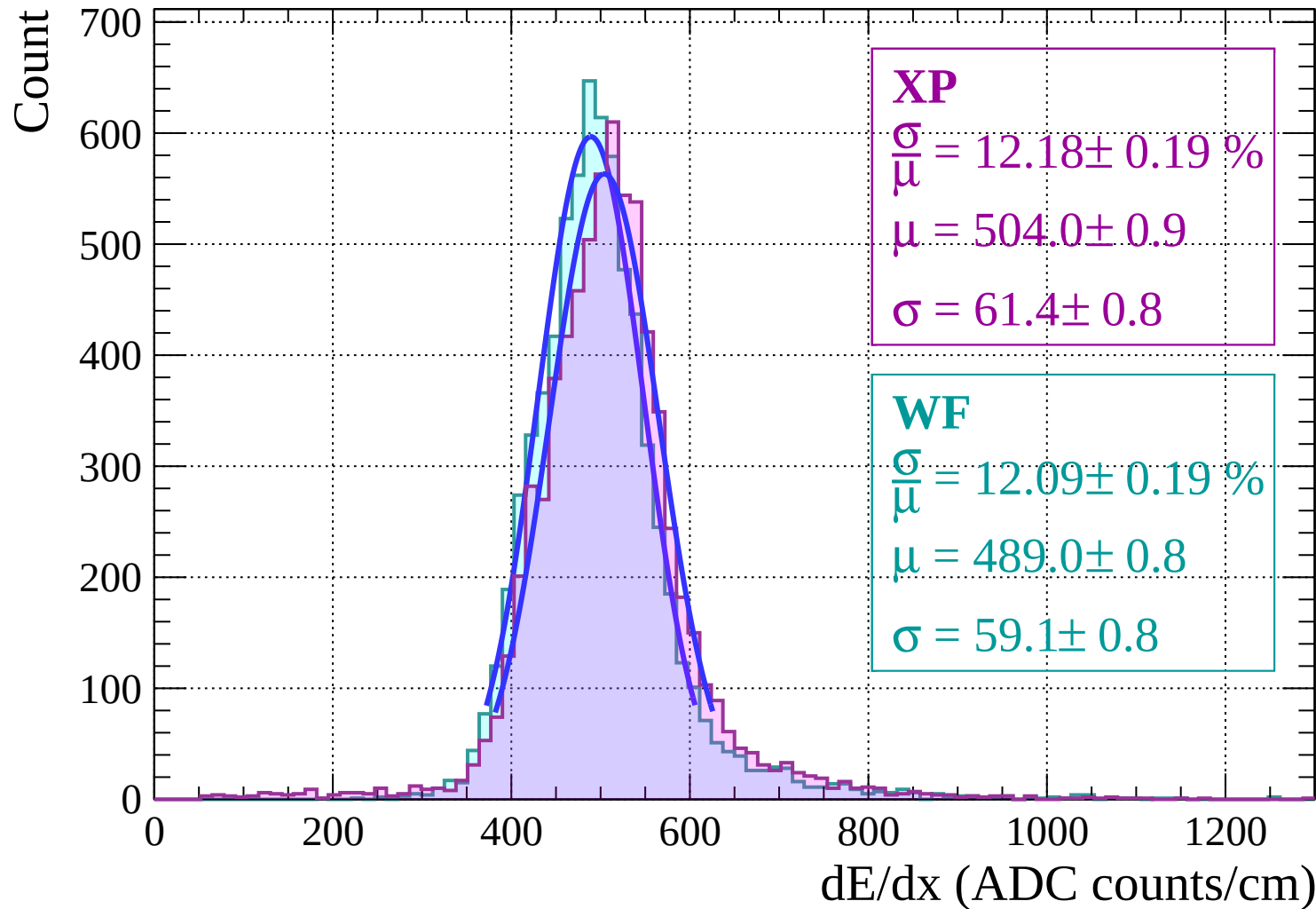
Count



Energy loss | $6 < \theta < 8$

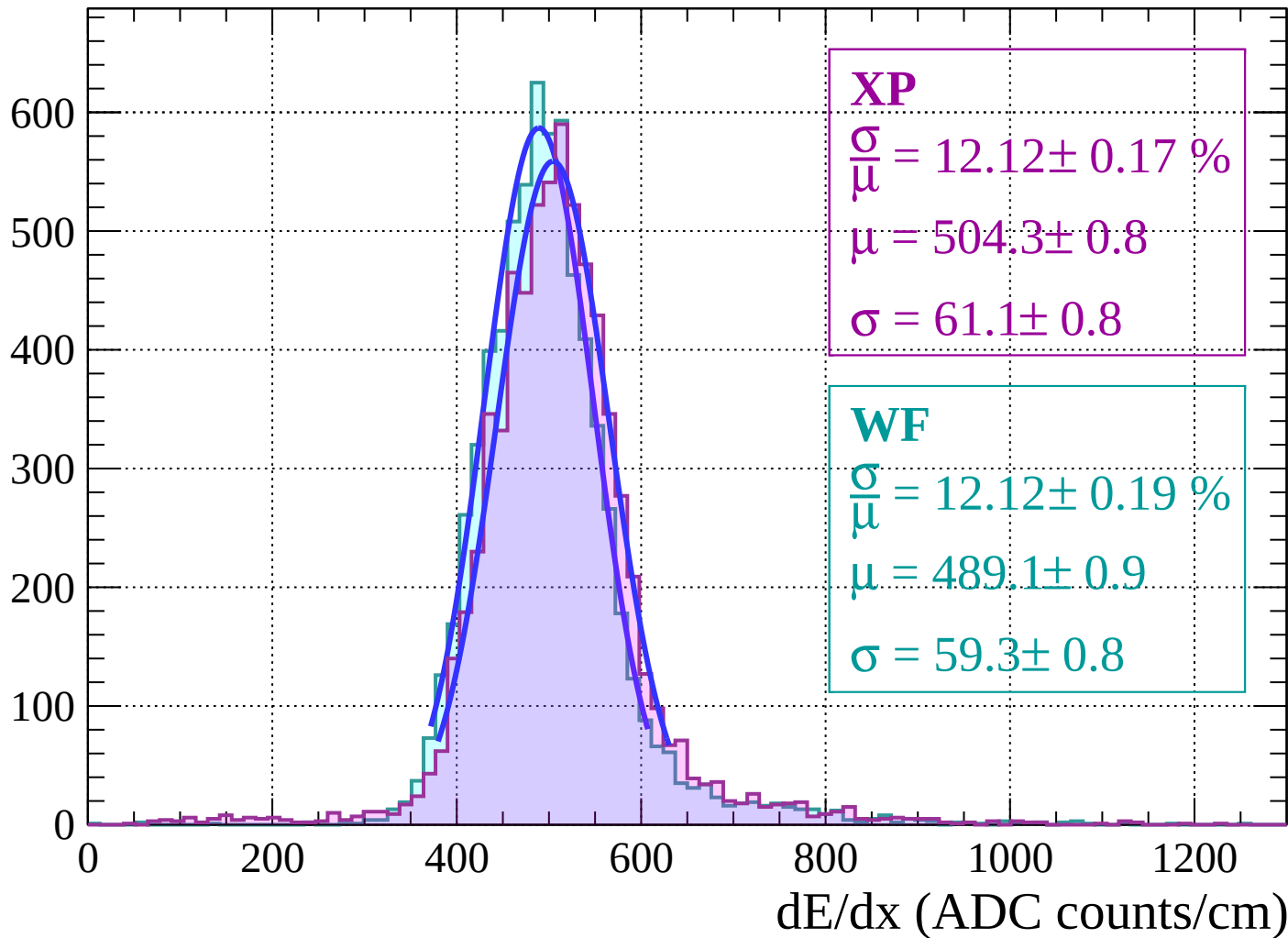


Energy loss | $8 < \theta < 10$

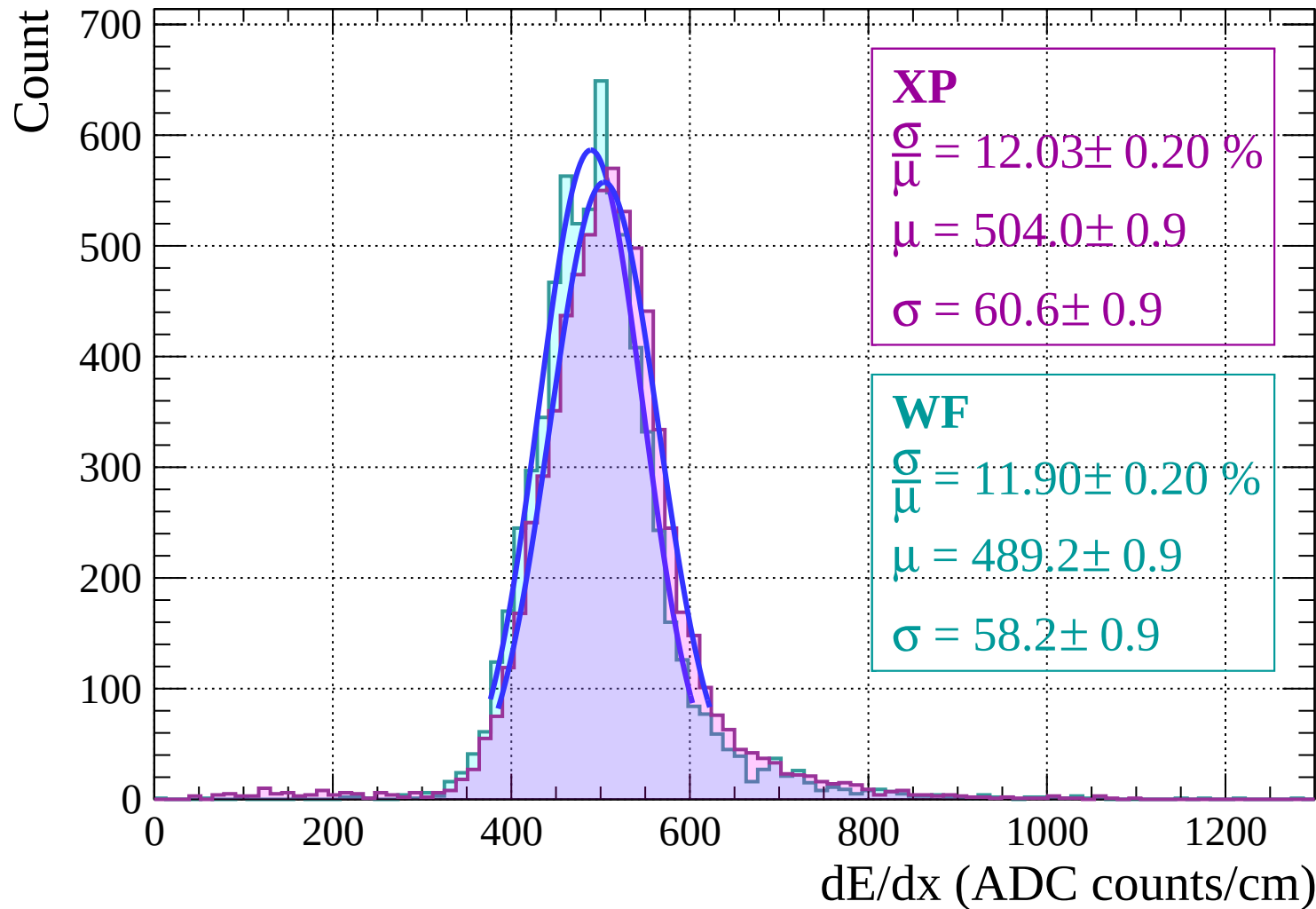


Energy loss | $10 < \theta < 12$

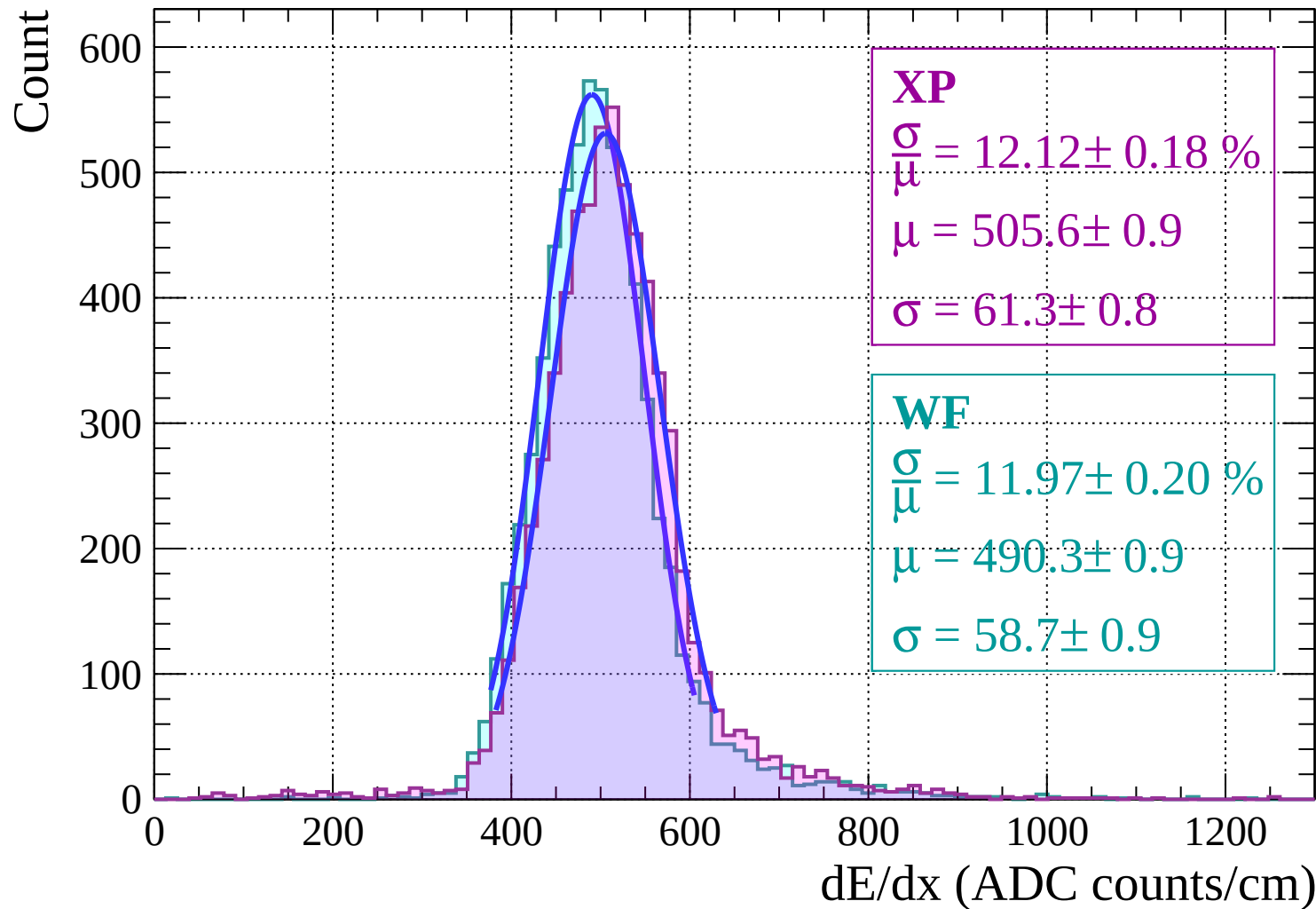
Count



Energy loss | $12 < \theta < 14$

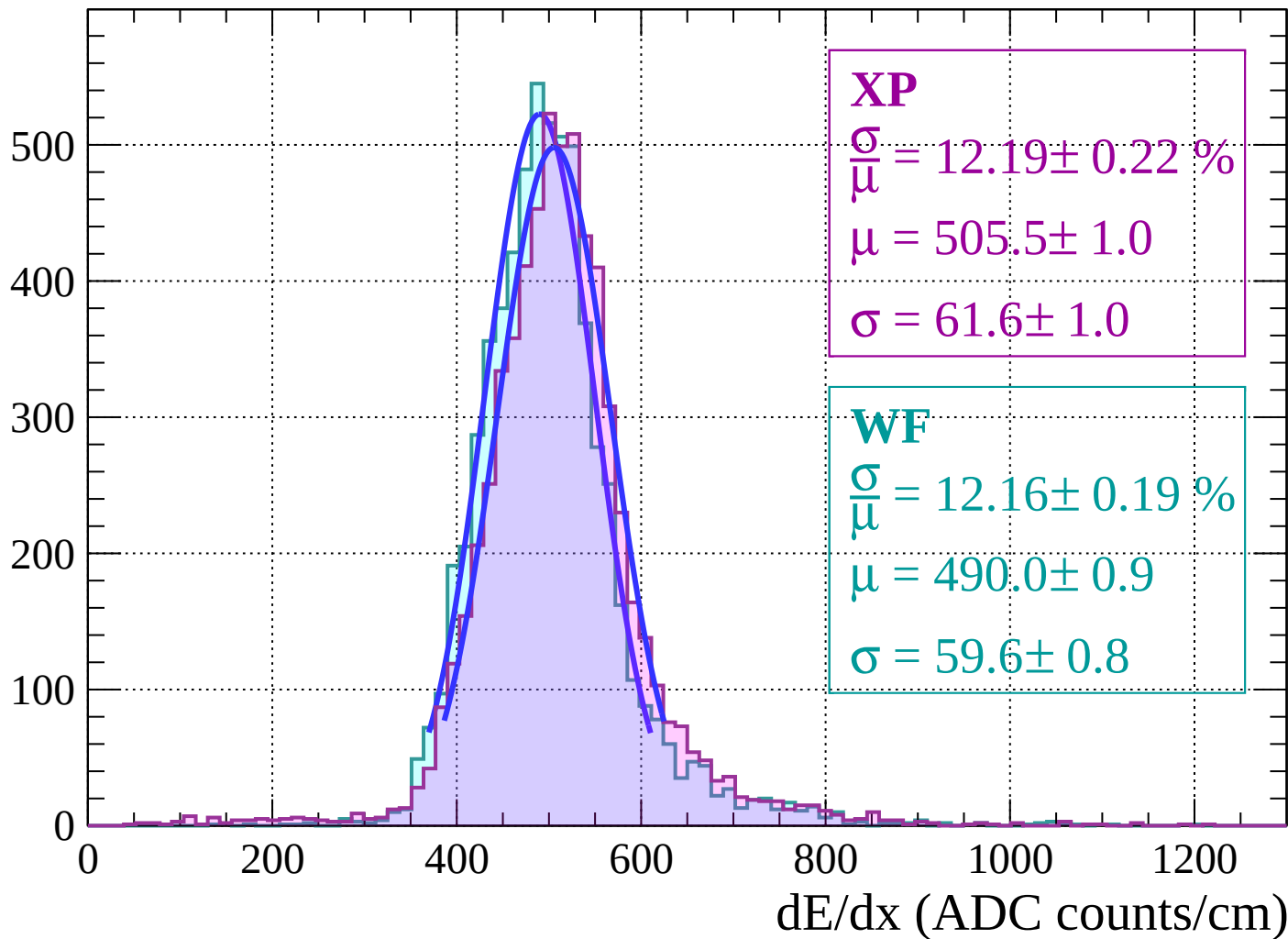


Energy loss | $14 < \theta < 16$



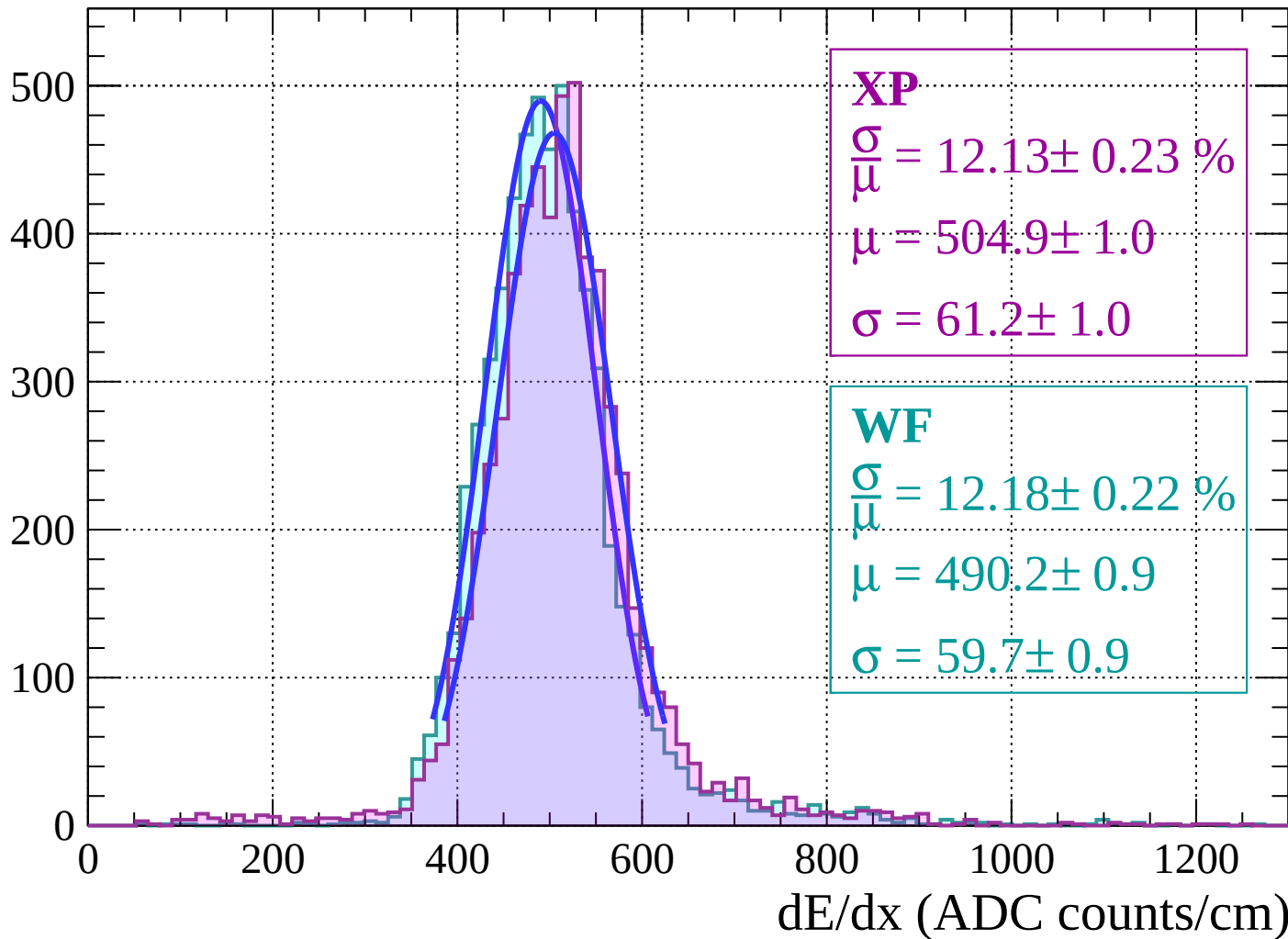
Energy loss | $16 < \theta < 18$

Count



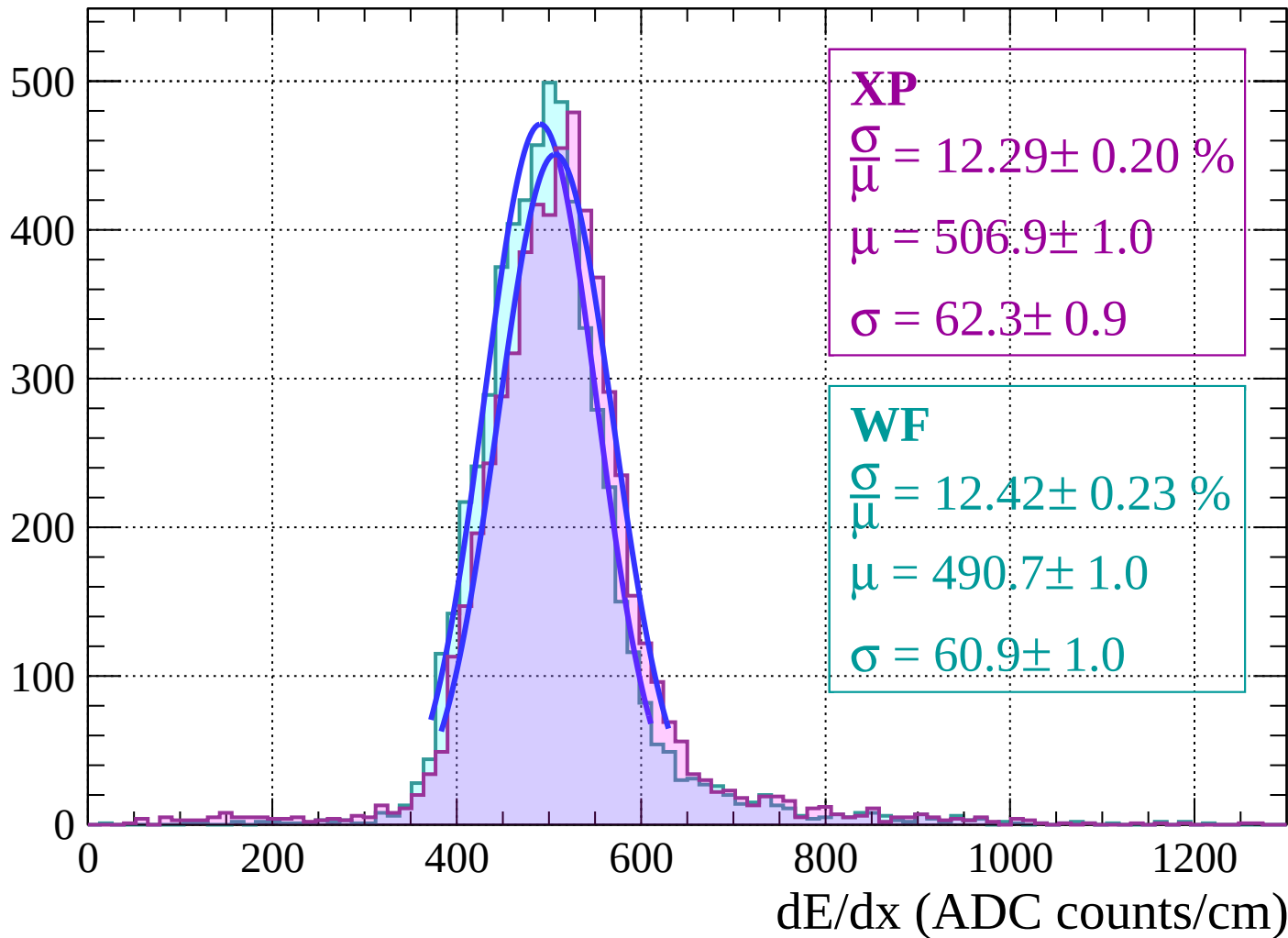
Energy loss | $18 < \theta < 20$

Count

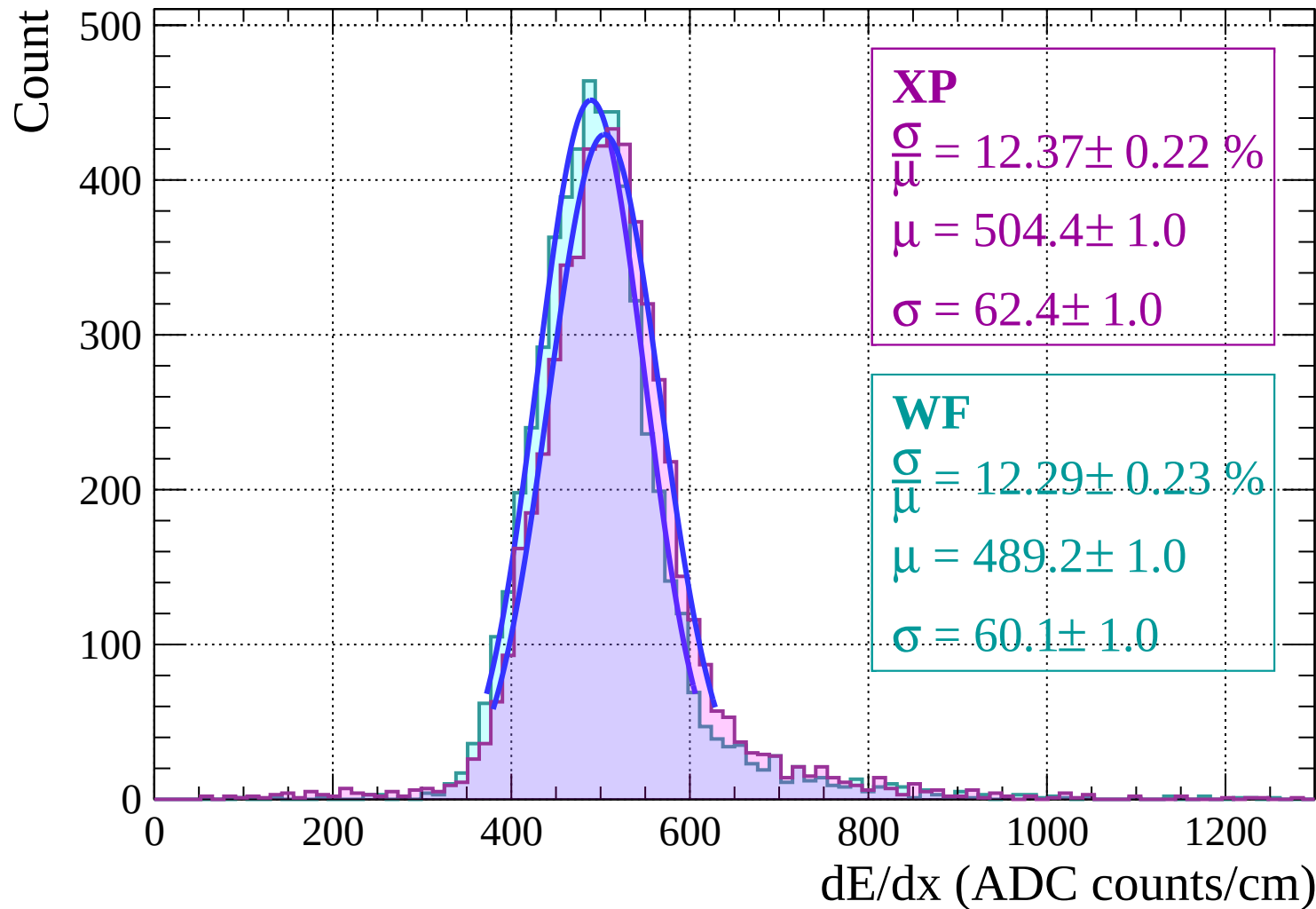


Energy loss | $20 < \theta < 22$

Count

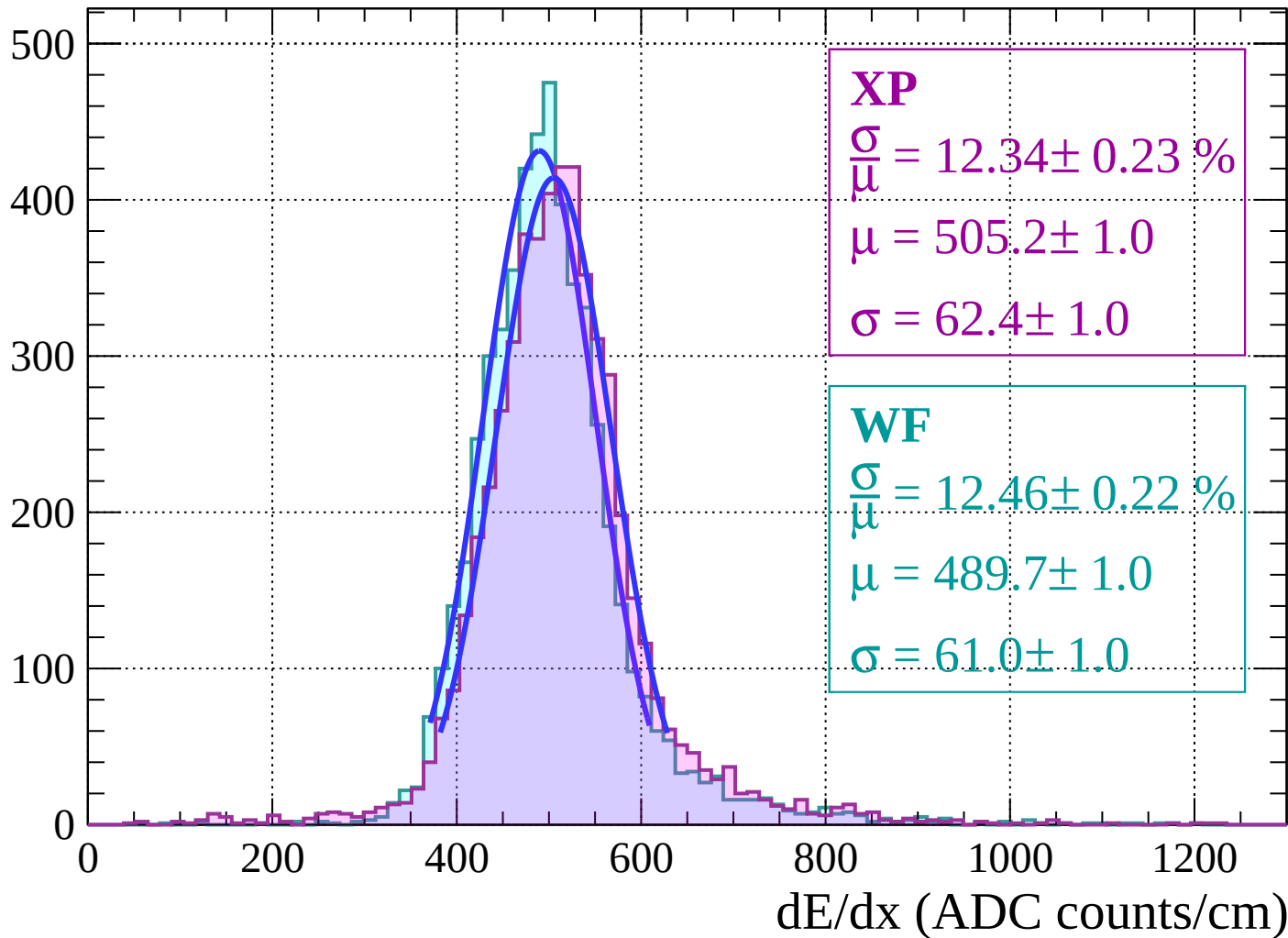


Energy loss | $22 < \theta < 24$



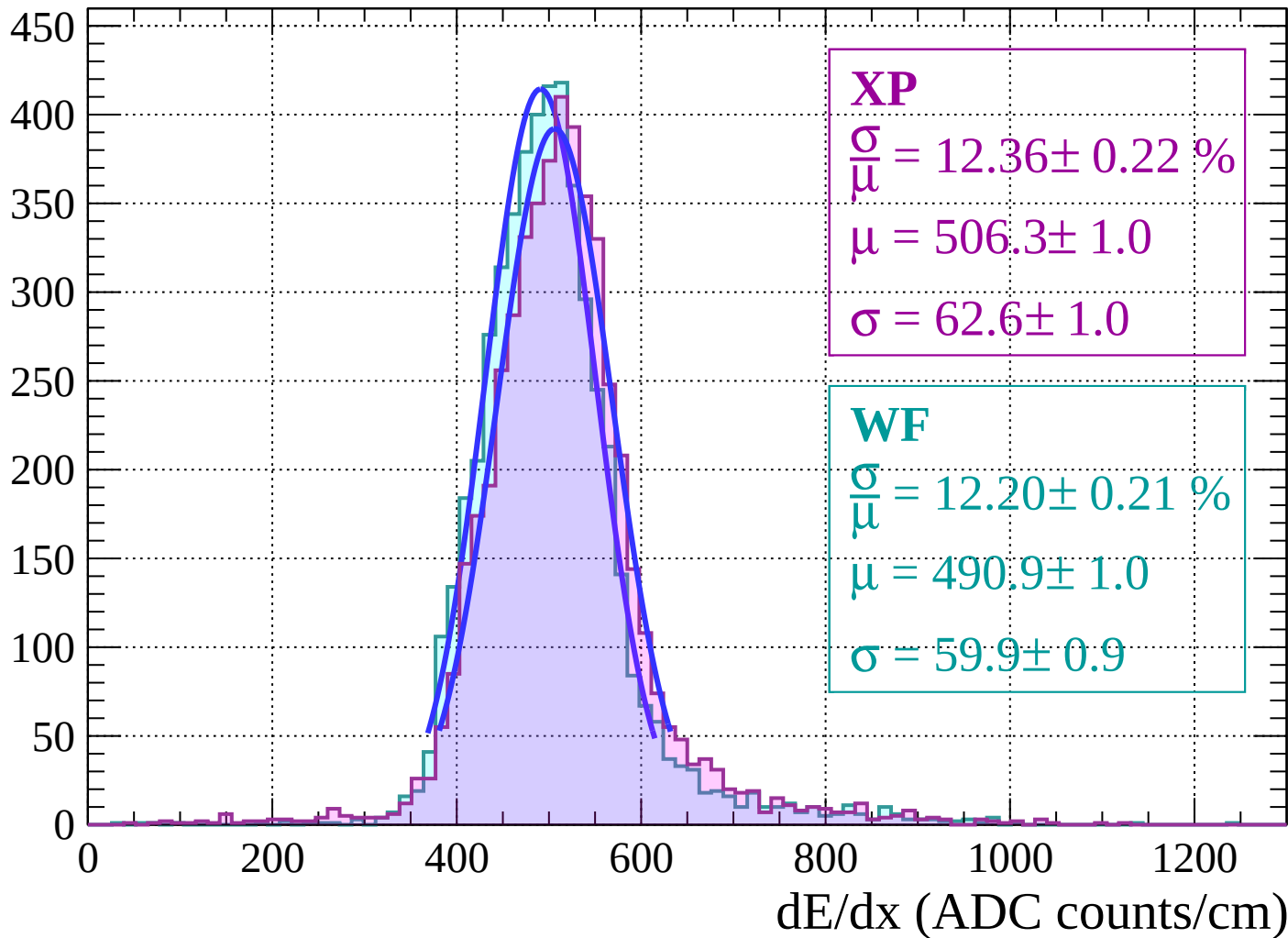
Energy loss | $24 < \theta < 26$

Count



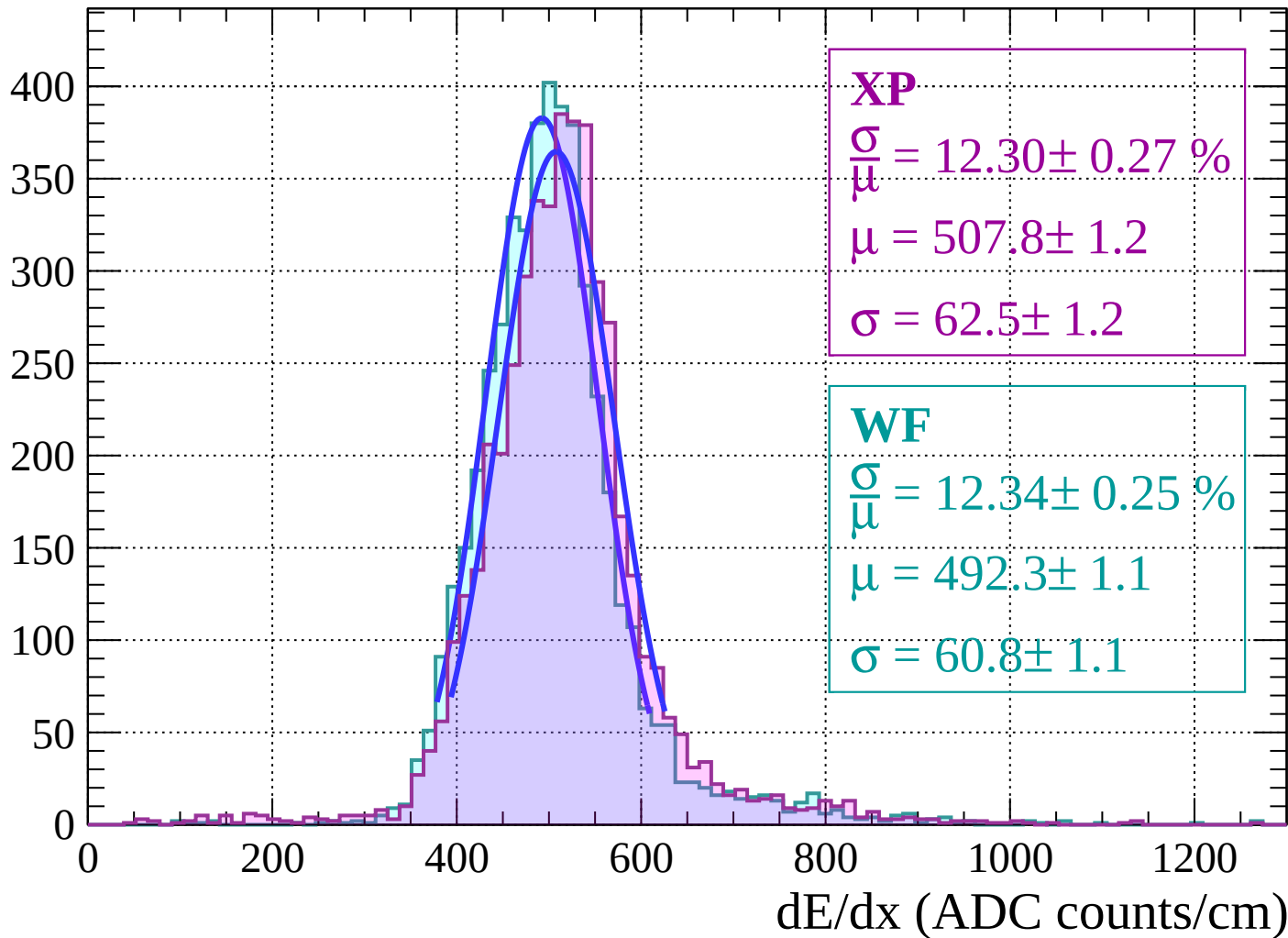
Energy loss | $26 < \theta < 28$

Count

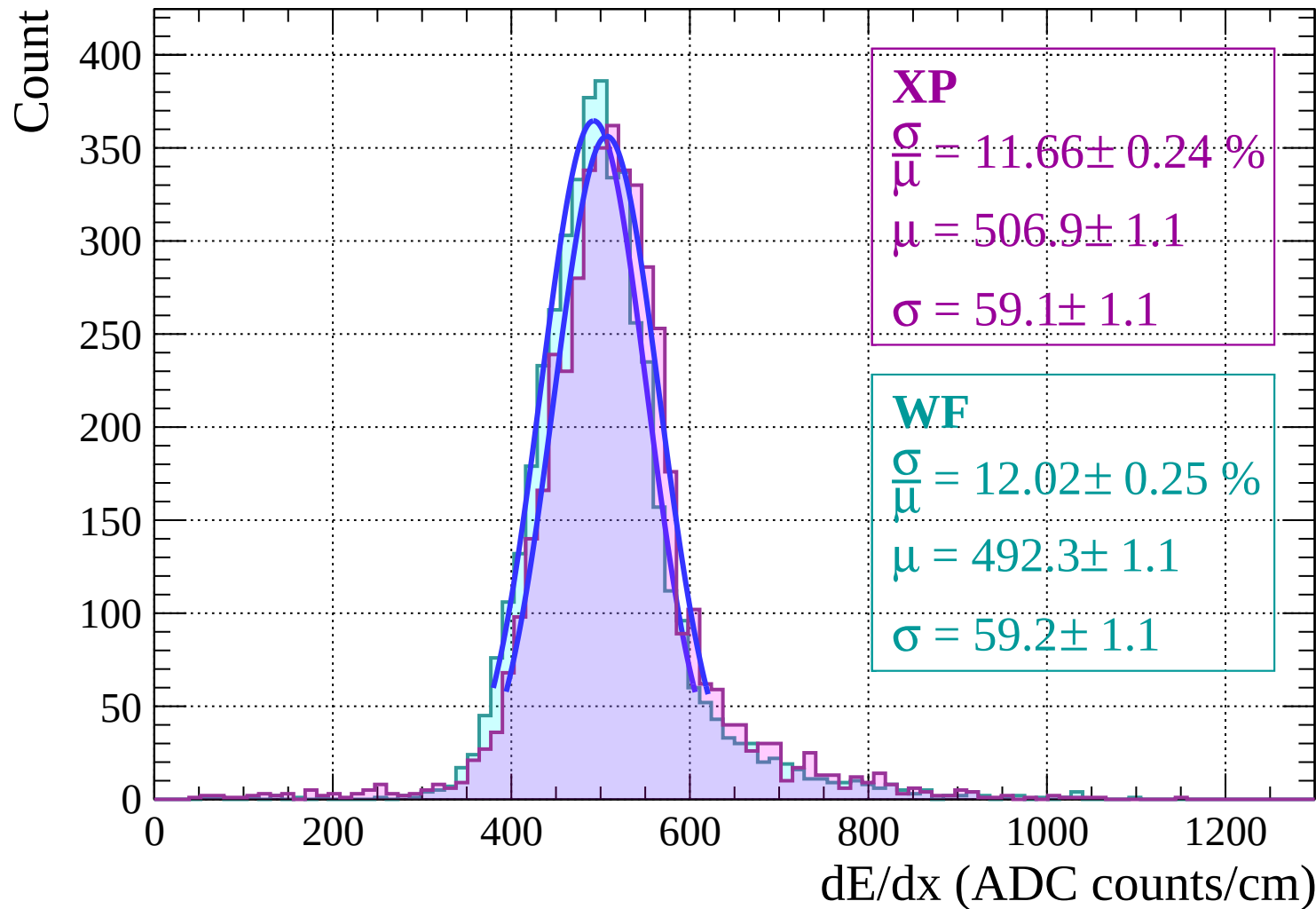


Energy loss | $28 < \theta < 30$

Count

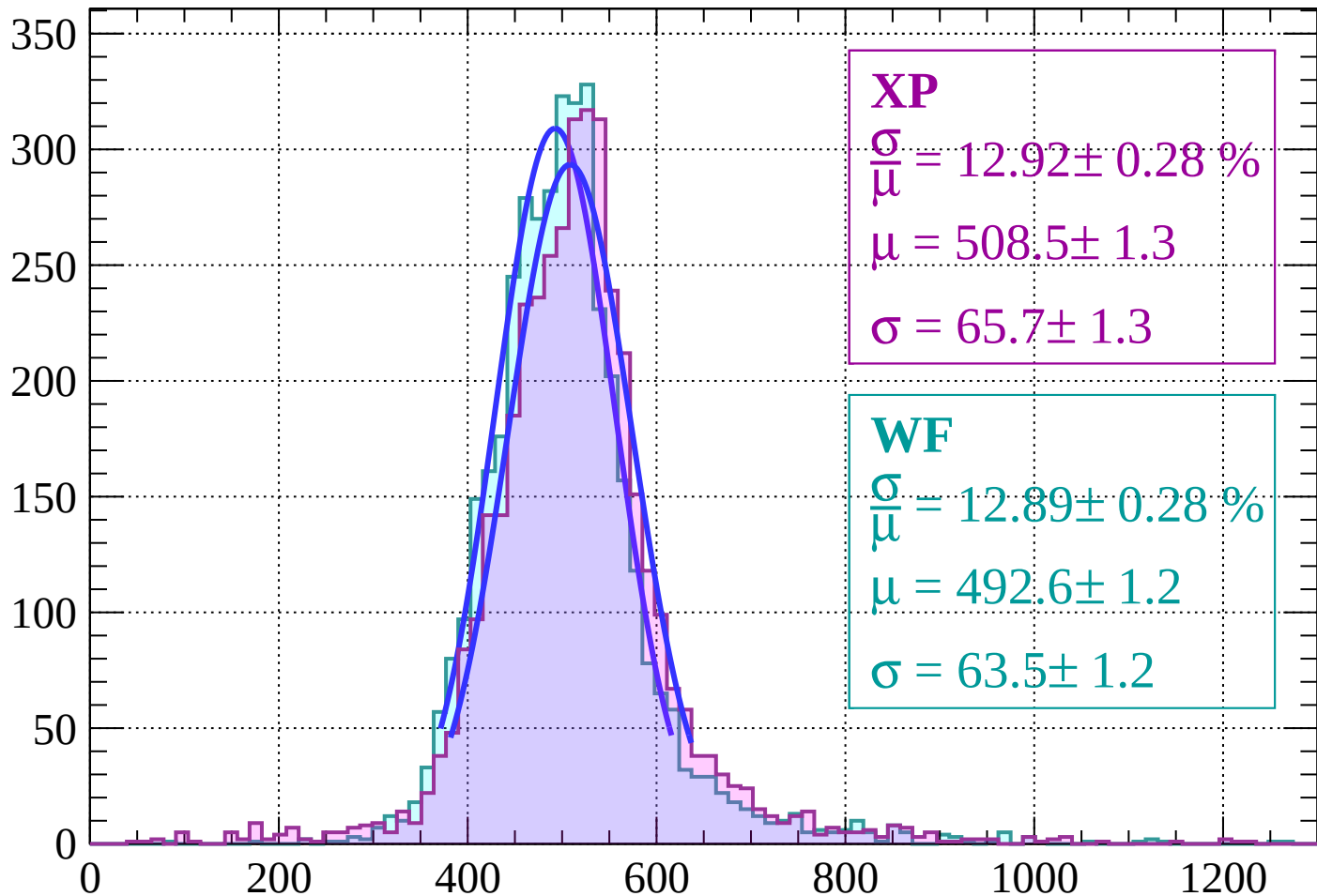


Energy loss | $30 < \theta < 32$



Energy loss | $32 < \theta < 34$

Count



XP

$$\frac{\sigma}{\mu} = 12.92 \pm 0.28 \%$$

$$\mu = 508.5 \pm 1.3$$

$$\sigma = 65.7 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 12.89 \pm 0.28 \%$$

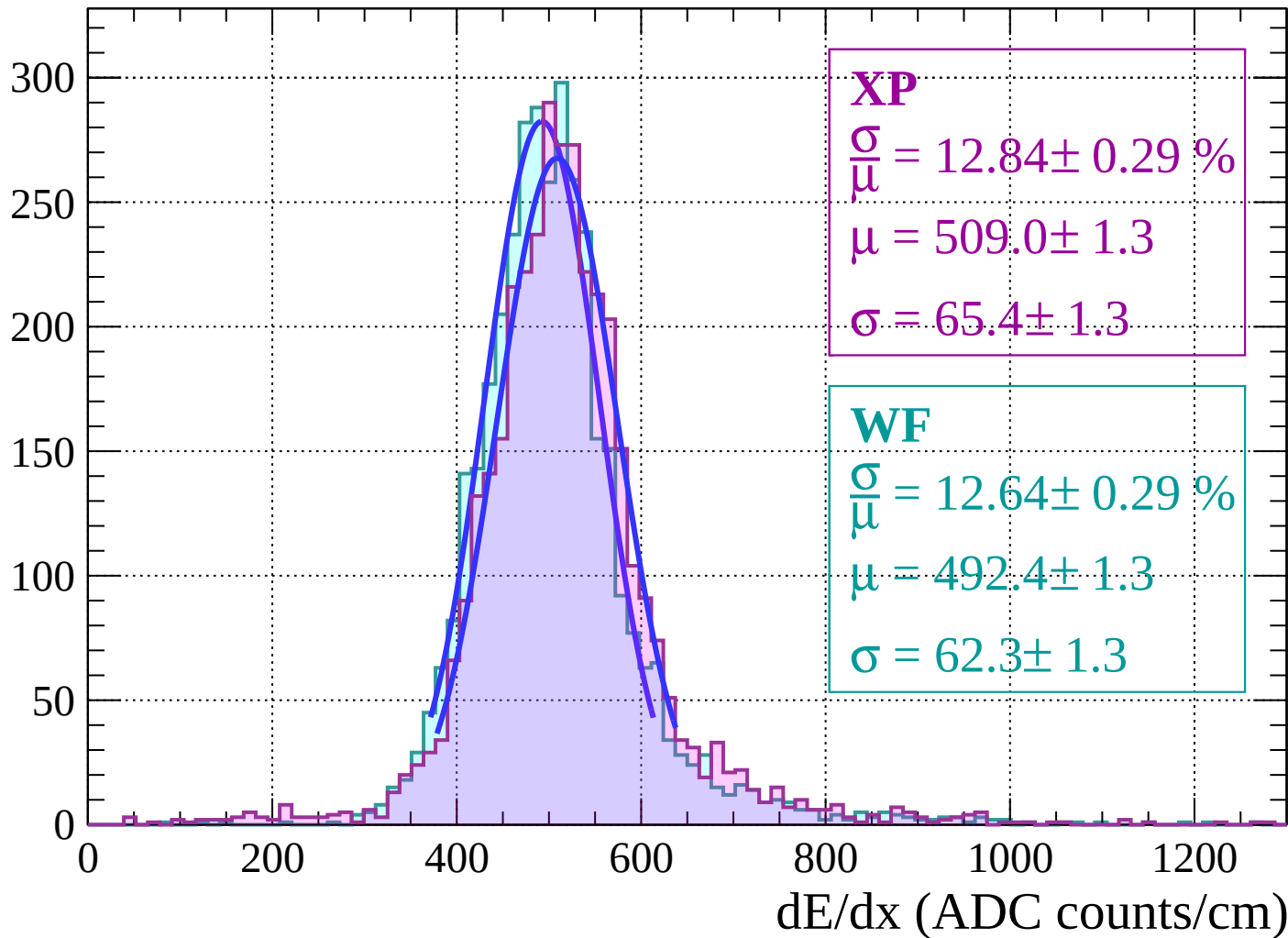
$$\mu = 492.6 \pm 1.2$$

$$\sigma = 63.5 \pm 1.2$$

dE/dx (ADC counts/cm)

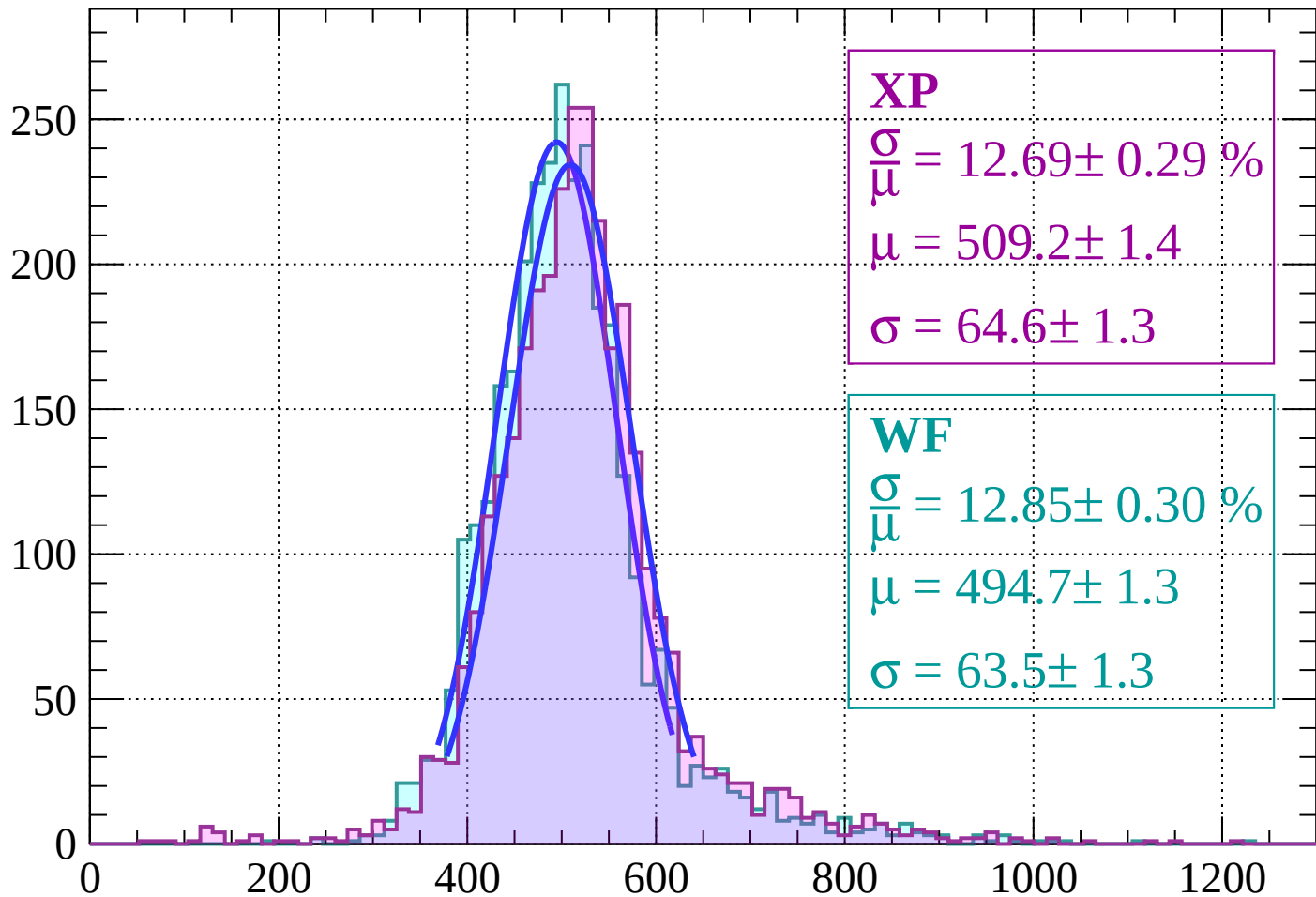
Energy loss | $34 < \theta < 36$

Count



Energy loss | $36 < \theta < 38$

Count



XP

$$\frac{\sigma}{\mu} = 12.69 \pm 0.29 \%$$

$$\mu = 509.2 \pm 1.4$$

$$\sigma = 64.6 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 12.85 \pm 0.30 \%$$

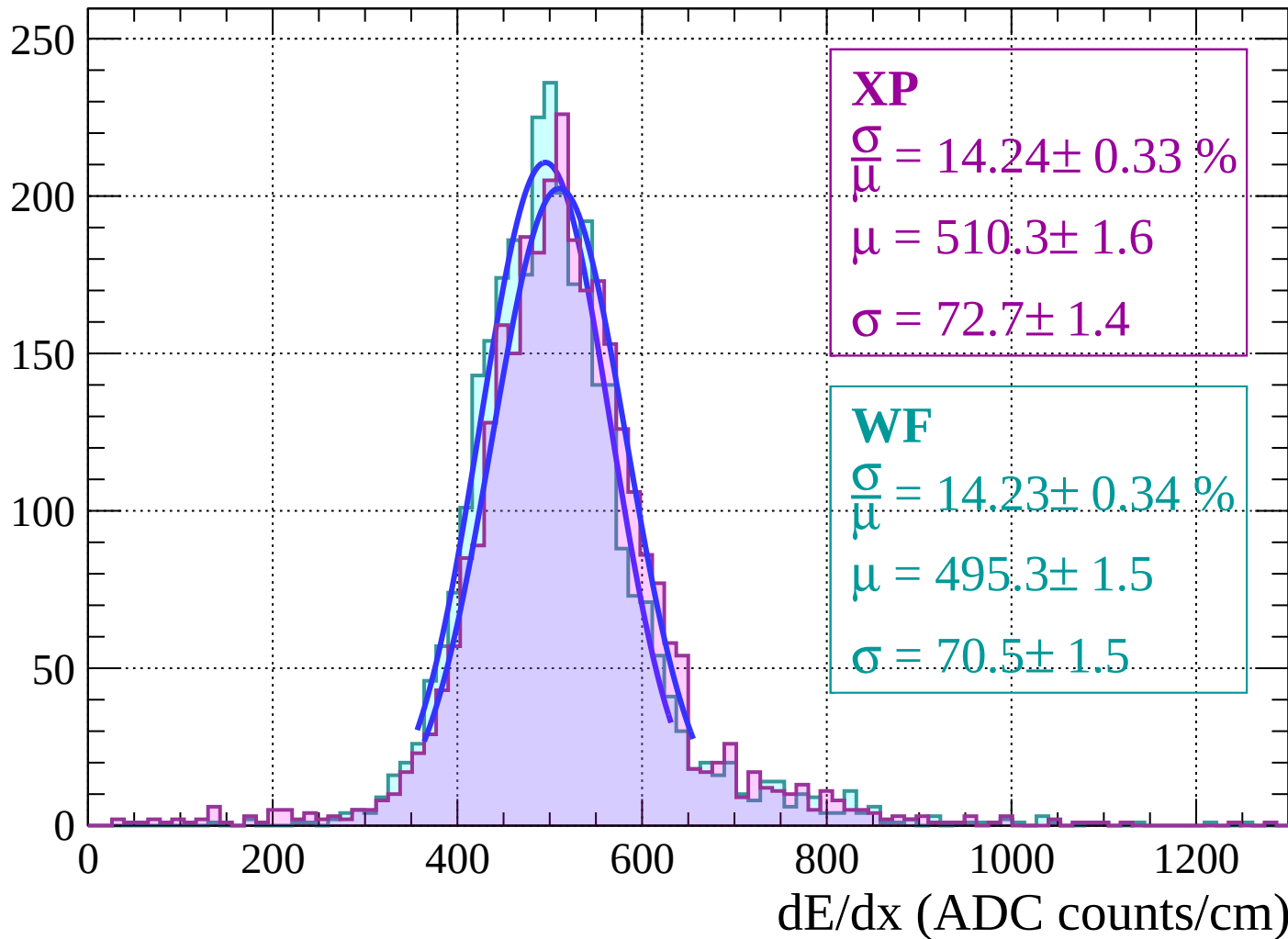
$$\mu = 494.7 \pm 1.3$$

$$\sigma = 63.5 \pm 1.3$$

dE/dx (ADC counts/cm)

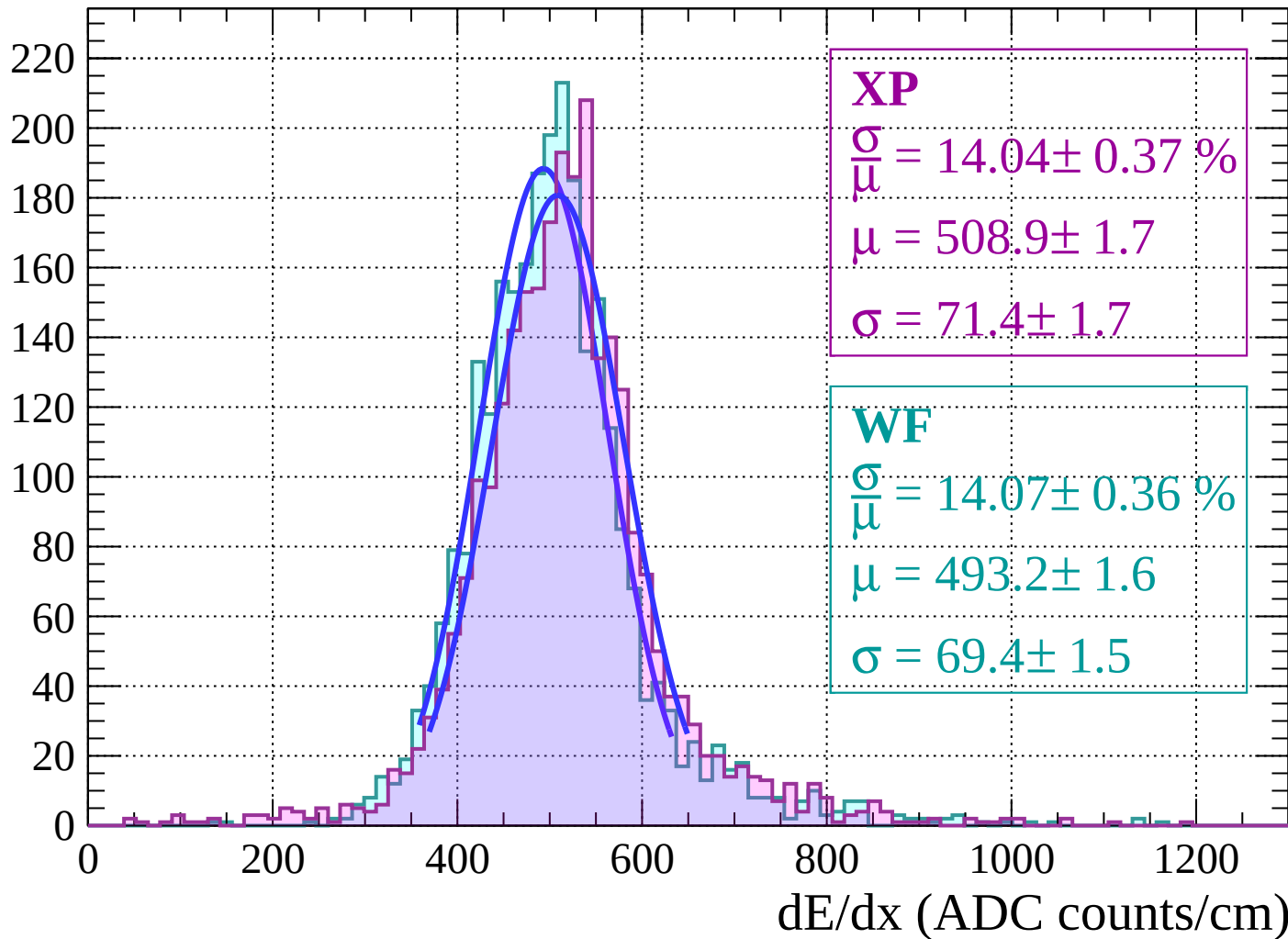
Energy loss | $38 < \theta < 40$

Count



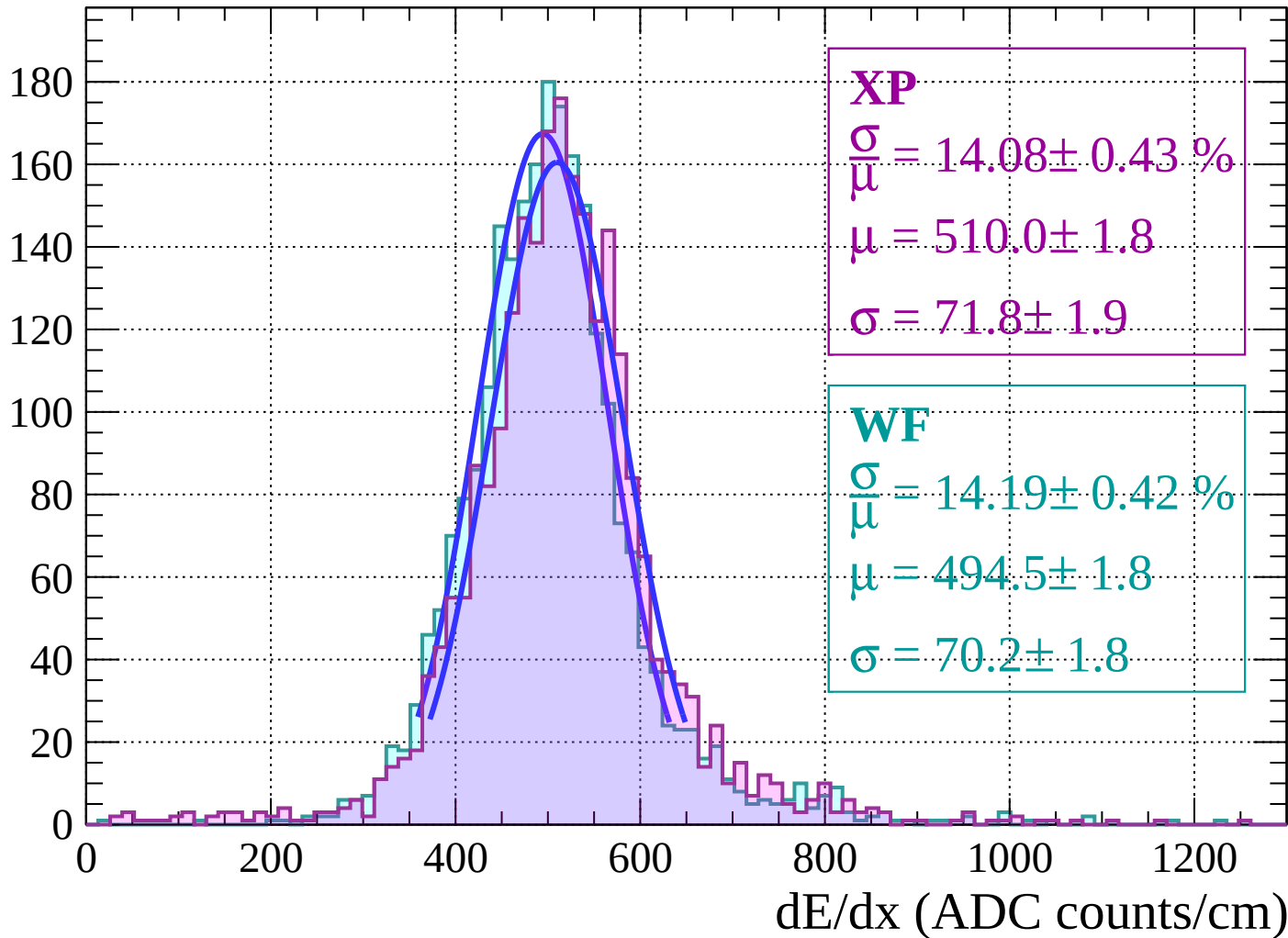
Energy loss | $40 < \theta < 42$

Count

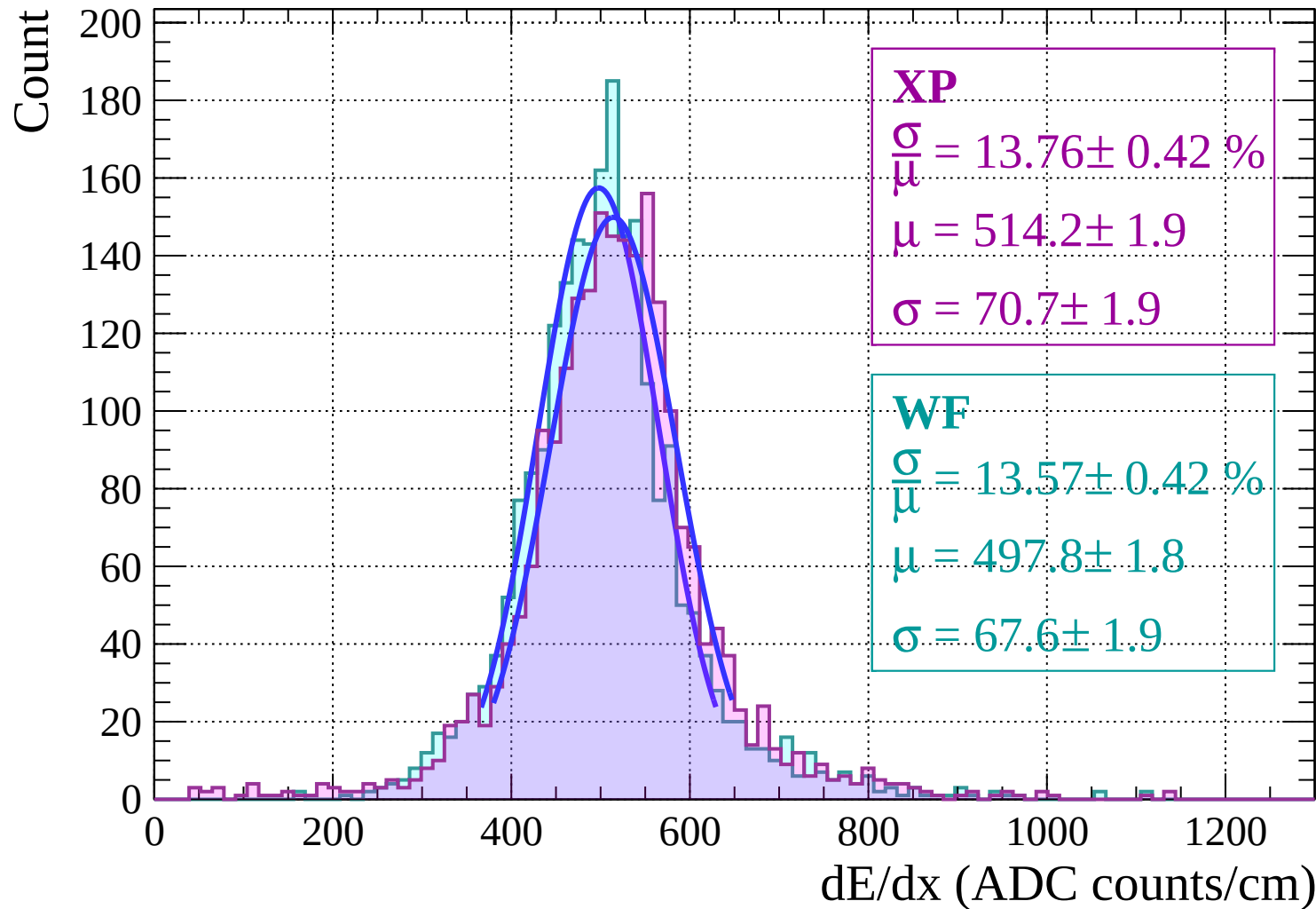


Energy loss | $42 < \theta < 44$

Count

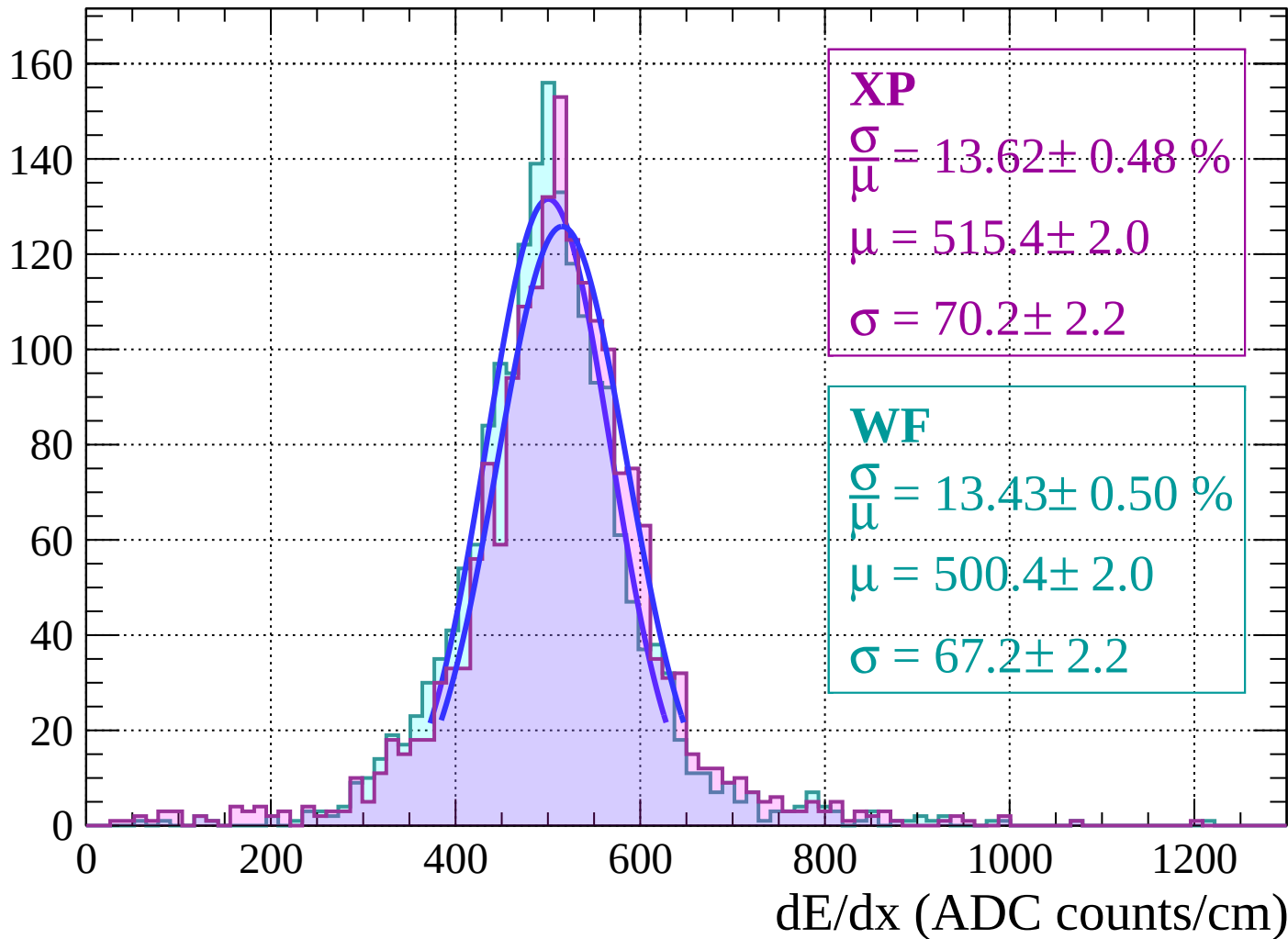


Energy loss | $44 < \theta < 46$



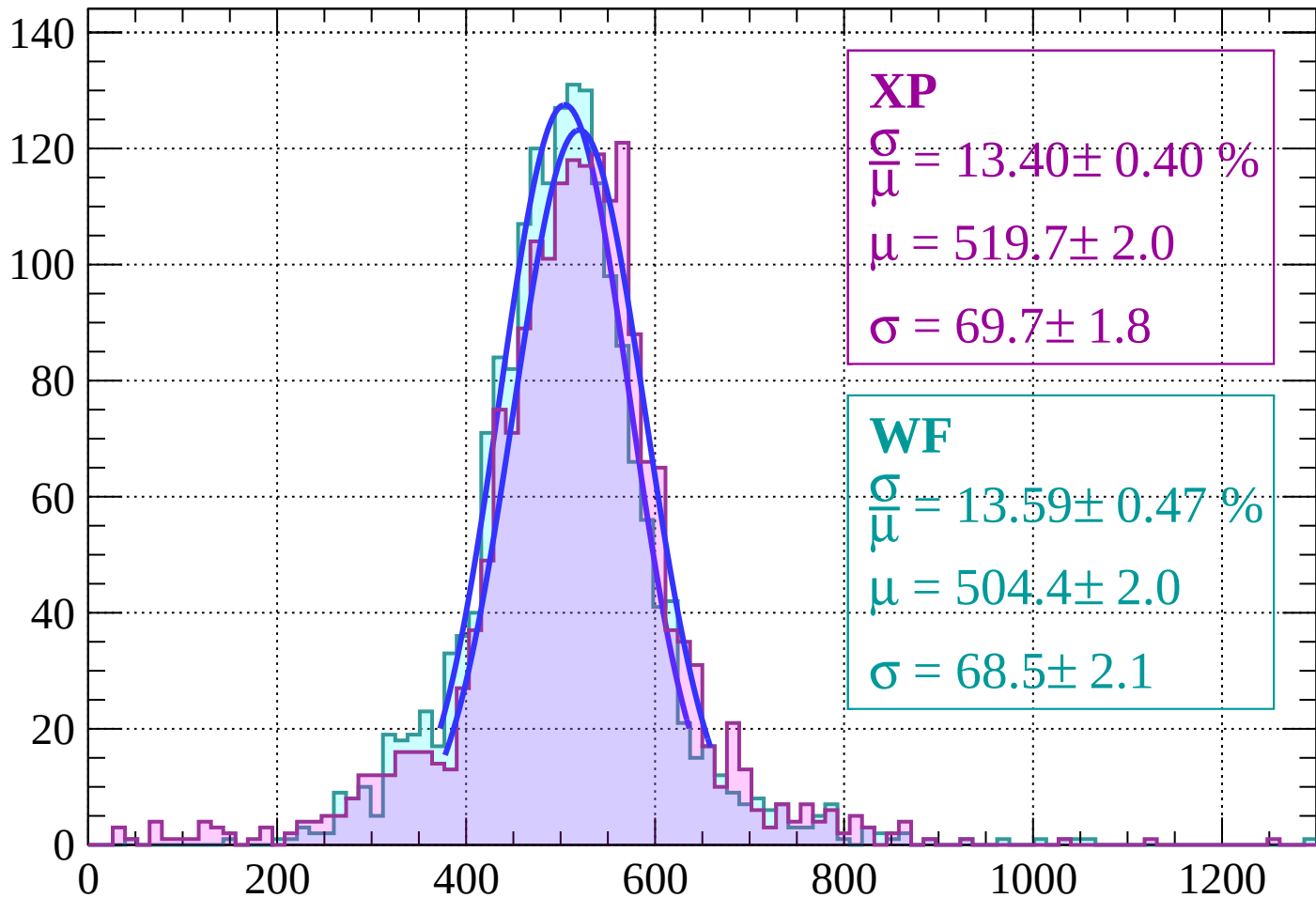
Energy loss | $46 < \theta < 48$

Count



Energy loss | $48 < \theta < 50$

Count



XP

$$\frac{\sigma}{\mu} = 13.40 \pm 0.40 \%$$

$$\mu = 519.7 \pm 2.0$$

$$\sigma = 69.7 \pm 1.8$$

WF

$$\frac{\sigma}{\mu} = 13.59 \pm 0.47 \%$$

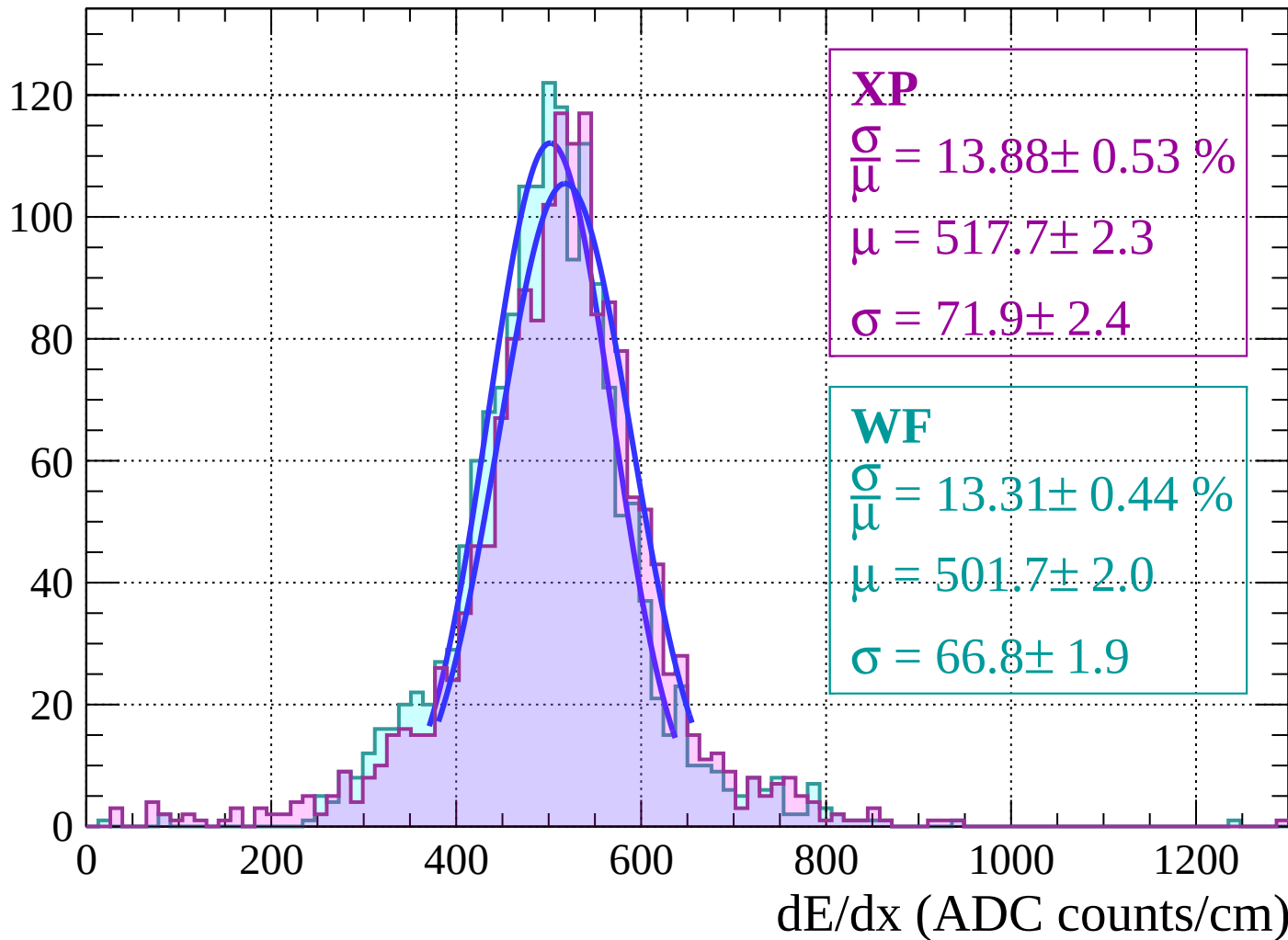
$$\mu = 504.4 \pm 2.0$$

$$\sigma = 68.5 \pm 2.1$$

dE/dx (ADC counts/cm)

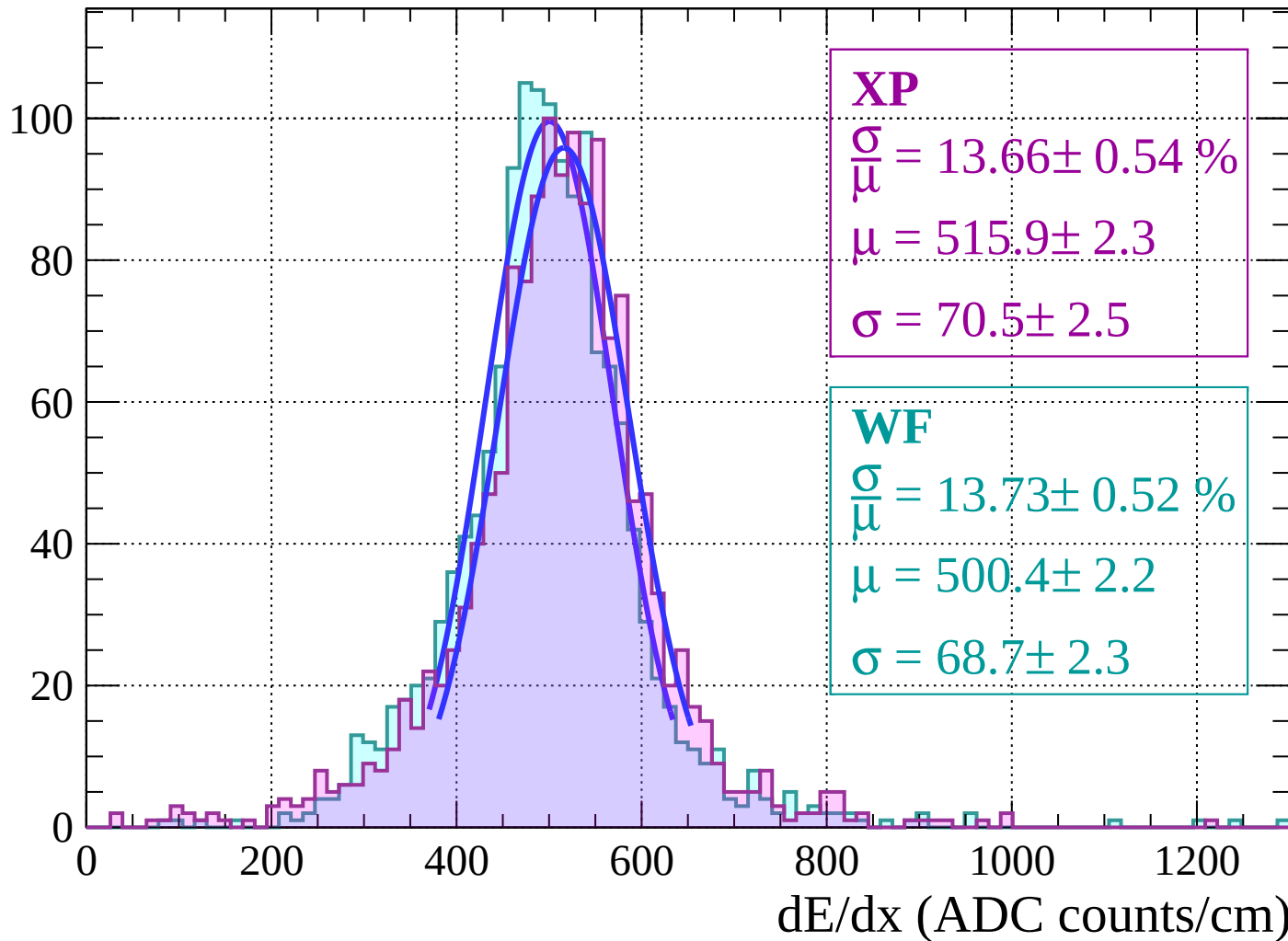
Energy loss | $50 < \theta < 52$

Count



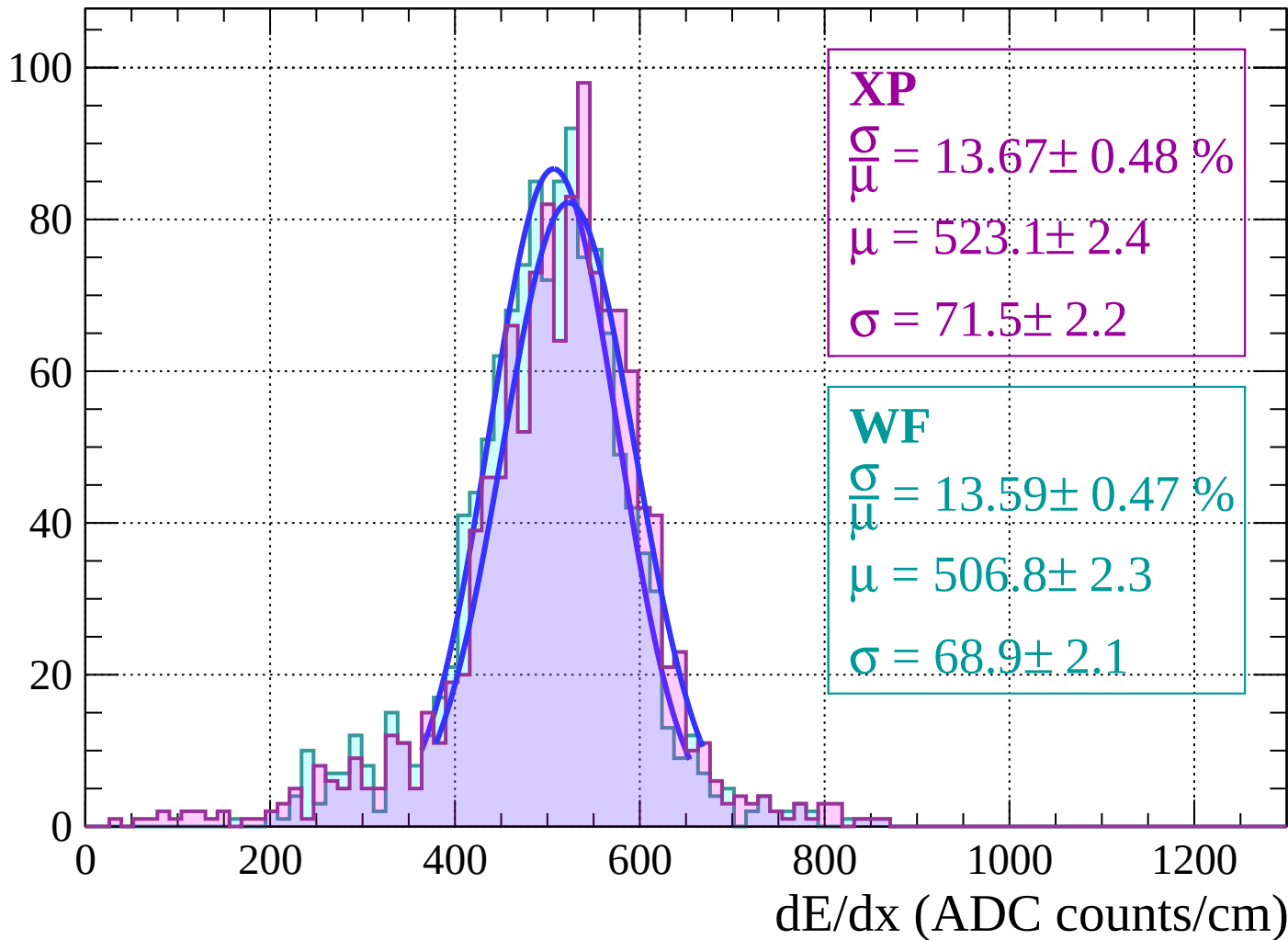
Energy loss | $52 < \theta < 54$

Count



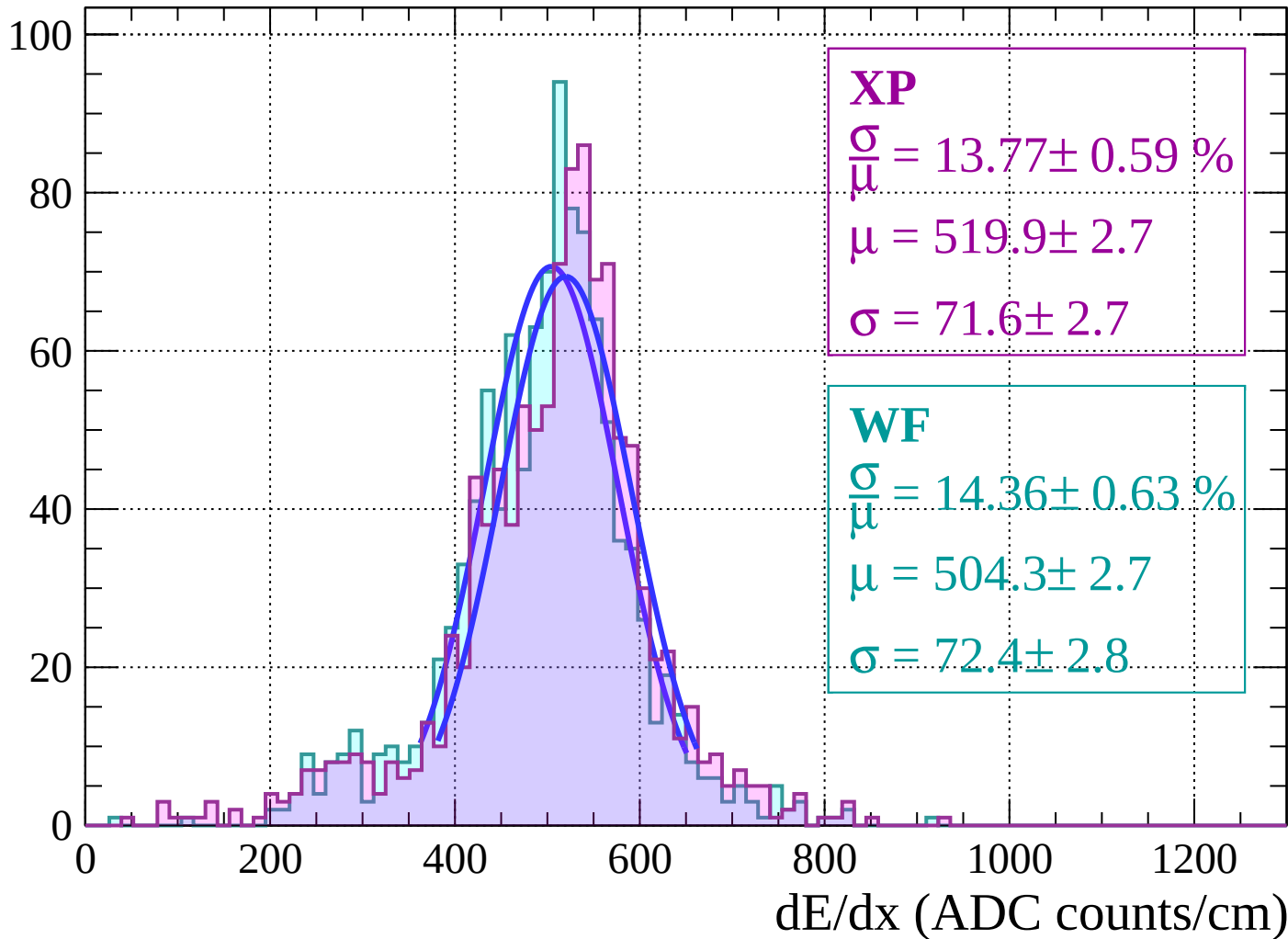
Energy loss | $54 < \theta < 56$

Count



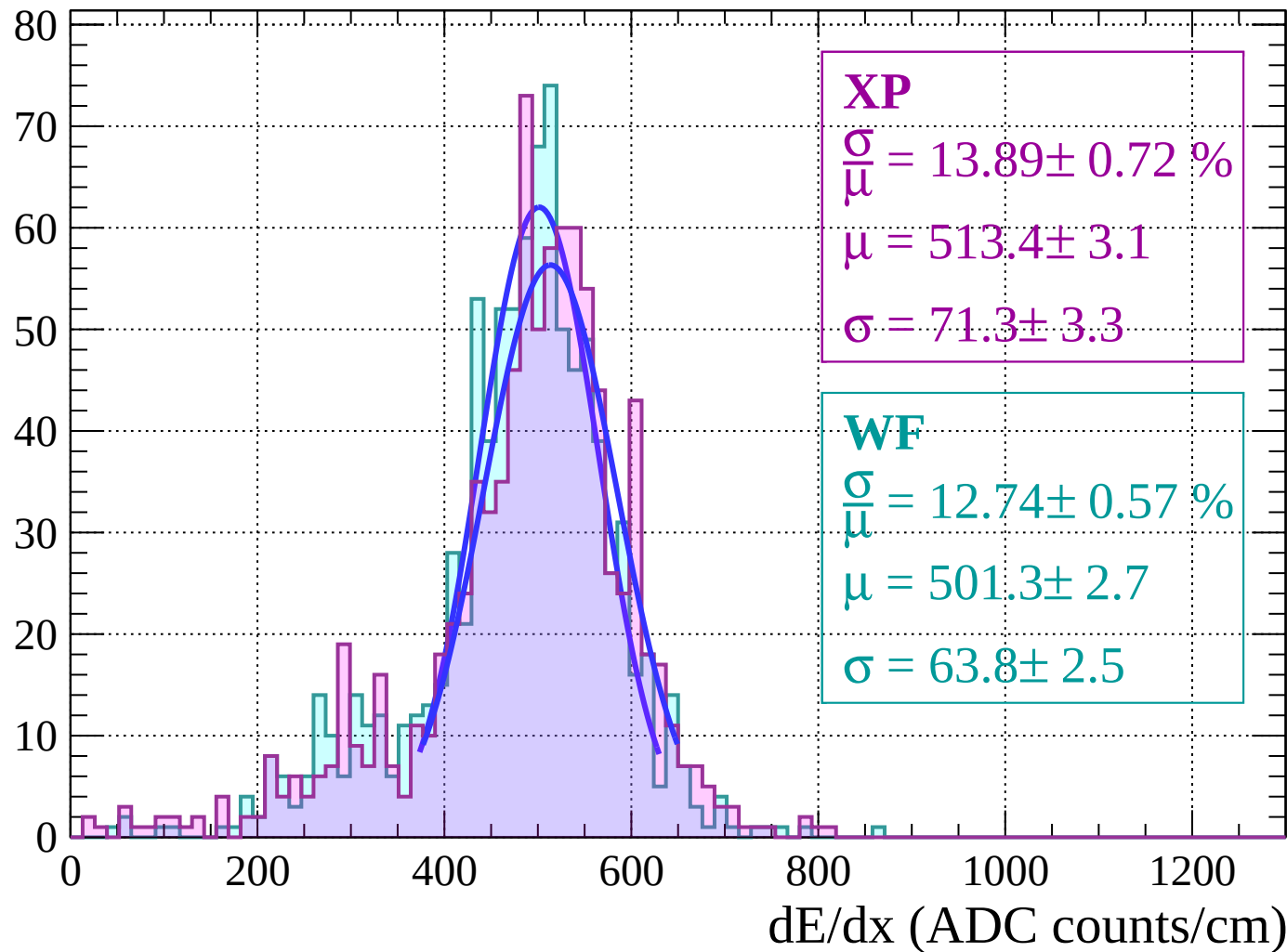
Energy loss | $56 < \theta < 58$

Count



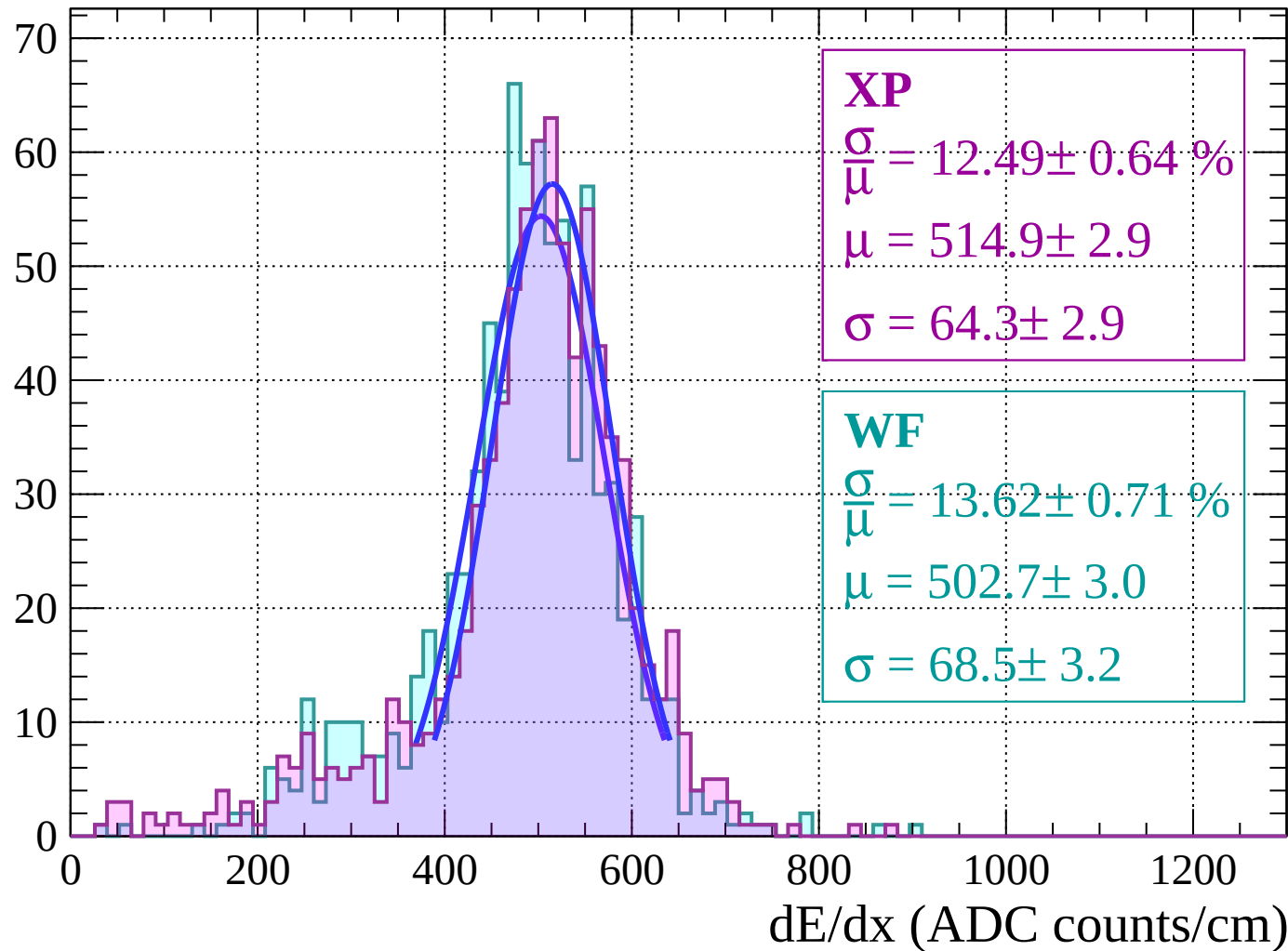
Energy loss | $58 < \theta < 60$

Count



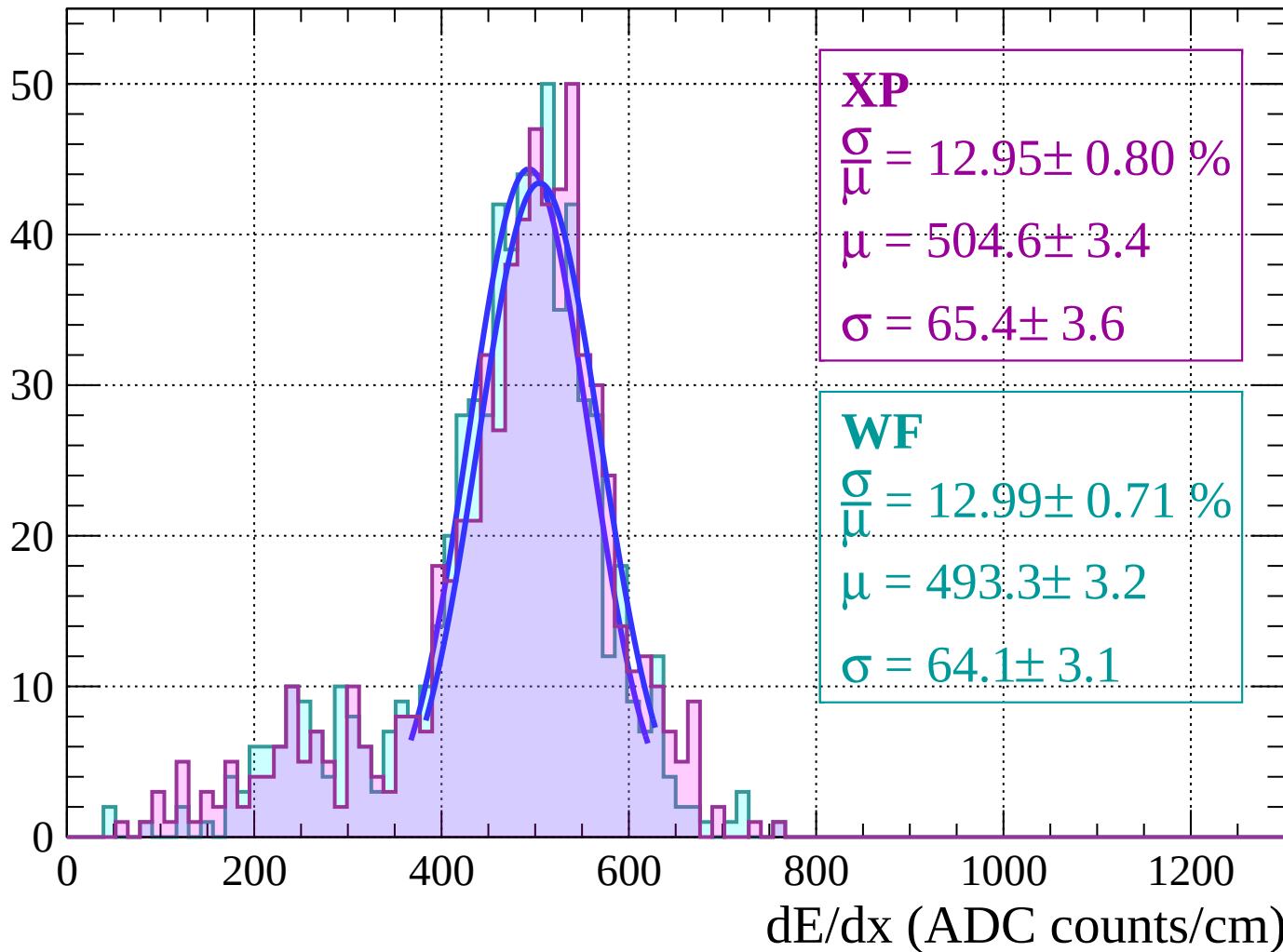
Energy loss | $60 < \theta < 62$

Count



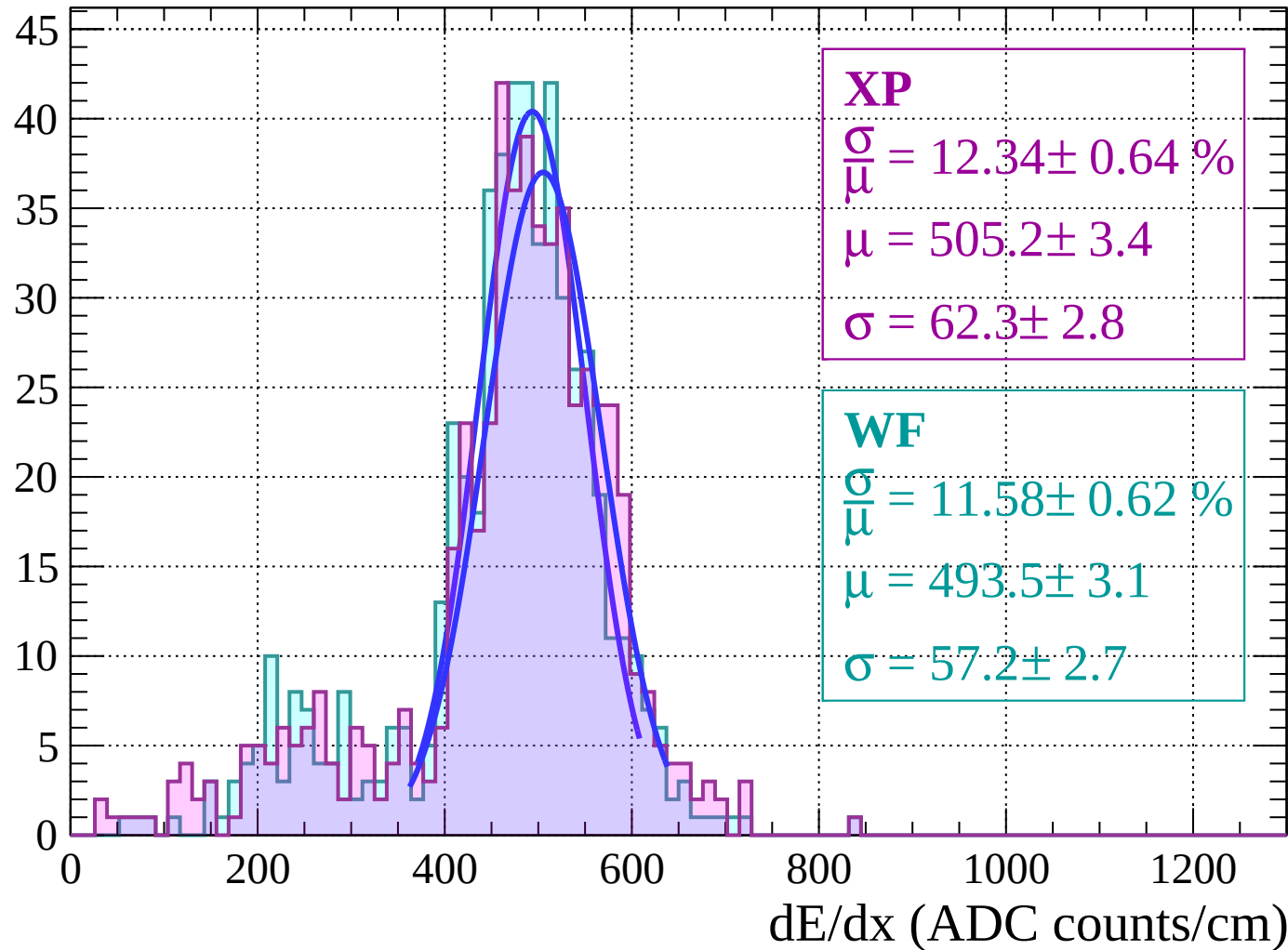
Energy loss | $62 < \theta < 64$

Count



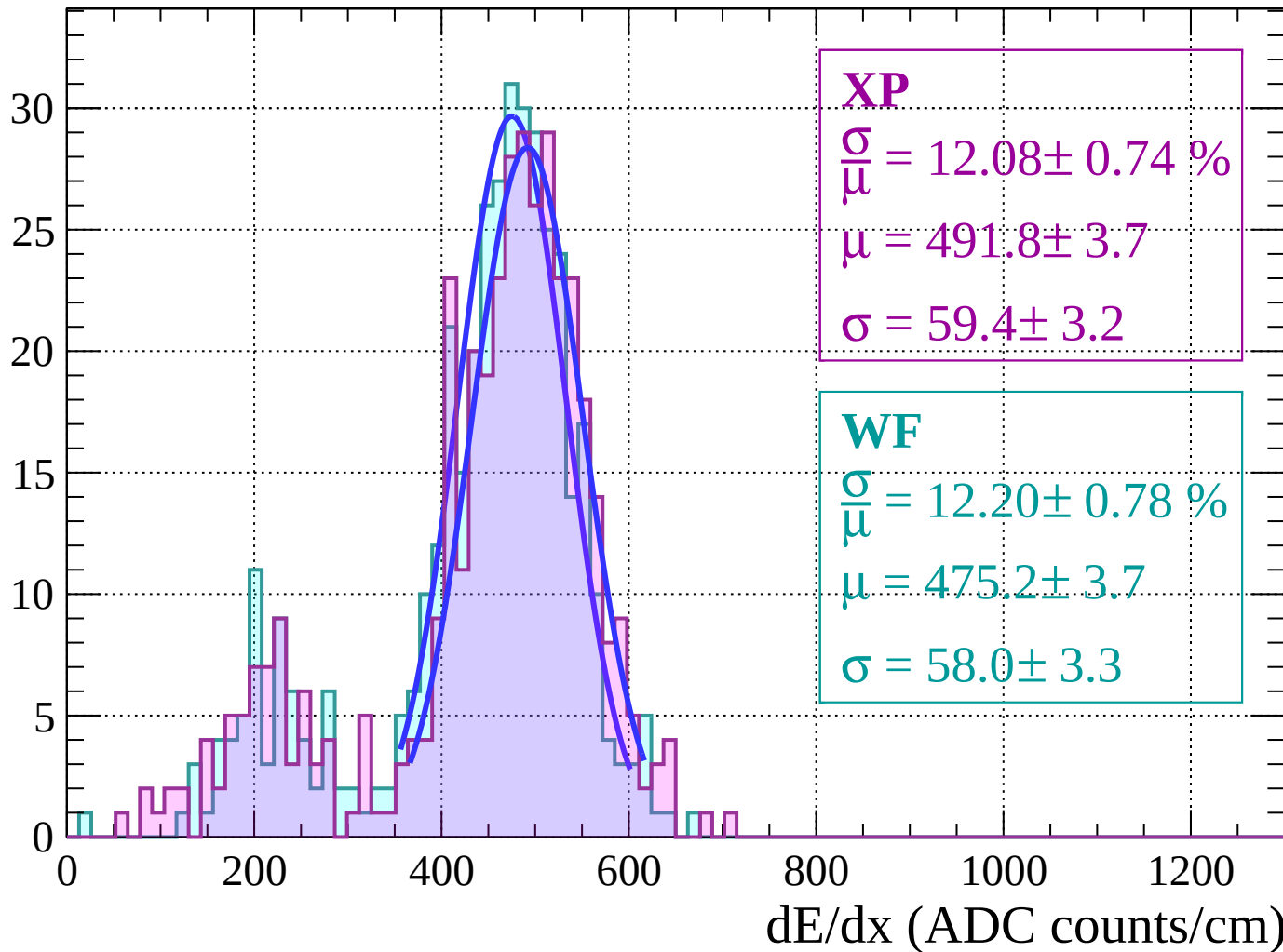
Energy loss | $64 < \theta < 66$

Count



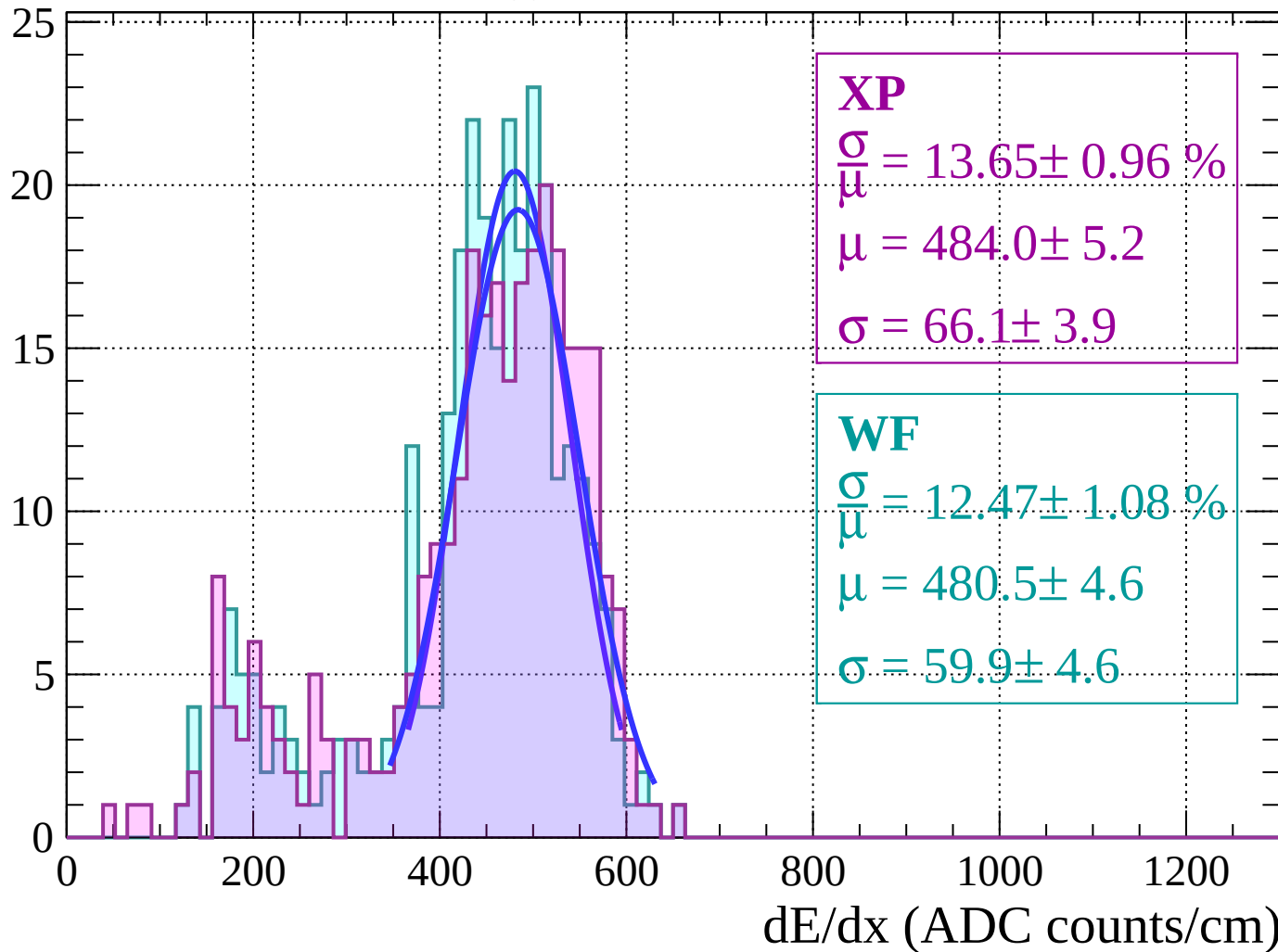
Energy loss | $66 < \theta < 68$

Count



Energy loss | $68 < \theta < 70$

Count



Energy loss | $70 < \theta < 72$

Count

XP

$$\frac{\sigma}{\mu} = 11.09 \pm 1.53 \%$$

$$\mu = 470.4 \pm 5.9$$

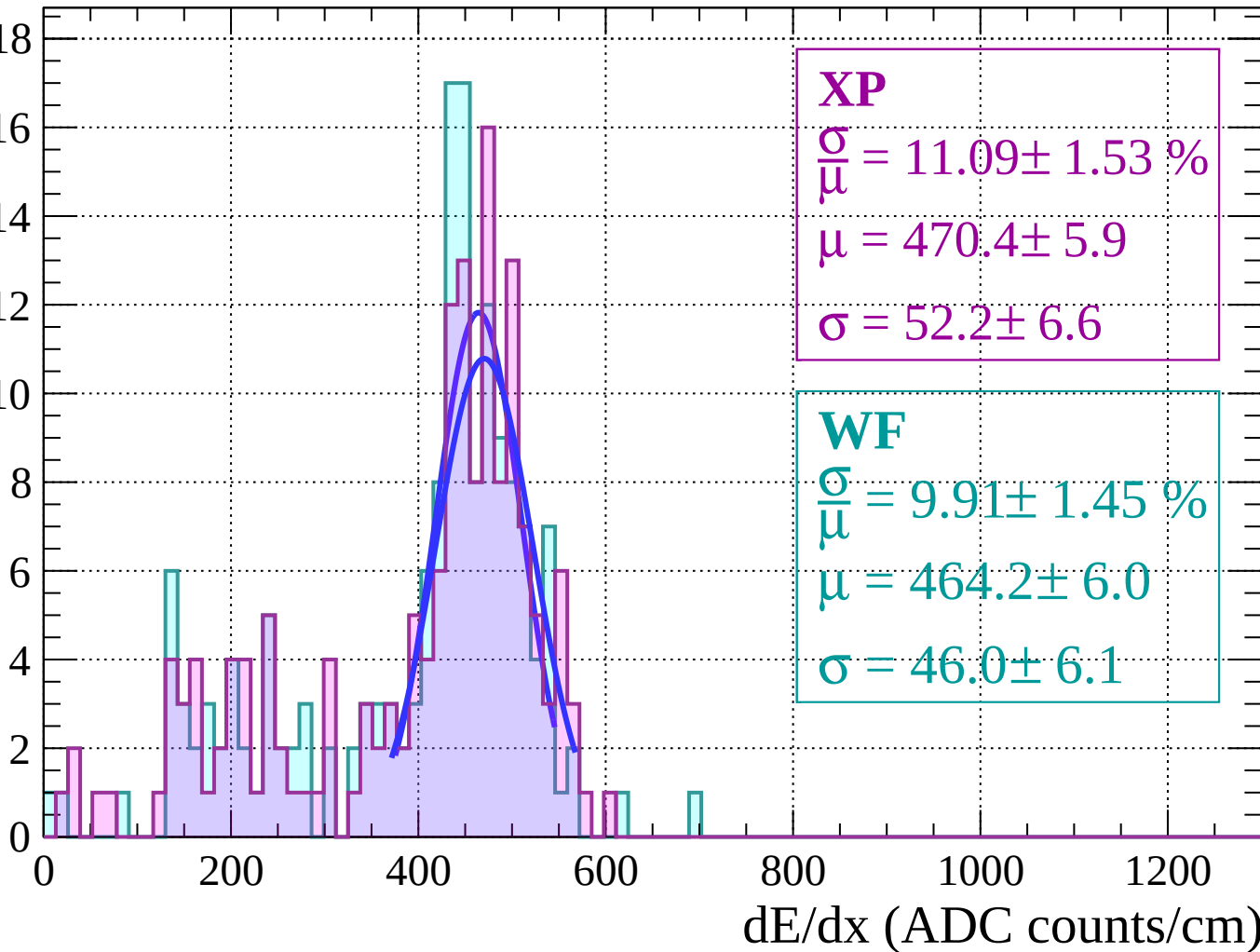
$$\sigma = 52.2 \pm 6.6$$

WF

$$\frac{\sigma}{\mu} = 9.91 \pm 1.45 \%$$

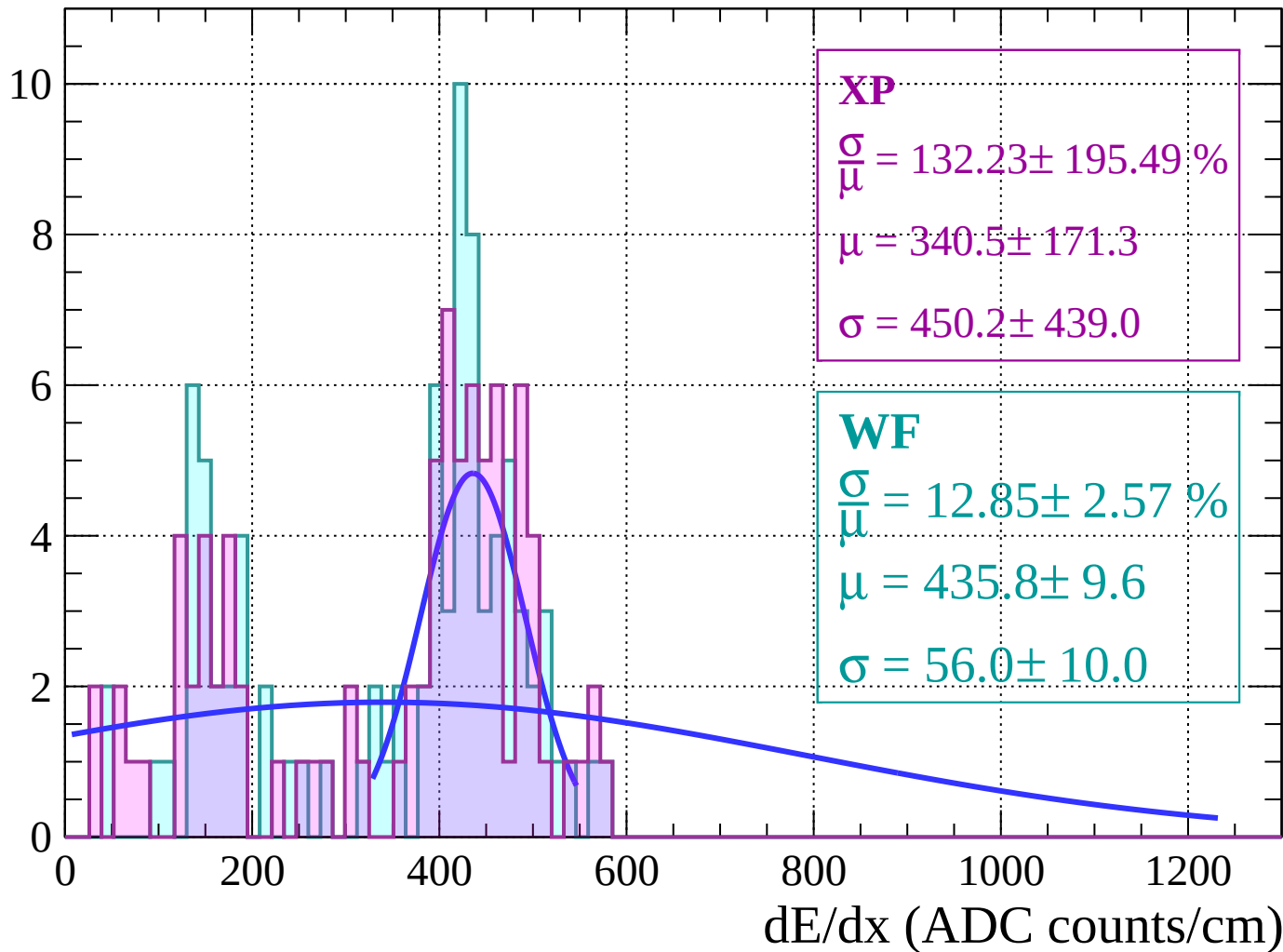
$$\mu = 464.2 \pm 6.0$$

$$\sigma = 46.0 \pm 6.1$$



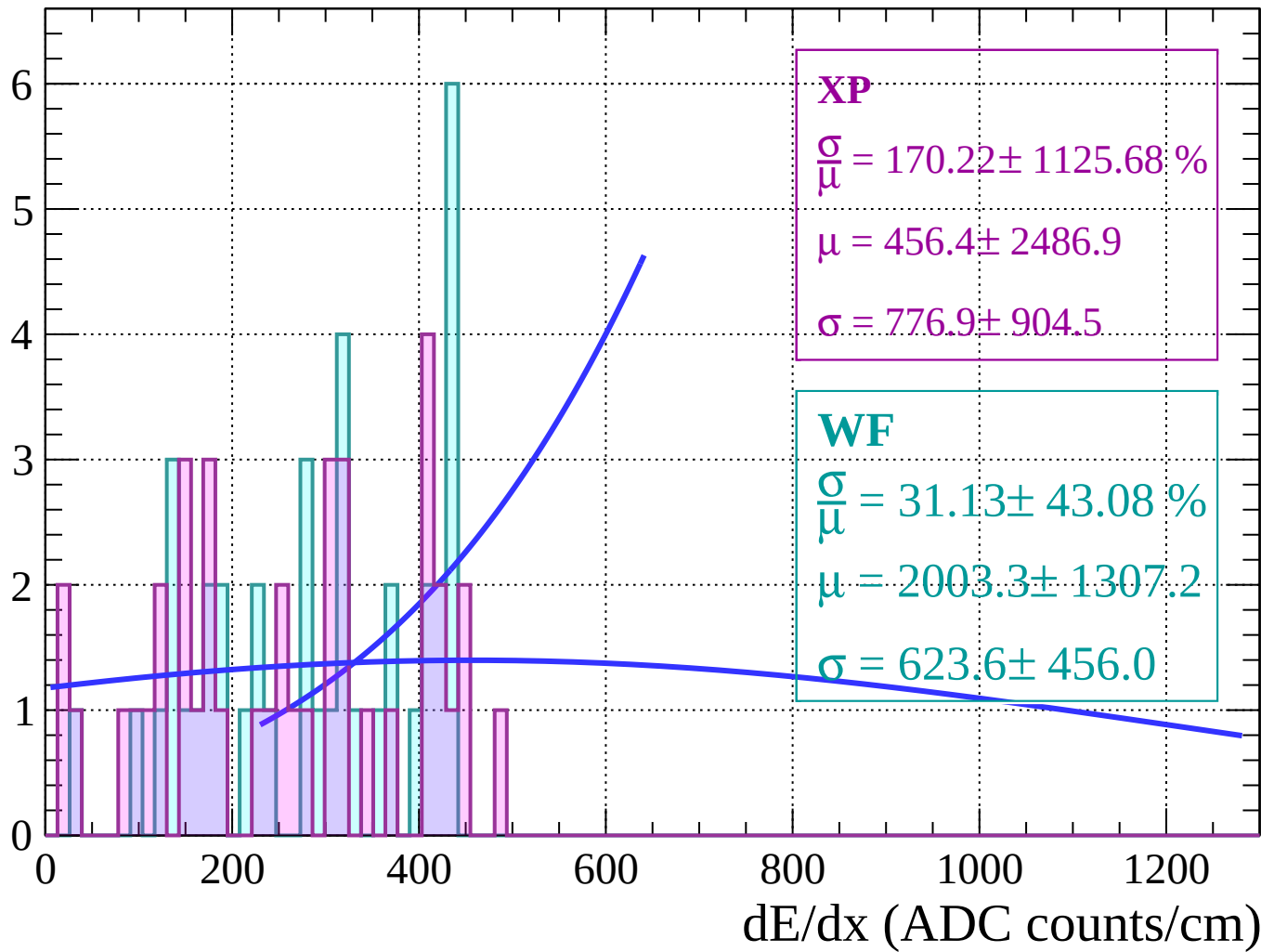
Energy loss | $72 < \theta < 74$

Count



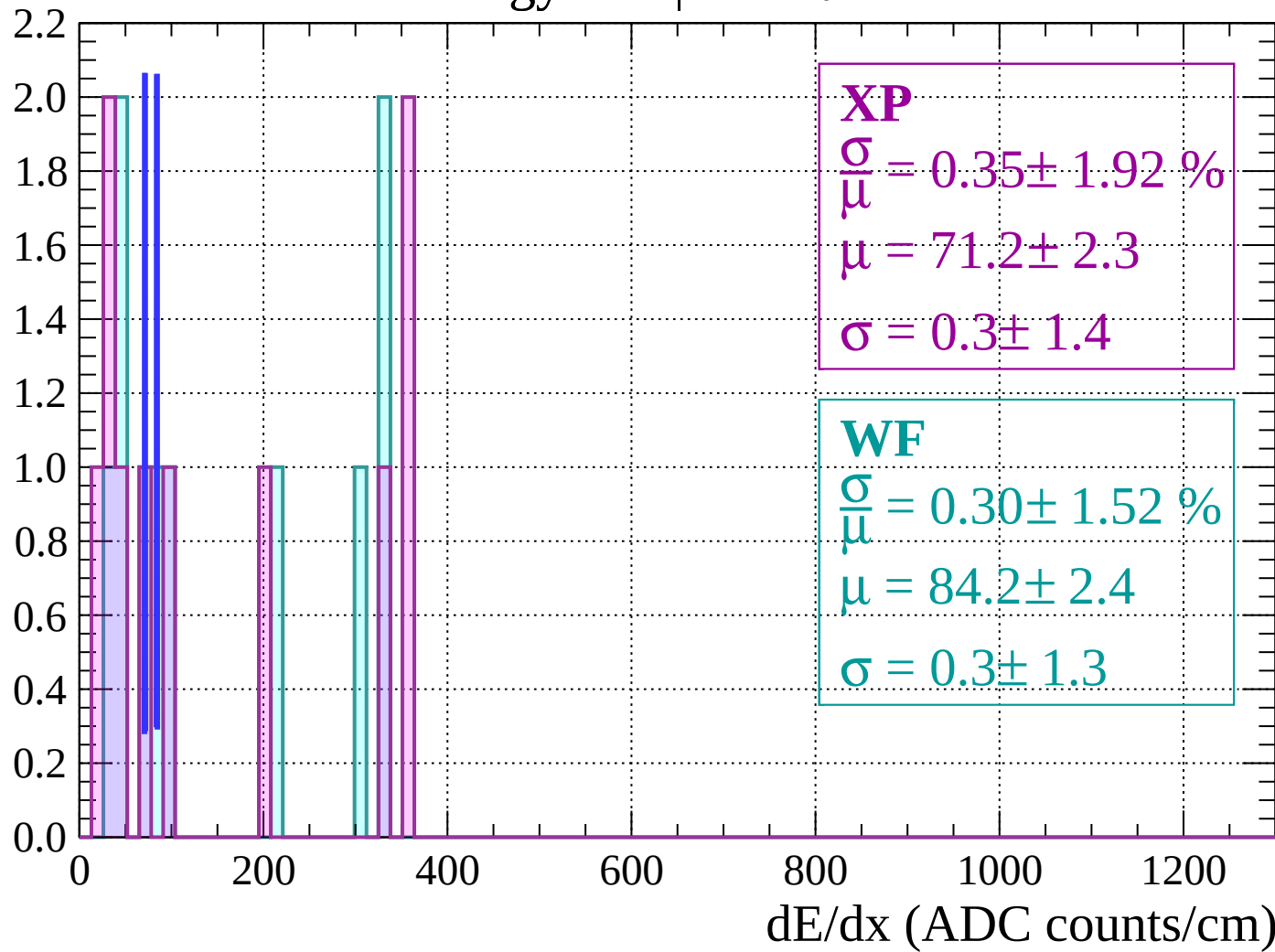
Energy loss | $74 < \theta < 76$

Count



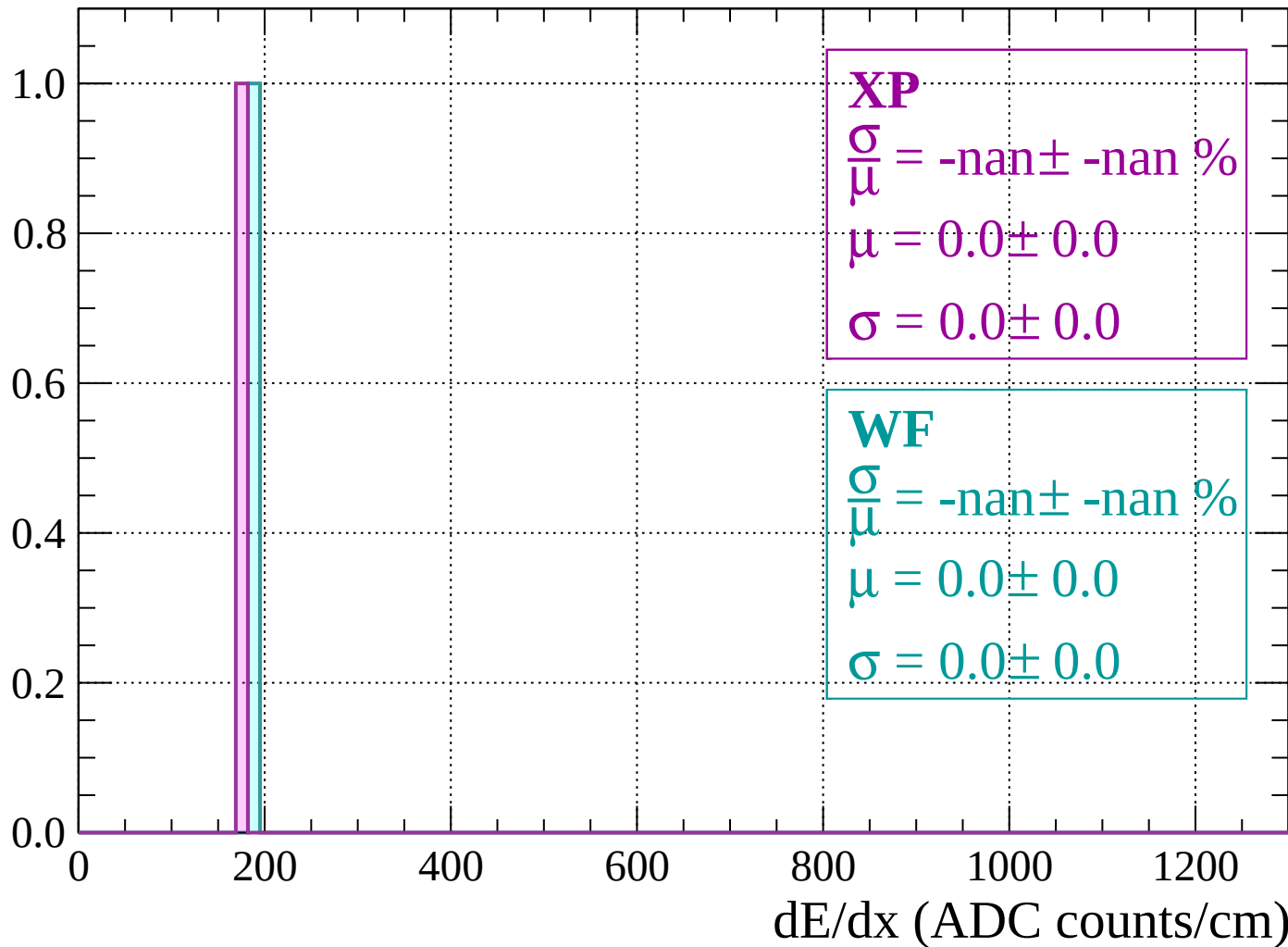
Energy loss | $76 < \theta < 78$

Count



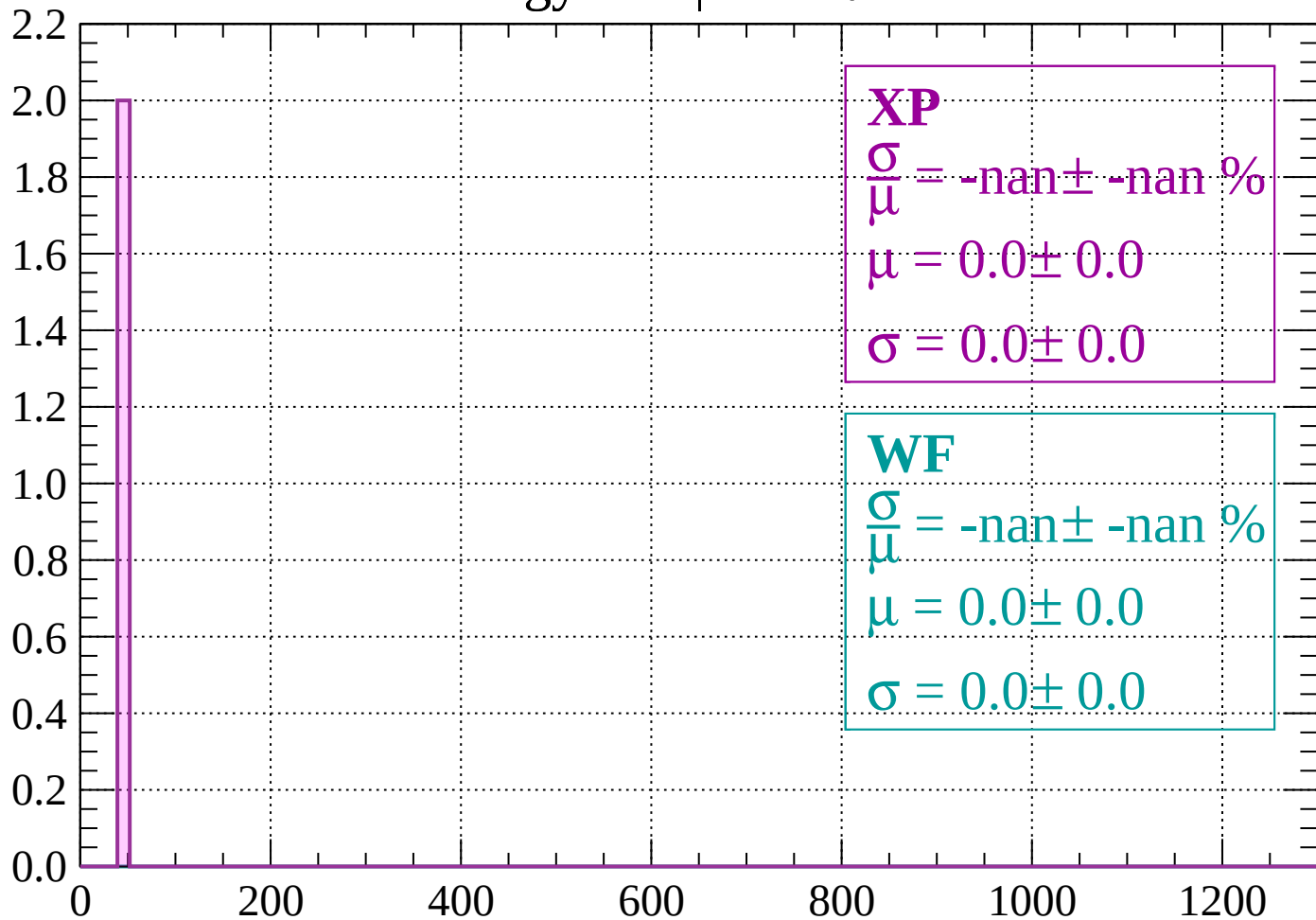
Energy loss | $78 < \theta < 80$

Count



Energy loss | $80 < \theta < 82$

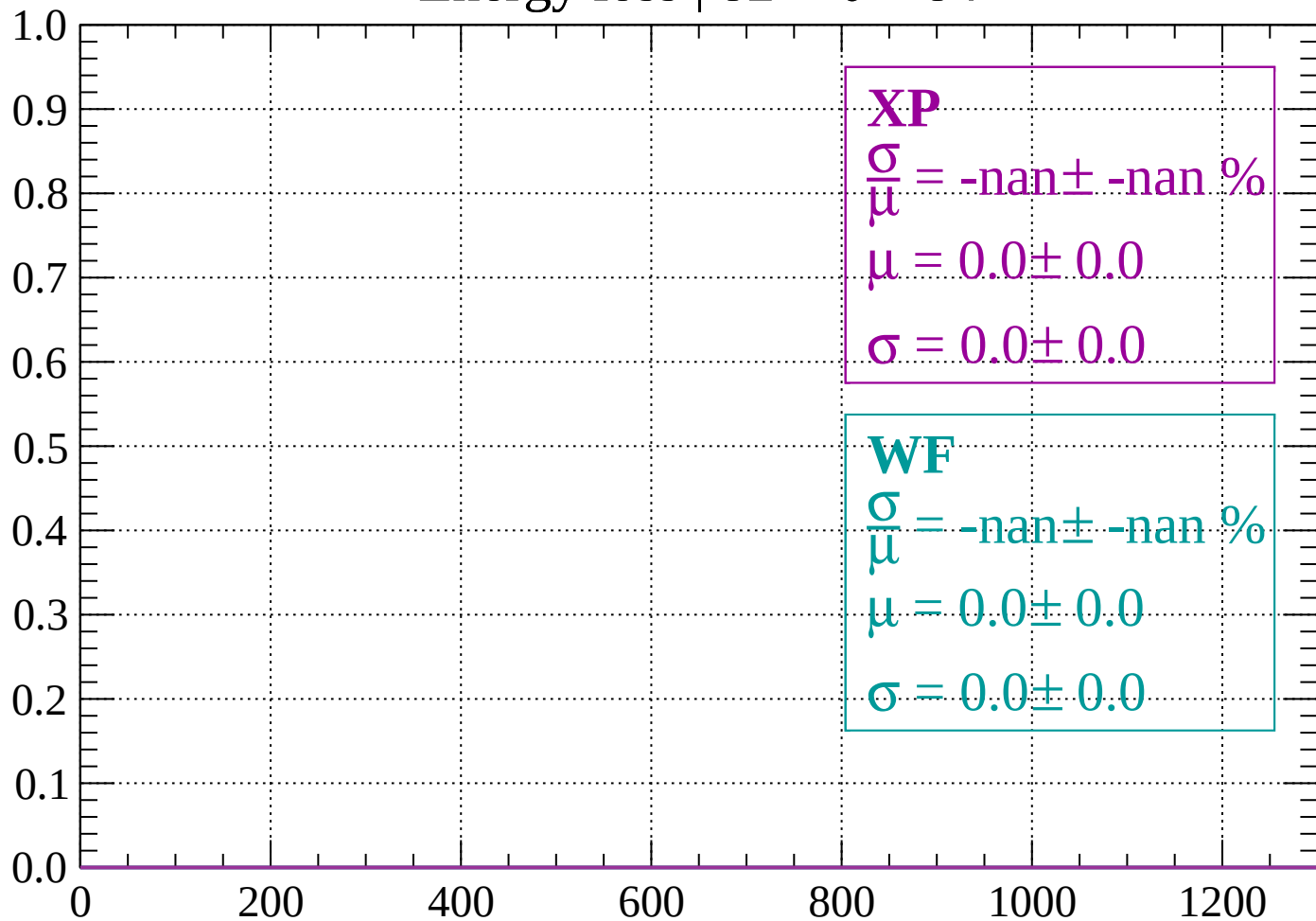
Count



dE/dx (ADC counts/cm)

Energy loss | $82 < \theta < 84$

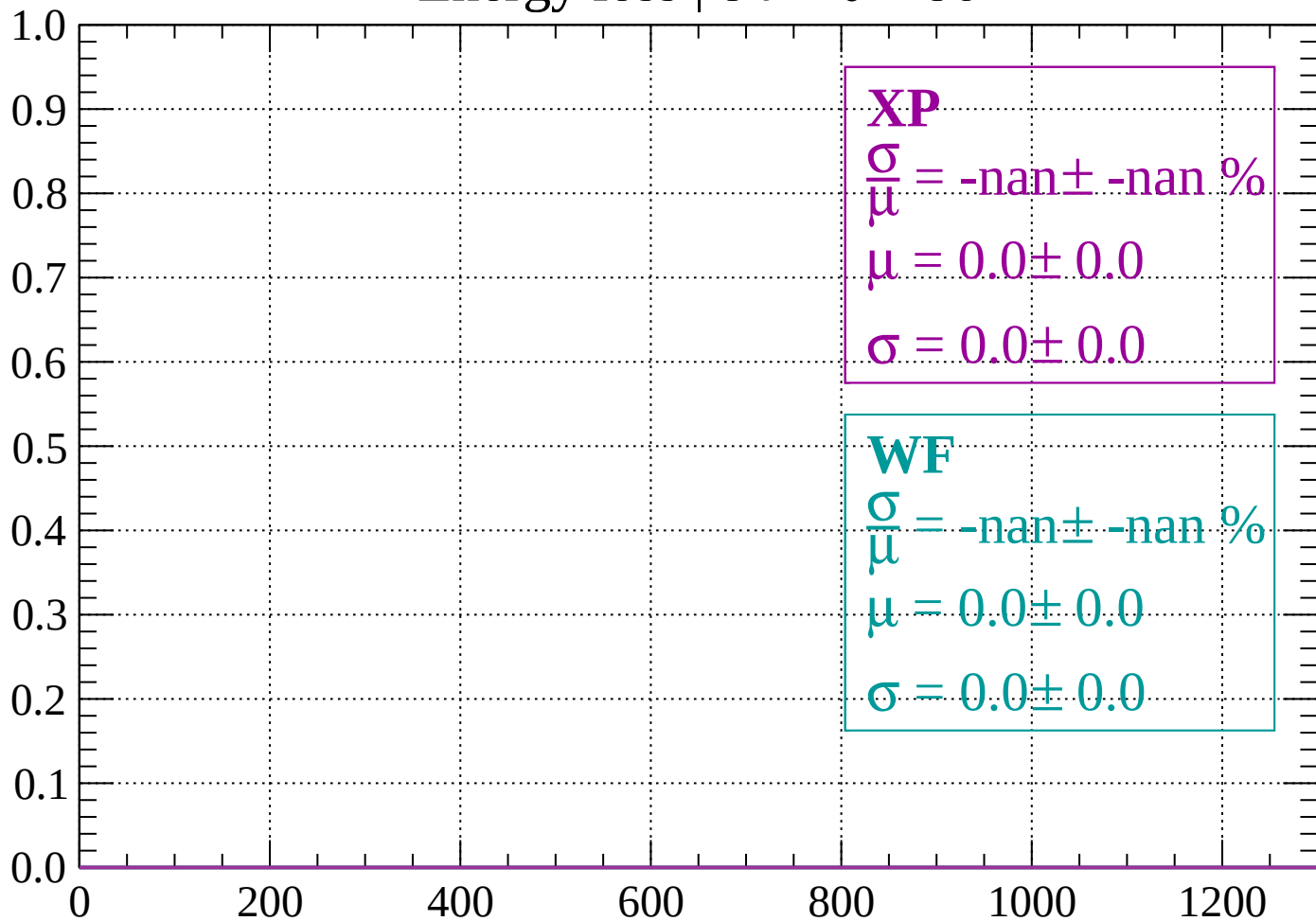
Count



dE/dx (ADC counts/cm)

Energy loss | $84 < \theta < 86$

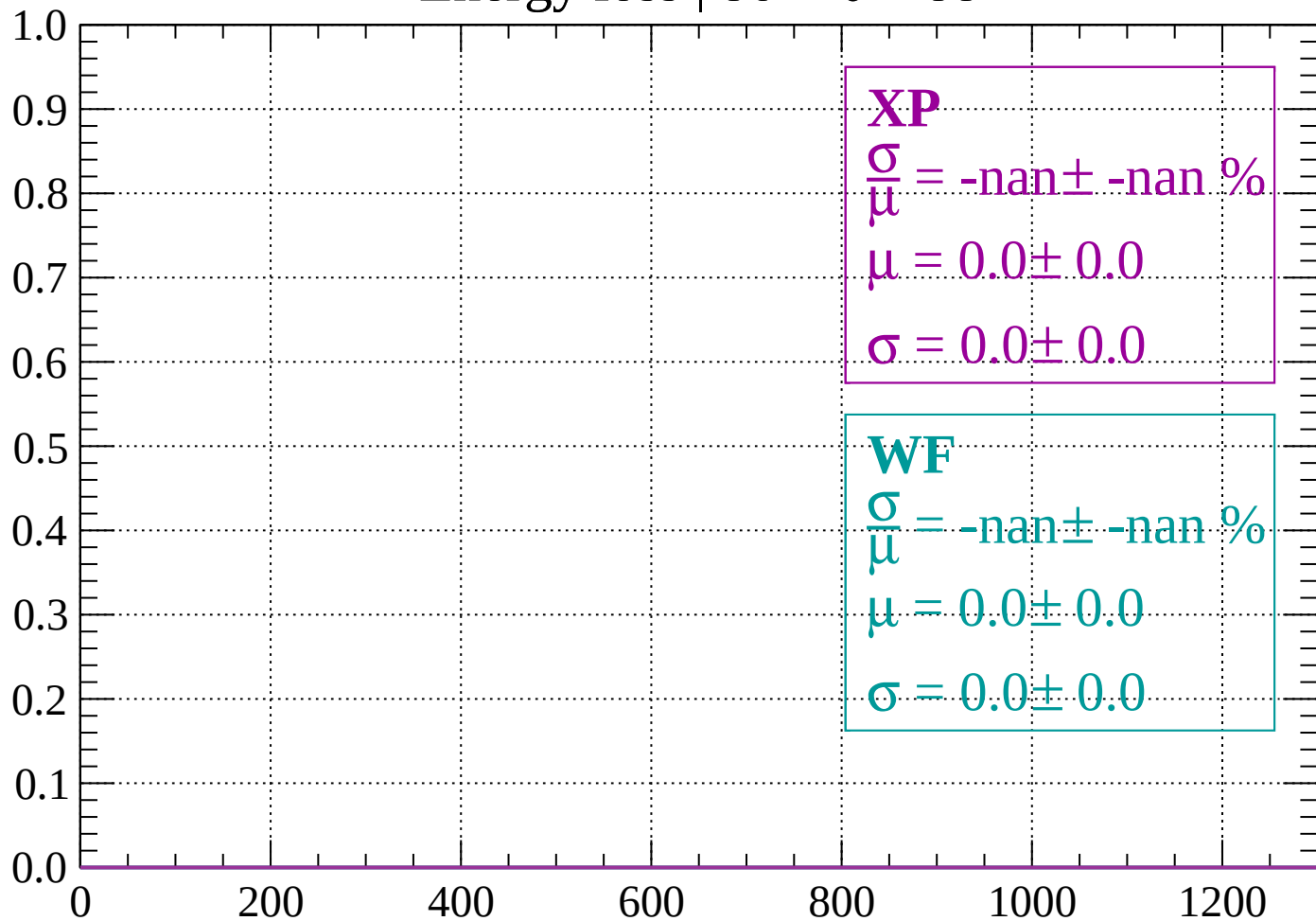
Count



dE/dx (ADC counts/cm)

Energy loss | $86 < \theta < 88$

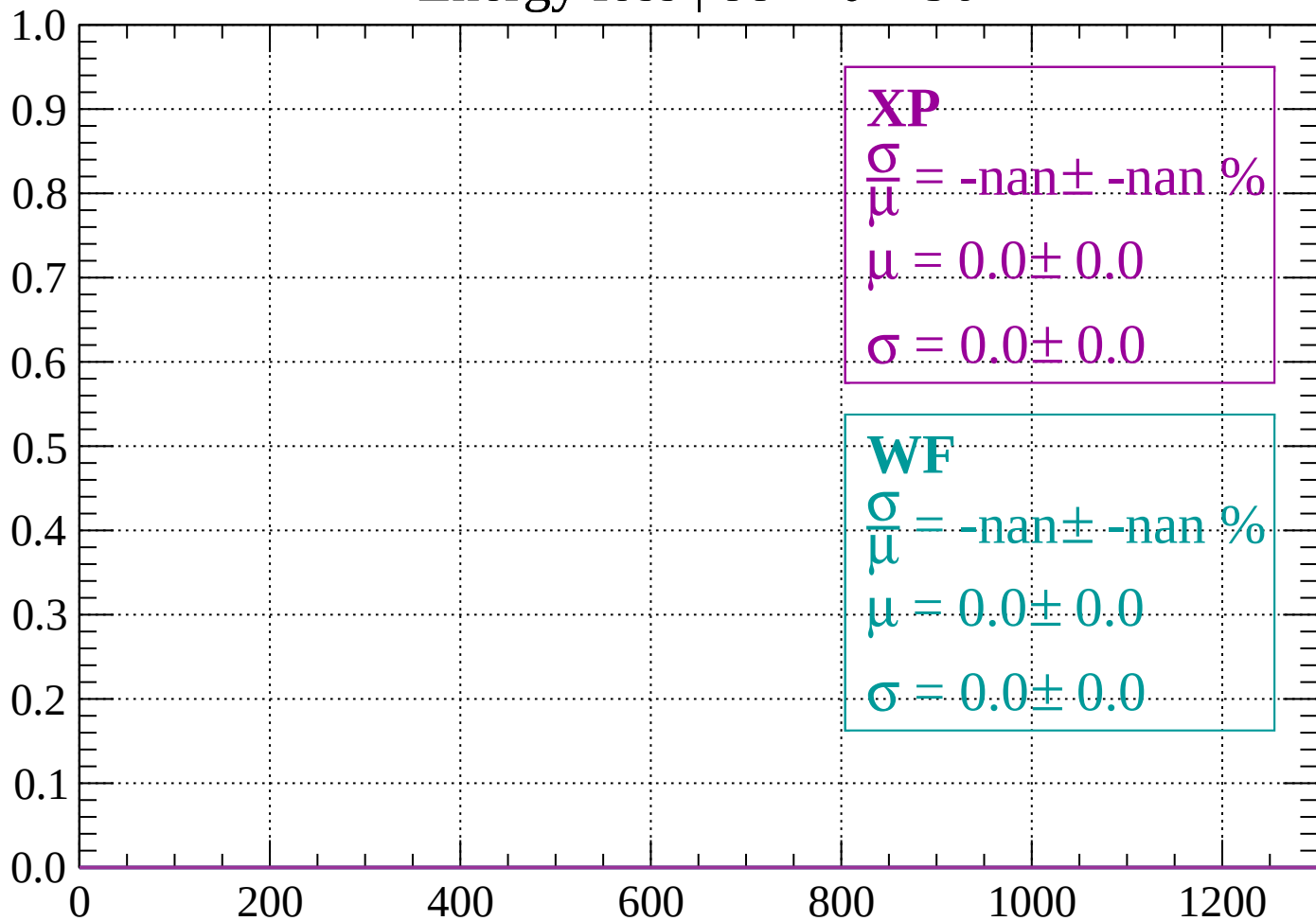
Count



dE/dx (ADC counts/cm)

Energy loss | $88 < \theta < 90$

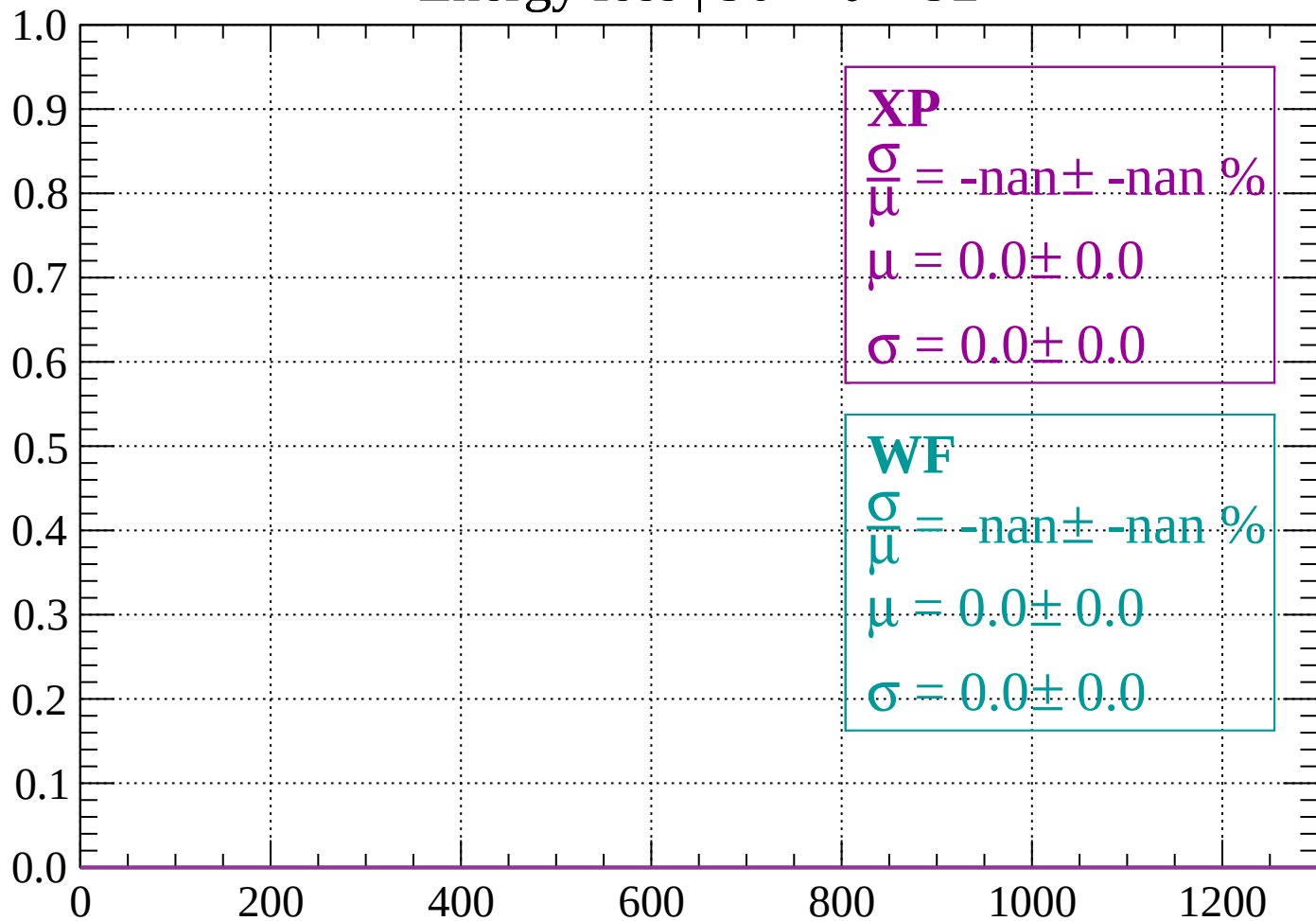
Count



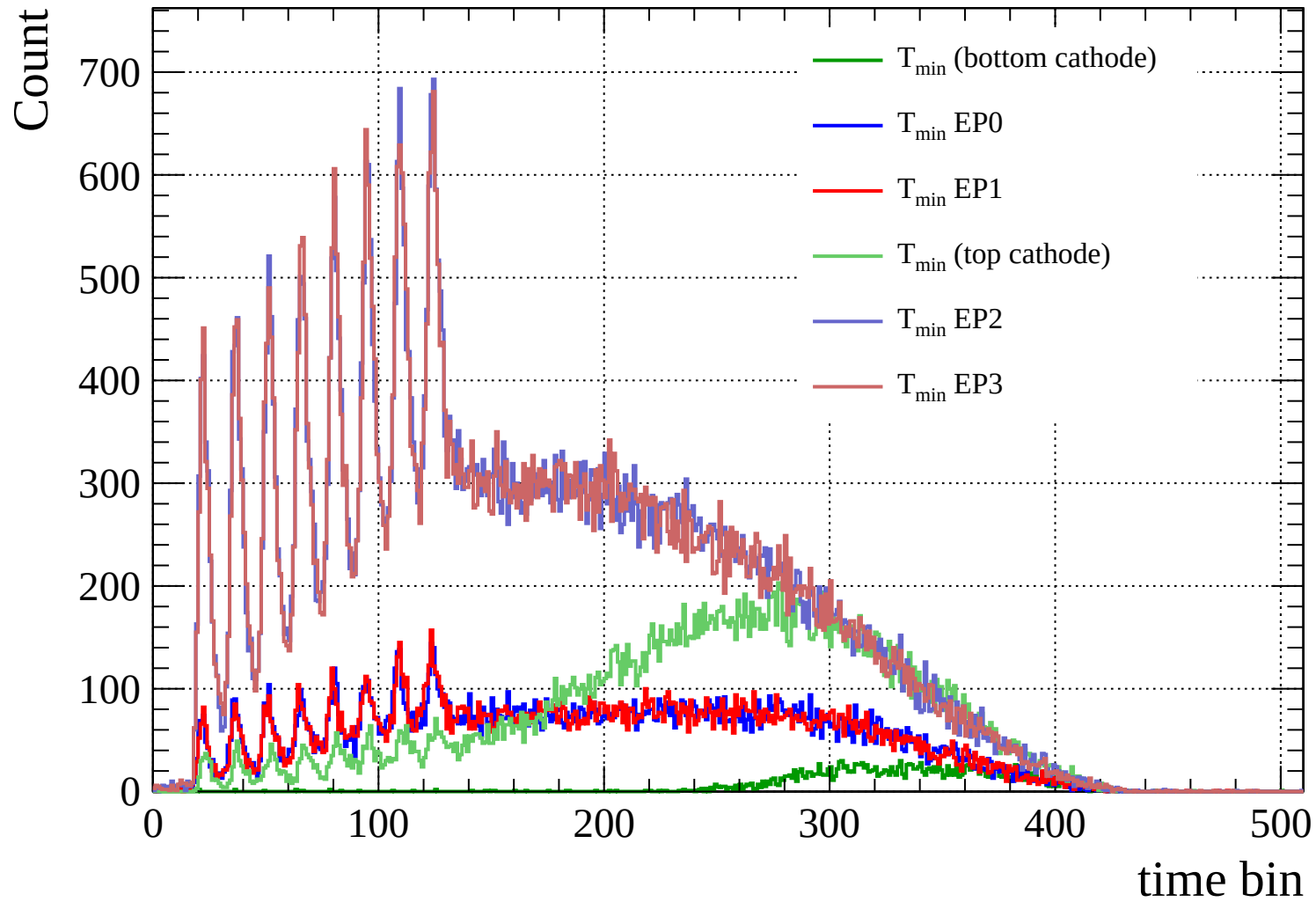
dE/dx (ADC counts/cm)

Energy loss | $90 < \theta < 92$

Count



dE/dx (ADC counts/cm)



Count

